Tempus Menu Reference

Product Version 26.10
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Contents

About This Manual

The Cadence® Tempus™ Timing Solution provides a signoff solution for Timing and Signal Integrity. This manual describes information specific to the forms and commands available for the Tempus graphical user interface.

Audience

This manual is written for experienced designers of digital integrated circuits. Designers must also have a solid understanding of UNIX and Tcl/Tk programming.

How This Manual Is Organized

The chapters in this manual are organized to correspond to the menus in the Tempus graphical user interface (GUI). Each chapter contains one of more of the following sections. A particular section is omitted from a chapter if there is no significant information related to the topic.

Form Overview	Introduces the features available through the various menu selections and provides an illustration of the forms and screens used to set options and run the features.
Forms and Options	Lists and describes the fields and options on each screen.
Related Topics	Provides links to related tasks and concepts in the Encounter User Guide.
Related Commands	Lists the text commands that correspond to the GUI forms for this feature.

Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

`text`	Indicates text that you must type exactly as shown. For example: `analyze_connectivity -analyze all`
`text`	Indicates information for which you must substitute a name or value. In the following example, you must substitute the name of a specific file for *`configfile`*: `wroute -filename` *`configfile`*
text	Indicates the following: ■ Text found in the graphical user interface (GUI), including form names, button labels, and field names ■ Terms that are new to the manual, are the subject of discussion, or need special emphasis ■ Titles of manuals
`[ ]`	Indicates optional arguments. In the following example, you can specify none, one, or both of the bracketed arguments: `command [-arg1] [arg2 value]`
`[ | ]`	Indicates an optional choice from a mutually exclusive list. In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one: `command [arg1 | arg2 | arg3 | arg4]`
`{ | }`	Indicates a required choice from a mutually exclusive list. In the following example, you must specify one, and only one, of the arguments: `command {arg1 | arg2 | arg3}`

`{[ ] [ ]}`	Indicates a required choice of one or more items in a list. In the following example, you must choose one argument from the list, but you can choose more than one: `command {[arg1] [arg2] [arg3]}`
{ }	Indicates curly braces that must be entered with the command syntax. In the following example, you must type the curly braces: `command arg1 {x y}`
`...`	Indicates that you can repeat the previous argument.
. . .	Indicates an omission in an example of computer output or input.
Command – Subcommand	Indicates a command sequence, which shows the order in which you choose commands and subcommands from the GUI menu. In the following example, you choose *Floorplan* from the menu, then *Power Planning* from the submenu, and then *Add Rings* from the displayed list: *Floorplan – Power Planning – Add Rings* This sequence opens the Add Rings form.

Related Documents

For more information about Tempus, see the following documents. You can access these and other Cadence documents with the Cadence Help online documentation system.

- Tempus Known Problems and Solutions

  Describes important Cadence Change Requests (CCRs) for Tempus, including solutions for working around known problems.

- Tempus Text Command Reference

  Describes the Tempus text commands, including syntax and examples.

- Tempus User Guide

  Describes Tempus graphical user interface.

- What's New

Provides information about new features in this release of Tempus.

- Tempus Foundation Flows User Guide

  Describes the Cadence-recommended procedures for performing basic STA and SI signoff, MMMC signoff, and CPF-based MMMC signoff using Cadence® Tempus™ Timing Solution.

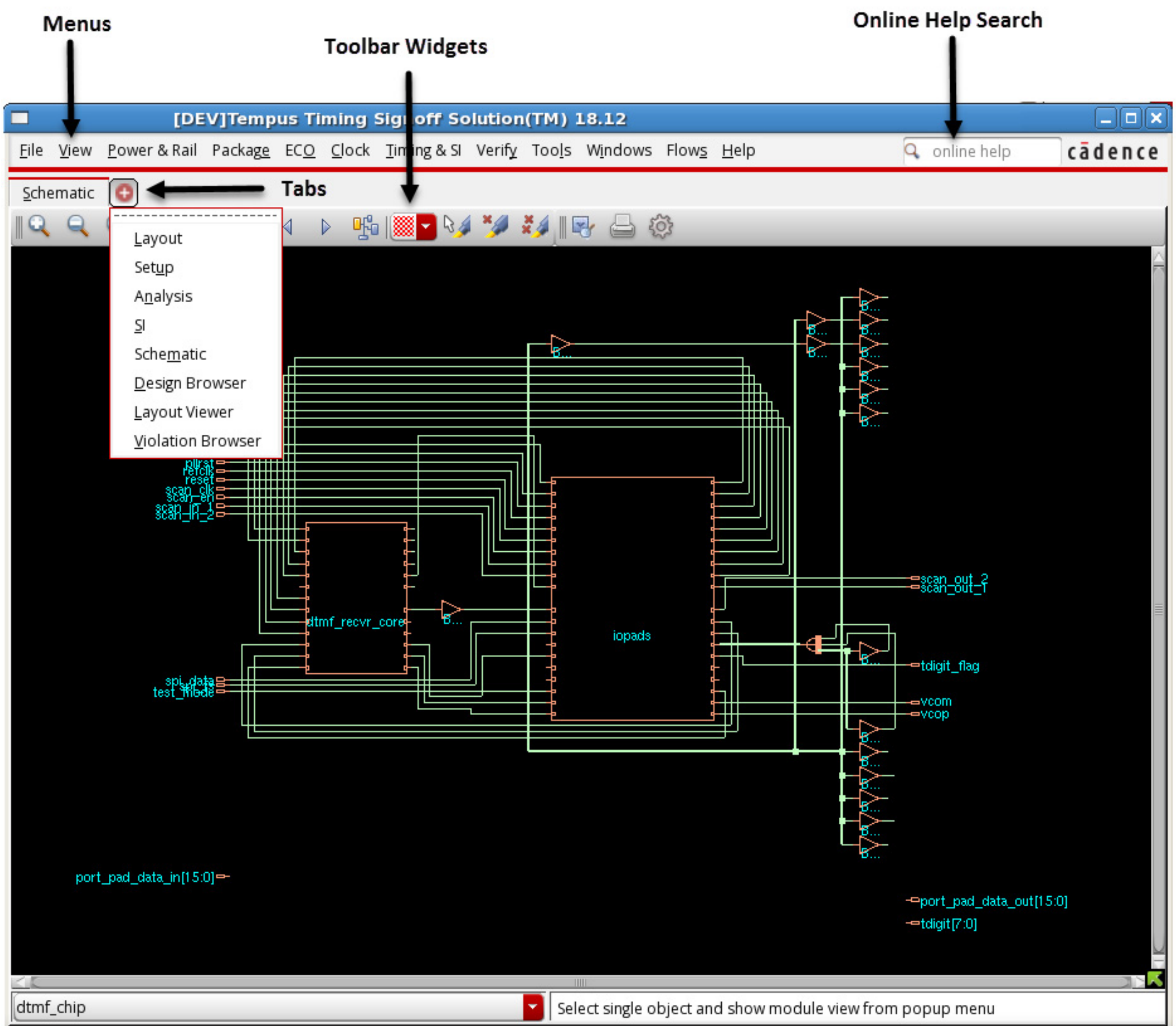
- README file
  Contains installation, compatibility, and other prerequisite information, including a list of Cadence change requests (CCRs) that were resolved in this release. You can read this file online at `downloads.cadence.com`.

1

The Main Window

Main Window Components

The components in the main window are identified in the following figure:



Menus

There are eleven pull-down menus:

- File Menu

- View Menu
- Power and Rail Menu
- Package Menu
- ECO Menu
- Clock Menu
- Timing & SI Menu
- Verify Menu
- Tools Menu
- Windows Menu
- Flows Menu
- Help

You can access the complete library of Tempus documentation through the *Help* menu. The *Help - Documentation Library* link opens the Cadence Help viewer with all the books in the Tempus documentation suite.

You can also view the individual books through their respective links from the *Help* menu. For example, to view the *Tempus Text Command Reference*, select *Help - Text Command Reference*.

Tabs

There are eight tabs:

- Layout Tab
- Setup Tab
- Analysis Tab
- SI Tab
- Schematic Tab
- Design Browser
- Layout Viewer
- Violation Browser

Mouse Operation

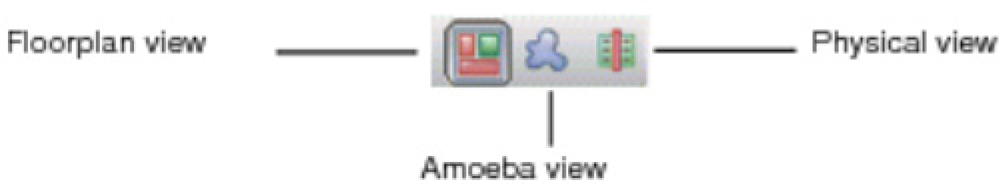
You can use the mouse buttons to perform various actions.

- Left-click the mouse to select a button or check box.
- To select multiple files in a directory, select the first item, then select the last item while holding the `Shift` key.

Design Views

The Tempus main window includes three different design views that you can toggle during a session:

- Floorplan View
- Amoeba View
- Physical View



Selecting a design view also changes the display of the available tool widgets.

Floorplan View

The Floorplan view displays the hierarchical module and block guides, connection flight lines, and floorplan objects, including block placement, and power and ground nets.

Use the `Shift+G` keys or click the *Ungroup* widget to display the submodules for a selected module guide. Each time you use the `Shift+G` keys, you move further down the hierarchy. Use the `G` key or click the *Group* widget to move up the hierarchy.

Amoeba View

The Amoeba view displays the outline of the modules and submodules after placement, showing physical locality of the module.

Use the `Shift + G` keys or click the *Ungroup* widget to display the submodules for a selected module guide. Each time you use the `Shift+G` keys, you move further down the hierarchy. Use the `G` key or click the *Group* widget to move up the hierarchy.

Physical View

The Physical view displays the detailed placements of the module's blocks, standard cells, nets, and interconnects. You can move standard cells, blocks, and power and ground objects in the Physical and Floorplan views.

Viewing Floorplan Objects in Physical View

You can view the floorplan objects even in the physical view. To enable this feature:

1. Select *View* - *Set Preference*.
2. Select the Display page in the *Preferences* form.
3. Select the option, *Show Floorplan Objects in Physical View*.

You can use this feature together with color groups to create a custom view. For example, you can create a color group that has a custom setting for block colors and use this color group while viewing floorplan objects in the physical view.

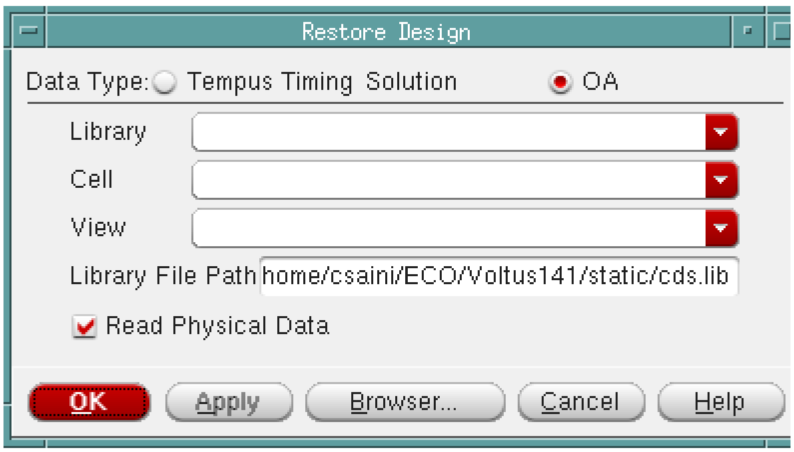
2

File Menu

Read Design

Use the *Restore Design* form to read design data.

- Choose *File - Read Design*.



The *Restore Design* form has two options:

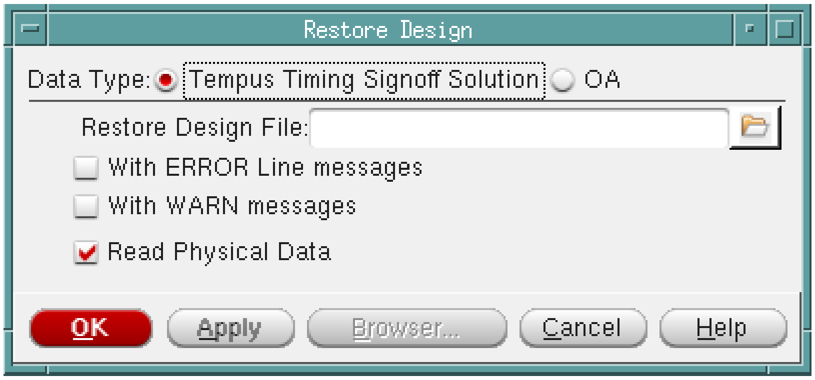
- Tempus Timing Solution
- OA

These are explained below.

- Restore Design - Tempus Timing Solution
- Restore Design - OA

Restore Design - Tempus Timing Solution

Use the *Restore Design* form to load saved data from a previous design session.



Restore Design - Fields and Options

Restore Design File	Specifies the name of the design file to load.
With ERROR Line messages	Reads the file, including error messages.
With WARN Line messages	Reads the file, including warnings.
Read Physical Data	Reads physical data.

Related Text Commands

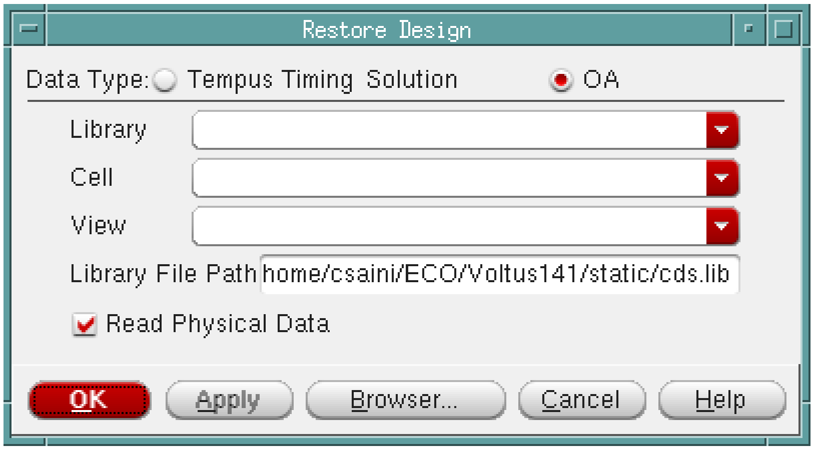
The following text command provides equivalent or additional functionality:

- `read_design`

For more information, see the "Design Import Commands" in *Tempus Text Command Reference*.

Restore Design - OA

The OA option within the Restore Design form is used to restore all the design information from the previous design session, using the OpenAccess database format.



Restore Design - OA Fields and Options

Library	Specifies the location of the OpenAccess database to be restored. It contains the cell and the view to be restored. The *Setup* tab contains the *Verilog* and *OA* radio buttons that changes the form from the current LEF based style (default) to the OpenAccess-based style.
Cell	Specifies the design cell from the library that is to be restored.
View	Specifies the view name of the cell that is to be restored. *Default:* The layout view.
Library File Path	Specifies the full path to the library definitions file (cds.lib).
Read Physical Data	Reads physical data.
Browser button	Opens the *Library Browser* form, which enables you to browse available libraries, cells, or views.

Save Design

The *Save Design* form enables you to save design files from a Tempus session or save all the design information using the OpenAccess (OA) database format.

The *Save Design* form contains the following two options:

- Tempus Timing Solution
- OA

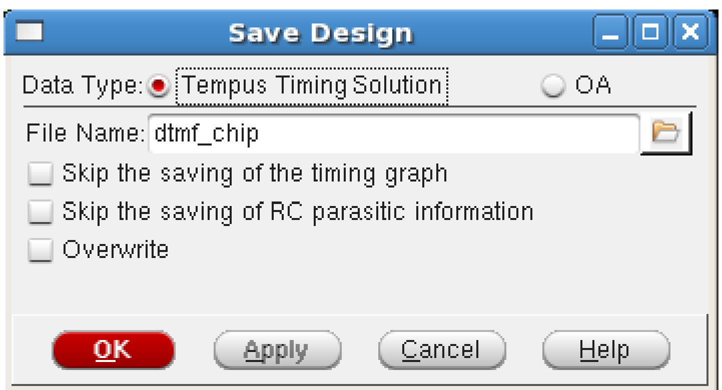
These are explained below.

- Save Design - Tempus Timing Solution
- Save Design - OA

Save Design - Tempus Timing Solution

Use the Tempus Timing Solution option of the *Save Design* form to save design files in a user-specified directory in a design session. Only files that exist during the save are copied to the directory.

† Choose *File - Save Design* and click the *Tempus Timing Solution* option button.



Save Design - Fields and Options

File Name	Enter the name of the file to which you want to save the design. Click the button next to the field to choose the directory where you can save your file. You can select one of more of the sub-options to skip saving of timing graph, RC parasitic information, or to overwrite the previous file.

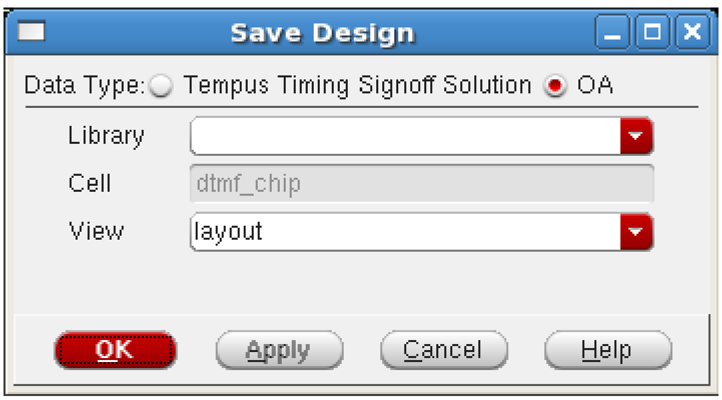
Related Text Commands

- save_design

Save Design - OA

Use the OA option in the *Save Design* form to save all the information from the current design session using the OA database format.

☨ Choose *File* - *Save Design*. The OA window opens by default.



Save Design - OA Fields and Options

Library	Specifies the target OpenAccess library name.
Cell	Specifies the top cell to be saved. This field is read-only.
View	Specifies the view to save.

Note:

- The OA Reference Libraries and OA Abstract View Names fields from the *File* - *Import Design* form are used when saving the design for the first time. If the design is loaded using *File* - *Restore Design* - *OA*, then the reference libraries and abstract names specified in the design are used.
- While saving a library, cell, or view to a different cell view, the overwrite is not allowed by default. So, you will be prompted to overwrite.

Related Text Commands

- save_design

Load Timing Constraints

Use the Load Timing Constraints form to read a SDC file.

- Choose *File - Read SDC*.

Fields and Options

Timing Constraint File	Specifies the name of the timing constraints (SDC) file.

Related Text Commands

The following text command provides equivalent or additional functionality:

- read_sdc

For more information, see the "Design Import Commands" in *Tempus Text Command Reference*.

Load SPEF File

Use the Load Spef File to read a SPEF file.

- Choose *File - Read SPEF.*

Fields and Options

SPEF File	Specifies the type of files that are displayed. *Default:* `SPEF files`
RC Corner	Specify the name of the corresponding RC Corner.

Related Text Commands

The following text command provides equivalent or additional functionality:

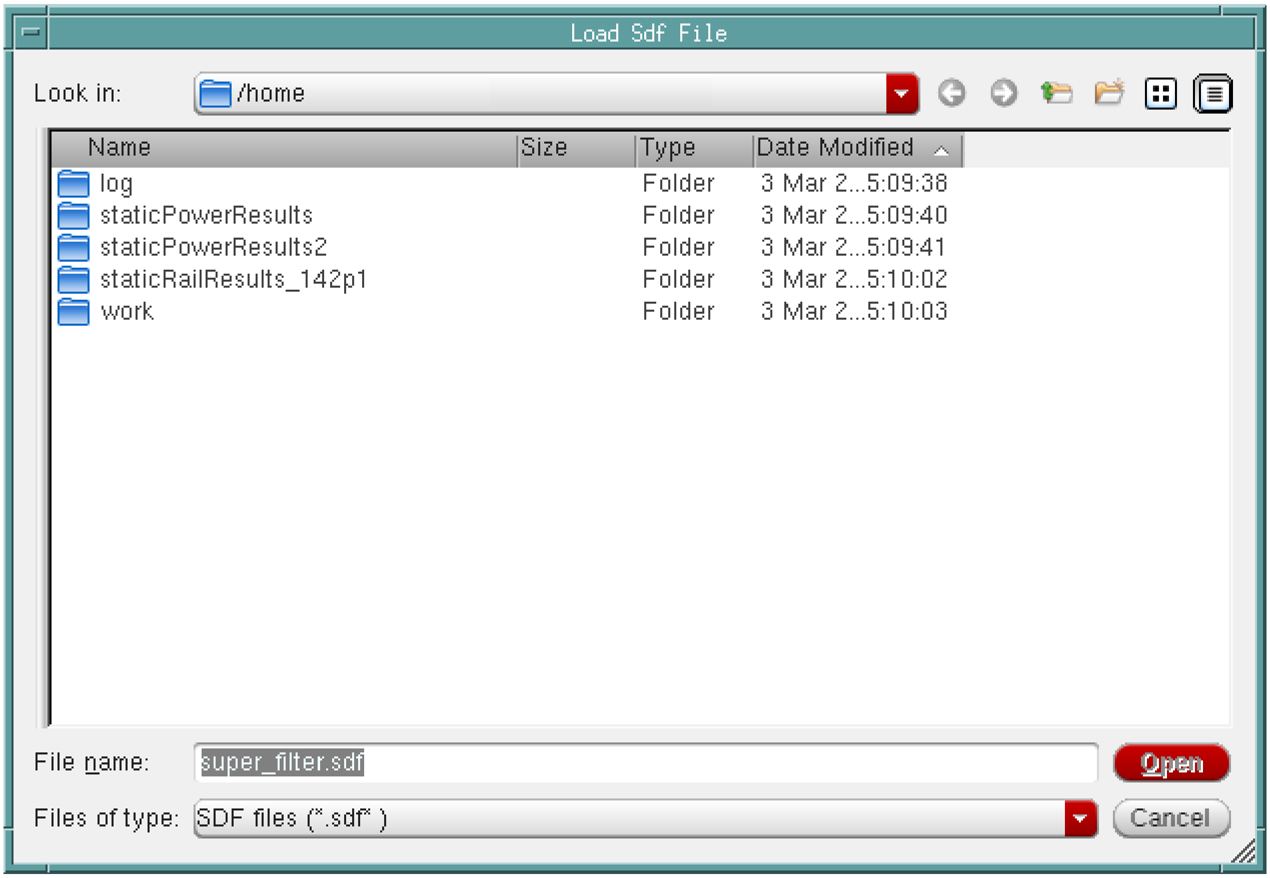
- `read_spef`

For more information, see Timing Analysis Commands in the *Tempus Text Command Reference*.

Load SDF File

Use the Load Sdf File form to read a SDF file.

☨ Choose *File - Read SDF*.



Fields and Options

Directory	Displays the current directory.
File name	Specifies the name of the SDF file.
Files of type	Specifies the type of files that are displayed. *Default:* `SDF files`

Related Text Commands

The following command provides equivalent or additional functionality:

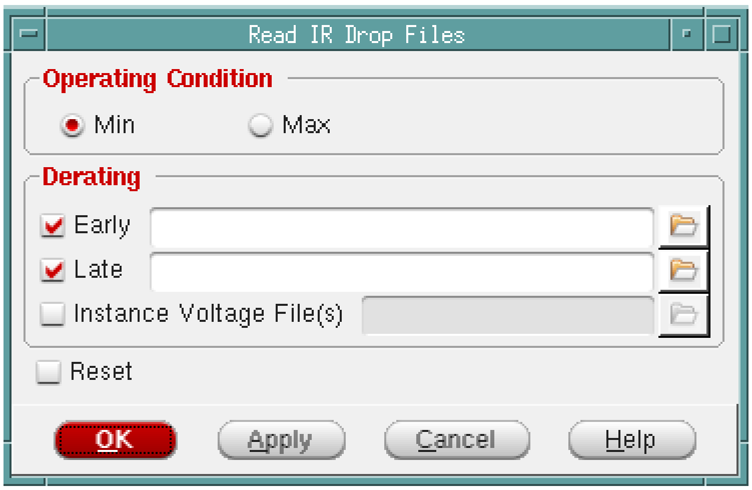
- `read_sdf`

For more information, see Timing Analysis Commands in the *Tempus Text Command Reference*.

Read IR Drop Files

Use the Read IR Drop Files form to read an IR drop file.

- Choose *File - Read IR Drop*.



Fields and Options

Operating Condition	Specify either Min or Max operating condition.
Derating	Applies the effects of IR-drop to the early path (data hold/clock setup) or late path (data setup/clock hold) in your design. Use the *Early* or *Late* parameter in conjunction with the *Min* or *Max* condition to account for the effects of early-max, early-min, late-max, or late-min IR-drop analysis.
Reset	Resets all IR-drop files, so a new one can be read.

Related Text Commands

The following text command provides equivalent or additional functionality:

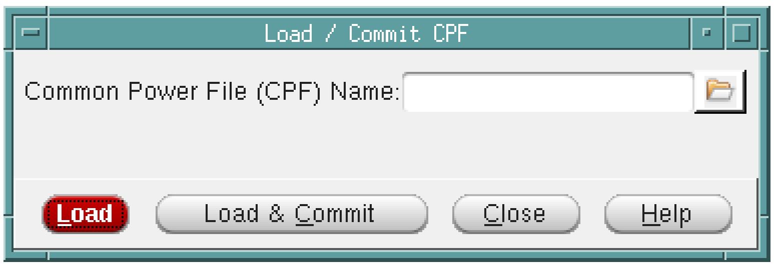
- `read_instance_voltage`

For more information, see the "Design Import Commands" in *Tempus Text Command Reference*.

Load/Commit CPF

Use the Load/Commit CPF form to load a Common Power Format (CPF) form into the design, or commit a CPF file that you have already loaded.

To open the *Load/Commit CPF* form, choose *File* - *Load/Commit CPF.*



Fields and Options

Common Power File (CPF) Name	Specifies the name of the CPF file to load or commit.

Related Text Commands

The following text command provides equivalent or additional functionality:

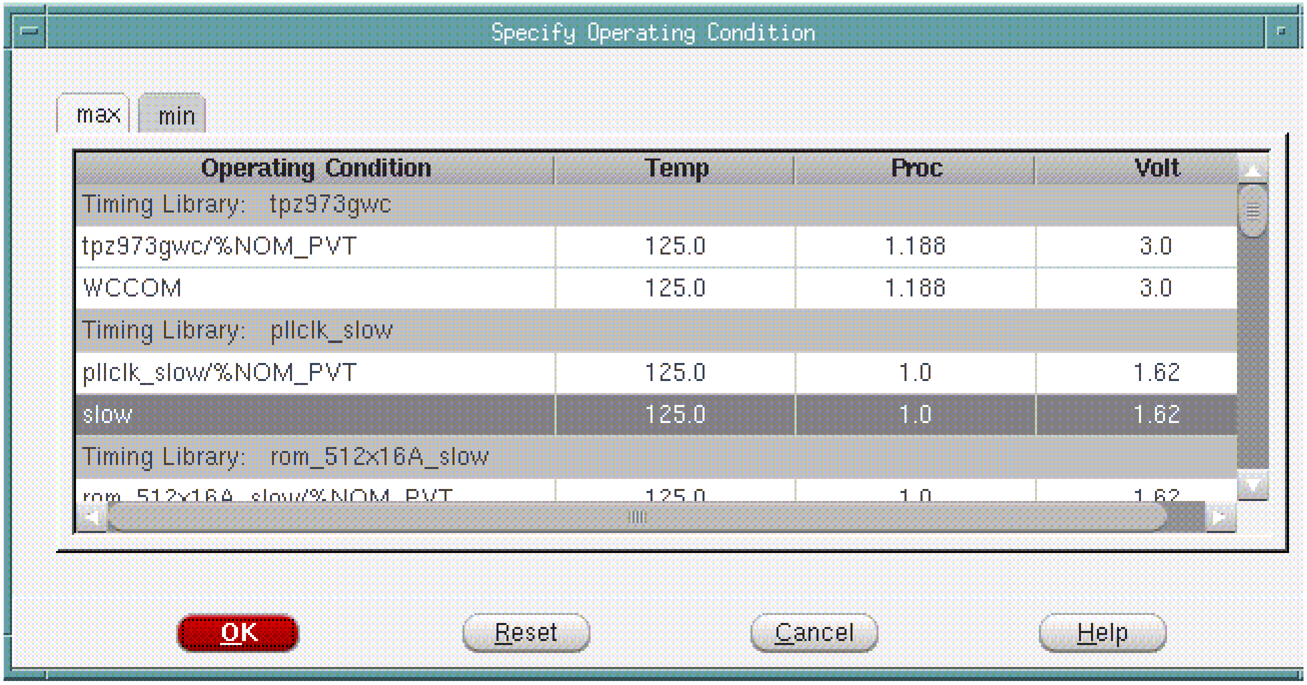
- `read_power_domain`

For more information, see the "Design Import Commands" in *Tempus Text Command Reference*.

Specify Operating Condition/PVT

Use the Specify Operating Condition form to select the operating temperature, process, or voltage conditions for the design. The operating conditions are contained in the timing library, which are normally read in during design import.

- Choose *File - Specify Operating Conditions / PVT* and click on either the Min or Max tab to select the operating condition you want to specify.



Fields and Options

Operating Condition	Displays the name of the operating condition. By default, the operating condition selected under the *Max* tab is used for setup analysis and the operating condition selected under the *Min* tab is used for hold analysis.
Temp	Displays either the minimum or maximum temperature for the operating condition.
Proc	Displays either the minimum or maximum process parameter for the operating condition.
Volt	Displays either the minimum or maximum voltage for the operating condition.

Related Text Commands

The following text commands provide equivalent or additional functionality:

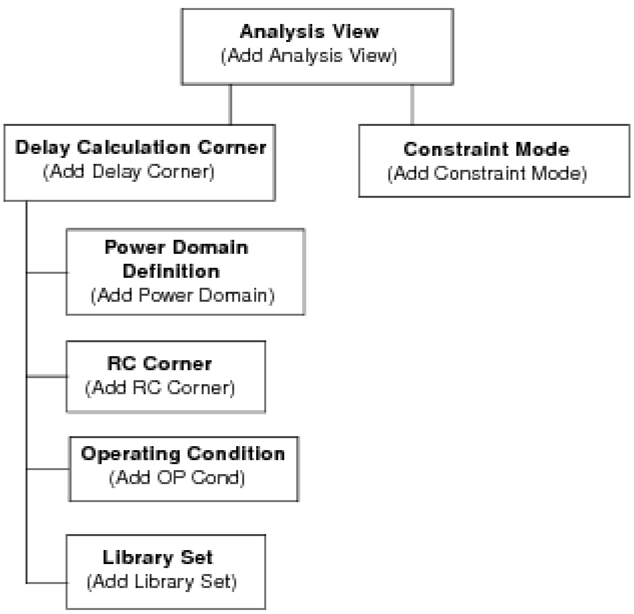
- `set_op_cond`

- set_timing_library

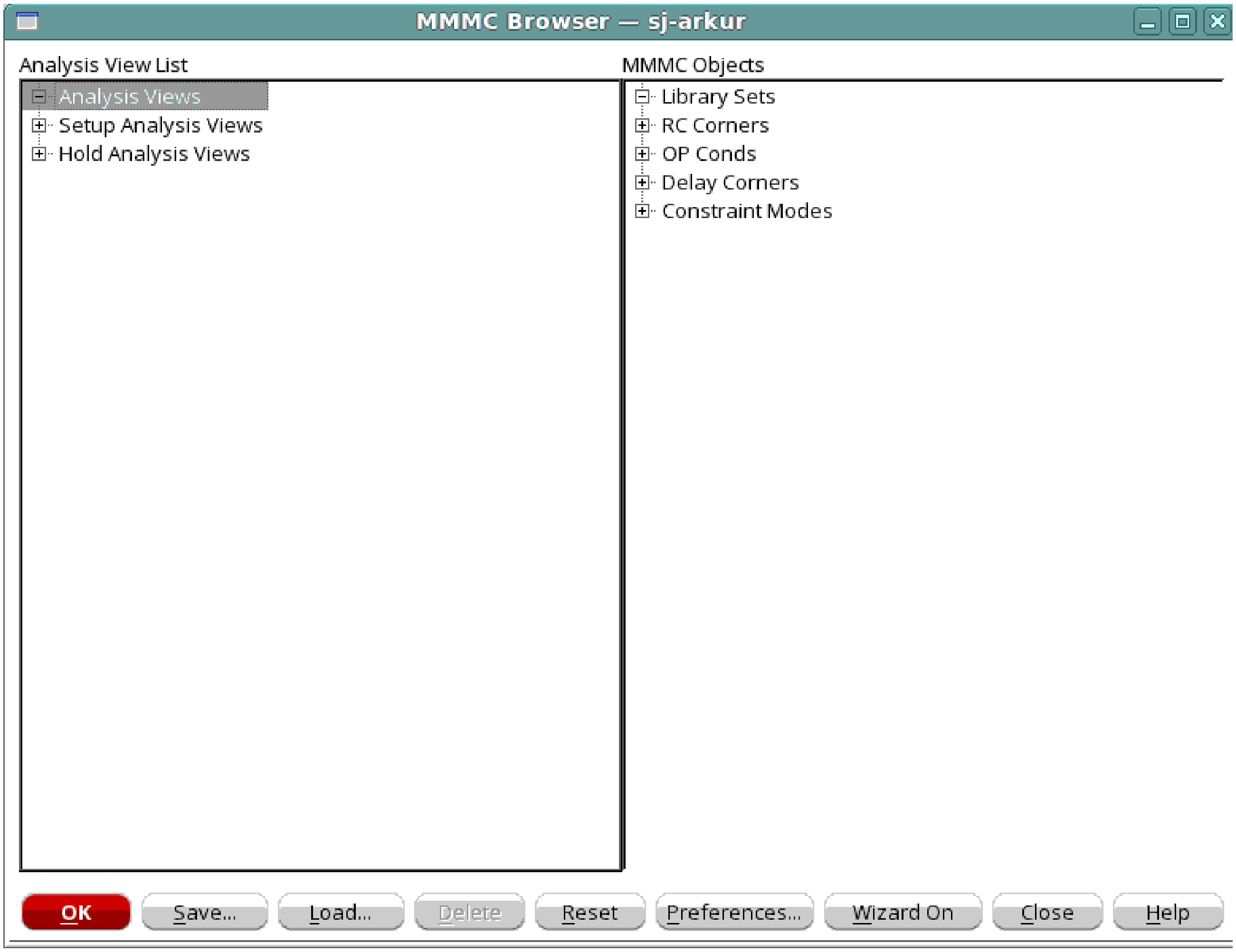
For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

MMMC Browser

Use the MMMC Browser to hierarchically create the view configurations necessary for multi-mode multi-corner analysis. Each top-level set of information is called an analysis view, and is composed of a delay calculation corner and a constraint mode. The active analysis views defined in the software represent the different design variations that will be analyzed.



✝ Choose *File - Configure MMMC*.



You can access the following forms from the MMMC Browser by double clicking on the generic object or analysis view name:

- Add Library Set
- Add OP Cond
- Add RC Corner
- Add Constraint Mode
- Add Delay Corner
- Add Power Domain
- Add Analysis View

- Add Hold Analysis View
- Add Setup Analysis View

You also can access the following editing forms from the MMMC Browser by double clicking on a specific object name:

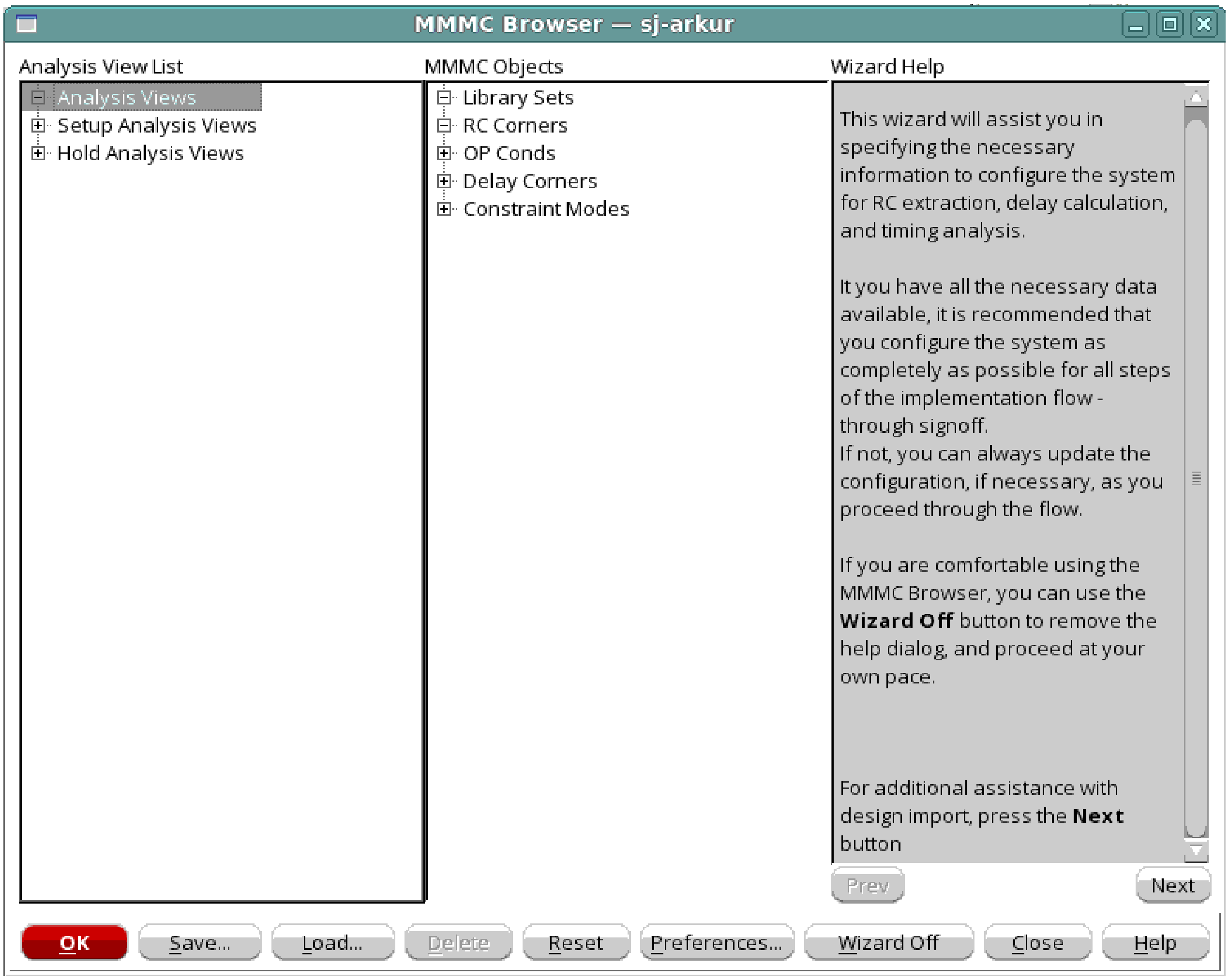
- Edit Library Set
- Edit OP Cond
- Edit RC Corner
- Edit Constraint Mode
- Edit Delay Corner
- Edit Power Domain
- Edit Analysis View

Additionally, you can access the following forms to change how the configuration is displayed and also view help on each of the options in the MMMC Browser:

- MMMC Preferences
- Wizard Help

Wizard Help

By default, the MMMC browser opens in Wizard On mode. Use the *Wizard Help* pane to view details on how to configure MMMC.



The wizard help will guide you through each of the recommended settings in the order shown below.

- Timing library
- RC extraction
- Operating conditions
- Delay corners
- Constraint modes

- Analysis views
- Setup and Hold View

For more information on configuring the MMMC settings and generating a MMMC view definition file, see MMMC Configuration.

MMMC Configuration

The MMMC environment is configured and assembled hierarchically. You must specify the lower-level information first, so that it can be referenced later on in the process.

The recommended configuration order is:

- Specifying timing related library information (Add Library Set)
- Specifying RC extraction corners (Add RC Corner)
- Specifying operating conditions (Add OP Cond)
- Specifying delay corners (Add Delay Corner and Add Power Domain)
- Specifying constraint modes (Add Constraint Mode)
- Creating analysis views (Add Analysis View)
- Selecting views for Setup (Add Setup Analysis View) and Hold analysis (Add Hold Analysis View)

The Wizard Help pane in the MMMC Browser takes you through each of these required steps in the order shown above. When you have finished configuring the analysis environment, click the *Save* button to save your configuration in a file and return to the main Design Import form. When the Design Import form is executed, the system will load the MMMC configuration for the active Setup and Hold views that you have specified.

You can also access the following editing forms from the MMMC Browser by double-clicking on an object name:

- Edit Library Set
- Edit OP Cond
- Edit RC Corner
- Edit Constraint Mode
- Edit Delay Corner

- Edit Power Domain
- Edit Analysis View

Additionally, you can access the following form to change how the configuration is displayed in the MMMC Browser:

- MMMC Preferences

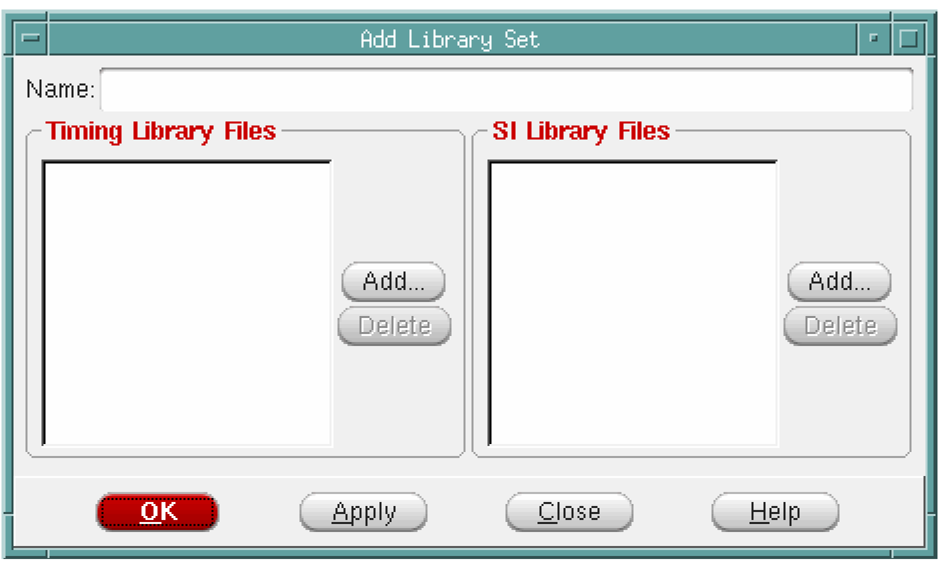
Add Library Set

Use the Add Library Set form to associate a list of timing and signal integrity (cdB) libraries with a library set name. The library sets allow a group of library files to be treated as a single entity so that higher-level descriptions (delay calculation corners) can simply refer to the library configuration by name. The same library set can be referenced several times by different delay calculation corners.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *Library Sets*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, and double click on *Library Sets*.



Fields and Options

Name	Specifies the name of the library set to be created. This name is used to associate the library list with a specific delay calculation corner.
Timing Library Files	Specifies the timing libraries to include in the library set.
SI Library Files	Specifies the cdB libraries to include in the library set.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `create_library_set`
- `get_library_set`
- `all_library_sets`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Creating Library Sets in the *Tempus User Guide*.

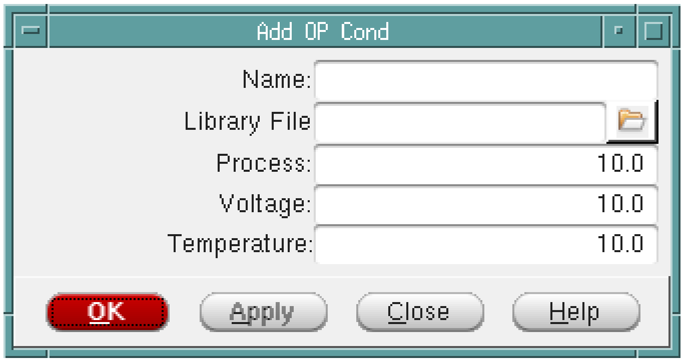
Add OP Cond

Use the Add OP Cond form to create a set of virtual operating conditions in the specified library without actually modifying the library. These virtual operating conditions can then be referenced by a delay calculation corner as if they actually exist in the library.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *OP Conds*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, and double click on *OP Conds*.



Fields and Options

Name	Specifies the name for the operating condition being created.
Library File	Specifies the library for which to create the virtual operating conditions.
Process	Specifies the process value for the operating condition.
Voltage	Specifies the voltage value for the operating condition.
Temperature	Specifies the temperature value for the operating condition.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `create_op_cond`
- `get_op_cond`
- `all_op_conds`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Creating Virtual Operating Conditions in the *Tempus User Guide*.

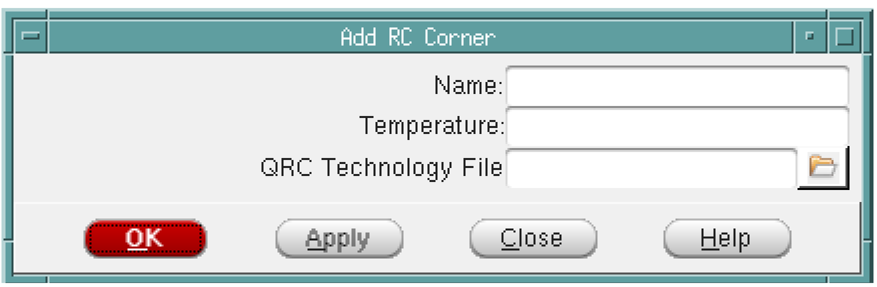
Add RC Corner

Use the Add RC Corner form to create a named RC corner object that can be referenced later when creating a delay calculation corner object. An RC corner object provides the software with all of the information necessary to properly extract, annotate, and use the RCs for delay calculation. In Tempus, you read the RC information using the SPEF files. A SPEF file contains information for a RC corner. In multi-mode multi-corner analysis, you can use a SPEF file for each view.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *RC Corners*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, and double click on *RC Corners*.



Fields and Options

Name	Specifies the name for the RC corner object being created. You can choose any name for an RC corner object, but it is recommended that you use a name that can be easily recognized as referring to a specific RC corner (for example, rc-min, rc-max, c-min, c-max, and so on).
Temperature	Specifies the operating temperature, in units of Celsius, to be used to derate extracted resistance values. The software does not require that the temperature specified by the PVT operating condition at the delay calculation corner level be the same as the RC nominal or user-specified temperature value. If you do not specify a temperature, the extracted resistance will not be derated.

QRC Technology File	Specifies the name of the Quantus QRC Technology file used by Standalone Quantus QRC extraction. For most technologies, you can get the Quantus QRC Technology file from your foundry. For technologies that are not available through the technology vendors, you can use the same ASCII-format interconnect technology (ICT) input file that is used to generate the captable. TechGen, the utility used to create the Quantus QRC Technology file, is provided with the Tempus software and does not require any additional license. For more information on running TechGen, see *Quantus QRC Documentation*.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `create_rc_corner`
- `all_rc_corners`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Creating RC Corner Objects in the *Tempus User Guide*.

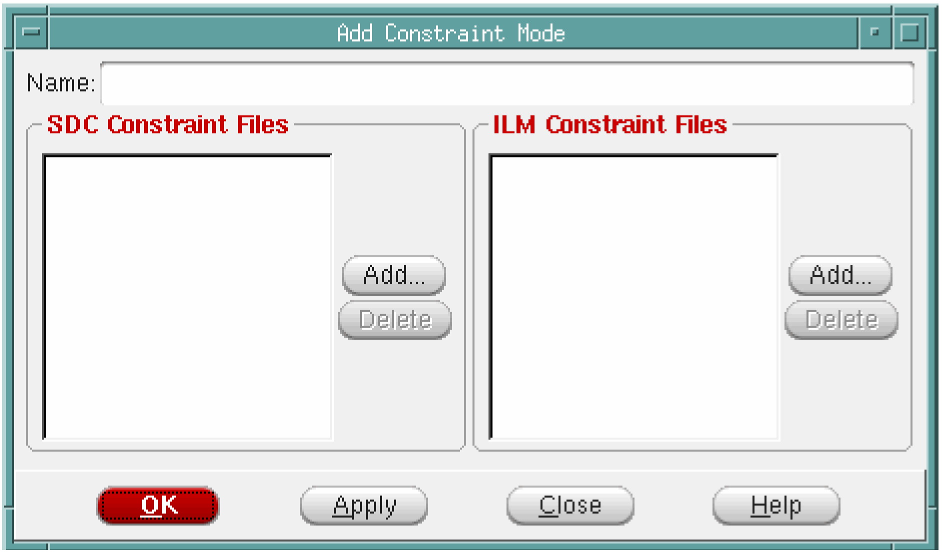
Add Constraint Mode

Use the Add Constraint Mode form to associate a list of SDC constraint files with a specified constraint mode name. This constraint mode name can be referred to later when creating analysis views. A constraint mode defines one of possibly many different functional, test behaviors, or Dynamic Voltage and Frequency Scaling (DVFS) modes of a design. The SDC files can be shared by many different constraint modes, and the same constraint mode can be associated with multiple analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *Constraint Modes*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, and double click on *Constraint Modes*.



Fields and Options

Name	Specifies the name of the mode to be created.
SDC Constraint Files	Specifies a list of SDC files to be included in the mode.
ILM Constraint Files	Specifies a list of ILM constraint files to be included in the mode.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `create_constraint_mode`
- `get_constraint_mode`
- `all_constraint_modes`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Creating Constraint Mode Objects in the *Tempus User Guide*.

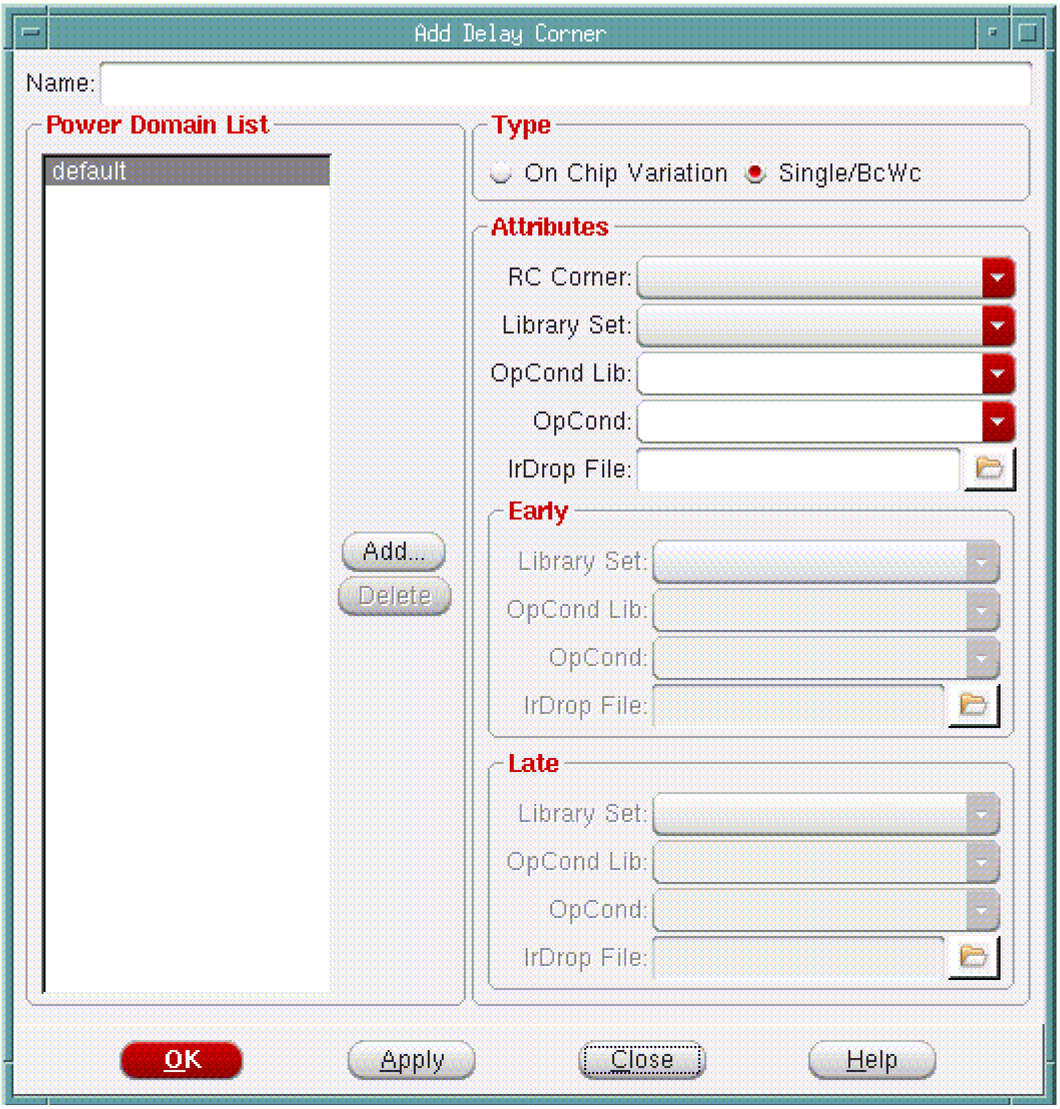
Add Delay Corner

Use the Add Delay Corner form to create a named delay calculation corner object that can be referenced later when creating an analysis view. A delay calculation corner provides all of the information necessary to control delay calculation for a specific view. Each corner contains information on the libraries to use, the operating conditions with which the libraries should be accessed, and the RC extraction parameters to use for calculating parasitic data. Delay corner objects can be shared by multiple top-level analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *Delay Corners*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, and double click on *Delay Corners*.



Fields and Options

Name	Specifies the name for the delay calculation corner object being created.

Type	Specifies the type of analysis for which you are configuring the multi-mode multi-corner environment. When running Single or Best-Case Worst-Case (BcWc) timing analysis, a delay calculation corner generally contains one library set and one RC corner object. When running on-chip-variation timing analysis, a delay calculation corner can contain early and late library sets and RC corners. If you use a delay calculation corner containing early and late attributes when the software is in BC-WC mode, the software uses the early information for analysis of hold views, and the late information for the analysis of setup views.
RC Corner	Specifies the RC corner object to associate with this delay calculation corner object.
Library Set	Specifies the library set to associate with this delay calculation corner object.
OpCond Lib	Specifies the internal library name for the library in which the operating condition is defined. This is *not* the library file name. If you do not specify a library name, the software searches the library set for the specified operating condition (`-opcond`), starting with the master library.
OpCond	Specifies the operating condition to use for setup and hold analysis. If you do not specify an operating condition, each library in the library set uses its own default operating condition.
IrDrop File	Specifies the IR drop files to apply when calculating early arrival times at a single delay corner.
Early RC Corner	Specifies the RC corner object to associate with this delay calculation corner object for calculating early arrival times at a single delay corner.
Early Library Set	Specifies the library set to associate with this delay calculation corner object for calculating early arrival times at a single delay corner.
Early OpCond Lib	Specifies the internal library name for the library in which the early operating condition is defined. This is *not* the library file name. If you do not specify a library name, the software searches the early library set for the specified operating condition (*Early OpCond*), starting with the master library.

Early OpCond	Specifies the operating condition to use for calculating early arrival times at a single delay corner. If you do not specify an operating condition, each library in the early library set uses its own default operating condition.
Early IrDrop File	Specifies the IR drop files to apply when calculating early arrival times at a single delay corner.
Late RC Corner	Specifies the RC corner object to associate with this delay calculation corner object for calculating late arrival times at a single delay corner.
Late Library Set	Specifies the library set to associate with this delay calculation corner object for calculating late arrival times at a single delay corner.
Late OpCond Lib	Specifies the internal library name for the library in which the late operating condition is defined. This is *not* the library file name. If you do not specify a library name, the software searches the late library set for the specified operating condition (*Late OpCond*), starting with the master library.
Late OpCond	Specifies the operating condition to use for calculating late arrival times at a single delay calculation corner. If you do not specify an operating condition, each library in the late library set uses its own default operating condition.
Late IrDrop File	Specifies the IR drop files to apply when calculating late arrival times at a single delay corner.
Power Domain List	Displays a list of the delay calculation corner's power domain definitions. To add a power domain definition to the delay corner, click the *Add* button to open the Add Power Domain form. To delete a power domain definition from the delay corner, choose the power domain in the list and click the *Delete* button.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `create_delay_corner`
- `get_delay_corner`
- `all_delay_corners`

- `update_delay_corner`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see the following sections in the *Tempus User Guide*.

- Creating Delay Corner Objects
- Adding Power Domain Definition to Delay Calculation Corners

Add Power Domain

Use the Add Power Domain form to add a power domain definition to the specified delay calculation corner. The instances contained within an MSMV power domain often require a different library or operating condition that is used for the default power domain. The power domain definitions within a delay calculation corner object can be used to assign specific libraries and PVT settings per power domain.

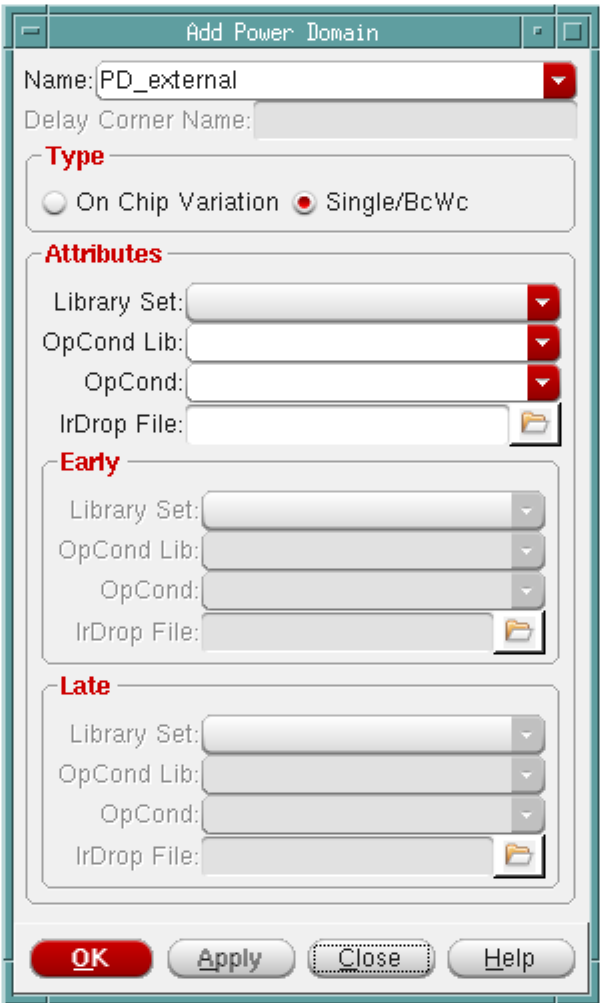
Note: Use separate delay calculation corners to define Best-Case Worst-Case differences. Use the *Early* and *Late* options within a single delay calculation corner to control on-chip-variation.

You must use the same power domain names that you read in using the power domain file to define the physical aspects of the domain.

- Choose *File - Configure MMMC* to open the MMMC Browser, double click on *Delay Corners*, and click the *Add* button next to the *Power Domain List*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, double click on *Delay Corners*, and click the *Add* button next to the *Power Domain List*.



Fields and Options

Name	Specifies the name of the power domain to add to the delay calculation corner.
Delay Corner Name	Displays the name of the delay calculation corner to which the power domain definition is being added.

Type	Specifies the type of analysis for which you are configuring the multi-mode multi-corner environment. When running Single or Best-Case Worst-Case (BcWc) timing analysis, a power domain generally contains one library set and operating condition. When running on-chip-variation timing analysis, a power domain can contain early and late library sets and operating conditions.
Library Set	Specifies the library set to associate with the power domain. All of the cells and macros in the power domain get their timing and power information from these libraries.
OpCond Lib	Specifies the internal library name for the library in which the operating condition is defined. This is *not* the library file name. If you do not specify a library name, the software searches the library set for the specified operating condition (*OpCond*), starting with the master library.
OpCond	Specifies the operating condition to use for the power domain. If you do not specify an operating condition, each library in the library set uses its own default operating condition.
IrDrop File	Specifies the IR drop files to apply to both early and late delay calculation for this power domain object.
Early Library Set	Specifies the library set to associate with the power domain for calculating early arrival times for this power domain.
Early OpCond Lib	Specifies the internal library name for the library in which the early operating condition is defined. This is *not* the library file name. If you do not specify a library name, the software searches the early library set for the specified operating condition (*Early OpCond*), starting with the master library.
Early OpCond	Specifies the operating condition to use for calculating early arrival times for this power domain. *Default:* If you do not specify an operating condition, each library in the early library set uses its own default operating condition.
Early IrDrop File	Specifies the IR drop files to apply when calculating early arrival times at a single power domain.

Late Library Set	Specifies the library set to associate with this delay corner object for calculating late arrival times for this power domain.
Late OpCond Lib	Specifies the internal library name for the library in which the late operating condition is defined. This is *not* the library file name. If you do not specify a library name, the software searches the late library set for the specified operating condition (*Late OpCond*), starting with the master library.
Late OpCond	Specifies the operating condition to use for calculating late arrival times for this power domain. *Default:* If you do not specify an operating condition, each library in the late library set uses its own default operating condition.
Late IrDrop File	Specifies the IR drop files to apply when calculating late arrival times at a single power domain.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `update_delay_corner`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Adding Power Domain Definition to Delay Calculation Corners in the *Tempus User Guide*.

Add Analysis View

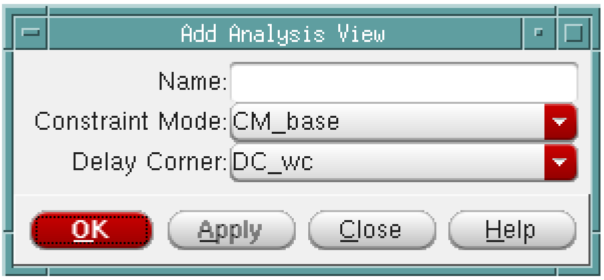
Use the Add Analysis View form to create an analysis view object that associates a delay calculation corner with a constraint mode. An analysis view object provides all of the information necessary to control a given multi-mode multi-corner analysis.

After creating analysis view objects, use the Add Hold Analysis View and Add Setup Analysis View forms to specify which views to use for setup and hold optimization or timing analysis.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *Analysis Views*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double click on *Analysis Views*.



Fields and Options

Name	Specifies the name of the analysis view being defined.
Constraint Mode	Specifies the name of the constraint mode to associate with this analysis view. A constraint mode groups a set of constraints to be used with any given analysis. To create a constraint mode, use the Add Constraint Mode form.
Delay Corner	Specifies the name of the delay calculation corner object to associate with this analysis view. A delay calculation corner object contains all of the information needed to control delay calculation for a specific analysis view. To create a delay calculation corner, use the Add Delay Corner form.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `create_analysis_view`
- `get_analysis_view`
- `all_analysis_views`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Creating Analysis Views in the *Tempus User Guide*.

Add Hold Analysis View

Use the *Add Hold Analysis View* form to define the analysis views to use for hold analysis. These "active" views represent the different design variations that will be analyzed. The active views can be changed throughout the flow to utilize different subsets of views. Libraries and data are loaded into the system, as required to support the selected set of active views. You must define at least one setup and one hold analysis view in a multi-mode multi-corner configuration.

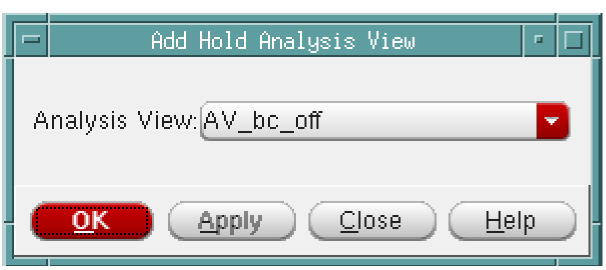
The order in which you specify hold views is important. By default, the first view defined is the default hold view. SI analysis in Tempus does not support multi-mode multi-corner. For SI analysis, the software can only process the data defined for a single view, and therefore uses the information defined for the default view.

Note: Use the `set_default_view` command, if you want to change the default analysis view. Using the `set_default_view` command does not affect software performance because it only uses views that are already active in the design. If you use the Add Hold Analysis View form to change the default views, the existing timing, delay calculation, and RC data is reset.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *Hold Analysis Views*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double click on *Hold Analysis Views*.



Fields and Options

Analysis View	Specifies the active view for hold analysis. Use the Add Hold Analysis View form multiple times to specify more than one active hold analysis view.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `set_analysis_view`
- `all_hold_analysis_views`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Setting Active Analysis Views in the *Tempus User Guide*.

Add Setup Analysis View

Use the Add Setup Analysis View form to define which views to use for setup analysis. These "active" views represent the different design variations that will be analyzed. Active views can be changed throughout the flow to utilize different subsets of views. The libraries and data are loaded into the system, as required to support the selected set of active views. You must define at least one setup and one hold analysis view in a multi-mode multi-corner configuration.

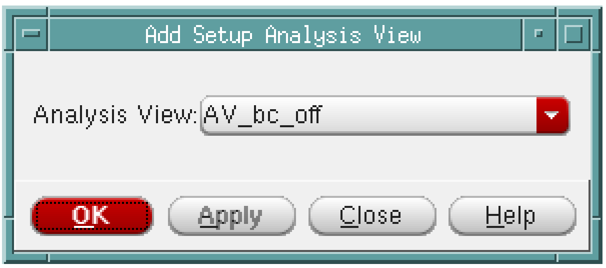
The order in which you specify setup views is important. By default, the first view defined is the default setup view. SI analysis in Tempus does not support multi-mode multi-corner. For SI analysis, the software can only process the data defined for a single view, and therefore uses the information defined for the default view.

Note: Use the `set_default_view` command, if you want to change the default analysis view. Using the `set_default_view` command does not affect software performance because it only uses views that are already active in the design. If you use the Add Setup Analysis View form to change the default views, the existing timing, delay calculation, and RC data is reset.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on *Setup Analysis Views*.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double-click on *Setup Analysis Views*.



Fields and Options

Analysis View	Specifies the active view for setup analysis. Use the Add Setup Analysis View form multiple times to specify more than one setup analysis view.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `set_analysis_view`
- `all_setup_analysis_views`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Related Topics

For more information, see Setting Active Analysis Views in the *Tempus User Guide*.

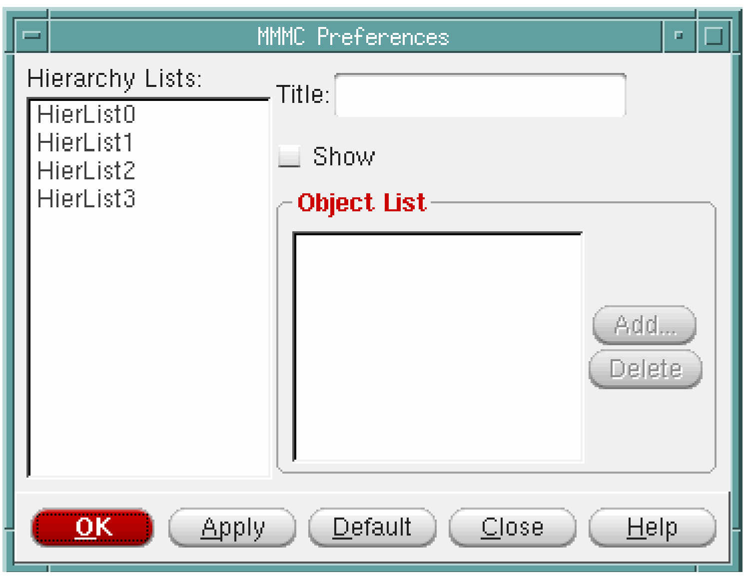
MMMC Preferences

Use the Preferences form to change how the MMMC Browser displays configuration information. By default, the MMMC Browser displays configuration information in two columns. You can change the number of columns displayed, rename the titles of the columns, change which objects are displayed in the columns, and rearrange the order in which the objects are listed.

- Choose *File - Configure MMMC* to open the MMMC Browser, and click the *Preferences* button.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then click the *Preferences* button.



Fields and Options

Hierarchy Lists	Lists the four columns that can be displayed, from left to right, in the MMMC Browser. When you select a column name in the *Hierarchy Lists* window, the form fields reflect the current settings for that column.
Title	Specifies the name of the selected column.
Show	Enables the display of the selected column in the Browser.

Object List	Lists the objects that are displayed in the selected column. To add an object to the column, click the *Add* button and choose an object from the drop down list on the Add Object form. To delete an object from the column, select the object and click the *Delete* button. You can rearrange the order of the objects in the list. Click and hold the left mouse button on the name of an object, and drag it to a different position in the list.

Related Text Commands

There is no equivalent text command for this function. You can only change how the Browser displays the configuration using the MMMC Preferences form.

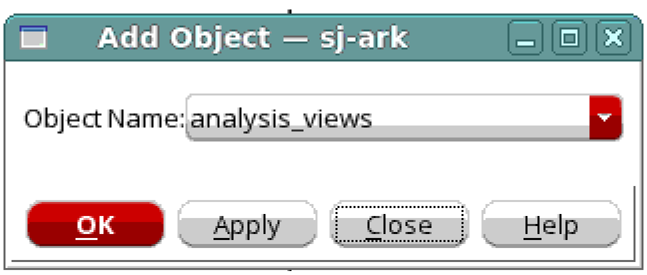
Related Topics

For more information, see Checking the Multi-Mode Multi-Corner Configuration in the *Tempus User Guide*.

Add Object

Use the Add Object form to add an object to the list of objects that you want displayed for a specific column on the MMMC Browser.

- Click the *Add* button on the Preferences form.



Fields and Options

Object Name	Specifies the object to add to the list.

Related Text Commands

There is no equivalent text command for this function. You only can add an object using the Add Object form.

Edit Library Set

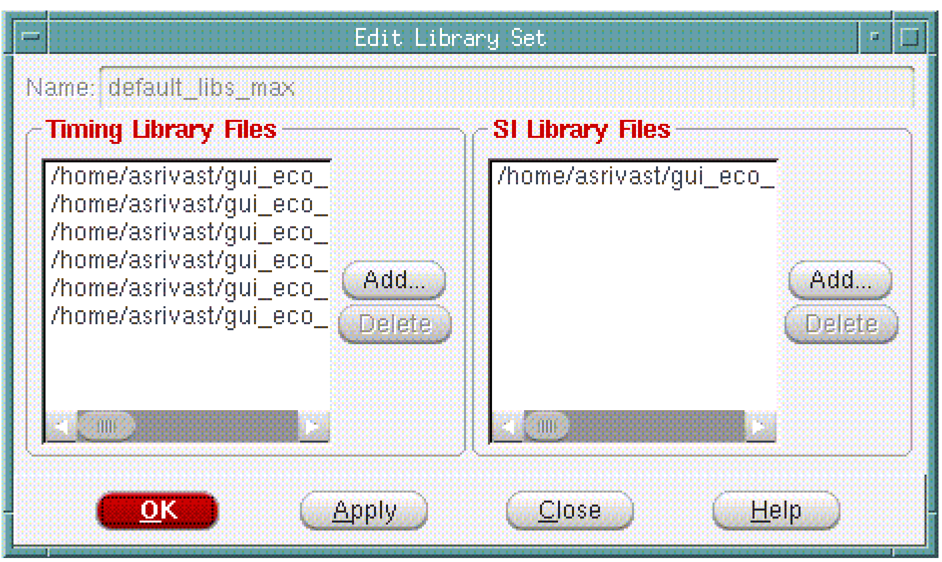
Use the Edit Library Set form to add or delete library files from an existing library set.

You can edit a library set using the Edit Library Set form before (or after) multi-mode multi-corner view definitions are loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to an existing analysis view results in the timing, delay calculation, and RC data being reset for all analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on the name of the library set you want to edit.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double click on the name of the library set that you want to edit.



Fields and Options

Name	Specifies the name of the library set being edited.

Timing Library Files	Displays the timing libraries that are included in the library set. Use the *Add* button to add libraries to the list. Use the *Delete* button to delete libraries from the list.
SI Library Files	Displays the cdB libraries that are included in the library set. Use the *Add* button to add libraries to the list. Use the *Delete* button to delete libraries from the list.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `update_library_set`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Edit OP Cond

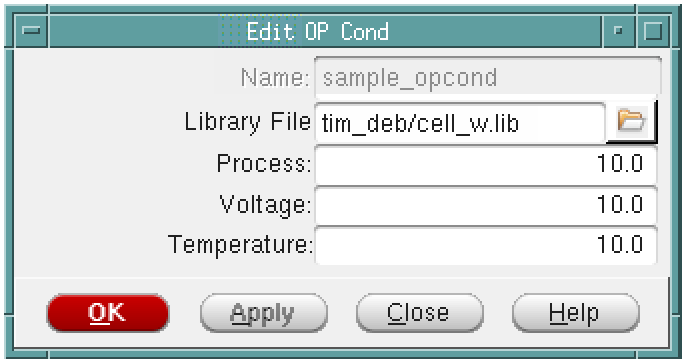
Use the Edit OP Cond form to add, delete, or change attributes for an existing operating condition.

You can edit an operating condition using the Edit OP Cond form before (or after) multi-mode multi-corner view definitions are loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to the existing analysis views results in the timing, delay calculation, and RC data being reset for all analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on the name of the virtual operating condition that you want to edit.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double-click on the name of the virtual operating condition that you want to edit.



Fields and Options

Name	Specifies the name of the operating condition being edited.
Library File	Specifies the library with which to associate the virtual operating conditions.
Process	Specifies the process value for the operating condition.
Voltage	Specifies the voltage value for the operating condition.
Temperature	Specifies the temperature value for the operating condition.

Related Text Commands

There is no equivalent text command for this function. You only can edit an operating condition using the Edit OP Cond form.

Edit RC Corner

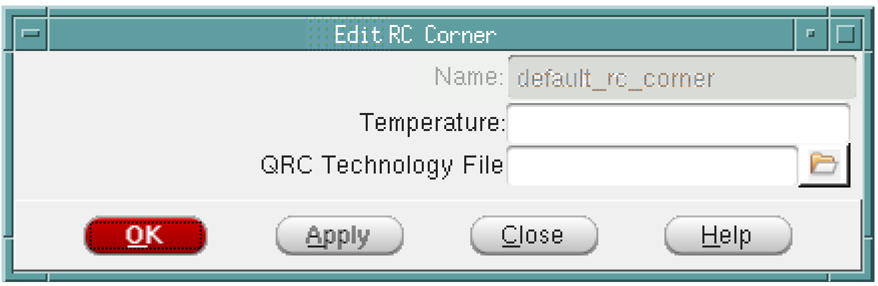
Use the Edit RC Corner form to add, delete, or change attributes for an existing RC corner object.

You can edit an RC corner object using the Edit RC Corner form before (or after) multi-mode multi-corner view definitions are loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to an existing analysis view results in the timing, delay calculation, and RC data being reset for all analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on the name of the RC corner object you want to edit.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double click on the name of the RC corner object that you want to edit.



Fields and Options

Name	Specifies the name of the RC corner object being edited.
Temperature	Specifies the Operating temperature, in units of Celsius, to be used to derate extracted resistance values. **Note:** The software does not require that the temperature specified by the PVT operating condition at the delay calculation corner level be the same as the RC nominal or user-specified temperature value. If you do not specify a temperature, the extracted resistance will not be derated.
QRC Technology File	Specifies the name of the Quantus QRC Technology file used by Standalone Quantus QRC extraction. For most technologies, you can get the Quantus QRC Technology file from your foundry. For technologies that are not available through the technology vendors, you use the same ASCII-format interconnect technology (ICT) input file that is used to generate the captable. TechGen, the utility used to create the Quantus QRC Technology file, is provided with the Innovus software and does not require any additional license. For more information on running TechGen, see Quantus *QRC Documentation*.

Edit Constraint Mode

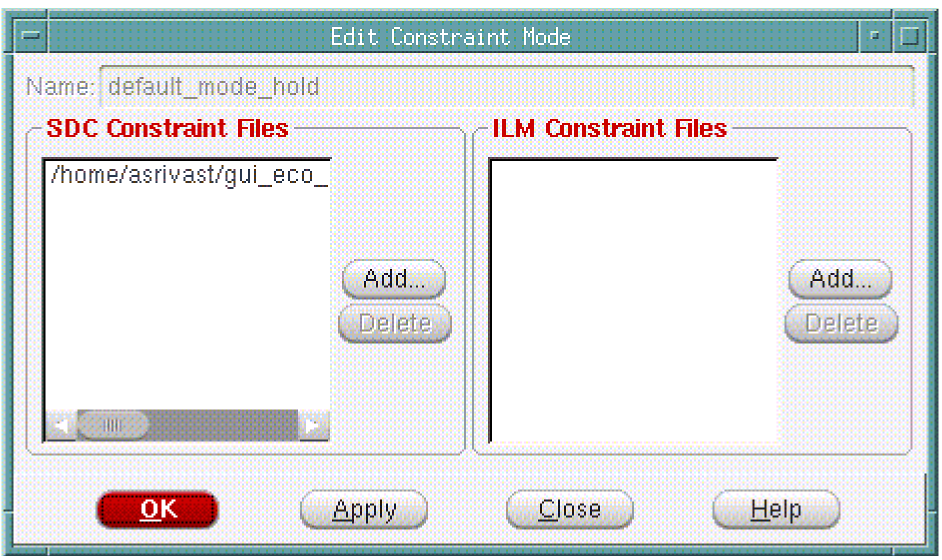
Use the Edit Constraint Mode form to add or delete SDC constraint files from an existing constraint mode object.

You can edit a constraint mode object using the Edit Constraint Mode form before (or after) multi-mode multi-corner view definitions are loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to the existing analysis views results in the timing, delay calculation, and RC data being reset for all analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on the name of the constraint mode object you want to edit.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double-click on the name of the constraint mode object you want to edit.



Fields and Options

Name	Displays the name of the constraint mode object being edited.
SDC Constraint Files	Displays the SDC constraint files that are associated with the constraint mode object. Use the *Add* button to add SDC files to the list. Use the *Delete* button to delete SDC files from the list.

ILM Constraint Files	Displays the SDC constraint files for the ILM flow that are associated with the constraint mode object. Use the *Add* button to add SDC files to the list. Use the *Delete* button to delete SDC files from the list.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `update_constraint_mode`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Edit Delay Corner

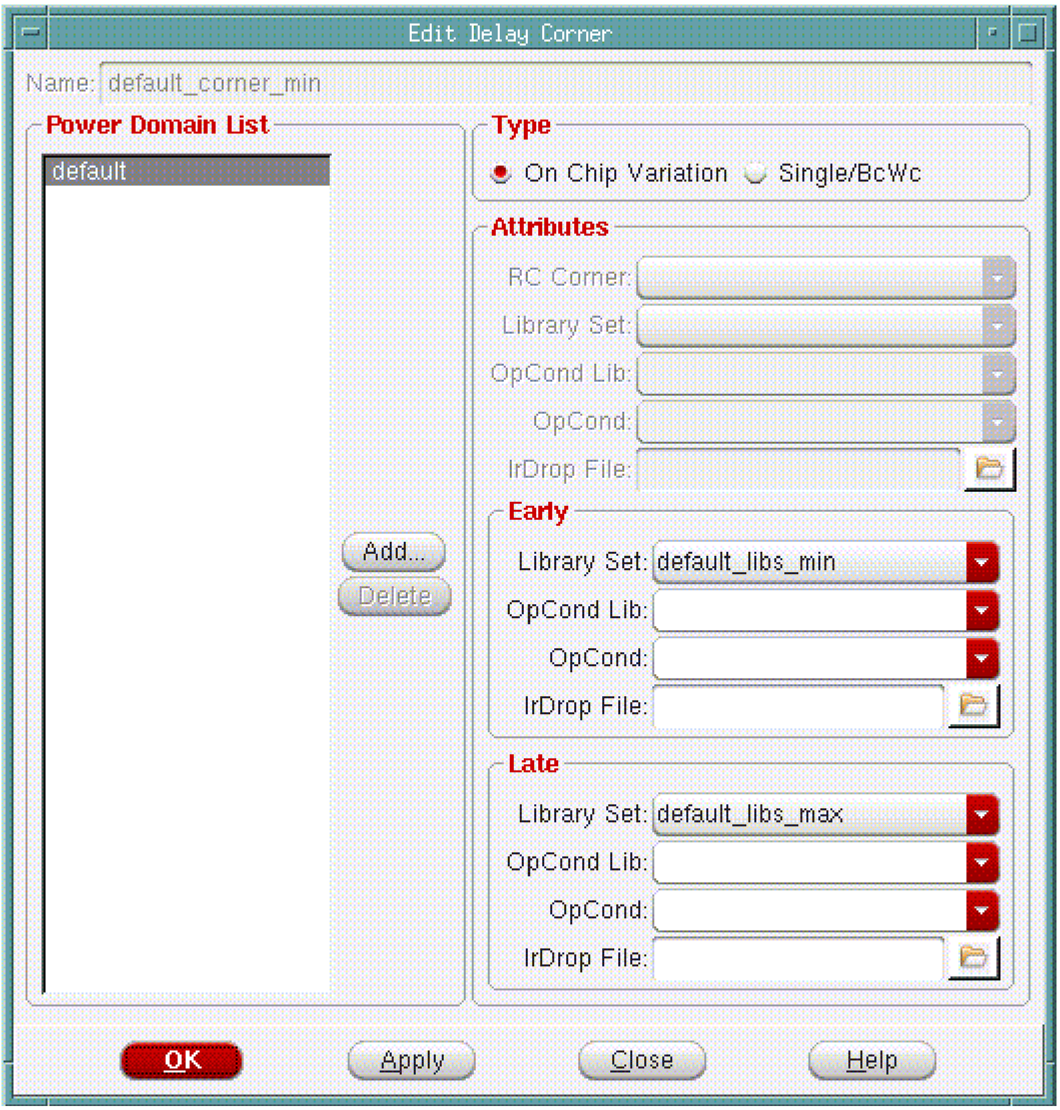
Use the Edit Delay Corner form to add, delete, or change attributes for an existing delay calculation corner object.

You can edit a delay calculation corner object using the Edit Delay Corner form before (or after) multi-mode multi-corner view definitions are loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to an existing analysis view results in the timing, delay calculation, and RC data being reset for all analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on the name of the delay calculation corner object you want to edit.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double-click on the name of the delay calculation corner object that you want to edit.



Fields and Options

Name	Displays the name of the delay calculation corner object being edited.

Type	Specifies the type of analysis for which you are configuring the multi-mode multi-corner environment. When running Single or Best-Case Worst-Case (BcWc) timing analysis, a delay calculation corner generally contains one library set and one RC corner object. When running on-chip variation (OCV) timing analysis, a delay calculation corner can contain early and late library sets and RC corners. **Note:** If you use a delay calculation corner containing early and late attributes when the software is in BC-WC mode, the software uses the early information for analysis of hold views, and the late information for the analysis of setup views.
RC Corner	Specifies an RC corner object to associate with this delay calculation corner object.
Library Set	Specifies a library set to associate with this delay calculation corner object.
OpCond Lib	Specifies the internal library name for the library in which the operating condition is defined. This is *not* the library file name.
OpCond	Specifies an operating condition to use for setup and hold analysis.
IrDrop File	Specifies the IR drop files to apply to both early and late delay calculation for this delay corner object.
Early RC Corner	Specifies an RC corner object to associate with this delay calculation corner object for calculating early arrival times at a single delay calculation corner. **Note:** The 8.1 version of the software does not support the use of early and late RC corners; therefore, this field is disabled.
Early Library Set	Specifies a library set to associate with this delay calculation corner object for calculating early arrival times at a single delay calculation corner.
Early OpCond Lib	Specifies the internal library name for the library in which the early operating condition is defined. This is *not* the library file name.
Early OpCond	Specifies an operating condition to use for calculating early arrival times at a single delay calculation corner.
Early IrDrop File	Specifies the IR drop files to apply when calculating early arrival times at a single delay corner.

Late RC Corner	Specifies an RC corner object to associate with this delay calculation corner object for calculating late arrival times at a single delay corner. **Note:** The 8.1 version of the software does not support the use of early and late RC corners; therefore, this field is disabled.
Late Library Set	Specifies a library set to associate with this delay calculation corner object for calculating late arrival times at a single delay calculation corner.
Late OpCond Lib	Specifies the internal library name for the library in which the late operating condition is defined. This is *not* the library file name.
Late OpCond	Specifies an operating condition to use for calculating late arrival times at a single delay calculation corner.
Late IrDrop File	Specifies the IR drop files to apply when calculating late arrival times at a single delay corner.
Power Domain List	Displays a list the delay calculation corner's power domain definitions. To add a power domain definition to the delay calculation corner, click the *Add* button to open the Add Power Domain form. To delete a power domain definition from the delay calculation corner, choose the power domain in the list and click the *Delete* button.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `update_delay_corner`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Edit Power Domain

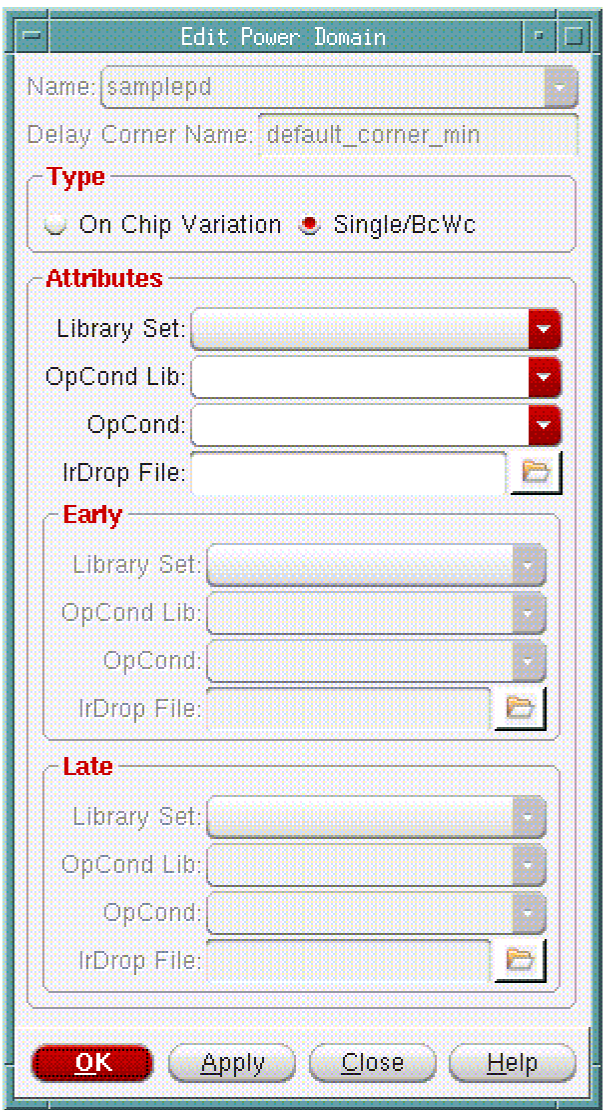
Use the Edit Power Domain form to add, delete, or change attributes for an existing power domain definition.

You can edit a power domain using the Edit Power Domain form before (or after) multi-mode multi-corner view definitions are loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to the existing analysis views results in the timing, delay calculation, and RC data being reset for all analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, click the + next to *Delay Corners* to list the available delay corners, click the + next to the delay corner to which the power domain definition belongs, and double click on the name of the power domain definition.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, click the + next to *Delay Corners* to list the available delay corners, click the + next to the delay corner to which the power domain definition belongs, and double click on the name of the power domain definition.



Fields and Options

Name	Displays the name of the power domain being edited.
Type	Specifies the type of analysis for which you are configuring the multi-mode multi-corner environment. When running Single or Best-Case Worst-Case (BcWc) timing analysis, a power domain generally contains one library set and operating condition. When running on-chip-variation timing analysis, a power domain can contain early and late library sets and operating conditions.
Delay Corner Name	Displays the delay calculation corner to which the power domain definition belongs.
Library Set	Specifies a library set to associate with the power domain. All of the cells and macros in the power domain get their timing and power information from these libraries.
OpCond Lib	Specifies the internal library name for the library in which the operating condition is defined. This is *not* the library file name.
OpCond	Specifies an operating condition to use for the power domain.
IrDrop File	Specifies the IR drop files to apply to both early and late delay calculation for this power domain object.
Early Library Set	Specifies a library set to associate with the power domain for calculating early arrival times at a single delay corner.
Early OpCond Lib	Specifies the internal library name for the library in which the early operating condition is defined. This is *not* the library file name.
Early OpCond	Specifies an operating condition to use for calculating early arrival times at a single delay corner.
Early IrDrop File	Specifies the IR drop files to apply when calculating early arrival times at a single power domain.
Late Library Set	Specifies a library set to associate with this delay corner object for calculating late arrival times at a single delay corner.
Late OpCond Lib	Specifies the internal library name for the library in which the late operating condition is defined. This is *not* the library file name.
Late OpCond	Specifies an operating condition to use for calculating the late arrival times at a single delay corner.

Late IrDrop File	Specifies the IR drop files to apply when calculating the late arrival times at a single power domain.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `update_delay_corner`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Edit Analysis View

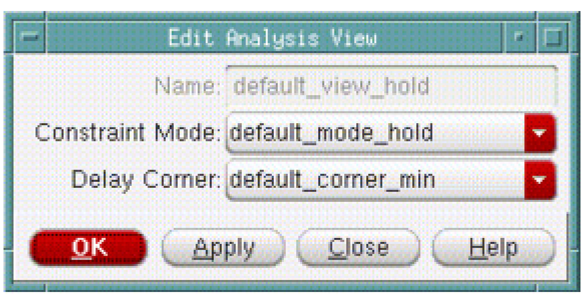
Use the Edit Analysis View form to add or delete constraint mode and delay corner objects from an existing analysis view.

You can edit an analysis view using the Edit Analysis view form before (or after) multi-mode multi-corner view are definitions loaded into the design. However, after the software is in multi-mode multi-corner analysis mode, any change to an existing analysis view results in the timing, delay calculation, and RC data being reset for all the analysis views.

- Choose *File - Configure MMMC* to open the MMMC Browser, and double click on the name of the analysis view that you want to edit.

or:

- Click the *Setup Data* tab, click *Multi-Mode Multi-Corner*, and click the *MMMC Browser* button to open the MMMC Browser, then double-click on the name of the analysis view that you want to edit.



Fields and Options

Name	Displays the name of the analysis view being edited.
Constraint Mode	Specifies the name of the constraint mode to associate with this analysis view.
Delay Corner	Specifies the name of the delay corner to associate with this analysis view.

Related Text Commands

The following text command provides equivalent or additional functionality:

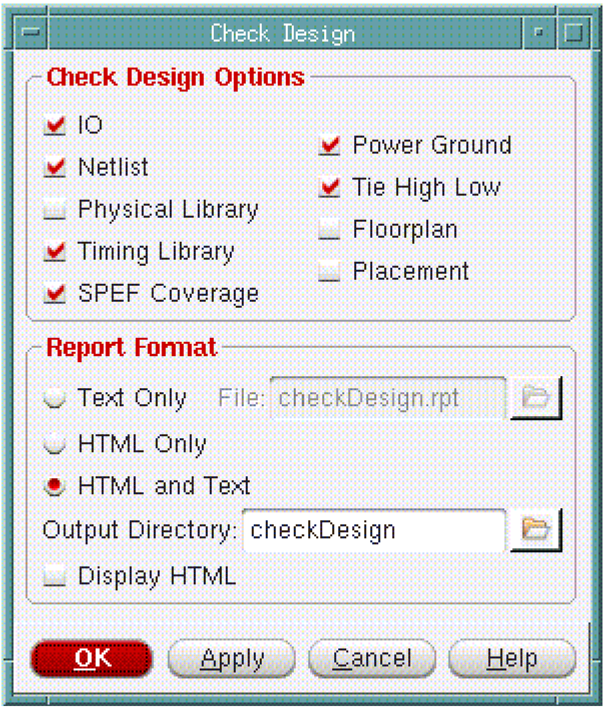
- `update_analysis_view`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Check Design

Use the Check Design form to check for missing or inconsistent library and design data at any stage of the design.

- Choose *File - Check - Check Design.*



Fields and Options

IO	Checks I/O pads, generates warning messages, and reports the following information: ■ Floating pins that belong to I/O pads (warning) ■ I/O pins connected to non-I/O instances (directly connected to core cells) ■ Floating I/O pins
Netlist	Checks the netlist, generates error or warning messages, and reports the following information: ■ Output pins tied to power/ground nets (for example, `BUFX1 U1 (.A(net1),.Y1(1'b0)))` ■ Input pins floating (warning) ■ Multiple driver nets (warning)

Physical Library	Checks the physical libraries and reports whether all the cells have LEF views. Issues error messages for the following conditions: ■ Cells not defined in LEF ■ Cells with missing power ground pins ■ Cells with missing dimensions ■ Pins with missing direction ■ Cells and pins with missing geometry ■ Cell dimensions are not an integer multiple of the core site dimensions ■ Cells and power ground pins with missing geometry
Timing Library	Checks whether the cells used in the design have been defined in the timing library. This option does not check for the presence of timing arcs. All physical cells are excluded from this check.
SPEF Coverage	Reports the following SPEF information: ■ SPEF coverage ■ Number of nets in the design that are not in the SPEF ■ List of nets
Power Ground	Checks power and ground connections, generates warning messages, and reports the following information: ■ Power terminals connected to ground net ■ Ground terminals connected to power net ■ Floating power and ground terminals ■ Power and ground terminals connected to non-power and ground nets.
Tie High Low	Reports unconnected tie-high or tie-low terminals, and generates a warning message.

Floorplan	Checks the floorplan, generates error or warning messages, and reports the following information: ■ Off-grid horizontal and vertical tracks ■ Instances not snapped to row site ■ Unplaced I/O pins ■ Off Grid Power/Ground Pre-routes ■ Instances not on the manufacturing grid ■ Pre-routes not on the manufacturing grid (error) ■ Regular pre-routes not on tracks
Placement	Checks the floorplan, generates error or warning messages, and reports the following information: ■ Off-grid horizontal and vertical tracks ■ Instances not snapped to row site ■ Unplaced I/O pins ■ Off Grid Power/Ground Pre-routes ■ Instances not on the manufacturing grid ■ Pre-routes not on the manufacturing grid (error) ■ Regular pre-routes not on tracks
Text Only	Generates only a text version of the report. *Default:* Generates both an HTML and a text version of the report.
File	Writes the report information to the specified file. *Default:* `checkDesign.rpt`
HTML Only	Generates only an HTML version of the report. *Default:* Generates both an HTML and a text version of the report.
HTML and Text	Generates both an HTML version and a text version of the report.

Output Directory	Saves both the HTML version and the text version of the report to the specified directory. If you select the *HTML Only* option, the software saves the HTML version of the report to the specified directory. *Default:* `checkDesign`
Display HTML	Displays the HTML version of the report in the Report menu.

Related Text Commands

The following text command provides equivalent or additional functionality:

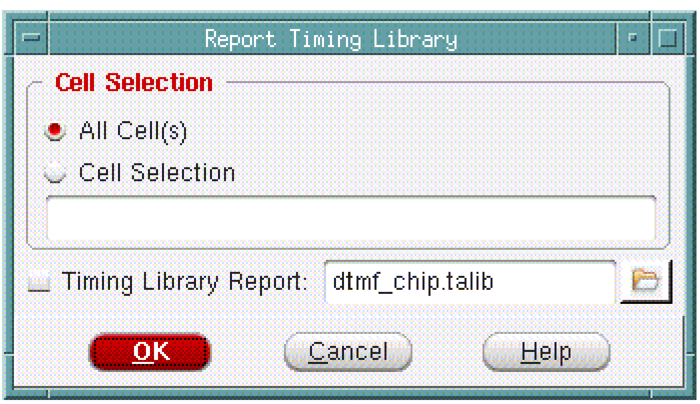
- `check_design`

For more information, see the "Design Import Commands" in the *Tempus Text Command Reference*.

Report Timing Library

Use the Report Timing Library form to output the cell information to a report file after importing the design and loading the timing library. The report file is generated in ASCII format.

☨ Choose *File - Check - Timing Library*



Fields and Options

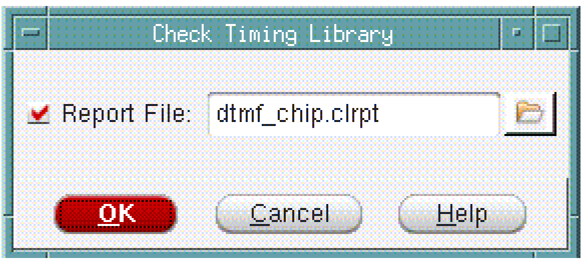
All Cell(s)	Reports timing information for all the cells. This is the default option.

Cell Selection	Reports timing information on the cells specified in the text box.
Timing Library Report	Specifies a filename or you can click the file folder icon to choose a file. If you do not select this option, the report is displayed on the Tempus console. The default filename is `TopCellName.talib`.

Check Timing Library

Use the Check Timing Library form to check the contents of the timing library and reports any inconsistencies in a report file. Use this form after importing the design and timing library.

- Choose *File - Check - Library*



Fields and Options

Report File	Specifies the name of the report file. The default filename is `TopCellName.clrpt`.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `check_library`

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Summary Report

Use the Summary Report form to create a report that contains statistics for the entire design, or a selected object in the design.

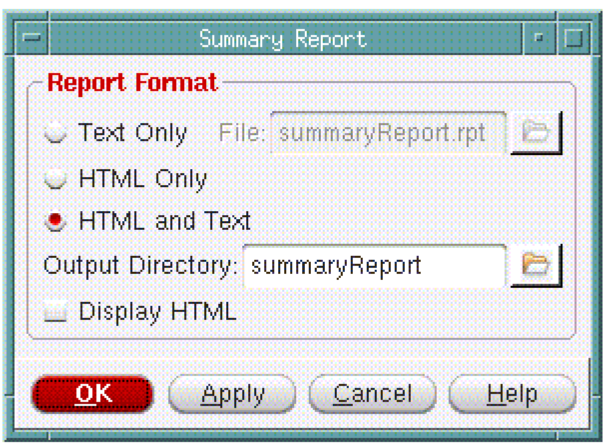
A report on the entire design includes statistics for the following categories:

- General Design Information
- General Library Information
- Netlist Information
- Timing Information

The information contained in a report on a selected object varies depending on the object you select. You can select a module, an instance, or a net.

Select an object in the design display area. (If you want to display information about the entire design, skip Step 1.)

- Choose *File - Report - Summary*.



Fields and Options

Text Only	Generates a text version of the summary report. *Default:* `Off`	
	File	Writes the report information to the specified file. *Default:* `summaryReport.rpt`

HTML Only	Generates an HTML version of the summary report. Some statistics in the sections include links to more detailed HTML reports. For example, if you click on the *# Layers* link under *General Library Information* section of the HTML report, the software displays a detailed HTML report on the layers in the design. *Default:* `Off`
HTML and Text	Generates both a text version and an HTML version of the summary report. *Default:* `On`
Output Directory	Saves the HTML version of the report to the specified directory. If you select the *HTML and Text* option, the software saves both the HTML and text versions of the report to the specified directory. *Default:* `summaryReport`
Display HTML	Opens a browser and displays the HTML version of the report. *Default:* `Off`

Netlist Stats

Use the *Netlist Statistics* menu command to create a list of design statistics, such as number of cells, nets, pins, and instances. A summary of the netlist is printed to the console and a log file, and saved in the run directory.

- To print a list of design statistics, choose *File - Report - Netlist Statistics*.

Recent Actions

Records the actions that you perform recently in the current session.

Hide GUI

Use *Hide GUI* to hide the GUI. Type the `start_gui` command from command line to re-display the GUI.

Exit

To exit Tempus Timing Solution.

3

View Menu

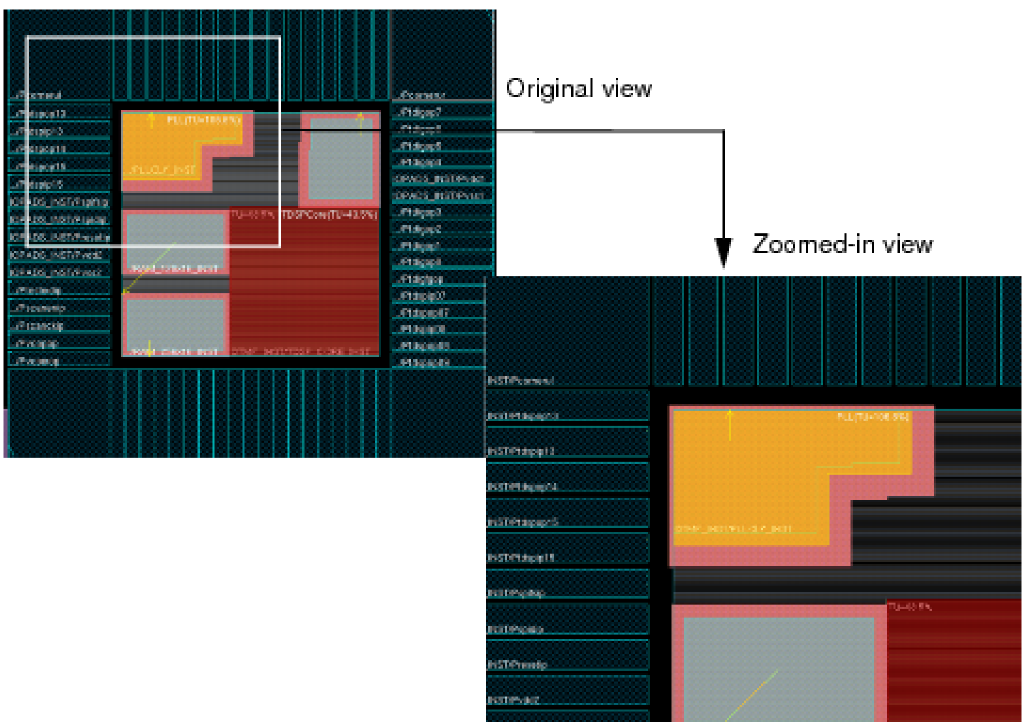
Zoom

The *View - Zoom* menu comprises the following items:

In

Use the *Zoom - In* menu command to zoom into a portion of the design and view it in greater detail. Each click zooms in two levels. The equivalent key is `z`.

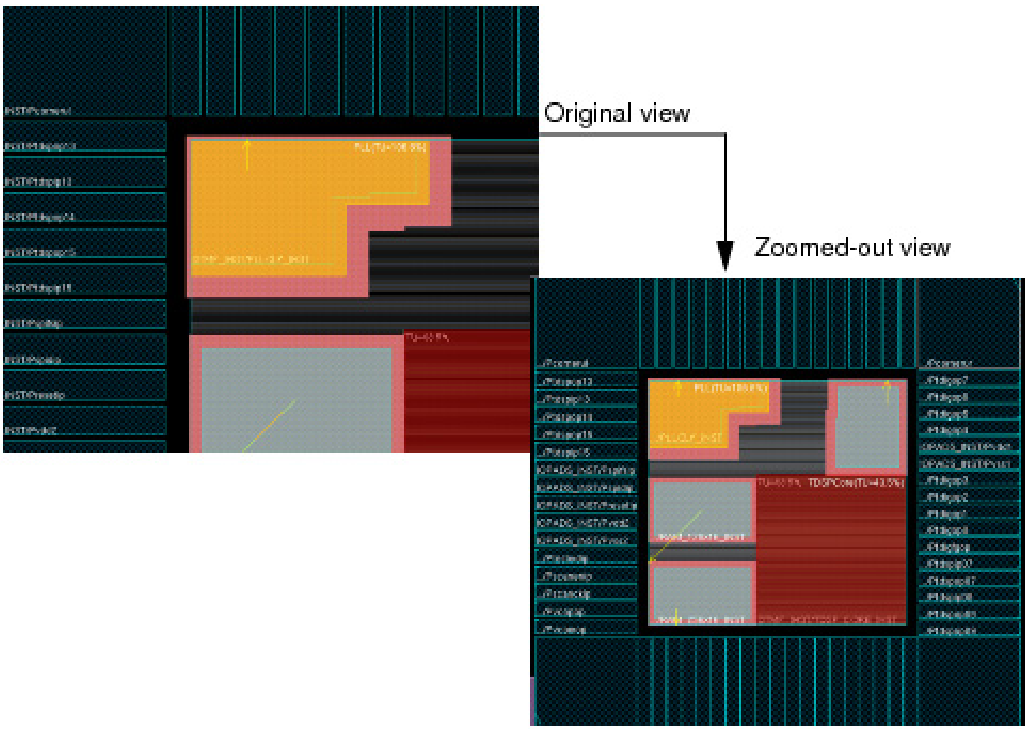
You can also use the scroll wheel of your mouse to zoom into a portion of the design. Each forward scroll of the scroll wheel zooms in one level.



Out

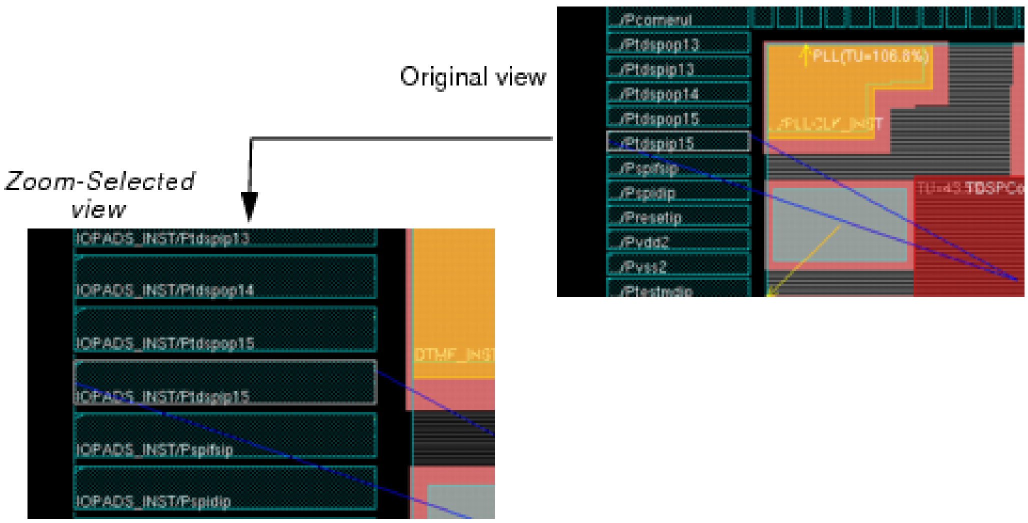
Use the *Zoom - Out* menu command to display a larger area of the design in less detail. Each click zooms out two levels. The equivalent key is `Z`.

You can also use the scroll wheel of your mouse to zoom out of a detailed view of the design. Each backward scroll of the scroll wheel zooms in one level.



Selected

Select a design object and then use the *Zoom-Selected* menu command to zoom into the area that contains the selected object.



Previous

Use the *Zoom-Previous* menu command to toggle the display between the previous location/zoom level and the current location/zoom level. The equivalent key is `w`.

Pan

The View - Pan menu comprises the following items:

- Up on page 76
- Down on page 77
- Left on page 77
- Right on page 77

Up

Pans up the viewable window in pages. The size of each page is equal to the size of the current viewable window. The equivalent key is the `Up` arrow key.

You can also use the scroll wheel of your mouse to pan up the viewable window. Press `Shift` and move the scroll wheel forward to pan upwards.

Down

Pans down the viewable window in pages. The size of each page is equal to the size of the current viewable window. The equivalent key is the `Down` arrow key.

You can also use the scroll wheel of your mouse to pan down the viewable window. Press `Shift` and move the scroll wheel backward to pan downwards.

Left

Pans the viewable window to the left in pages. The size of each page is equal to the size of the current viewable window. The equivalent key is the `Left` arrow key.

You can also use the scroll wheel of your mouse to pan the viewable window to the left. Press `Ctrl` and move the scroll wheel forward to pan to the left.

Right

Pans the viewable window to the right in pages. The size of each page is equal to the size of the current viewable window. The equivalent key is the `Right` arrow key.

You can also use the scroll wheel of your mouse to pan the viewable window to the right. Press `Ctrl` and move the scroll wheel backward to pan to the right.

Fit

Use the *Fit* menu command to fit the entire design within the design display area. The equivalent key is `f`.

Redraw

Use the *Zoom-Redraw* menu command to refresh the display in the viewable window. The equivalent keyboard shortcut is `Ctrl+r`.

Set Preferences

Use the *View - Set Preferences* command to access the Preferences form. This form allows you to set the design settings for the Tempus session.

You can save custom preference settings to a file, load a previously saved file, or load default settings. The default settings are saved in the ssv`.pref.tcl` file; custom settings can be saved with a different filename.

The Preferences form also allows you to create, edit, apply, and save design keyboard shortcut commands.

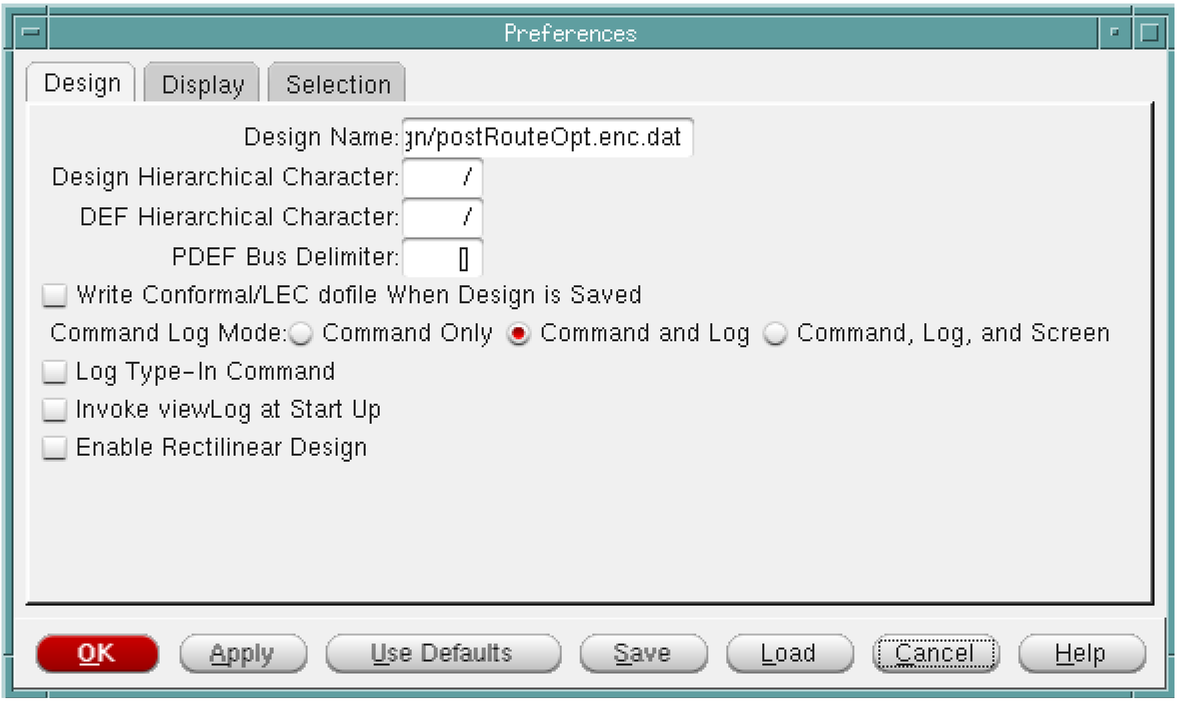
The Preferences form contains the following pages:

- Preferences - Design on page 78
- Preferences - Display on page 81
- Preferences - Selection on page 85

Preferences - Design

Use the *Design* tab of the Preferences form to set hierarchy levels in the netlist and DEF files, specify bus delimiters for PDEF files, denote output of command logs, and create key shortcuts. The default settings are saved in the `ssv.pref.tcl` file; custom settings can be saved with a different filename.

- Choose *View - Set Preference* and click the *Design* tab.



Preferences - Design Fields and Options

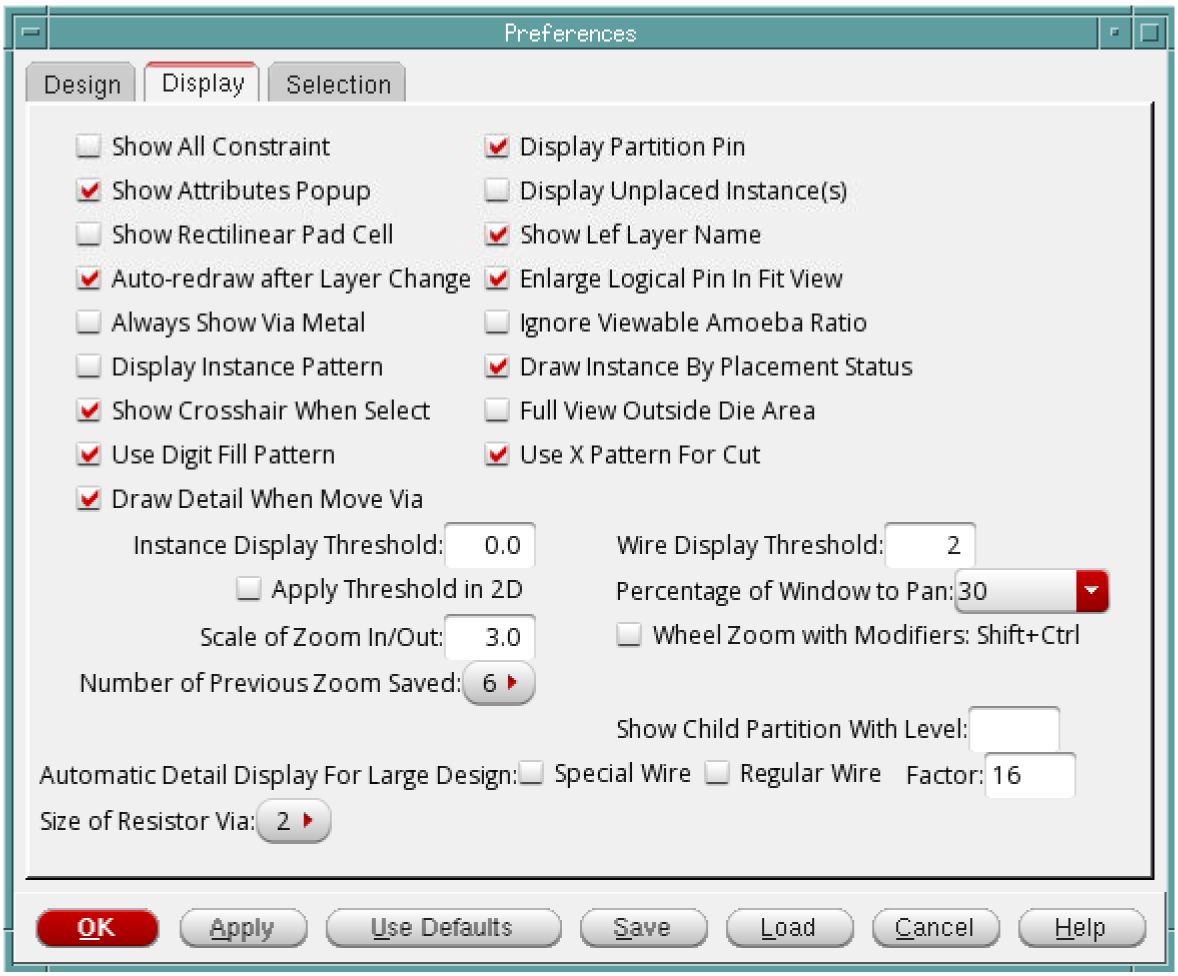
Design Name	Specifies the name of the design for which to set the preferences.
Design Hierarchical Character	
	Denotes the character being used as the hierarchy delimiter in a netlist. If the netlist uses another character, enter it in the text box and import the design. *Default: /*
DEF Hierarchical Character	Denotes the character being used as the hierarchy delimiter in a DEF file. If the DEF file uses another character, enter it in the text box and load the DEF file. When writing a DEF file, the software uses the same character. *Default: /*

PDEF Bus Delimiter	Denotes the character being used as the hierarchy delimiter in a PDEF file. If the PDEF file uses another character, enter it in the text box and load the PDEF file. When writing a PDEF file, the software uses the same character. *Default:* []	
Write Conformal/LEC dofile When Design is Saved		
	When saving a netlist, this option saves a dofile (a set of commands contained in a specified file) for the Conformal® Equivalence Checking capabilities. *Default:* Off	
Command Log Mode	Specifies where the command log is written. *Default: Command and Log*	
	Command Only	Writes to the command file only.
	Command and Log	Writes to the command file and log files.
	Command, Log, and Screen	Writes to the command file, log files, and screen.
	Note: When writing to the log file and screen, the commands are preceded with <CMD>.	
Log Type-In Command	Logs manually entered (typed) commands to the command file. In the case of source commands, only the source command is logged, and not the commands in the sourced file. *Default:* Off	
Invoke viewLog at Start Up		
	Displays log viewer at startup. If you select this check box and click Save in the Preferences form, `setPreference logviewer 1` is written to the user specified preference file. *Default:* Off	
Enable Rectilinear Design		
	Enables you to set a rectilinear object or cut a rectilinear area in a design with IO cells.	

Preferences - Display

Use the *Display* tab of the Preferences form to set viewing preferences for standard cells, wires, route congestion ranges, modules, and objects. The default settings are saved in the `ssv.pref.tcl` file; custom settings can be saved with a different file name.

- Choose *View - Set Preference* and click the *Display* tab.



Preferences - Display Fields and Options

Show All Constraint	Displays all constraints, such as fences, guides, regions, and softguides, regardless of the current hierarchy. When this option is selected, you can see the child guides of each fence as you shape the fence. *Default:* Off

Display Partition Pin	Displays partition pins. *Default:* On
Show Attributes Popup	Specifies that the Context Pop-up Attribute Viewer, which is displayed when you place the mouse cursor over an object, should be displayed.
Display Unplaced Instance(s)	Displays unplaced cells in the Physical view. *Default:* Off
Show Rectilinear Pad Cell	Displays bounding box if the pad cell has rectilinear shape. *Default:* Off
Show Lef Layer Name	Specifies that the layer name as specified in the LEF file should be displayed in the color bar. *Default:* On
Auto-redraw after Layer Change	
	Enables or disables auto-redraw when layers are changed. *Default:* On
Enlarge Logical Pin In Fit View	Displays larger symbols for logical pins in Fit view. Select this option to check the pin distribution in a block design. *Default:* On
Always Show Via Metal	Keeps the additional metal of a via visible even if you turn off the via cut layer. With the *Always Show Via Metal* option selected: ■ If you turn off the metal layer and keep via on, only the cut via is visible in the main window. ■ If you turn off the via and keep metal layer on, the top/bottom layer of the via is visible. This option makes it easy for you to view the top/bottom layer of the via by turning off the via. If this option is selected, you can select the via metal and view its attributes in the *Attribute Editor*. *Default:* Off

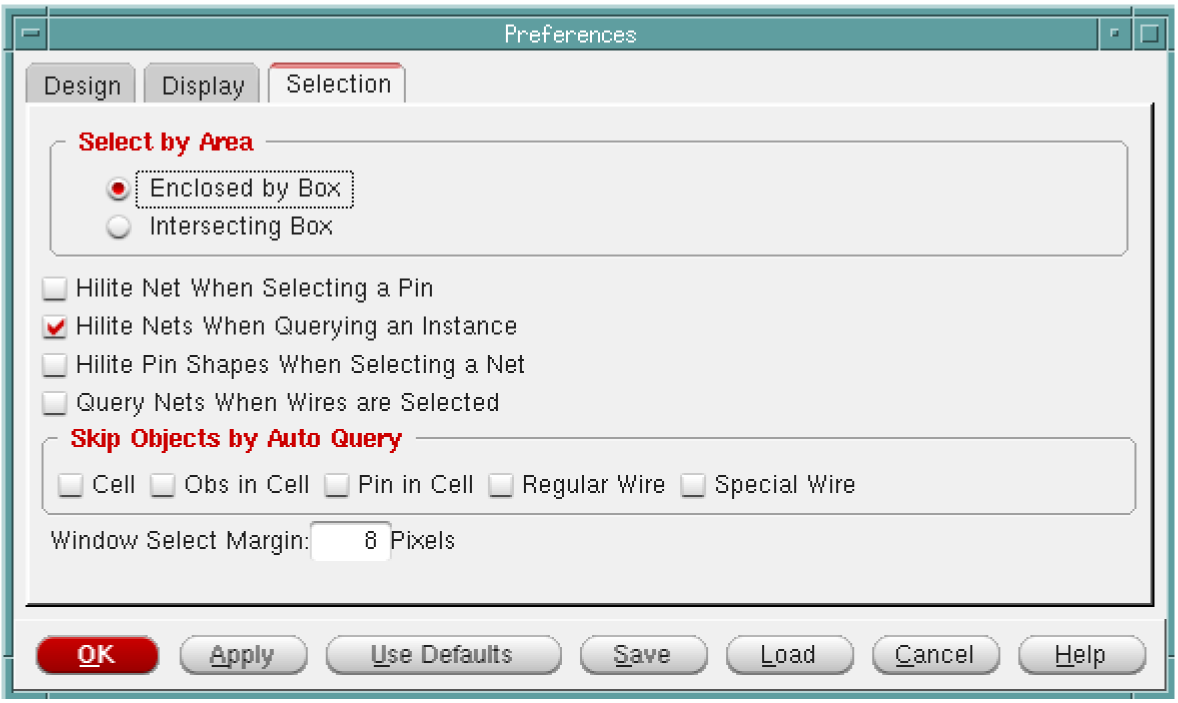
Ignore Viewable Amoeba Ratio	Displays all modules in the *Amoeba View* when selected, irrespective of their display size. By default, if the display size of a module is smaller than 1/3rd the area of the current module or 1/20th of the fplan box, it is not displayed in the *Amoeba View*. *Default:* Off
Display Instance Pattern	Marks overlapping instances with X in the display area. This makes it easier to identify overlapping instances as compared to the default fill pattern. *Default:* Off
Draw Instance by Placement Status	Controls the thickness of instance borders on the basis of their placement status. Select the new *Draw Instance by Placement Status* option if you want fixed instances to be marked with a thick border. *Default:* On
Instance Display Threshold	Sets the draw size of the standard cell instance in the physical view. The default value of 0.0 draws all standard cells. A value of 2.0 draws fewer standard cells, thereby yielding faster draw speed. If you change this value, you must redraw the view by choosing *View - Redraw* from the menu.
Wire Display Threshold	Sets the minimum number of pixels used to display a wire or via. This value works in conjunction with the *Apply Threshold in 2D* option. At the chip level, set 1 for the *Wire Display Threshold* and turn off *Apply Threshold in 2D* to display all of the wires/vias in the design. If the display speed is too slow, then change either or both variables appropriately.
Apply Threshold in 2D	Checks both width and height of a wire or via when used with *Wire Display Threshold*. When off, the wire or via is displayed if either its width or height meets the threshold. When on, both width or height must meet the threshold for the wire or via to be displayed. *Default:* Off
Percentage of Window to Pan	Specifies the percentage of display area that will be panned when you pan the display using the right or the left arrow keys. *Default:* The default value is 95%.

Scale of Zoom In/Out	Specifies the scale for the zoom in and zoom out feature. You can use a decimal number for the scale.	
Wheel Zoom with Modifiers: Shift+Ctrl	Enables use of scroll wheel to zoom in and out, when used along with the `Shift` and `Ctrl` keys.	
Number of Previous Zoom Saved	Specifies the number of zoom views that are saved. You can then view the previous zoom views by clicking the Zoom Previous icon or the key W. For example, if you set the number to 4, you can view four previous zoom views by clicking the Zoom Previous icon or the key W. *Default:* 1	
Show Child Partition With Level	Displays child partitions from the specified level. *Default:* 0	
Automatic Detail Display for Large Design	Controls the performance of redraw in large designs. If you select *Special Wire* and/or *Regular Wire*, the tool only draws these wires when you zoom in to the specified level (*Factor*). By default, the wires are drawn in full view (no zoom) itself.	
	Special Wire	Draws special wires only when you zoom in to the specified level. *Default:* Off
	Regular Wire	Draws regular wires only when you zoom in to the specified level. *Default:* Off
	Factor	Specifies the zoom level at which wires will be drawn in large designs. *Default:* 16
Size of Resistor Via	Controls the resistor via size displayed in GUI. The default resistor size is `1`. This field is useful for debugging purpose when viewing and analyzing vias only.	

Preferences - Selection

Use the *Selection* tab of the Preferences form to select and display objects and determine their behavior within a selected area. The default settings are saved in the ssv`.pref.tcl` file; custom settings can be saved with a different filename.

- Choose *View - Set Preference* and click the *Selection* tab.



Preferences - Selection Fields and Options

Select by Area		
	Enclosed by Box	Selects objects that are fully enclosed inside the specified area. Use the left mouse button to draw and define an area or bounding box. Any objects fully enclosed inside the drawing area are selected. *Default:* On

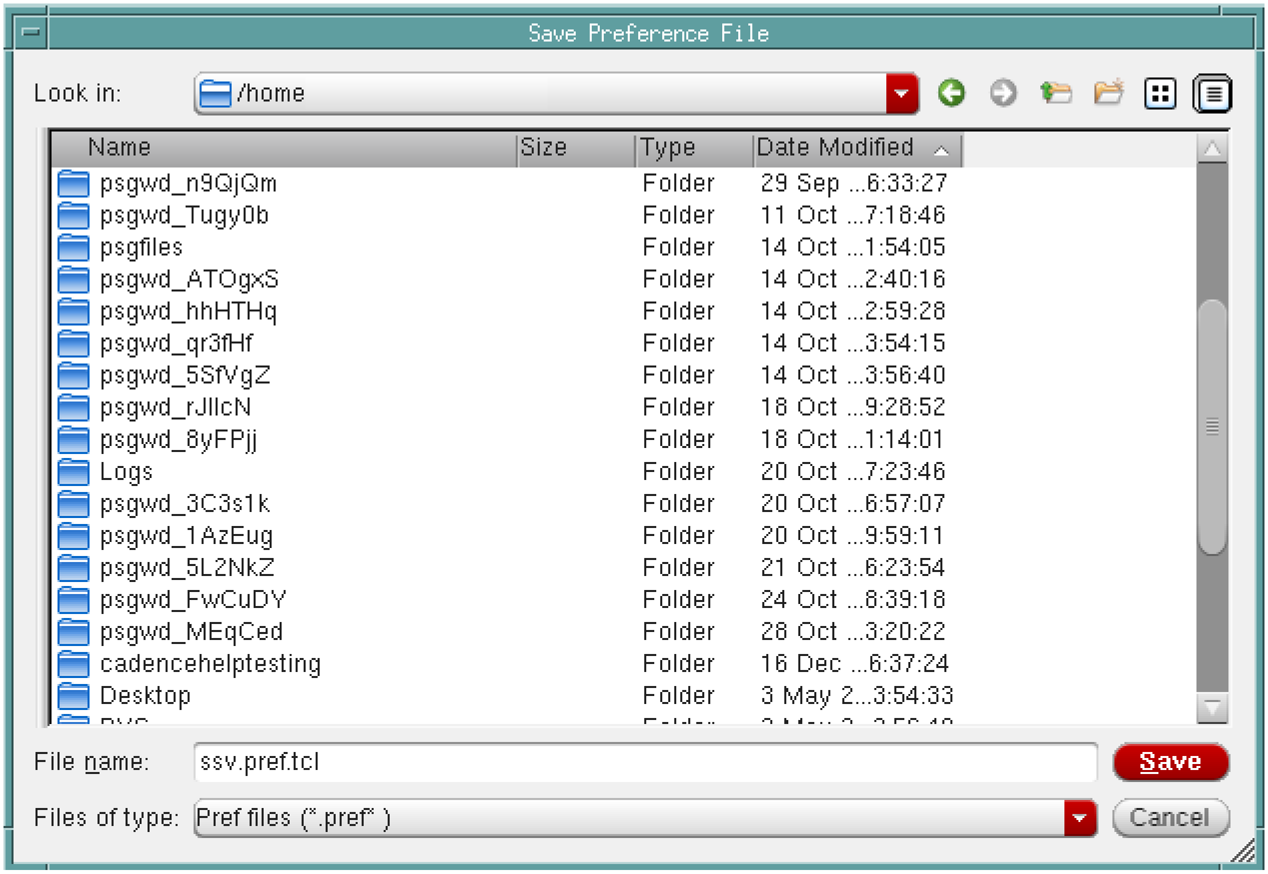
	Intersecting Box	Selects objects that are fully enclosed or intersected within the specified area. Use the left mouse button to draw and define an area or bounding box. Any objects enclosed inside or intersected with the area bounding box are selected. This mode should be used to select wire or row. *Default:* Off
Hilite Net When Selecting a Pin		
	Highlights a net through a pin selection. If this option is set when you select a pin, the corresponding net is highlighted. *Default:* Off	
Hilite Nets When Querying an Instance		
	Highlights nets that are connected to an instance when this instance is queried. For example, if you query a selected instance and this option is set, all nets that connect to this instance pin are also highlighted. *Default:* On	
Hilite Pin Shapes When Selecting a Net		
	Highlights connected cell term shapes when you select a net. *Default:* Off	
Query Nets When Wires are Selected		
	Queries all nets that connect to the selected wires. *Default:* Off	
Skip Objects by Auto Query		

	Skips querying of specified objects. You can choose to skip Auto Query of: ■ *Cell* ■ *Obs in Cell* ■ *Pin in Cell* ■ *Regular Wire* ■ *Special Wire*
Window Select Margin	Specifies the number of pixels for the window margin. *Default:* 8

Save Preference File

Use the Save Preference File form to save custom preference design settings to a file. By default, settings are saved in the `ssv.pref.tcl` file; however, custom settings can be saved to a different file.

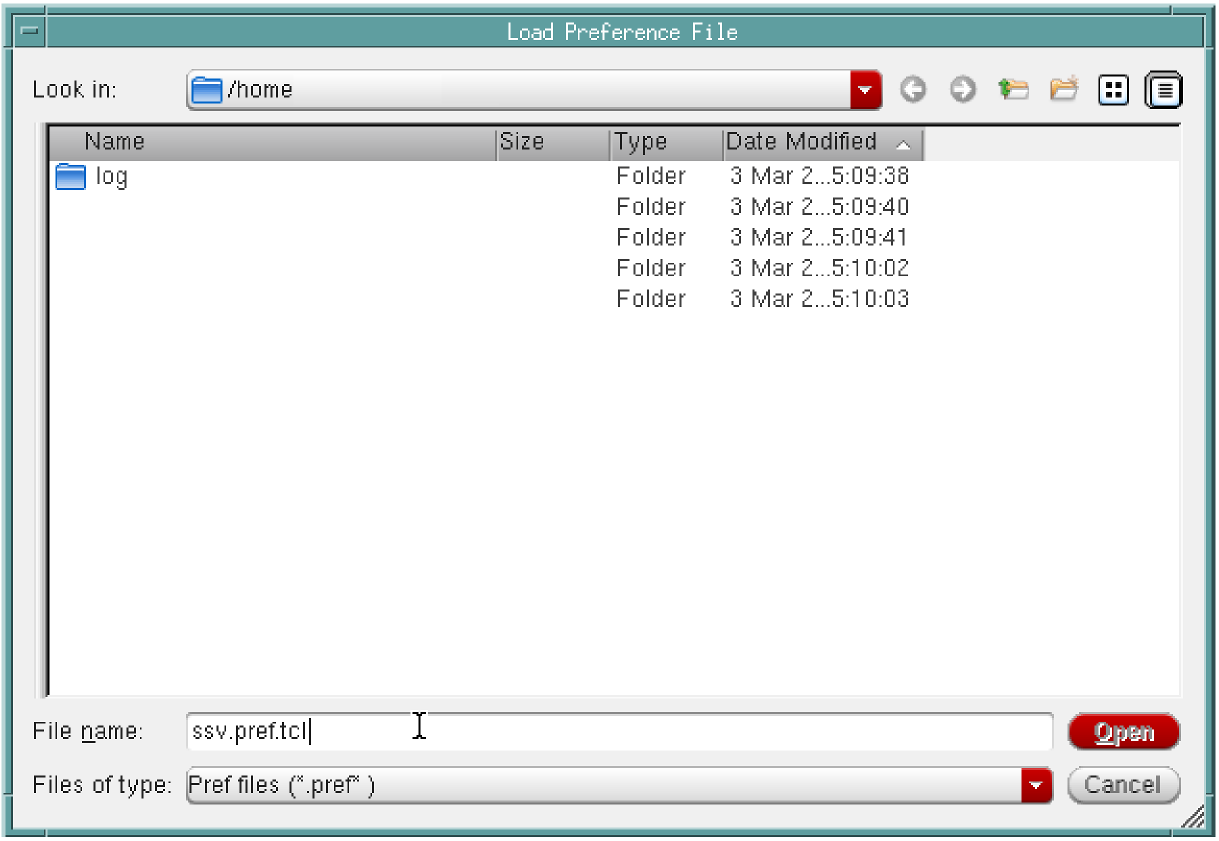
- Choose *View - Set Preference* to open the Preferences form and click the *Save* button.



Load Preference File

Use the Load Preference File form to load a previously saved file. By default, settings are saved in the ssv`.pref.tcl` file; however, custom settings can be saved to a different file.

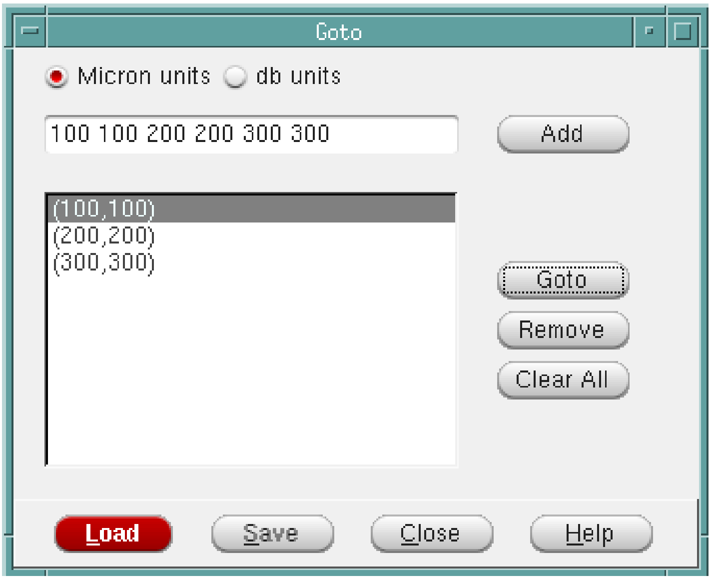
- Choose *View - Set Preference* to open the Preferences form and click the *Load* button.



Go To

Use the Goto form to go to a specific point in the main window display. To open the Goto form:

- Select *View - Go To*.



Goto Fields and Options

Micron units	Specifies the coordinates in micron or db units.
db units	
Add	Adds the specified coordinates to the list box below.
Goto	Zooms in to the selected location in the main window display and places a marker on the location. Click this button after selecting a coordinate from the list.
Remove	Removes the selected coordinate from the list and the corresponding marker from the main window display.
Clear All	Clears the list and removes all *Goto* markers from the main window display.
Load	Loads the coordinates from a file.
Save	Saves the coordinates in the list to a file.
Close	Closes the Goto form.

Find/Select Object

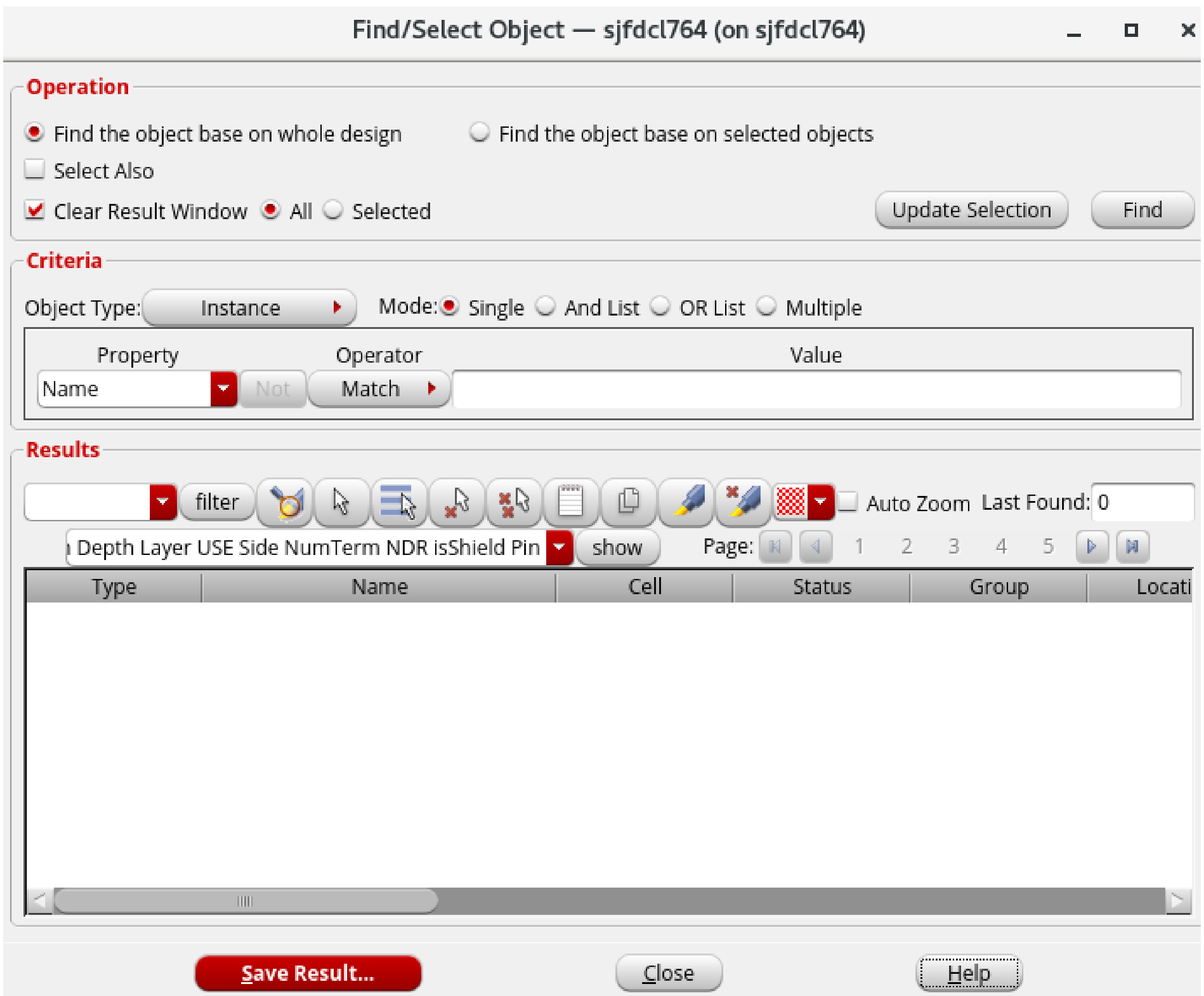
Use the *Find/Select Object* form to find and select specific object types in the design display area.

Note: The Select operations are recorded in the command file and the log file.

- Choose *View - Find/Select Object*.

Or

- Press `Shift+F`.

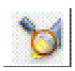
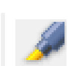


Find/Select Object Fields and Options

Object Type	Specifies the type of object to find. Choose from: *Instance*, *Pin, Net, Module, Instance Group*, or *Bump*. The Pin object type pin can be used for I/O pins, partition pins, and instance pins. Names of instance pins and/or partition pins should be specified with the full path name. For example, pin A of instance `DTMF_CHIP/alu` should be specified as: `DTMF_CHIP/alu/A` You can use the *Net* option to find/display bus names.
Mode	Specifies the type of find/select operation. Select one of the following: ■ *Single*: Performs a simple search for a Property/Value combination. ■ *And List*: Performs a find/select operation that matches an AND list of two or more multiple Property/Value combinations. ■ *OR List*: Performs a find/select operation that matches an OR list of two or more multiple Property/Value combinations. ■ *Multiple*: Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND lists and OR lists. For more information, see Mode Settings on page 96.

Property	Displays the available properties for the object that you selected in the Object Type field. You can use the pre-defined property Connection to find/select objects by connectivity. The Connection property supports connectivity search for the following objects: ■ Instance ■ Pin ■ Net ■ Module ■ InstanceGroup ■ Bump ■ Wire
NOT	Inverses the value of the operator in the Operator field. For example, If the value in the Operator field is <, clicking the NOT button effectively changes the value to NOT <.
Operator	Specifies the relationship between the property and its value. This field is dynamically populated based on the specified object type.
Value	Specifies the value of the property. This field is dynamically populated based on the specified object type. In some cases, you may have to manually specify a value, for example, the name of a cell.
Add	Adds the search criteria based on the values of Property, Operator, and Value fields. The search criteria are displayed in the form. This button is not available if you select the *Single* mode.
Delete	Deletes the currently selected search criteria. This button is not available in the *Single* mode.
Update	Updates the currently selected search criteria based on the values of Property, Operator, and Value fields. This button is not available in the *Single* mode.

Add List	Adds a blank list to the search criteria. This button is available only in the *Multiple* mode. If the And button is selected, an AND list is added If the OR button is selected, an OR list is added. You can then add search criteria to the list by specifying values in the Property, Operator, and Value fields, selecting the list in the search criteria, and clicking the *Add* button. You can create a NOT AND list or a NOT OR list by selecting the *Not* button before you click the *Add List* button.
And	Specifies that the list added through the *Add List* button is an AND list. This button is available only in the *Multiple* mode.
OR	Specifies that the list added through the Add List button is an OR list. This button is available only in the *Multiple* mode.
Not	Creates a NOT AND list or a NOT OR List when the Add List button is clicked.
Find the object base on whole design	Searches for objects that match the specified criteria in the entire design and reports them in the *Results* section. This is the default operation if no objects are selected in the design display area when you open the Find/Select Object form.
Select Also	Selects the objects matching the criteria in the design display area. *Default:* This check box is turned off by default. Objects are reported in the results window but are not selected in the design display area.
Find the object base on selected objects	Searches for the required object within the selected object list, instead of the entire design. This is the default operation if you have selected objects in the design display area before opening the Find/Select Object form.

Clear Result Window	Clears objects previously listed in the *Results* list, then finds the objects that meet the specified criteria and reports them in the Results list. ■ *Default:* The objects previously listed are retained in the results window along with the results of the current search. ■ Additionally, you can choose one of the following clear options: ■ *All* - Clears all objects previously listed in the *Results* list. ■ *Selected* - Clears selected entries from the *Results* list.
Find Only	Finds all objects that meet the specified criteria and reports them in the *Results* window.
Zoom in Highlighted 	Zooms in the highlighted object(s). You can highlight object(s) by selecting them in the results window and clicking the *Highlight* button
Select 	When you click *Select*, the objects selected in the Results window are selected in the design display area.
Select All 	When you click *Select All*, all the objects in the Results window are selected in the design display area.
Deselect 	When you click *Deselect*, the objects selected in the Results window are deselected in the design display area.
Deselect All 	When you click *Deselect All*, all the objects in the Results window are deselected in the design display area.
Highlight 	When you click *Highlight*, the objects selected in the Results window are highlighted in the design display area.
Dehighlight 	When you click *Dehighlight*, the objects selected in the Results window are dehighlighted in the design display area.
Highlight Color 	Specifies color for highlighting objects in the design display area.

Auto Zoom	Automatically zooms in to the selected object in the design display area as you browse through the search results in the Find/Select Object form. The *Auto Zoom* option is off by default.

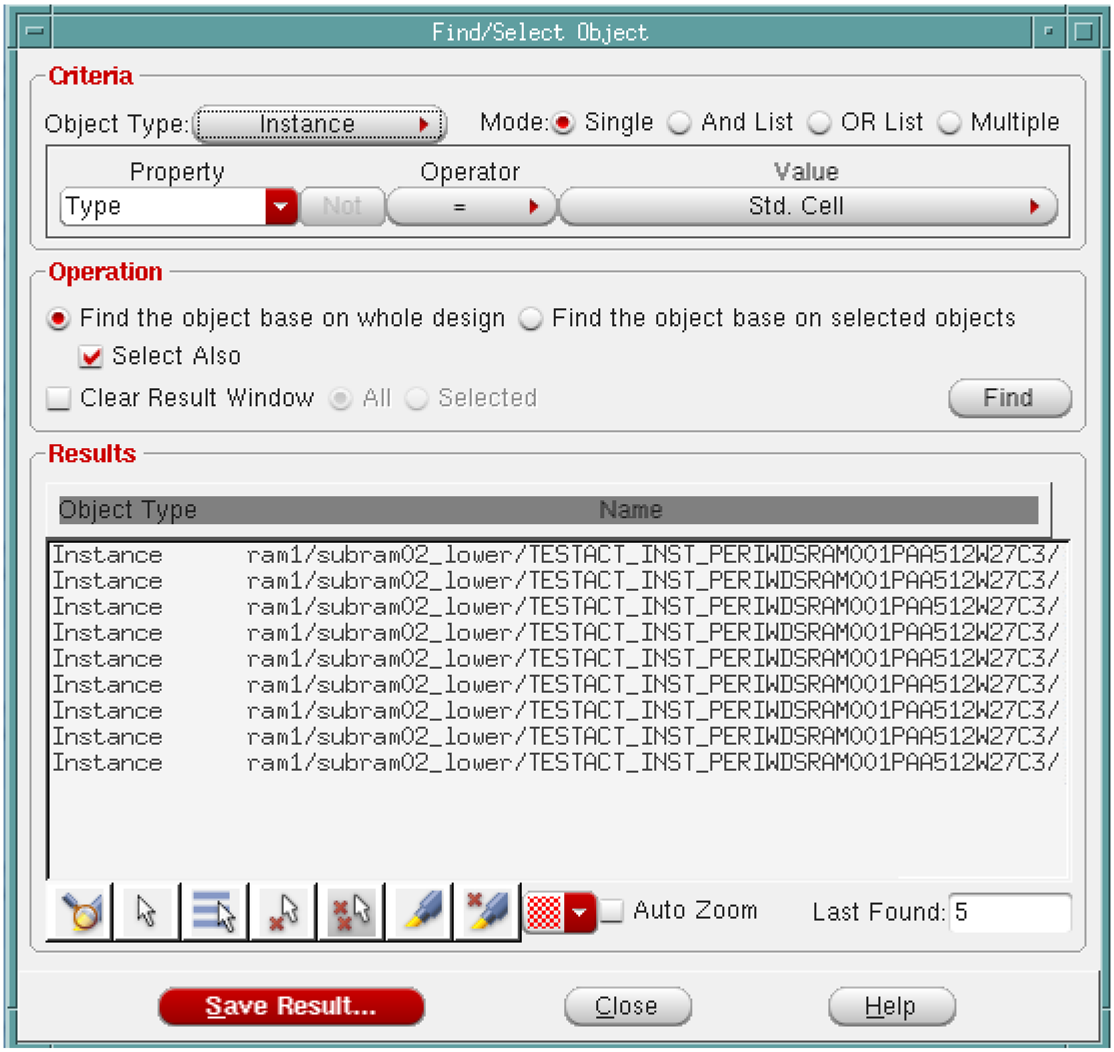
Mode Settings

The *Mode* field in the *Find/Select Object* form provides the following options:

- Single on page 97
- And List on page 97
- OR List on page 99
- Multiple on page 99

Single

Performs a simple search for a Property/Value combination. The following example illustrates the search for all instances of the type Standard Cell.

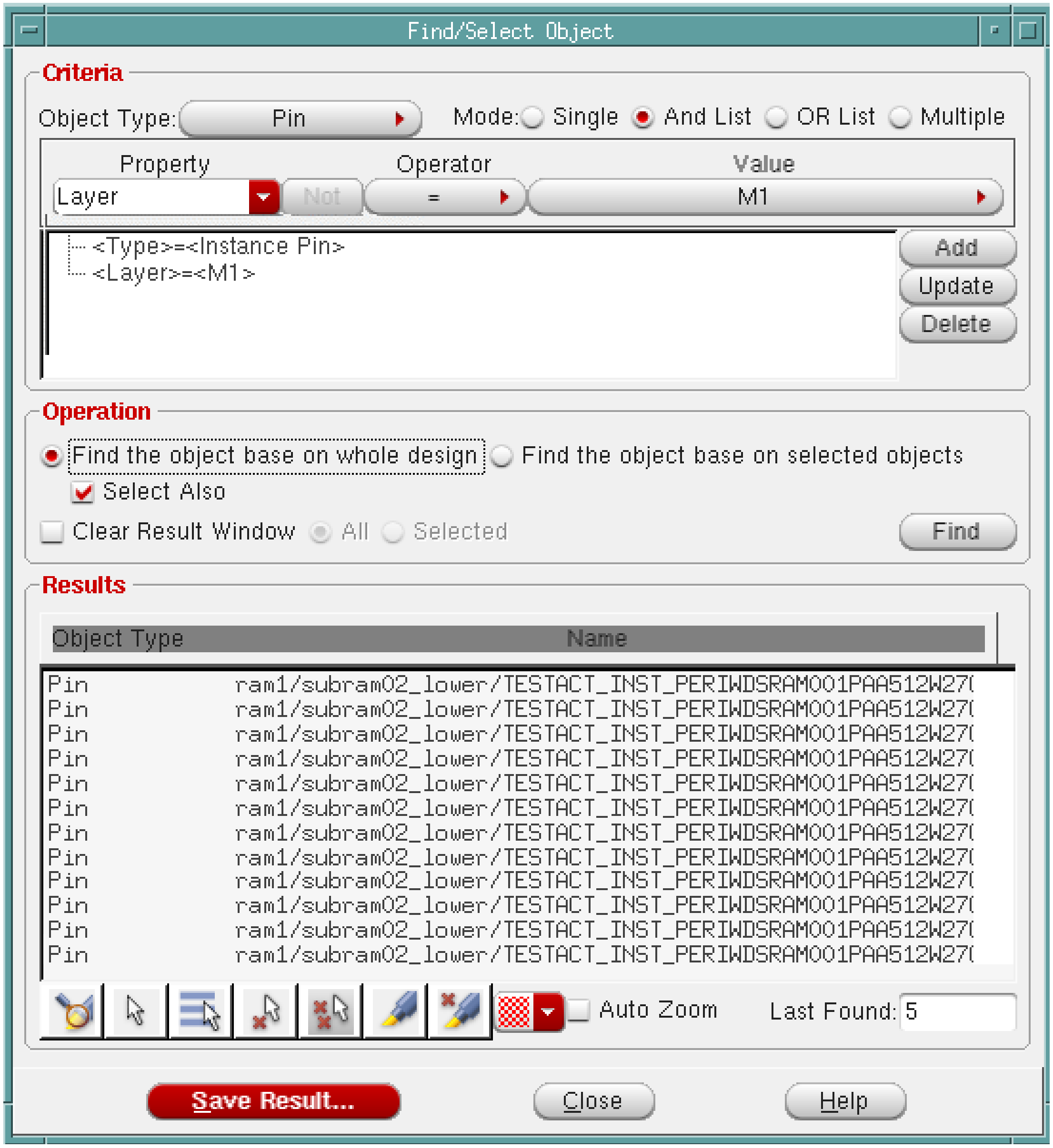


And List

Performs a find/select operation that matches an AND list of two or more Property/Value combinations. The following example illustrates the search for all pins that match *all* of the following criteria:

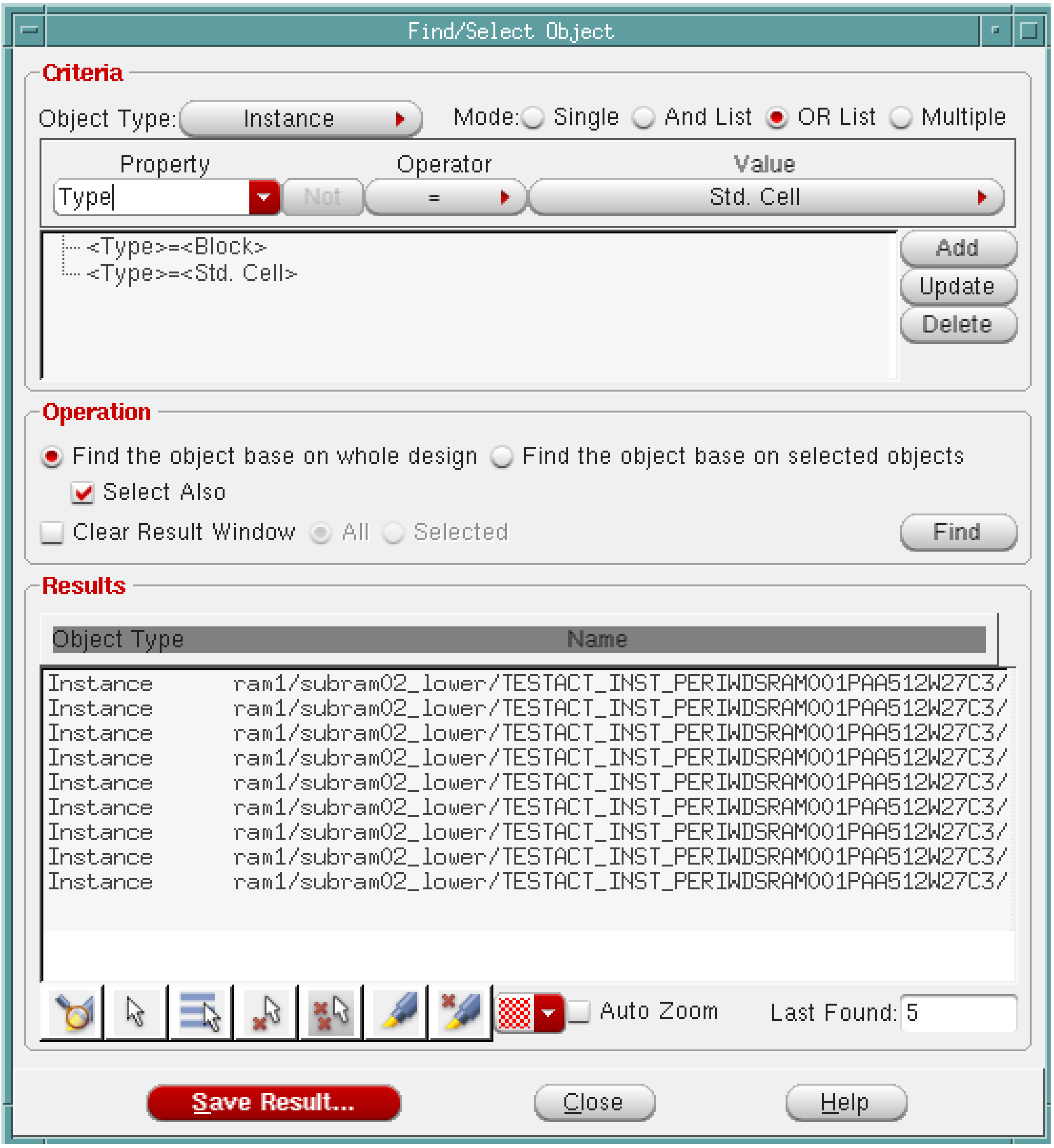
- Type is Instance Pin

- Layer is M1


Find/Select Object
Criteria
Object Type: Pin
Mode: Single And List OR List Multiple
Property Operator Value
Layer Not = M1
<Type>=<Instance Pin>
<Layer>=<M1>
Add
Update
Delete
Operation
Find the object base on whole design
Find the object base on selected objects
Select Also
Clear Result Window All Selected
Find
Results
Object Type Name
Pin ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27(
Auto Zoom
Last Found: 5
Save Result...
Close
Help

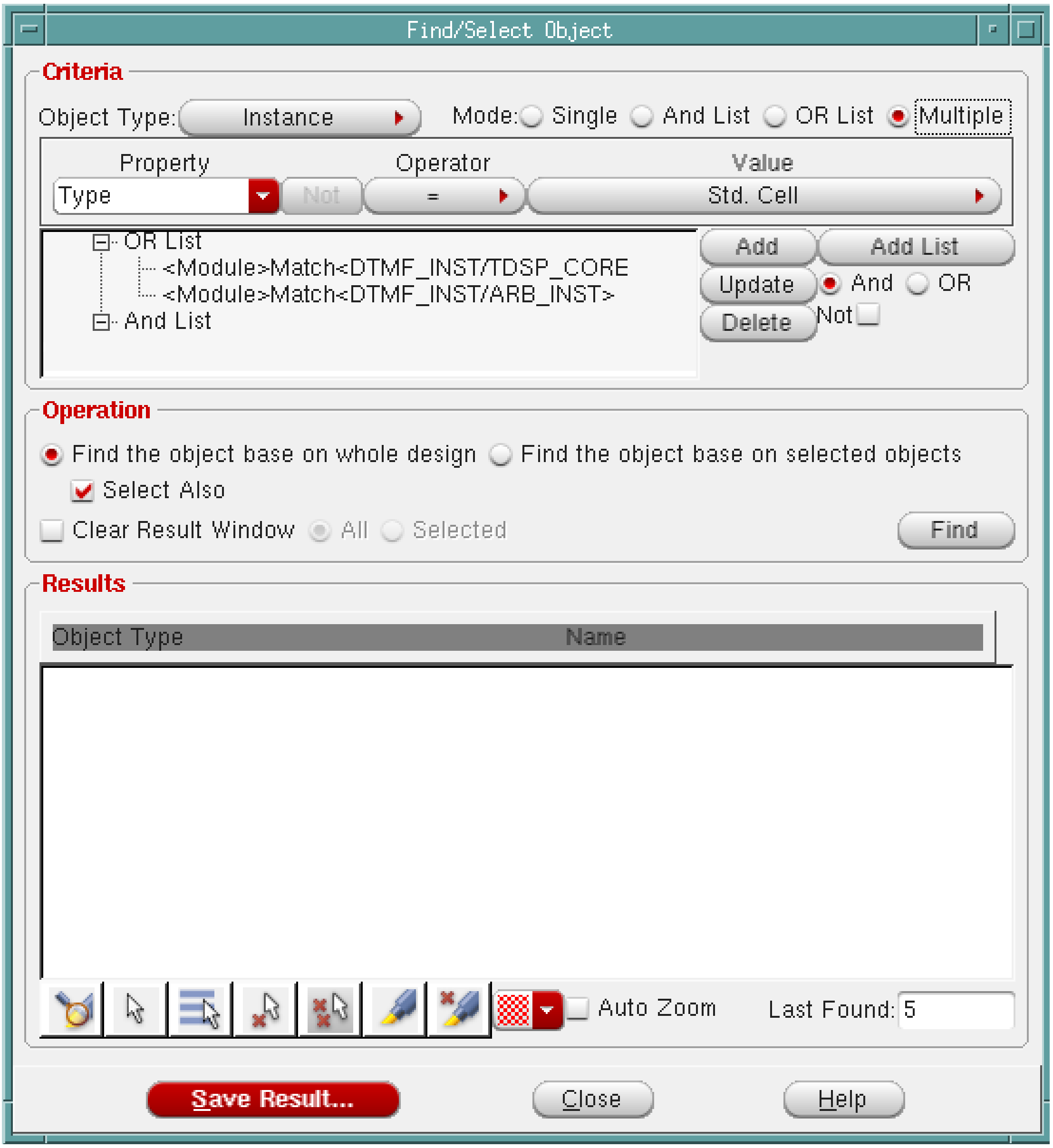
OR List

Performs a find/select operation that matches an OR list of two or more Property/Value combinations. The following example illustrates the search for all instances whose type is Block or Standard Cell.



Multiple

Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND and OR lists. The following example illustrates the find/select operation for all pins of the modules `DTMF_INST/TDSP_CORE_INST` and `DTMF_INST/ARB_INST` that are on the top side and have a status of *Placed*.



Deselect All

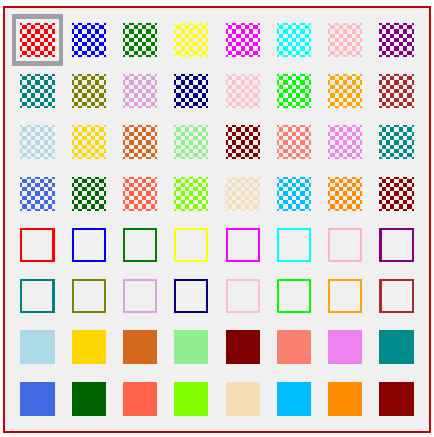
Use the *Deselect All* menu command to deselect all selected nets.

- Choose *View - Deselect All*.

Highlight Selected

Use the *Highlight Selected* menu command to highlight the object or object set you have selected in the design display area. You can select object sets using the Find/Select Object form.

- Select an object or object set.
- Click *View - Highlight Selected* and select any one of the colors to highlight the selected area as shown below:

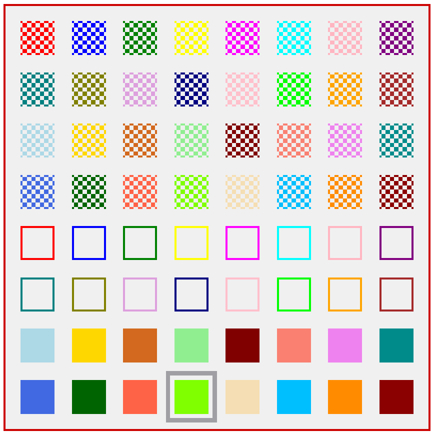


Note: The Highlight option is also available from the context menu, which you can open by right-clicking an object in the design display area.

Clear Highlight

Use the *Clear Highlight* submenu to clear highlights.

- Choose *View - Clear Highlight* and select any one of the color to clear the highlights from the object selected in the design display area as shown below:

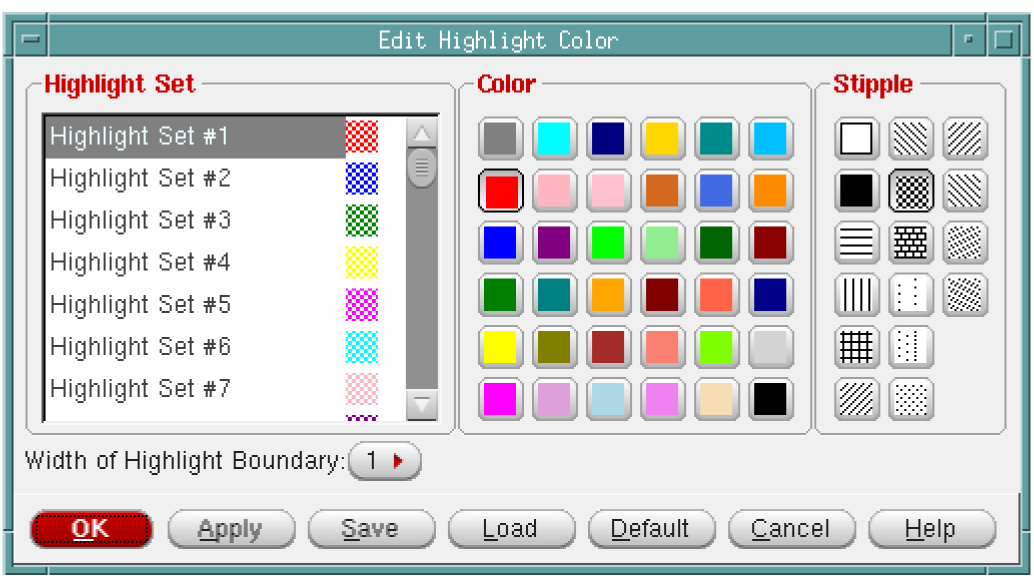


Note: The Dehighlight option is also available from the context menu, which you can open by right-clicking an object in the design display area.

Edit Highlight Color

Use the Edit Highlight Color form to further customize the highlight color and pattern of selected object sets in the design display area.

- Choose *View - Edit Highlight Color*.



Edit Highlight Color Fields and Options

Highlight Set	Lists the available highlight sets. To customize a highlight set: ■ Select the set. ■ Choose the appropriate highlight color and pattern from the *Color* and *Stipple* area, respectively. ■ Click *Apply*.
Color	Specifies the highlight color for the selected highlight set.
Stipple	Specifies the highlight pattern for the selected set.
Width of Highlight Boundary	Specifies the width of highlight boundary. You can set the number as 1, 2, 3, 4, or 5. *Default:* 1
Save	Saves the highlight settings. The default filename is enc.hilite.tcl.
Load	Loads highlight settings previously saved in a file.
Default	Reverts to the default settings for all highlight sets.

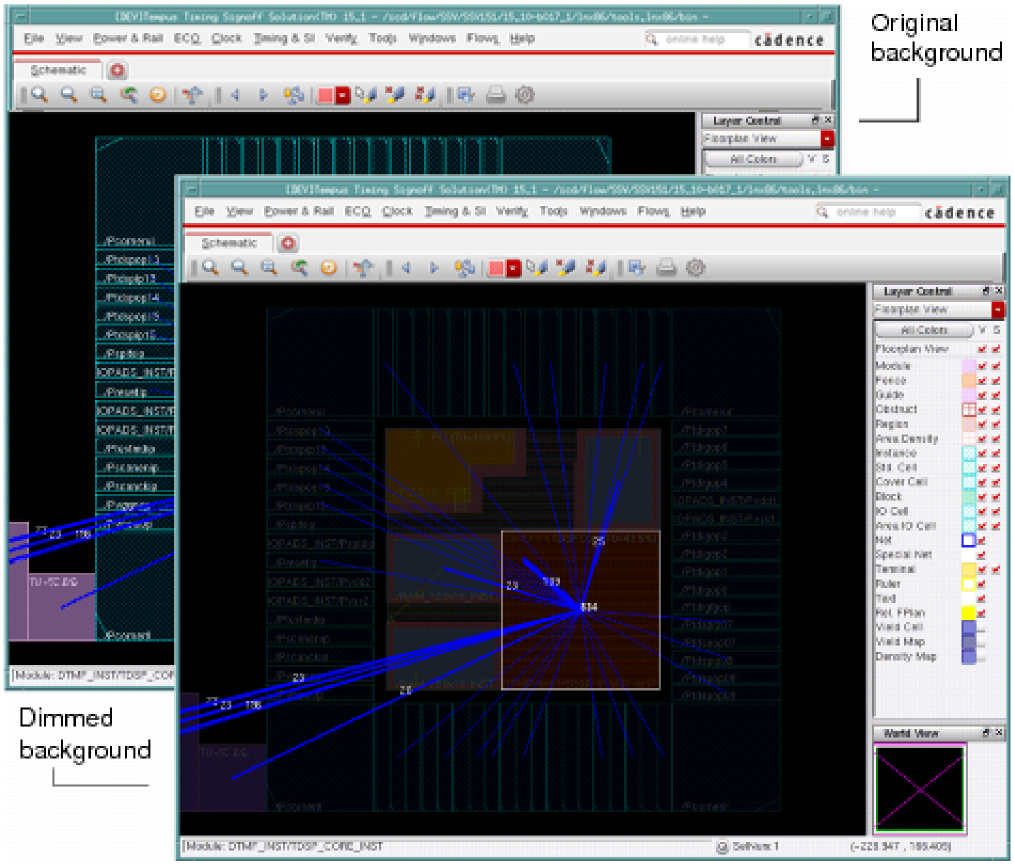
Dim Background

Use the *View - Dim Background* menu command to dim the background of the Tempus display. This enhancement helps you view selected and highlighted objects more clearly. This is especially useful when you are debugging.

To enable the dim mode:

☨ Select the *View - Dim Background* menu command.

The equivalent key is `F12`. The `F12` key is used as a three-way toggle; so if you want to quit dim mode, press the `F12` key again twice.



4

Power and Rail Menu

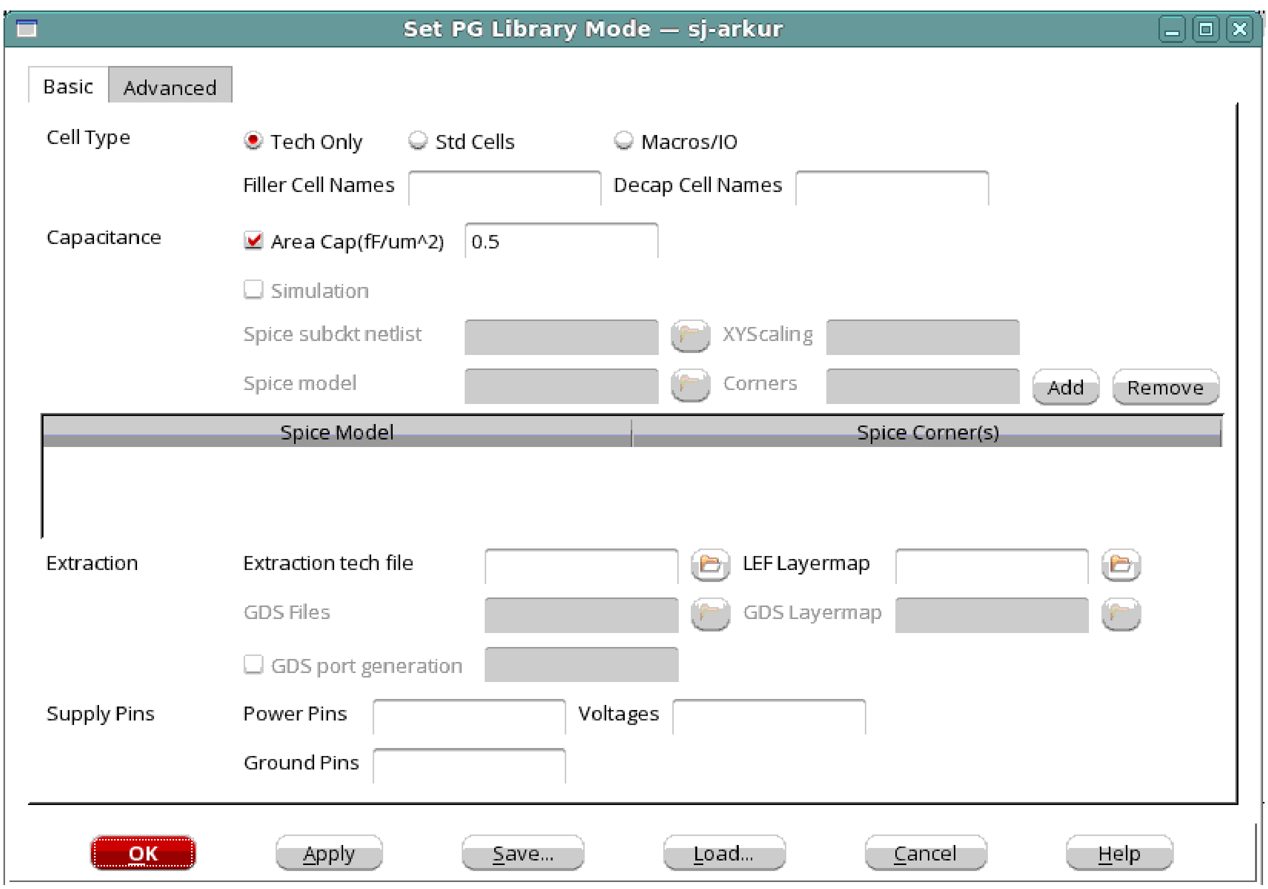
Power Grid Library

The *Power Grid Library* submenu contains the following items:

- Set PG Library Mode - Basic
- Set PG Library Mode - Advanced
- Generate PG Library

Set PG Library Mode - Basic

Specifies what kind of power-grid library will be created. This form is used to define the type of input and configuration for generating power-grid libraries.



Fields and Options

<table>
<tr><td>Cell Type</td><td>Specifies the cell type used for library creation. The choices are Tech Only, Std Cells, Macros/IOs.

The default is Tech Only.

■ Tech Only
Creates a technology Library that contains the tech file, area capacitance specification, decap/filler/powergate cells, and Tech view for all the cells. A technology library is a requirement for running rail analysis.

■ Std Cells
Generates a power-grid library for all the standard cells using the Spice netlist. The stdcells power-grid library does not contain the tech file. Power-grid is connected at the LEF pin, and all the capacitance are derived from simulation. The power-grid library contains three kind of power-grid views (PGVs), Early/IR/EM, and the views are selected based on the accuracy mode of rail analysis.

■ Macros/IOs
Generates a power-grid library for memory, I/Os and macros using the Spice and GDS netlist. The GDS is mandatory for the cell type macros. The macros power-grid library does not contain the tech file. Power-grid is connected down to contact/specified via, and all the capacitance are derived from simulation. The power-grid library contains three kind of PGVs, Early/IR/EM, and the views are selected based on the accuracy mode of rail analysis.</td></tr>
<tr><td>Filler Cell Names</td><td>Lists the names of the filler cells that have no GDS data. Wildcards (*) can be used. These cells can be swapped out for decaps during decap optimization.</td></tr>
<tr><td>Decap Cell Names</td><td>Specifies the names of cells that have explicit decoupling capacitance for use in decap optimization. These cells are nearly equivalent to filler cells, except that there is typically an active device between the power and ground nets to provide extra decoupling capacitance. Power-grid view library generation uses this list of cells to ensure that a capacitance value is associated with each cell. If no capacitance is associated with a cell, a warning message will be issued.

You can use wildcards (*).</td></tr>
<tr><td colspan="2">Capacitance</td></tr>
</table>

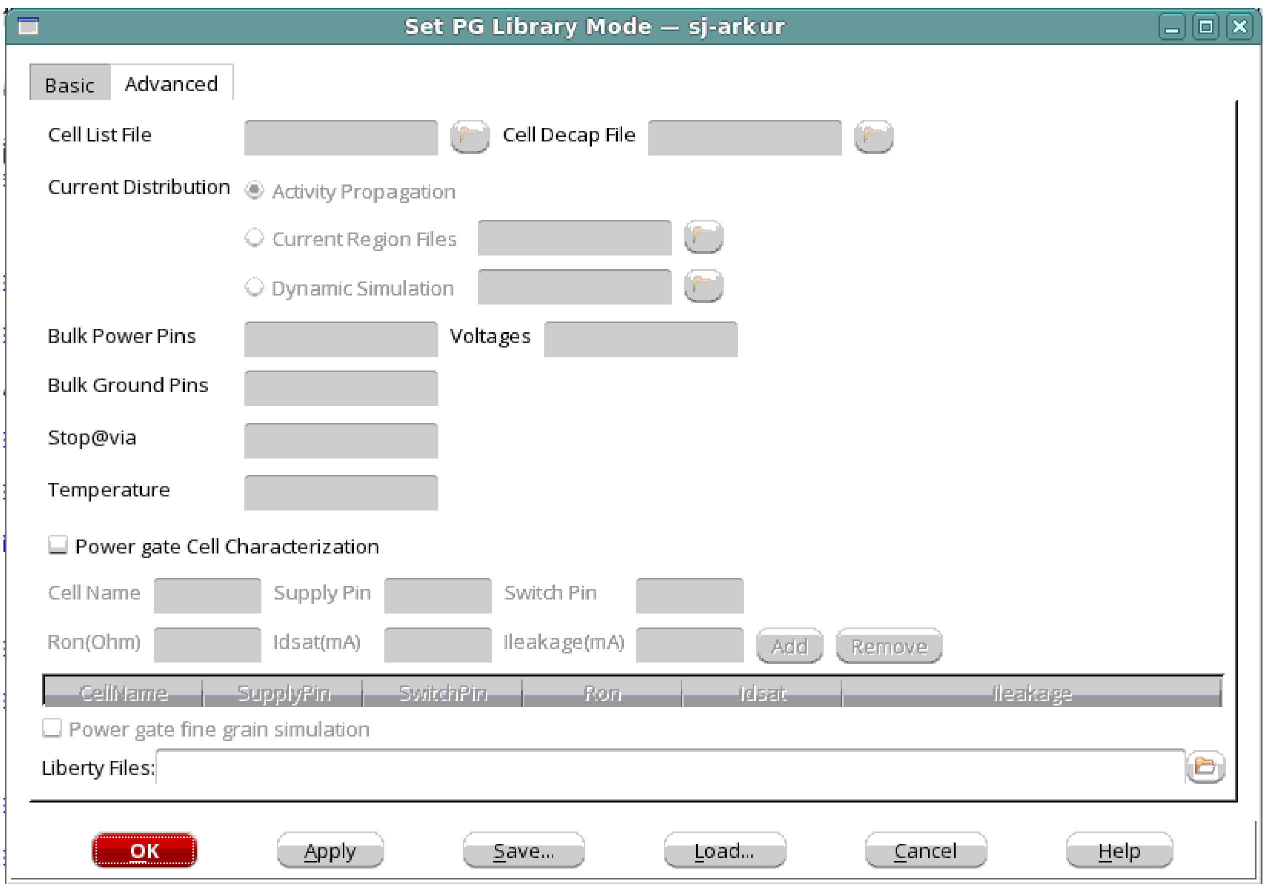
Area Cap (fF/ um^2)	Specifies the default amount of area based decoupling capacitance in a cell. This option is enabled for the cell type *Tech Only*.
Simulation	Specifies that the capacitance is derived from netlist simulation. This option is enabled for the cell types *Std Cells and Macros/IOs*.
Spice subckt netlist	Specifies the names of the SPICE netlist files containing the subcircuits corresponding to the cells being processed during power-grid library generation. These subcircuit netlists will be used to calculate pin capacitances using Voltus library simulator.
XY Scaling	Specifies the current region scale factor for the Netlist-based flow. The current region scale factor is the factor by which the netlist coordinates are scaled before finding the closest node to attach the tap currents.
Spice Model	Specifies the names of the SPICE model files used by the SPICE netlist file.
Corners	Specifies a list of SPICE corners. This field is optional. If you specify SPICE corner(s), the number of corner items should be equal to the number of SPICE model files specified in the *Spice Mod*el field.
Extraction	
Extraction Tech File	Specifies the name of the technology file that will be used for extraction.
LEF Layermap	Specifies a file that provides the mapping of layers between the LEF and technology file.
GDS Files	Specifies the names of the GDSII files used during power-grid library generation.
GDS Layermap	Specifies a file that provides the mapping of layers between the LEF and GDS files.
GDS Port Generation	Identifies the layer name that library generation uses to compute the placement bounding box for GDSII cells that are placed by the DEF file.
Supply Pins	
Power Pins	Specifies the names of the nets considered power nets for all cells for which the power nets are not specified.
Voltages	Specifies the voltages for the power pins.
Ground Pins	Specifies the names of the nets considered power nets for all cells for which the power nets are not specified.

Related Text Commands

See `set_pg_library_mode` and `set_advanced_pg_library_mode` command in the *Voltus Text Command Reference*.

Set PG Library Mode - Advanced

Specifies the advanced power-grid library generation features.



Fields and Options

Cell List File	Specifies a file containing a list of cells that will be processed. The *Cell List File* field is required when celltype *Macros/IOs* is selected. The program spawns the library generation job in serial mode for each macro in the cell list file. A separate library for each macro is generated. This facilitates debugging when library generation of any macro fails. It also helps in improving the overall performance.
Cell Decap File	Specifies the name of a file containing a list of cells and the decoupling capacitance value for each.
Current Distribution	Specifies the method of current distribution inside the cell. ■ **Activity Propagation** uses Voltus library simulator to determine the propagation activity to distribute the current. ■ **Current Region Files** specifies the amount of current to distribute within a cell region. ■ **Dynamic simulation** uses a trigger file to run detailed analysis to determine current distribution.
Bulk Power Pins	Specifies the names of nets that connect transistor bulk terminals to the power net. This information is used to prevent any current from being assigned to these nets. If a net is connected to any transistor terminals other than a bulk terminal, you must specify that net in the *Power Pins* field.
Voltages	Specifies the voltages for the bulk power pins.
Bulk Power Pins	Specifies the names of nets that connect transistor bulk terminals to the ground net. This information is used to prevent any current from being assigned to these nets. If a net is connected to any transistor terminals other than a bulk terminal, you must specify that net in the *Ground Pins* field.
Stop@via	Stops the extraction of the power-grid network inside a cell at the specified via layer.
Temperature	Specifies the temperature in degree celcius used in library characterization for thermal aware EM/IR analysis.

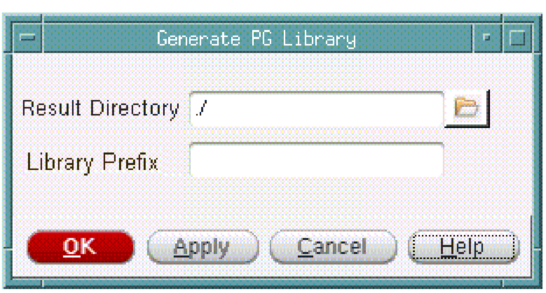
Power Gate Cell Characterization	Specifies the information that power-grid library generation needs to create a port power gate view. Specifies the cell name, the unswitched power pin, the switched power pin, the on resistance in ohms, the saturation current in milliamps, and the leakage current in milliamps. If Ron, Idsat, orIleakage, values are specified, they will be used to override the data extracted by Thunder.
Cell Name	Specifies the power gate cell name.
Supply Pin	Specifies the power gate supply pin (unswitched power pin).
Switch Pin	Specifies the power gate switched power pin.
Ron (ohms)	Specifies the power gate on resistance.
Idsat (mA)	Specifies the power gate saturation current.
Ileakage (mA)	Specifies the power gate leakage current.
Power gate fine grain simulation	Specifies to enable finegrain powergate simulation.

Related Text Commands

See `set_pg_library_mode` and `set_advanced_pg_library_mode` command in the *Voltus Text Command Reference*.

Generate PG Library

Generates the output for power-grid library generation.



Fields and Options

Result Directory	Generates the output for power-grid library generation in the specified directory.
Library Prefix	Specifies a prefix for the power-grid library. When this option is specified, the default PGV for each cell type is as follows: ■ techonly - `prefix_techonly.cl` (default: `techonly.cl`) ■ stdcells - `prefix_stdcells.cl` (`default: stdcells.cl`) ■ macros - `prefix_cellname.cl` (default: `macros_cellname.cl`)

Related Text Commands

See `generate_pg_library` command in the *Voltus Text Command Reference*.

Power Analysis

The Power Analysis sub-menu contains the following items:

- Set Power Analysis Mode
  Opens the Set Power Analysis Mode form
- Run Power Analysis
  Opens the Run Power Analysis form

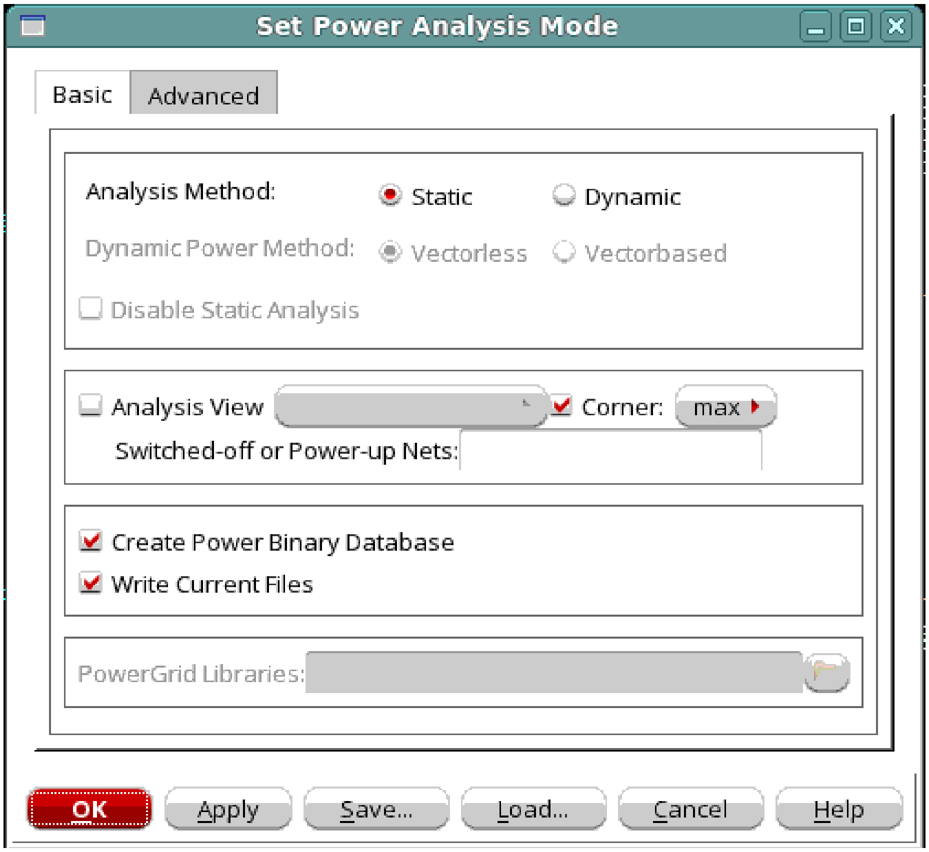
Set Power Analysis Mode

The Set Power Analysis Mode form contains the following pages:

- Set Power Analysis Mode - Basic
- Set Power Analysis Mode - Advanced

Set Power Analysis Mode - Basic

Specifies whether static or dynamic power analysis should be run and some additional settings.

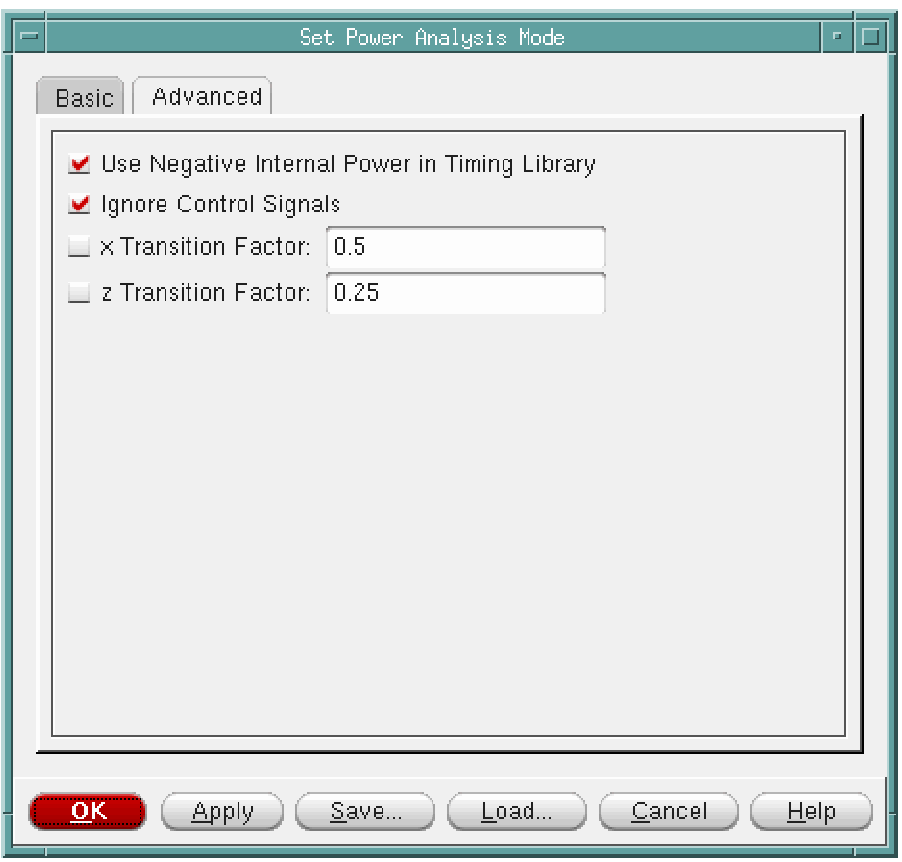


Fields and Options

Analysis Method	Specifies either static or dynamic power analysis
Dynamic Power Method	If dynamic power analysis is set this specifies if the method of analysis is vector less or vector-based.
Disable Static Analysis	When selected in the dynamic vector-based or dynamic vectorless flows, the command performs only dynamic analysis and turns-off static power calculation. *Default*: Not selected for Static analysis method, selected for Dynamic analysis method.
Analysis View	Specifies the Analysis view.

Corner	Specifies the corner via a pull-down list. The choices are min, or max. *Default*: `max`
Switch off Power Nets	Specifies if power nets should be switched off.
Create Power Binary Database	Specifies that a binary power database should be created. This can be used for viewing the power data graphically.
Write Current Files	Specifies to generate the current data files, one current data file per net. *Default*: `true`
PowerGrid Libraries	Specifies the name of the Cadence power cell libraries (.cl).

Set Power Analysis Mode - Advanced



Fields and Options

Use Negative Internal Power in Timing Library	Specifies to keep negative internal energy numbers from the .lib internal power table. When not set, negative power is treated as 0. *Default*: `true`
Ignore Control Signals	If set specifies that control signals will be ignored when propagating activity. *Default*: `true`
X Transition Factor	When using VCD one can specify how transitions to and from X are counted. These transitions are defined as shown in Table. Transitions to and from the X state will be treated as full transitions multiplied by the specified factor. *Default*: `0.5`
Z Transition Factor	When using VCD one can specify how transitions to and from Z are counted. These transitions are defined as shown in Table. Transitions to and from the Z state will be treated as full transitions multiplied by the specified factor. *Default*: `0.25`

Transitions from/to X and Z Values

Transition	Definition	Default
to/from X	[0 or 1 or Z] <-> X	0.5
to/from Z	[0 or 1] <-> Z	0.25

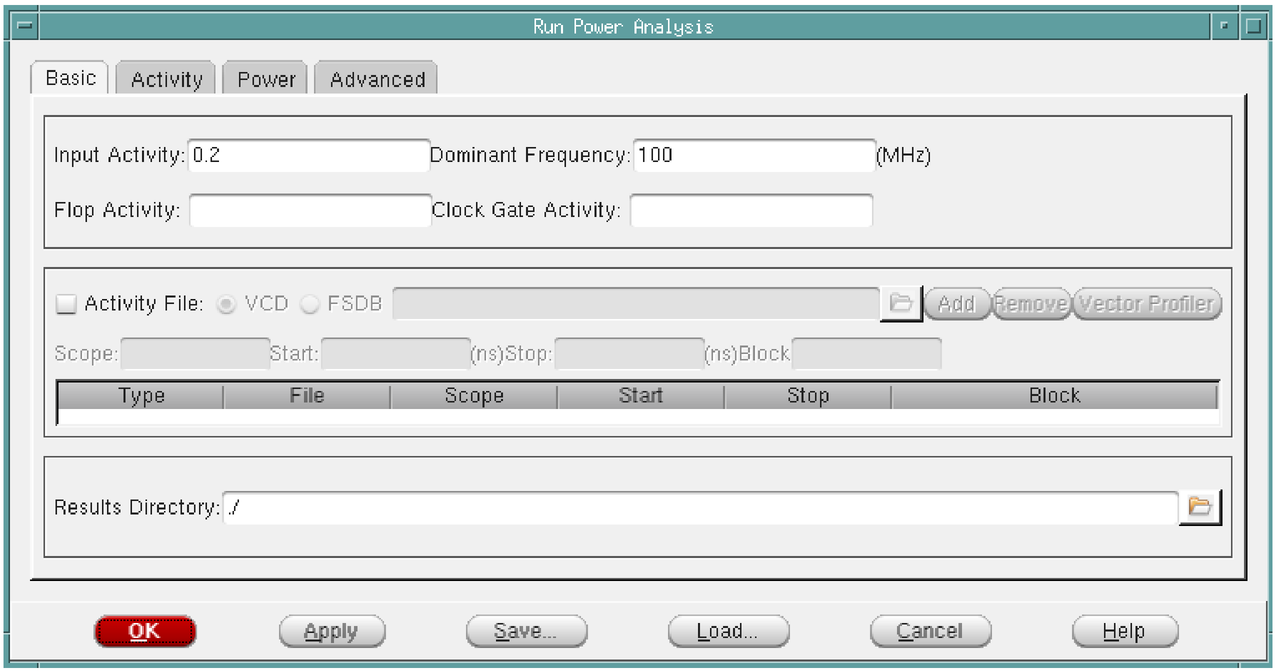
Run Power Analysis

The Run Power Analysis form has five pages:

- Run Power Analysis - Basic
- Run Power Analysis - Activity
- Run Power Analysis - Power
- Run Power Analysis - Advanced

Run Power Analysis - Basic

Runs the power analysis with the provided parameters.



Fields and Options

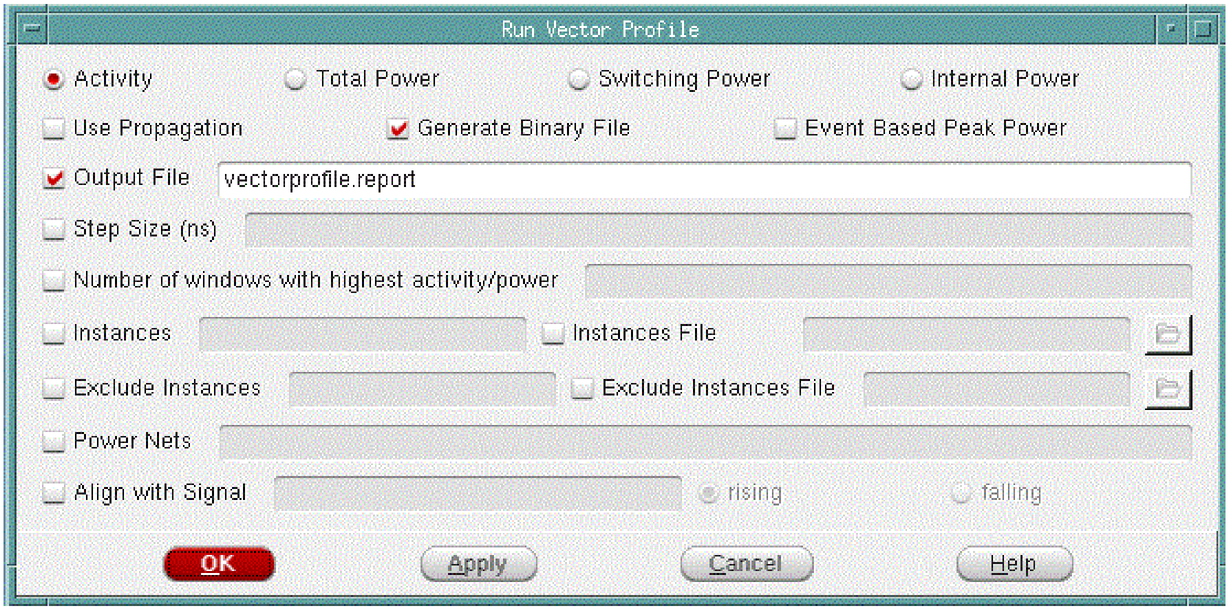
Input Activity	Specifies the average number of times as a (Specified as a fraction) that a primary input switches in a clock cycle. It can be any number between 0 and 1. *Default*: `0.2`
Dominant Frequency	Specifies the dominant frequency of the design.
Flop Activity	Specifies the activity of outputs of sequential logic. There is no default.

Clock Gate Activity	Specifies the average number of times that a clock-gating cell switches in a clock cycle at the output of the pin. Enter any positive number. There is no default. **Note:** The value that you specify in GUI will overwrite the value in `set_default_switching_activity` `-clock_gates_output` parameter.
Activity File	If selected, activates the fields for entering VCD/FSDB information. A VCD/FSDB file must be specified.
Vector Profiler	When clicked, displays the *Run Vector Profile* window.
Scope	Specifies the scope of the VCD file.
Start	Specifies the start time in nanoseconds (ns) of the VCD file that will be used for power analysis.
Stop	Specifies the stop time in nanoseconds (ns) of the VCD file that will be used for power analysis.
Block	Specifies the block name.
Results Directory	Specifies the directory where the power analysis output will be placed.

Run Vector Profile

Use this form to create Vector profile report for the specified VCD file.

- Choose *Power & Rail - Run Power Analysis - Basic* tab, and then click the *Vector Profiler* button.



Fields and Options

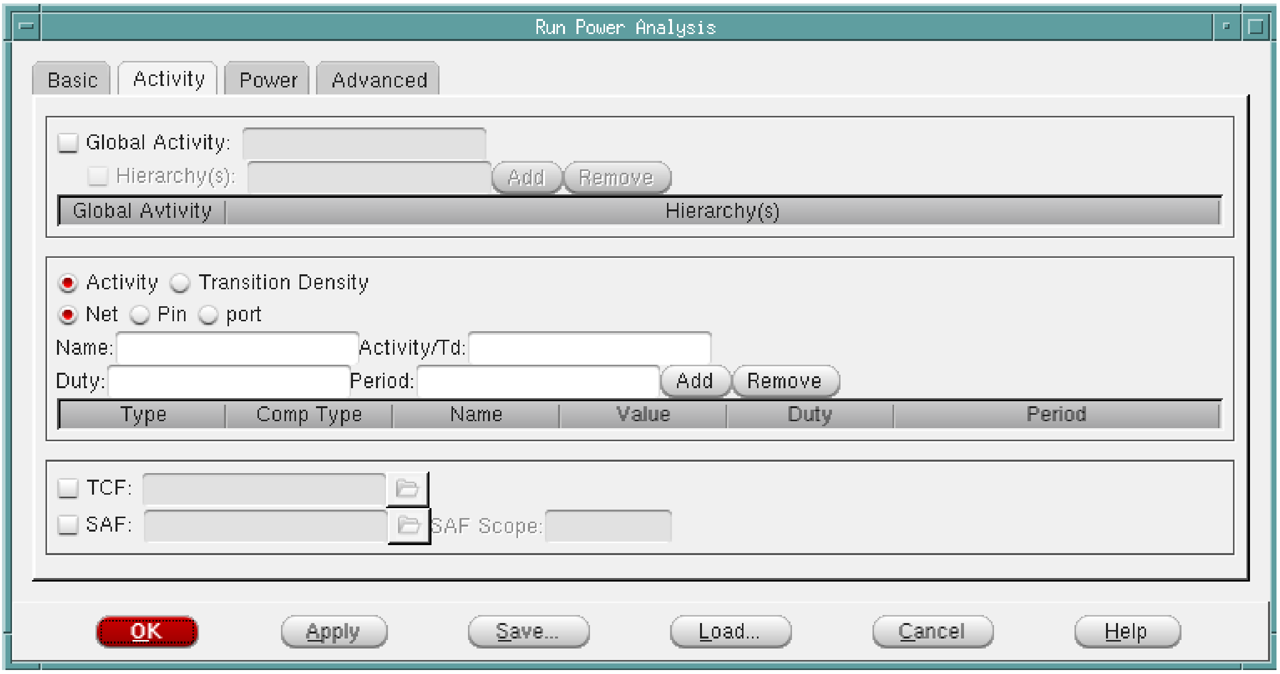
Activity	Enables activity profiling. Reports the number of toggles for every window.
Total Power	Enables total power profiling. Reports total power in milliWatts (mW).
Switching Power	Enables switching power profiling. Reports switching power in milliWatts (mW).
Internal Power	Enables internal power profiling. Reports internal power in milliWatts (mW).
Use Propagation	Enables propagation when partial VCD is provided.
Event Based Peak Power	Specifies to enable event-based vector profiling. This method computes power profile of every event on each net. The event-based peak power profiling enables you to accurately capture vectors that could produce peak power using very small resolution (1ps) and with better performance than average power profiling which uses larger resolution to compute average toggle density.
Output File	Writes out the Vector profiling report into the file that you specify. By default, the report is written out into vcdprofile.report file.

Step Size (ns)	Specifies the step size, in nanoseconds (ns). Activity / Power profiling is carried out for every step size that you specify.
Number of windows with highest activity/ power	Provides n top windows with maximum activity or power. *Default*: 3
Instances	Enables profiling of the instances that you specify.

Instances File	Enables profiling of the instances from the instance file that you specify.
	For example, instances in an instance file can be specified as follows:
	`-------------------`
	`inst_1`
	`inst_2`
	`inst_3`
	`.`
	`.`
	`.`
	`inst_n`
	`-------------------`

Exclude Instances	Disables counting of toggles or calculating power for the specified instances while reporting activity or calculating power for the entire design.
Exclude Instances File	Disables counting of toggles or calculating power for the specified instances file while reporting activity or calculating power for the entire design.
Power Nets	Enables power profiling for the specified power net(s).
Align with Signal	Starts profiling at the first rise or fall of the signal. Select *rising* to start profiling at the first rising edge. Select *falling* to start profiling at the first falling edge.

Run Power Analysis - Activity

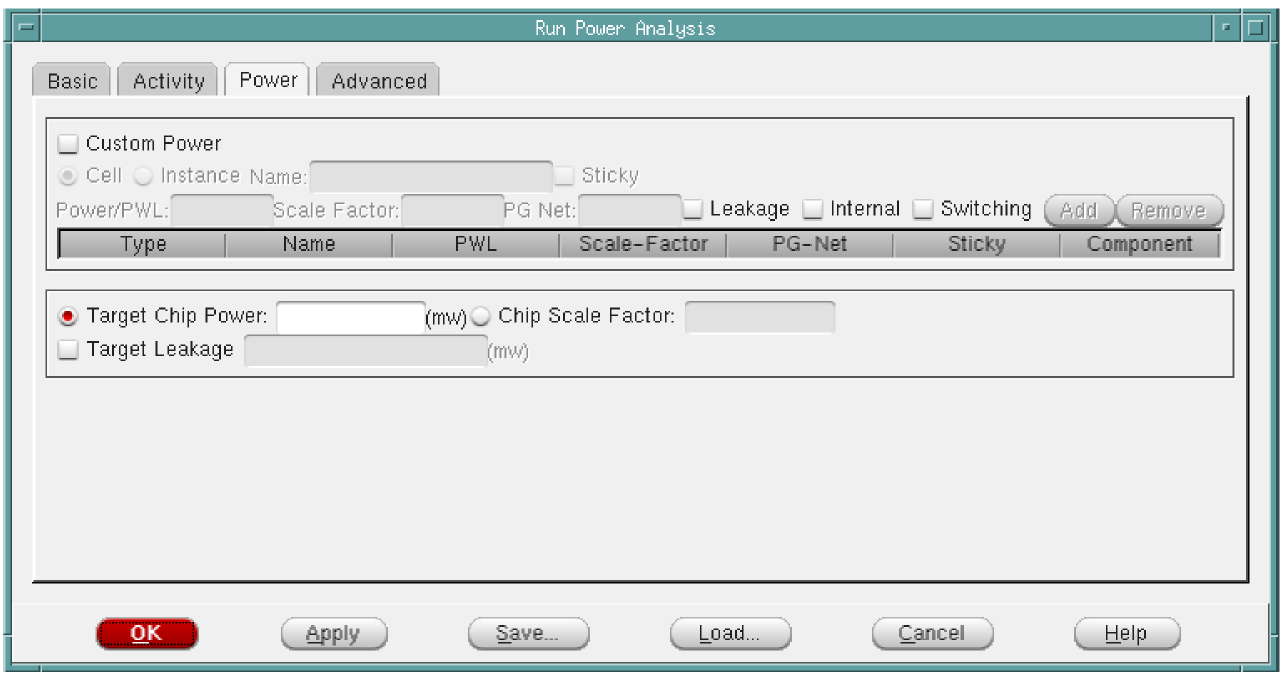


Fields and Options

Global Activity	Specifies the average number of times that all unset nodes switch in a clock cycle. If not specified the fastest domain frequency will be used unless one turns off clock domains. There is no default.
Hierarchy(s)	A hierarchical name that the Global Activity is being assigned to. If not provided the whole design is assumed. Selecting the Add button adds hierarchical activity information to a list. The Remove button removes a selected item in the list. To highlight multiple items, click them while pressing either the *Shift* or *Control* key.
Activity Transition Density	Specifies whether *Activity* or *Transition* Density numbers will be used when supplying *Net*, *Pin*, or *Port* activities.
Net, Port, Pin	Specifies that the activity numbers are for a Net, Pin, or Port.
Name	Name of the *Net*, *Pin*, or *Port*.
Activity/Td	The *Activity* or *Transition Density* value based on which of these two was selected.

Duty Period	Duty cycle and Period for the named *Net*, *Pin*, or *Port*. Duty is the fraction of time that a net, pin, or port has a value of 1.0 A duty cycle of 0.25 has a value of one 25% of the time. By selecting *Add* adds the supplied information to the list. Selecting Remove deletes selected items on the list. To highlight multiple items, click them while pressing either the Shift or Control key.
Period	Period for the named *Net*, *Pin*, or *Port*.
TCF	TCF file.
SAF	SAF file
SAF Scope	Scope of SAF file, if SAF is selected.

Run Power Analysis - Power

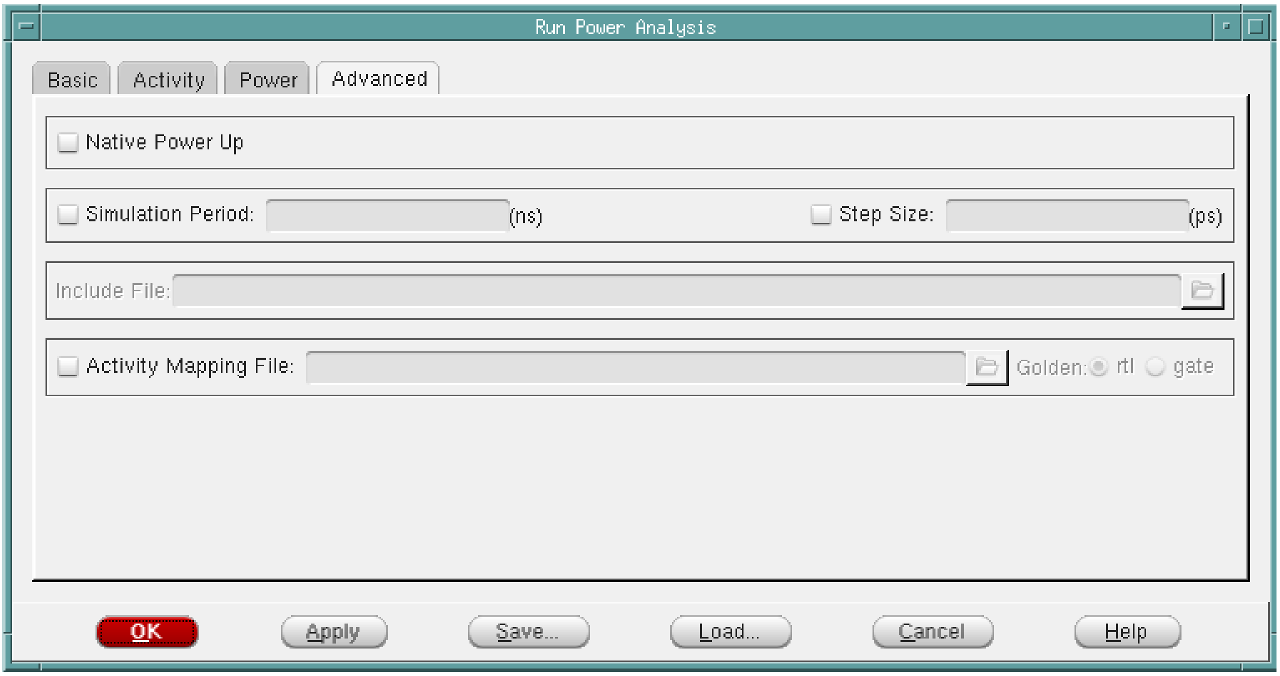


Fields and Options

Custom Power	When selected it activates the Power/PWL and PG Net fields for adding Cells.
Cell	Specifies that the Custom Power information is for a cell.

Instance	Specifies that the Custom Power information is for an Instance.
Name	The name of the cell or instance. Must be provided.
Sticky	Applies the simulation-based PWL waveform as is to the specified cell/instance.
Power/PWL	After selecting Custom Power, Cell or Instance, the Power/PWL and PG Net fields can be entered. By selecting the Add button this information is added to the list. The Remove button deletes any selected items that were previously added to the list. To highlight multiple items, click them while pressing either the Shift or Control key.
Scale Factor	Specifies to scale the power value by the specified scale factor. You can use this option to specify the scale factor for specific cells and instances.
PG Net	Specifies the name of the power rail.
Leakage	Specifies the target leakage power of the design, cell, or instance. To specify leakage power for cells and instances, you can use the Leakage option in conjunction with the Cell or Instance option.
Internal	Specifies the target internal power of the design, cell, or instance. To specify internal power for cells and instances, you can use the Internal option in conjunction with the Cell or Instance option.
Switching	Specifies the target switching power of the design, cell, or instance. To specify switching power for cells and instances, you can use the Switching option in conjunction with the Cell or Instance option.
Target Chip Power	Specifies the power for the entire design or rail, in milliwatts (mW).
Chip Scale Factor	Scales the power value by the specified scale factor.
Target Leakage	Sets the total target leakage power of the entire design in milliwatts (mW).

Run Power Analysis - Advanced



Fields and Options

Native Power Up	Specifies to enable the native power up flow.
Simulation Period	The simulation period that dynamic rail analysis will be performed.
Step Size	Specifies the transient time step size or resolution.
Include File	Adds the Power Analysis options that do not have an equivalent option in Tempus. For more information, refer to "Power Analysis Options" in *Voltus Text Command Reference*.
Activity Mapping File	Specifies the rtl2gate mapping file which contains instance name mapping between RTL netlist and GATE level netlist. RTL level VCD or TCF are obtained from simulation on RTL Verilog netlist. Power analysis uses this instance name mapping to annotate the output nets of this instance from the pin based RTL, VCD, or TCF. By default, this option is always selected when either VCD/TCF is specified.

Golden rtl gate	Specifies whether rtl or gate is Golden. The one not specified as golden will be considered as Revised. Currently the Conformal MatchPoint file does not specify this. *Default*: rtl

Rail Analysis

The Rail Analysis section has the following items:

- Set Rail Analysis Mode form
- Set Power Network Optimization Mode form
- Run Rail Analysis form
- Run Resistance Analysis form
- Run ESD Analysis form
- Run ESD Optimization form

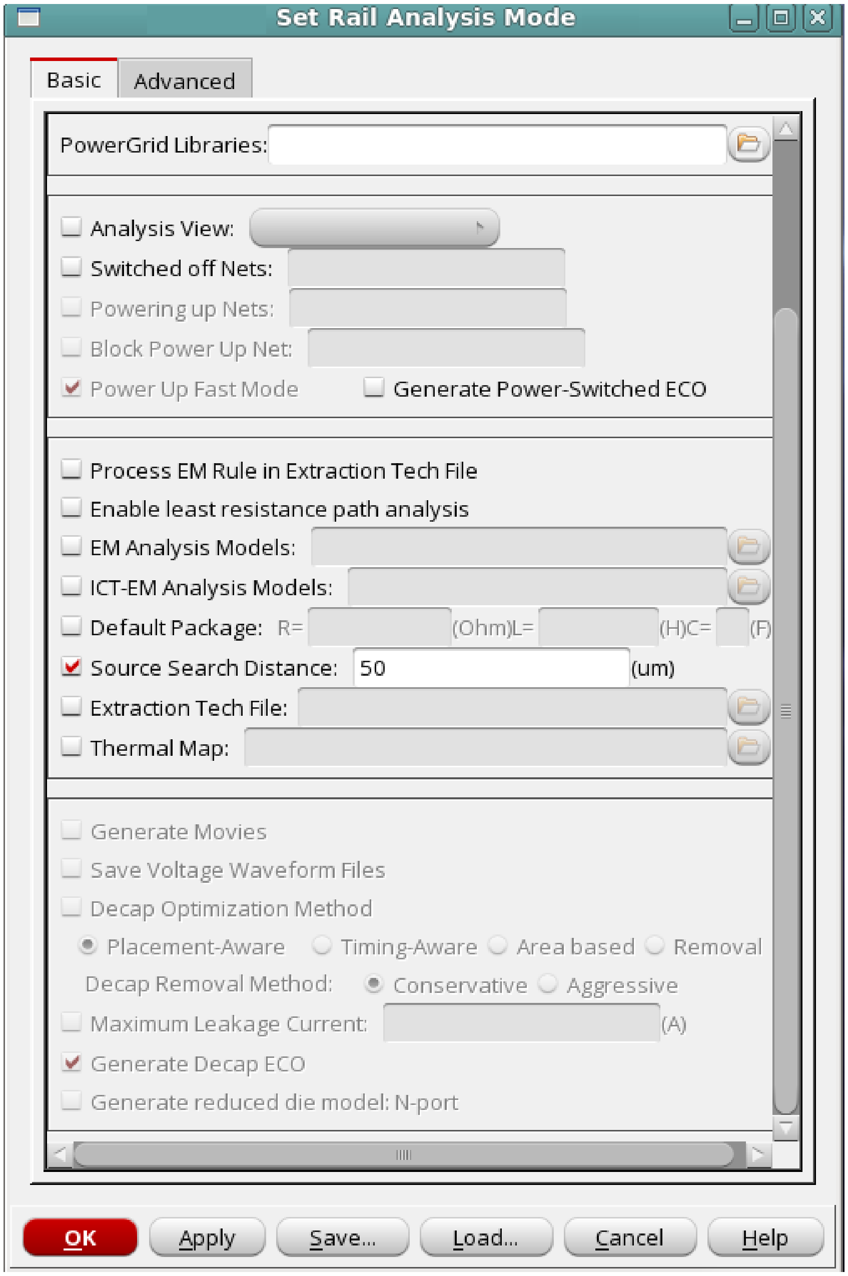
Set Rail Analysis Mode

The Set Rail Analysis Mode form includes the following:

- Set Rail Analysis Mode - Basic
- Power Grid Library Directories
- Set Rail Analysis Mode - Advanced
- Create Current Region

Set Rail Analysis Mode - Basic

Sets whether static or dynamic rail analysis should be performed along with some related parameters.



Fields and Options

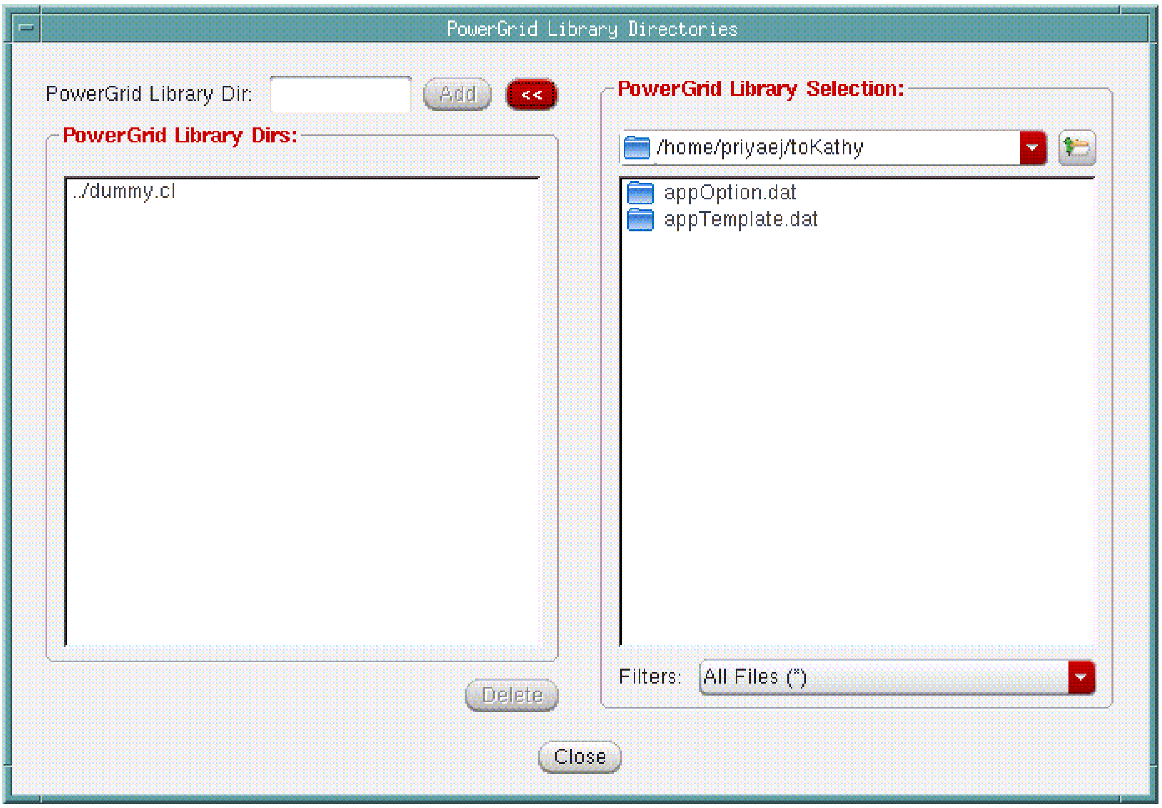
Analysis Stage	Specifies whether early rail analysis (*Early*) or signoff (*Sign Off*) rail analysis will be done. If you select *Early*, the *XD* mode is selected by default and *HD* mode is disabled. *Default*: Sign Off
Analysis Method	Specifies whether static or dynamic analysis will be done. Default: static
Accuracy	Specifies the accuracy mode of analysis. ■ XD (accelerated definition) – use for early implementation stage IR/EM analysis ■ HD (high definition) – use for final verification stage IR/EM analysis *Default*: XD
PowerGrid Libraries	Specifies the name of the power-grid view library directories. If you select the folder icon, the Set Rail Analysis form appears, which will be used to select and add the appropriate files.
MMMC View	Specifies the Mult-Mode Multi-Corner (MMMC) view.
CPF mode	Specifies Common Power Format (CPF) mode.
Switched off Power Nets	Specifies a list of nets to exclude from analysis.
Powering up Nets	Specifies a list of powering up nets.
Block Power Up Net	Specifies a net for block power-up analysis.
Generate Power-Switched ECO	Specifies that a power switch ECO file should be created. *Default*: false
Process EM Rule in Extraction Tech File	Processes the EM rules defined in the extraction technology file. When *Process EM Rule in Extraction Tech File* is specified with *Extraction Tech File*, the extraction technology file will be processed. If *Process EM Rule in Extraction Tech File* is specified without *Extraction Tech File*, the .cl model will be processed.

Enable Least Resistance Path Analysis	Enables least resistance path analysis for all the instances in the design.
EM Analysis Models	Specifies the name of a file that includes the electromigration (em) models.
ICT-EM Analysis Models	Specifies the ICT-EM file which only contains the EM rules and excludes the RC extraction data in ICT. The software will use the EM rules in the specified ICT-EM file to calculate EM limits for power net EM analysis even if there are embedded EM rules in the Quantus tech file.
Default Package	Specifies the default resistance, inductance and capacitance values of the package to be used for RLC model.
Source Search Distance	Specifies the search distance in microns(u) for adding voltage sources at power pin locations. *Default*: 50um
Extraction Tech File	Specifies the extraction technology file to be used for top-level power-grid extraction. If you do not specify this field, the software will use the extraction technology file stored inside the power-grid view library. *Extraction Tech File* is a mandatory input file for early rail analysis if PGV is not specified.
Thermal Map	Specifies to process the thermal map file generated for Quantus to calculate resistance based on uniform temperature. *Default*: Not selected
Generate Movies	Used to specify that dynamic movies will be created. For dynamic only. *Default*: `true`
Save Voltage Waveforms Files	
	Saves voltage waveform files in the state directory of analyzed net(s) to feed to SPICE critical path analysis and Substrate Noise Analysis. *Default*: Not selected This option is disabled in the *Static* mode.

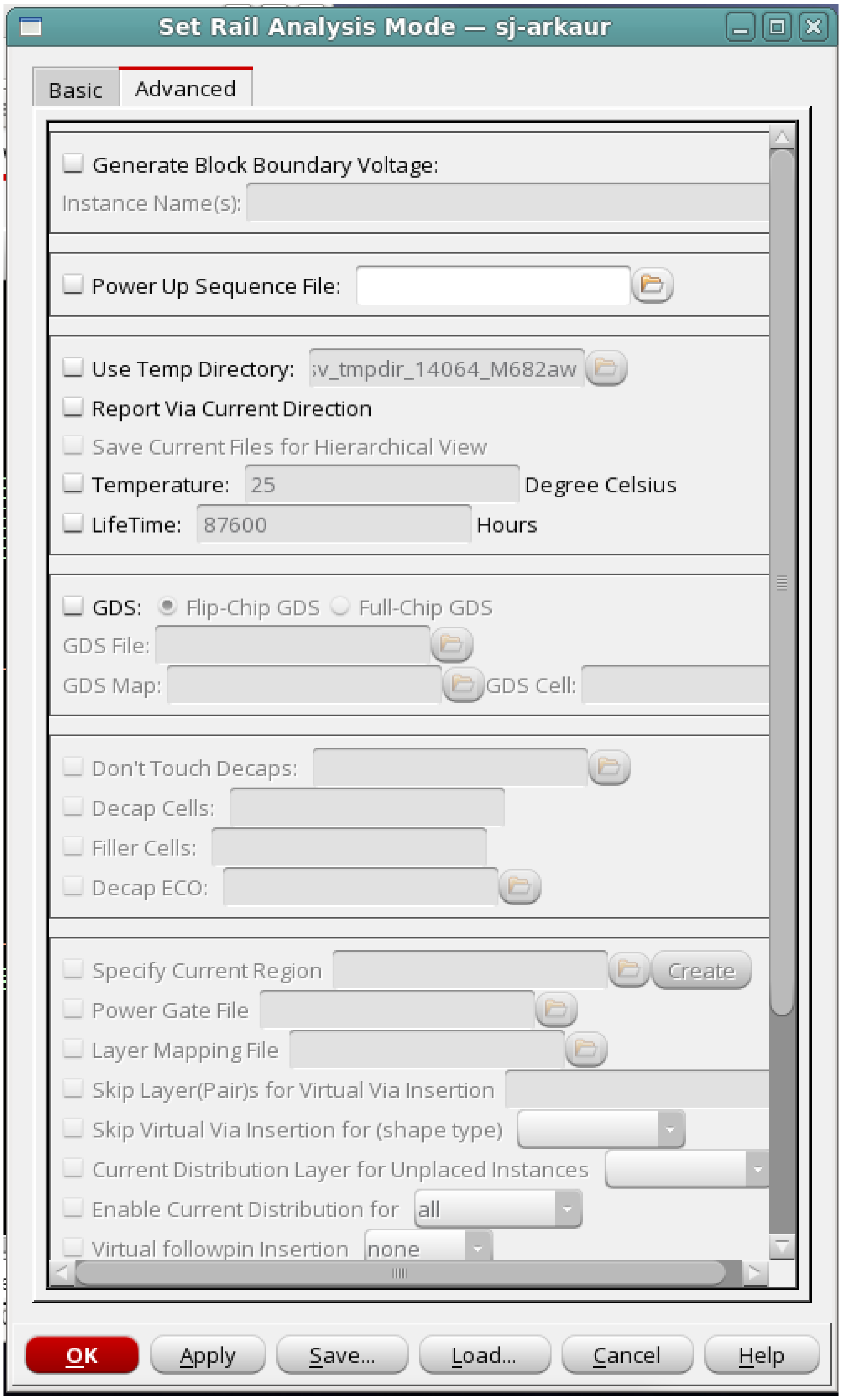
Decap Optimization Method	Specifies the decap optimization method to use. For dynamic only. Choices are Placement-Aware and Timing-Aware, Area-based, and Removal. *Default*: Area-based
Decap Removal Method	Choices are Conservative and Aggressive. Used with Removal option of Decap Optimization Method. *Default*: Aggressive
Maximum Leakage Current	Specifies the maximum leakage current in A, for placement-aware and timing-aware decap optimization methods.
Generate Decap ECO	Specifies that a decap eco file should be created.
Generate reduced die model:N-port	
	Creates `n_port_freq_domain` as well as `n_port_time_domain` die-models simultaneously.

Power Grid Library Directories

A form such as the one shown below appears when the folder icon is selected from a menu. Selecting the folder icon on this form will toggle the Selection part of the form. To add items, select a file, and then select the *Add* button. To add multiple items from the selection list, select the first item, then move the cursor to the last item and hold down the shift key while selecting the last item. The *Delete* button can be used for deleting items. Similarly the shift key can be used for deleting multiple items.



Set Rail Analysis Mode - Advanced



Fields and Options

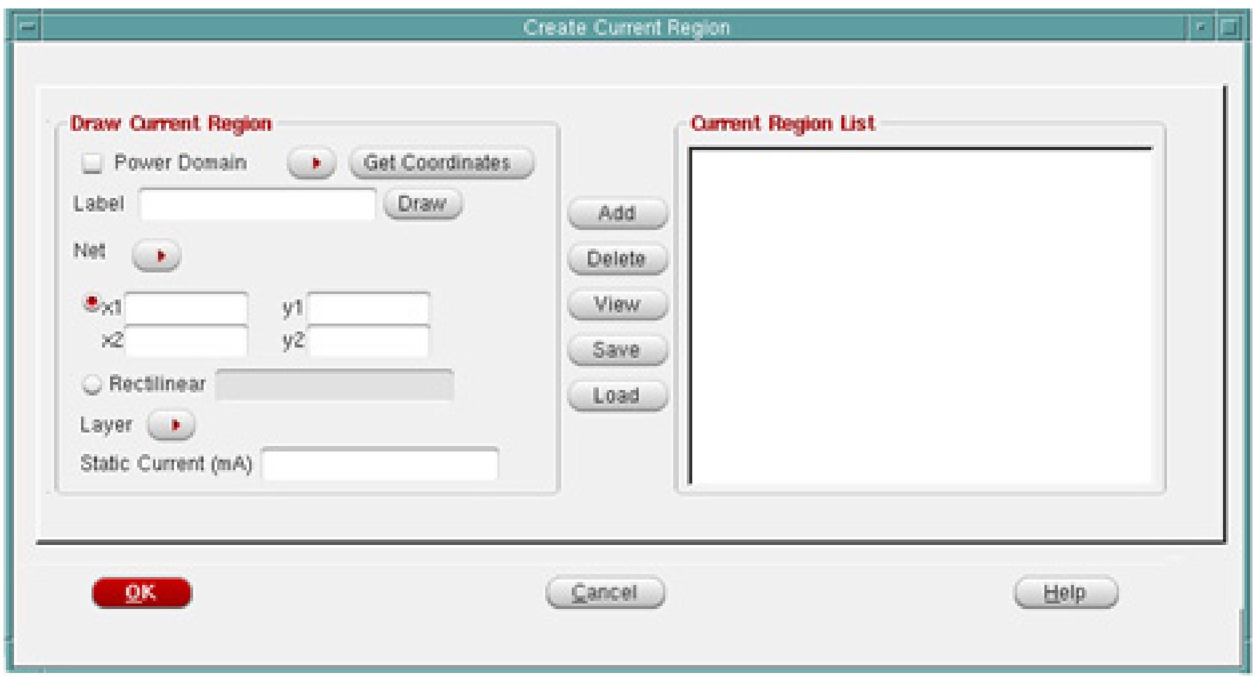
Generate Block Boundary Voltage	Generated the voltages on the block pins. This data can be used for doing rail analysis at the next level up in hierarchy.
Instance Name(s)	The name of the instance that the block boundary voltages will be generated for.
Use Temp Directory	Specifies a directory to use for temporary files.
Report Via Current Direction	Determines that the current direction of vias will be reported. *Default*: false
Save Current Files for Hierarchical View	Determines whether Voltus automatically saves current files. *Default*: false
Temperature	Specifies the temperature in degree celcius during rail analysis. The default temperature is 25 degree celsius.
Lifetime	Specifies the lifetime value in hours. *Default*: 87,600 hours
GDS	Specifies that a GDS file will be used.
Flip-Chip GDS	Specifies that the gds data is for flipchip.
Full-Chip GDS	Specifies that the gds data is for full chip.
GDS File	Specifies the name of the gds file.
GDS Map	Specifies the name of the gds map file.
GDS Cell	Specifies the name of the gds top cell.
Don't Touch Decaps	Specifies the decaps that will not be removed. For dynamic only and used if option: `-decap_opt_method` removal is set.
Decap Cells	Specifies the decap cells that will be used during decap optimization. This option is used when the decap cells are not tagged in the power-grid library. For dynamic only.

Filler Cells	Specifies the filler cells that will be used during Placement- Aware or Timing-Aware decap optimization. This option is used when the filler cells are not tagged in the power-grid library. For dynamic only.
Decap ECO	Specifies the name of the decap ECO file for domain based analysis that will be created after decap analysis is performed. This file is generated in decap_opt/addcap.cmd in the state directory. For dynamic only.
Specify Current Region	Specifies a file that includes a list of regions and the amount of current to be distributed within them for the power-grid. This field will be enabled only if you select *Early* analysis stage in the Set *Rail Analysis Mode - Basic* tab. If you select this checkbox, the *Create* button is enabled that lets you create current regions for Static and Dynamic analysis. Click *Create* to view the *Create Current Region* form. Default units in file are mA. You can also specify rectilinear current regions in the file.
Power Gate File	Specifies the name of the power gate (power switch) file that will be used to perform power gate steady-state analysis. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.
Layer Mapping File	Specifies the layer map file to generate a port view. If a layer map file is not provided, it would be automatically inferred by the tool. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.
Skip Layer (Pair)s for Virtual Via Insertion	Specifies to skip via insertion between stripes and non-stripes, on the specified LEF layer pairs. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.

Skip Virtual Via Insertion for Shape Type	Specifies to skip a given via type. By default, ERA generates all virtual via layer types. ■ *whatif* - vias are virtual vias that have connectivity to user-defined what if shapes. ■ *def* - vias are virtual vias between two metal shapes defined in DEF. ■ *all* - will skip all virtual via generation.
Current Distribution Layer for Unplaced Instances	Specifies the layer name for distributing unplaced current in the early rail analysis mode. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.
Enable Current Distribution for	Controls the behavior of era current distribution. set_power_data area based power, or current_region_file need to be specified for ERA current distribution to work. Placed instanced without uti or ascii based power can also be considered for era current distribution. ■ *Unplaced*: Enable current distribution only for unplaced instances. If set_power_data area based power is specified, -era_current_distribution_layer will be required. ■ *Placed*: Enable current distribution for placed instances without any power specified. ■ *All*: Both unplaced and placed instances without power specified; will have ERA current. ■ *None*: Disable ERA current distribution. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.
Virtual Followpin Insertion	Generates virtual followpins. The extended followpins will create followpins that extend from one stripe to another. The standard followpins may extend to previous stripe but does not reach the next stripe. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.

Current Distribution Factor for Placed Macros	When you use Distribute Current for Placed Instance or Generate Current Region File for Unplaced Area, the ratio of power allocated to each macro/block is calculated based on its total area. This field controls the current distribution factors for the placed instances, hence power allocated for area-based power calculation. For example, if you specify 0.5, the software will assume all placed instances to be 50% of its actual size and distribute current accordingly. This field will be enabled only if you select Early analysis stage in the *Set Rail Analysis Mode - Basic* tab.
Ignore Shorts	Specifies to perform sign-off mode extraction while ignoring shorts, extract each net per geometry defined in LEF/DEF and honor manufacturing effects. This option allows you to ignore shorts and continue sign-off mode extraction.
Enable Manufacturing Effects	Specifies to honor manufacturing effects during static or dynamic analysis.

Create Current Region



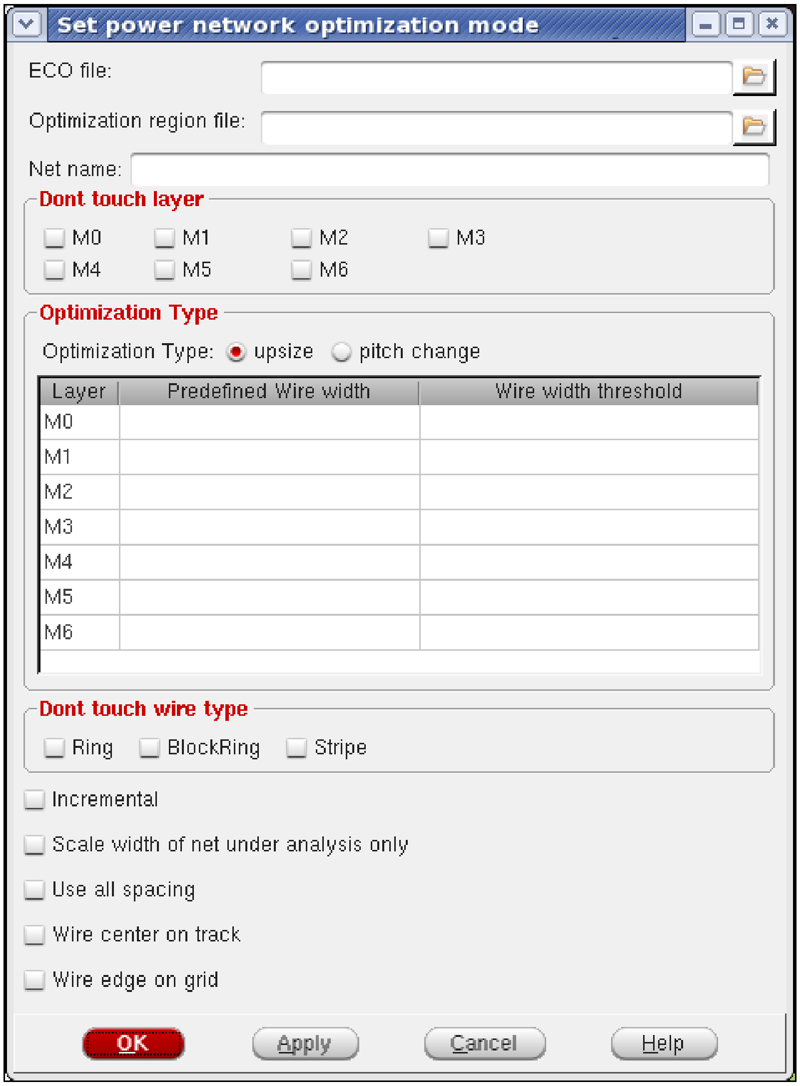
Fields and Options

Draw Current Region	Specifies the coordinates of the region, the layer, and the static current. The *Add* button can be selected to add the region to the Current Region List. The Delete button will delete a selected item on the list. Label specifies a name for the region. If not specified, Innovus will provide a name (region1, region2...) Net specifies the power/ground net name for domain-based analysis. This option allows you to specify current region values for a specific net. x1, y1, x2, and y2 specifies a rectangular region that the current will be distributed within. Rectilinear specifies the rectilinear current region that the current will be distributed within. **Note:** In the static mode, ERA splits the rectilinear region into several rectangular regions and distributes current to the rectangular regions based on the area. In the dynamic mode, you can specify rectilinear regions interactively in the layout canvas by selecting the Draw button. You can select a rectlinear region by clicking each vertex of the rectlinear in sequence (clockwise or counter-clockwise), and then pressing the ESC key on the keyboard. This will quit the point selection mode, and the selected points appear in the Rectlinear field. Current in rectangular region = Area of the rectangle / Area of the rectilinear Layer specifies the metal layer that the current sink will be placed on. If you select *Draw*, then select a box in main window, the coordinate of this box will be automatically populated in the x1 y1 x2 y2. If you select View after selecting an item in the list, the selected region will be displayed in the main window.
Dynamic PWL (0ns 0mA...)	Specifies the dynamic PWL for regions without any placed instance, in time (in ns) and current (in mA) format. For example, 0ns 0mA 0.5ns 1mA 1ns 2mA 3ns 1mA 4ns 0. This option is displayed only if you select the analysis method Dynamic.

Loading Cap (pf)	Specifies loading capacitance in pf (picoFarads) waveform for the current region(s). This option is displayed only if you select the analysis method Dynamic.
Estimate	Estimates Gate Capacitance (C1) and Loading Capacitance (C2) values for the specified current regions. These estimated values of C1 and C2 are used along with the PWL current waveform for the regions to perform dynamic rail analysis. This option is displayed only if you select the analysis method Dynamic.
Gate Cap (pf)	Specifies gate capacitance in pf (picoFarads) waveform for the current region(s). This option is displayed only if you select the analysis method Dynamic.
Current Region List	Displays the current region list. This is created by specifying the Draw Current Regions information and selecting Add. Selecting an item on the list and then Delete will delete the selected item.

Set Power Network Optimization Mode

- Use the *Set Power Network Optimization Mode* form, which specifies how the power network optimization flow will be performed.



Fields and Options

ECO File	Specifies the ECO file for power networks. The ECO file contains the Innovus power-planning commands. If this parameter is not specified, the software creates the default ECO file name pno_eco.tcl.

Optimization region file	Specifies the region of IRdrop that should be optimized to fix IRdrop violations.
Net name	Specifies the net to be optimized in MSMV analysis.
Dont touch layer	Specifies the layers on which the wires are not be optimized.
Optimization Type	Specifies the optimization type of power-grids. This parameter works only when the wire_group resolution is selected. The default value is *upsize*.
Dont touch wire type	Specifies the wire shapes that are not be optimized. The options are: ■ *RING* ■ *BLOCKRING* ■ *STRIPE*
Incremental	Specifies to generate only the power planning commands for analyzing the net.
Scale width of net under analysis only	Specifies to scale the width of only those nets that are under analysis, while keeping the width of the other nets as is.
Use all spacing	Specifies to use every possible spacing between nets.
Wire center on track	Specifies that the PG wire center should be on either full or half routing track.
Wire edge on grid	Specifies that the PG wire edge should be on the snap grid.

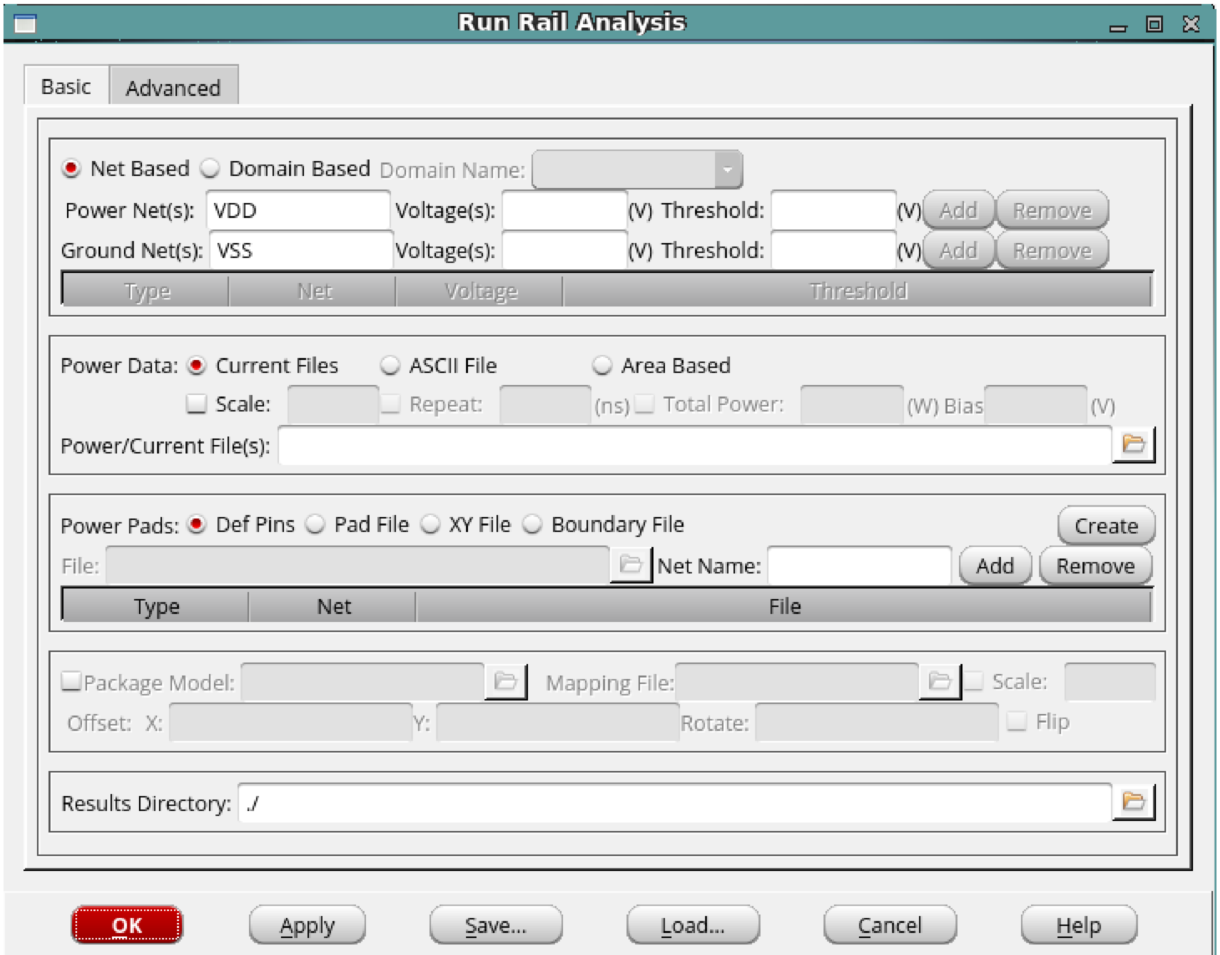
Run Rail Analysis

The Run Rail Analysis form has two pages:

- Run Rail Analysis - Basic
- Edit Pad Location
- Run Rail Analysis - Advanced

Run Rail Analysis - Basic

Runs rail analysis using the provided parameters.



Fields and Options

Net Based	Set if net based analysis is being run.
Domain Based	Set if domain based analysis is being run.
Domain Name	Name of the power domain that rail analysis will be performed on. Only used for domain based analysis.
Power Net(s)	Name of the power nets that the rail analysis will be performed on.
Voltages	The voltage of the power net.

V Threshold	The voltage threshold of the power net
Ground Net(s)	Name of the ground nets that rail analysis will be performed on.
Voltage(s)	The voltage of the ground net.
V Threshold	The voltage threshold of the ground net.
Add	Click the *Add* button to add details of a specific net to the power/ ground nets table for domain-based analysis.
Remove	Click the *Remove* button to remove details of a specific net from the power/ground nets table for domain-based analysis.
Power Data	The power data will be supplied by Current Files, an ASCII file or will be area based.
Scale	Specifies a scale factor to apply to the current data. You can use this parameter to scale currents to a more optimistic or pessimistic power-consumption estimate. This parameter is only for `-format current` and `-format ascii`.
Repeat	Repeats the dynamic current waveform for the specified time. You must ensure that the stop time for dynamic analysis is also specified and it matches the repeat duration. This option is only available for current format files during dynamic analysis. The repeat option can be used when multiple current files have different simulation periods or when IR drop analysis needs to be run for longer duration. *Default*: ns
Total Power	Specifies the power per area in watts. Only required for option `-format area`.
Bias	The value of the bias voltage. This parameter is only for `-format ascii` and `-format area`.
Power File(s)	Specifies the name of the power data files. The file is optional if -format area is specified.

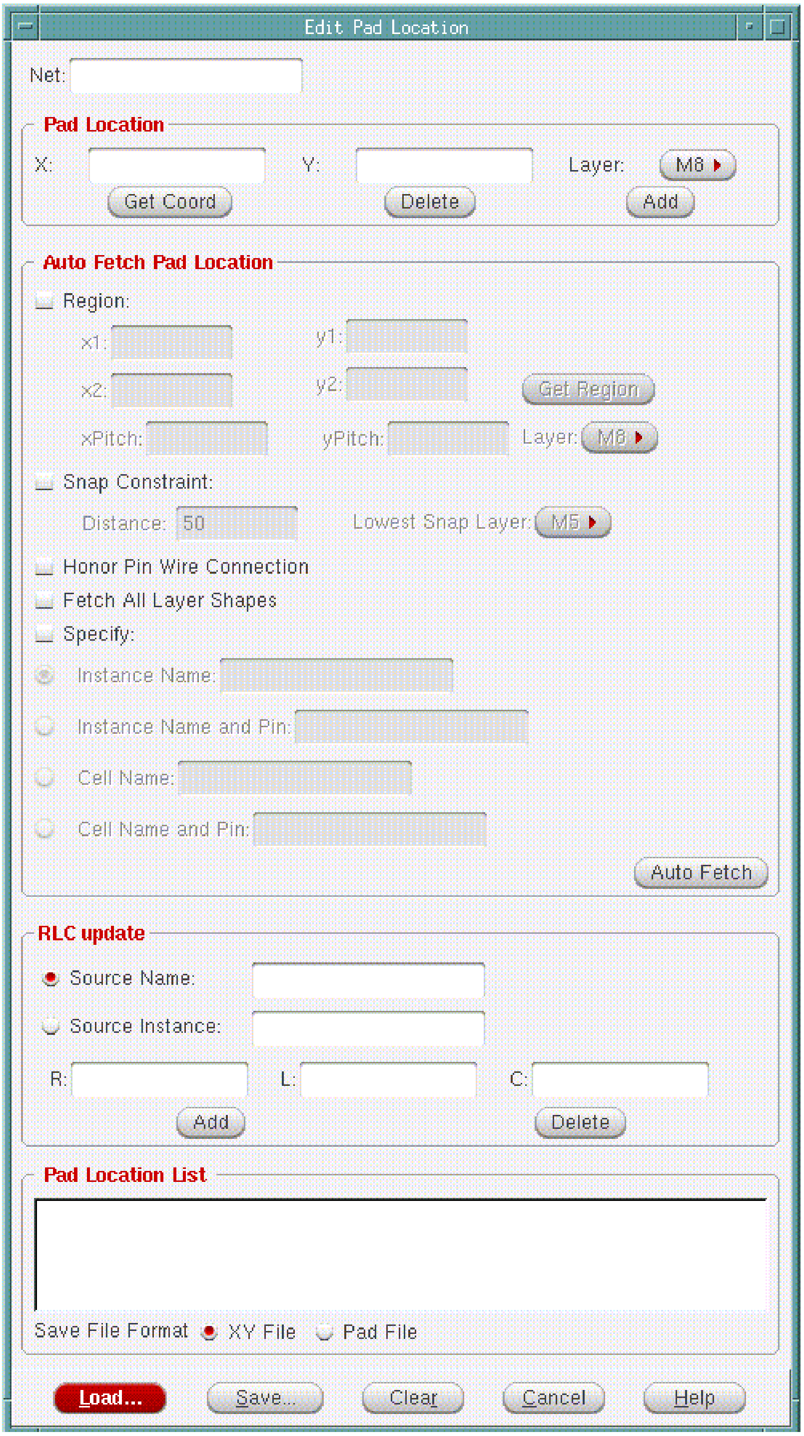
Power Pads	*XY*File specifies that voltage sources will be specified by x y coordinates. ■ *Def Pins* specifies that voltage sources will be determined by power pins. ■ *Pad File* specifies that voltage sources will be determined by pad cells given in the file. ■ *Boundary File*s specifies that voltage source locations will be specified by information in the boundary file. *Default: Def Pins*
File	The name of the power file.
Net Name	The name of the power net. The *Add* button adds the net to the list. The *Remove* button removes the net from the list. To highlight multiple items, click them while pressing either the *Shift* or *Control* key.
Package Model	Specifies a file containing a SPICE subcircuit describing the package model. This file is generated by Allegro Package SI (also known as APE3D from Cadence) or other package-modeling products.
Mapping File	Specifies the name of the package terminal mapping file, which contains the mapping of the terminals of the SPICE sub-circuit in the package model file (Specified by the *Package Model* field above) to the locations at the grid or top-level pins.
Offset X *Offset Y*	Specifies the lower left and upper right points coordinate of the die.
Rotate	Specifies the rotation angle of the die. The possible values are: 0/45/90/135/180/225/270/315. Rotation is always in the counter clockwise direction from the original point.
Flip	Specifies to flip the die along the Y axis.
Results Directory	The directory where the rail analysis output will be saved.

Edit Pad Location

Use the *Edit Pad Location* form to create or edit power pad location files (one for each power net) to mark points on the power structures as voltage source reference points. Power pad location files identify the voltage source reference points, such as VCC, VDD, and VSS. You can use these files during all stages of power analysis.

Note: Pad location displays are updated immediately after clicking *Auto Fetch*, *Delete*, or *Add*. The IR drops leading from the pad locations are displayed after running power analysis.

- To open the Edit Pad Location form, click the *Create* button in the *Run Rail Analysis* form:



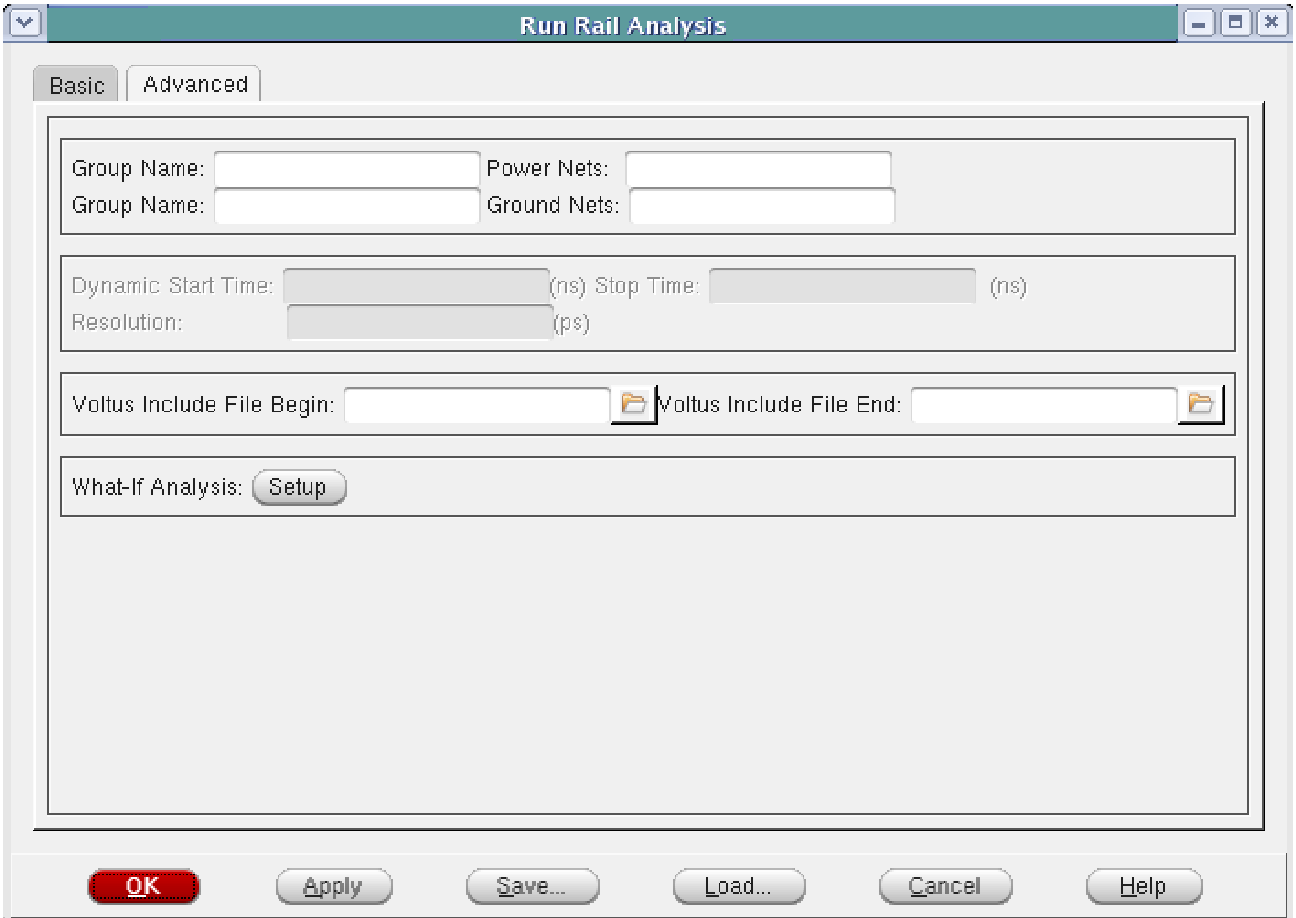
Fields and Options

Net	The power net or ground net name of the pad cells. This field is required only when the file format is *VS*. **Note:** You can only edit the pad location file for one net at a time.
Pad Location	The *X* and *Y* coordinates of a pad cell.
Layer	The layer that contains the pad cell.
Get coord	Populates the *X* and *Y* coordinates in the *Pad Location* field. Click this button, then click a location in the design display area.
Delete	Removes one or more pad location items from the *Pad Location List*. Highlight the item be removed, then click this button. To highlight multiple items, click them while pressing either the *Shift* or *Control* key.
Add	Adds the *X* and *Y* coordinates of a pad cell to the *Pad Location List*.
Region	*x1*, *y1*, *x2*, and *y2* specifies a rectangular region that the current will be distributed within. A grid is created by using the *xPitch* and *yPitch* within the specified region. The current will be specified at each of the intersecting pitch lines. *Layer* specifies the metal layer that the current sink will be placed on. If you select *Get Region*, then select a box in main window, the coordinate of this box will be automatically populated in the x1 y1 x2 y2.
Snap Constraint	Specifies whether to snap DC sources from I/O pad pins to some location on the power structure. If you select this option, the software performs auto fetch with snapping. For I/O pad pins that are connected to core ring, the software selects the highest layer and center of the pin as the DC source location. When I/O pad pins are not connected to core ring, the software snaps the location of the I/O pad to somewhere on the core ring. The DC source location should be within the distance of snapping distance constraint from I/O pad pins, and the layer should not be lower than the lowest layer snap constraint.
Distance	Specifies the distance of snapping distance constraint from IO/ pad pins. The default value is `50`. You can edit this option only if *Snap Constraint* is selected.

Lowest Snap Layer	Specifies the lowest layer used to fetch a DC source. Use a value such as M1 or M2. The default value is `M5`. You can edit this option only if *Snap Constraint* is selected.
Honor Pin Wire Connection	Honors pin wire connection when fetching voltage source location from a power pad. When you select this option, only pins with wire connection are output as voltage sources.
Fetch All Layer Shapes	Specifies to fetch all layer shapes as voltage sources. When a PG pin has only one port with many shapes, this parameter specifies to fetch all shapes as voltage sources.
Specify	Specifies an instance/cell and their pins to auto fetch DC sources.
Instance Name	Specifies the instances from which voltage sources are determined.
Instance Name and Pin	Specifies the instances and their pins from which voltage sources are determined.
Cell Name	Specifies the cells from which voltage sources are determined.
Cell Name and Pin	Specifies the cells and their pins from which voltage sources are determined.
Auto Fetch	Automatically populates the *Pad Location List* with the pad locations based on properties defined in the LEF file. There are two ways to get the dc source. ■ auto Fetch + net ■ auto Fetch + region
Source Name	Specifies the name of the voltage source that has to be updated.
Source Instance	Specifies the instance name of the voltage source that has to be updated.
R	Specifies the resistance value of the package resistance-inductance-capacitance (RLC) model in Ohm.
L	Specifies the inductance value of the package resistance-inductance-capacitance (RLC) model in Henry (H).
C	Specifies the capacitance value of the package resistance-inductance-capacitance (RLC) model in Farad (F).
Add	Adds the modified RLC values in *Pad Location List*.

Delete	Removes the modified RLC values from *Pad Location List.*
Pad Location List	The list of coordinates for pad cells. To populate this field manually, click the *Add* button. To populate this field automatically, click the *Auto Fetch* button.
Save File Format	The format of the saved Pad Location File. The choices are *XY File* (xy coordinates) and *Pad File* (VoltageStorm). You can create and save the voltage sources location file with a unique name for the voltage sources (both VDD and VSS).
Load	Opens the Load Pad Location File form. Select a file from this list, then click *Open* to display the pad locations in the *Pad Location List*.
Save	Opens the Save Pad Location File form. Specify a location for the pad location file, then click *OK*. This saves the pad location file, clears the *Pad Location List*, and closes the Save Pad Location form.
Clear	Clears the *Pad Location List*. Once the *Pad Location List* is cleared, you can begin creating a pad location file for a new net. **Note:** Cadence recommends that you use the *Clear* button after you save the pad location file before you use *Auto Fetch* again.

Run Rail Analysis - Advanced



Fields and Options

Group Name	Specifies the name for a group of power or ground nets.
Power Nets *Ground Nets*	Replace power domain by group.
Start Time	Specifies the start time in nanoseconds (ns) for dynamic analysis.
Stop Time	Specifies the stop time in nanoseconds (ns) for dynamic analysis.
Resolution	Specifies the transient time step size or resolution in picoseconds (ps).

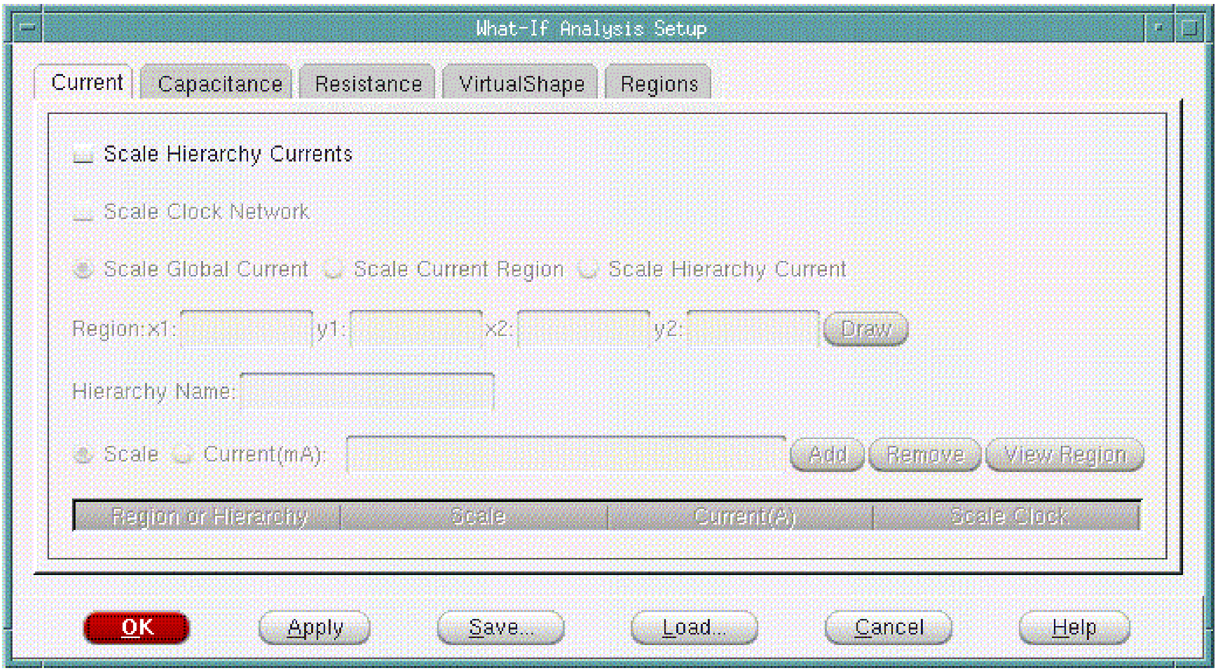
Voltus Include File Begin	This field provides the ability to provide a file that includes vstorm2 commands that will be run prior to the `analyze` command. This can be used for those VoltageStorm commands that do not have a Voltus GUI equivalent.
Voltus Include File End	This field provides the ability to provide a file that includes vstorm2 commands that will be run after the vstorm2 command. This can be used for those VoltageStorm commands that do not have a Voltus GUI equivalent.
What-If Analysis Setup	Clicking the *Setup* button displays the GUI for ■ What-If Analysis Setup - Current ■ What-If Analysis Setup - Capacitance ■ What-If Analysis Setup - Resistance ■ What-If Analysis Setup - Virtual Shape ■ What-If Analysis Setup - Regions

What-If Analysis Setup

- What-If Analysis Setup - Current
- What-If Analysis Setup - Capacitance
- What-If Analysis Setup - Resistance
- What-If Analysis Setup - Virtual Shape
- What-If Analysis Setup - Regions

What-If Analysis Setup - Current

Use this form to scale current to do what-if analysis for a hierarchical partition and assess its effect on IR drop.



Fields and Options

Scale Hierarchy Currents	Specifies to scale hierarchical currents for all instances within the given hierarchy.
Scale Clock Network	Specifies to scale instances that belong to the clock network and scale currents for the rest of the instances within the given hierarchy to the specified value. You cannot specify this option with `-region` *x1 y1 x2 y2* or `-global` options. *Default*: Not selected
Scale Global Current	Specifies current scaling to be applied to the entire design.
Scale Current Region	Specifies current scaling to be applied to the specified current region in the design.

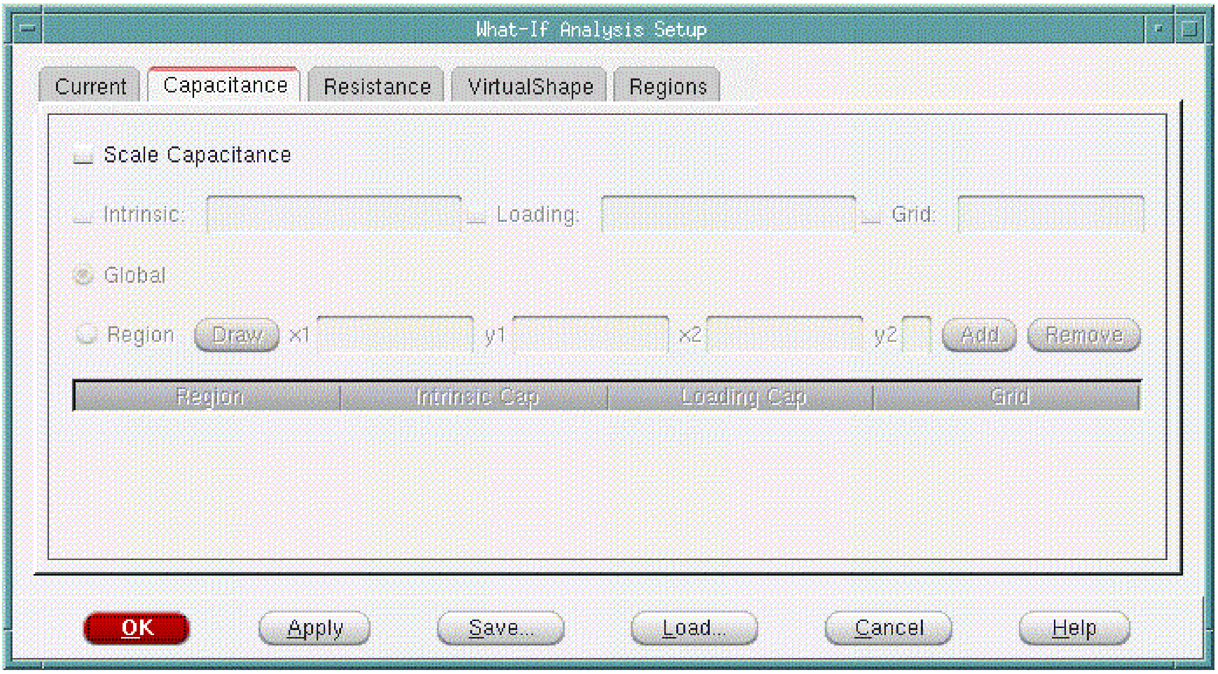
Scale Hierarchy Current	Specifies current scaling to be applied to the specified hierarchical block.
Region	Specifies a region where current scaling is to be applied. The default unit of x and y coordinates is in microns (um).
Hierarchy Name	Specifies the name of the hierarchical block. For example, `-hierarchy top/blockA`. To specify global scaling for the design use "`/`". For example, `-hierarchy /` This is a required option.
Scale	Specifies any value greater than 0. For example, a value of 0.5 will scale the resistance in half. This is a required option.
Current (mA)	Specifies current value in milliAmperes. This is used as target peak current for the given hierarchy and is then scaled appropriately. You cannot specify this option with `-region` *x1 y1 x2 y2* or `-global` options.

Related Topics

See command in chapter in the *Text Command Reference*.

What-If Analysis Setup - Capacitance

Use this form to scale the intrinsic cell capacitance value to determine how much additional capacitance can be added in a region to reduce dynamic IR drop.



Fields and Options

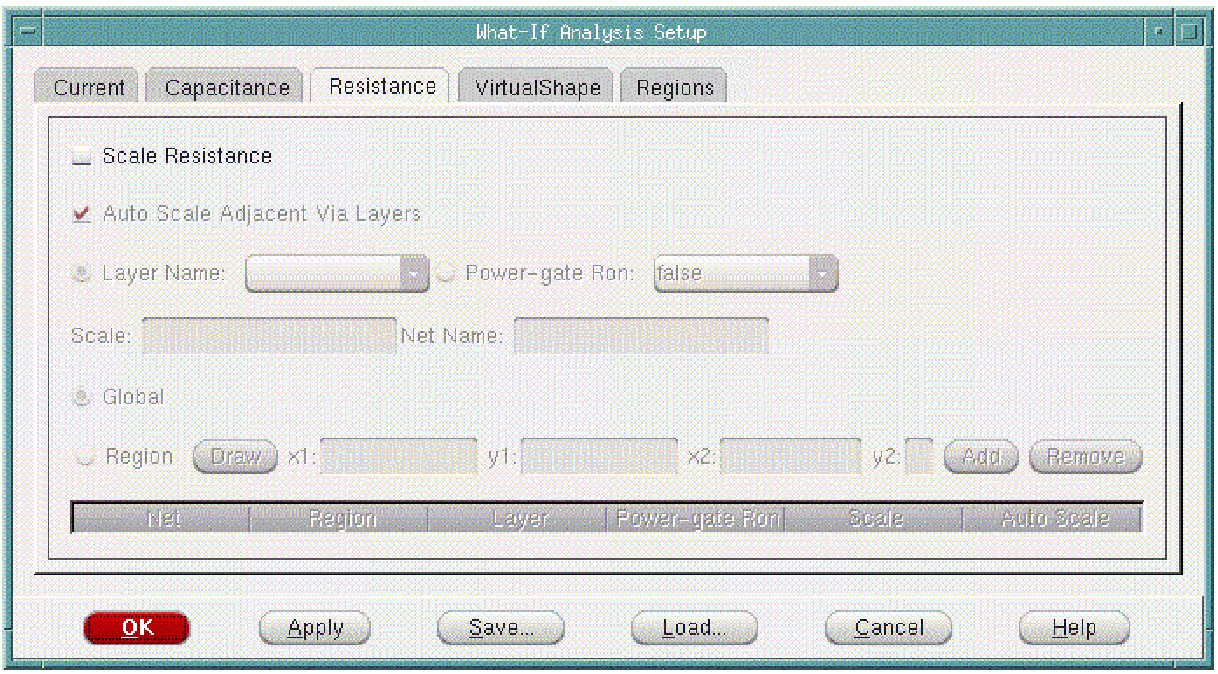
Scale Capacitance	Specifies to scale capacitance.
Intrinsic	Scales instance internal gate capacitance to the specified scale factor. The default scale value is 1.
Loading	Scales loading capacitance of the instance to the specified scale factor. The default scale value is 1.
Grid	Scales power-grid capacitance of the design to the specified scale factor. The default scale value is 1.
Global	Scales the specified capacitances for the entire design.
Region Draw	Scales the specified capacitances in the region. The default unit of *x1 y1 x2 y2* is in database units.

Related Topics

See command in chapter in the *Text Command Reference*.

What-If Analysis Setup - Resistance

Use this form to scale resistance globally or in a region to determine how much resistance can be modified in a region to reduce the entire IR drop.



Fields and Options

Scale Resistance	Specifies to scale resistance.
Auto Scale Adjacent Via Layers	Specifies to use auto via scaling for what-if resistance analysis.
Scale	Specifies any value greater than 0. For example, a value of 0.5 will scale the resistance in half. This is a required option.

Layer Name	Specifies the layer name (LEF layer or technology layer) on which the scaling is performed. Multiple commands can be issued to scale resistance on multiple layers. This is a required option.
Net Name	Scales resistance of the specified net in the domain which is being analyzed. This is a required option.
Global	Scales the resistance to the specified value for the entire design for a given layer.
Region *Draw*	Scales the resistance in the specified region. The default unit of *x1 y1 x2 y2* is in database units.

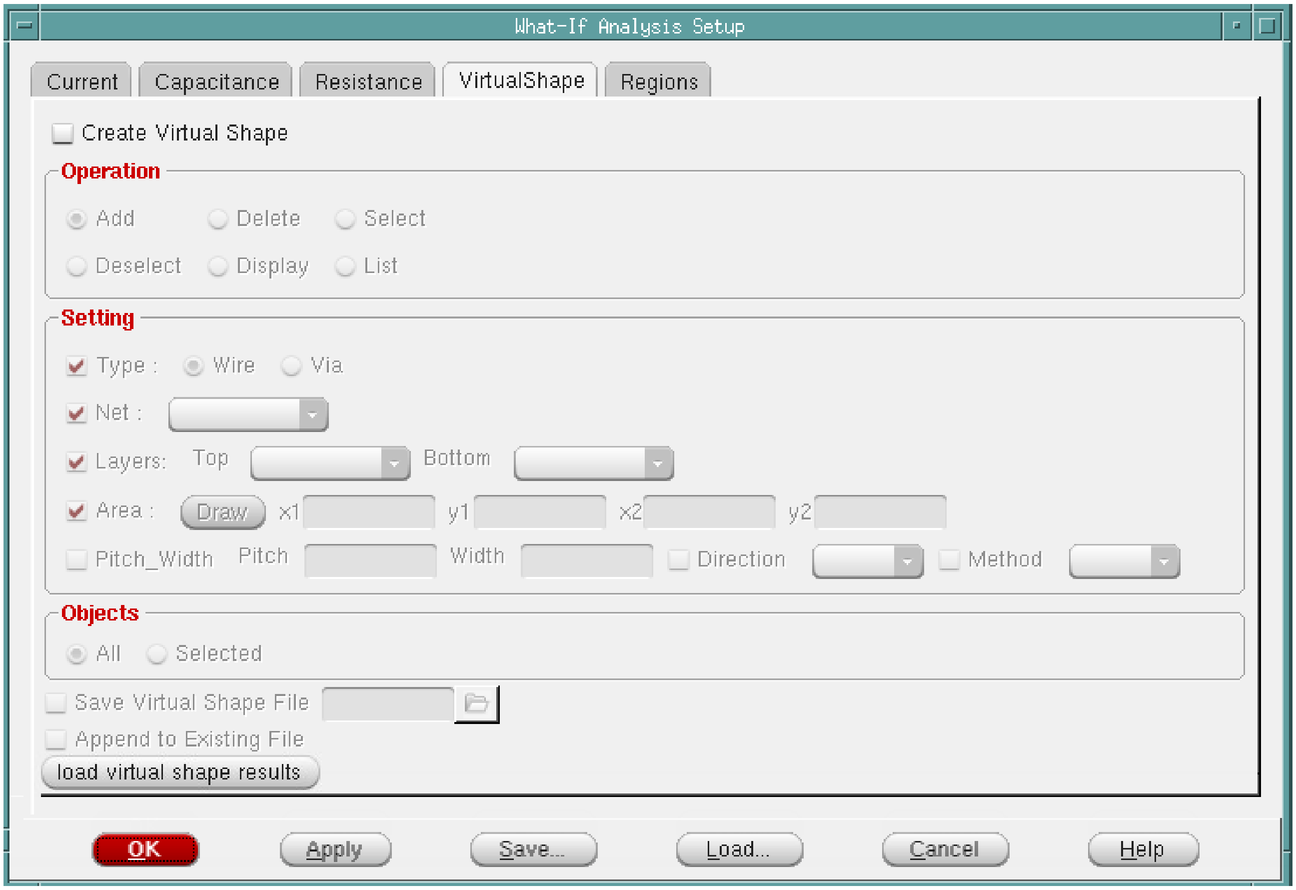
Related Topics

See command in chapter in the *Text Command Reference*.

What-If Analysis Setup - Virtual Shape

Use this form to add, select, deselect, delete, list, and display the what-if wires/vias to a design.

- Choose *Power & Rail > Run Rail Analysis > Advanced* form, click on the *What-If Analysis - Setup* button > *Virtual Shape* tab.



Fields and Options

Create Virtual Shape	Creates virtual shapes (wires/vias) to a design.

Operation	■ *Add* - Specifies to add what-if wires/vias. ■ *Delete* - Specifies to delete all or selected what-if shapes by constraint *Net/Layers/Type*. You can use this field in conjunction with the *Objects* field. ■ *Select* - Specifies to select all what-if shapes by constraint *Net/ Layers/Type*. Use this field in conjunction with *Area* to select what-if shapes overlapping with the specified area. ■ *Deselect* - Specifies to deselect all what-if shapes by constraint *Net/Layers/Type*. Use this parameter in conjunction with *Area* to deselect what-if shapes overlapping with the specified area. ■ *Display* - Invokes the GUI to show what-if shapes and zooms to the selected shapes. ■ *List* - Specifies to list all or selected what-if shapes by constraint *Net/Layers/Type*. You can use this field in conjunction with the *Objects* field.
Type	Specifies what kind of shape is to be created.
Net	Specifies the name of the net for which what-if wires/vias is to be created.
Layers	Specifies the LEF layer name for wires, or layer pair for via to be added.
Area	Specifies the rectangular region in which the what if wires/vias are to be generated.
Pitch Width	Specifies the width of what-if wires and the distance between the centre line of two adjacent what-if wires (pitch) to be drawn in the specified *Area* in the preferred direction of the specified layer. The direction of created wires can also be controlled using the *Direction* field. If this field is not provided, a single rectangular shape of size {llx,lly,urx,ury} is created.
Direction	Controls the direction of the wires added with the *Pitch Width* field.

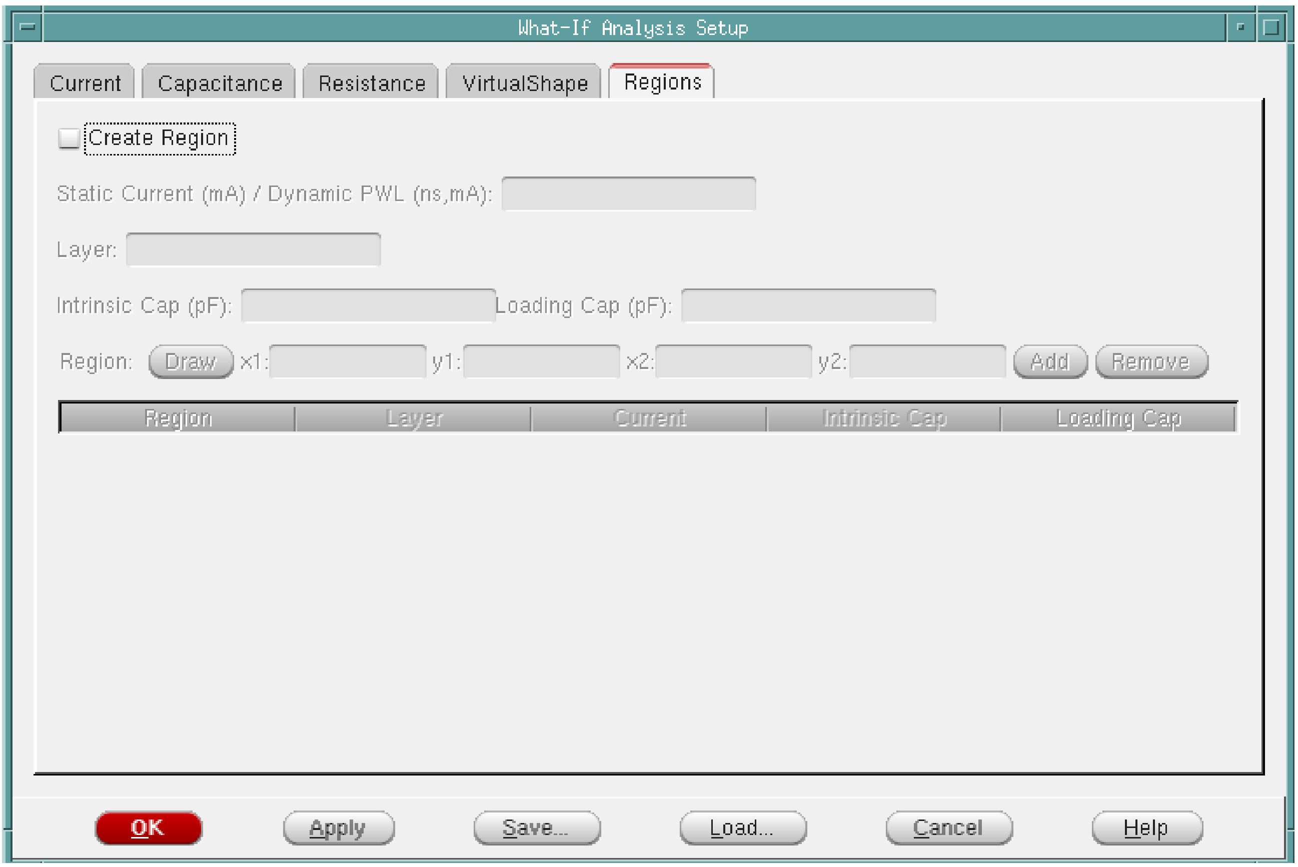
Method	Specifies the method used to add vias: *Auto*: When area and layer pair are specified, the tool automatically adds vias on all intersecting points on the specified layer pairs in the given area. *Exact*: When an area and layer pair are specified, the tool adds a via of exact size of the specified area between the specified layer pair.
Objects	Specify *Objects* in conjunction with the *Delete* or *List* fields to delete or list the selected what-if shapes. You can also delete or list all what-if shapes. The default value is *All*.
Save Virtual Shape File	Specifies to save all the in-memory what-if wires/vias to the specified file.
Append to Existing File	Specifies to append all what-if wires/vias to the existing file specified with the *Save Virtual Shape File* field.
Load Virtual Shape Results	Loads the saved files of all vias/wires from the *What-If Analysis Setup* form and results are displayed in the design layout.

Related Topics

See `create_what_if_shape` command in the *Rail Analysis Commands* chapter in the *Innovus Text Command Reference.*

What-If Analysis Setup - Regions

Use this form to specify static or dynamic current regions on global power-grids over the placed macro to accurately capture high and low current consuming regions during rail analysis.



Fields and Options

Create Region	Creates current regions on global power-grids.
Static Current / Dynamic PWL	*Static Current*: Specifies a single current value in mAmp. *Dynamic PWL*: Specifies a PWL current waveform in time (in ns) and current (in mA) format. For example, 0ns 0mA 0.5ns 1mA 1ns 2mA 3ns 1mA 4ns 0.
Layer	Specifies a layer name on which the scaling will be performed. This is a required option.

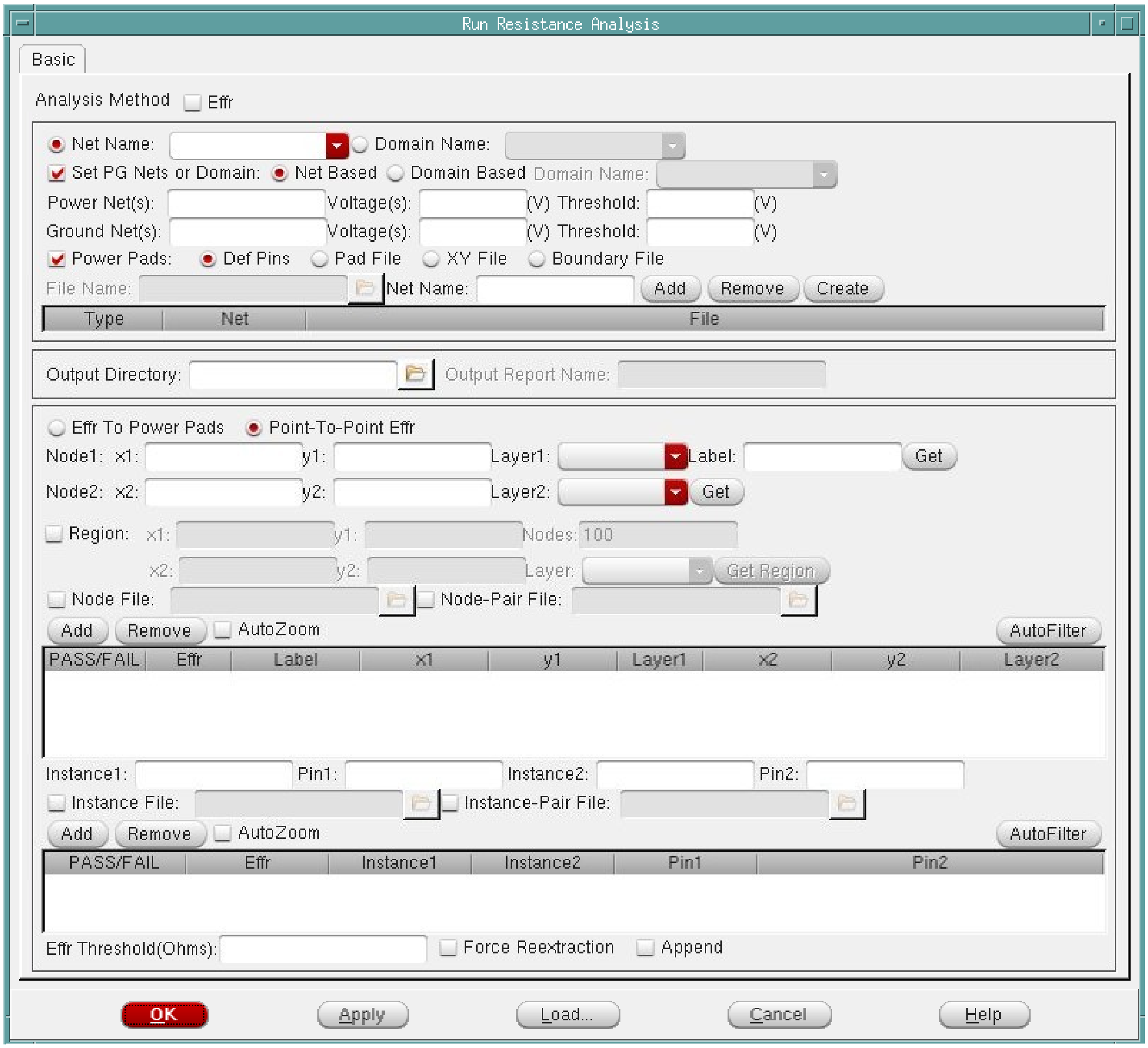
Intrinsic Cap	Specifies the cell intrinsic capacitance in pico farads (pf). This is required option when set in the dynamic mode (`set_rail_analysis_mode -method dynamic`). This option is disabled in static mode.
Loading Cap	Specifies the cell loading capacitance in pico farads (pf). This is required option when set in the dynamic mode (`set_rail_analysis_mode -method dynamic`). This option is disabled in static mode.
Region *Draw*	Specifies the region on the layer across which the specified current and capacitance (in dynamic mode) is distributed by the rail analysis engine. This is a required option.

Related Topics

See command in chapter in the *Text Command Reference*.

Run Resistance Analysis - Basic

Performs effective resistance calculation for all instances in a design. Effective resistance computes effective resistance between two nodes on the power-grid, or from a node to all the voltage sources in the design, or from an instance to all the voltage sources in the design.



Fields and Options

Analysis Method	Specifies the resistance analysis method as *Effr*.

Net Name	Specifies the net name that is to be analyzed.
Domain Name	Specifies the domain name that is to be analyzed.
Set PG Nets or Domain	*Net Based* - is equivalent to the `set_pg_nets` command. *Domain Based* - is equivalent to the `set_rail_analysis_domain` command. If you have used Tcl command in your scripts, you do not have to use these GUI fields to set PG nets or rail analysis power domain.
Power Net(s)	Specifies names of the power nets for which resistance analysis will be performed.
Voltage(s)	The voltage of the power net.
Threshold	The voltage threshold of the power net
Ground Net(s)	Specifies names of the ground nets for which resistance analysis will be performed.
Voltage(s)	The voltage of the ground net.
Threshold	The voltage threshold of the ground net
Power Pads	■ *Def Pins* - specifies that voltage sources will be determined by power pins. ■ *Pad File -* specifies that voltage sources will be determined by pad cells given in the file. ■ *XY File* - specifies that voltage sources will be specified by x y coordinates. ■ *Boundary File* - specifies that voltage source locations will be specified by information in the boundary file. *Default: Def Pins*
File Name	The name of the power file containing the locations of the power rail voltage sources (VDD, VSS). You can automatically generate this file by clicking the *Create* button. The Edit Pad Location form appears.
Net Name	The name of the power net. The *Add* button adds the net to the list. The *Remove* button removes the net from the list.
Output Directory	Specifies the name of the directory in which the output report file is saved.

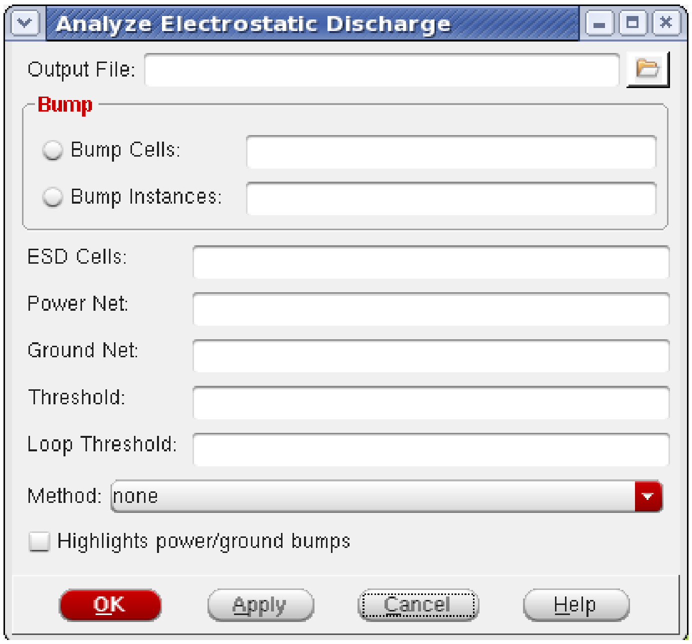
Output Report Name	Specifies the name of the output report file that contains the report header and the sorted list for resistance values.
Effr To Power Pads	Specifies to check effective resistance of a node to the voltage sources or power pads. This option is enabled only when *Power Pads* is selected. When this option is selected, *Node2* and *Instance2* options are disabled.
Point-To-Point Effr	Specifies to check effective resistance between two nodes on the power-grid. When this option is selected, *Node2* and *Instance2* options are enabled.
Node1	Specifies the list of user-defined nodes for effective resistance calculation between the specified user-defined node and all voltage sources, or between two nodes on the power-grid. This option is enabled for both *Effr To Power Pads* and *Point-To-Point Effr*. You can add a list of nodes based on the region on a given layer. ■ *x1* and *y1* are the coordinates of the node. ■ *Layer1* is the generic metal routing layer name on which the node is located. ■ *Label* is any name chosen by you to identify node entries in the list. This is optional but helps during debugging. ■ *Get*
Node2	Specifies the second node for effective resistance calculation between two nodes on the power-grid. This option is enabled only for *Point-To-Point Effr*. ■ *x2* and *y2* are the coordinates of the second node. ■ *Layer2* is the generic metal routing layer name on which the second node is located. ■ *Get*

Region	Analyzes resistance in the specified region. You can draw a box on a specific layer, and specify the number of nodes to be selected (maximum limit of 100 nodes). This field allows you to compute effective resistance on a given layer across the chip and determine how the resistance is changing for the design from top to bottom layer. You can assign a name for the region. ■ *x1*, *y1*, *x2*, and *y2* are the coordinates of the box. ■ *Nodes* are the number of nodes to be selected. ■ *Layer* is the generic metal routing layer name on which the nodes are located. ■ Get Region
Node File	Specifies the name of the node list file. This file contains the list of user-defined nodes in the same format as mentioned for the `-node_list` parameter of the `analyze_resistance` command. The file allows you to specify a large number of nodes.
Node-Pair File	Specifies the name of the node pair list file. This file contains the node pair list in the same format as mentioned for the `-node_pair_list` parameter of the `analyze_resistance` command. The file allows you to specify a large number of node pairs.
Instance1	Specifies the list of instances for which effective resistance is to be calculated. This option is enabled for both *Effr To Power Pads* and *Point-To-Point Effr*.
Pin1	Optionally, you can specify instance pin name. By default, the tool will use the instance pin associated with the net being analyzed. Pin name can be used in special cases when two different pins of an instance are connected to the same net.
Instance2	Specifies the second instance to enable effective resistance calculation between two instances in the design. This option is enabled only for *Point-To-Point Effr*.
Pin2	Specifies the pin name for the second instance.
Instance File	Specifies the name of the instance list file. This file contains the list of instances in the same format as mentioned for the `-instance_list` parameter of the `analyze_resistance` command. The file allows you to specify a large number of instances.

Instance-Pair File	Specifies the name of the instance pair list file. This file contains the instance pair list in the same format as mentioned for the `-instance_pair_list` parameter of the `analyze_resistance` command. The file allows you to specify a large number of instance pairs.
Effr Threshold (Ohms)	Specifies the minimum allowed effective resistance to display node/node-pair/instance/instance-pair that have a resistance value above this threshold value.
Force Re-extraction	Specifies to force a re-extract of resistances.
Append	When you select *Append* and click the *Load* button, the content of the effective resistance analysis result file is appended to the previous list in the node/instance list box. If *Append* is not selected, the previous list is cleared.
Add	Adds the specified node/node-pair/instance/instance-pair to the list box.
Remove	Removes the selected node/node-pair/instance/instance-pair to the list box.
Auto Zoom	Specifies to automatically zoom in to the selected node/instance in the list box.
AutoFilter	Specifies to show the nodes or instances with eight color filter. When you specify *Effr Threshold (Ohms)* and click *AutoFilter*, the layout displays the nodes or instances with eight color filter. When you change the threshold value, you will get different plots on the layout.

Analyze ESD

Use the *Analyze Electrostatic Discharge* form to report an effective resistance from the power or ground bump to the nearest ESD cell.



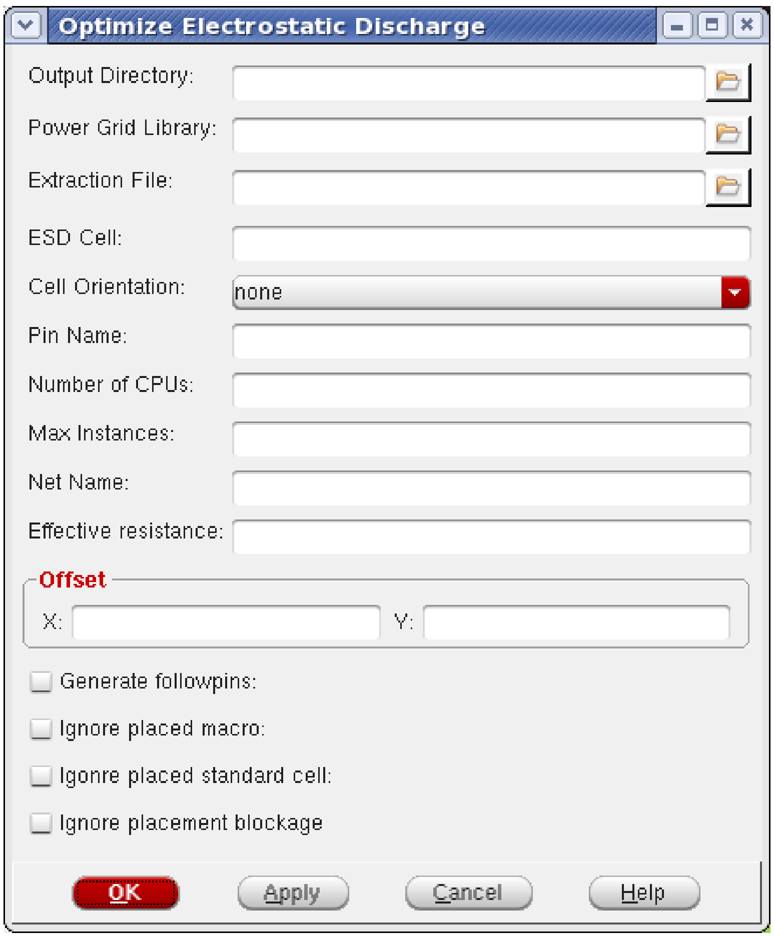
Fields and Options

Output File	Specifies the name of the report file in which details of resistance between power/ground bumps to ESD is written.
Bump File	*Bump Cells:* Specifies to account for the connected bump instances only of the specified cells. If this parameter is not specified, all connected bump instances will be taken into account. *Bump Instances:* Specifies to account only for the listed connected bump instances. If this parameter is not specified, all connected bump instances will be taken into account. This parameter is mutually exclusive to the `-bump_cell_list` parameter.
ESD Cells	Specifies the ESD cell names.
Power Net	Specifies the power net for which ESD protection has to be checked.
Ground Net	Specifies the ground net for which ESD protection has to be checked.
Threshold	Specifies the minimum allowed effective resistance between a power or ground bump and an ESD device pair.

Loop Threshold	Specifies the minimum loop resistance that an ESD clamp cell would have between power and ground bump cells. This is the effective shunt resistance that current would pass through during an ESD event when the ESD device is on.
Method	Specifies the ESD calculation method. When *bump_to_esd_resistance* is specified, the software checks effective resistance between the ESD clamp cell and the power or ground bump depending upon what nets are specified. In this case, you must specify either power or ground nets. When *bump_to_esd_loop_resistance* is specified, the software calculates effective resistance between the power bump, ESD clamp cell and ground bump, and assumes that the specified threshold is for loop resistance. In this case, you must specify both power and ground nets.
Highlights power/ ground bumps	Selecting this checkbox highlights the power or ground bump cells.

Optimize ESD

Use the *Optimize Electrostatic Discharge* form to perform ESD cell optimization that enables you to determine optimal number and location of ESD cells given a set of bump locations in the design.



Fields and Options

Output Directory	Specifies the output directory for ESD optimizer.
Power Grid Library	Specifies the cell library for ESD and bump cells.
Extraction File	Specifies the extraction tech file name for power-grid view generation.
ESD Cell	Specifies the cell name that is to be placed.
Cell Orientation	Specifies the direction in which ESD cell is to be placed.

Pin Name	Specifies the name of the pin.
Number of CPUs	Specifies the number of CPUs on the local machine. This parameter is required for multi-threading.
Max Instances	Specifies the maximum number of locations to be tried for searching optimal ESD locations. If a blockage, macro, or placed instance is found, then that location is discarded. If you do not specify this parameter, the maximum number of locations are searched.
Net Name	Specifies the net that needs to be processed.
Effective resistance	Specifies the effective resistance from the power or ground bump to the nearest Electrostatic Discharge (ESD) cell.
Offset	Specifies the offset values in the X and Y directions.
Generate followpins	Specifies to generate followpins for routing.
Ignore placed macro	Specifies to ignore placed macros.
Ignore placed standard cell	Specifies to ignore placed standard cells.
Ignore placement blockage	Specifies to ignore placed placement blockages.

Report

The Report section contains the following items:

- Library Cell Viewer form
- Power and Rail Plots form
- Power Library Report form
- Power Report form
- Power Histograms form
- Dynamic Movies form
- Dynamic Waveforms form

Cell Viewer

The Cell Viewer provides an artwork window for viewing the following objects:

- any LEF library cell
- any OpenAccess database cell view
- abstract view of any partition in the design
- power grid library cell

The Cell Viewer displays the content from the library database. It provides a convenient way to view the content of a cell, cell view or partition without having to zoom into the design. You can access this form without loading a design.

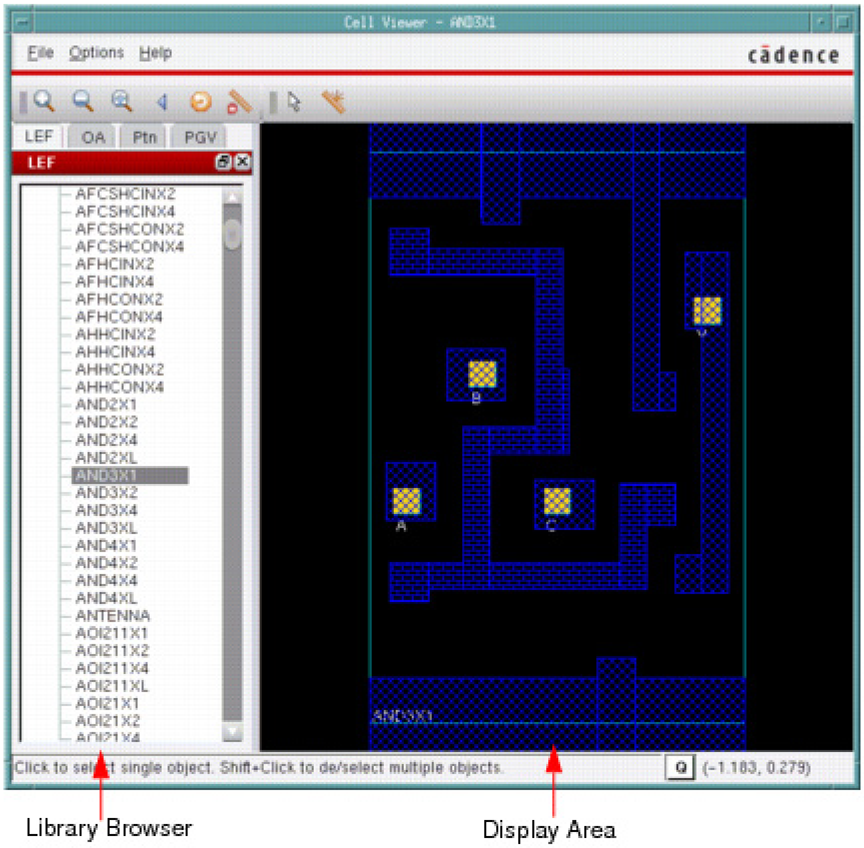
The Cell Viewer has the following tabs:

- Cell Viewer - LEF
- Cell Viewer - OA
- Cell Viewer - Ptn
- Cell Viewer - PGV

Cell Viewer - LEF

Use the LEF page to view the details of library cells using the information in the LEF file. You can navigate to a cell using the Library Browser, which is in the left side of the Cell Viewer. The details of the cell are displayed in the Display Area, which is in the right side of the Cell Viewer.

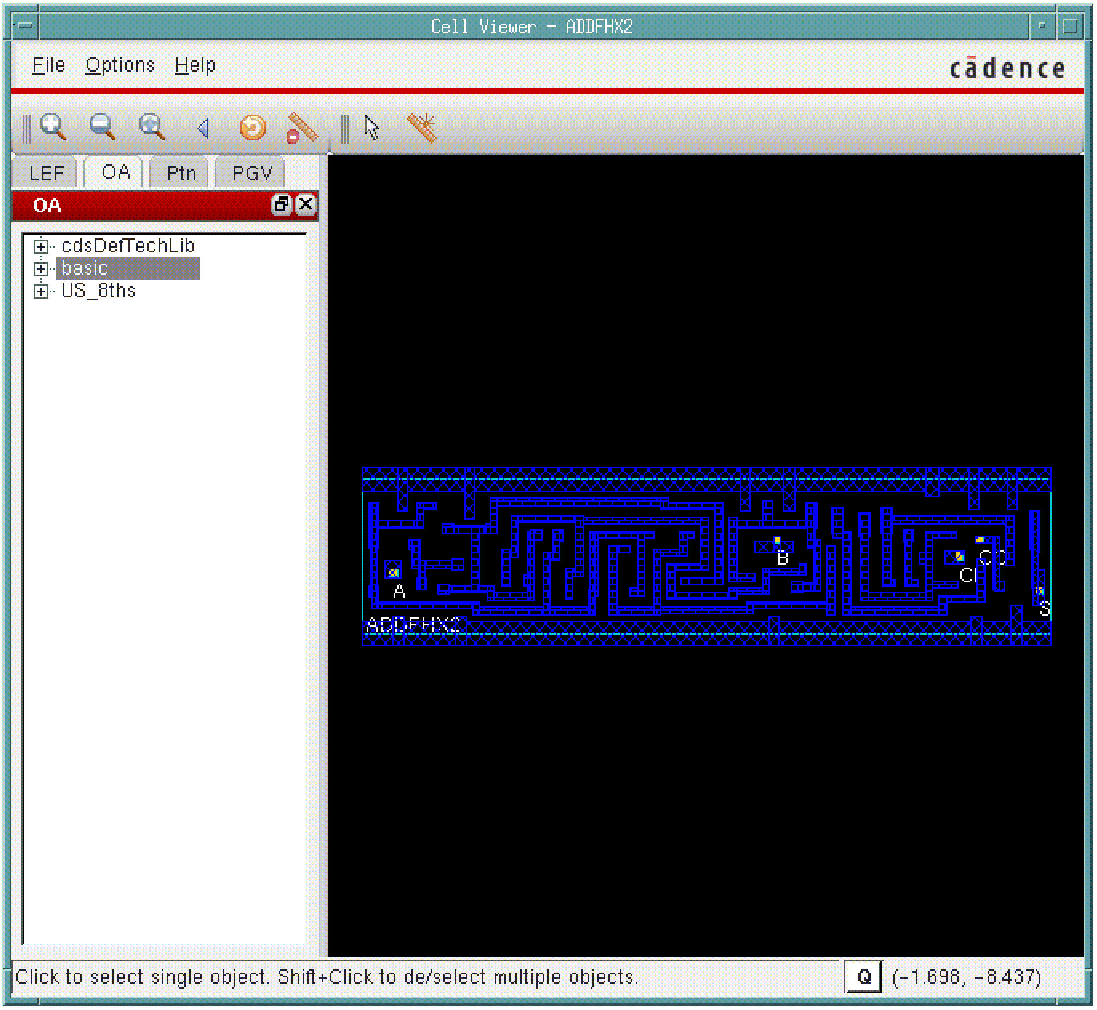
- Choose *Tools - Cell Viewer*, and click on the LEF page if it is not already selected.



Cell Viewer - OA

Use the OA page to view the details of OpenAccess cells using the information in the OA library.

- Choose *Tools - Cell Viewer*, and click on the OA page.

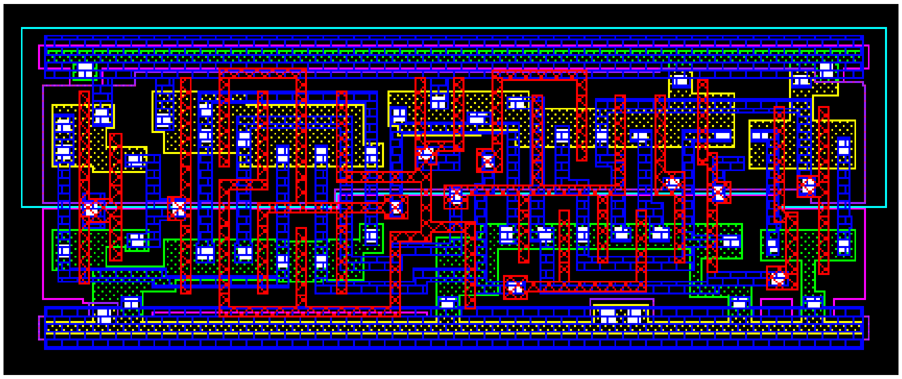


You can view an OA cell in two modes:

- OA-Layout View
- OA-Abstract View

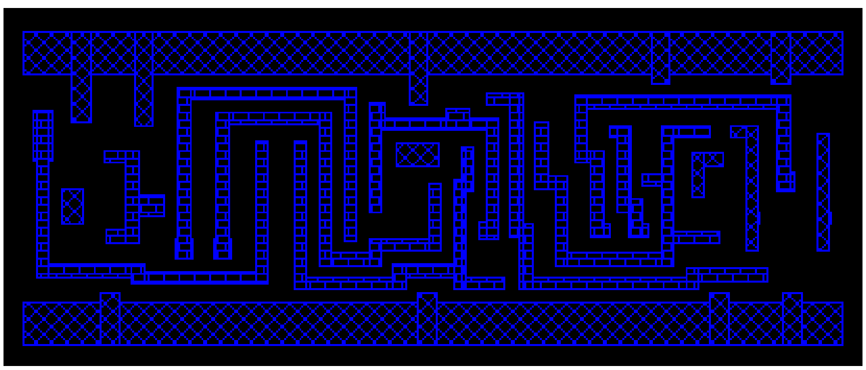
OA-Layout View

- In the Library Browser, select the OA cell, and then click *Layout*.



OA-Abstract View

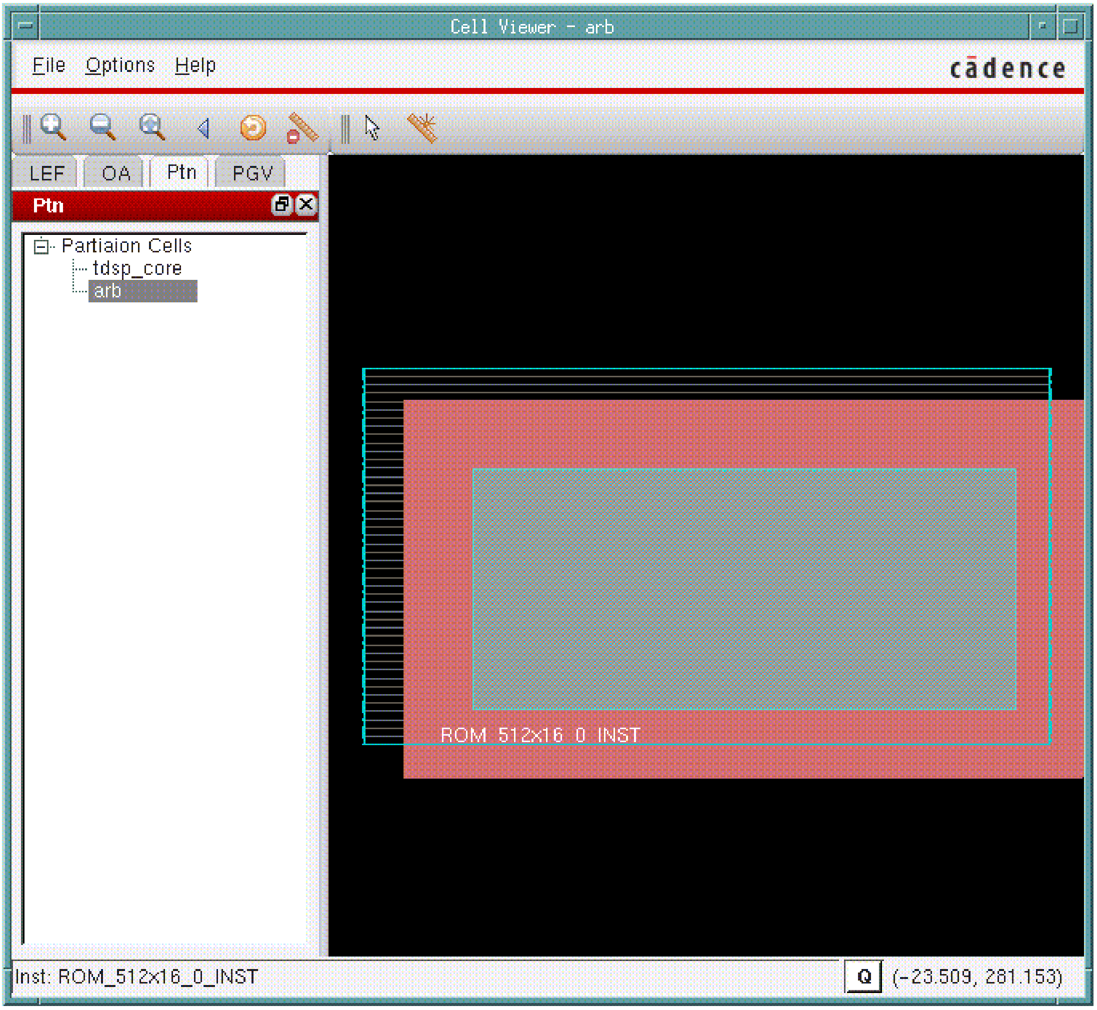
- In the Library Browser, select the OA cell, and then click *Abstract*.



Cell Viewer - Ptn

Use the Ptn page to display the abstract of any partition in the design. You can navigate to a partition through the Library Browser in the left side of the Cell Viewer.

- Choose *Tools - Cell Viewer*, and click on the Ptn page.

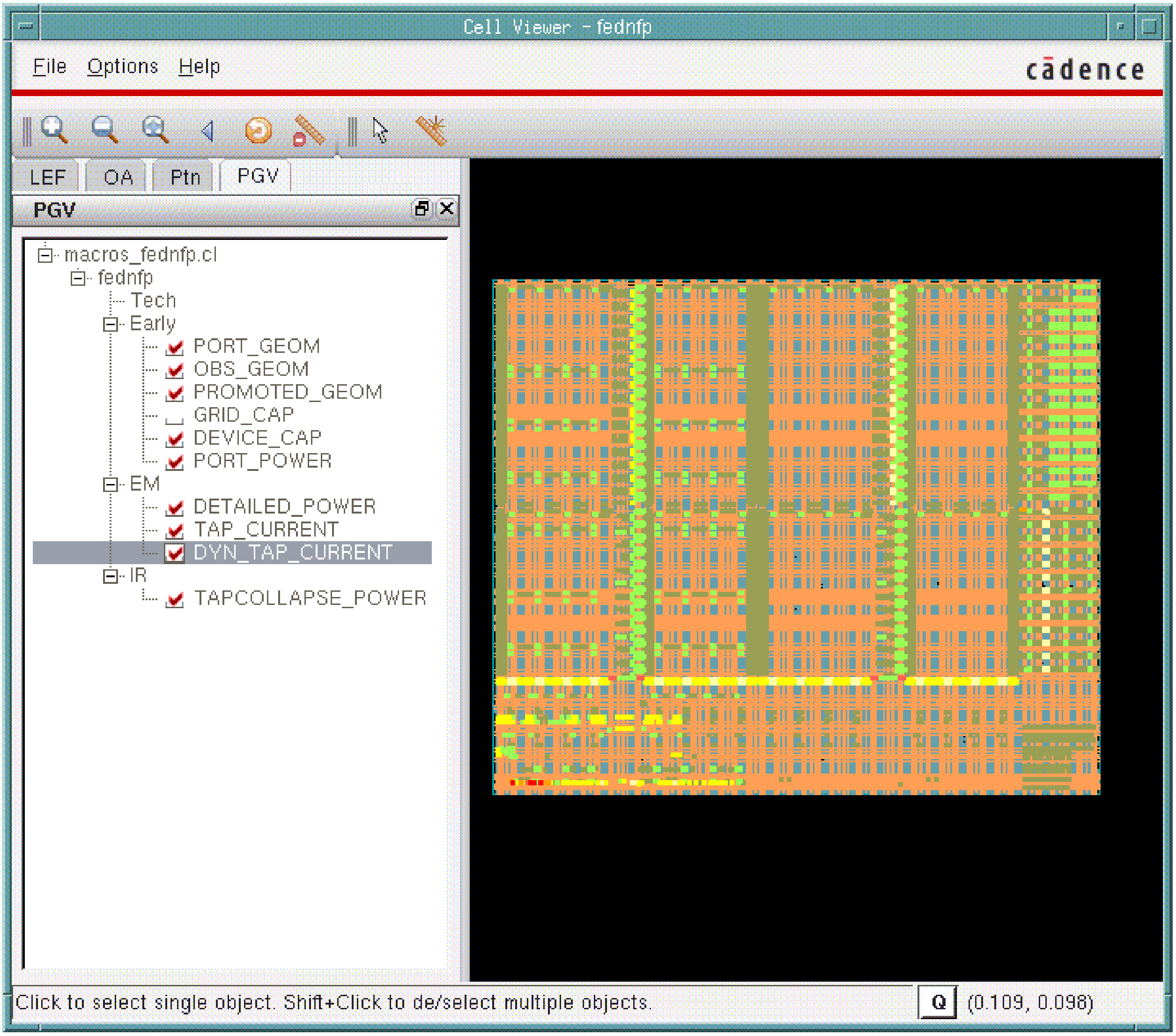


Cell Viewer - PGV

Use the PGV page to view and debug power-grid libraries. To load multiple libraries, select *Open Power Grid Library* from the *File* menu.

You can navigate to a cell using the Library Browser, which is in the left side of the Cell Viewer. The details of the cell are displayed in the Display Area, which is in the right side of the Cell Viewer. You can expand a cell to view the default views corresponding to each PGV type (Tech, Early, EM, and IR).

- Choose *Tools - Cell Viewer*, and click on the PGV page.



This form allows you to take a physical layout and overlay it with a power-grid resistor network extracted by Power-Grid Library Generator.

Fields and Options

File	The *File* menu provides the following options:
Open Power Grid Library	Opens multiple power grid libraries.

Free PGV Database	Specifies to free all the loaded power-grid libraries.
Quit	Closes the Cell Viewer window.
Options	
Net Based View - Show Popup Automatically	
	If this menu item is selected, clicking on a view displays a dialog box automatically. This dialog box allows you to display the power-grid of a single net. By default, this menu item is not selected. The dialog box displays the power, ground, and signal nets. You can select a net and click *Plot* to view a single net. Alternatively, you can view this dialog box using the following method: Right-click on a view to see the short-cut menu. Click the view name option on the shortcut menu. **Note:** Clicking on the cell name returns to the whole view mode, and displays the selected views for the cell name.

PGV Cell Views

You can click the + button to expand a cell and view the default cell views. Each view has a check box for displaying the content specific to the selected view. You can view either one or multiple views overlapped over each other.

The default views for different power-grid view types are:

- Tech
  Displays the basic port information of a cell.
- Early
  Geometrical View - displays the LEF of a cell. The types of geometrical views are: PORT_GEOM, OBS_GEOM, PROMOTED_GEOM and OPTIMIZED_GEOM views.

  Port View - displays more accurate port and tap information of a cell when . This information is available in the PORT_POWER view of the library.

  Capacitance Views - displays the grid and device capacitance information of a cell. The types of capacitance views are: GRID_CAP and DEVICE_CAP views. These views are net specific.

- EM
  Detailed View - displays the detailed power-grid information for all nets. This information is available in the DETAILED_POWER view of the library.

  Tap Current View - displays tap information of a cell. All the tap currents are shown in the viewer drawn in 8 colors for 8 linear filters according to the min max value. This information is available in the TAP_CURRENT view of the library.
  Dynamic Tap Current View - displays dynamic tap currents for each interval. The total current plot is net based, and supports multiple trigger view in a single PGV. This information is available in the DYN_TAP_CURRENT view of the library.

  Powergate View - displays the power switch information of a power-gate cell. This information is available in the POWERGATE view of the library.
- IR
  Collapsed View - displays the collapsed current taps for the bit-cells inside the memory. This information is available in the TAPCOLLAPSE_POWER view of the library.

Cell Viewer Widgets

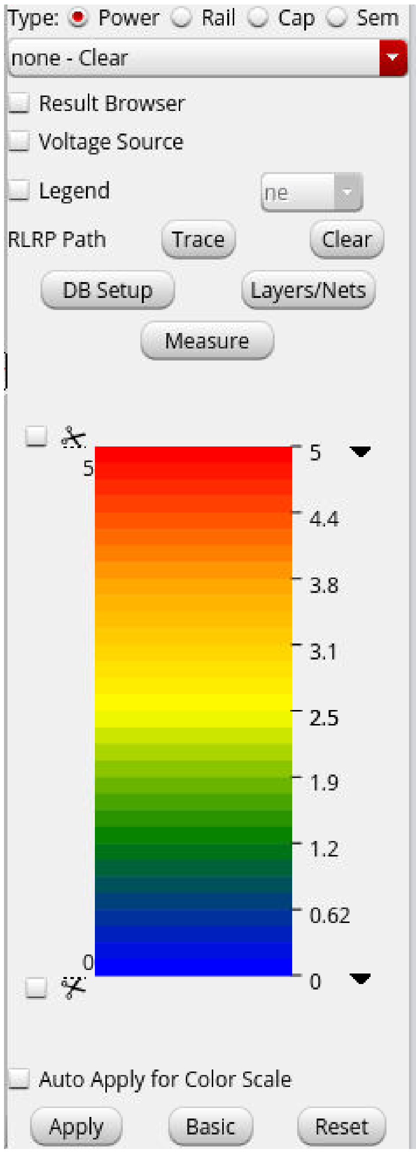
The Cell Viewer supports the following widgets: Zoom In, Zoom Out, Fit, View Last, Redraw, Clear All Rulers, Select and Ruler.

Auto Query

The Cell Viewer window includes an Auto Query (Q) button located at the bottom of the design display area that enables and disables the auto query. When you enable auto query, text information displays for any object that is under the cursor. You can use auto query for information on cell instances, instance pins, and resistors.

Power and Rail Plots

The Power & Rail Plots form is the main console window for displaying the power and rail analysis results. The data can be loaded into GUI using the *Power & Rail Setup* form accessed by clicking the *DB Setup* button. The displayed nets and layers selection can be changed using the *Power & Rail Layers/Nets* form accessed by clicking the *Layers/Nets* button. The rest of the display can be configured on the main window.



Fields and Options

<table>
<tr><td>Type</td><td>Power - Plots the power analysis plot type.
Rail - Plots the rail analysis plot type.
Cap - Plots the capacitance plot type.
Sem - Plots the signal electromigration (EM) plot type.
You can select a power, rail, capacitance, or signal EM plot type from the drop-down list. For a description of the various plot types, see the <code>set_power_rail_display</code> command.</td></tr>
<tr><td>Result Browser</td><td>Displays the Results Browser form.</td></tr>
<tr><td>Voltage Source</td><td>Displays voltage sources (power pads) using white pixels in the layout canvas.</td></tr>
<tr><td>RLRP Path</td><td>You can select an object (instance or resistor) and click the Trace button to highlight the least effective resistance path from the object to the nearest voltage source, and to open the Resistance Path form. The rail analysis must include RLRP analysis (<code>set_rail_analysis_mode</code> <code>-enable_rlrp_analysis true</code>) in order to have the required data for this functionality. You can click the Clear button to clear the path.

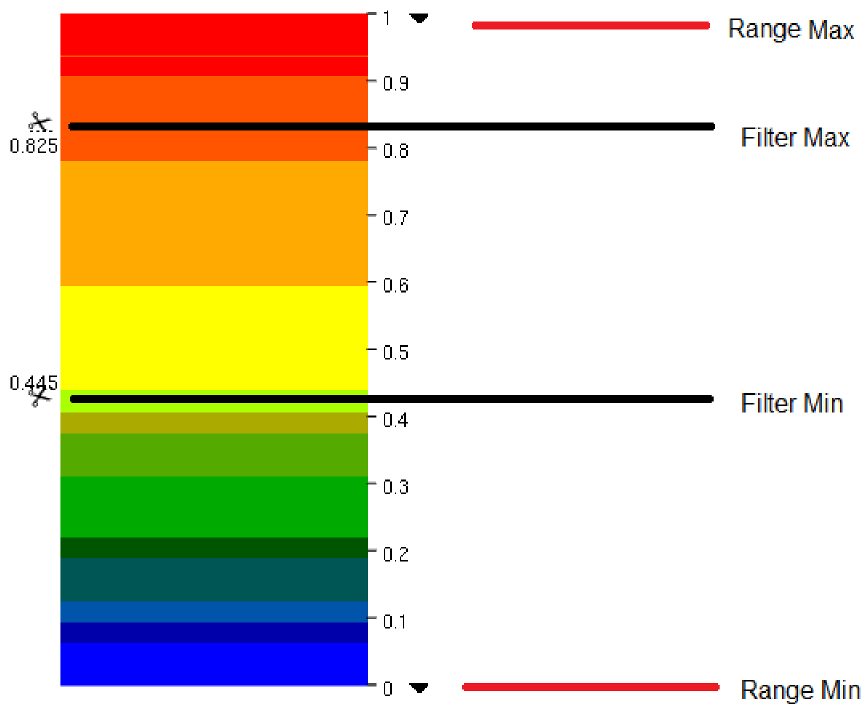
The Result Browser will list each instance’s total resistance, and the Resistance Path form will list the break down of each resistance in the rlrp path. You can select a resistor in this form, single click to select/zoom, and double click to query.

Note: If the Resistance Path form is closed, you can open it again by first selecting the object that you want to trace, and then clicking the Trace button.</td></tr>
<tr><td>Legend</td><td>Displays the filter color and range details. When you select a location (ne nw se sw), this opens the legend at the specified location on the design layout. You can clear the checkbox to close the legend.

Note: Legend is useful when dumping the screen capture to save the color range information.</td></tr>
<tr><td>DB Setup</td><td>Opens the Power & Rail Setup form that allows you to load power/rail/signal EM analysis results into GUI.</td></tr>
</table>

Layer/Nets	Opens the Power & Rail Layers/Nets form that allows you to control the visibility of the layers, the opacity of the rail analysis or design database, the visibility of power/ground nets, and visibility of the signal EM layers.
Measure	Provides the ability to measure tap current and capacitance of the specified region. The use model is as follows: Click the *Measure* button. Select a region in the main window. Look for the results in the console window.

<table>
<tr>
<td>Range</td>
<td>When a plot type is selected, the filter values automatically get filled in.

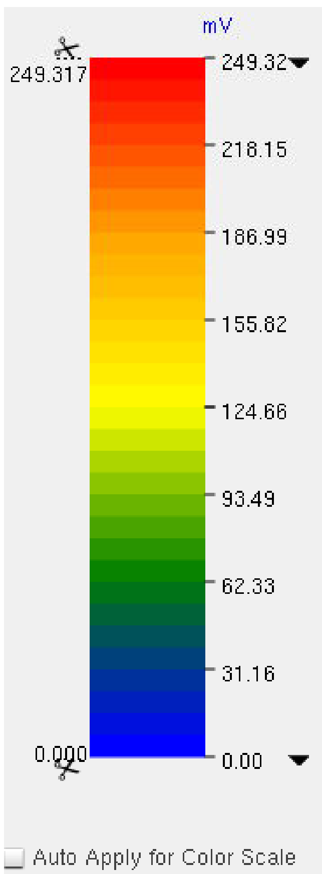


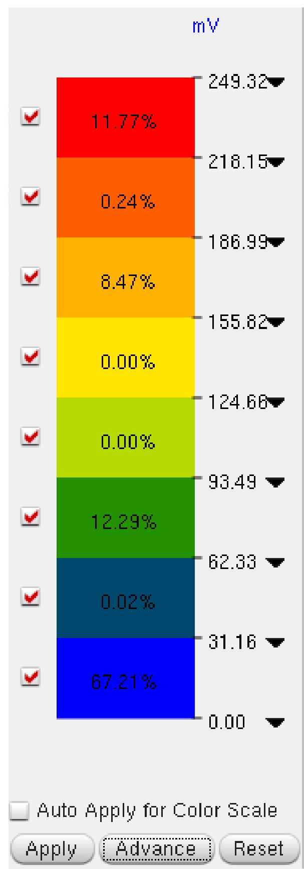
Range Min and Max - Click the icon to specify the minimum and maximum value to report. When the filter is at the default (max/min) position, any values greater than “Range Max” will take on the range max color, and vice-versa for “Range Min”.

Filter Min and Max - Click the icon to specify the minimum and maximum value to include in the report/display. Use the scissor icon to further filter out the data within the range specified. This is used to obtain statistic of filter range. “Filter Max” will hide any data point greater than the filtered value, and vice-versa for “Filter Min”.</td>
</tr>
<tr>
<td>Auto Apply for Color Scale</td>
<td>Allows you to automatically redraw the layout after modifying the range values. If you do not select this checkbox, you can click the Apply button to manually redraw the layout.</td>
</tr>
</table>

<table>
<tr>
<td>Basic/Advanced</td>
<td>
The Basic/Advanced toggle button allows you to alternate between the color gradient (Basic) and color range (Advanced) mode when it is clicked. By default, the Basic button and the color gradient based filter are displayed. When you click the Basic button, the button text changes to Advanced, and the form shows the color range based filter. The color gradient based filter allows you to modify the minimum and maximum values, and the color range based filter allows you to modify 8 equal filters between the minimum and maximum values.

Basic (Color Gradient) Advanced (Color Range)





Selecting Apply will accept the changes, and Reset will set it to default values.
</td>
</tr>
</table>

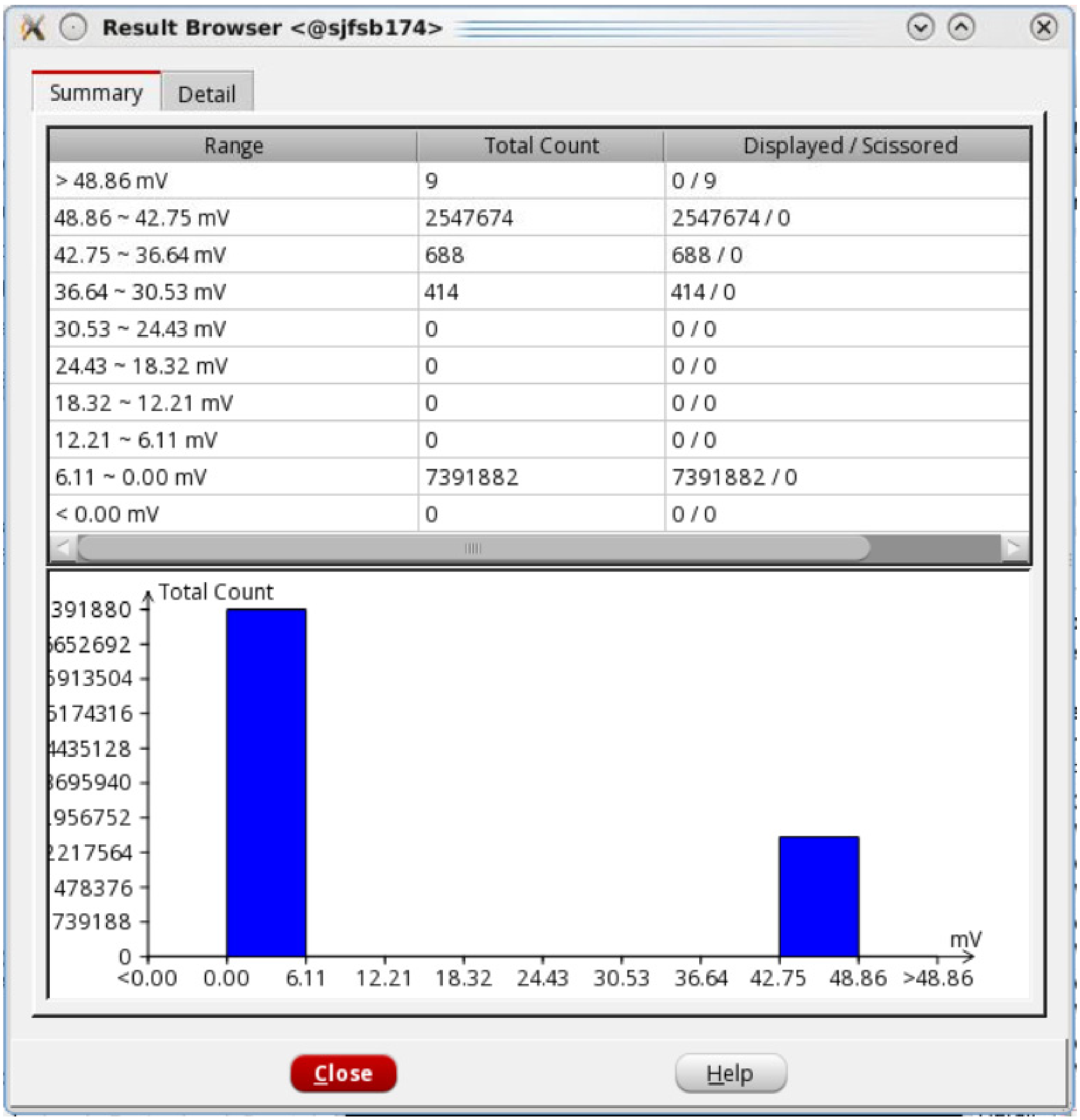
Related Text Commands

See `report_power_rail_results` and `set_power_rail_display` commands in *Voltus Text Command Reference.*

Result Browser

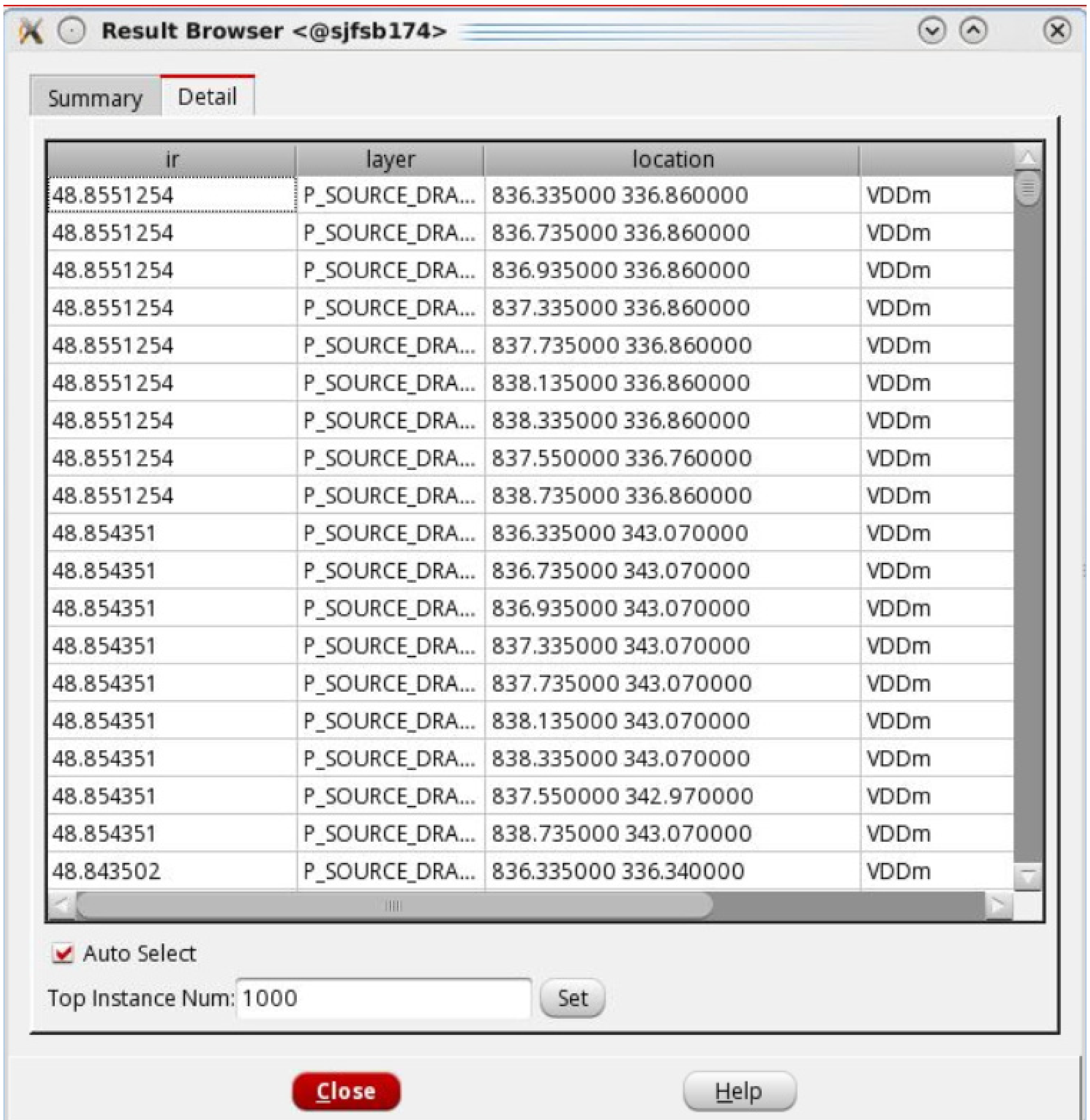
The Result Browser form has the following two tabs:

Summary - provides summary information about the plot displayed. The data points are divided into buckets based on the range specified, and the summary information provides the statistic of each bucket. The tab also displays the rail analysis results in the histogram form.



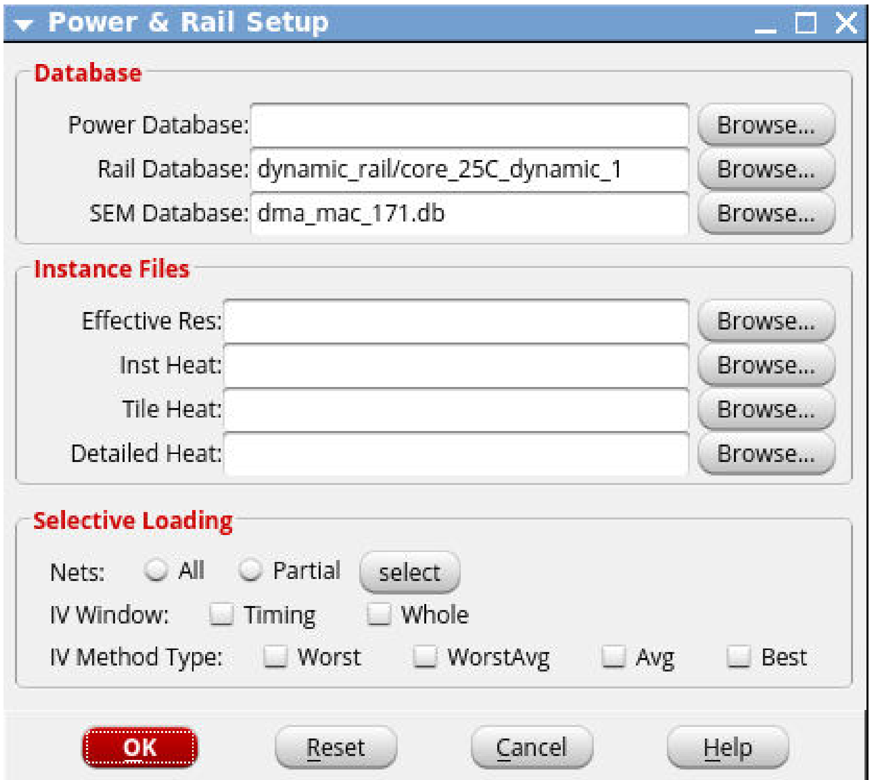
Range	Total Count	Displayed / Scissored
> 48.86 mV	9	0 / 9
48.86 ~ 42.75 mV	2547674	2547674 / 0
42.75 ~ 36.64 mV	688	688 / 0
36.64 ~ 30.53 mV	414	414 / 0
30.53 ~ 24.43 mV	0	0 / 0
24.43 ~ 18.32 mV	0	0 / 0
18.32 ~ 12.21 mV	0	0 / 0
12.21 ~ 6.11 mV	0	0 / 0
6.11 ~ 0.00 mV	7391882	7391882 / 0
< 0.00 mV	0	0 / 0

Detail - provides details of the individual data points of the plot displayed. By default, the first 1000 data points are displayed.



Power and Rail Plots - DB Setup

Loads power/rail analysis results into GUI.



Fields and Options

Power Database	Specifies the power database file. This file was created by using the `-create_binary_db` option of the `set_power_analysis_mode` command.
Rail Database	Specifies the rail analysis directory. You can load all the domain results by selecting the automatically generated analysis directory under the output directory (<domain>_<temp>_<dynamic/avg>_#), or load individual nets by going one level further and selecting the net directory.
SEM Database	Specifies the database file generated by the `verify_AC_limit -report_db` command. This file contains the information required for loading and displaying signal EM data in GUI.

Effective Res	Specifies the result file from effective resistance analysis (only instance-based result is supported). When the rail analysis plot type `effr` is selected and a result file is specified, the layout shows effective resistance analysis result for each instance. This field supports display of both net and domain based result file.
Inst Heat	Specifies the name of the instance-based delta temperature file from self-heating analysis. This is the device instance temperature.
Tile Heat	Specifies the name of the tile-based delta temperature file of self-heating analysis. This is the worst instance temperature among all wire segments in a tile. This is also layer-based. The FEOL or cell layer is used to show the device instance temperature.
Detailed Heat	Specifies the name of the detailed wire-based delta temperature file from self-heating analysis.
Custom Value Instance Files	Specifies the name of the custom value instance file (CVIF) that can be used to display a user-specified instance voltage file in GUI. After specifying the CVIF file, you can select the *cvil Rail* plot in the *Power & Rail Plots* form for displaying the EIV map.
Selective Loading	*Nets* - Specifies whether all nets or selected nets are to be loaded into the GUI. *All* - loads all the nets in the domain-based state directory to the GUI. This option is selected by default. *Partial* - Loads only the specified nets in the domain-based state directory to the GUI. To specify nets: Select *Partial* and click the *Select* button. The *Selective Nets* form appears. Select the required power and ground nets, and click *OK*. *IV Window* - Loads the instance voltage data for the selected EIV window. By default, the instance voltage results for both the timing and switching windows, and the entire rail simulation are loaded. *IV Method Type* - Loads the instance voltage data for the specified EIV computing method. By default, the instance voltage results for all the four methods are loaded.

Related Text Commands

See `read_power_rail_results` command in *Voltus Text Command Reference.*

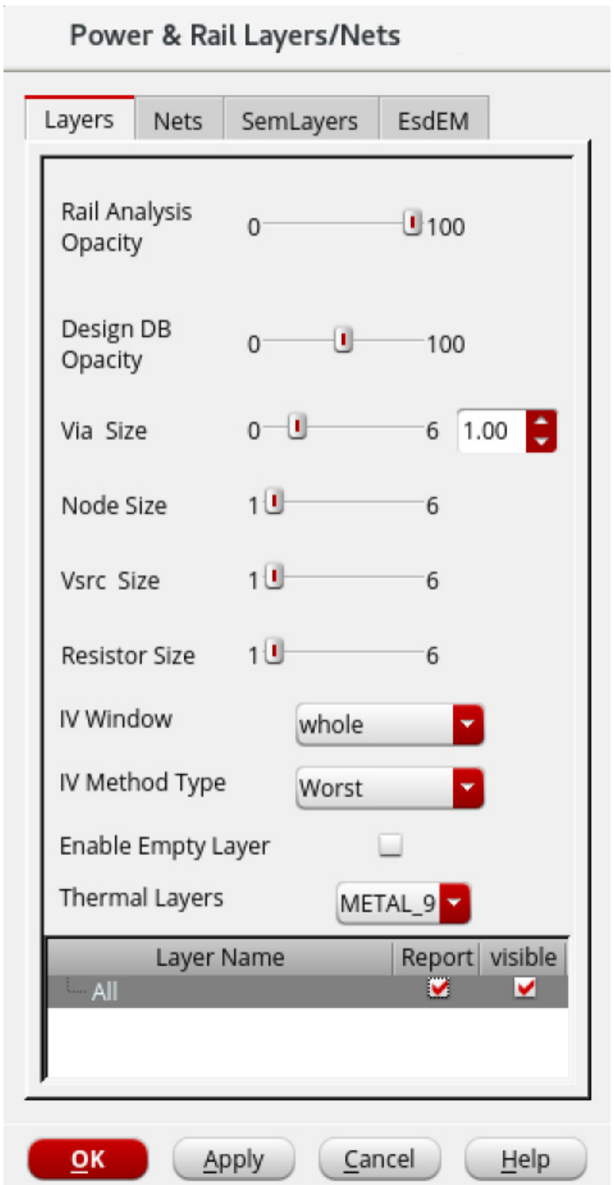
Power and Rail Plots - Layers/Nets/Sem Layers

Choose *Power Rail* - *Power Rail Plots*, and then click the *Layers/Nets* button.

Layers	Provides the ability to set the visibility of the layers and the opacity of the rail analysis or design database. This makes it easier to view when overlaying the rail analysis and the design database.
Nets	Controls the visibility of power/ground nets.
Sem Layers	Controls the visibility of the signal EM layers.
Esd EM	Provides the ability to selectively load the required ESD EM result directories if there are multiple result directories under the ESD directory.

Layers

Provides the ability to set the visibility of the layers and the opacity of the rail analysis or design database. This makes it easier to view when overlaying the rail analysis and the design database.



Fields and Options

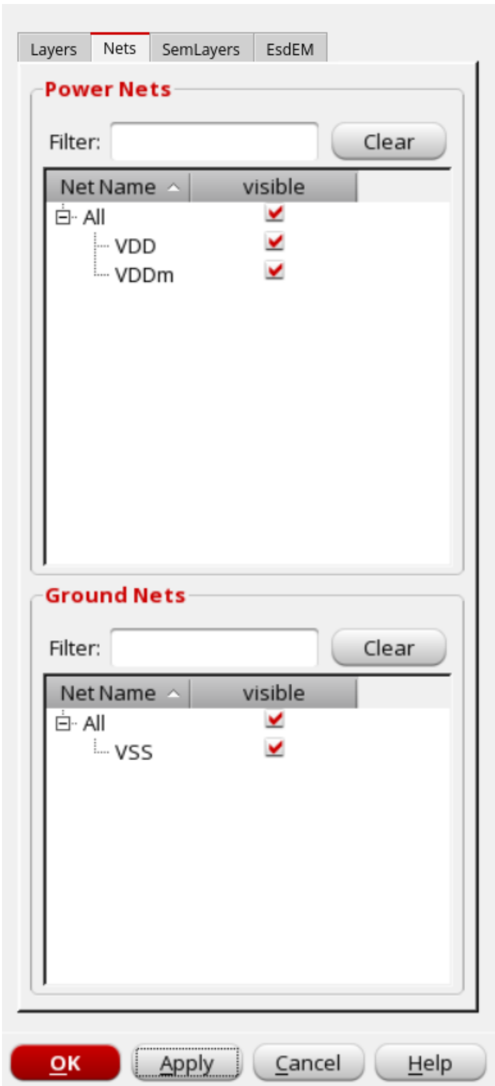
Rail Analysis Opacity	Sets the opacity of the rail analysis output. These will be the resistor/node database load from rail analysis.
Design DB Opacity	Sets the opacity of the design database. These will be the shapes coming from DEF (main control panel).
IV Window	Controls the window where effective instance voltage (EIV) is evaluated. EIV is obtained by processing the voltage waveforms and finding the worse-case effective voltage between the power and ground pins during a specific window. There are two possible options to evaluate EIV: ■ *timing/switching* – both switching activity (current is non-leakage) and timing window ■ *whole* – entire rail simulation not filtered to a specific window
IV Method	Allows different methods of computing effective instance voltage during rail analysis. Using this option, you can write out the *Best*, *Worst*, or *Average* effective instance voltage between power and ground nets. The *WorstAvg* method computes the average instance voltage during each evaluation window defined by the *IV Window* field, and reports the worst average instance voltage out of all the windows. If *IV Method* is *WorstAvg* and *IV Window* is *whole*, then this method will report average voltage across all time steps in simulation. *Default*: `Worst`
Layers	You can enable or disable the reporting and visibility of the displayed layers.
Report	Specifies to select the user-defined layers to be used for processing.
Visible	Specifies which layers should be made visible when plotting.

Related Text Commands

See `set_power_rail_layers_nets` command in *Voltus Text Command Reference.*

Nets

Controls the visibility of power/ground nets.



Fields and Options

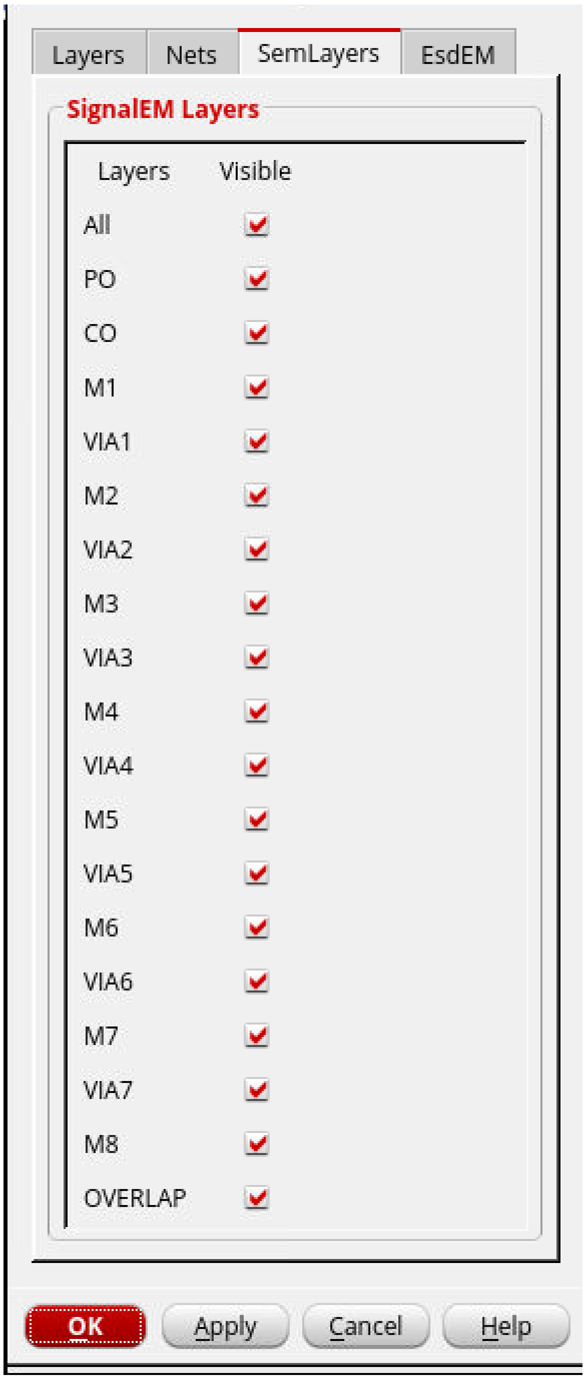
Power Nets	Displays the selected power nets.
Ground Nets	Displays the selected ground nets.

Related Text Commands

See set_power_rail_layers_nets command in *Voltus Text Command Reference.*

Sem Layers

Controls the visibility of the signal EM layers.



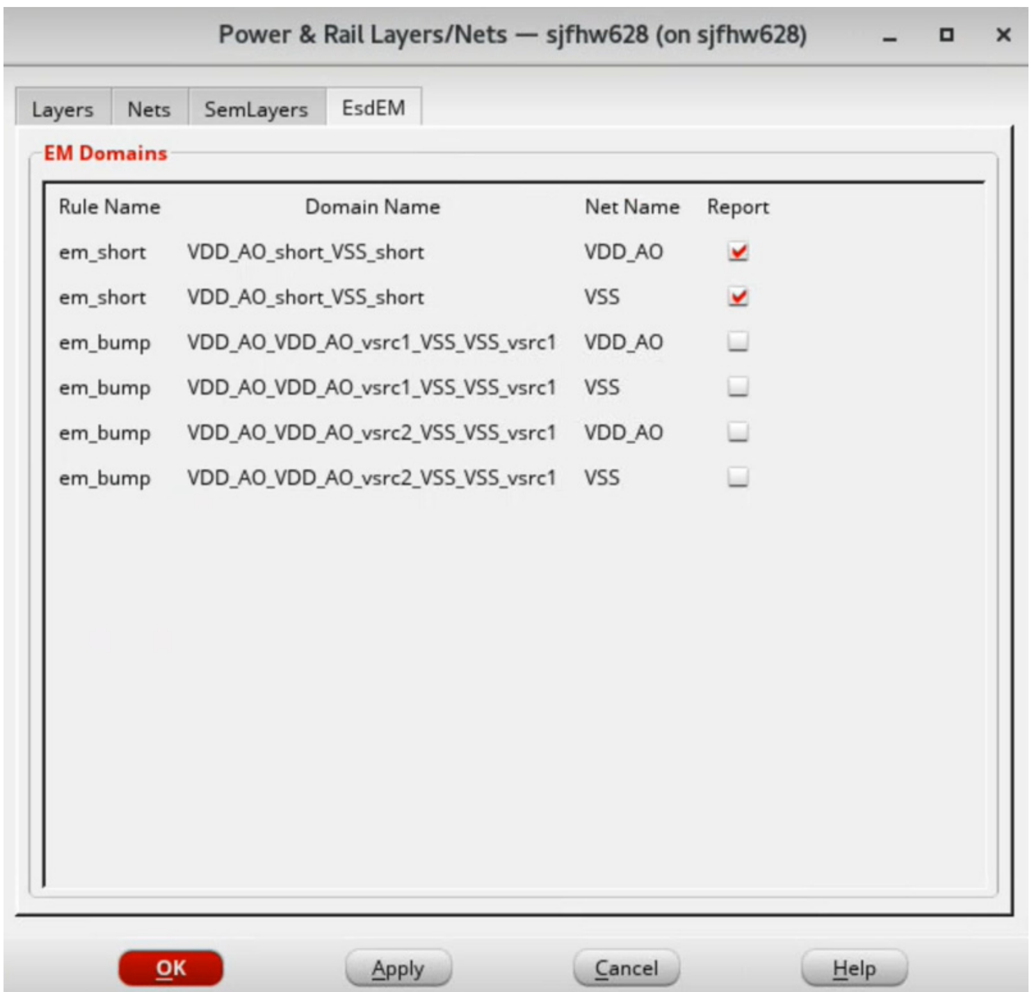
Fields and Options

Layers	You can enable or disable the visibility of the displayed signal EM layers.
Visible	Specifies which layers should be made visible when plotting.

Related Text Commands

See `set_power_rail_layers_nets` command in *Voltus Text Command Reference.*

EsdEM



Fields and Options

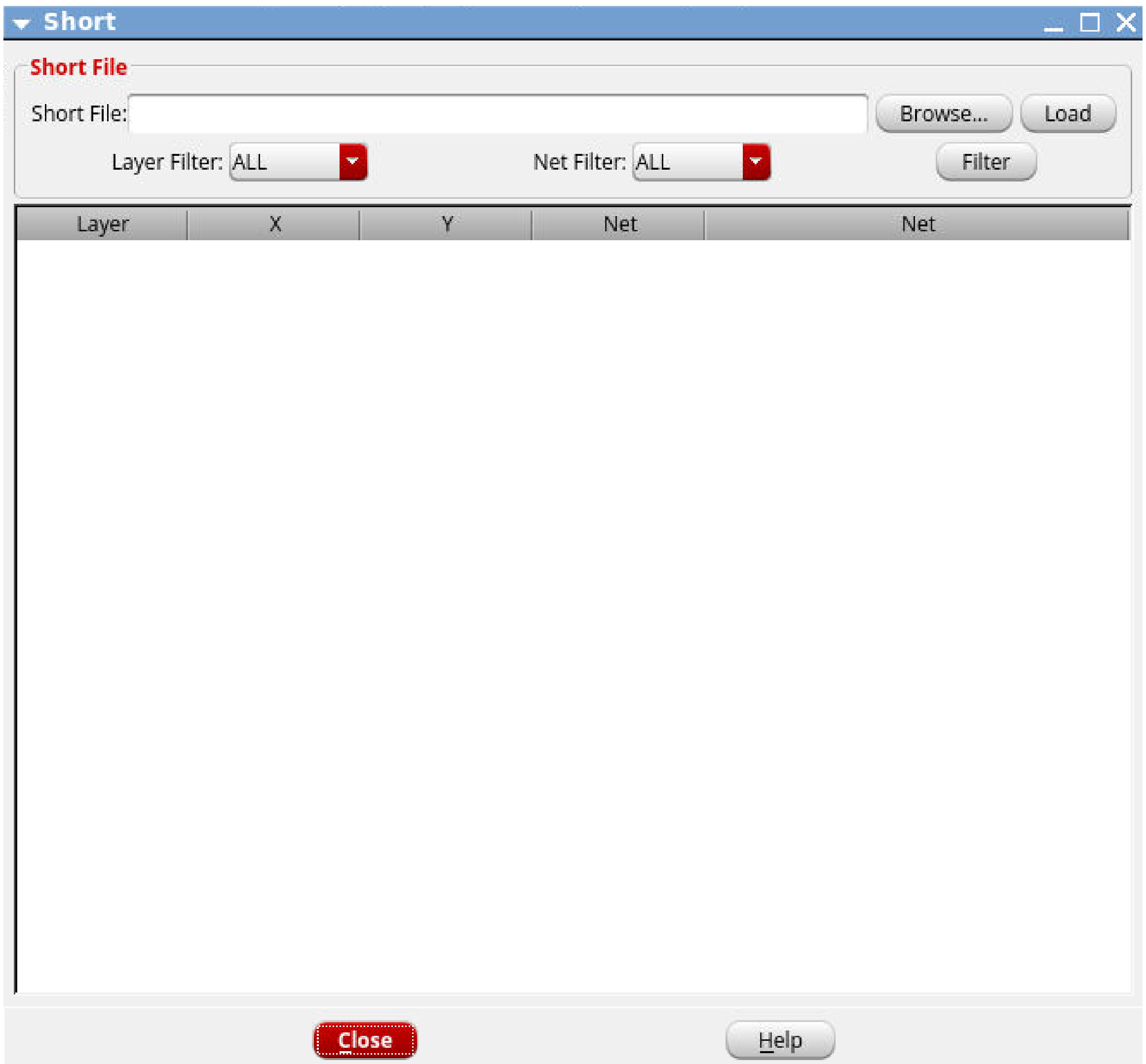
Rule Name	Displays the list of EM rule name.
Domain Name	Displays the list of domain name associated with the EM rule.
Net Name	Displays the list of nets associated with the EM rule. **Note:** You cannot select the same net from different rule or domain.
Report	Specifies which ESD EM result directory should be loaded.

Related Text Commands

See `set_power_rail_layers_nets` command in *Voltus Text Command Reference.*

Short

Loads the short violations report (`voltus_short.report`) in the GUI. This file is generated using the `set_rail_analysis_mode` `-report_shorts true` command parameter, and is located in the work directory.



Fields and Options

Short File	Specifies the path to the short violations report.
Browse button	Opens the Short File browser form, which enables you to browse to the short violations report.

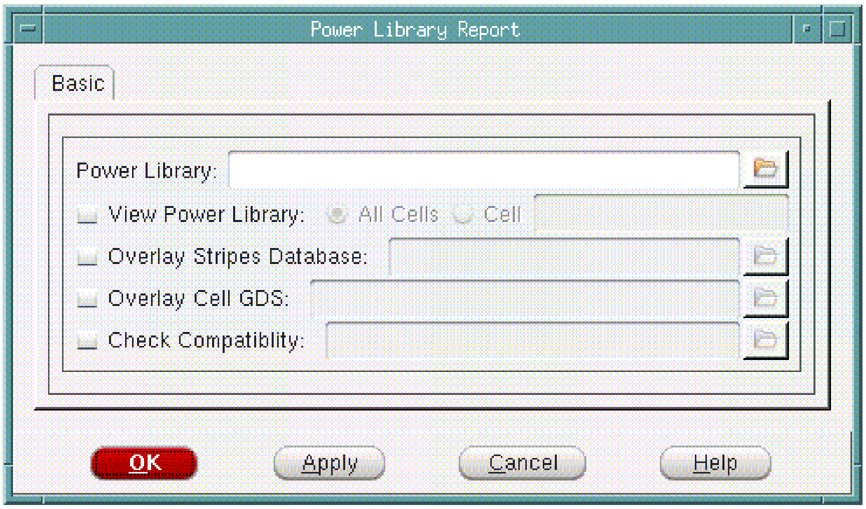
Load button	Loads the report file and displays the list of short violations on the power/ground nets.
Layer Filter	Allows you to filter the records based on the selected layer.
Net Filter	Allows you to filter the records based on the selected net.
Filter button	Displays the list of short violations based on the selected options from the *Layer Filter* and *Net Filter* drop-down lists.
Layer	The layer on which the short is detected.
X	The x coordinate where the short is detected.
Y	The y coordinate where the short is detected.
Net	The short violations report lists shorts between two geometries belonging to different nets. This field specifies the name of the first power or ground net.
Net	The short violations report lists shorts between two geometries belonging to different nets. This field specifies the name of the second power or ground net.

Related Text Commands

See `set_rail_analysis_mode` and `read_power_rail_results` commands in the *Voltus Text Command Reference*.

Power Library Report

Reports contents of the already generated power library and physical view of the library power-grid network.



Fields and Options

Power Library	Specifies the power-grid library to generate the power library report.
View Power Library	Specifies the following options to view the power library: ■ *All Cells* ■ *Cells*
Overlay Stripes Database	Specifies the stripes database to be overlaid during the psviewer session.
Overlay Cell GDS	Specifies the cell gds design file to be overlaid, during the psviewer session.
Check Compatibility	Checks the compatibility between the specified cell libraries.

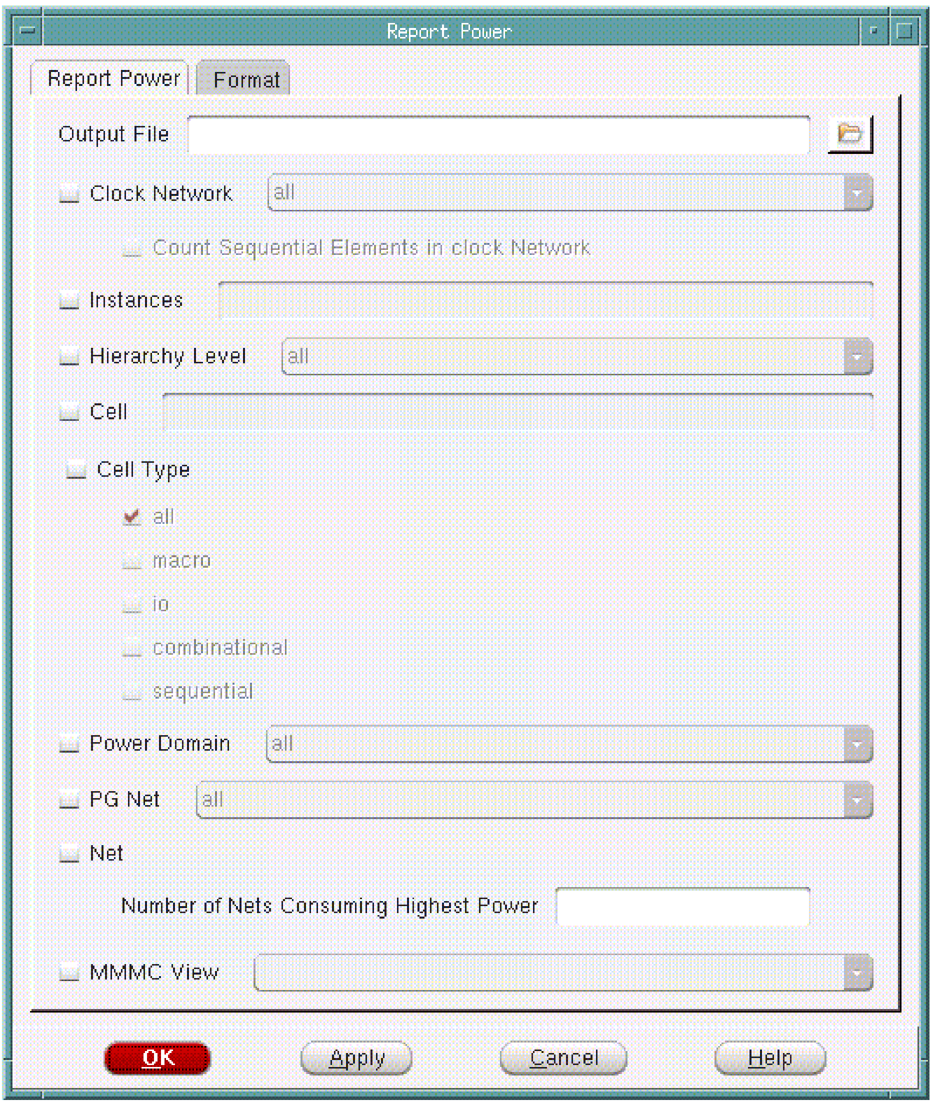
Related Topics

See *Control Panel - Cell Views* and *Viewing stripes data* in the "Menus and Forms" chapter of the *Power Systems Viewer Manual*.

Power Report

Provides the ability to provide reports of power information in various ways.

Power Report - Report Power

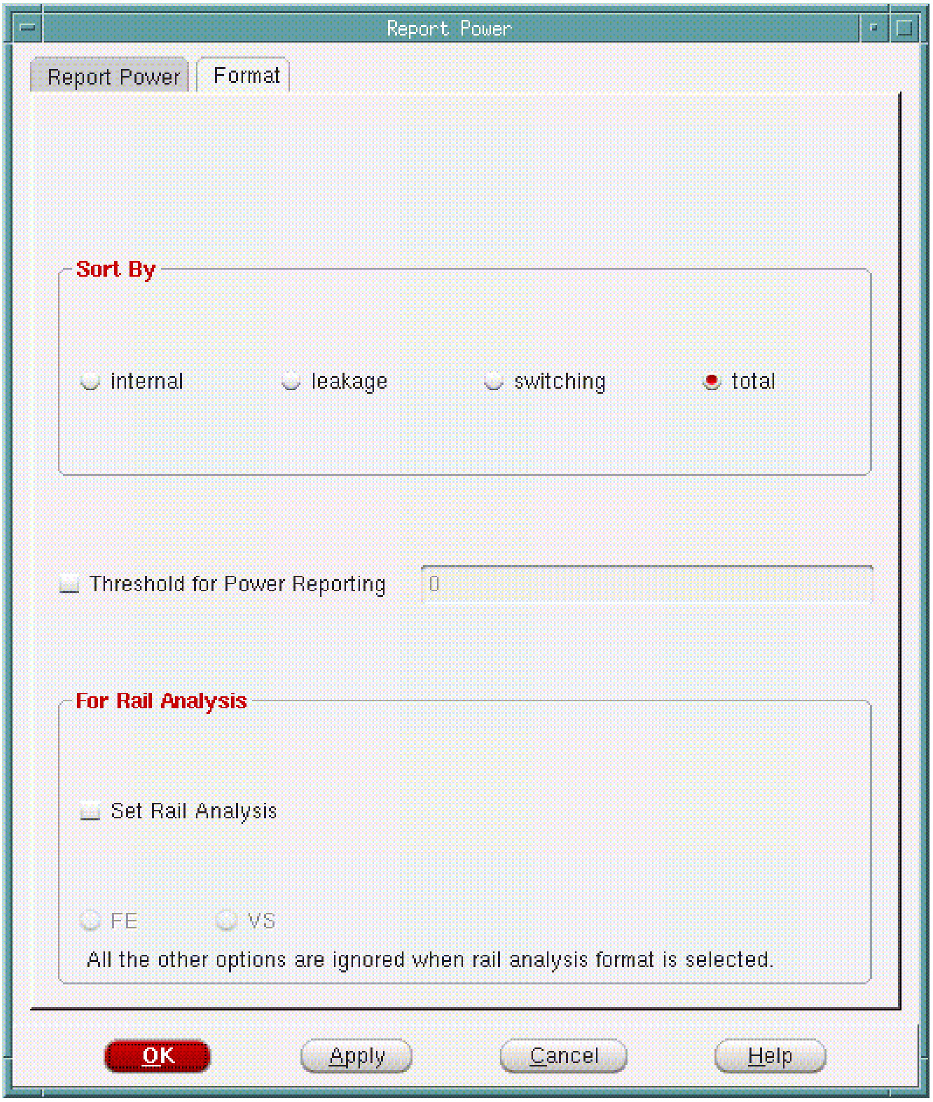


Fields and Options

Output File	Specifies the name of the report file in which power analysis output information is written. You can include the directory name. **Note:** If you do not specify this parameter, the software directs the power report to the command line, as well as the log file, and no power report is written to a separate file.
Clock Network	Reports the power consumption of the clock network, including the power for generated clocks.Specify a list to report the power consumed by specific clocks.Specify `all` to report all clock networks. *Default*: `all`
Count Sequential Elements in clock Network	If selected, report will include the leaf flip-flop power in the clock network power report. *Default*: The leaf flip-flop power will not be included in the clock network power report.
Instances	Specifies a list of instances to include in the power report. The power report provides the amount of power consumed by a specified list of hierarchical or leaf-level instances. This field accepts wildcards (*) for leaf-level instances only.
Hierarchy Level	Specifies the hierarchy level that is included in the power report. You can specify the *Hierarchy_level* as a number that corresponds to a specific hierarchy level in the design. A *Hierarchy_level* of 0 specifies the top level. *Default*: `all`
Cell	Specifies the cells to include in the power report. Accepts wildcards (*)
Cell Type	Specifies the cell type to included in the power report. The choices are `all` or any combination of macro, io, combinational, or sequential. *Default*: `all`
Power Domain	Specifies all power domains or a list of power domains to be included in report.

PG Net	Specifies `all` power nets or those of a particular hierarchy level to be included in power report.
Net	Reports the net switching power, as well as the load capacitance, toggle rate, and switching values for each net in the design.
Number of Nets Consuming Highest Power	Reports the number of nets with the highest net switching power. **Note:** You must use this parameter in conjunction with the *Net* parameter.
MMMC View	Specifies the view that was created using the multi-mode multi-corner (MMMC) `create_analysis_view` command. Power analysis is performed for the specified view. *Default*: Uses current view.

Power Report - Format



Fields and Options

Sort By: *internal* *leakage* *switching* *total*	Sorts power report based on the power (internal / leakage / switching / total) option that you select.

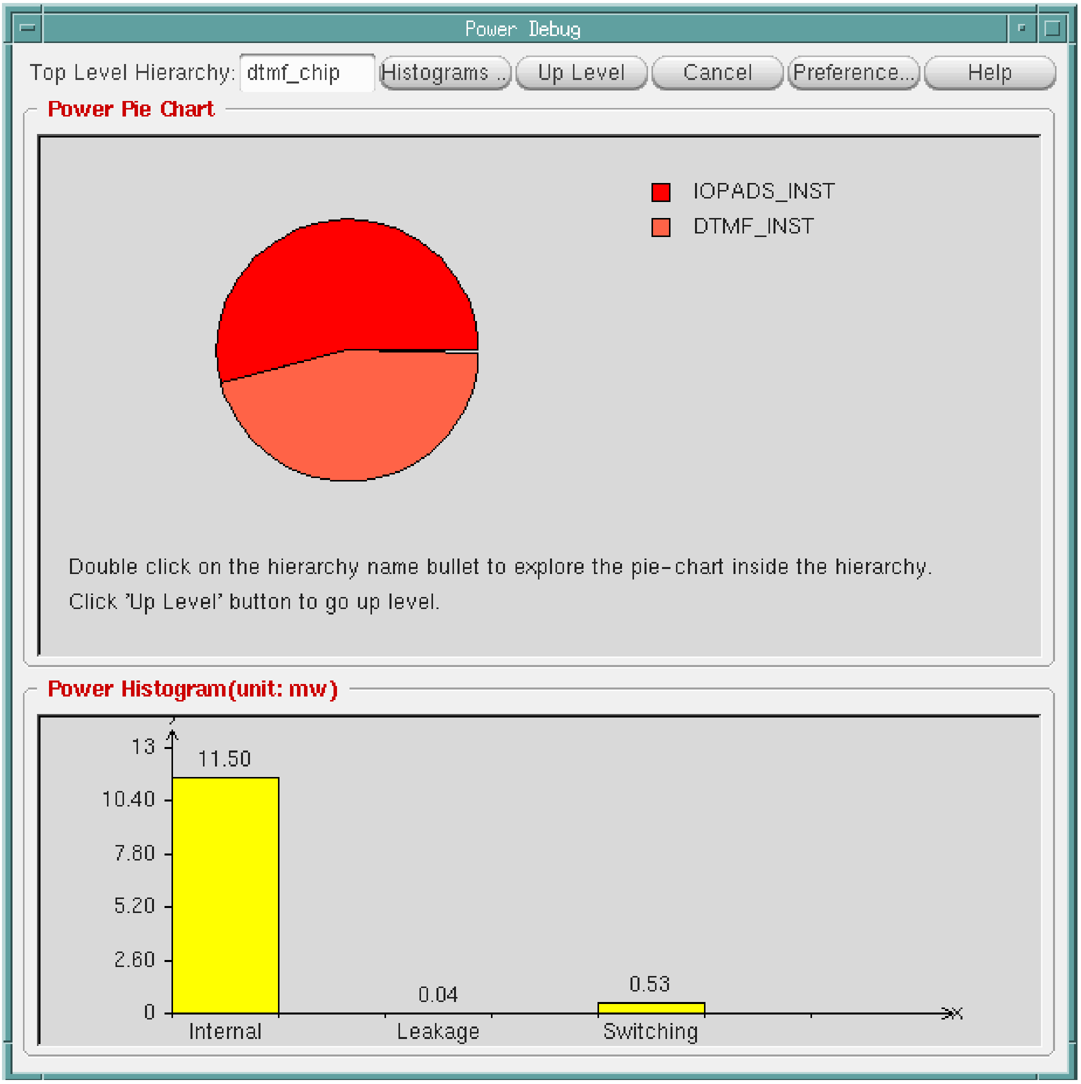
Threshold for Power Reporting	
	Specifies the threshold value for power reporting.
For Rail Analysis	*Set Rail Analysis: Specifies to report power in rail analysis format.* *FE:* *VS:*

Power Histograms

The Power Debug form displays power histograms in various ways.

- To open the Power Debug form, choose .
- To display information from one level down in the hierarchy, double-click on a listed instance.

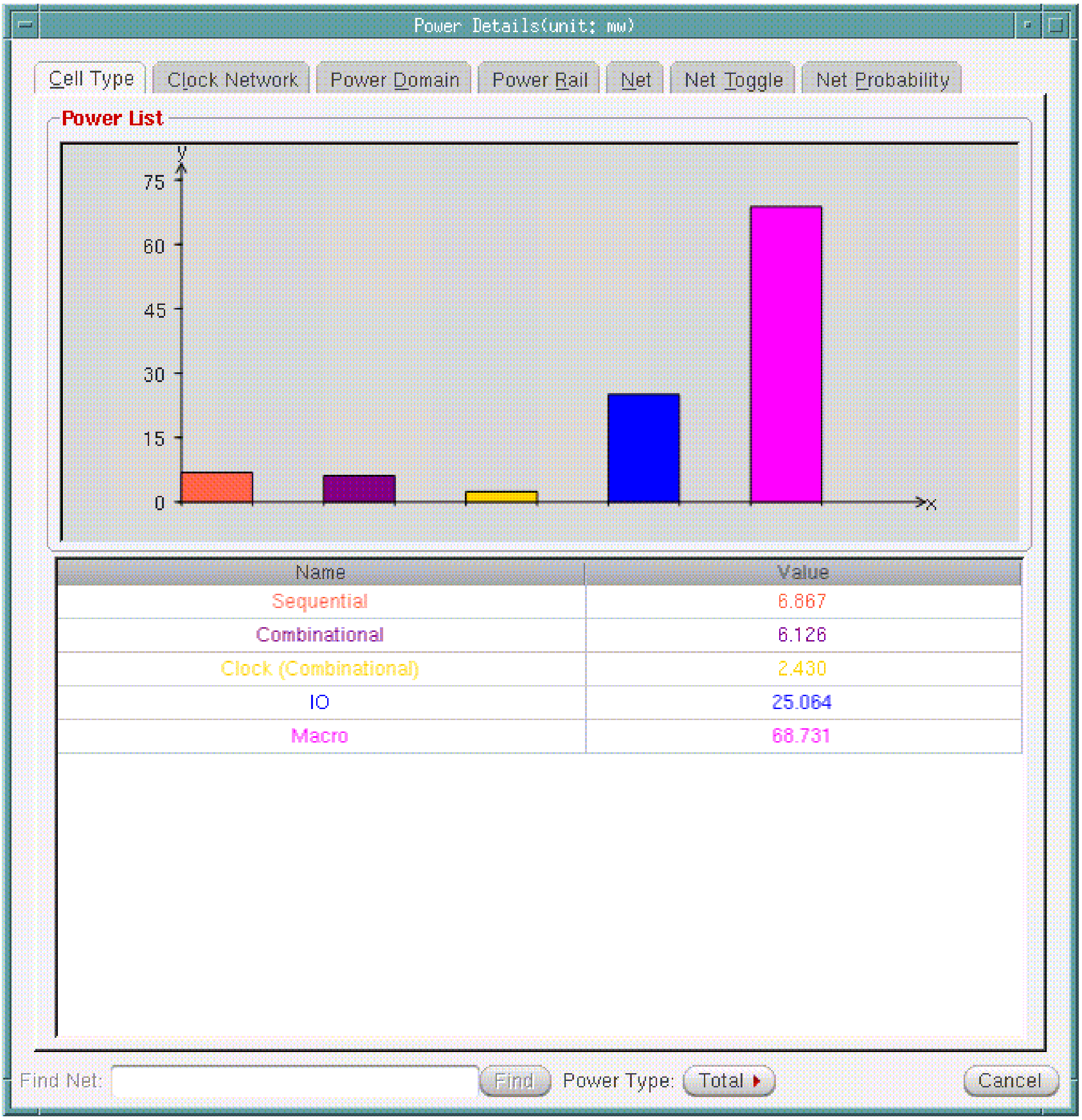
■ To display information from one level higher in the hierarchy, select *Up Level.*



Power Histograms - Power Details

The Power Details form displays histograms by cell type, clock domain, power domain, power rail, net, net toggle, or net probability.

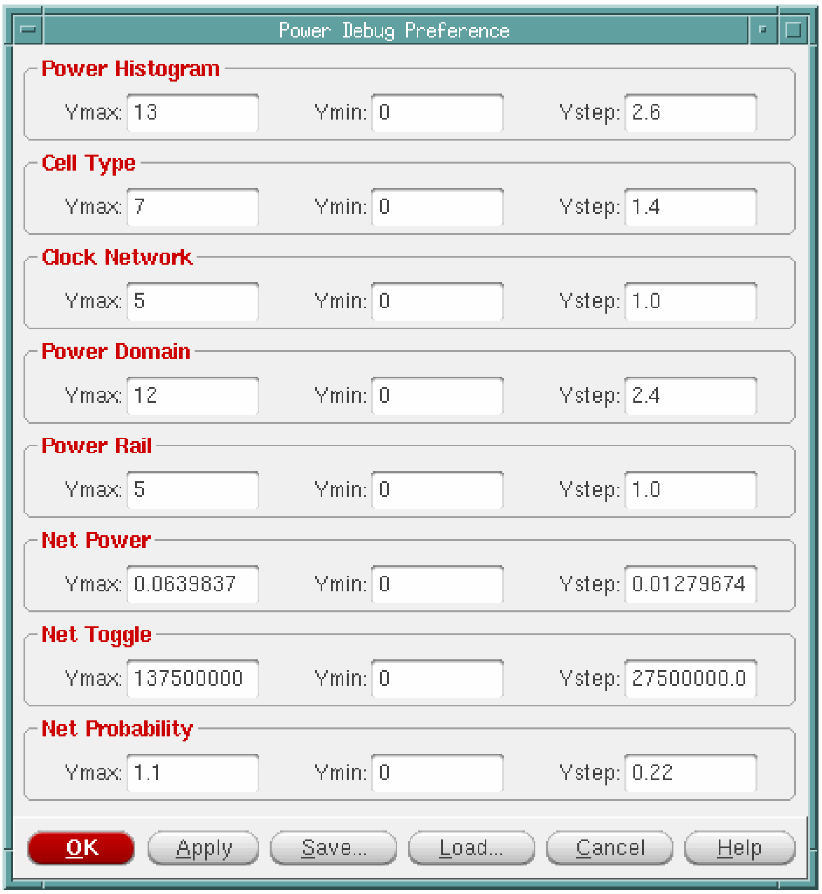
- To open the Power Details form, click on the *Histograms* button on the Power Debug form.



Power Histograms - Power Debug Preference

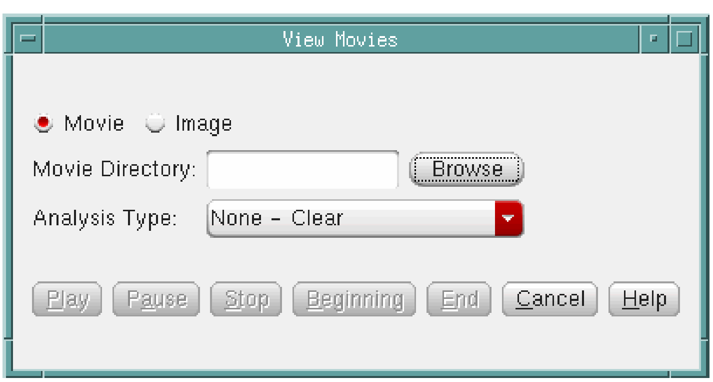
The Power Debug Preference form is used to specify Y-max/Y-min/Y-step for each of the power debug histogram, such as Cell Type, Clock Network, and so on. You can also save and load these setting for all the histograms as a .tcl file. The default file name is pdg.pref.tcl.

- To open the Power Debug Preference form, click on the *Preference* button on the Power Debug form or right-click on any of the power debug histograms in the Power Details form.



Dynamic Movies

Provides the ability to view dynamic movies.

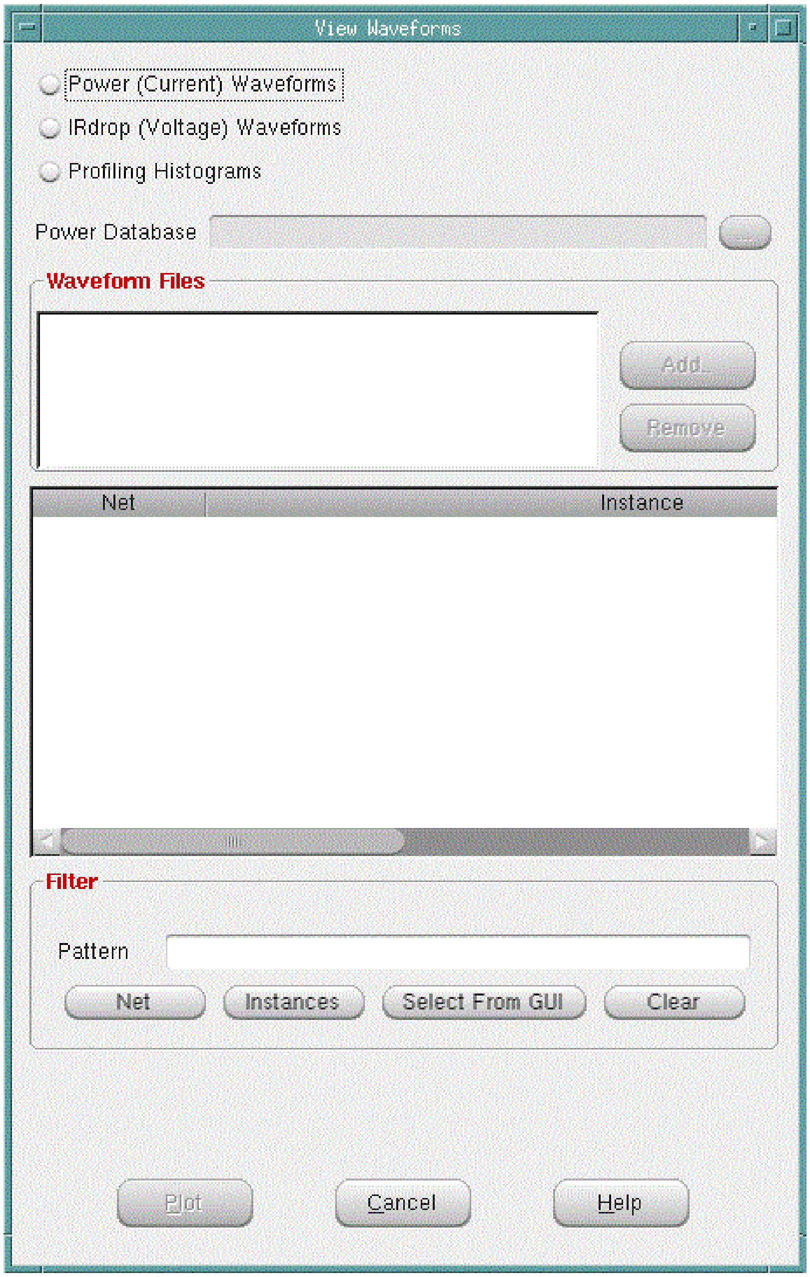


Fields and Options

Movie Directory	Specifies the directory that includes the movie files.
Analysis Type	Specifies the analysis type that will be viewed.
Movie	Specifies that a movie will be viewed.
Image	Specifies that an image or plot will be viewed.
Play	Plays the movie.
Pause	Pauses the movie.
Stop	Stops the movie.
Beginning	Goes to the beginning of the movie.
End	Goes to the end of the movie.
Cancel	Cancels the movie.

Dynamic Waveforms

Provides the ability to display dynamic waveforms. This functionality requires a Simvision license.



Fields and Options

Power (Current) Waveforms	Specifies Current waveforms will be displayed.
IRdrop (Voltage) Waveforms	Specifies Voltage waveforms will be displayed.
Profiling Histograms	Specifies to profile Power Histograms.
Power Database	Specifies the power database.
Waveform Files	Specifies the waveforms that will be viewed. Current or Voltage files can be loaded or removed using the *Add* and *Remove* buttons respectively. To highlight multiple items, click them while pressing either the *Shift* or *Control* key.
Filter - Pattern	The items on the list can be filtered. This is done by entering a value in the *Pattern* field and selecting either *Net* or *Instance*.
Net	Specifies that the nets will be filtered by the text in the *Pattern* field
Instance	Specifies that the instances will be filtered by the text in the *Pattern* field
Select From GUI	This button can be used to plot a waveform for an instance that is selected in the window.
Clear	The Pattern field can be emptied by selecting the *Clear* button.
Plot	To view a current or voltage waveform, select an item on the list, and select the *Plot* button. This will result in Simvision being run and a Simvision window appearing with the selected current waveform displayed.
Cancel	Closes the View Waveforms form.

5

Package Menu

The items in this menu require Cadence® Voltus IC Power Integrity Solution license. See the *Voltus IC Power Integrity Solution Menu Reference* for details.

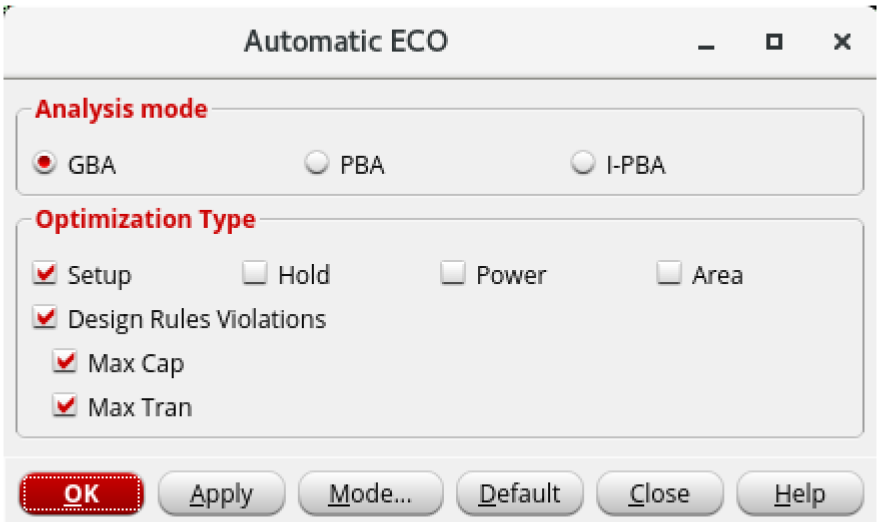
6

ECO Menu

Automatic ECO

Use the *Automatic ECO* form that lets you choose between the various optimization engines (DRV/Setup/Hold/Area/Power) and different timing analysis modes.

- Choose *ECO - Automatic ECO*



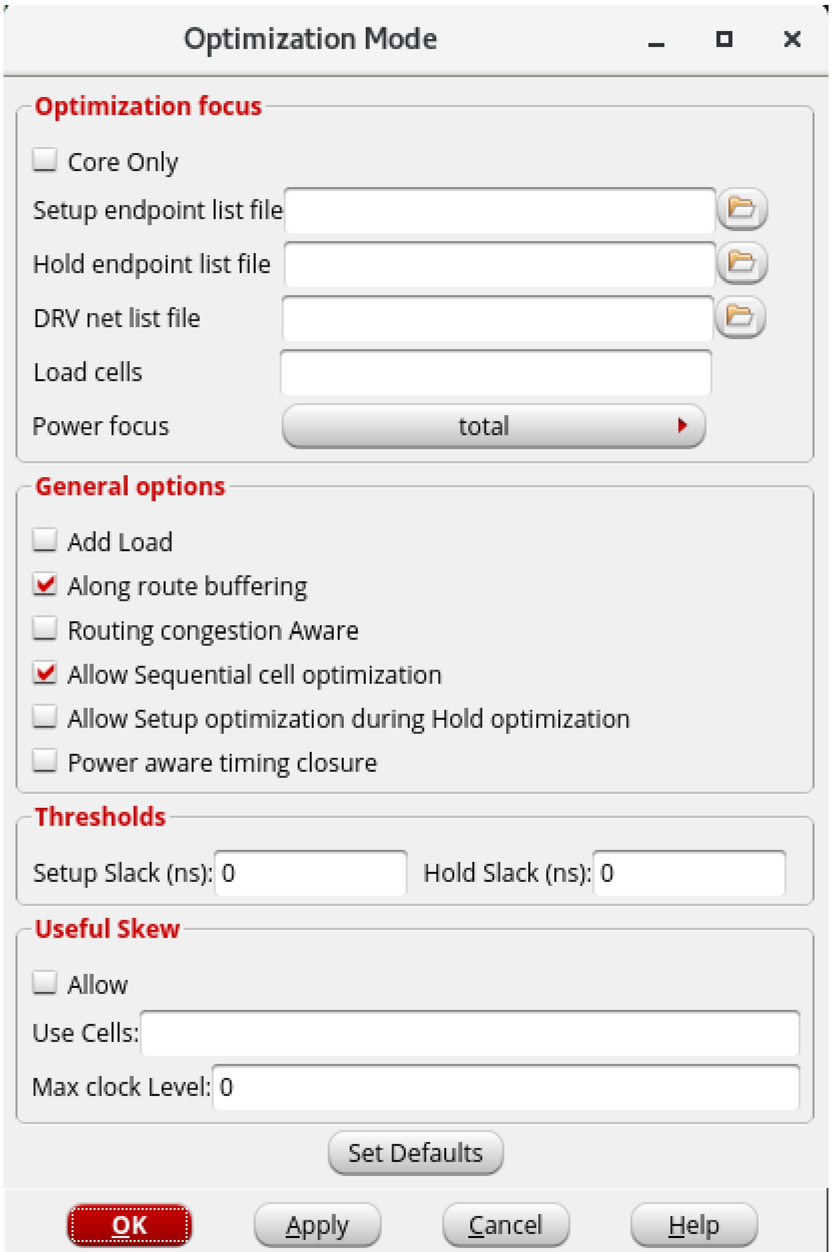
Fields and Options

Analysis	Specifies the following analysis modes for path retiming: ■ GBA: Specifies the Graph-Based Analysis mode. ■ PBA: Specifies the Path-Based Analysis mode. ■ I-PBA: Specifies the Infinite-depth Path Based Analysis mode.

Optimization Type	Choose the type of optimization to fix the violations: ■ *Setup*: Fixes the setup timing violations. By default, this type is selected. ■ *Hold*: Fixes the hold timing violations. By default this type is not selected. ■ *Power*: Reduces the leakage and dynamic power . By default, this type is not selected. ■ *Area*: Reduces the design area. By default, this type is not selected.
	Design Rules Violations: Specifies the type of design rule violation to fix. ■ *Max Cap*: Fixes the maximum capacitance violations in the design based on the threshold specified in the SDC constraints file or in the timing libraries. By default, this field is selected. This field is available only if the *Design Rule Violations* field is selected. ■ Max Tran: Fixes the maximum transition violations in the design based on the threshold specified in the SDC constraints file or in the timing libraries. By default, this field is selected. This field is available only if the *Design Rule Violations* field is selected.

Optimization Mode

Use the *Mode* button on the *Automatic ECO* form to open the *Optimization Mode* form.



Fields and Options

Optimization Focus

Core Only	Specifies that the tool should automatically skip the timing optimization on the IO paths in order to focus only on the internal register to register paths. However, when enabled for leakage and area optimization, the tool doesn't touch any instances, which would be part of an IO path in order to avoid creating any new timing violations on the IO paths.
Setup endpoint list file	Specifies a file containing the list of endpoints for setup fixing.
Hold endpoint list file	Specifies a file containing the list of endpoints for holdfixing.
DRV net list file	Specifies a file containing the list of nets to be optimized during DRV fixing.
Load cells	Specifies the list of load cells to be inserted during Hold fixing, regardless of whether those cells have a dontUse attribute.
Power focus	Specifies whether a given netlist should be optimized only for leakage or dynamic power reduction, or more for leakage or dynamic power while still reducing the overall total power during total power optimization. The options are described below: ■ `total`: The power optimization engine focuses on getting the optimal total power in the design. ■ `leakage`: The power optimization engine optimizes total power but focuses more on leakage power reduction than on dynamic power reduction. ■ `dynamic`: The power optimization engine optimizes total power but focuses more on dynamic power reduction than on leakage power reduction. ■ `leakageOnly`: The power optimization engine focuses only on leakage power reduction. ■ `dynamicOnly`: The power optimization engine focuses only on dynamic power reduction. *Default*: `total`
General options	
Add Load	Specifies whether Hold optimizer is allowed to add dummy cell instances.

Along route buffering	Specifies that the tool should use nodes location provided through the parasitic data in order to insert buffers along the routing topology not only at the start or end of the net. This helps to better balance the slew or the load for long nets, which is typically needed for Setup timing optimization or DRV fixing.
Routing congestion Aware	Specifies that the tool should avoid doing ECO changes in the design areas where routing congestion is very high. This will avoid causing any new routing DRC violations due to ECO buffering/ resizing and routing detour for new ECO nets.
Allow Sequential cell optimization	Specifies that the timing optimizer can optimize the sequential cells. In addition, it performs VT swapping and resizing of sequential cells. By default, this field is selected.
Allow Setup optimization during Hold optimization	Specifies that the tool should perform incremental Setup fixing during Hold fixing in order to create more Setup margins on Hold violated nodes that could not be fixed earlier due to Setup timing conflict.
Power aware timing closure	Specifies that the tool should spend extra effort finding lesser power moves in all optimizers and transforms.
Thresholds	
Setup Slack (ns)	Specifies the threshold value for setup slack.
Hold Slack (ns)	Specifies the threshold value for hold slack.
Useful Skew	
Allow	Specifies that the tool is allowed to create useful skew to either reduce the launch clock delay or increase capture clock delay.
Use Cells	Specifies a restricted list of buffers/inverters/clock gating cells or any other cells that are allowed to be used by ECO on the clock tree.
Max clock Level	Specifies the maximum levels in clock skewing thus limiting the amount of clock tree levels and avoiding the creation of an unbalanced clock tree. This field sets an upper boundary when adding a buffer or pair of inverters to the clock nets. The software counts the amount of levels in the clock tree branch that is being modified to ensure that the maximum level will be equal or below the value set by this field.

Related Text Commands

The following text command provides equivalent or additional functionality:

- set_eco_opt_mode

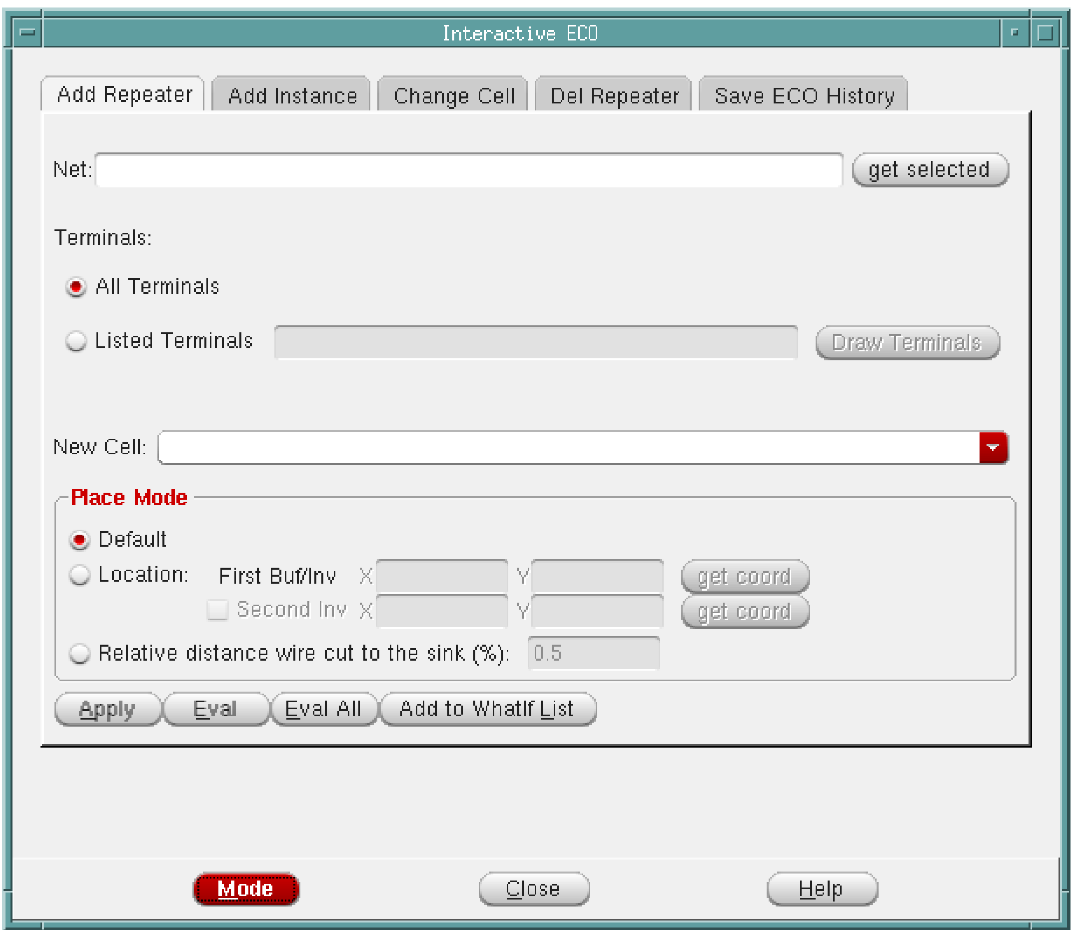
For more information on this command, see the "Signoff ECO Commands" chapter of the *Tempus Text Command Reference*.

Add Repeater

Use the *Add Repeater* page of the *Interactive ECO* form to add either a single buffer or two inverters to a net.

To use the GUI to add repeaters to a net, complete the following steps:

- Choose the *ECO - Interactive ECO*. Click on the *Add Repeater* tab.



Note: The *Eval* and *Eval All* options are active only when the SPEF file is read-in.

Interactive ECO - Add Repeater Fields and Options

Net	Type the net name, or click on a net in the *layout* tab and click *get selected*.
Get Selected	Displays the name of a selected net. If you click on the *Get Selected* button on this form without first selecting a net, the software issues an error message.
Terminals	Provides two methods for specifying the buffer connection.
All Terminals	Connects the added buffer to drive all the receivers. Use this to reduce the delay and output transition time of a weak driver driving a large capacitive load.
Listed Terminals	Connects the added buffer to drive the listed receivers, and provides full flexibility for building an arbitrary buffer connection.
Draw Terminals	Allows you to draw an area covering the terminals to which you want to add the buffer.
New Cells	Enter the cell type name of the buffer cell to add, or click on the arrow to right of the field and select a buffer from the list.
Place Mode	The following options specify various aspects for ECO placement.
Default	The software automatically determines a location and places the new cell.
Location	Specifies where the buffer should be inserted.
First Buf/Inv X Y	Specifies the X and Y locations for the buffer or first inverter. You can use the automatically assigned location, enter the location, or click on an area in the design display window and click *get coord.* If the cell is a buffer and location is specified, the newly created buffer is placed at the specified location. If the cell is an inverter and only one location is specified, both inverters will be placed at the same location.
Second Inv X Y	Specifies the X and Y locations for the second inverter.You can use the automatically assigned location, enter the location, or click on an area in the *layout* tab and click *get coord.*
get coord	Retrieves and displays the X and Y coordinates for a specified location in the layout window.
Relative distance wire cut to the sink (%):	

	Specifies the location of the buffer based on its distance from the sink or the driver pin. The value is a number between 0 and 1. A low value (0.1) places the buffer near the sink; a high value (0.9) places the buffer near the driver.
Eval	Evaluates the effect on timing if you add *New Cell*. These values are reported but not applied, so you can experiment with different cell types. This option is not supported for non-physical mode or incremental SI analysis.
Eval All	Evaluates the effect on timing for all the cell types available for *New Cell*. The timing report shows the effects of all the cell types, enabling you to select the best cell for your design. This option is not supported for non-physical mode or incremental SI analysis.
Apply	Applies your selections.
Add to WhatIf List	Allows you to add your specifications with ECO commands in series.

Related Text Command

The following text command corresponds to *Add Repeater* GUI command:

- `add_repeater`

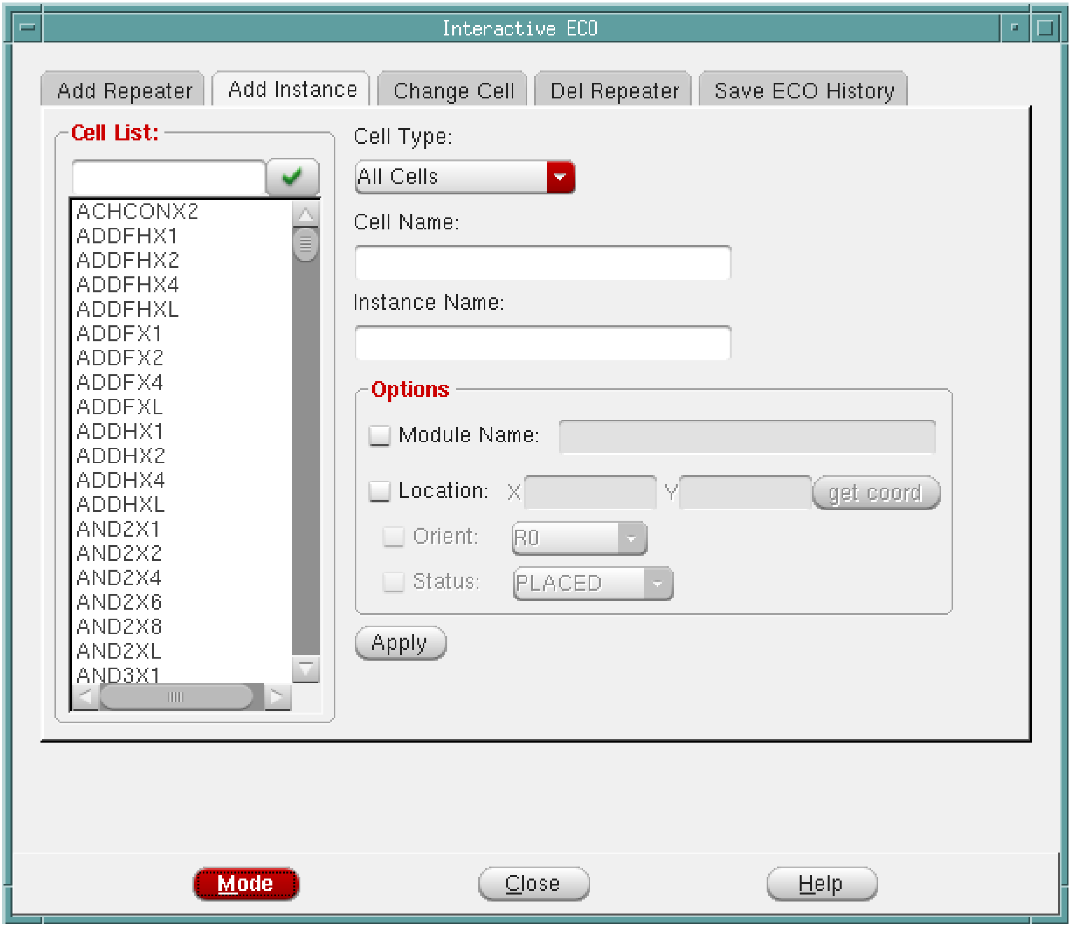
For more information on this command, see the "Manual ECO Commands" chapter of the *Tempus Text Command Reference*.

Add Instance

To use the GUI to add instances to a net, complete the following steps:

- Choose *ECO - Interactive ECO*.

Click on the *Add Instance* tab.



Interactive ECO - Add Instance Fields and Options

Cell List	Specifies the names of a collection of cells used as a reference for the new instances added. To search for a cell, enter the cell name in the text box and click the tick button.

Cell Type	Specifies the type of cell used as a reference for the new instances added. Allows you to select the following options: ■ *All Cells* ■ *Standard Cell* ■ *Physical Cell* ■ *IO Cell* ■ *TIE Cell* ■ *Macro Cell*
Cell Name	Specifies the librarycell name of the instance.
Instance Name	Specifies the name of the instance to add and place. If the instance name contains special characters, you need to start the part of the instance name, which contains special characters, with a backlash '\' and end it with a space.
Options	
Module Name	Specifies a module name.
Location	Specifies the X and Y coordinates for the instance.
get coord	Retrieves and displays the X and Y coordinates for a specified location in the design window.
Orient	Specifies the orientation of the instance.
Status	Specifies the placement status of the added instance. You can specify the following as the placement status: ■ *placed*: The status of the added instance is placed. ■ *fixed*: The status of the added instance is fixed. ■ *covered*: The status of the added instance is covered. Default: *placed*
Apply	Applies your selections.

Related Text Commands

The following text command provide equivalent or additional functionality:

- `add_inst`

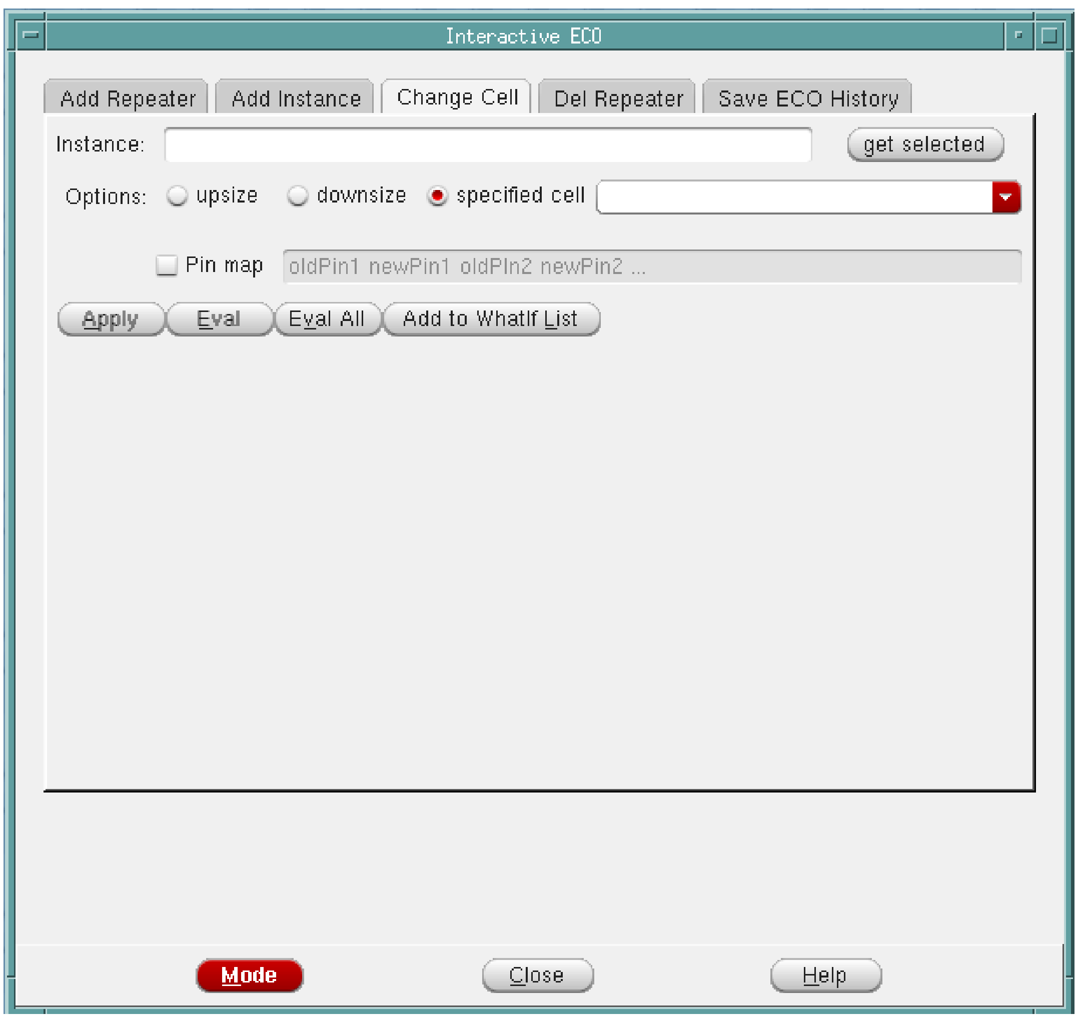
For more information on this command, see the "Manual ECO Commands" chapter of the *Tempus Text Command Reference*.

Change Cell

Use the *Change Cell* page of the Interactive ECO form to upsize or downsize an instance. Upsizing an instance using the `change_cell` command enables driving a larger load and improves the driver delay and the transition time at the receivers. Downsizing an instance on the noncritical path reduces the loading of its driver on the critical path.

To use the GUI to upsize or downsize an instance, complete the following steps:

- Choose *ECO - Interactive ECO*. Click on the *Change Cell* tab.



Note: The *Eval* and *Eval All* options are active only when the SPEF file is read-in.

Interactive ECO - Change Cell Fields and Options

Instance	Enter the hierarchical instance name to be changed.
Get Selected	Displays the name of a selected instance. If you click on the *Get Selected* button on this form without first selecting an instance, the software issues an error message.
Options:	Select from the following options
upsize	Upsizes the specified instance with the next available cell in the same footprint unless the specified cell is already of the highest driving strength. Upsizing an instance that drives a large load can improve the driver delay and the transition time at the receivers.
downsize	Downsizes the specified instance with the next available cell in the same footprint unless the specified cell is already of the lowest driving strength. Downsizing an instance on the noncritical path can reduce the loading of its driver on the critical path.
specified cell	Enter the name of the placement cell, or use the pull-down menu to select a cell.
Pin map	Specifies the pin mapping for the new cell based on the old cell.
Eval	Evaluates the effect on timing if you upsize or downsize a cell, or change to a different cell. These values are reported but not applied. This option is not supported for non-physical mode or incremental SI analysis.
Eval All	Evaluates the effect on timing for all the options available. This option is not supported for non-physical mode or incremental SI analysis.
Apply	Applies your selections.
Add to WhatIf List	Allows you to add your specifications with ECO commands in series.

Related Text Commands

The following text command corresponds to *Change Cell* GUI command:

- `change_cell`

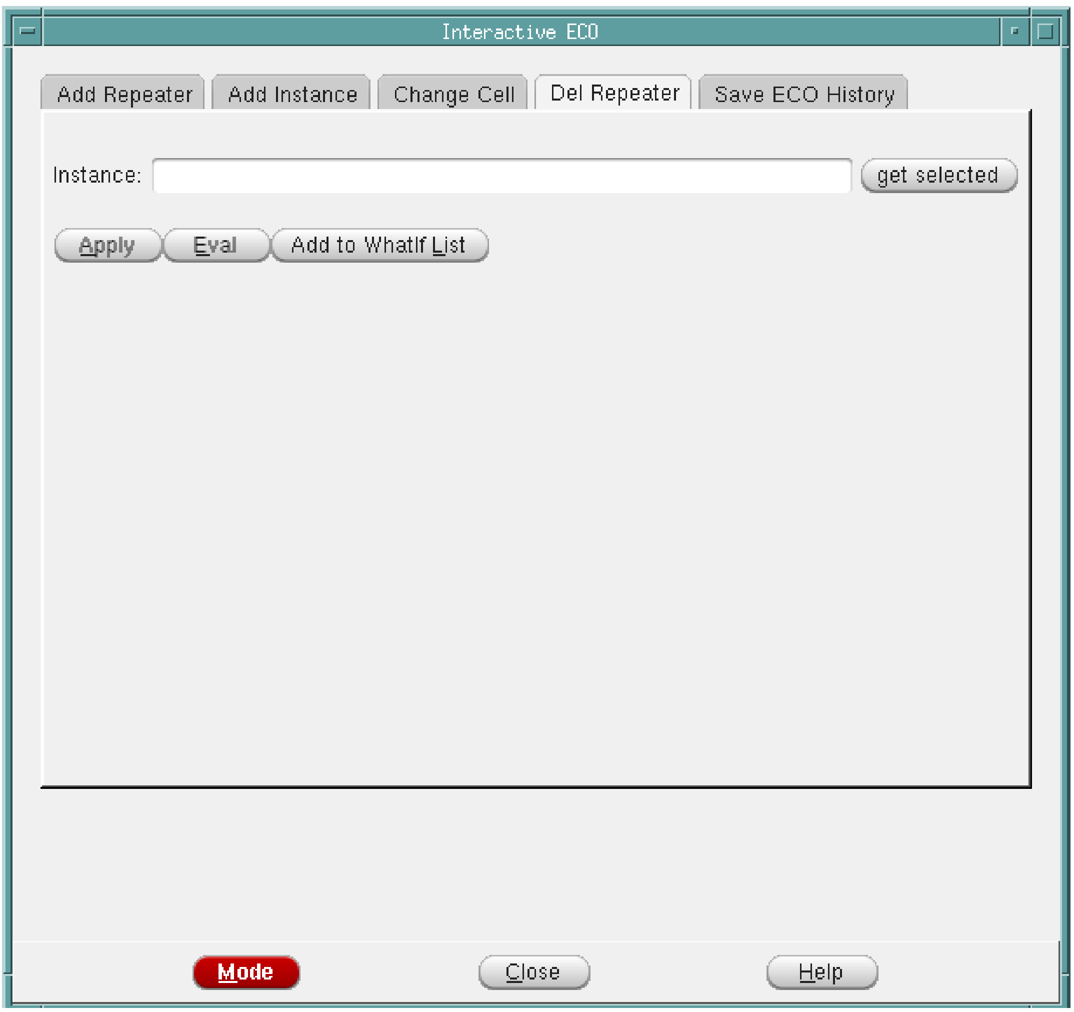
For more information on this command, see the "Manual ECO Commands" chapter of the *Tempus Text Command Reference*.

Delete Repeater

Use the *Del Repeater* page of the Interactive ECO form to delete redundant buffers that cause extra delay. Buffers are typically over-added by synthesis tools based on wireload models.

To use the GUI to delete repeaters from a net, complete the following steps:

- Choose *ECO - Interactive ECO*. Click on the *Delete Repeater* tab.



Note: The *Eval* option is active only when the SPEF file is read-in.

Interactive ECO - Del Repeater Fields and Options

Instance	Enter the name of the buffer instance to be removed.
Get Selected	Displays the name of a selected instance. If you click on the *Get Selected* button on this form without first selecting an instance, the software issues an error message.
Eval	Evaluates the effect on timing if you delete the specified instances or the buffer tree. These values are reported but not applied until you click *Apply*. This option is not supported for non-physical mode or incremental SI analysis.
Apply	Applies your selections.
Add to WhatIf List	Allows you to add your specifications with ECO commands in series.

Related Text Command

The following text command corresponds to *Delete Repeater* GUI command:

- `delete_repeater`

For more information on this command, see the "Manual ECO Commands" chapter of the *Tempus Text Command Reference*.

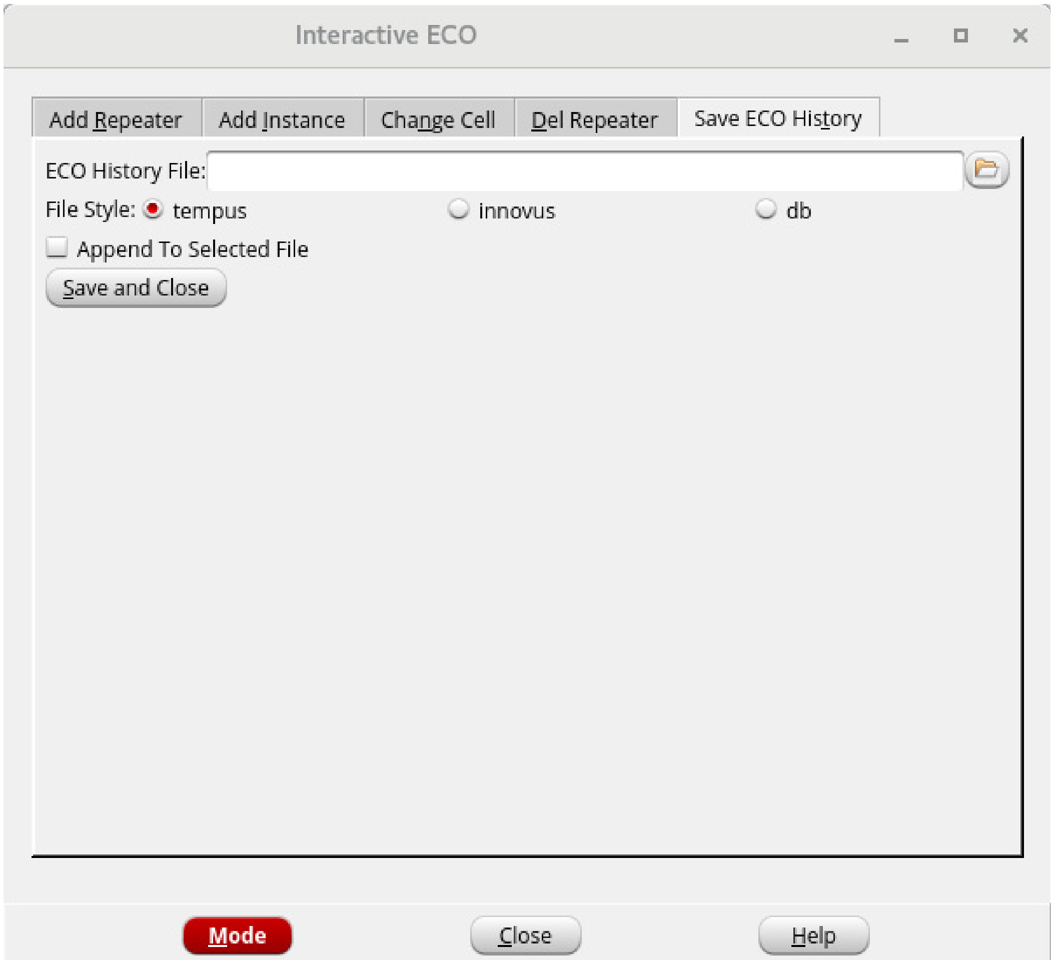
Save ECO History

Use the *Save ECO History* page of the Interactive ECO form to save the ECO history in an output file. You can save this file in Tempus, Innovus, or db.

To use the GUI to write an ECO file, complete the following steps:

† Choose *ECO - Interactive ECO*.

- Click on the *Save ECO History* tab.



Interactive ECO - Save ECO History Fields and Options

ECO History File	Specifies the name of the file that this menu command produces.
File Style *tempus* *innovus* *db*	Generates an ECO file in the specified format. The db format is supported for backward compatibility only. You can specify the *Append to Selected File* option to append db ECO changes to a db format file, but you cannot regenerate or convert db changes into Tempus or SOCE-format file.
Append To Selected File	
	Appends changes to an existing ECO file.

Related Text Commands

The following text command corresponds to *Save ECO History* GUI command:

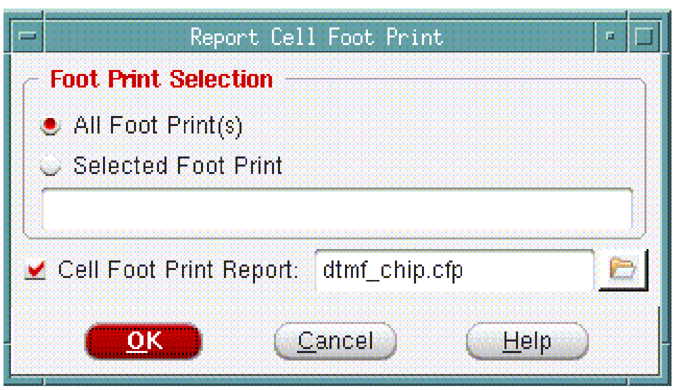
- `write_eco`

For more information on this command, see the "Manual ECO Commands" chapter of the *Tempus Text Command Reference*.

Report Cell Foot Print

Use the Report Cell Foot Print form to output the cell footprint information to a report file after importing the design and reading the timing library. This report file contains the cell names, footprint names, the rising and falling drive resistance values, and the functions.

- Choose *ECO - Cell Footprint*



Fields and Options

All Foot Print(s)	Reports information for all cells. This is the default option.
Selected Foot Print	Reports information on the cells specified in the text box.
Cell Foot Print Report	Specify a filename, or click the file folder icon to choose a file. If you do not select this option, the report is displayed on the Tempus console. The default filename is *TopCellName*.cfp.

7

Clock Menu

CCOpt Clock Tree Debugger

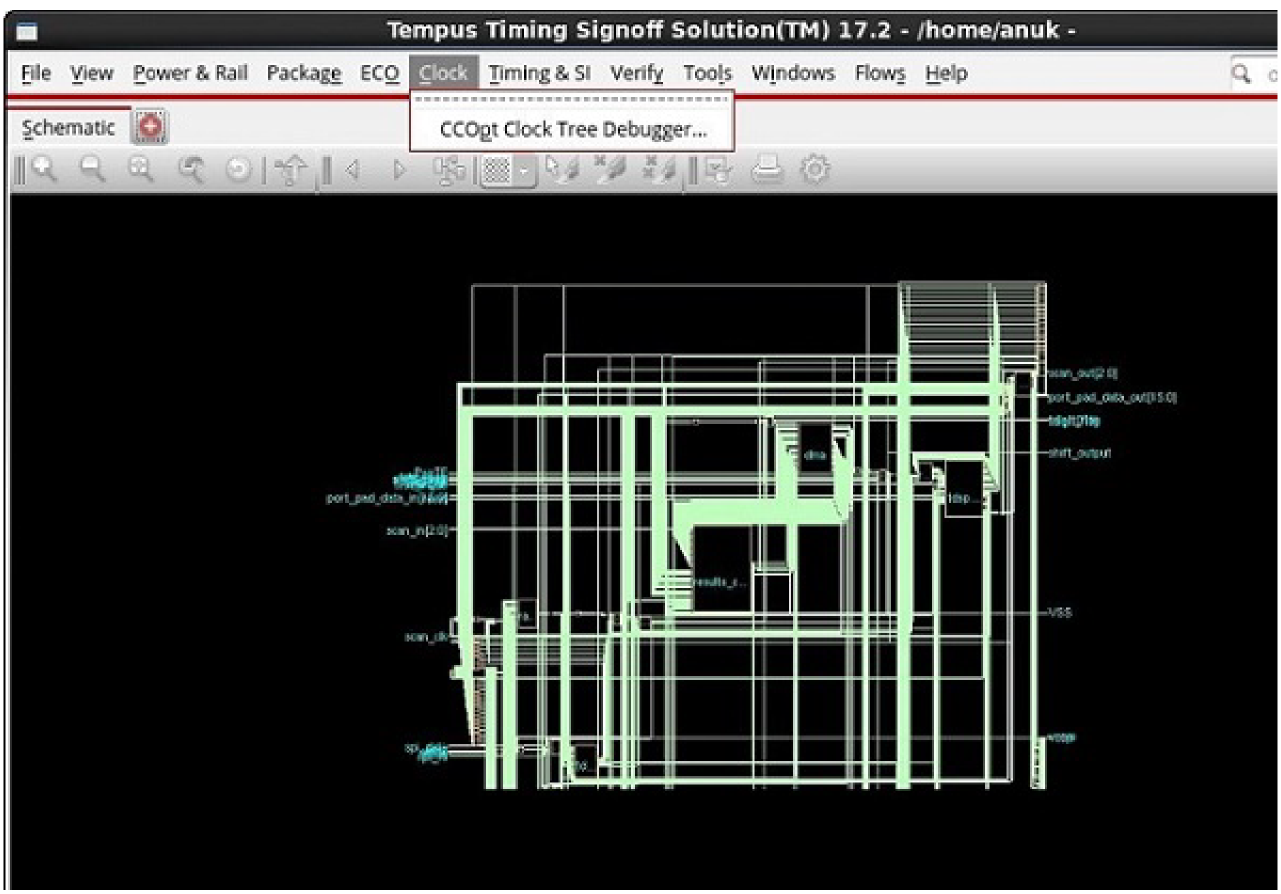
The *CCOpt Clock Tree Debugger* provides clock tree debugging facilities that will help you understand the quality of the CCOpt results better. The software provides the following three basic views for clock tree visualization:

- **Clock Tree Debugger**: Displays a top-down tree view of the CCOpt clock trees.
- **Clock Path Browser**: Displays the clock path data in a table and provides the option for bringing up a clock path analyzer either from its context menu or by double-clicking on a row in the table.
- **Clock Path Analyzer**: Provides a view to examine a single clock path.

The three views are detailed in sections below.

The *CCOpt Clock Tree Debugger* submenu appears only when you have created a CCOpt clock tree specification file.

Choose *Clock – CCOpt Clock Tree Debugger*



The *Clock Tree Debugger* tool displays a top-down tree view of the CCOpt clock trees with the vertical axis representing delays. It has the following features:

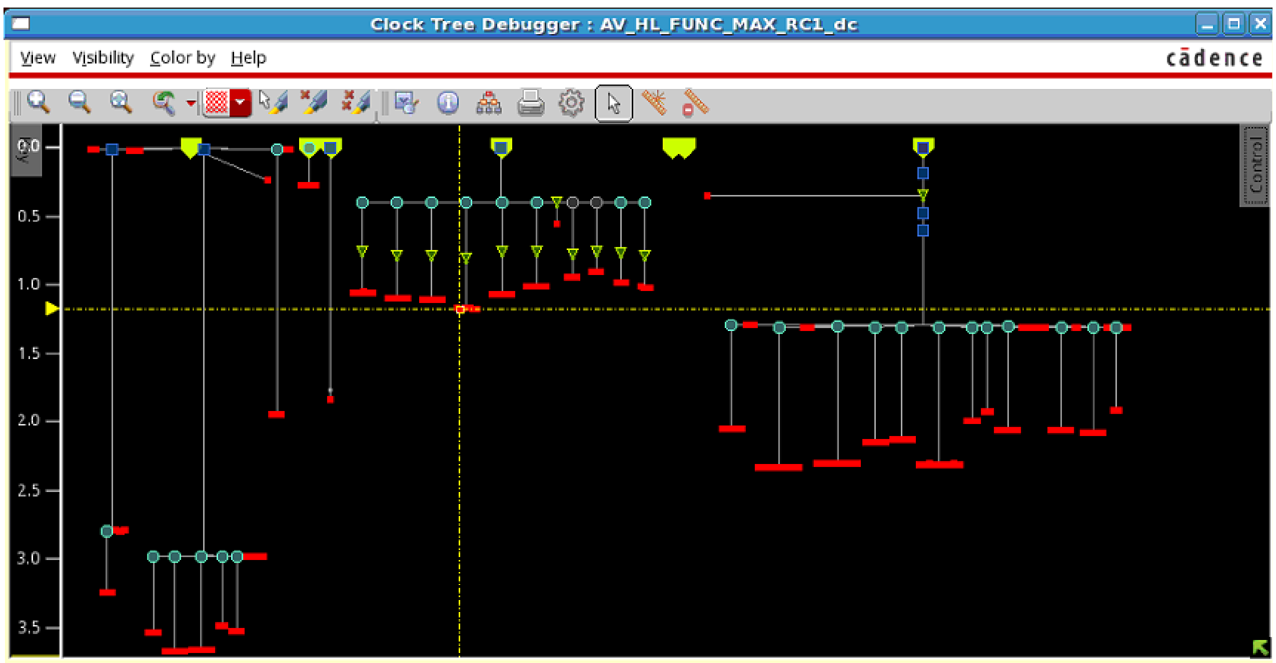
The tool display shows a schematic-like representation of the CCOpt clock trees. You can control the level of details and the elements that you want to view.

The results shown in the tool are based on the CCOpt clock tree specification file.

The units on the vertical scale of the display represent time, so that distance between two elements indicates delays that you can measure.

A comprehensive search mechanism lets you find instances, paths, and reconvergent clocks easily.

The tool displays several clocks at the same time. The software can display reconvergent and crossover clocks.



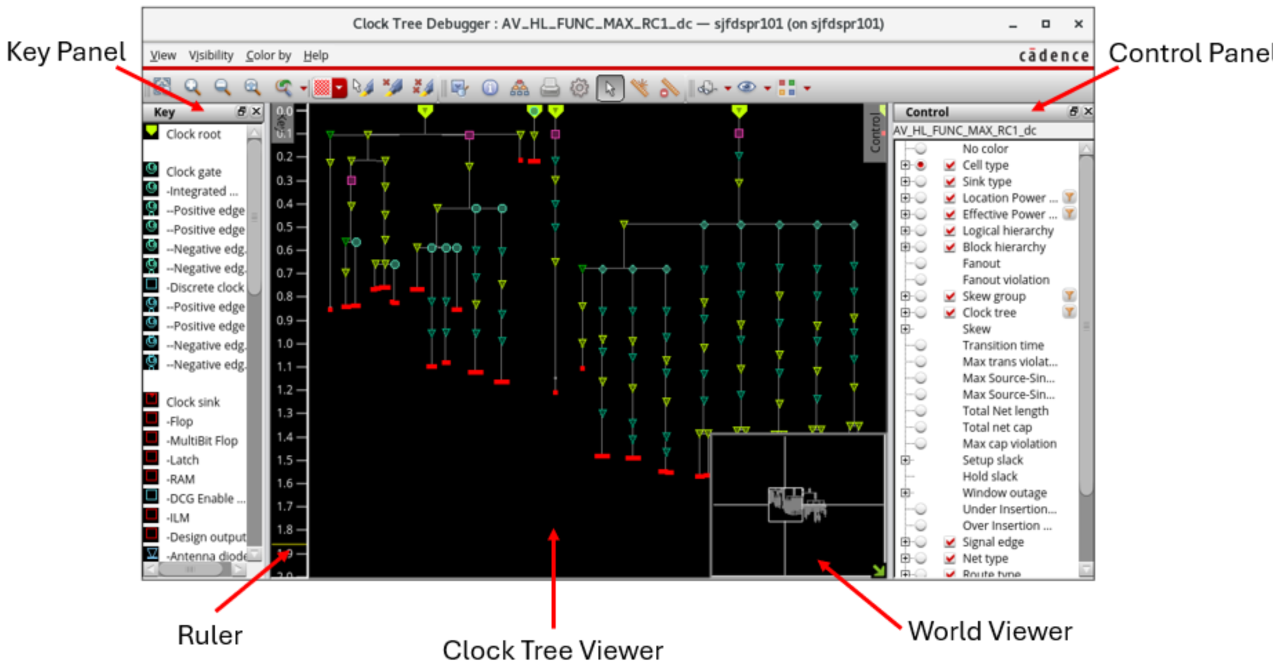
Commands to Invoke, Name, and Close CTD Windows

Use the below commands to invoke, name and close CTD windows.

- `ctd_win`: Opens a CTD window with the user-defined window ID and title. Use the `-unit_delay` parameter of this command to open the CTD in unit delay mode.
- `close_ctd_win`: Closes open CTD windows.
- `get_ctd_win_id`: Retrieves the IDs of the CTD windows.
- `get_ctd_win_title`: Retrieves the titles of the CTD windows.
- `set_ctd_win_title`: Specifies the title of the CTD window.
- `ctd_trace`: Highlights the clock tree path from the clock root to the pin or from one pin to the other if the two pins are on the same path using the specified color or color index.
- `ctd_save_histogram`: Saves the CTD histogram to a file.
- `ctd_save_view`: Saves the current snapshot of the clock tree viewer into a file.

Clock Tree Debugger - Main Window Features

The following options are available in the main window of the *Clock Tree Debugger*:



Options	Descriptions
Ruler	Displays time units (in nanoseconds) that measure the delay between elements.
Browser	A text-based browser that shows information about the different analysis views.
Display area	Displays the clock tree views.
World viewer	

<table>
<tr><td></td><td>It is a rectangular area at the right-bottom corner of the Clock Tree Debugger canvas that shows you a satellite view, or the world view, of the clock trees. The canvas of the world viewer is demarcated by a white bounding box. You can right-click on any region in the world viewer to navigate to that region with the current zoom level. Click the button again to hide the world viewer.

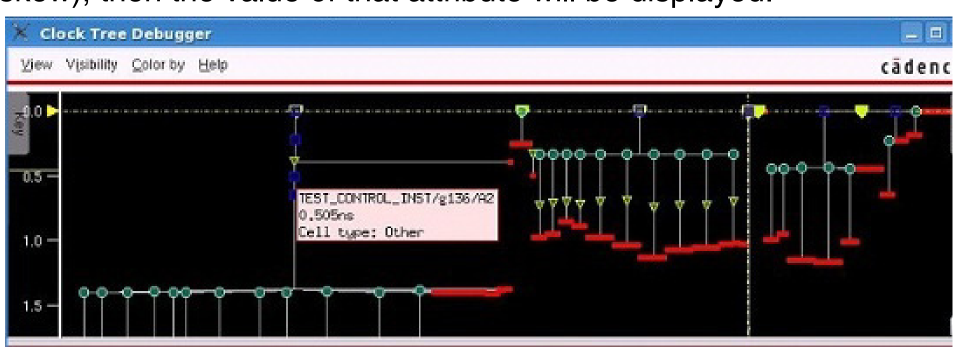
The button is highlighted using a red textbox, as shown in the image below.

You can increase or decrease the size of the world viewer from the Set Preference form. Click the Preference submenu in the View menu to open this form. Click Small, Middle, or Large to change the size of the world viewer window. For details, see "Set Preferences for CTD GUI Configuration".</td></tr>
<tr><td colspan="2">The Context Pop-up Attribute Viewer</td></tr>
<tr><td></td><td>When you position the pointer over an object for a particular time, an attribute pop-up window will appear. The context pop-up attribute viewer displays the basic properties of an object in a pop-up window right next to the location of the object.

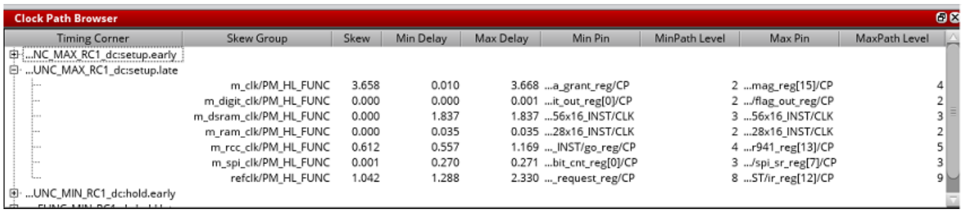
It shows the following information about the object, if available:

■ Name
■ Insertion delay
■ Skew groups
■ Clock trees

If the visualization is being colored by some attribute (for example, skew), then the value of that attribute will be displayed.

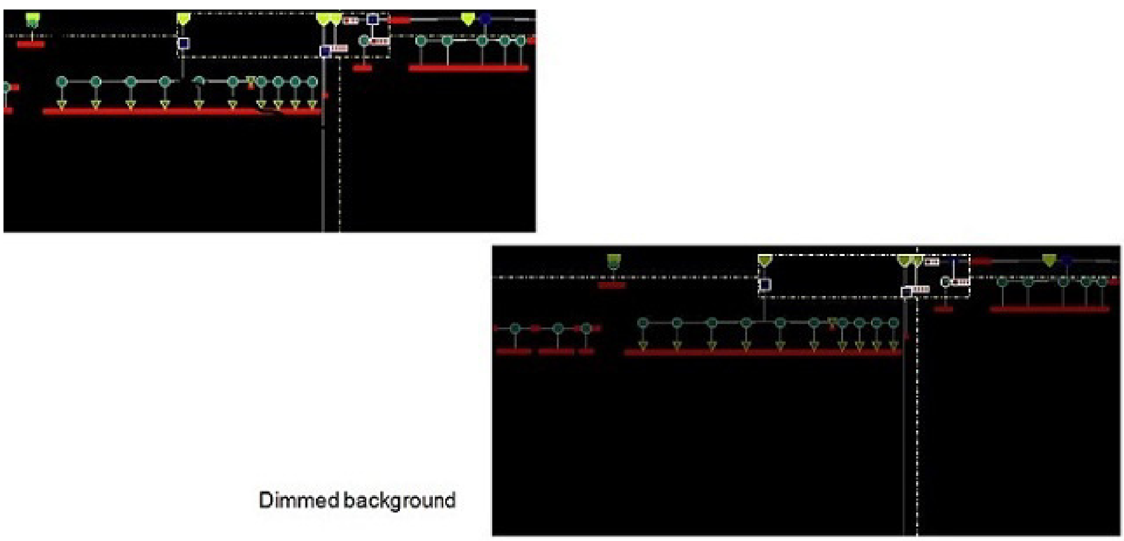


</td></tr>
</table>

<table>
<tr>
<td>Browser</td>
<td>A text-based browser that shows information about the different timing corners. When you open the CTD main window, the browser does not open by default. To view the browser in the Clock Tree Debugger main window, select Enable clock path browser in the View menu. The browser opens as a new window.

</td>
</tr>
</table>

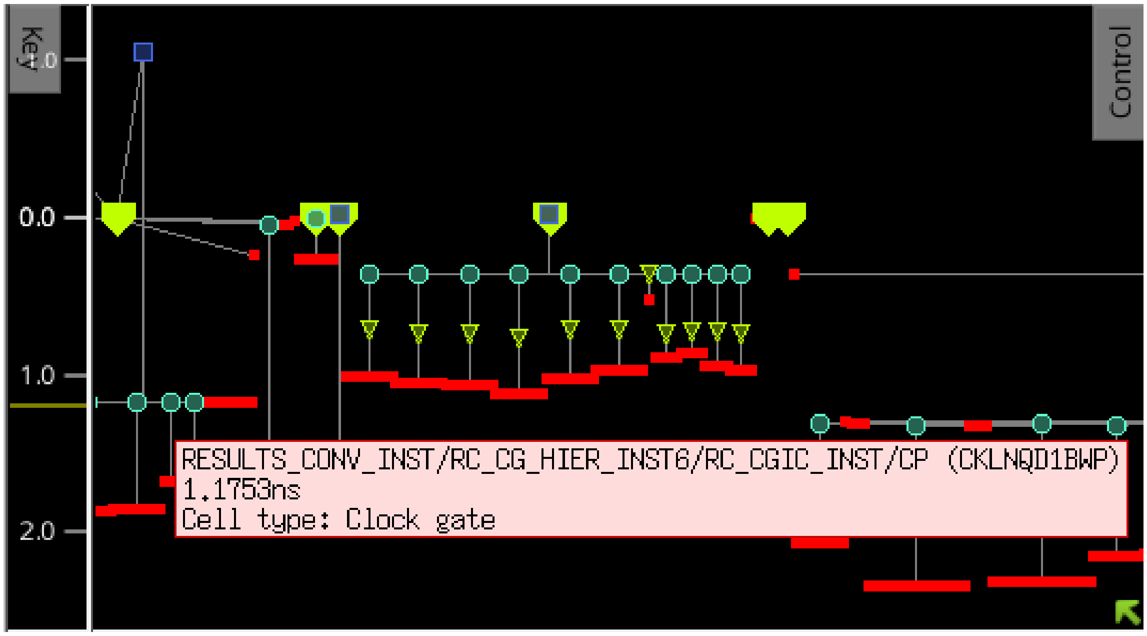
Dim Background of Clock Tree Debugger Display

In addition to the above features, the clock tree debugger allows you to dim the background of the display area. This allows you to view selected and highlighted objects more clearly. This is useful when you are debugging clock trees. To dim the display, press F12. The F12 key is used as a three-way toggle. So, to quit the dim mode, press the F12 key again twice.



Clock Tree Debugger - Tooltips

The tooltips are provided to let you view relevant information about any object by hovering your mouse over that object in the display area.



Clock Tree Debugger - Menu Bar

The *Clock Tree Debugger* has the following menus:

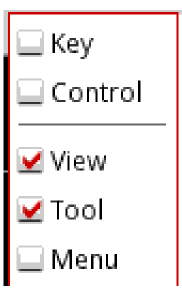
- View
- Visibility
- Color by

You can tear off any menu or submenu and place it anywhere in the layout window. To tear off a menu or submenu, click on the dotted line that appears at the top of the menu or submenu, and drag the menu to where you want it displayed in the design area. The menu remains on the screen as a window when you release the mouse button.

Clock Tree Debugger - Toolbar Widgets

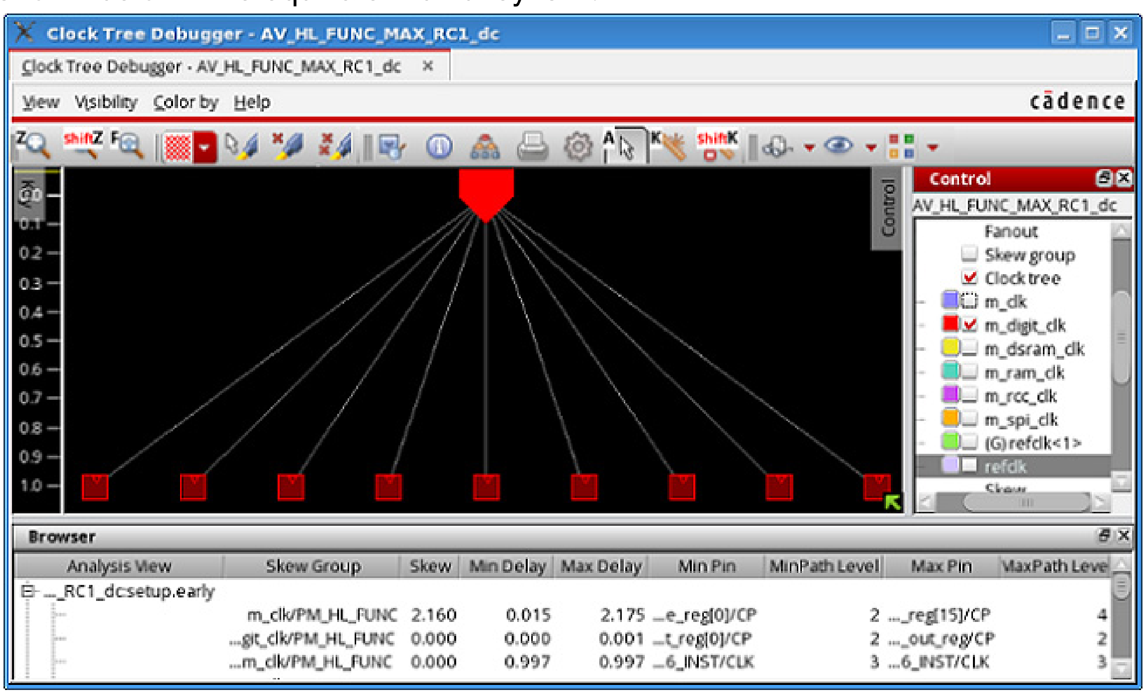
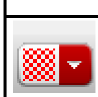
The following row of widgets, located below the menus and above the display area, includes shortcuts for some commonly used commands, forms, and menus in CTD.

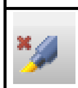
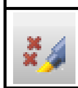
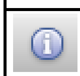
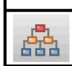
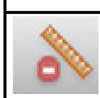
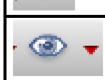
Right-click on the toolbar to enable or disable all or a particular group of widgets. The following options are available.

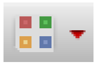


- The options, *Key* and *Control* allow you to toggle the visibility of the Key Panel and Control Panel, respectively.
- The *View* widgets provide options to customize the clock tree display.
- The *Tool* widgets open up forms in CTD.
- The *Menu* widgets display all the submenus of the CTD main menus.

The widgets enabled by the *View*, *Tool*, and *Menu* options are detailed below.

Widget	**Description**
	Zoom Selected: Click this widget to zoom into the selected item in the Clock Tree Viewer. The equivalent bindkey is Ctrl+Z.
	*Zoom In*--Click this widget to zoom into a smaller area of the clock tree to view it in greater detail. Each click zooms in one level. The equivalent bindkey is `z`.
	*Zoom Out*--Click this widget to zoom out of the clock tree. Each click zooms out one level. The equivalent bindkey is `Shift+Z`.
	*Fit*--Click this widget to fit the clock tree within the display area. You can also select one clock tree or specific clock trees in the control panel. In this case when you click the widget, only the selected trees will be fit in the display area. This is shown below. The equivalent bindkey is `f`. 
	*Edit Highlight Color*–Allows you to select a highlight color from the available choices in the drop-down menu.

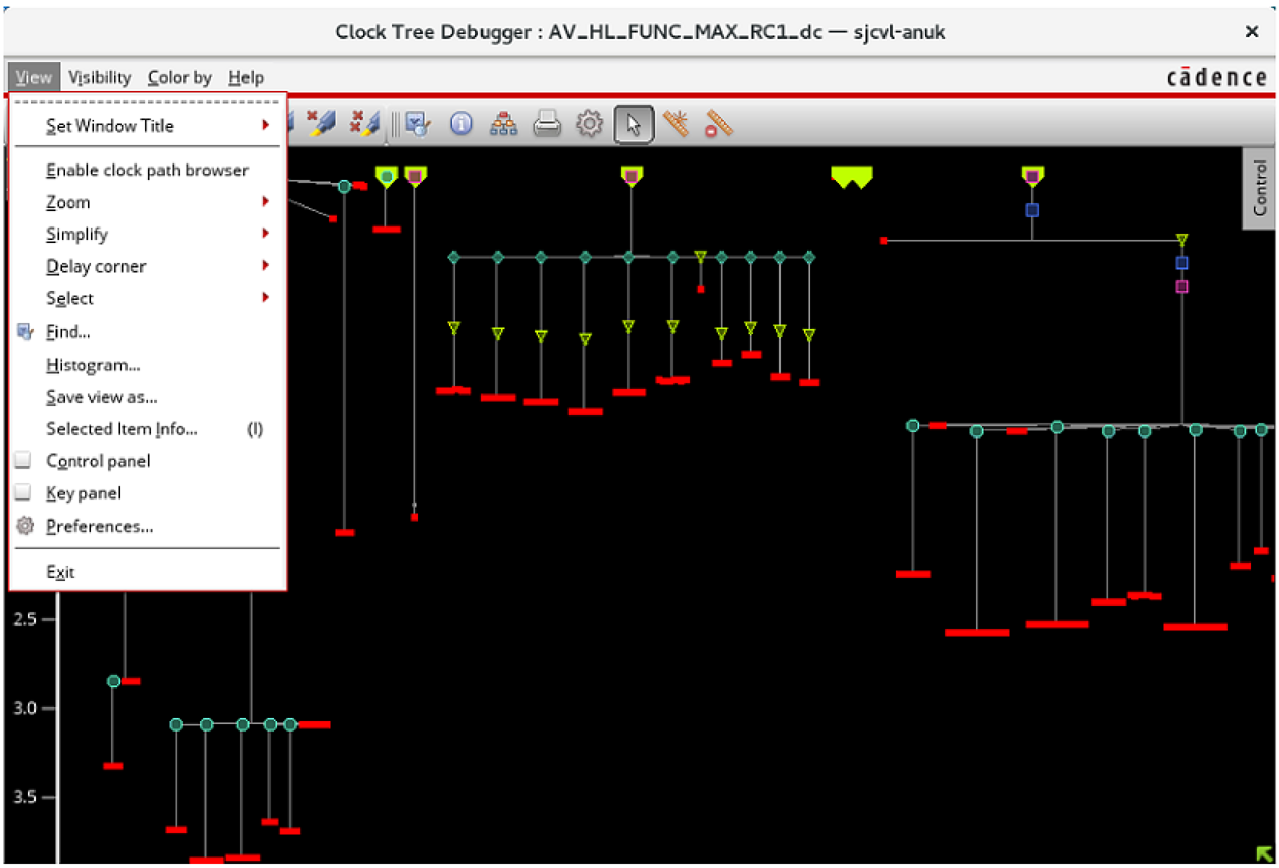
	Zoom Previous–Allows you to save a current display and view previously-saved displays. When you click this widget, a drop-down menu is available. In this menu: ■ Click Save Current Area to save the current display. ■ Click Go To Saved Area to view a previously-saved display. You can save multiple displays. They are numbered 1,2,3...and so on. You can select the display you want to view again. ■ Click Delete All Saved Area to delete all previously-saved displays.
	*Highlight Selected*--Highlights the objects selected in the GUI with the current highlight set, as displayed in the *Edit Highlight Color* widget. Use the drop-down menu to select a new highlight set.
	*Clear Highlight*--Clears the highlight from those objects that are highlighted with the current highlight set, as displayed in the *Edit Highlight Color* widget.
	*Clear All Highlight*--Clears all highlights from the GUI.
	*Find*--Opens the Find/Select Object form.
	*Attribute Editor*--Opens the Attribute Editor for the selected object.
	*Design Browser*--Opens the Design Browser form.
	*Print*–Opens the Dump View form.
	Opens the Set Preference form.
	*Select* - Enables the select arrow. The equivalent bindkey is A.
	*Create Rulers* - Click this widget, then left-click in the display area and drag the mouse to add a ruler (in micrometers), and left-click again to end the ruler. The equivalent bindkey is k.
	*Clear All Rulers*--Clears all rulers from the display area. The equivalent bindkey is Shift+`K`.
	*View*–Provides all the options of the View menu of the CTD in a drop-down menu.
	*Visibility*–Provides all the options of the Visibility menu of the CTD in a drop-down menu.

	*Color by*–Provides all the options of the Color by menu of the CTD in a drop-down menu.

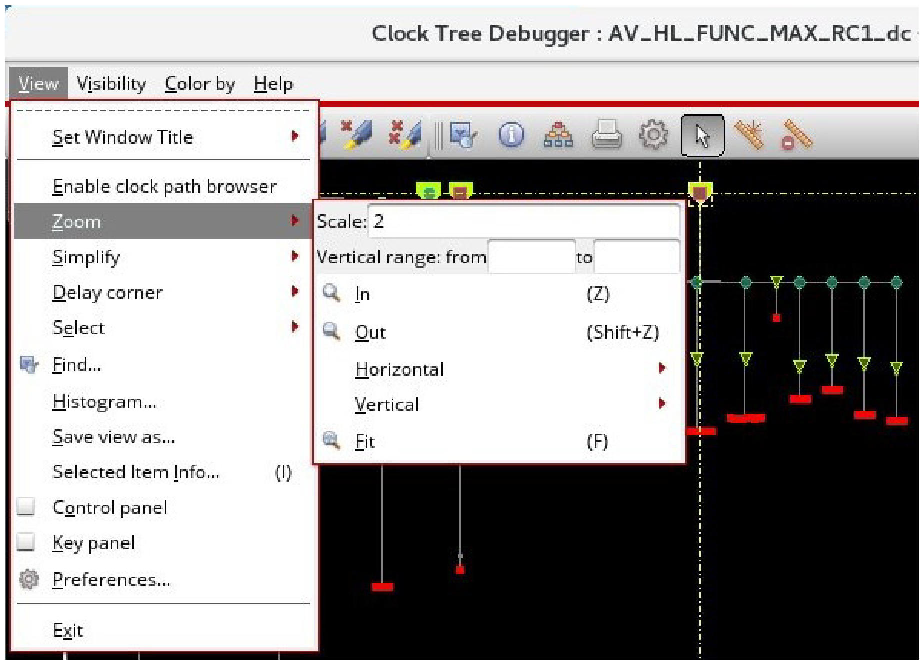
Clock Tree Debugger - View Menu

Use the *View* Menu to customize the clock tree display.

Choose *Clock Tree Debugger – View*.

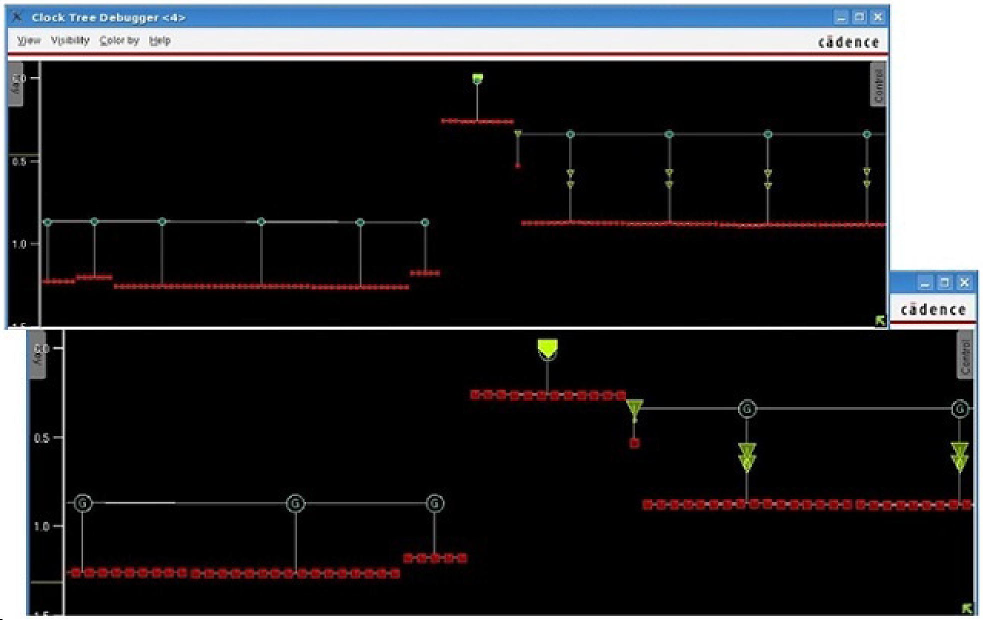


Options	Descriptions
Set Window Title	Specifies the window title for the CTD session. After you type the title and press enter, the new title appears in the top bar of the CTD.
Enable clock path browser	Opens the Clock Path Browser in a separate window. For more information, see Clock Path Browser.

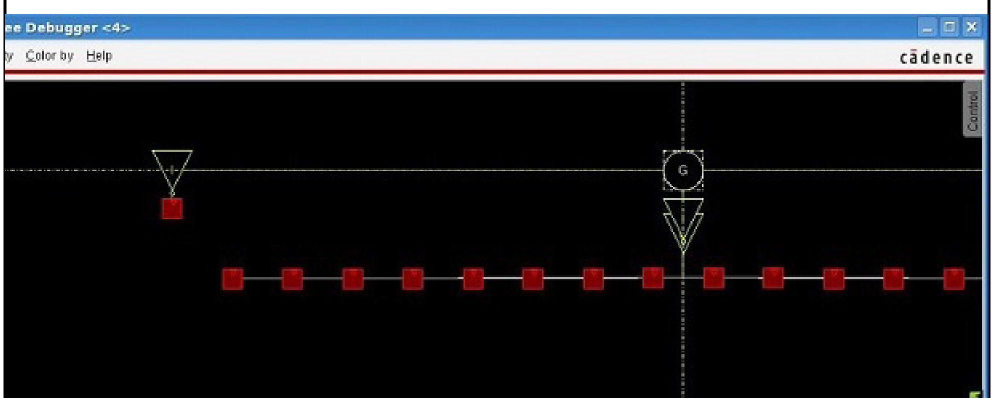
Zoom	Allows you to zoom in and out of the clock tree view. 
	Scale Allows you to specify the scale of zoom in or zoom out of the clock tree view.
	Vertical Range: from Allows you to zoom from and to a specific vertical range. When CTD is in timing-based mode, the view zooms to parts of the clock tree parts where latency is between ymin and ymax. The unit is the same as the global timing unit. When CTD is in Unit Delay mode, the view zooms to the portions of the clock tree where the stage depth is between ymin and ymax. Zooming is applied only in the vertical direction.
	In Zooms into the clock tree view.
	Out Zooms out of the clock tree view.

Horizontal

Zooms in or out of the clock tree view horizontally. In this submenu, you can select In or Out. You can also use the keys, "x" and "shift x" to zoom in and out of the display horizontally or along the x-axis. By default, the display will zoom with respect to its center point.

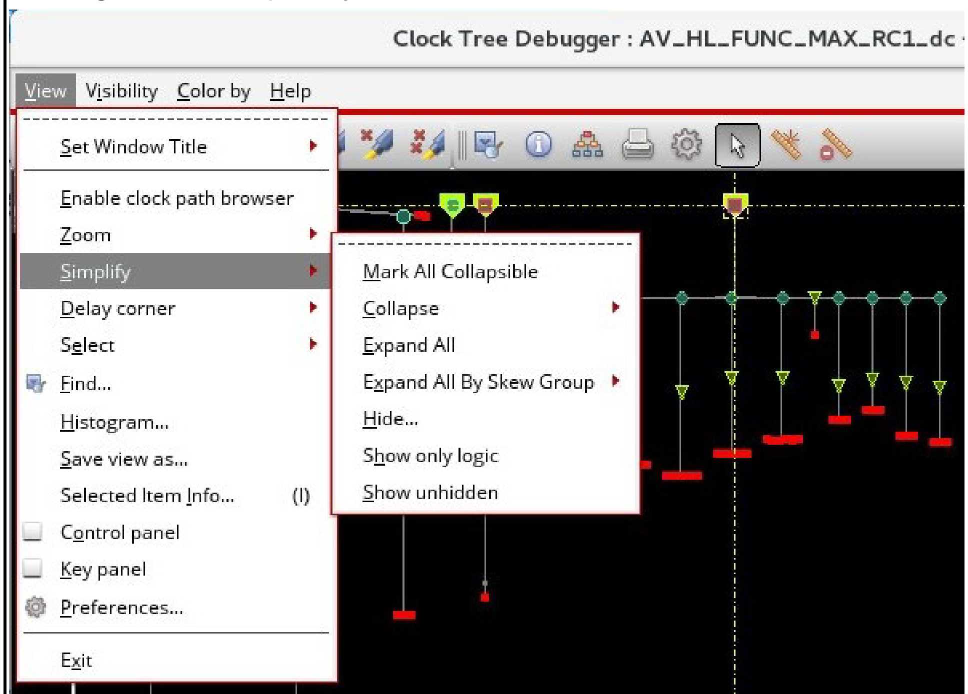


To zoom into specific objects, select the object and keeping the left mouse button pressed, press "x".



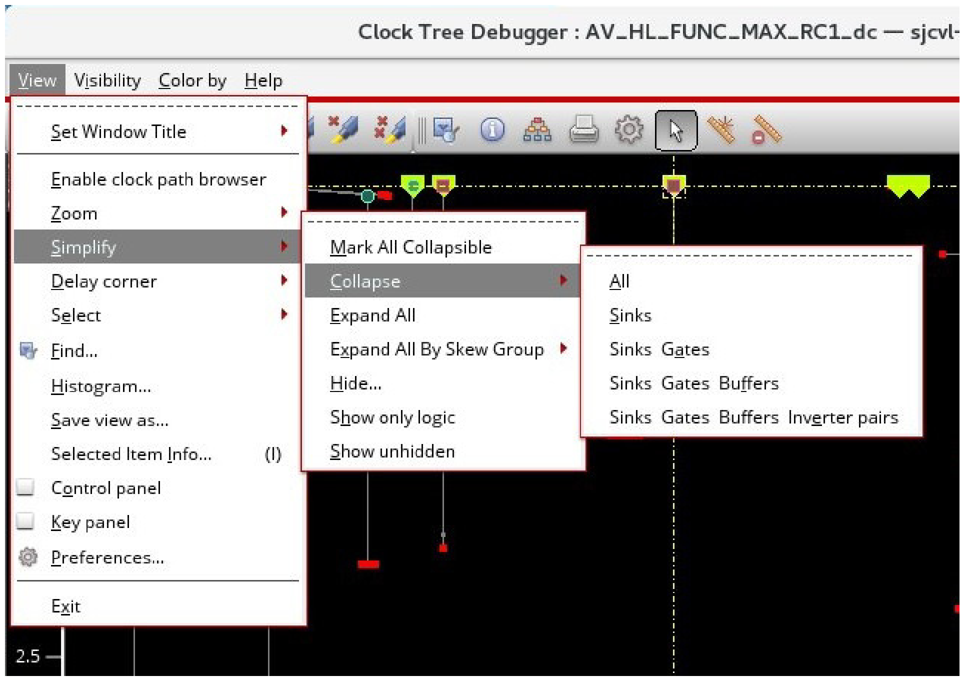
Vertical

Zooms in and out of the clock tree view vertically. In this submenu, you can select *In* or *Out*. You can also use the keys, "y" and "shift y" to zoom in and out of the display vertically or along the y-axis. By default, the display will zoom with respect to its center point.

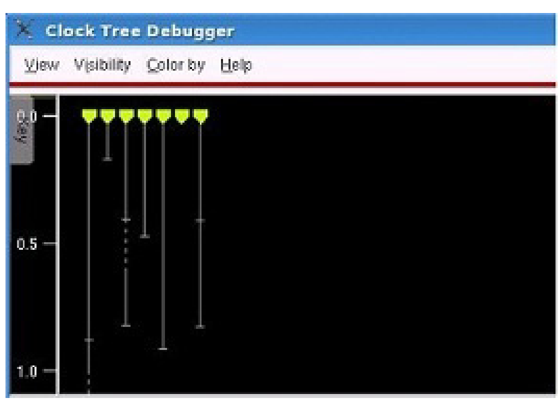
	Fit Zooms in or out to fit the *Clock Tree Debugger* window.
Simplify	Manages the complexity of the clock tree visualization. 
	Mark All Collapsible Marks all nodes as collapsible. This includes all nodes that may have been marked un-collapsible using the "Mark Uncollapsible" option available in the context menu of the CTD.

Collapse

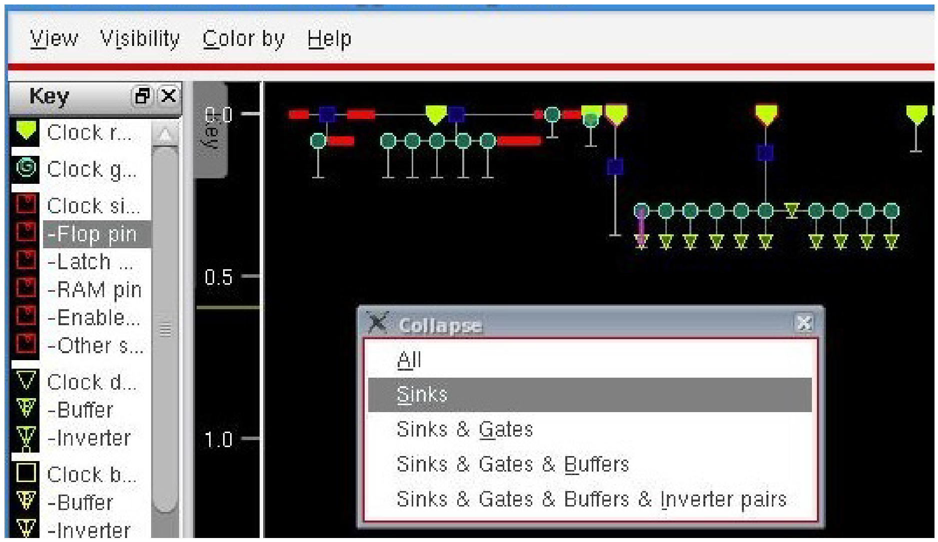
Provides options to collapse subtrees that consist of clock gates and flops in their fanout. There are different options available for collapsing subtrees in the drop-down menu of this option.



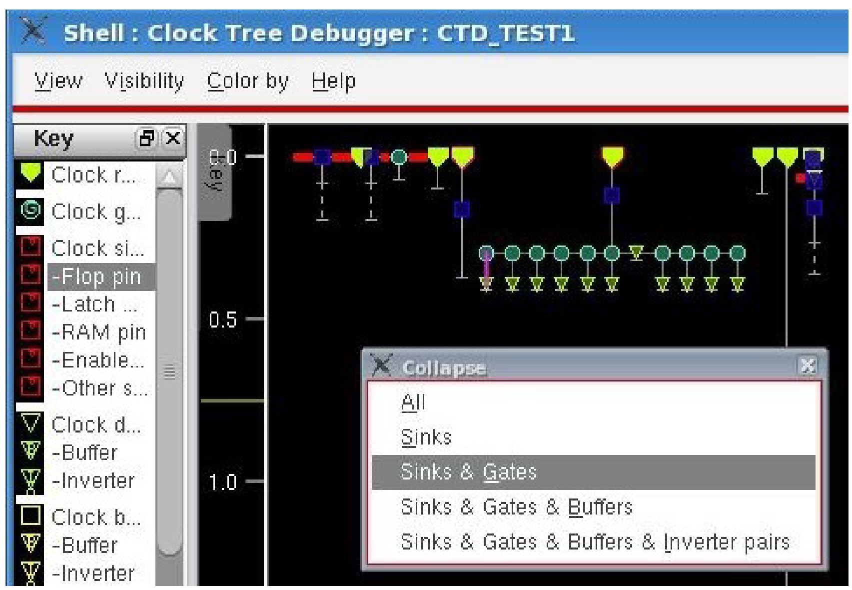
All - Collapses everything up to the top-level clock tree roots, so that the visualization shows only a set of clock trees represented by vertical summary bars.



Sinks - collapses subtrees with sinks.



Sinks & Gates - collapses subtrees with clusters of gates directly driving sinks.



Sinks & Gates & Buffers - collapses subtrees with clusters of gates driving sinks connected by buffers

Sinks & Gates & Buffers & Inverter pairs - collapses subtrees with clusters of gates driving sinks connected by buffers or inverters where the edge is not changed

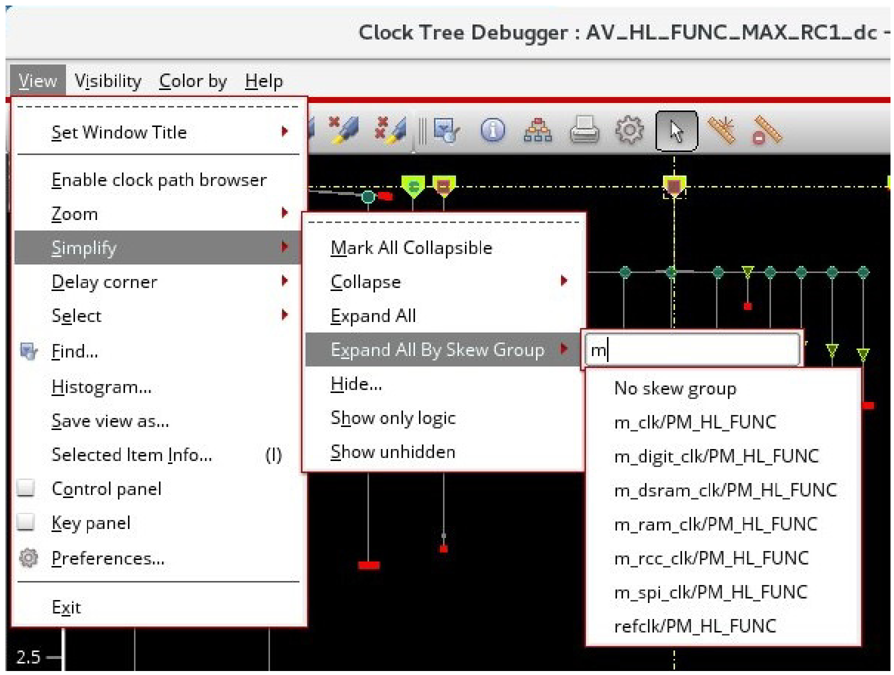
You can also double-click to collapse the subtrees of any node of the clock tree.

	Expand All Expands everything so that no subtree is collapsed. You can also double-click to expand the subtrees of any node of the clock tree.

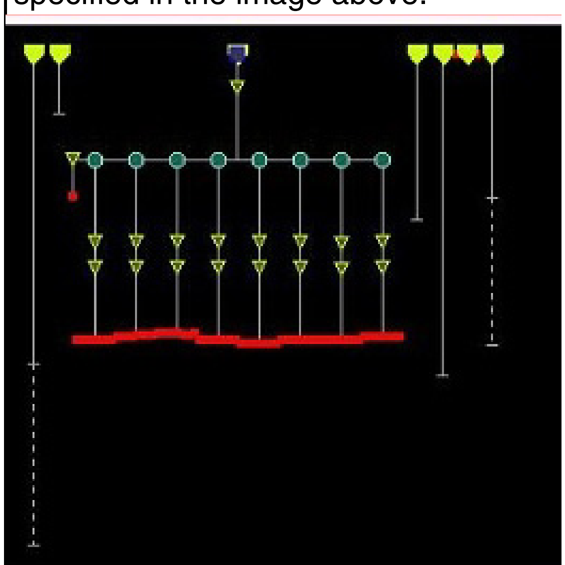
Expand All By Skew Group

Expands all subtrees that pass through the selected skew group, choosing parents of re-convergent nodes that are in the skew group.

You can choose a skew group as shown below.

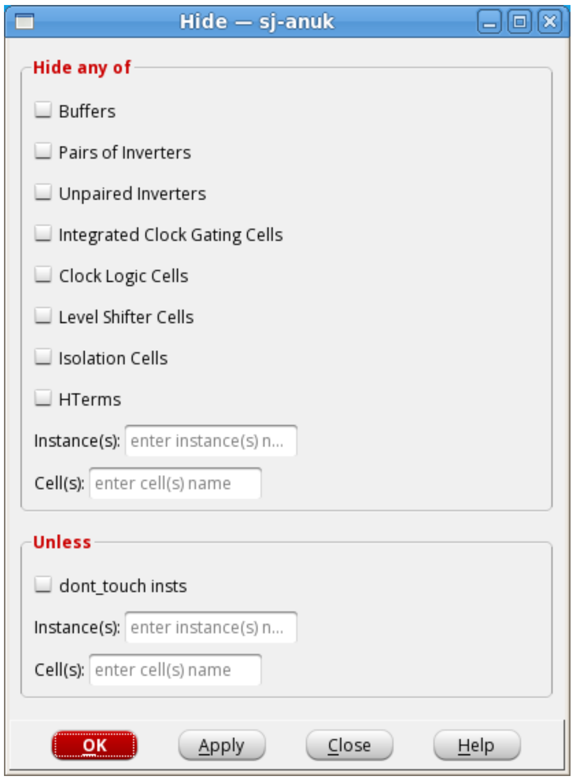


The image below shows the expanded subtrees for the skew group specified in the image above.



Hide

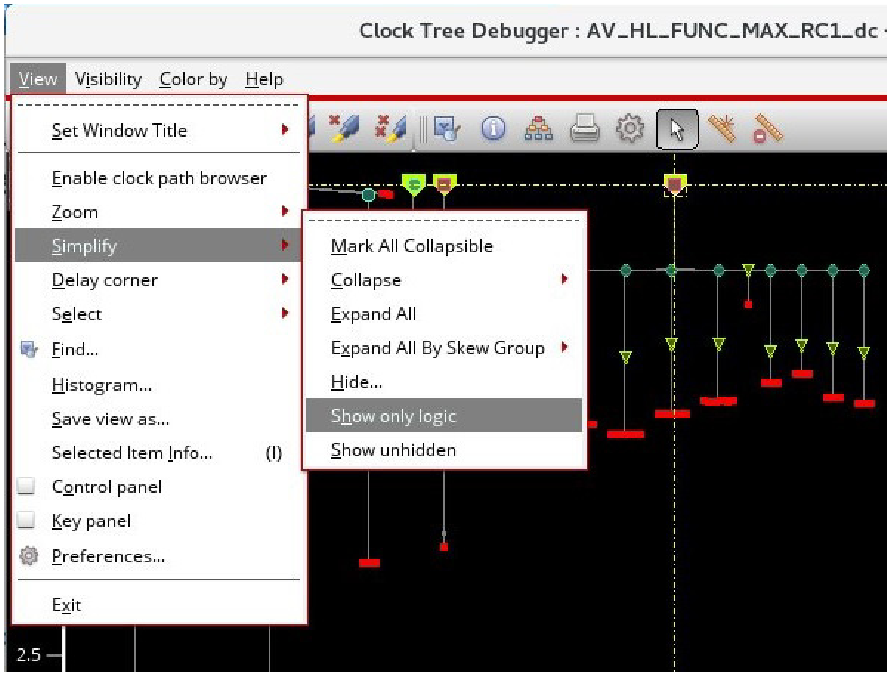
Hides specified instances and cells from view in the Clock Tree Viewer. The hidden objects are shown as small icons in the visualization. When you click this option, the form below opens.

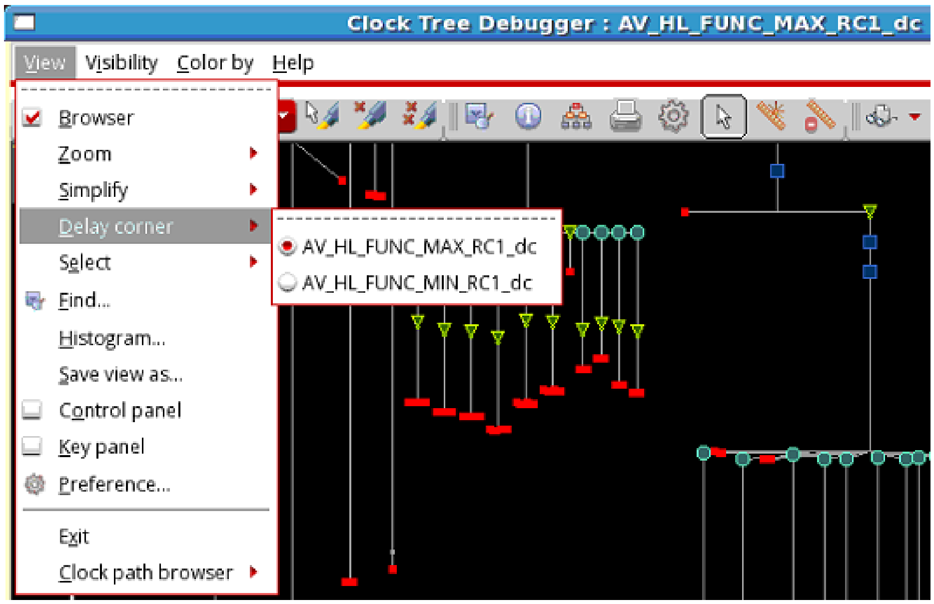


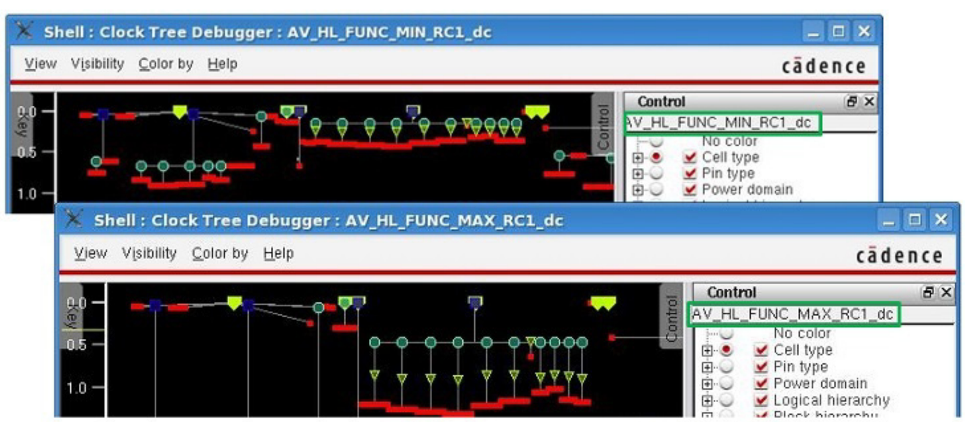
You can select the following options:

- *Buffers:* Hides all buffers of the clock tree.
- *Pairs of Inverters:* Hides all pairs of inverters of the clock tree.
- *Unpaired Inverters:* Hides all unpaired inverters of the clock tree.

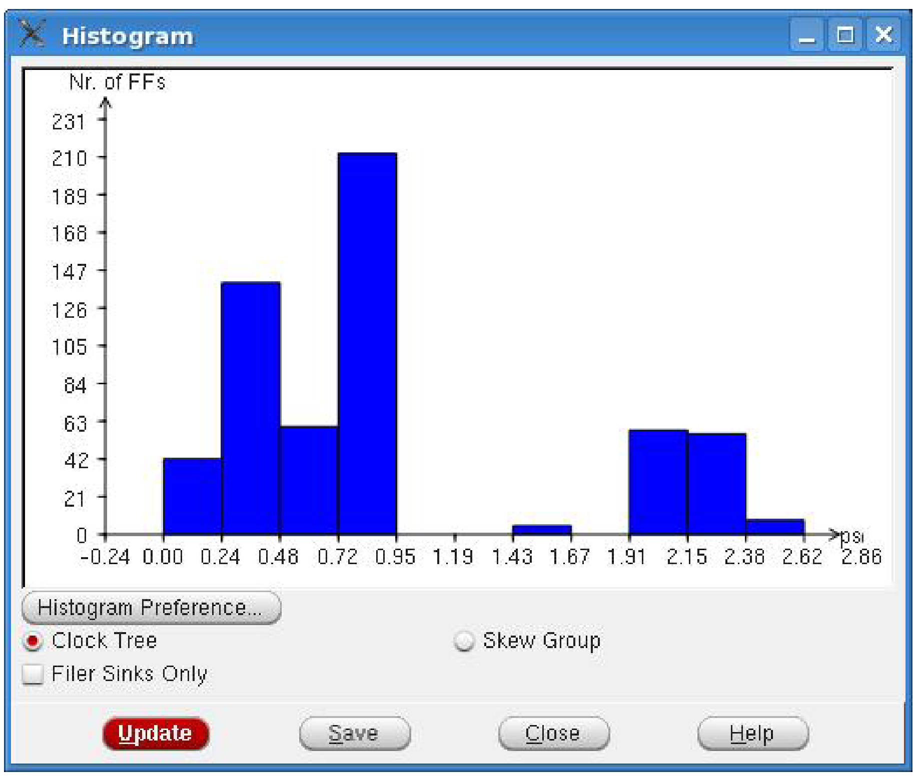
	■ *Integrated Clock Gating Cells:* Hides all integrated clock gating cells of the clock tree. ■ *Clock Logic Cells:* Hides all clock logic cells of the clock tree. ■ *Level Shifter Cells*: Hides all level shifter cells of the colck tree. ■ *Isolation Cells*: Hides all isolation cells of the clock tree. ■ *Instance(s):* Names of instances to be hidden. You can provide multiple instance names separated by a space. ■ *Cell(s):* Names of cells to be hidden. You can provide multiple cell names separated by a space. ■ *dont_touch insts:* Specifies that any clock tree objects that have the dont_touch constraint in the SDC and are specified, should not be hidden. You can specify the following: ■ *Instance(s)* - You can provide multiple instance names separated by a space. ■ *Cell(s)* - You can provide multiple cell names separated by a space. All options are unselected by default. For details, see the "The Hide Function in CTD".

	Show only logic Hides all the buffers, inverters, clock gates, and collapses subtrees consisting entirely of sinks, buffers, inverters or clock gates in a single step. Only logic cells are displayed. 
	Show unhidden Expands all the hidden instances and cells in the Clock Tree Viewer.

<table>
<tr>
<td>Delay corner</td>
<td>
Selects the delay half-corner you want to use to produce the visualization. This delay corner is mainly used to calculate timing for delays, as shown on the y axis, but also determines the set of available clock trees and available skew groups.
Note: When you select a delay corner, the name of the delay corner appears in the title bar of the CTD. This is shown below.


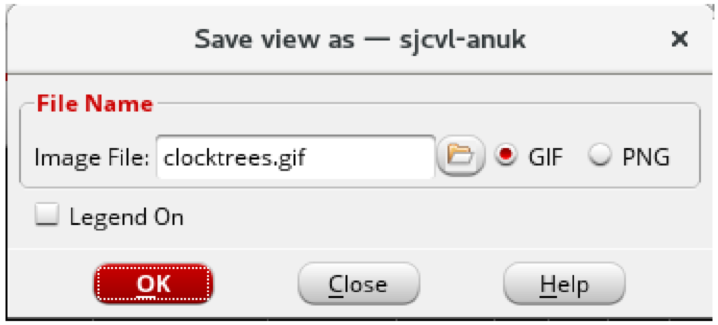
You can also switch the delay corner by using the bindkeys. Use the “d” key to switch to the next delay corner and use "D" or "shift + d" to display the default delay corner. The name of the current delay corner, which is displayed both in the information bar on the top of the Control Panel and in the title bar of the CTD, will be updated every time the delay corner is switched using the bindkey.
</td>
</tr>
<tr>
<td></td>
<td>


</td>
</tr>
</table>

<table>
<tr><td>Select</td><td colspan="2">Selects the specified object.</td></tr>
<tr><td></td><td>Select minimum</td><td>Selects the object with the minimum value for the current coloring.</td></tr>
<tr><td></td><td>Select maximum</td><td>Selects the object with the maximum value for the current coloring.</td></tr>
<tr><td></td><td>Enable crossprobing</td><td>Allows you to turn on/off cross-probing - locating objects in the Tempus layout window when the corresponding object is selected in the Clock Tree Debugger, and vice versa. For an example of cross-probing of a clock root, see the "Cross-probing of Clock Root" section.</td></tr>
<tr><td></td><td>Enable layout zoom selected</td><td>Allows you to turn-off zoom selection in the layout while cross probing. By default, this option is enabled.</td></tr>
<tr><td>Find...</td><td colspan="2">Finds and selects specific object types in the design display area.</td></tr>
<tr><td>Histogram</td><td colspan="2">Displays histogram of number of leaf pins versus clock insertion delay for clock trees and skew groups.


For details, see "Clock Tree Debugger Histogram".</td></tr>
</table>

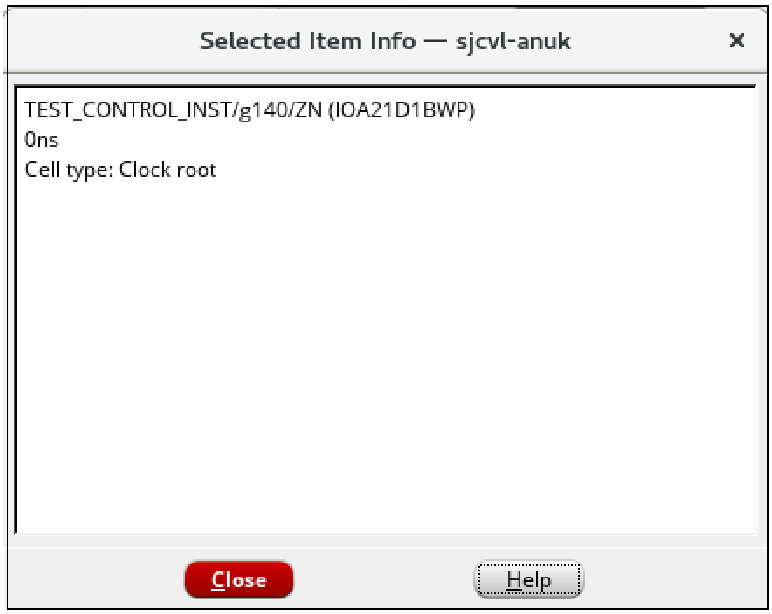
<table>
<tr>
<td>Save view as...</td>
<td>
Saves the current snapshot of the clock tree viewer into a file. When you click this option, the Save view as form opens.

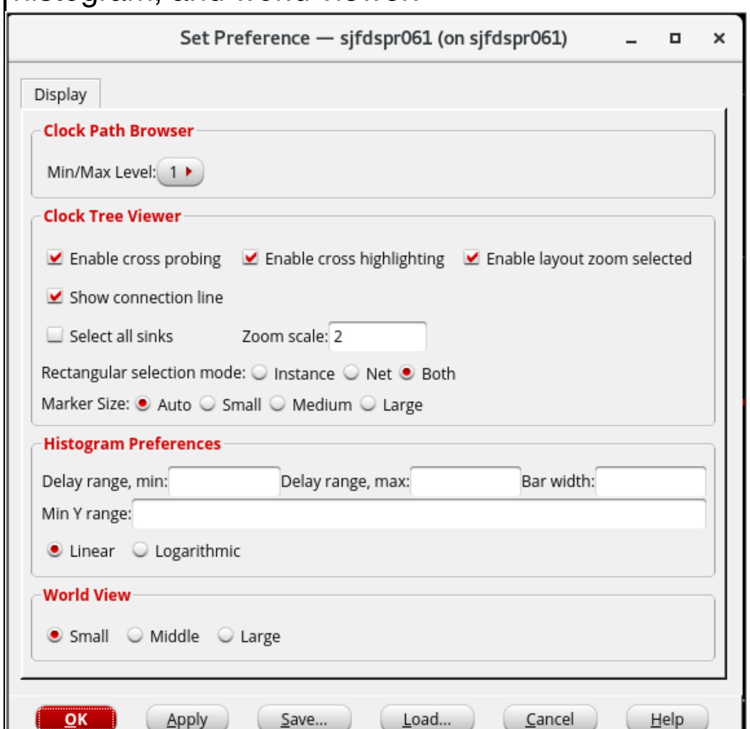


Th options are described below:

Image File: Specifies the name of the file where you want to dump the display image.
Default: <code>dump.gif</code>
GIF: Specifies that the format of the image should be GIF.
PNG: Specifies that the format of the image should be PNG.
Legend On: When checked, specifies to save the whole window including the panels that are open.

Note: You can also save the view by using the <code>ctd_save_view</code> command.
</td>
</tr>
</table>

Selected Item Info...	Displays information related to the object selected in the display area of the CTD. Select an object in the display area before clicking the Selected Item Info... option. The dialog box shown below opens. The content displayed in the dialog box is similar to that displayed in the tooltip. If you select another object, the content in the window is updated accordingly.  **Note:** ■ If no item is selected in CTD display area before clicking Selected Item Info...an error message, "No Object Selected" will be displayed. ■ When multiple things are selected, the content is displayed for the most recently selected item. ■ The text in the window is selectable. You can copy it for later reference.

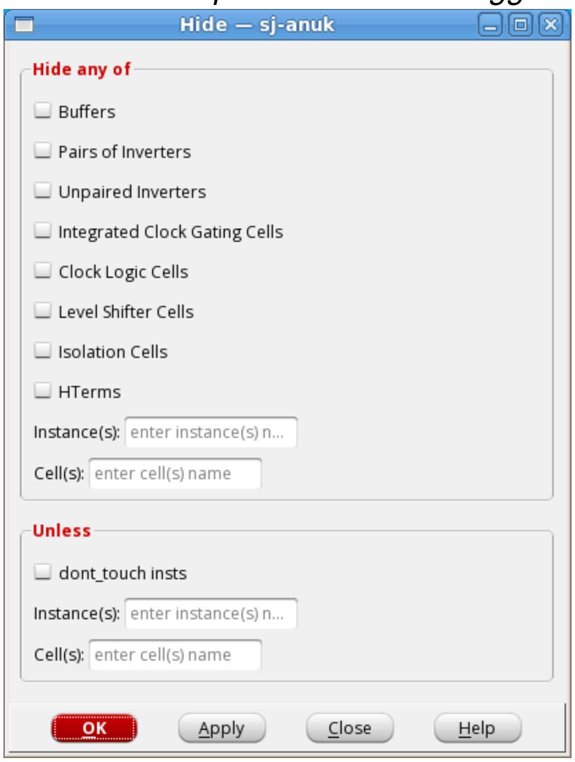
Control panel	Controls the visibility of the control panel in the *Clock Tree Debugger* window. When checked, you can drag and view the control panel on the right side of the *Clock Tree Debugger* main window and the *key panel* on the left side of the main window, as shown below. The Control panel shows the current selections for No color, Cell type, Pin Type, Power domain, Logical hierarchy, Block hierarchy, Fanout, Fanout Violation, Skew group, Clock tree, Skew, Transition time, Max trans violations, Net length, Net length violations, Total net cap, Max cap violations, Setup slack, Hold slack, Window outage, Signal edge, Net type, Route type, Constraints, and Timing windows. For details, see "Control Panel of CTD".
Key panel	When enumeration or subset coloring is in effect, a window showing a key is available in the Clock Tree Debugger. The key contains a list of enumeration categories or subsets and their corresponding colors. For details, see "Key Panel of CTD".
Preference	Opens the Set Preference form. In this form, you can specify the GUI configuration settings for clock path browser, clock tree viewer, histogram, and world viewer. 

Exit	Allows you to exit the *Clock Tree Debugger* tool.

The Hide Function in CTD

Use the Hide form to specify the buffers, inverters, instances, and cells that you want to hide in the CTD displays.

Choose - *CCOpt Clock Tree Debugger* - *View* - *Simplify* - *Hide*



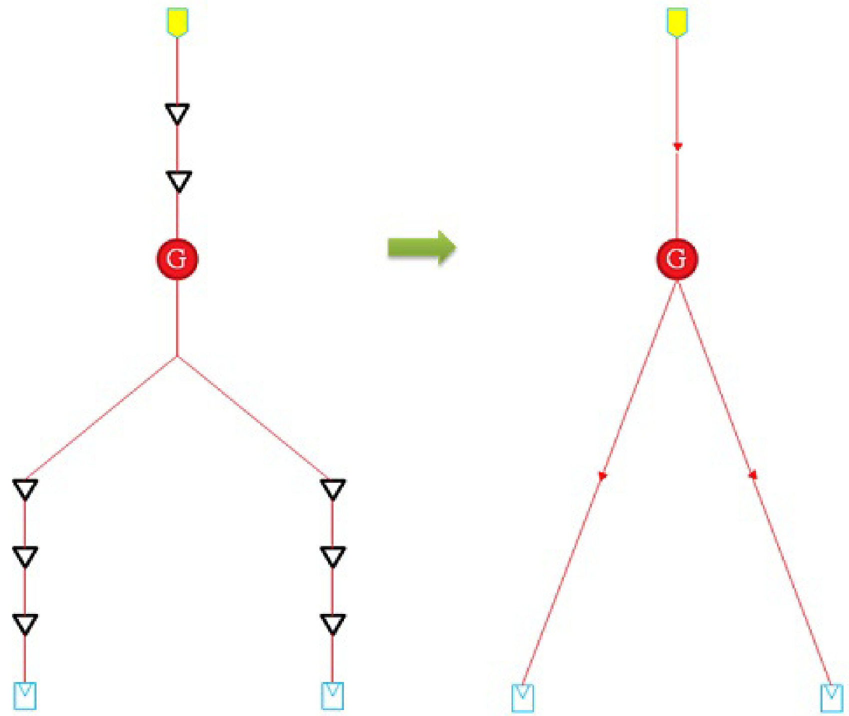
Options	**Descriptions**
Buffers	Hides all buffers of the clock tree.
Pairs of Inverters	Hides all pairs of inverters of the clock tree.
Unpaired Inverters	Hides all unpaired inverters of the clock tree.
Integrated Clock Gating Cells	Hides all integrated clock gating cells of the clock tree.
Clock Logic Cells	Hides all clock logic cells of the clock tree.
Level Shifter Cells	Hides all level shifter cells of the clock tree.
Isolation Cells	Hides all isolation cells of the clock tree.
HPins	Hides all the preserved ports. The preserved ports are visible only in the Unit Delay mode. Use this option to hide the port bits.

Instance(s)	Names of instances to be hidden. You can provide multiple instance names separated by a space.
Cell(s)	Names of cells to be hidden. You can provide multiple cell names separated by a space.
dont_touch insts	Specifies that any clock tree objects that have the dont_touch constraint in the SDC and are specified, should not be hidden. You can specify the following: ■ Instance(s) - You can provide multiple instance names separated by a space. ■ Cell(s) - You can provide multiple cell names separated by a space.

Hiding Buffers and Inverters

The figure below illustrates the behavior of only hiding buffers and inverters of clock trees. When you specify only the Buffers or Pairs of Inverters or Unpaired Inverters options in the Hide form, then all cascaded logic will be turned into one small icon, which will be a small red triangle. This is shown in the figure below.

After hiding, the hidden part of clock tree will no longer care about the delays, and the hidden structure will show the simplified clock tree logic structure. The structure will turn into a simplified structure in which the basic element is a path from one “non-hidden” point to another “non-hidden” point or to a sink pin. This path contains a start point, an end point, a small icon that stands for the hidden clock tree logic, and a flight line that connects the start point and end point, and feeds through the small icon. A “non-hidden” point is the clock tree object which is not included in user-specified list.

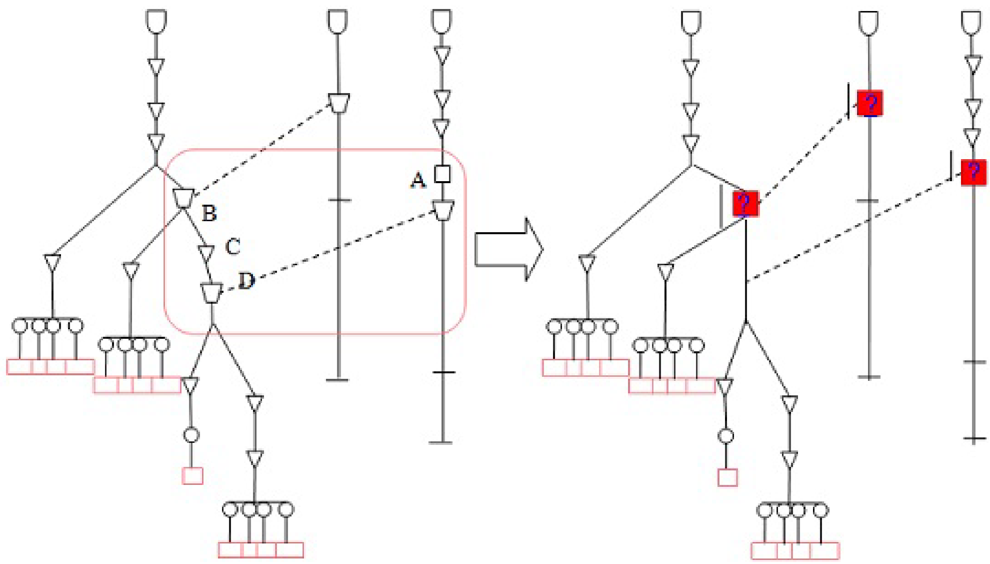


Hiding Clock Logic Cells

Clock logic cells, such as clock gate cells, can be specified for hiding. If the cells, which are connected directly, contain any clock logic (excluding buffers and inverters), then they are collapsed as large red square icons in the Clock Tree Viewer. The red square has a question mark "?", in the center, which indicates that the content is unknown or complex. This size of the red square icon is larger than the symbol shape of the normal clock objects. This square icon is placed at the earliest point at which it is encountered in each clock tree.

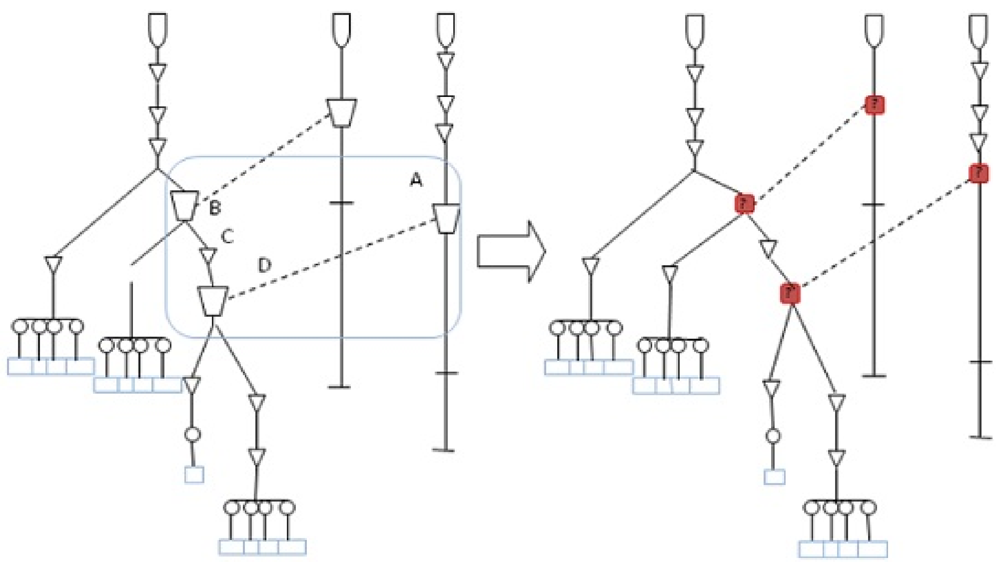
In the sample scenario 1 shown below, the points A, B, C, and D are user-specified points. The points B, C, and D are on the same clock tree so the square icon is placed only on point B, which is encountered first. The vertical line below the hidden logic is used to show the total delay across the logic, and connect offshoots in and out at the appropriate time offsets.

Sample Scenario 1



Now consider another scenario in which point C is not specified by the user for hiding. In this case, the hide will be as shown below. The red square icon will be placed on both points B and D.

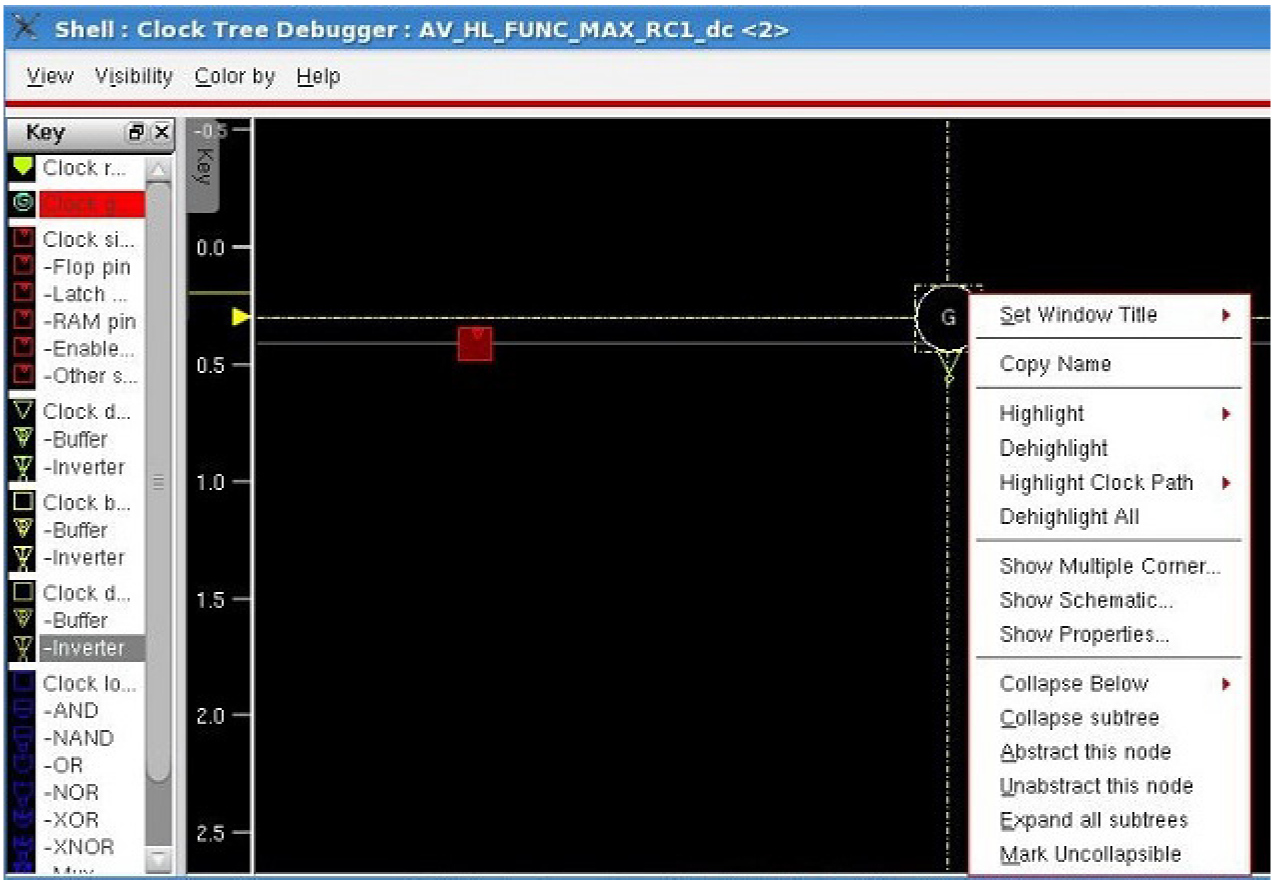
Sample Scenario 2



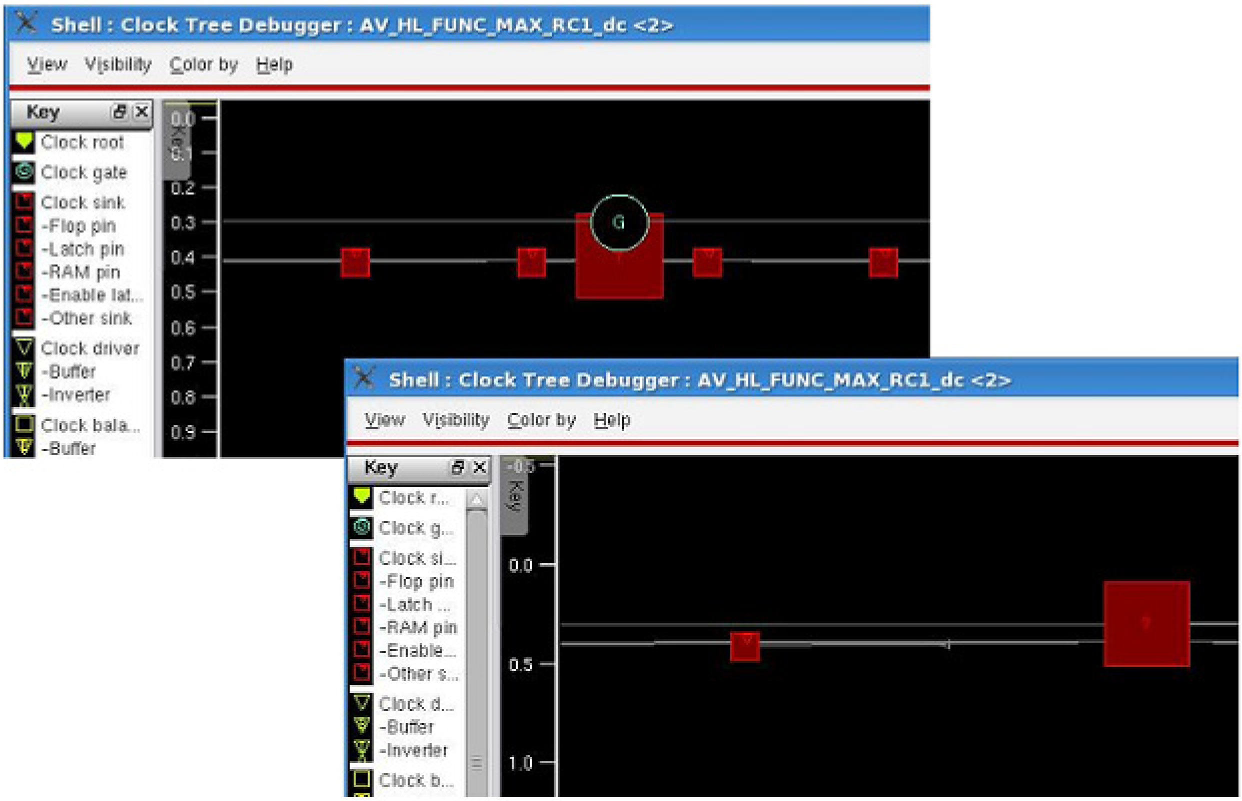
Note:

- To view the hidden objects again, select the *Show unhidden* option in the *Simplify* submenu of the View menu.

- The hide function also allows you to hide and unhide selected nodes by clicking options, *Abstract this node* and *Unabstract this node*, provided in the content menu of the Clock Tree Viewer.



Below is a screenshot showing the hiding of a selected clock gate.


Shell : Clock Tree Debugger : AV_HL_FUNC_MAX_RC1_dc <2>
View
Visibility
Color by
Help
Key
Clock root
Clock gate
Clock sink
-Flop pin
-Latch pin
-RAM pin
-Enable lat...
-Other sink
Clock driver
-Buffer
-Inverter
Clock bala...
-Buffer
0.0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
G
Shell : Clock Tree Debugger : AV_HL_FUNC_MAX_RC1_dc <2>
View
Visibility
Color by
Help
Key
Clock r...
Clock g...
Clock si...
-Flop pin
-Latch ...
-RAM pin
-Enable...
-Other s...
Clock d...
-Buffer
-Inverter
Clock b...
0.0
0.5
1.0

Cross-Probing of Clock Root in CTD

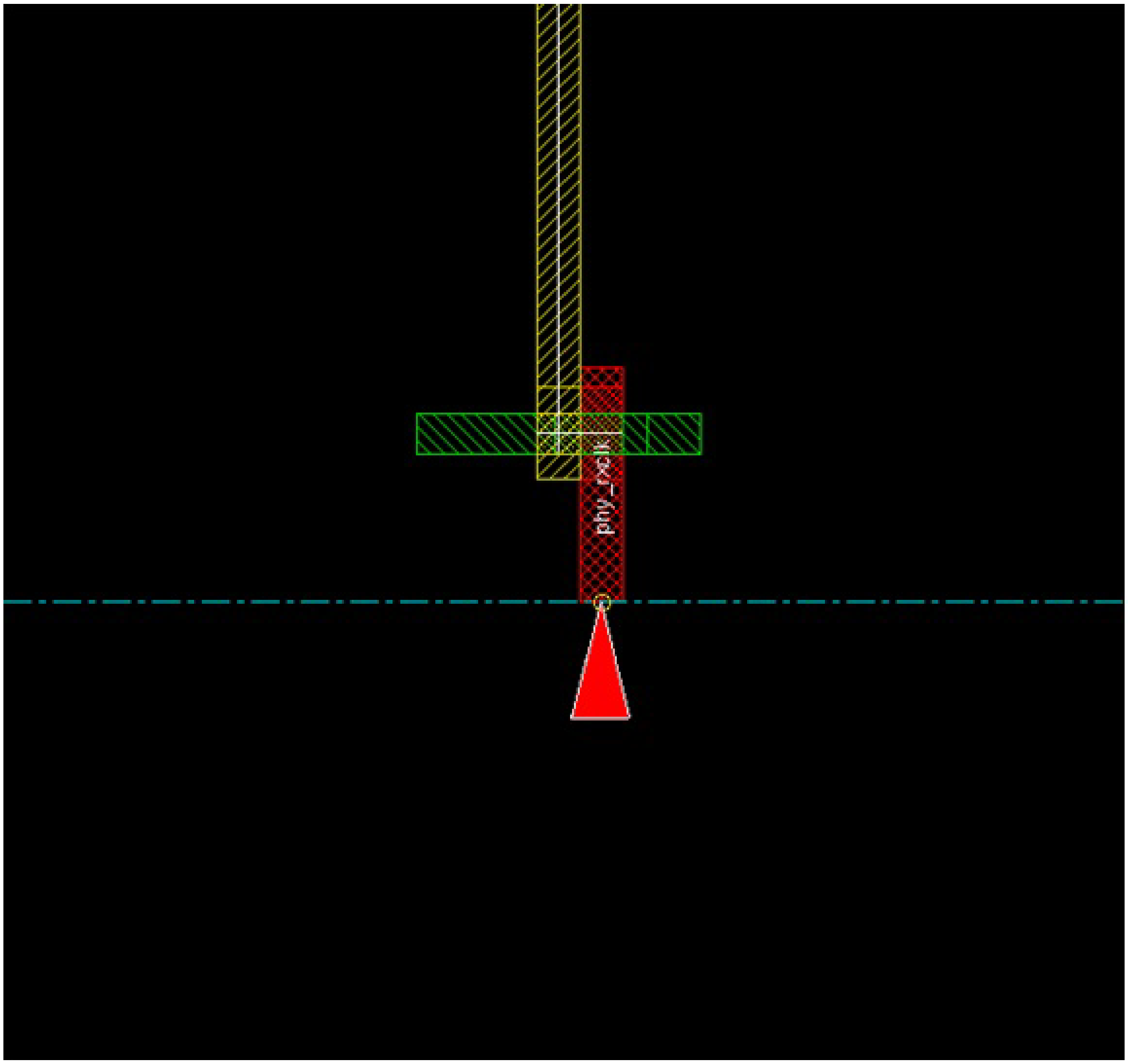
To cross-probe a clock root, select the clock root symbol in the display area of the CTD. Then select the *View* menu and check the *Enable crossprobing* option in the *Select* submenu. The layout window will zoom into the object in which that clock root is defined. The object of clock root could be a cell pin, an I/O port, or an I/O pad. If the clock root is a cell pin, layout window will zoom in the area of that pin and select that cell. This is shown in the figure below.

Note: Alternatively, you can select the clock root pin, port, or pad - depending on where the clock root is defined - in the layout window and the clock root will be selected in the CTD.

Clock Root in Cell Pin

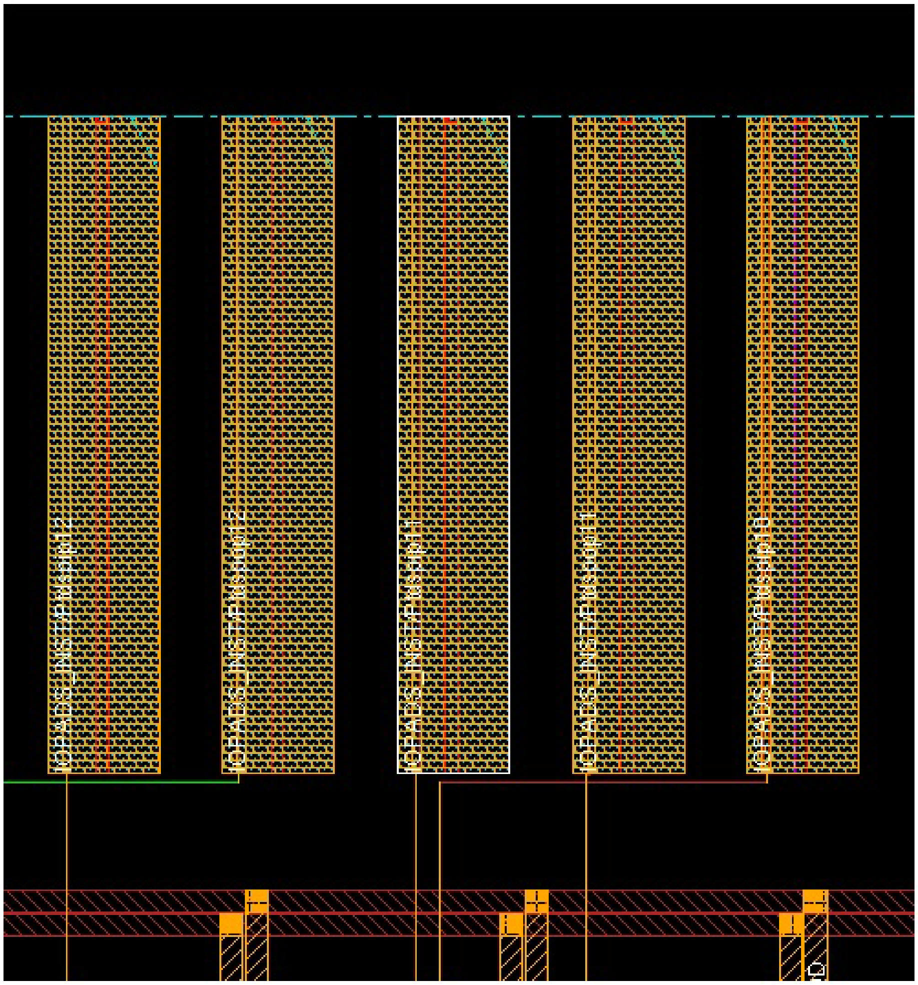
If the clock root is defined in an I/O port, the layout window will zoom to the area where the port is and select that port.

Clock Root in I/O Port



If the clock root is defined in an I/O pad, the layout window will zoom to the area where the pad is and select that pad.

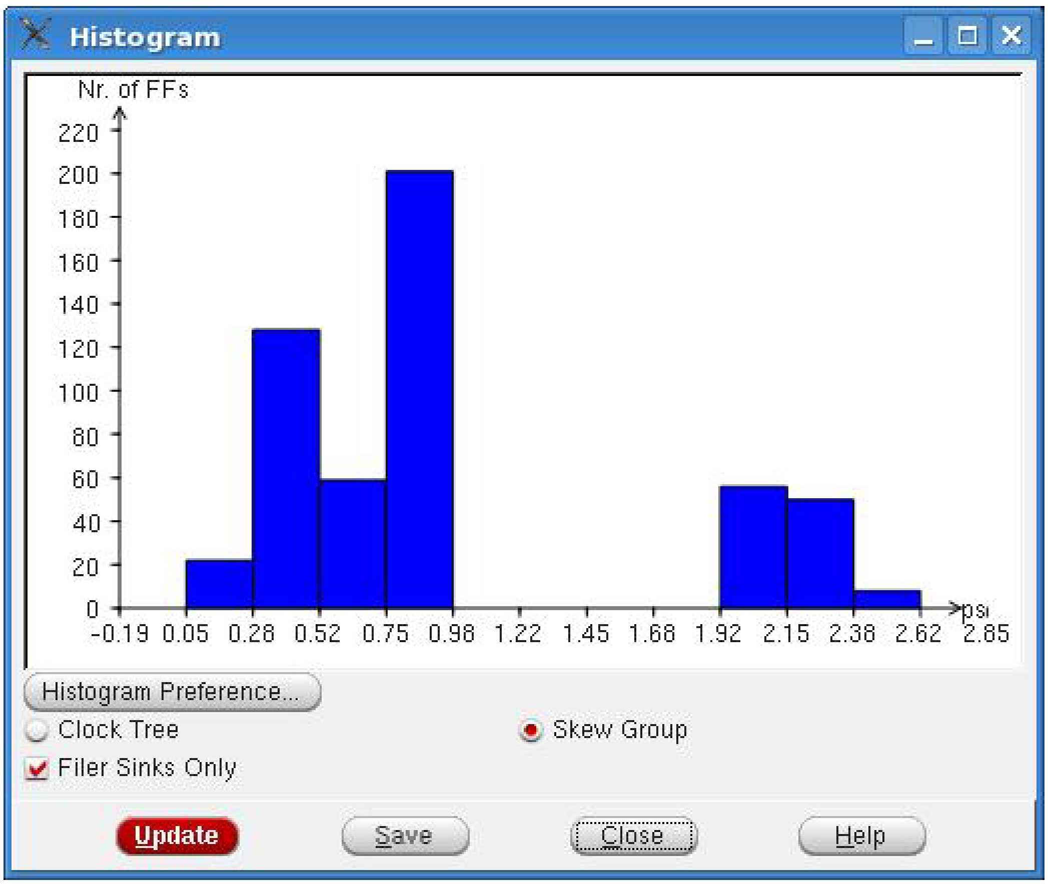
Clock Root in I/O Pad



Clock Tree Debugger Histogram

Displays histogram of number of leaf pins versus clock insertion delay. In the *View* menu, choose *Histogram*. The following form opens. The vertical axis of the histogram shows the number of leaf pins that have the delay shown (in picoseconds) on the horizontal axis of the histogram. You can view histogram for multiple corners at once in different colors. You can also Export the data displayed in the histogram in a CSV format.

You can also save the histogram in a file by using the `ctd_save_histogram` command. However, before using this command, first open and configure the histogram in the GUI, including skew group and delay corners.



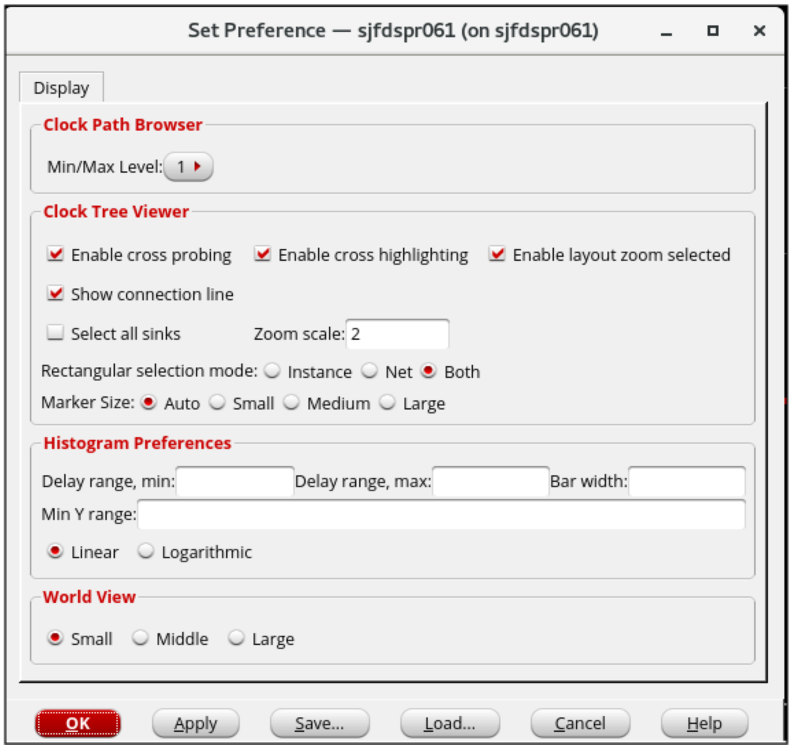
Options	Descriptions
Skew group	Specifies the skew group to be displayed in the histogram. You can select different skew groups from the dropdown list. You can also select the delay corners for which to display the graph. The graphs for different delay corners is displayed in different colors.

Export CSV	Exports the underlying data in a CSV format.
Prefs...	Allows you to select preferences for the histogram.
Save	Saves the histogram information to a file.

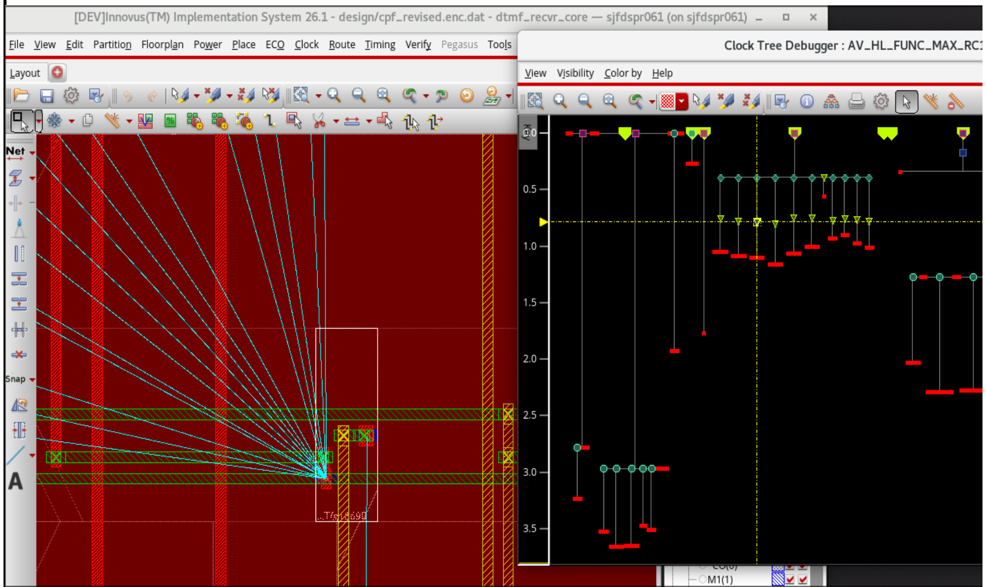
You can also click the Preference submenu in the View menu of the CTD to open this form. For details, see "Set Preferences for CTD GUI Configuration".

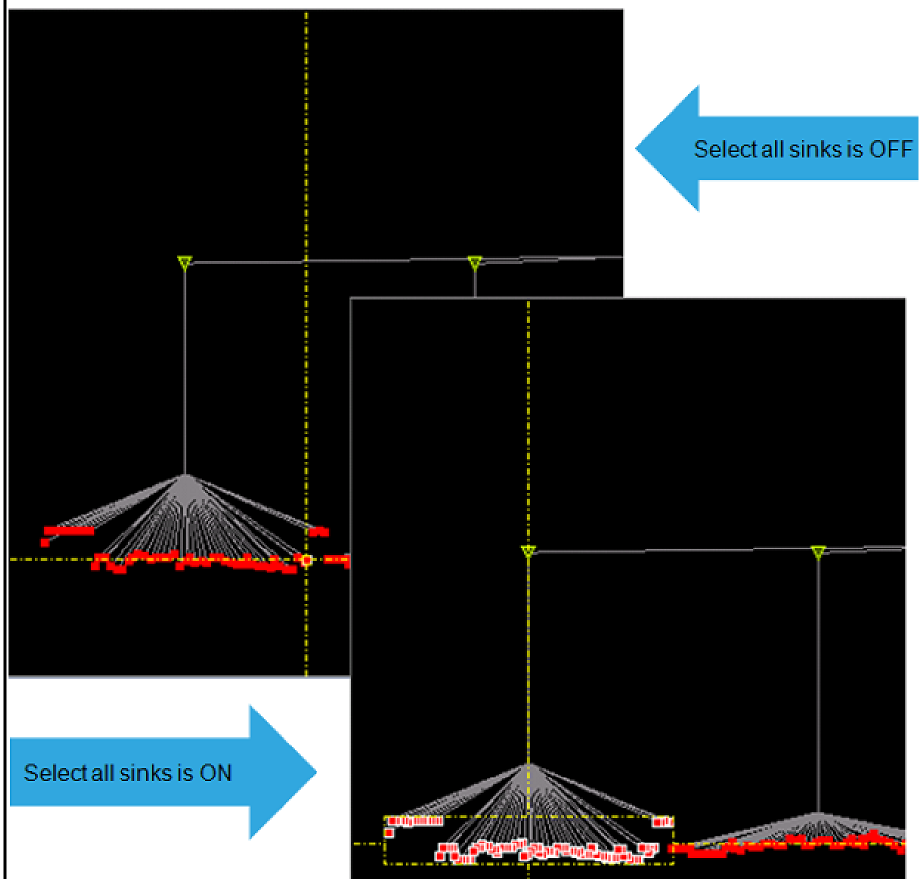
Set Preferences for CTD GUI Configuration

Displays settings for Clock Path Browser, Clock Tree Viewer, Histogram, and World Viewer. In the *View* menu, choose *Preference*. The following form opens:



Options	**Descriptions**	
Clock Path Browser	*Min/Max Level*: Specifies the number of min/max levels for the skew groups in the Browser. You can choose 1, 2, 3, or 4 to view the first, second, third, or fourth level of min/max paths for skew groups. By default, 1 is selected. When you choose an option in the above submenu, the Clock Path Browser is updated to show the specified number of min/max paths.	
Clock Tree Viewer	Displays the settings for the clock tree viewer.	
Enable cross probing	Allows you to turn on/off cross-probing - locating objects in the Tempus layout window when the corresponding object is selected in the CTD, and vice versa. By default, cross probing is enabled.	
Enable cross highlighting	Allows you to turn on/off cross-highlighting -highlighting objects in the layout window when the corresponding object is highlighted in the CTD and vice versa. By default, cross highlighting is enabled.	

Enable layout zoom selected	Allows you to enable or disable zoom-on-selection in the layout when an item is selected in the Clock Tree Viewer. The equivalent bindkey is Ctrl+E. When enabled, selecting any clock net or node in the CTD automatically zooms the layout window to that object. This option is enabled by default. 

Select all sinks	Allows you to select clock sinks. When this option is disabled, you can select a single sink. When this option is enabled and if you select one sink under a node (maybe a buffer or a gate), the CTD will select all sinks under that node. This is shown in below image. 
Show connection line	Toggles the display of flight lines in the Clock Tree Viewer. By default, the option is turned ON.
Zoom scale	Specifies the scale of zoom into the clock tree view. By default, the zoom scale value is 2.

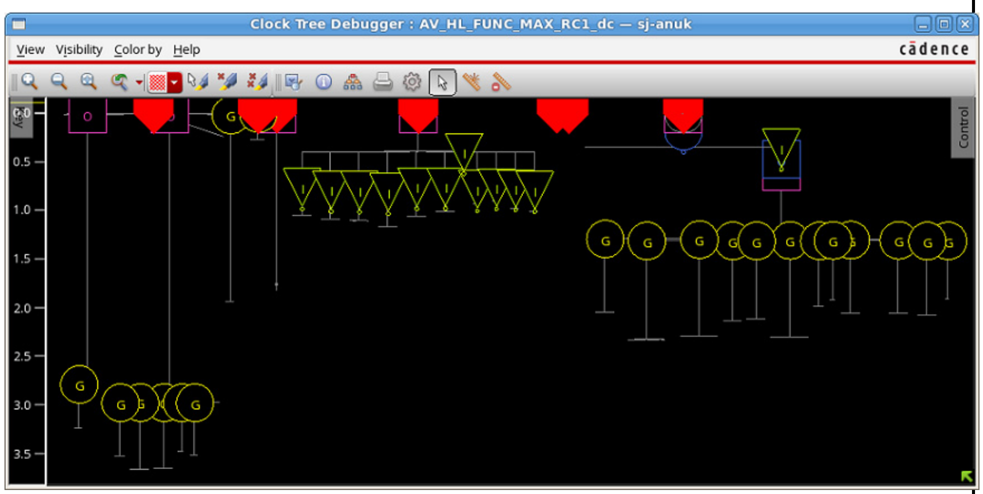
<table>
<tr>
<td>Rectangular selection mode</td>
<td>Specifies the object that is to be selected in a rectangular selection. You can specify any one of the following:

Instance: Specifies that instances should be selected
Net: Specifies that nets should be selected
Both: Specifies that both instances and nets should be selected. This option is ON by default.

</td>
</tr>
<tr>
<td>Marker Size</td>
<td>Specifies the size of the markers displayed in the CTD. The setting is applied to all markers. You can choose from the following options:

Auto
Small
Medium
Large

The default setting is Auto. The CTD display for large marker size setting is shown below.



</td>
</tr>
</table>

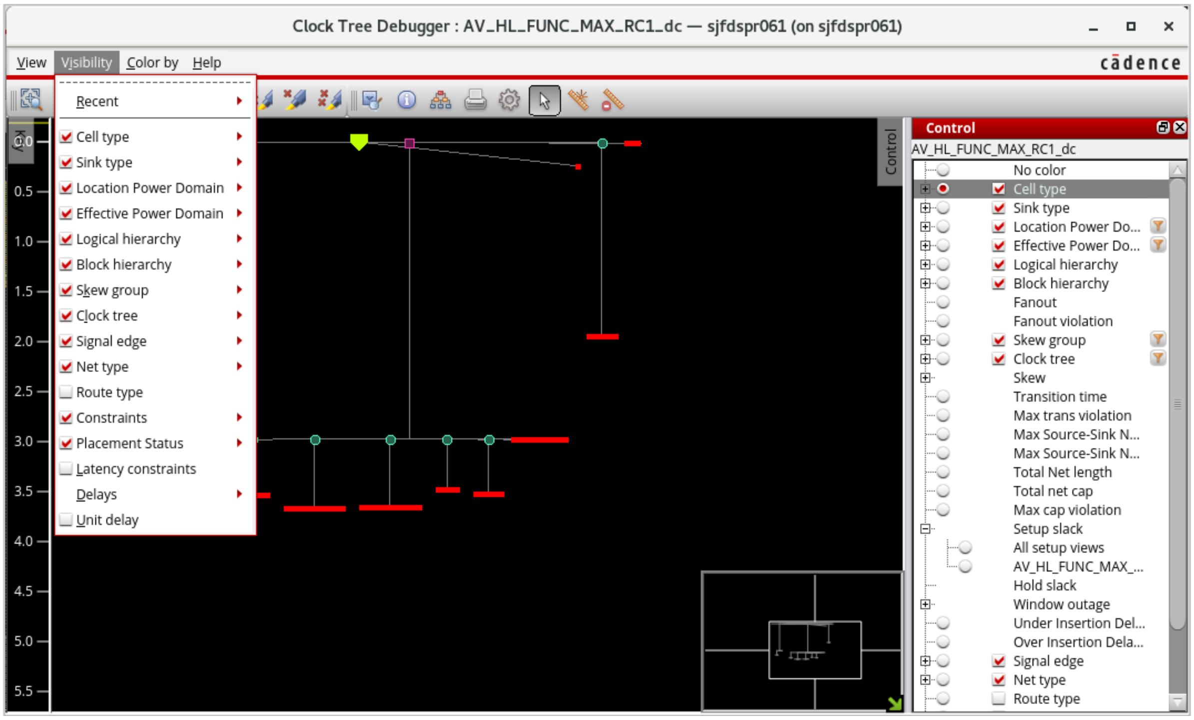
<table>
<tr><td>Histogram Preference</td><td colspan="2">Displays the settings for the histogram.
Delay Range, min: The minimum value of the delay range to be displayed for the X-axis.
Delay Range, max: The maximum value of the delay range to be displayed for the X-axis.
Bar width: Sets the width of each bar displayed in the histogram. For example, if the range is from 0 to 1000ps, and you want 10 bars, you can specify a width of 100ps to accommodate all 10 bars in the histogram. A smaller width lets you display more bars.
Min Y Range: Specifies the minimum range for the Y-axis. The histogram automatically scales the vertical axis to accommodate the tallest bar that it is showing. However, you can specify this to make sure that it is at least a certain height to make it easier to compare with other histograms. This is similar to setting a Y range, but if you set a value smaller than the tallest bar, it will be ignored because it does not make sense to chop the bar that it has to display.

For example, if you set a value of 1000, but the tallest bar is 487, then the tallest bar will be a little less than half the height of the histogram. If you do not specify one, then that bar will be the whole height of the histogram. So, if you are comparing histograms between different runs, it is easier to view them if they are all showing the same ranges.
Linear/Logarithmic: Select Linear or Logarithmic for the type of histogram you want to view.</td></tr>
<tr><td>World View</td><td colspan="2">Displays the settings for the size of the world viewer.</td></tr>
<tr><td></td><td>Small</td><td>Select for a small world viewer window.</td></tr>
<tr><td></td><td>Middle</td><td>Select for a medium world viewer window.</td></tr>
<tr><td></td><td>Large</td><td>Select for a large world viewer window.</td></tr>
</table>

Clock Tree Debugger - Visibility Menu

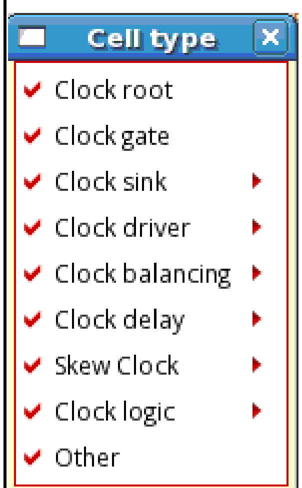
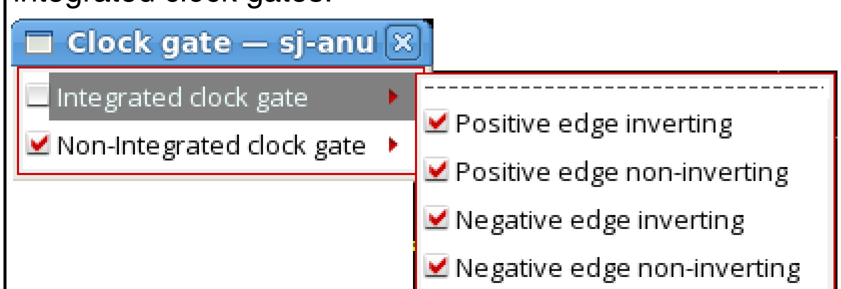
Use the *Visibility* menu to control the visibility of items in the clock tree view.

Choose *Clock – CCOpt Clock Tree Debugger – Visibility.*



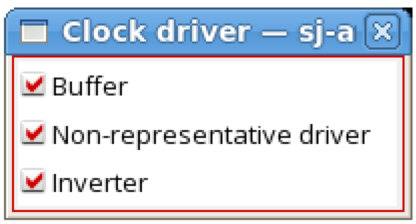
In the Visibility menu, for items with both a check box and a submenu, checking the box also checks all items in the submenu and its descendants, and unchecking it unchecks those items. If none of the descendants are checked, the box is also shown as unchecked. In addition, double-clicking a check box selects that check box and unselects all of its peers underneath the same top-level menu item. For example, double-clicking the check box for a single skew group will select that skew group, and unselect all other skew groups, while leaving other menus such as *Cell type* unchanged.

Options	Descriptions

Recent	Allows you to view the the five most recently used items from the *Visibility* menu. Select or deselect the check boxes against each option in the drop-down menu depending on what you want to view. You can choose to view or not view the Cell type, Cell type/Clock root, Cell type/Clock gate, Cell type/clock sink, and Cell type/clock sink/Flop pin, and so on. These are full copies of the original items, and can be checked and unchecked in the same way. The items are also still present in their usual locations.
Cell type	Specifies the visibility of various elements of visualization. Select or deselect the check boxes against each option in the drop-down menu depending on what you want to view. You can choose to view or not view the *Clock root*, *Clock gate*, *Clock sink*, *Clock driver*, *Clock balancing*, *Clock delay*, *Clock logic,* and *Other*. 
	For Clock gate, you can choose to view positive/negative edge inverting/non-inverting subcategories for integrated and non-integrated clock gates: 

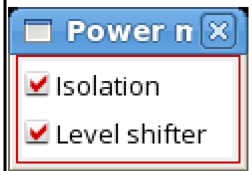
<table>
<tr>
<td></td>
<td>For *Clock sink*, you can choose to view one or more of the following:

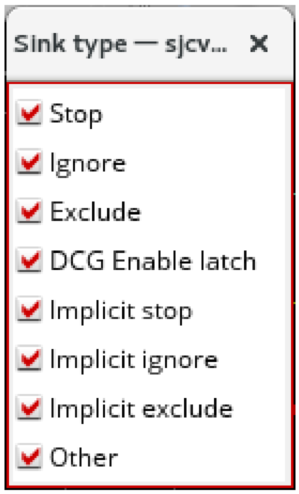
</td>
</tr>
<tr>
<td></td>
<td>For Clock driver you can choose to turn on or off the display of buffers and paired inverters, and non-representative drivers.



</td>
</tr>
<tr>
<td></td>
<td>For *Clock balancing*, and *Clock delay* you can choose to turn on or off the display of buffers and paired inverters.</td>
</tr>
<tr>
<td></td>
<td>For *Clock logic*, you can select or deselect any of the following:

</td>
</tr>
</table>

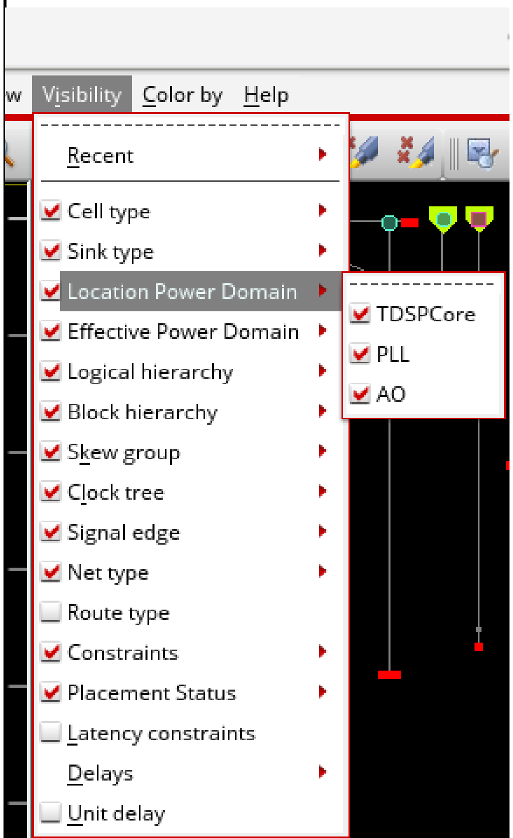
	For Power management, you can choose to turn on or off the display of Isolation cells or Level shifter. 

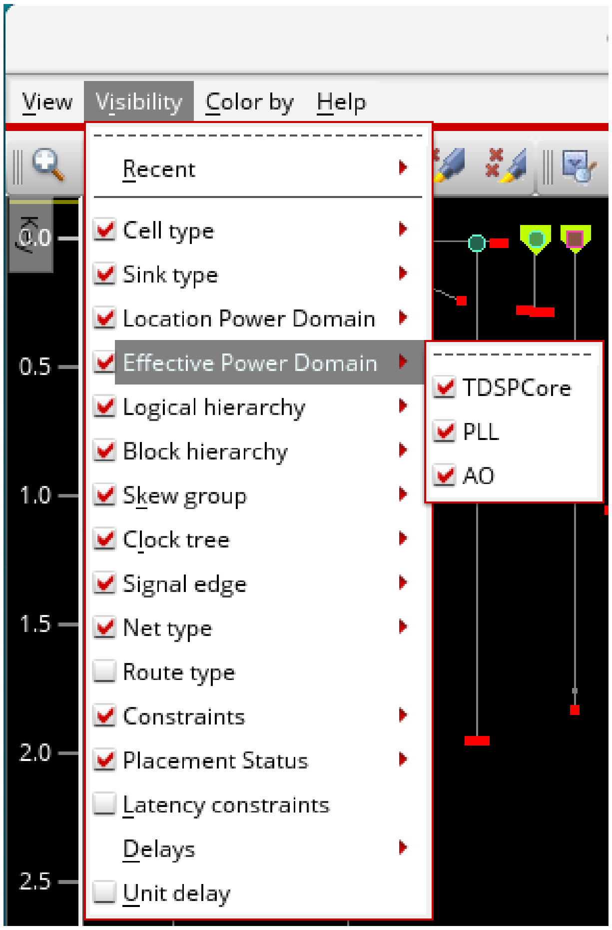
<table>
<tr>
<td>Sink type</td>
<td>
Specifies the visibility of different types of pins. Select or deselect the check boxes against each option in the drop-down menu depending on what you want to view. By default, all options are selected.


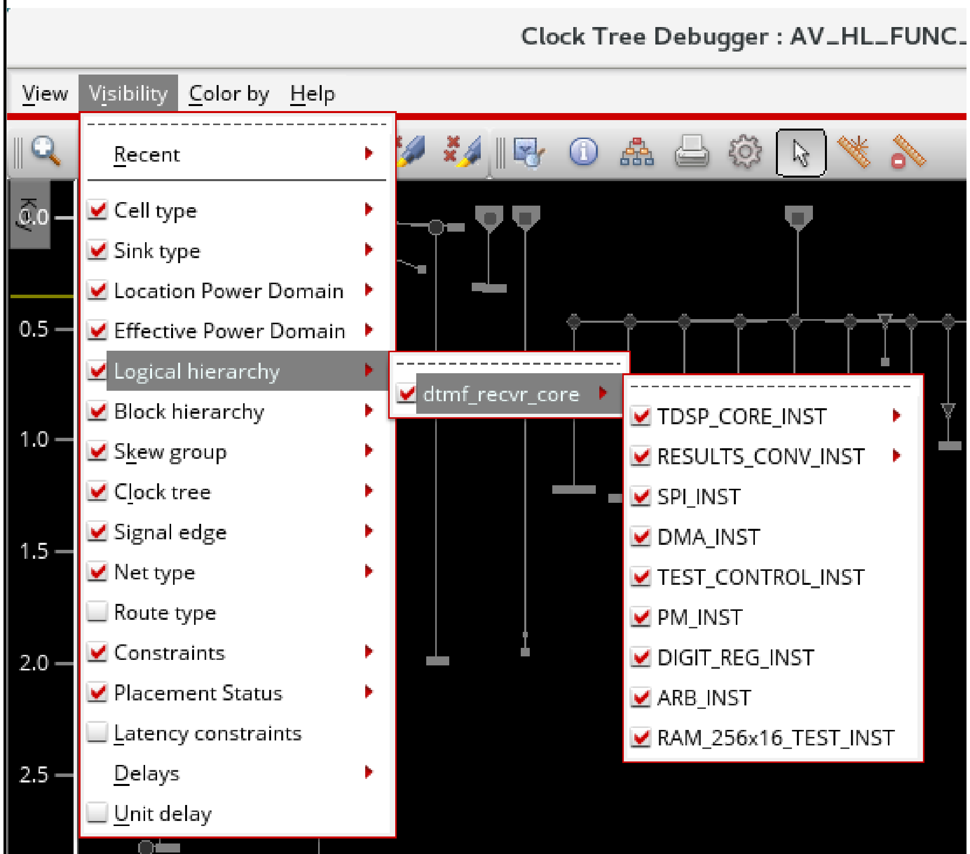
An implicit stop pin is created on a pin where there is no combinational arc across the cell, but where the pin is marked as being a clock pin in the liberty data or it is listed as having a trigger or a check arc. These pins behave the same as explicit or user-defined stop pins and will be delay balanced against the other sinks in the appropriate skew groups. An implicit ignore pin is created at a cell where there is no combinational arc across the cell, and there is no trigger or check arc present. These pins will not be delay balanced, but will have clock tree design rule violations (DRVs) enforced on the path to them.

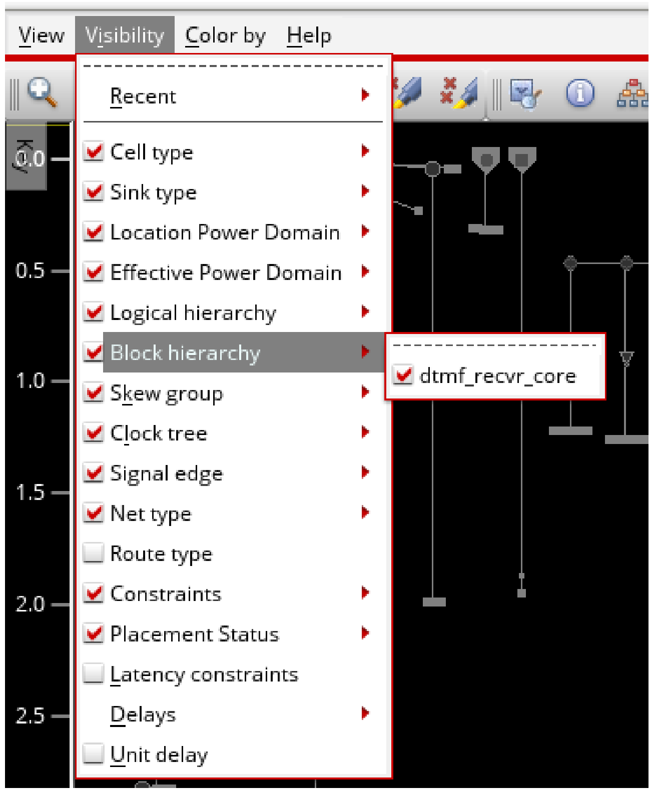
Each option has a default color.
■ Stop pin – red
■ Ignore pin – green
■ Exclude pin – blue
■ Latch pin – yellow
■ Implicit stop pin – pink
■ Implicit ignore pin – green
■ Other pin – gray

When this option is selected, all other objects in the clock tree are displayed as gray.
</td>
</tr>
</table>

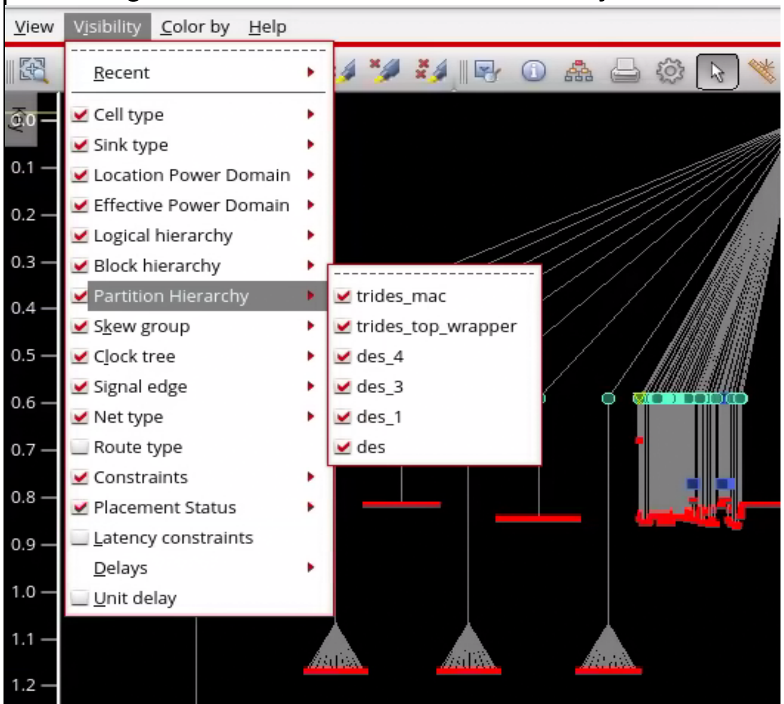
Location Power domain	The coloring or visibility of the marker indicates the location power domain to which the cell corresponding to the marker belongs. 

Effective Power Domain	The coloring or visibility of the marker indicates the effective power domain to which the cell corresponding to the marker belongs. 

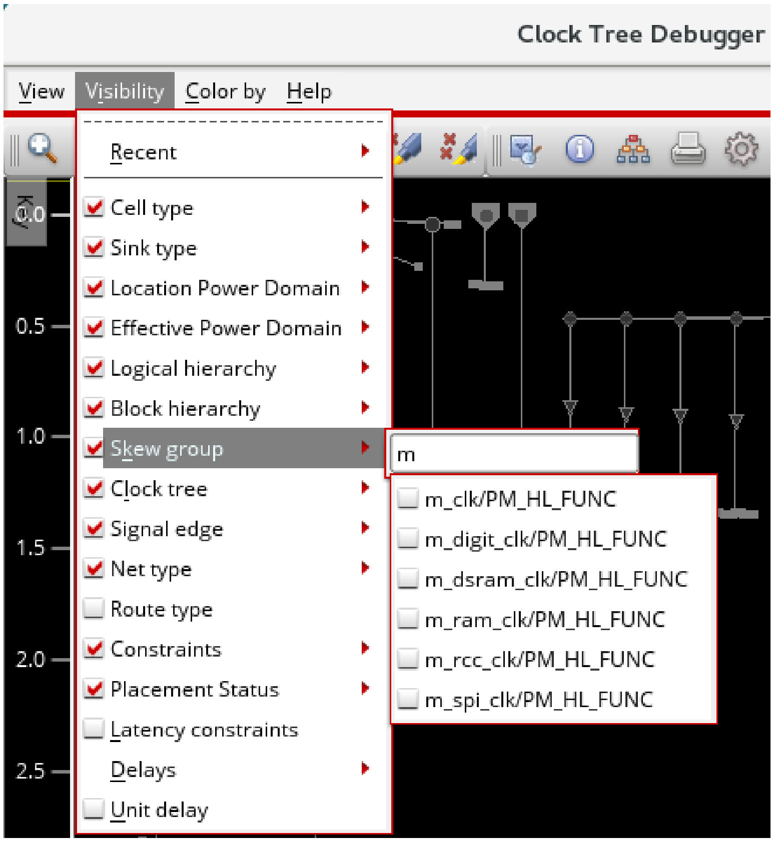
Logical hierarchy	The coloring or visibility of the marker indicates the Verilog module containing the cell corresponding to the marker. 

Block hierarchy	The coloring or visibility of the marker indicates modules that represent ILM and other sub-blocks. 

<table>
<tr><td>Partition Hierarchy</td><td>The coloring or visibility of the marker indicates different partitions in the design.

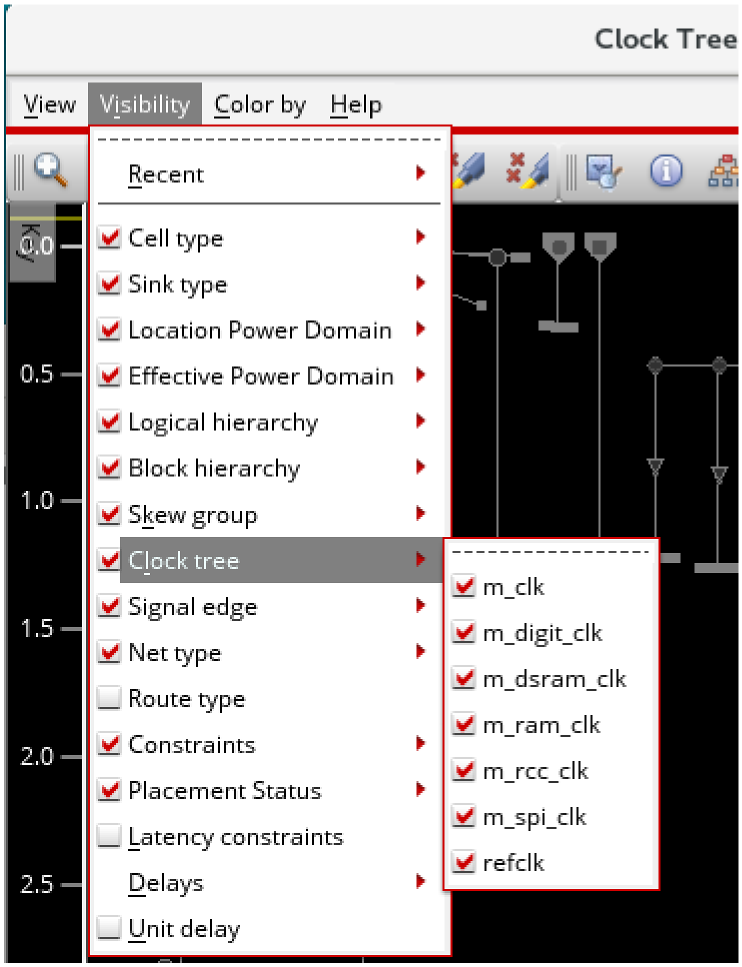
Note: This option is displayed only when partitions are configured in the design. It is available in both unit delay and timed modes.
</td></tr>
</table>

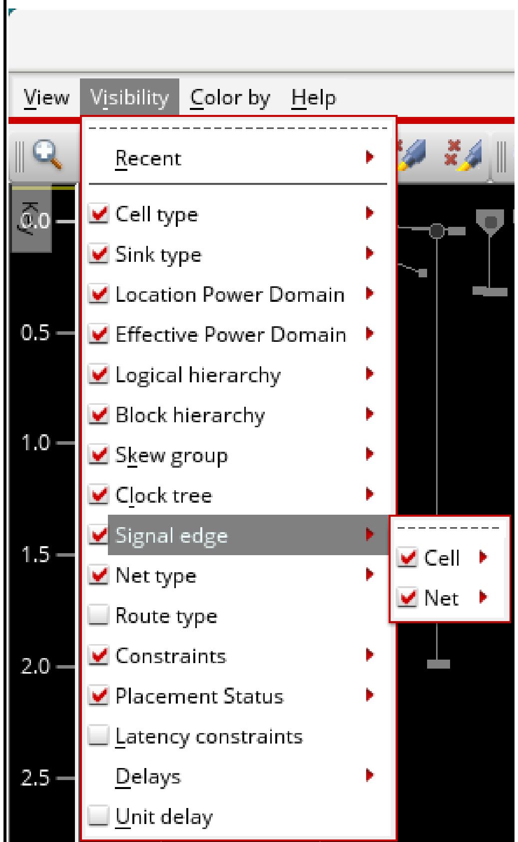
<table>
<tr>
<td>Skew group</td>
<td>
Selectively shows, hides, and highlights the specified skew groups in your clock tree view. The list of skew groups is presented as a filtered hierarchy. You can select the skew group from the drop-down list.



If you do not type any letters in the search field, you will get a list of all skew groups in the drop-down menu. Type "No" in the search field to view the No skew group option. This option lets you view nets and buffers that do not belong to any skew group.

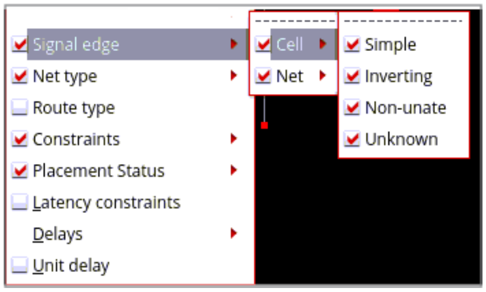
Note: Markers may be in more than one skew group at one time.
</td>
</tr>
</table>

Clock tree	Selectively shows, hides, and highlights individual clock trees - including generated clocks - in your design. You can select the clock tree from the drop-down list.  **Note:** Markers may be in more than one clock tree at one time.

Signal edge	Provides options for turning on the visibility of signal edges for cells and nets.  *Cell subgroup*: Cell subgroup: When selected, turns on the visibility of the instances based on the timing arcs found by the Common Timing Engine (CTE). The following options are available in the Cell subgroup: ■ *Simple:* Simple cells do not change the edge. ■ *Inverting:* Inverting cells have toggled the edge that was input. ■ *Non-unate:* Non-unate cells, such as XOR gates, do not deterministically set the output net edge, so it will be represented as rising. ■ *Unknown:* These include the non-generated root cells. This option is generally used when a timing arc cannot be found or the CTD marker does not represent something with a timing arc (roots, sinks).

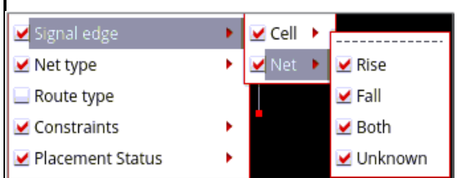
<table>
<tr>
<td></td>
<td>

Note: The generated root cells are included in one of these types;

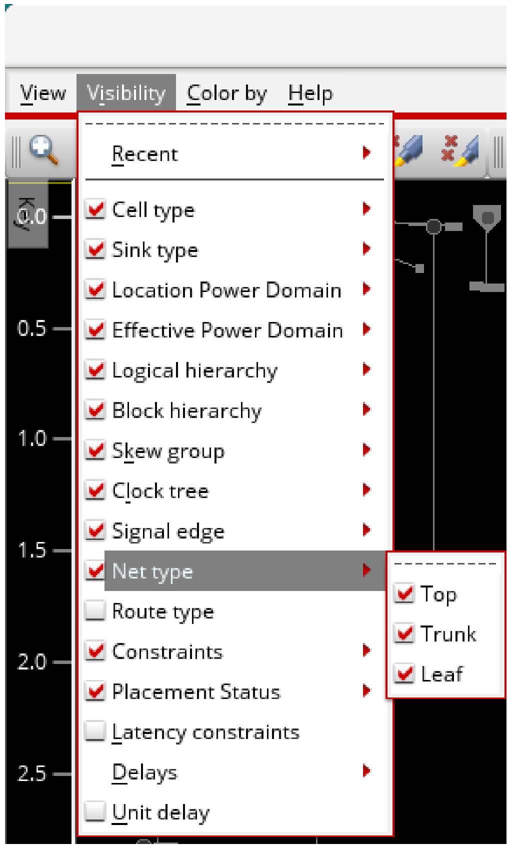


Simple, Inverting, or Non-unate.

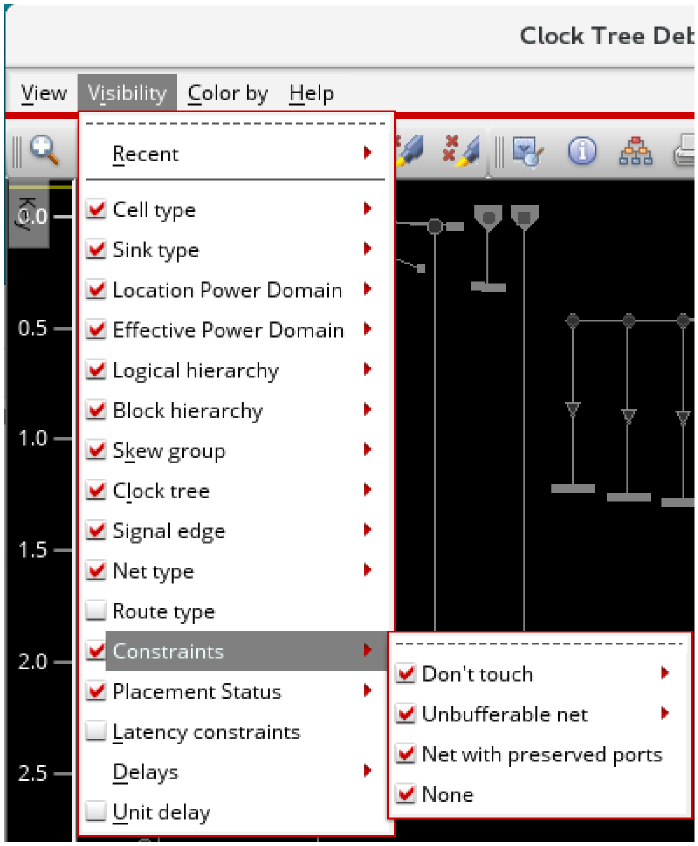
Net subgroup: When selected, turns on the visibility of connectors representing the nets as Rise or Fall edge, Both, or Unknown. Unknown is used when there is no skew group present.



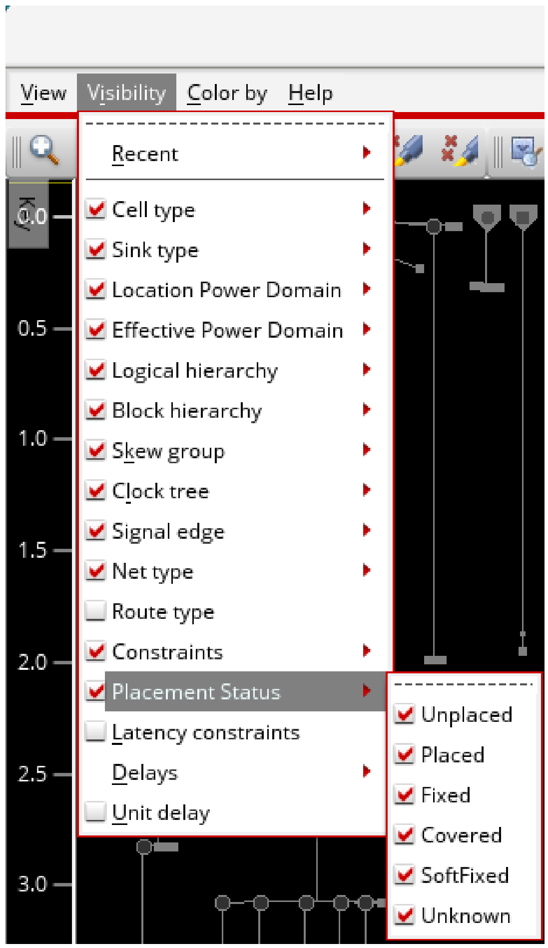
</td>
</tr>
</table>

Net type	When enabled, visualization shows all the net types; Top, Trunk, and Leaf. Use this option to view a specific net type. 
Route type	When enabled, visualization shows all the route types.

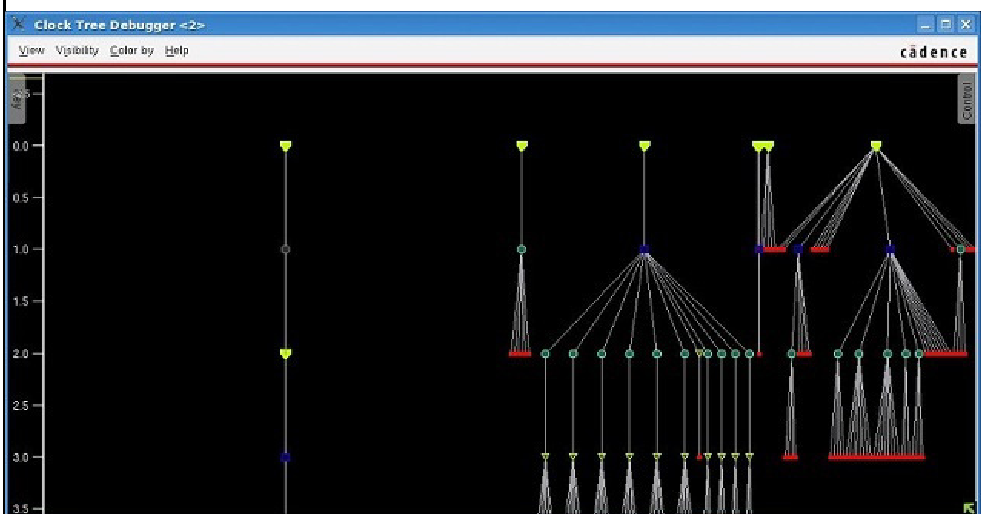
<table>
<tr>
<td>Constraints</td>
<td>
Shows the don't touch, unbufferable clock nets, and nets with preserved ports in the Clock Tree Viewer. By default, all options are selected.



A clock net may be unbufferable or Don't touch for a number of reasons. When you select these options, a list of possible reasons for why the net is unbufferable or don't touch are provided. This is shown below. You can choose to view clock nets based on these. By default, all sub options or reasons are selected.
</td>
</tr>
</table>

Placement Status	When selected, highlights instances that are Unplaced, Placed, Fixed, Covered, SoftFixed, and Unknown. This option is also available in the Control panel. 
Latency Constraints	When enabled, visualization shows the target delays that the CCOpt algorithm is aiming for as a range. For a user, these target delays are useful only after running the `ccopt_design` command. This option is also available in the Control panel. When this option is selected, hovering the pointer anywhere over the timing window will show the timing window information in the tooltip.

Delays	Specifies the delays that you want to view in the clock tree view. Select or deselect the check boxes against each option in the drop-down menu depending on what you want to view. You can choose to view or not view the Gate delay, Wire delay, Virtual delay, Pin insertion delay, or Gate level delay. **Gate delay** is the delay within the gate between the clock pin and the output pin. It is shown as a vertical line below the marker. **Wire delay** is the delay on the wire, between the output pin and the pins it drives. This is shown as diagonal lines from the bottom of the gate delay, or the marker, if the gate delay is not shown, to the markers for each of the fanout. **Virtual delay** is the delay that CCOpt has scheduled but not yet implemented. It will be present if a trial run has been performed. It is shown as a double vertical line. **Pin insertion delay** is the additional user-specified delay. It is shown as a dotted line. **Note:** If a component of the delay is unchecked, it is not used to calculate the y-axis coordinates. So markers below that type of delay will appear higher. Markers are shown at the arrival time for the clock signal at the clock pin.

Unit delay	Unit delay is the delay between the input to the gate and the output. The unit delay lets you view the depth of clock tree objects easily. This is useful when you have large designs with deep gate-level trees. When debugging clock trees, you can use this view to check how the clock trees are balanced by looking at the depth of the clock trees. When this option is selected, the ruler by the side of the Key panel is updated to display the number of units. All nodes of a tree are aligned by its level. The level starts from 0 and increases by one unit for each level. This is shown in the image below. **Note:** When this option is selected, the *Timing windows* and *Delays* options are disabled.  **Note:** You can also open the CTD directly in the unit delay mode by using the `-unit_delay` parameter of the `ctd_win` command. This can be done even for unplaced designs without running CTS. For details, see the `ctd_win` command.

Clock Tree Debugger - Color by Menu

Use the *Color by* menu to control the coloring of the various elements in the Clock Tree Viewer. Colorings are mutually exclusive – selecting one deselects all others. The items for enumerations also have a submenu to allow the color used for individual categories to be customized.

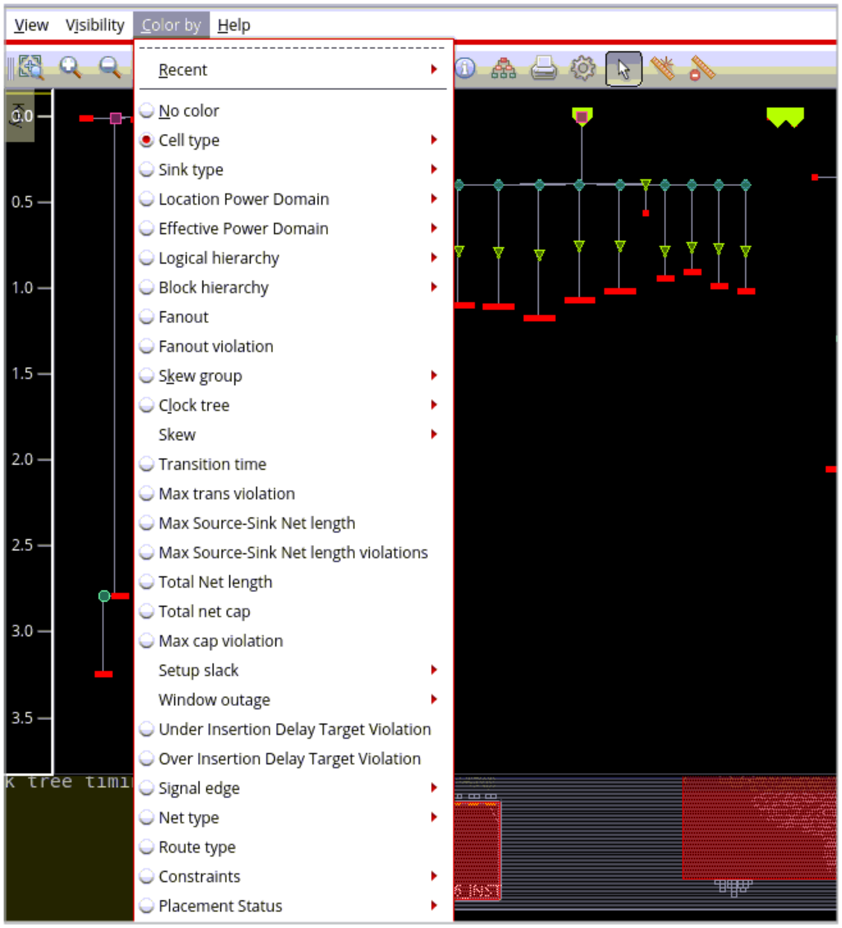
Additionally, the visibility options in the CTD are independent of the coloring options. You can choose to color an item, say Cell type, from the Color by menu or the Control panel, while at the same time hide the skew groups by unchecking the Skew group option in the Visibility menu or by unchecking the option just before the items of interest in the Control panel.

Types of Colorings

Three types of colorings are supported:

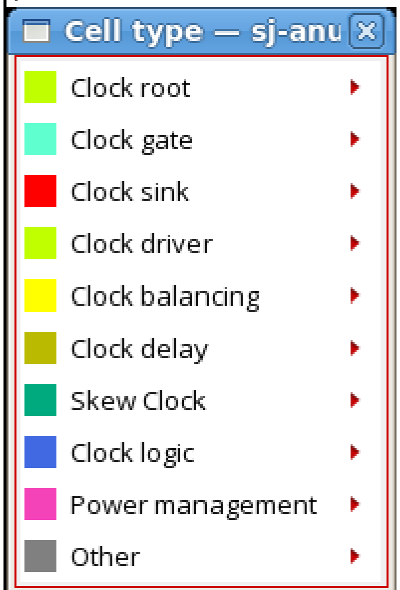
- **Enumerations**, which partition the objects in the visualization into a number of non-overlapping categories. For example, *Cell type* categorizes cells into various types. Each category in an enumeration, for example *Clock gate* within *Cell type*, has a distinct color.
- **Continuous color scales**, which color each applicable element according to the value of some numerical value. For example, *Transition time*.
- **Collections of subsets of objects** in the design, which color objects within the subset in a single color. For example, a skew group is one subset.

Choose *Clock – CCOpt Clock Tree Debugger – Color by.*

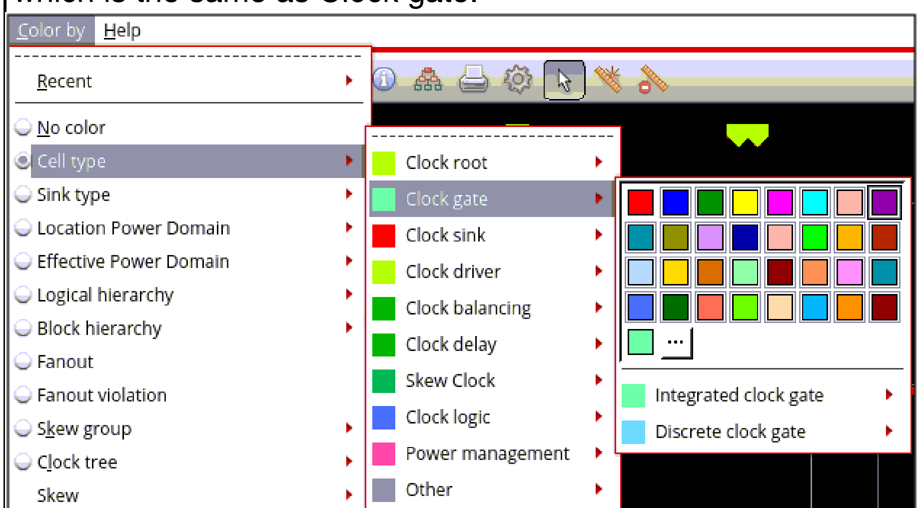


Options	Descriptions
Recent	Shows a list of the five most recently used items from the *Color by* menu. Select the radio button for each option to view the most recently done coloring for that option.
No color	Removes all colors from the display.

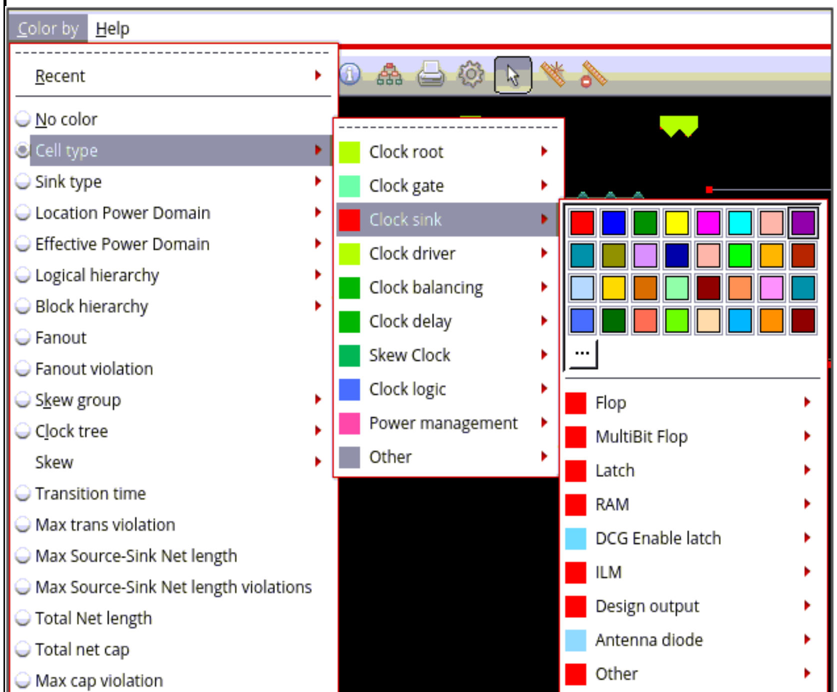
<table>
<tr>
<td>Cell type</td>
<td>
Specifies the colors for different cell types. You can choose colors based on your preference form the color options available in the color picker.

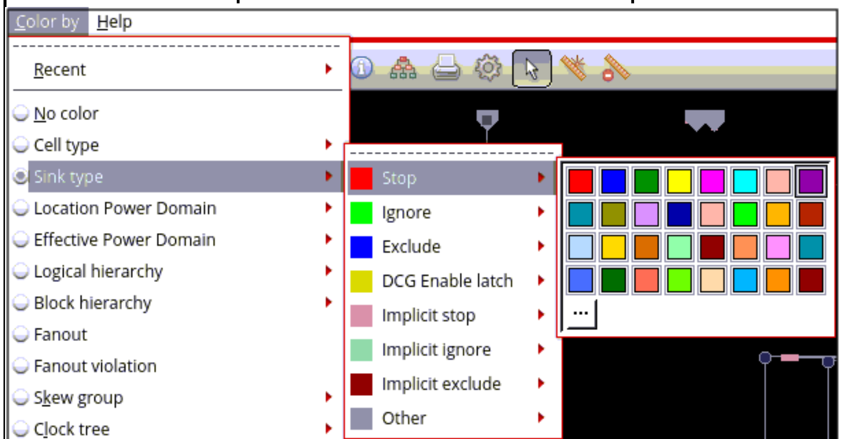
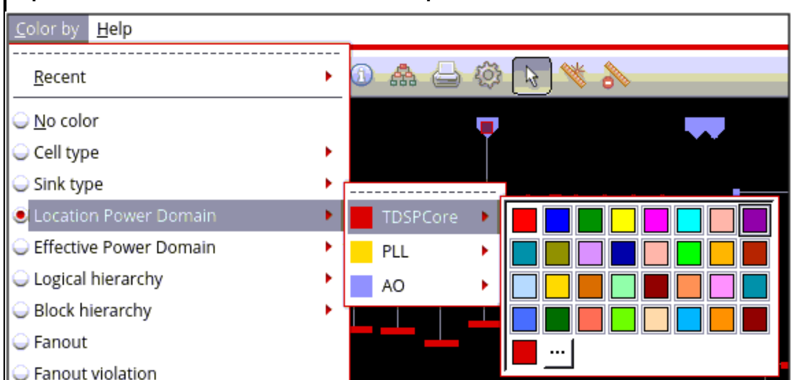


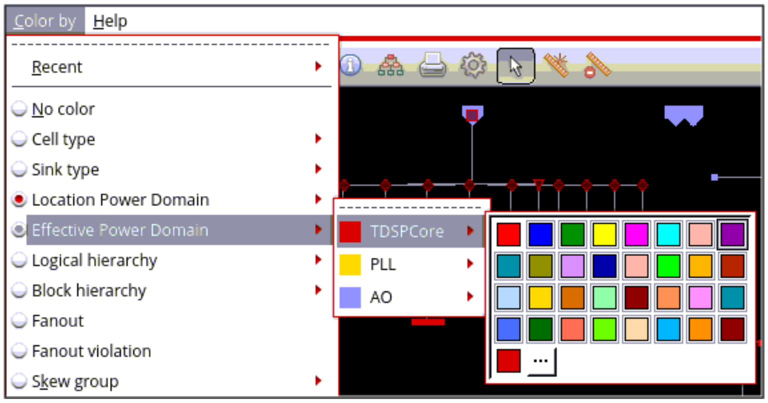
For Clock gate, you can choose to display all the categories and sub-categories in colors other than the default color assigned to them, which is the same as Clock gate.



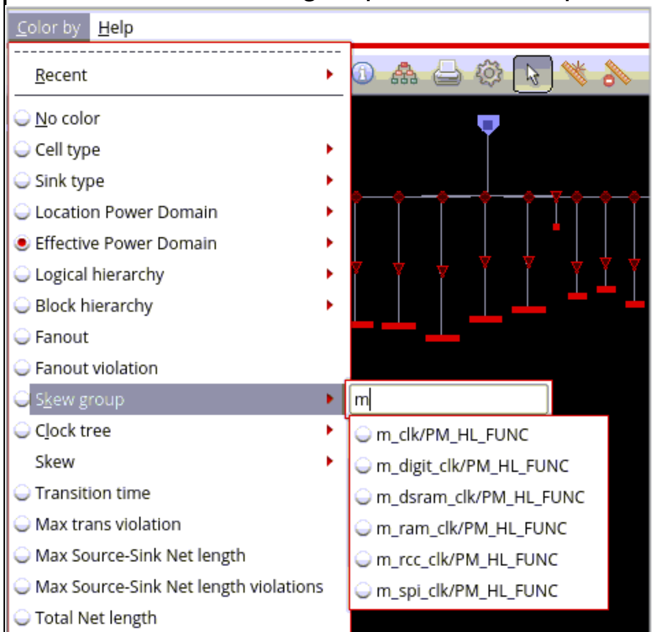
</td>
</tr>
</table>

<table>
<tr><td></td><td>For Clock sink, you can choose to display all the categories and sub-categories listed below in colors other than the default color assigned to them.

</td></tr>
<tr><td></td><td>For Clock driver you can choose to display of buffers and paired inverters, and non-representative drivers in colors other than the default colors assigned to them. The non-representative driver is colored gray by default. The representative driver is colored as a normal driver, which is greenish yellow by default.</td></tr>
</table>

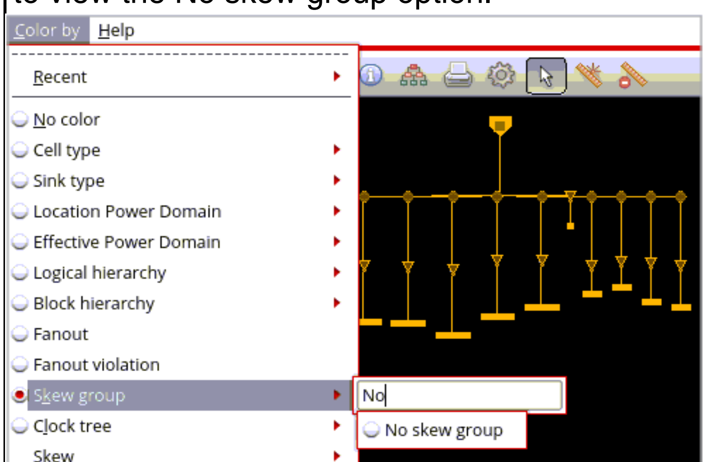
<table>
<tr><td>Sink type</td><td>Specifies the colors for different pin types. If this option is selected, the rest of clock tree objects are displayed as gray, and the CTS pins, such as ignore pins and stop pins are shown in the colors defined in the control panel. You can choose colors based on your preference form the color options available in the color picker.

</td></tr>
<tr><td>Location Power Domain</td><td>Specifies the colors for different location power domains of the clock tree. You can choose colors based on your preference from the color options available in the color picker.

</td></tr>
</table>

Effective Power Domain	Specifies the colors for different effective power domains of the clock tree. You can choose colors based on your preference from the color options available in the color picker. 
Logical hierarchy	Specifies the colors for the different hierarchy levels in the clock tree view.
Block Hierarchy	Specifies the colors for the block hierarchy levels in the clock tree view.
Partition Hierarchy	Specifies the colors for the different partitions in the clock tree view. **Note**: This option only appears in the menu for designs in which partitions are configured. It is available in unit delay and timed modes.
Fanout	Colors each cell based on its fanout, which is the number of cells it drives.
Fanout Violation	When selected, nodes are colored based on how much they exceed the maximum fanout constraint. The tooltip displays both the violation amount and the target value. In *Unit Delay* mode, the fanout shown in Fanout coloring counts preserved ports as one, so the displayed fanout matches the visual display. However, the constraint itself applies only to flat-hierarchy fanout. In Fanout Violation coloring, fanout is always reported using flat-hierarchy counts. In this case, the tooltip explicitly notes (not including preserved ports) for the target.

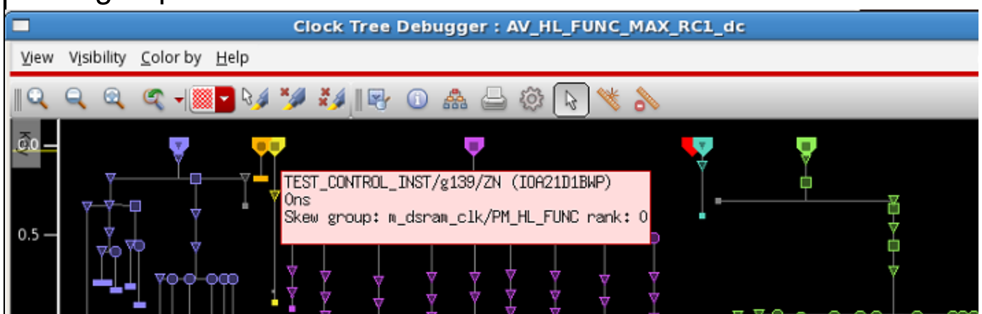
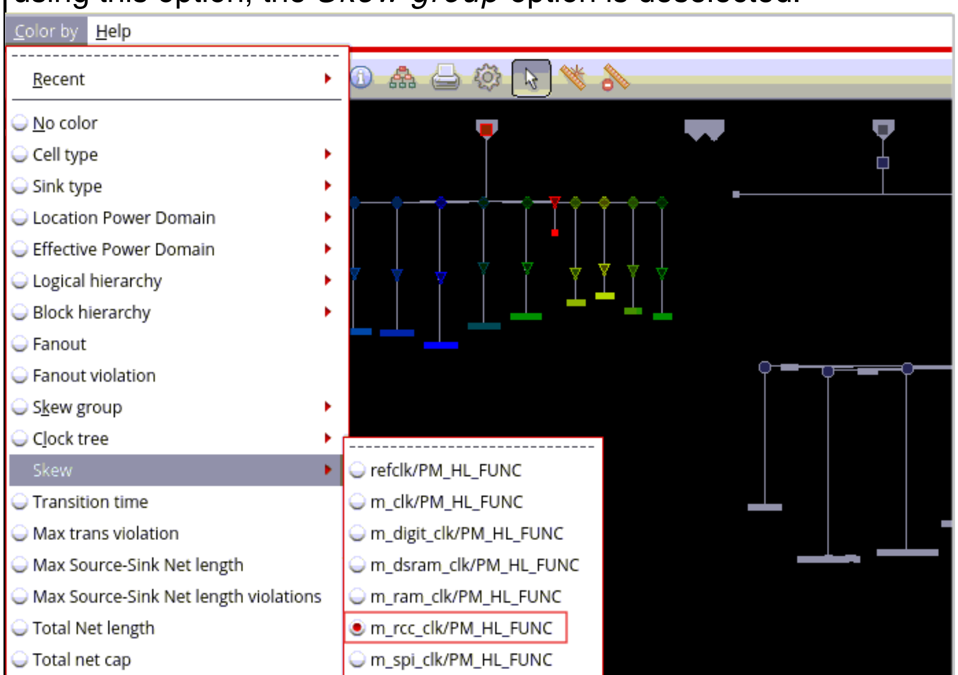
<table>
<tr>
<td>Skew group</td>
<td>
Specifies the colors for the different skew groups in the clock tree view. The list of skew groups is presented as a filtered hierarchy. You can select the skew group from the drop-down list.



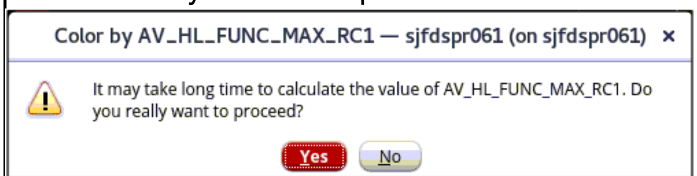
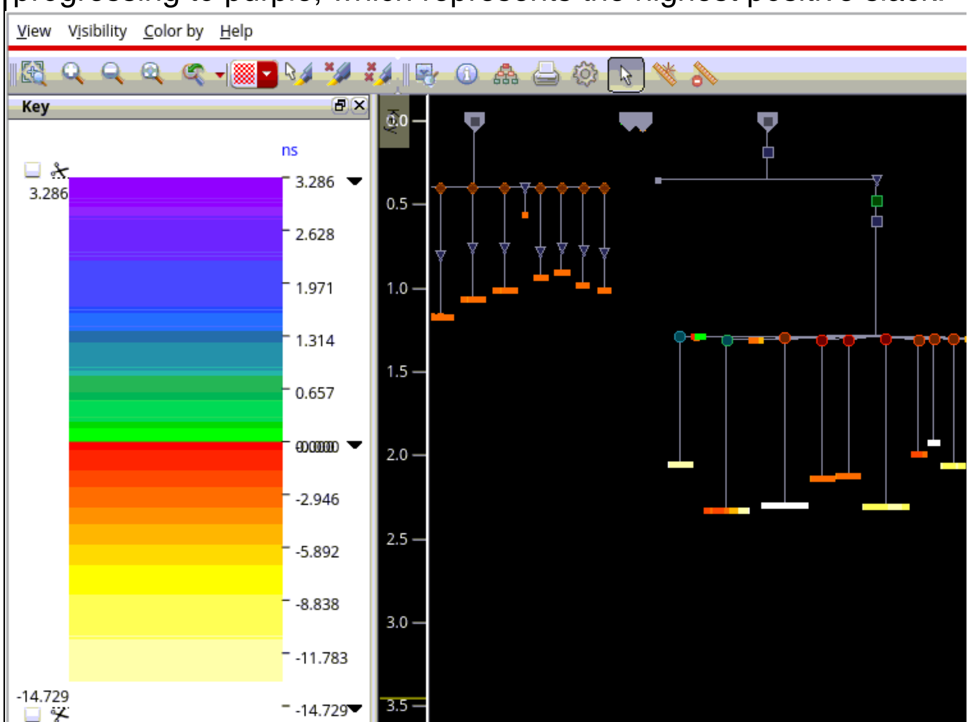
If you do not type any letters in the search field, you will get a list of all skew groups in the drop-down menu. Type "No" in the search field to view the No skew group option.

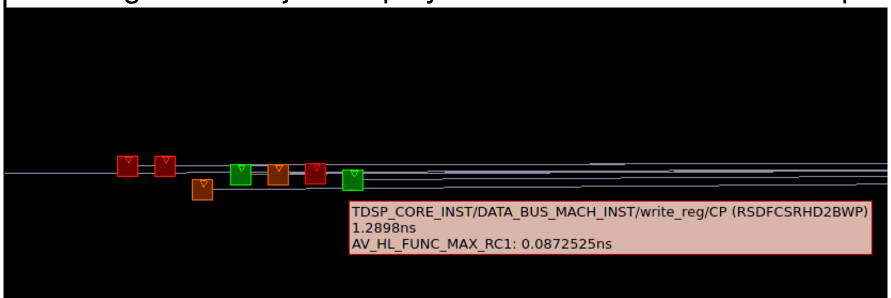
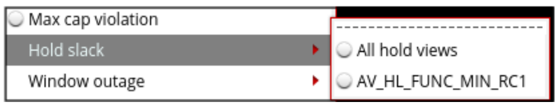
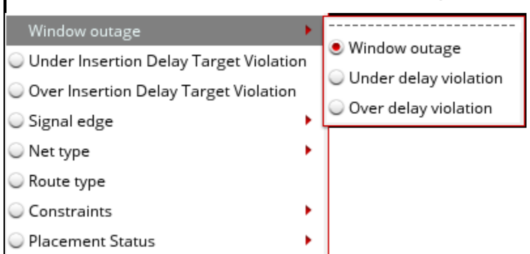


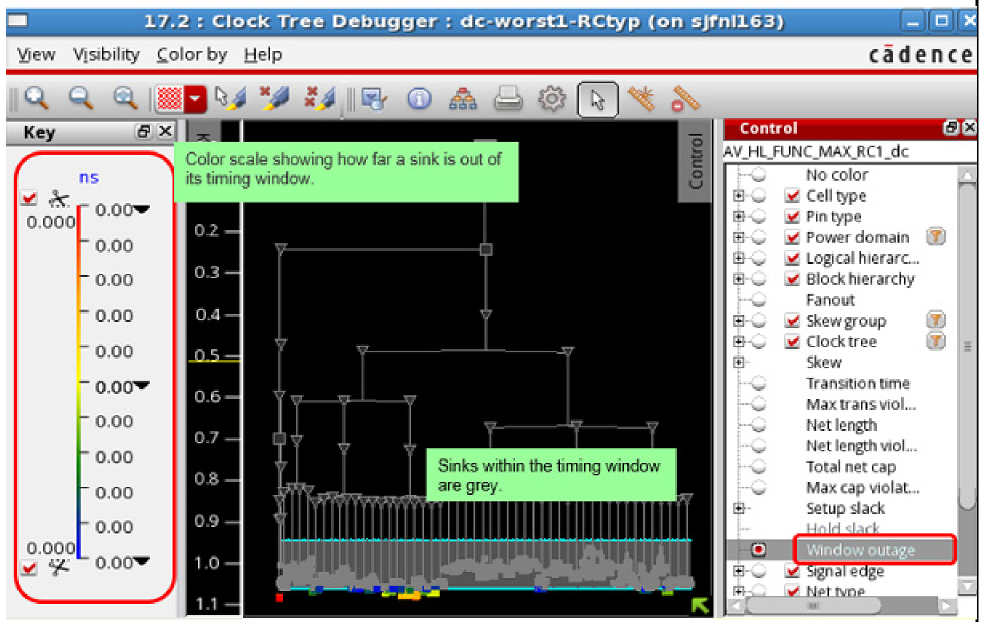
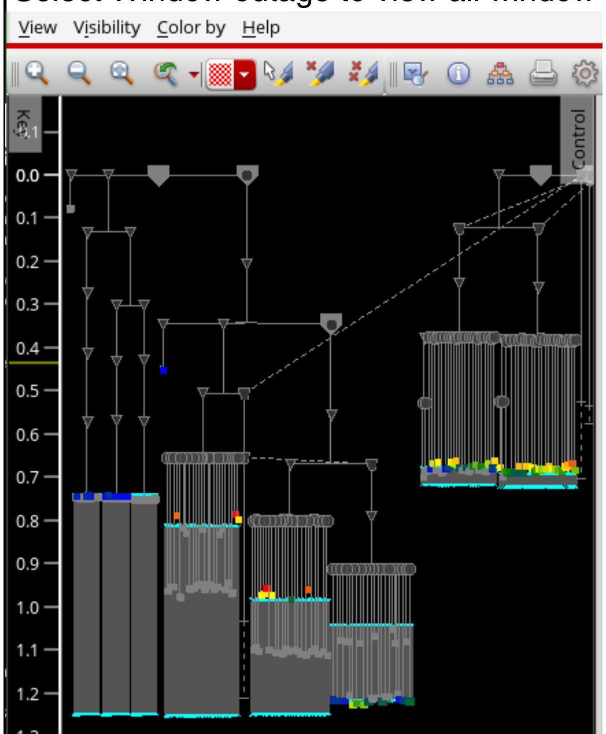
</td>
</tr>
</table>

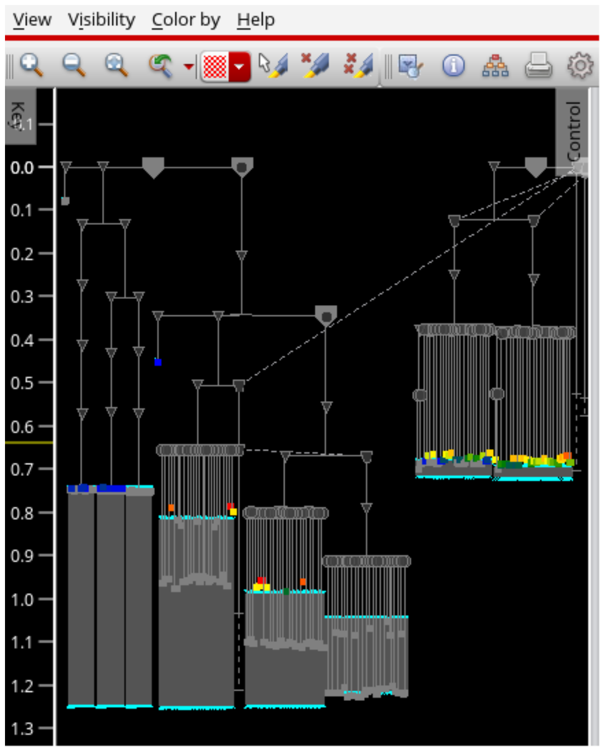
	When you select this option, hovering the pointer above a clock node/sink will show a pop-up or a tooltip that displays the skew group name and rank. Non-constraining - reporting only - skew groups are not shown in the tooltips. The rank is only shown for constraining skew groups. 
Clock tree	Specifies the color for the clock tree. You can *Clear all* to clear the color selections for all clock trees. You can select the clock tree from the dropdown list.
Skew	Shows the skew group specified in the drop-down menu. The list of skew groups is presented as a filtered hierarchy. Select the radio button for each skew group to view it in the clock tree view. When you select the radio button to view the colors for the specific skew group using this option, the *Skew group* option is deselected. 
Transition time	Shows the transition time in the clock tree view using color.

Max trans violation	Colors each node in the clock tree based on the maximum transition violation. **Note:** The options - *Transition time*, *Total net cap*, and *Max trans violation* - are mutually exclusive.
Max Source-Sink Net length	Shows the maximum source-to-sink net length in the clock tree view using color.
Max Source-Sink Net length violations	Shows the maximum source-to-sink net length violations using color.
Total Net Length	Shows the total net length in the clock tree view using color.
Total net cap	Shows the total net cap in the clock tree view using color.
Max cap violation	Colors each node in the clock tree based on the maximum capacitance violation.

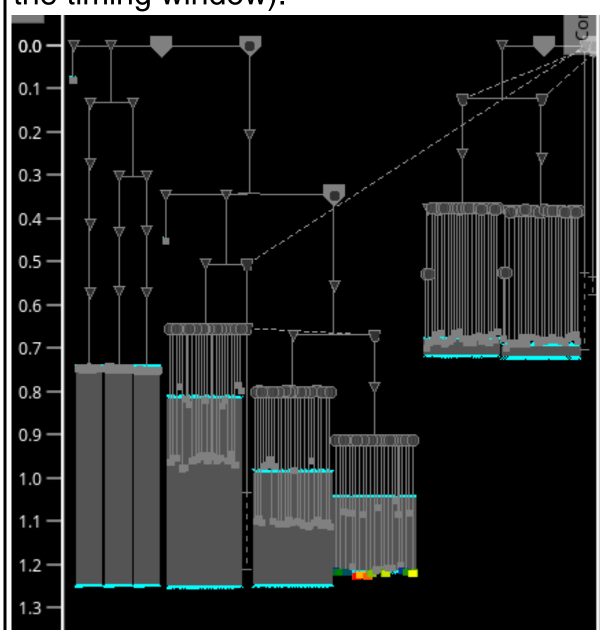
Setup slack	Colors nodes by setup slack, as measured by the Common Timing Engine or the CTE.  You can choose to color by setup slack for all setup views or specific setup views. When either option is selected, a pop-up window opens to confirm if you want to proceed.  The color map represents the slack range from negative to positive values. White indicates the most negative slack, transitioning to red at zero slack. Positive slack is shown starting in green and progressing to purple, which represents the highest positive slack. 

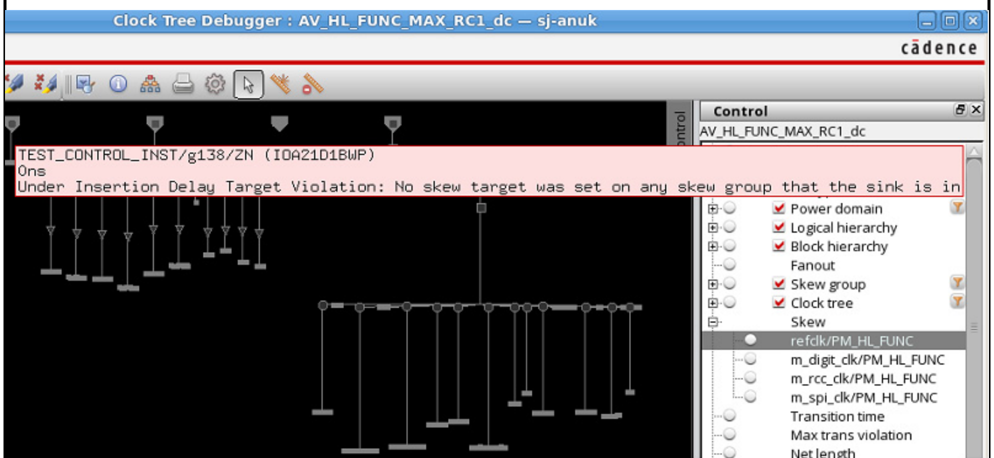
	Hovering on the object displays the slack value in the tooltip. 
Hold slack	Colors nodes by hold slack, as measured by the CTE.  **Note**: This option is available when hold views are configured. By default, the Setup slack option is displayed.
Window outage	This is a color scale that shows how far a sink is out of its timing window. A sink within its timing window is gray.  Select Window outage to view all window outages (early and late).

	 Select Window outage to view all window outages (early and late).  Select Under delay violation to view all early window outages (below the timing window).



Select Over delay violation to view all late window outages (above the timing window).



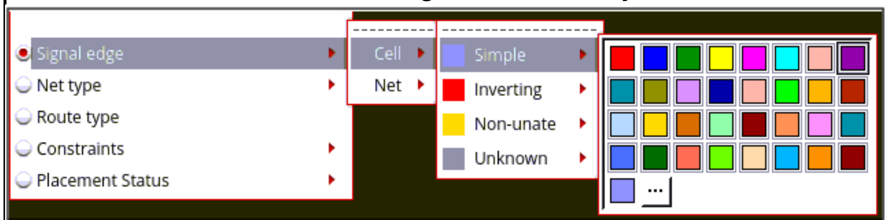
<table>
<tr>
<td>Under Insertion Delay Target Violation</td>
<td>When selected, colors sinks by how far under the insertion delay target they are. To see other skew groups, you can deselect the skew group visibility. The new coloring will only consider currently visible skew groups. Hovering over a sink shows the amount under the target, and the applicable skew group. If no insertion delay target is set, the following is displayed.


</td>
</tr>
<tr>
<td>Over Insertion Delay Target Violation</td>
<td>When selected, sinks are colored according to how far their insertion delay is above the target value. Coloring is applied only to the currently visible skew groups. Hovering over a sink shows the amount above the target and the associated skew group.</td>
</tr>
</table>

Signal edge

Provides options for signal edge colorings. The colorings are based on timing arcs found by Common Timing Engine (CTE).

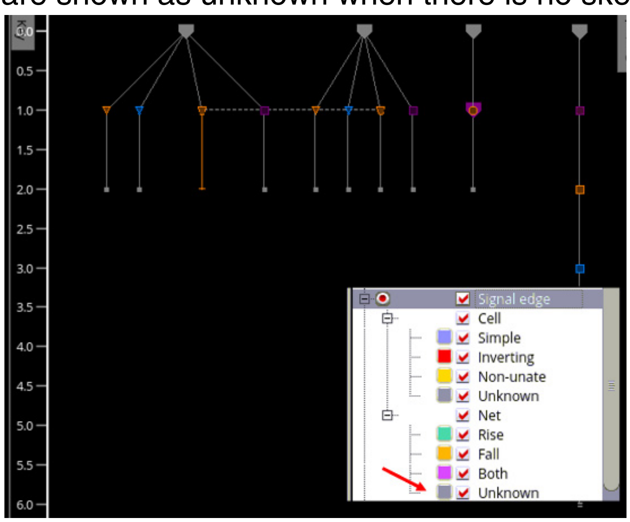
Cell: The Cell subgroup provides options to specify colors of instances based on the timing arcs found by CTE.



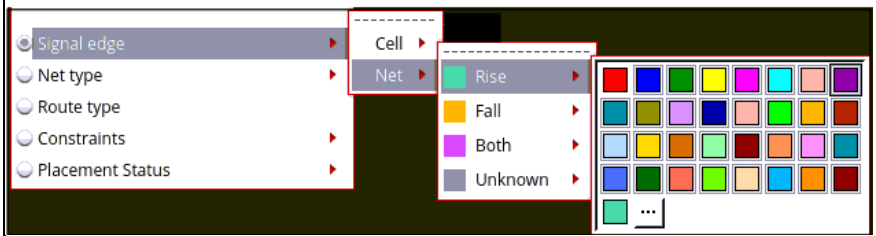
The following options are available for coloring:

- *Simple*: Specifies the color for simple cells.
- *Inverting*: Specifies the color for inverting cells, which are cells that have toggled the edge that was input.
- *Non-unate*: Specifies the color for non-unate cells, such as XOR gates.
- *Unknown*: Specifies the color for non-generated root cells. This option is used for cells when a timing arc cannot be found or the CTD marker does not represent something with a timing arc (roots, sinks).

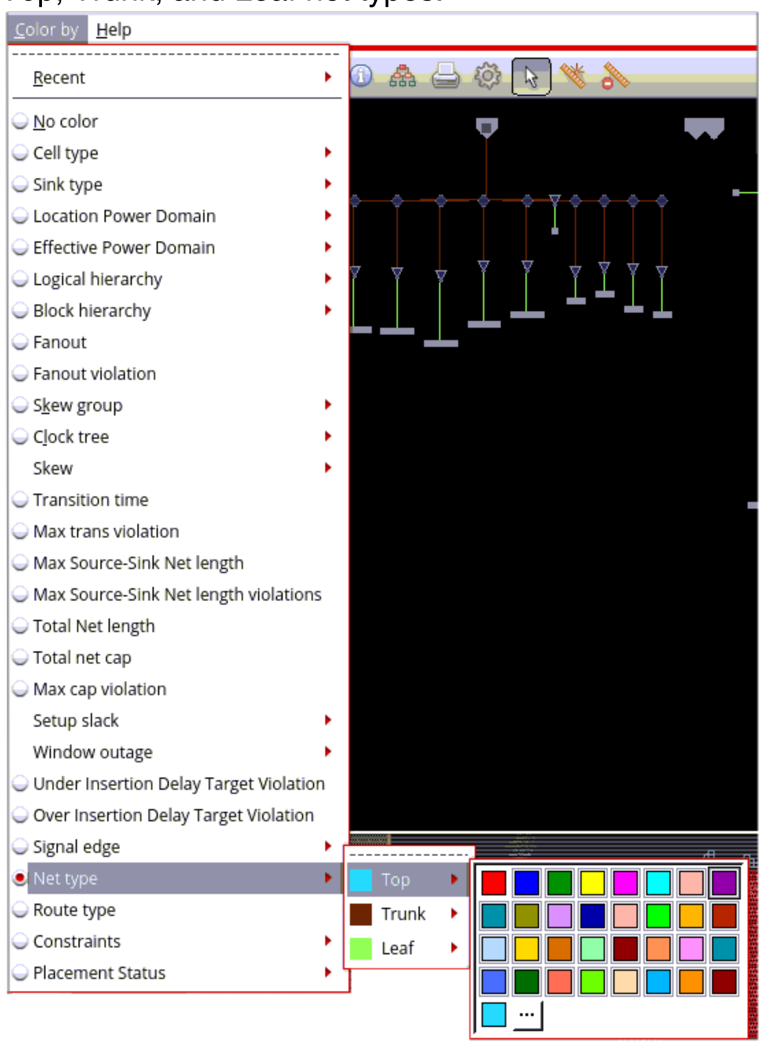
Net: The Net subgroup provides options to color the connectors representing the nets as Rise or Fall edge, Both, or Unknown. Nets are shown as unknown when there is no skew group present.



<table>
<tr>
<td></td>
<td>You can change the color from the color picker provided against each option.



Note: If the visibility of any of the above options is turned off, the specified coloring will only be visible once the option visibility is enabled from the Visibility menu.</td>
</tr>
<tr>
<td>*Net Type*</td>
<td>Specifies the color for net types. Use this option to specify colors for Top, Trunk, and Leaf net types.


</td>
</tr>
</table>

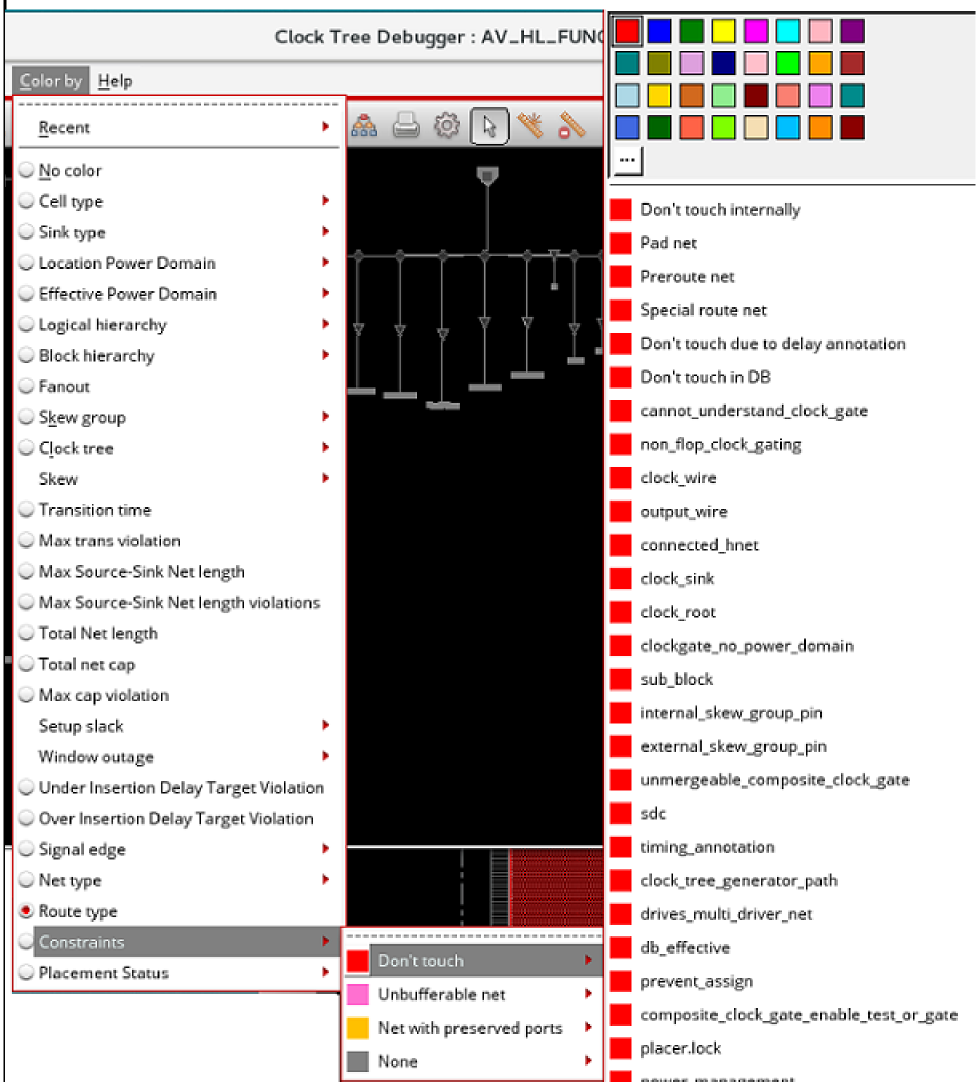
Route type	Specifies the color for route types.

Constraints

Specifies the color of nets with different types of constraints.

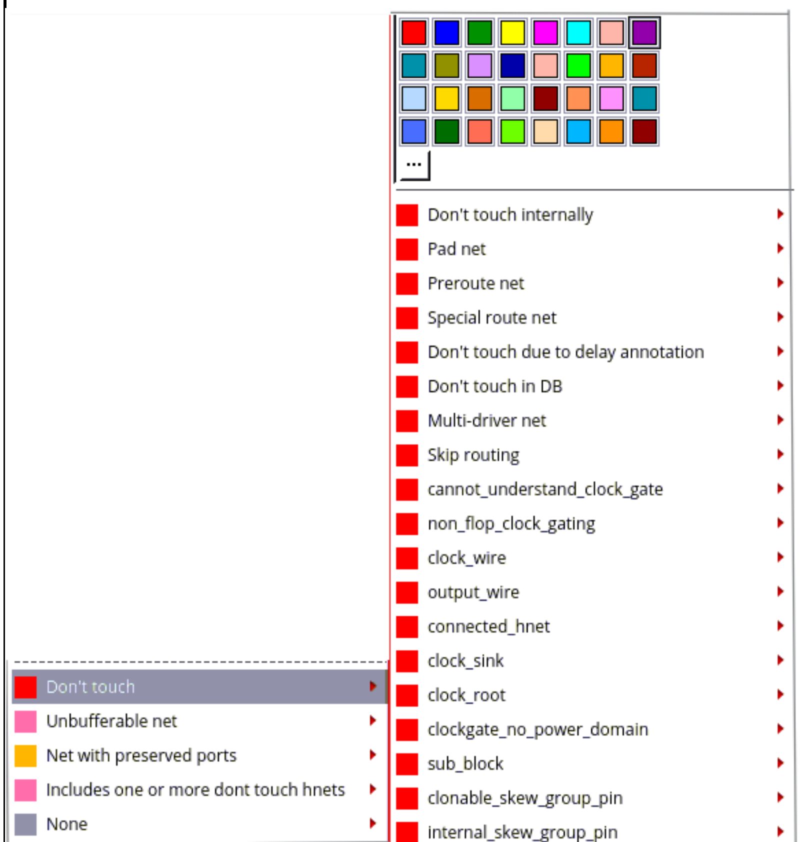
The following options are available:

- Don't touch
- Unbufferable net
- Net with preserved ports
- Includes one or more dont touch hnets: nets that include one or more hnets
- None: nets with no constraints

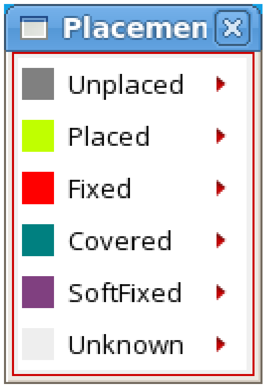


By default, the don't touch nets are colored red while the unbufferable nets are colored pink and the nets with preserved ports are colored yellow. You can change the highlight color from the color options available in the color picker as per your preference.

For the don't touch and unbufferable clock nets, you can choose to highlight nets based on the reasons why they are don't touch or unbufferable. The list of reasons is provided in the drop-down menu of these options.



You can change of the color of the reasons from the color picker provided against each reason. By default, a clock net may be don't touch or unbufferable for a number of reasons, but it is colored with the first applicable reason. However, if the visibility of that reason is disabled, then it will be colored with the next applicable reason.

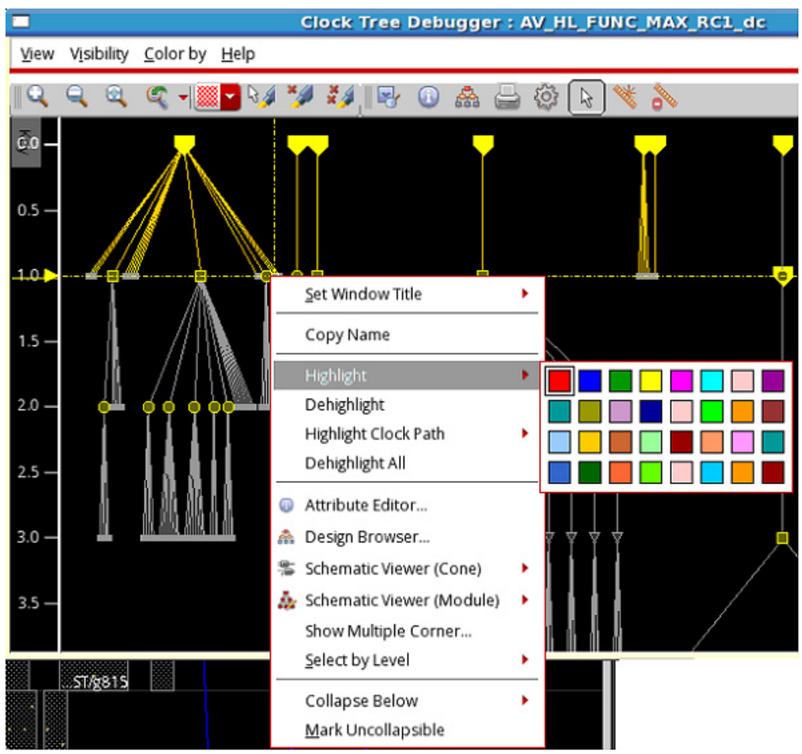
<table>
<tr>
<td>Placement Status</td>
<td>When selected, uses different colorings to highlight instances that are Unplaced, Placed, Fixed, Covered, SoftFixed, and Unknown.


</td>
</tr>
</table>

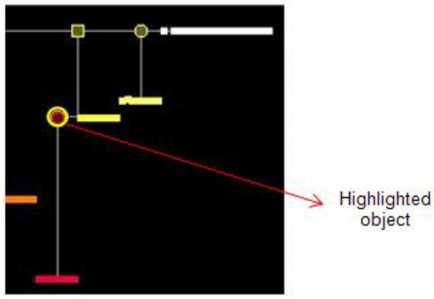
Context Menu of Clock Tree Debugger

The context menu of the Clock Tree Debugger provides the following functionalities, which you can access when you right-click any marker or line in the clock tree viewer.

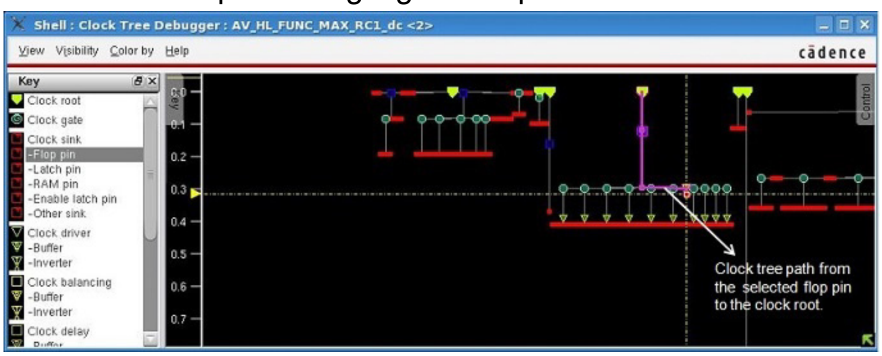
Options	Descriptions
Copy Name	Copies the name of the selected object to the clipboard.

<table>
<tr>
<td>Highlight</td>
<td>Highlights the selected markers or lines. The highlighting functionality is different from the coloring options described above. When showing highlights in the skew clock tree, the coloring is shown as a halo, rendered behind the highlighted objects but in front of non-highlighted objects. The halo is created by rendering the highlighted objects with a thick stroke, for example five pixels, to give a two-pixel border, and then rendering them again using a normal stroke. This allows highlighting to be viewed concurrently with other colorings. If multiple highlights cover the same objects, the colors in question will be blended for the objects where the highlights overlap. You can change the highlight color from the color options available in the color picker as per your preference.


The image below shows the highlighted object. A yellow border is seen around the highlighted object.</td>
</tr>
</table>

	
Dehighlight	Dehighlights the selected highlighted object.

<table>
<tr>
<td>Highlight Clock Path</td>
<td>Highlights the complete clock tree path from the selected clock tree object, which may be a flip flop, a clock gate, a clock buffer or an inverter, to the clock root in the Clock Tree Viewer. In addition, if cross-probing is enabled, then the physical layout window of Tempus will also either highlight or show flight lines for that full clock tree path. If there are wires, the layout will highlight the path, otherwise, it will show only flight lines.

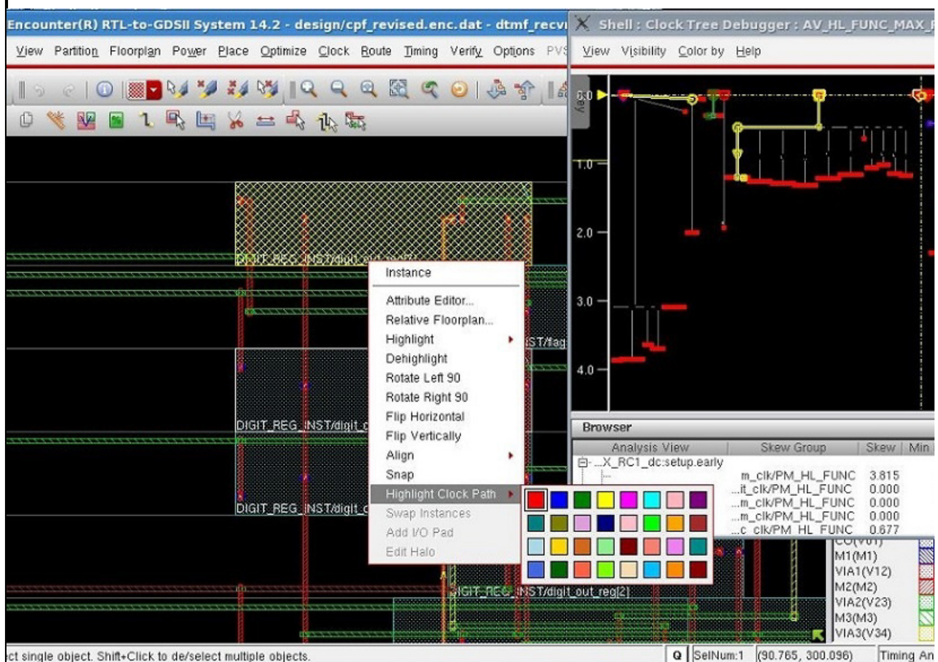
You can change the highlight color from the color options available in the color picker as per your preference. The image below shows the clock tree path from the selected flop pin to the clock root. The path is highlighted in pink.


This option only shows up in the context menu when you right-click any clock tree object. It does not show up when you right-click either a clock net or a blank area of the Clock Tree Viewer. For details, see the "Highlight Clock Tree Path" section.

You can also highlight the clock tree paths in the following ways:

Use the <code>ctd_trace</code> command to highlight clock tree paths in the specified color.

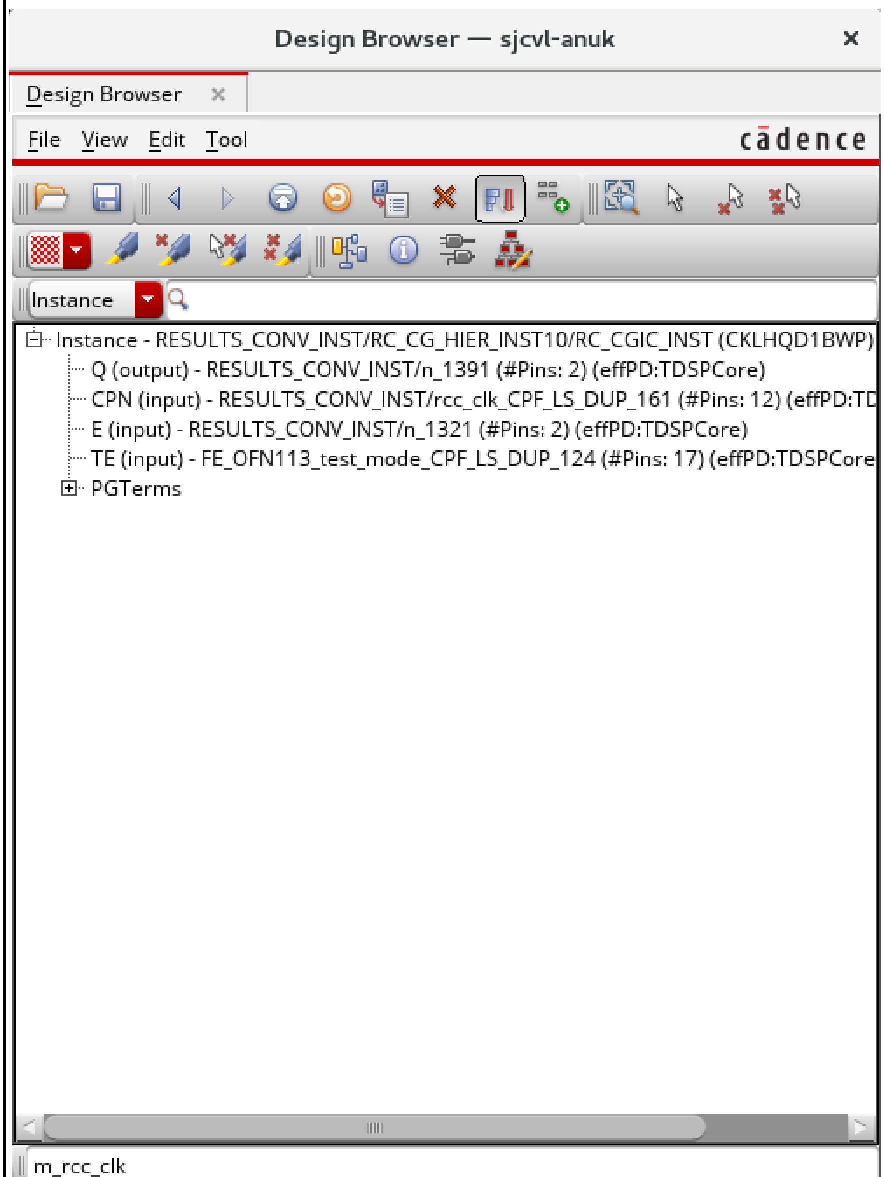
Use the Highlight Clock Path option from the context menu in the Tempus layout window. You can select an instance in the Tempus layout window. Right-click to view the context menu. In this menu, select Highlight Clock Path and select the color in which you want to highlight the clock path from the color picker. This will highlight the clock path from the clock root to that instance if it is in the clock tree. This is shown below.</td>
</tr>
</table>

<table>
<tr><td></td><td>
</td></tr>
<tr><td>Dehighlight All</td><td>Removes the highlights from all markers or lines.</td></tr>
<tr><td>Attribute Editor</td><td>Shows the properties of the selected object in the Attribute Editor.

</td></tr>
</table>

<table>
<tr>
<td>Design Browser</td>
<td>
Use the Design Browser to highlight specific modules, instances, or nets in the design display area. Select an object in the CTD and click on Design Browser in the context menu. The Design Browser window shown below opens.

Design Browser Window



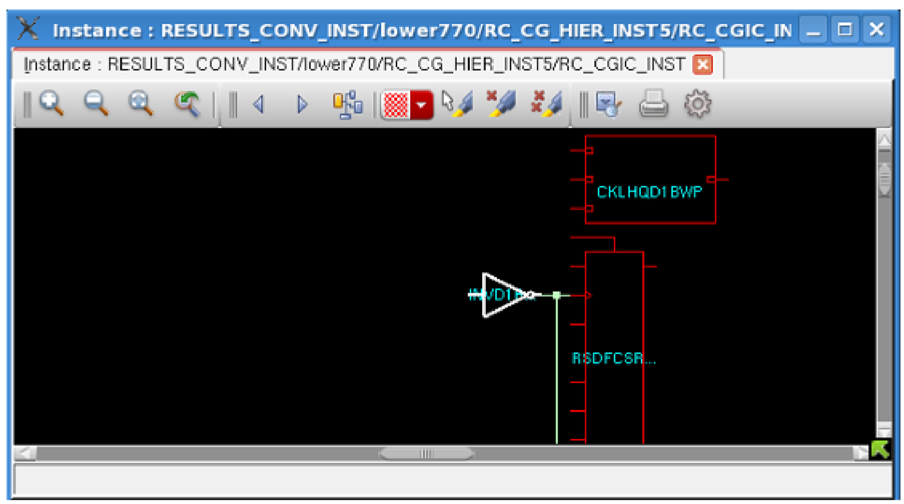
You can use the widgets in the Design Browser to open forms, to navigate through displays, and perform actions. For details of the Design Browser and the available tool widgets, see the "Design Browser " section in the Tools Menu chapter.
</td>
</tr>
</table>

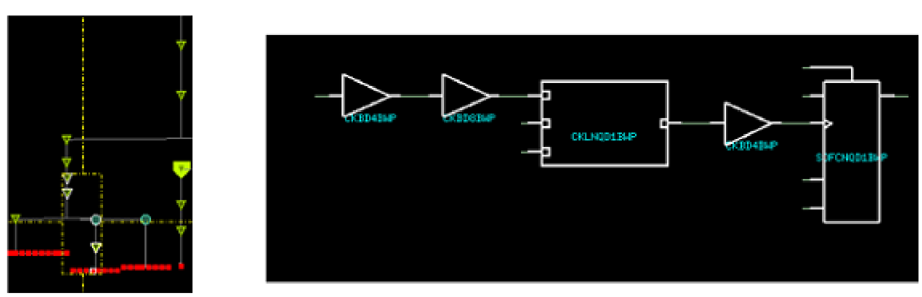
<table>
<tr>
<td>Schematic Viewer (Cone)</td>
<td>
Shows instance-based schematic of the selected objects in the Schematic Viewer. You can choose from the following submenus:

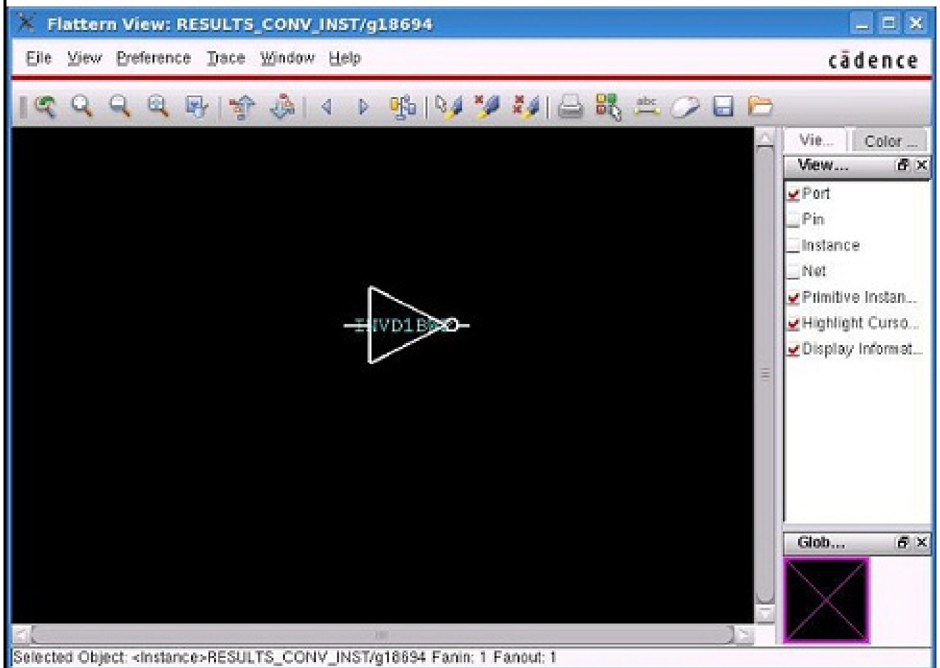
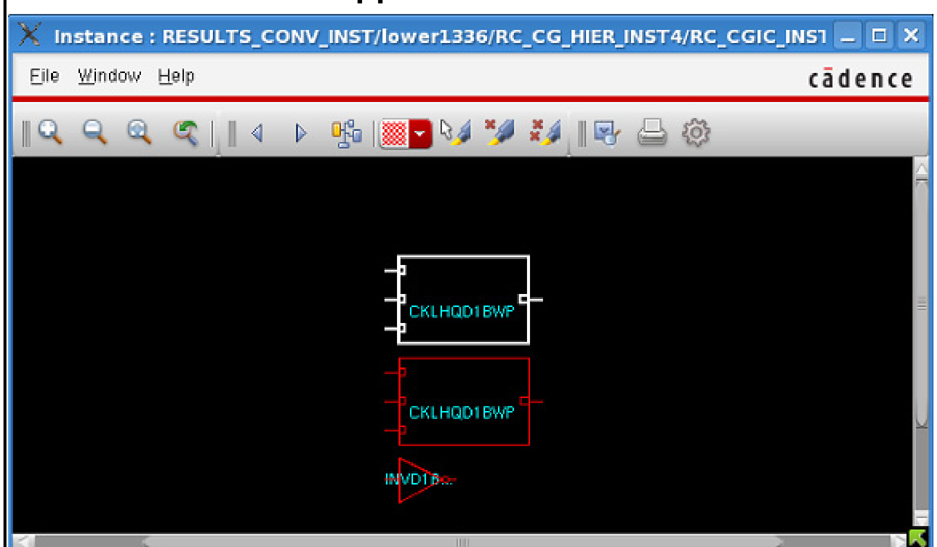
In Main....: opens the schematic view of the objects within the main window of the Schematic Viewer. For this, select multiple objects in the CTD GUI and then select Schematic Viewer (Cone) > In Main.

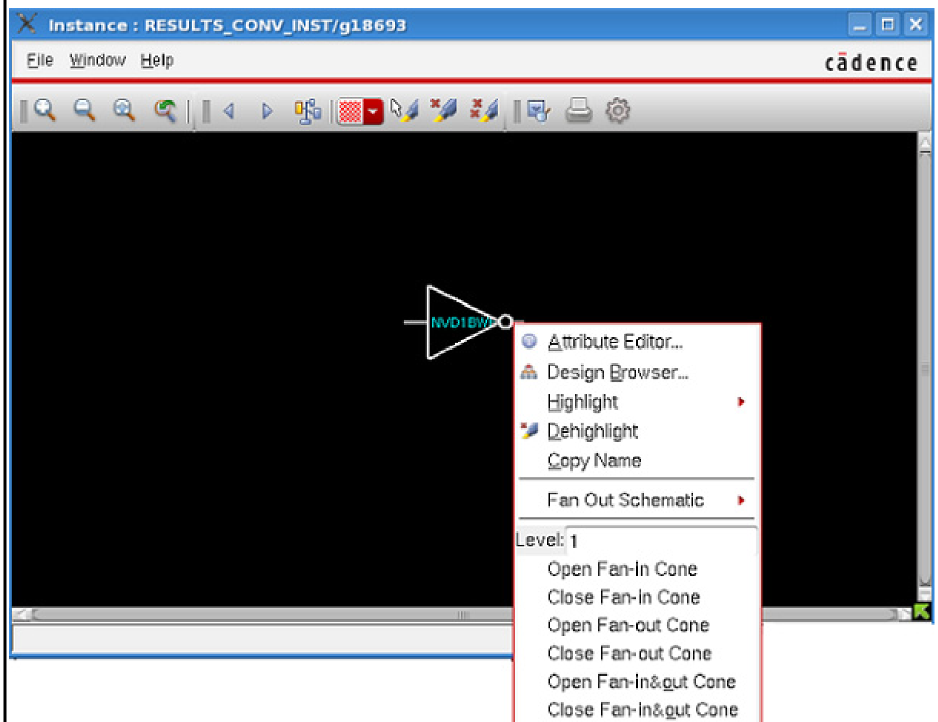
In New...opens the schematic view of the selected objects in a new Schematic Viewer window even if a window is already open.

Append: appends the schematic view of the selected objects in an existing open Schematic Viewer window of another object.

Schematic Viewer - In Main


Viewing Multiple Objects in the Same Schematic Viewer - In Main

</td>
</tr>
</table>

	Schematic Viewer - In New 
	Schematic Viewer - Append 

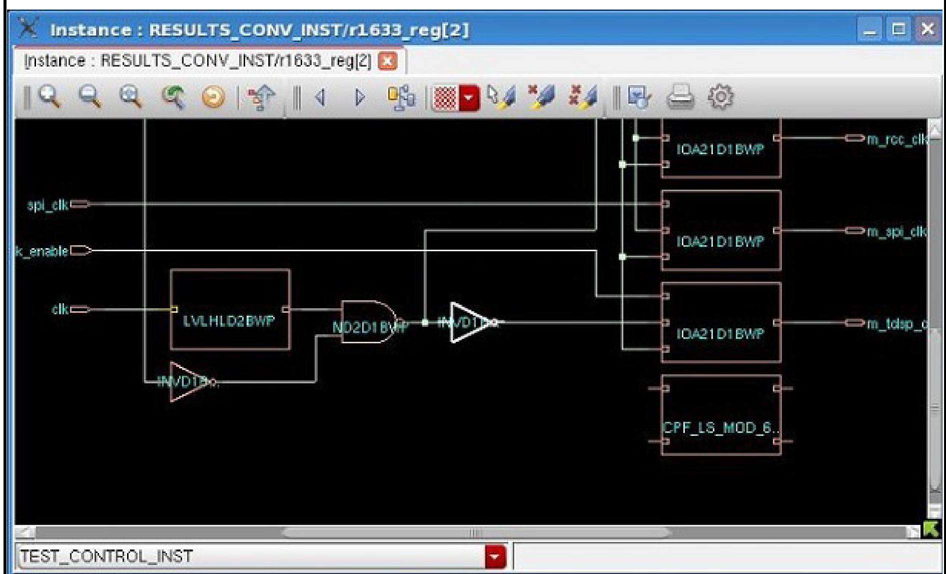
	For any instance open in the Schematic Viewer window, you can customize the view using the options available in the context menu. 

<table>
<tr>
<td>Schematic Viewer (Module)</td>
<td>
Shows hierarchical-based schematic of the selected object in the Schematic Viewer. You can choose from the following submenus:

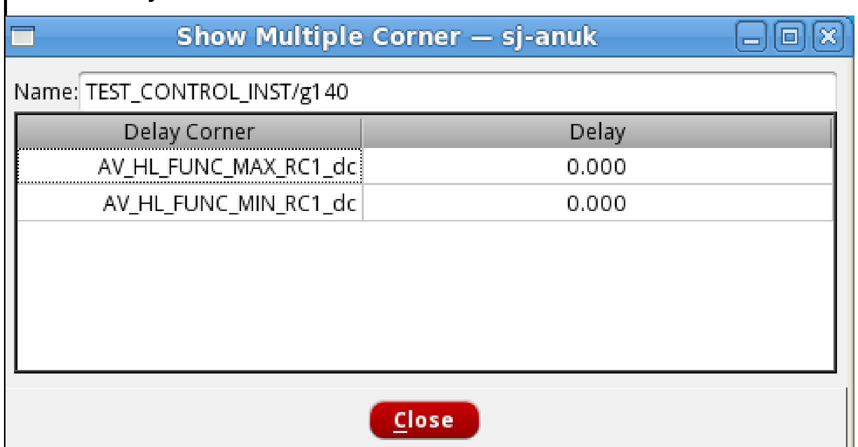
In Main....: opens the schematic view of the object within the main window of the Schematic Viewer.

In New...opens the schematic view of the selected object in a new Schematic Viewer window even if a window is already open.

Schematic Viewer (Module) - In Main



</td>
</tr>
<tr>
<td>Show Multiple Corner...</td>
<td>
Shows the delay values for multiple corners for the selected object in the Show Multiple Corner form. The form lists the name of the object and the delay values for all the corners. This lets you view delays for different corners at the same time.

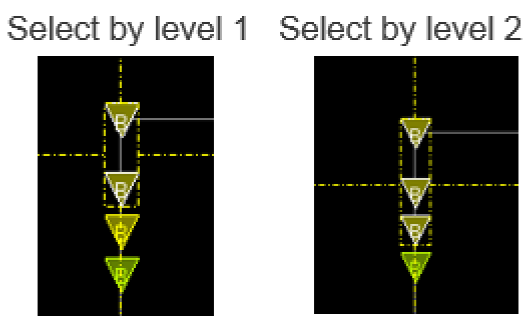


</td>
</tr>
</table>

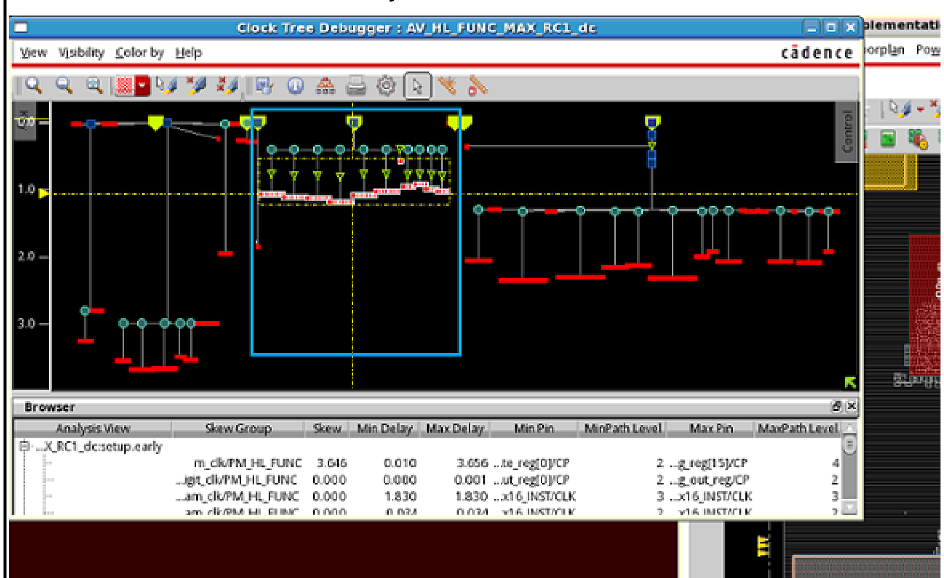
<table>
<tr>
<td>Select</td>
<td>
Provides options for selection. The following options are available:

- Select by Level
- Select transitive sink fanout
- Select subtree
- Select inst

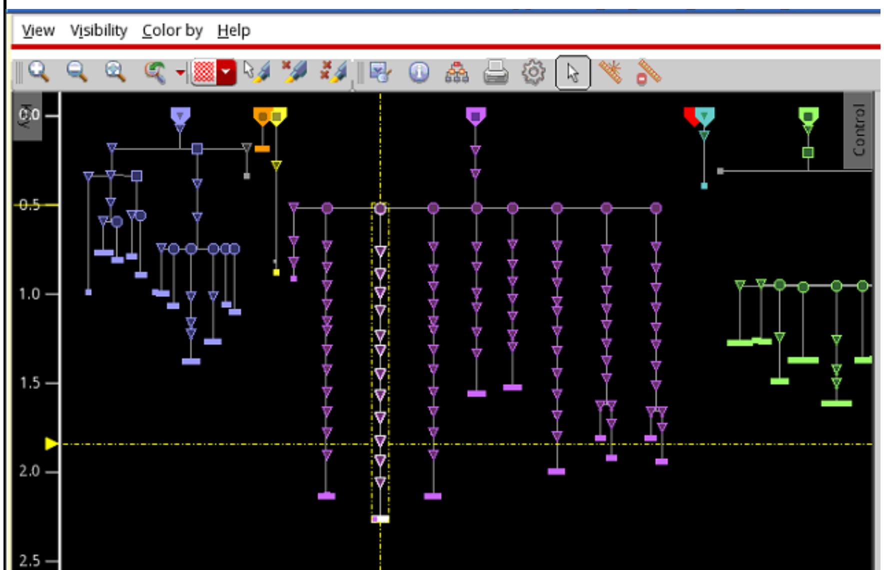
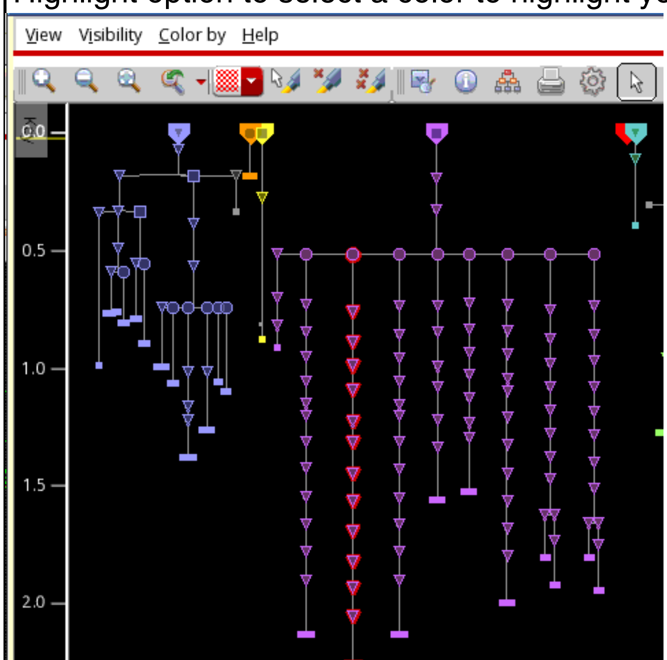
The *Select by Level* option lets you select clock trees from one depth level up to another depth level. The default level is 1. This is shown below.



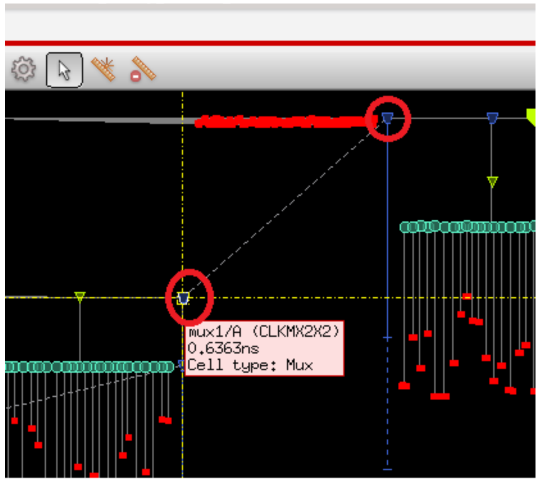
The *Select transitive sink fanout* option is available when you right-click objects other than sinks. It selects the transitive sink fanouts for the selected object.



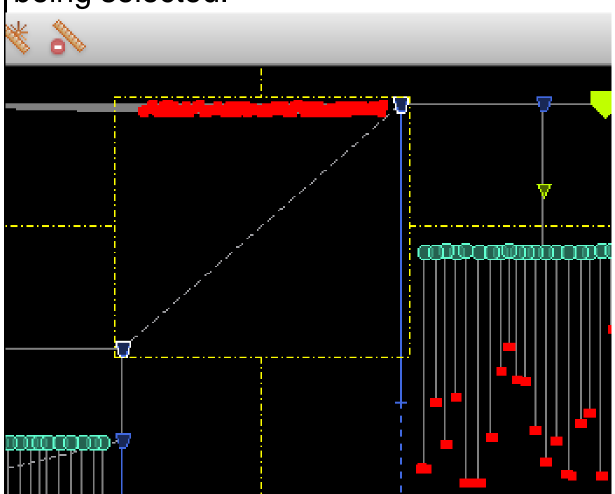
</td>
</tr>
</table>

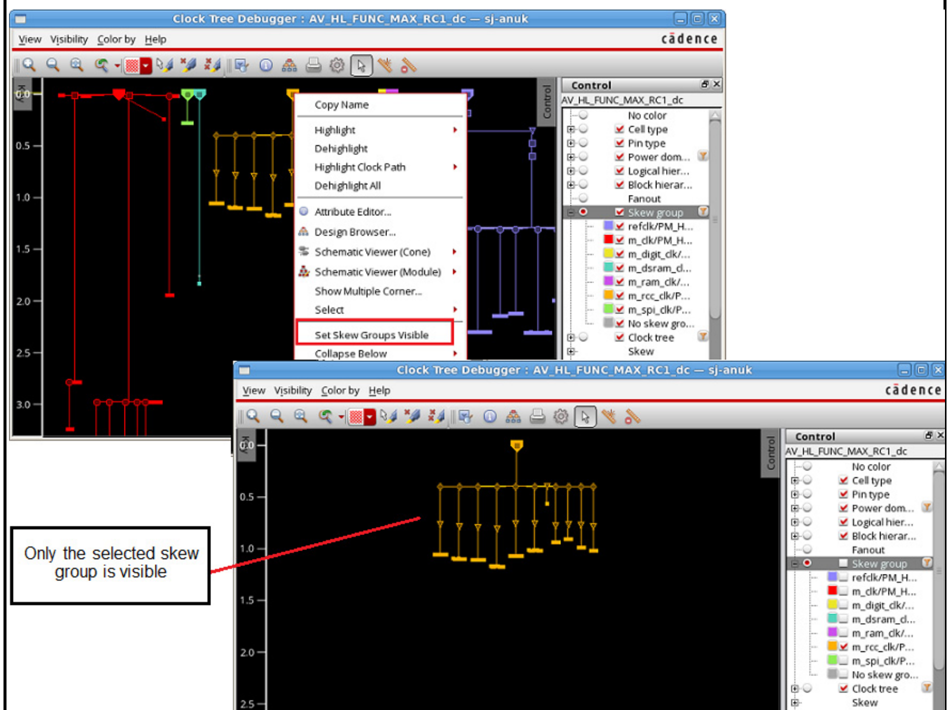
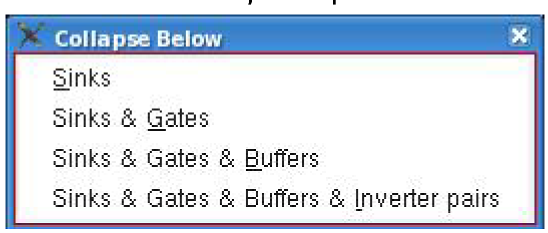
	The *Select subtree* option selects everything under the selected node.  To highlight the whole subtree, select the subtree and use the Highlight option to select a color to highlight your selection.  The *Select inst* option is available when an instance has multiple markers. When you select this option, the instance will be selected. This is especially useful in more complex designs where there are so many dashed lines that it is impossible to tell what they're connecting.

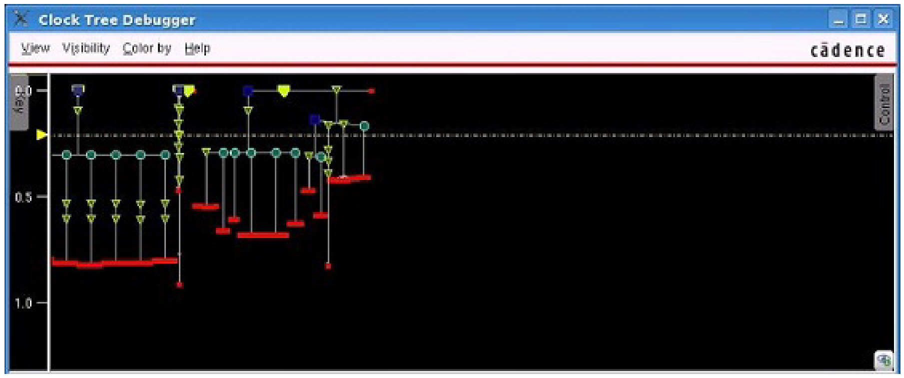
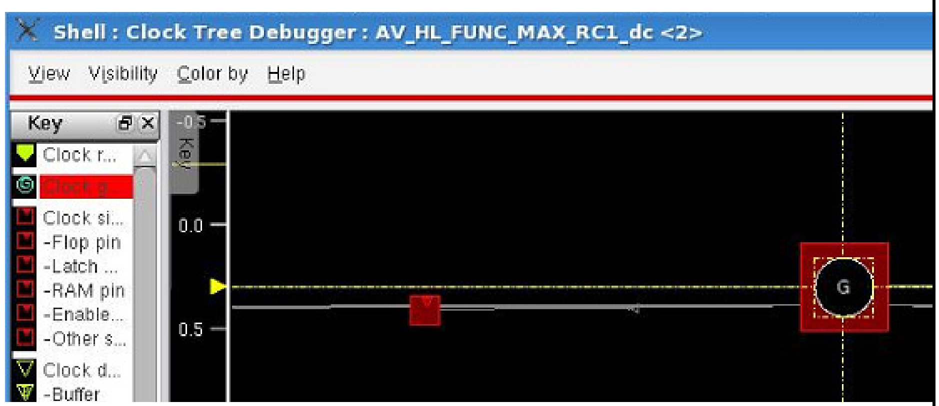
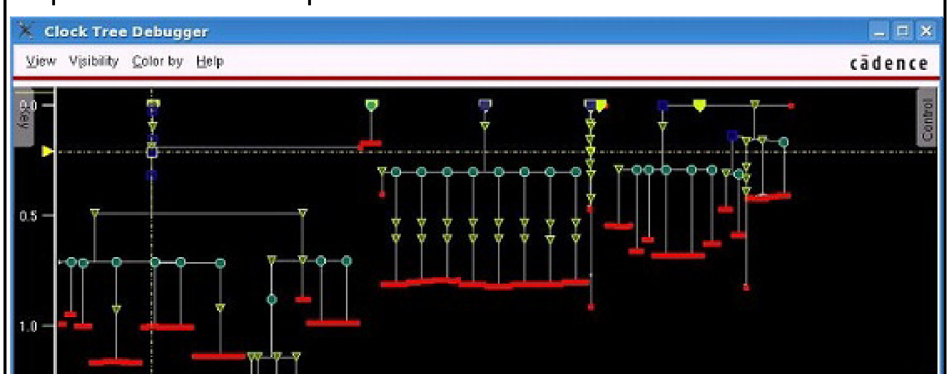
For example, the instance shown in below figure is a mux. It has two markers: one represents the A pin, the other represents the B pin. When you click on one to select it, it will select the A or B pin, respectively.

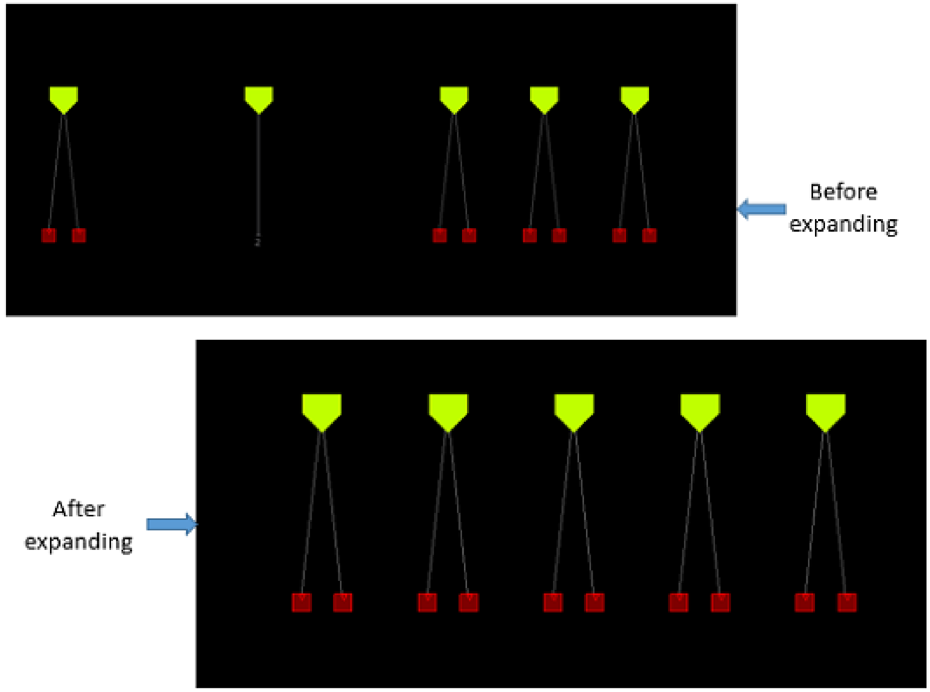


But, if you right click and in *Select* menu, choose the *Select inst* option, then selecting the instance will show all the pins on it as being selected.



Set Skew Groups Visible	Sets skew groups present at the selected marker to be visible while making all others invisible. 
Collapse Below	Provides options to collapse subtrees with clock gates and flops. Similar options are available in the *View* menu - *Simplify* submenu - *Collapse* option.  *Sinks* - collapses subtrees with clusters of sinks *Sinks & Gates* - collapses subtrees with clusters of gates directly driving sinks *Sinks & Gates & Buffers* - collapses subtrees with clusters of gates driving sinks connected by buffers *Sinks & Gates & Buffers & Inverter pairs* - collapses subtrees with clusters of gates driving sinks connected by buffers or inverters where the edge is not changed

Collapse subtree	Collapses the subtree.  You can also double-click to expand or collapse subtrees of any node of the clock tree.
Abstract this node	Abstracts the selected node.  This option only appears in the context menu if you select buffers, inverters, or clock gates. It does not appear if you select clock root.
Unabstract this node	Unabstracts the selected node.
Expand all subtrees	Expands all the collapsed subtrees. 

Expand all sinks	Expands all the collapsed sinks. This is shown in the image below. 
Mark Uncollapsible	Marks the selected nodes as uncollapsible. If you mark a node as uncollapsible, then when the parent node is collapsed, all the subnodes below that node will not be collapsed.
Mark Uncollapsible/ Collapsible	Marks the selected nodes as uncollapsible/collapsible. If you mark a node as uncollapsible, then when the parent node is collapsed, all the subnodes below that node will not be collapsed. To change this setting, select *Mark as Collapsible* from the context menu.

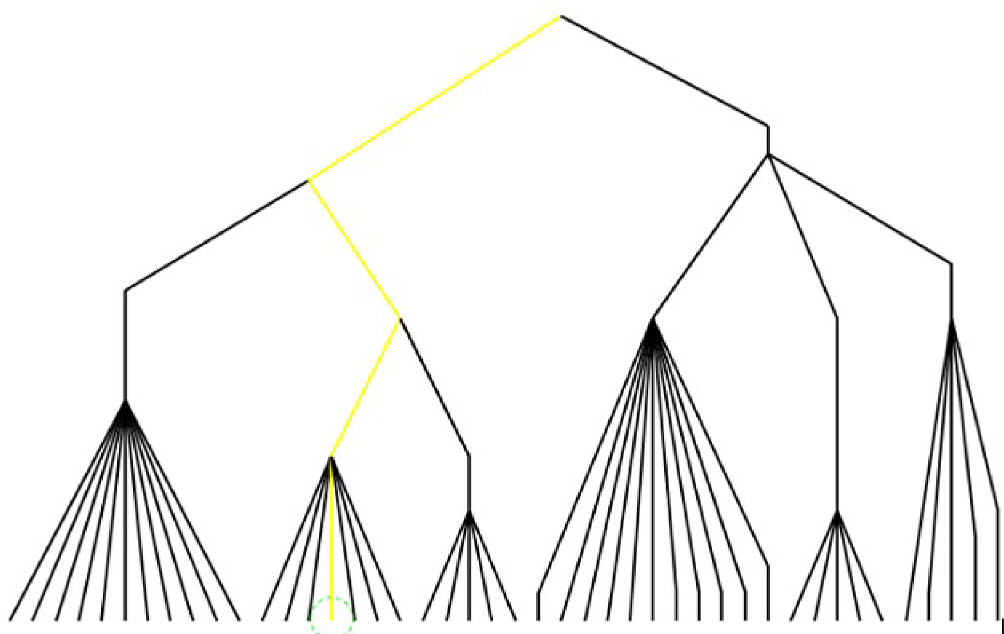
Highlighting the Clock Tree Path

The *Highlight Clock Path* option in the context menu of the Clock Tree Viewer highlights the complete clock tree path from the selected clock tree object to the clock root in the Clock Tree Viewer. When you select this option, the CTD traces back to find the path from the clock root to the selected node and then highlights the path in the Clock Tree Viewer. Two scenarios are detailed below.

You can also use the `ctd_trace` command to highlight clock tree paths in the specified color.

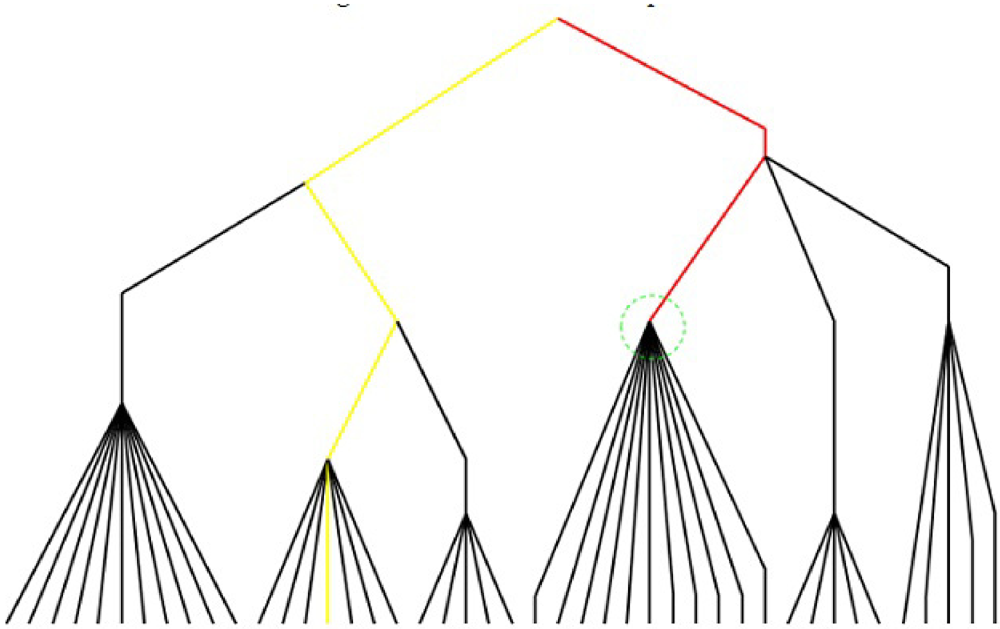
Scenario 1: When you select a pin

In this case, the path from the clock root to the sink will be highlighted. This is shown in the image below.

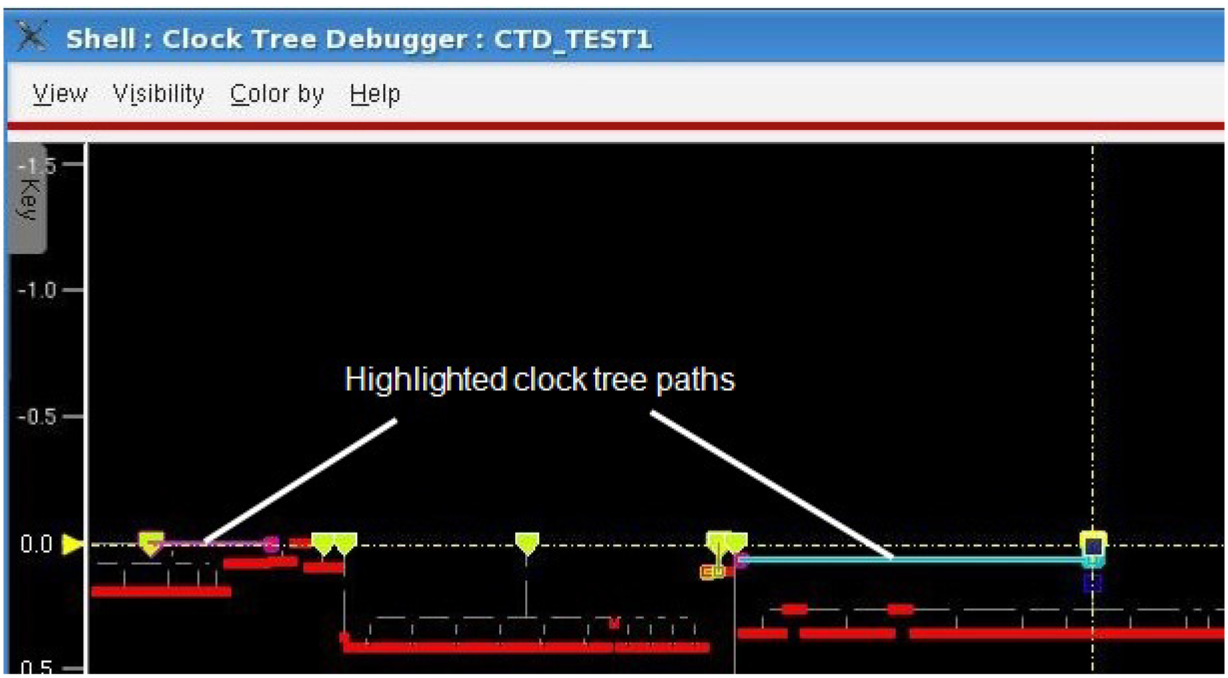


Scenario 2: When you select an intermediate node

In this case, the path from the clock root to the intermediate node (buffer, inverter, or clock gate) will be highlighted.



The highlighted path for a selected object remains highlighted even when you select another object and trace its path. So, you can view multiple paths at a time. In the image below, two paths are highlighted, one from a clock gate and the other from a flop pin.

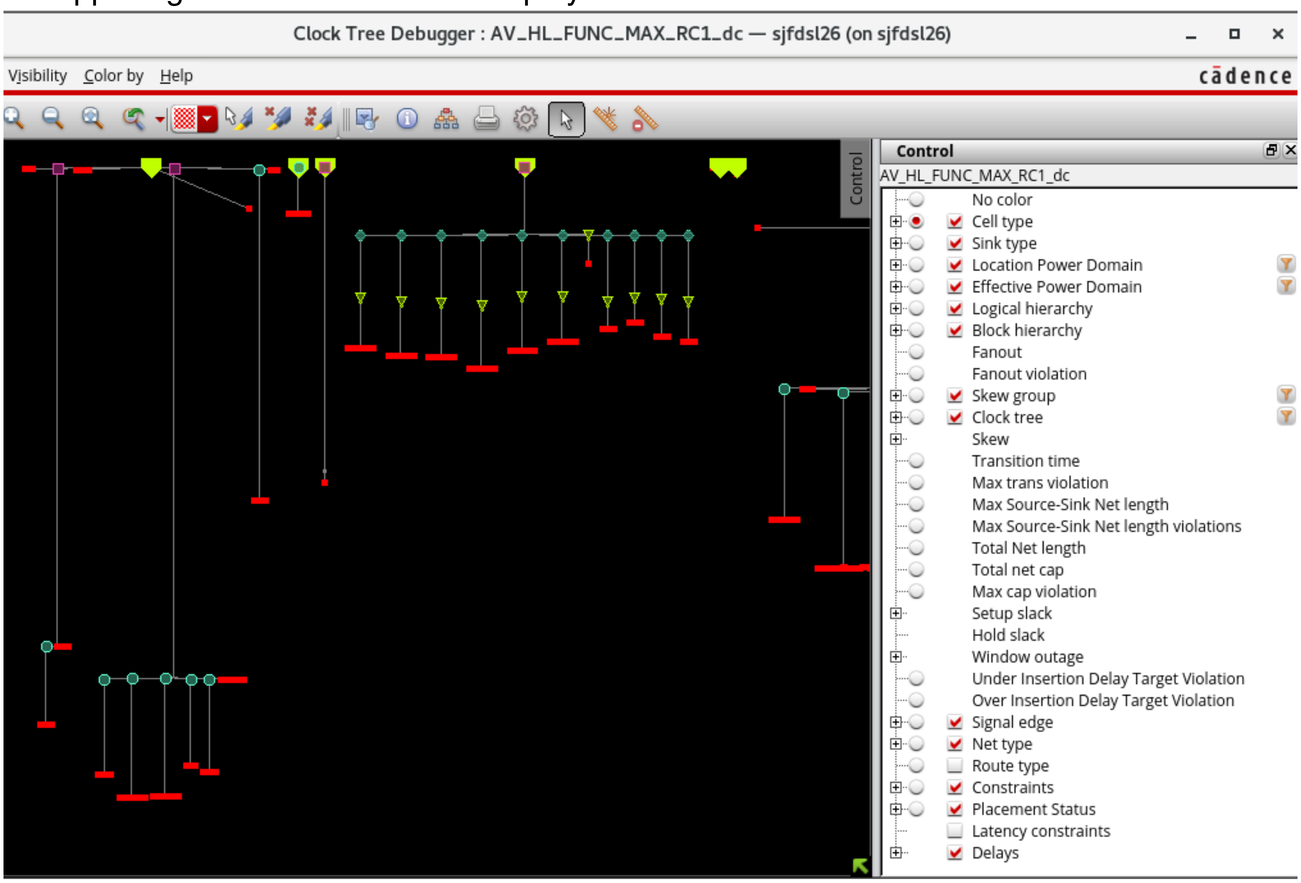


To erase all clock tree path highlights and flight lines from the Clock Tree Viewer and the physical layout window respectively, click the *Dehighlight All* option in the context menu.

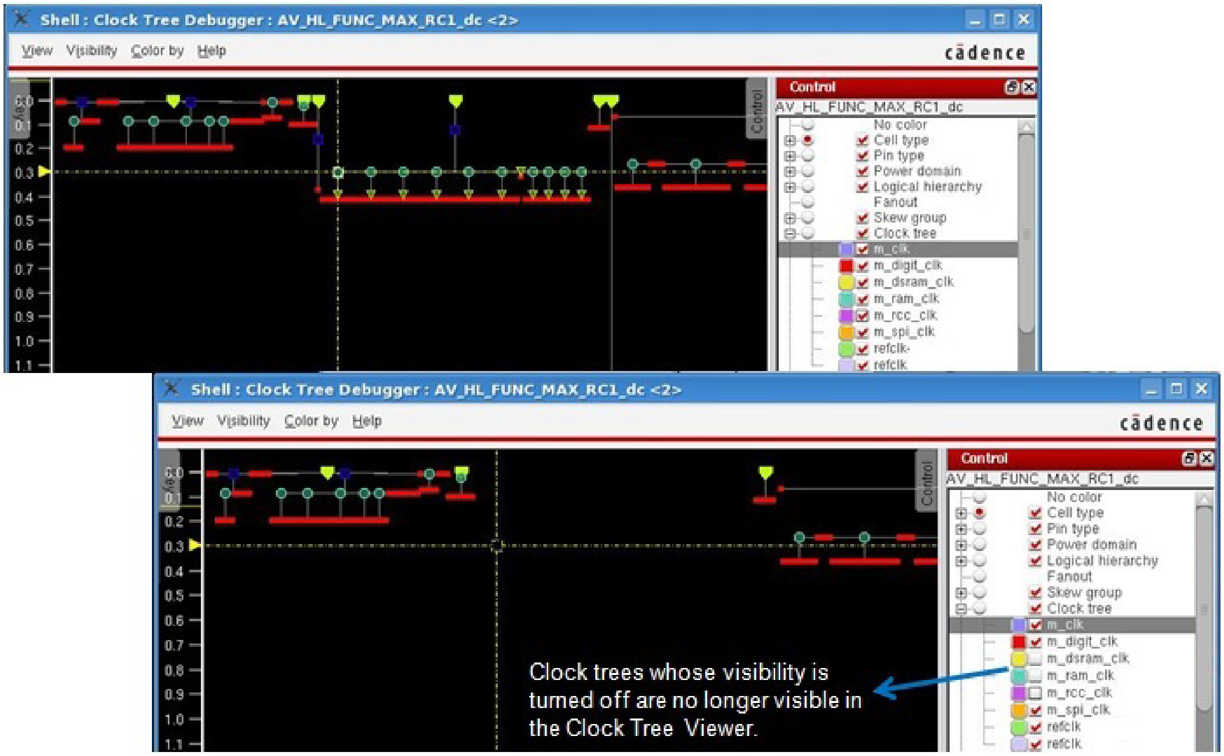
Control Panel of CTD

The Control panel displays and manages merged settings from the Visibility and Color by menus. It presents these items in a tree view that follows the same hierarchy as the two menus, with similarly named items combined. For example, the Cell type item includes both a radio button from the Color By menu and a submenu from the Visibility menu. In the Skew group submenu, individual items provide check boxes for both visibility and color.

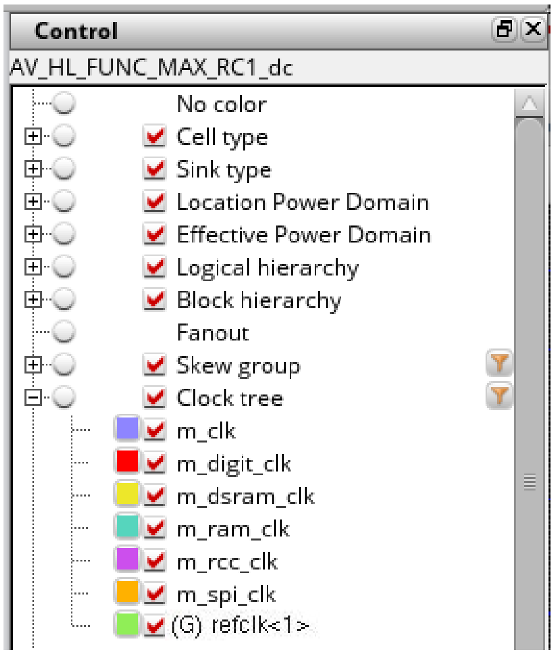
To open the *Control* panel, choose *View – Control* panel. Alternatively, double-click Control in the upper-right corner of the CTD display.



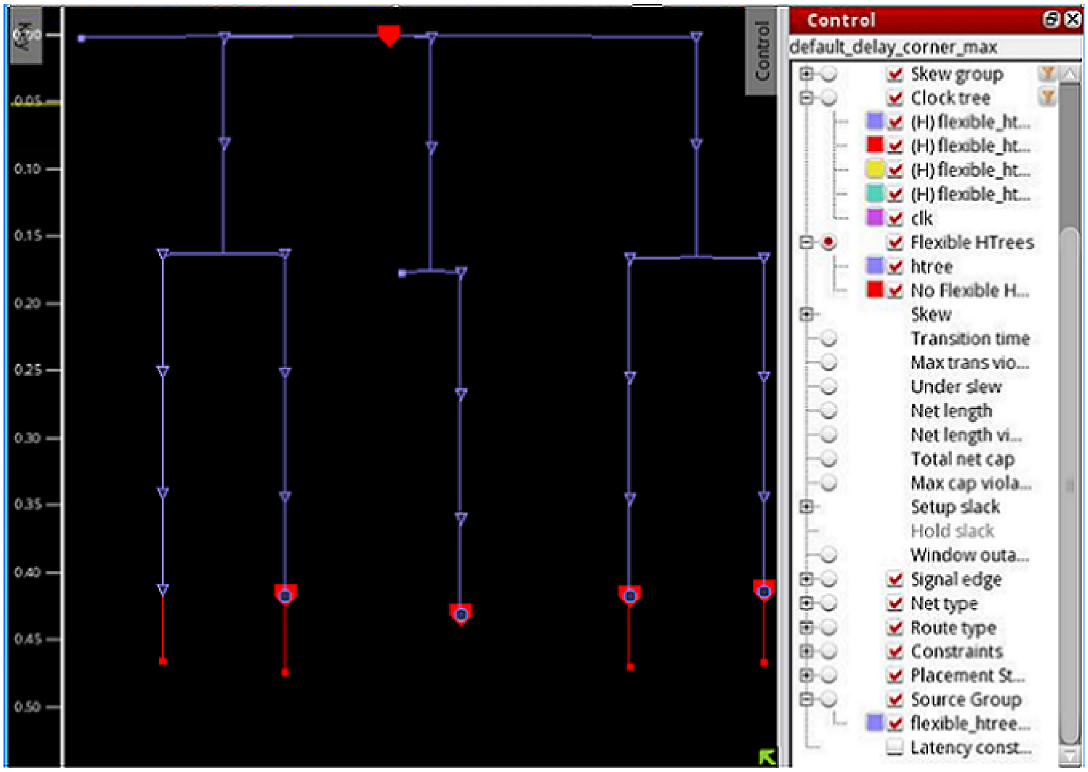
The Control panel allows you to turn off the visibility of clock trees in the Clock Tree Viewer. For example, when you turn off the visibility of some trees in the control panel, the trees will no longer be shown in the Clock Tree Viewer. This is useful if you have many clock trees and the display window is very tight. You can choose to turn off the display of some trees using the Control panel.



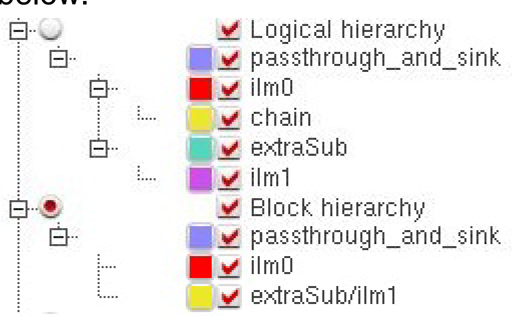
The Control panel also allows you to identify generated clock trees easily. In the Control panel, a mark, (G), appears before the names of all generated clock trees. This helps users to identify the generated clock trees easily and saves them time when debugging clock trees. The image below shows a generated clock marked by (G) in the Control panel.



The Control panel lets you identify flexible H-trees when the design has flexible H-trees. In the Control panel, a mark, (H), appears before the names of all flexible H-trees. A new coloring is added to show things in flexible H-trees in different colors. This helps users to identify the H-trees trees easily and saves them time when debugging flexible H-trees. A source group coloring is added so that everything in the same clock tree source group can be colored together. The source group appears only when the design has source groups. The image below shows flexible H-trees marked by (H) in the Control panel.



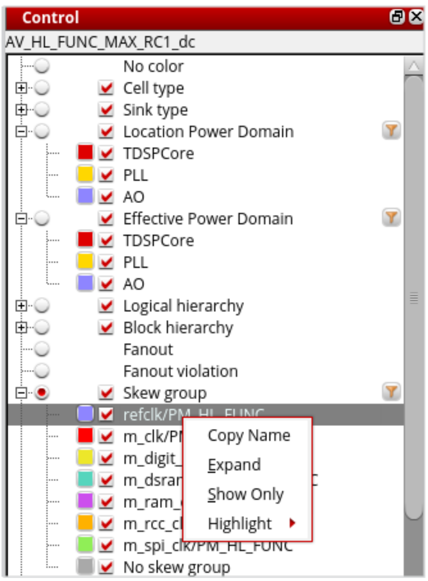
Note: The Logical hierarchy option in the Control panel shows the level hierarchy as a tree of modules or level names that can be toggled. The level hierarchy displays only those level names that are used by the elements in the CCOpt clock tree. This is shown in the image below.



The Block hierarchy option is used to color modules that represent ILM and other sub-blocks. For example, in the image above, you can see that there are four modules in the logical hierarchy but only two of these are ILM blocks, `ilm0` and `ilm1`. The Block hierarchy option lists both these blocks. Also, in order to distinguish names, the hierarchical name of the `ilm1` block (`extraSub`) is preserved in the listing.

View Skew Groups

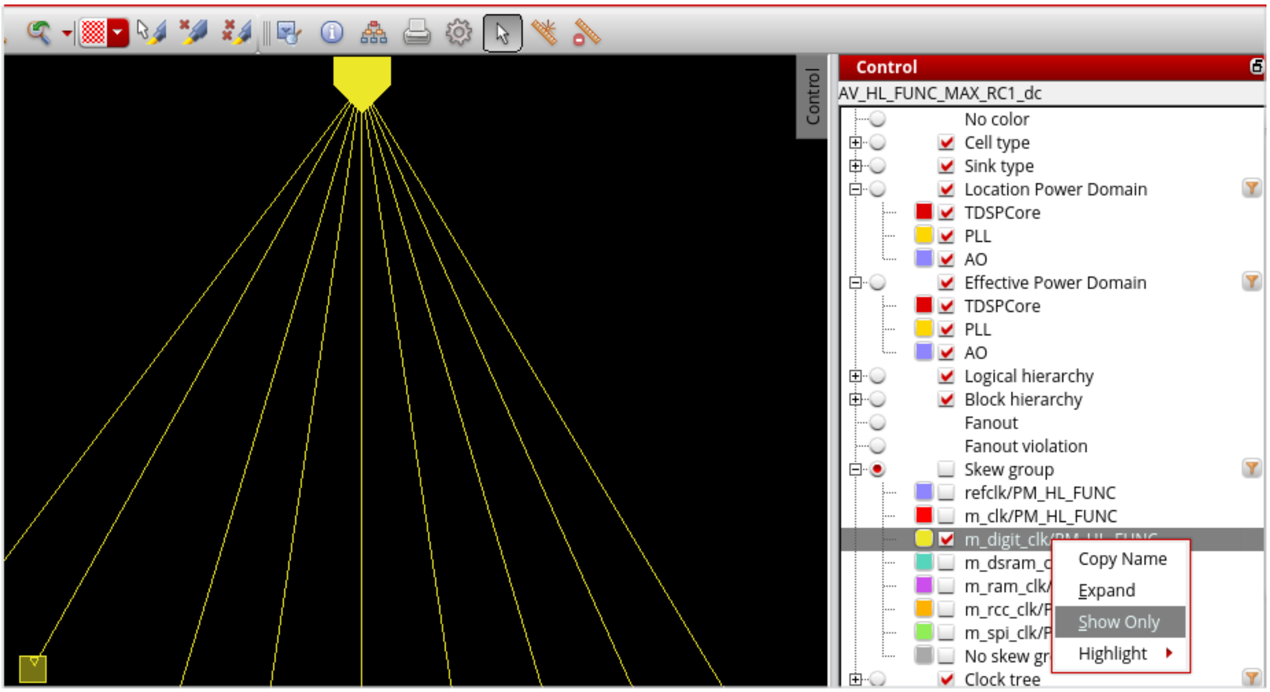
In the Control panel of the Clock Tree Debugger, a context menu is provided for the skew groups. When you right-click a skew group, the following options are available in the context menu for viewing, copying the name, expanding, and highlighting the selected skew group:



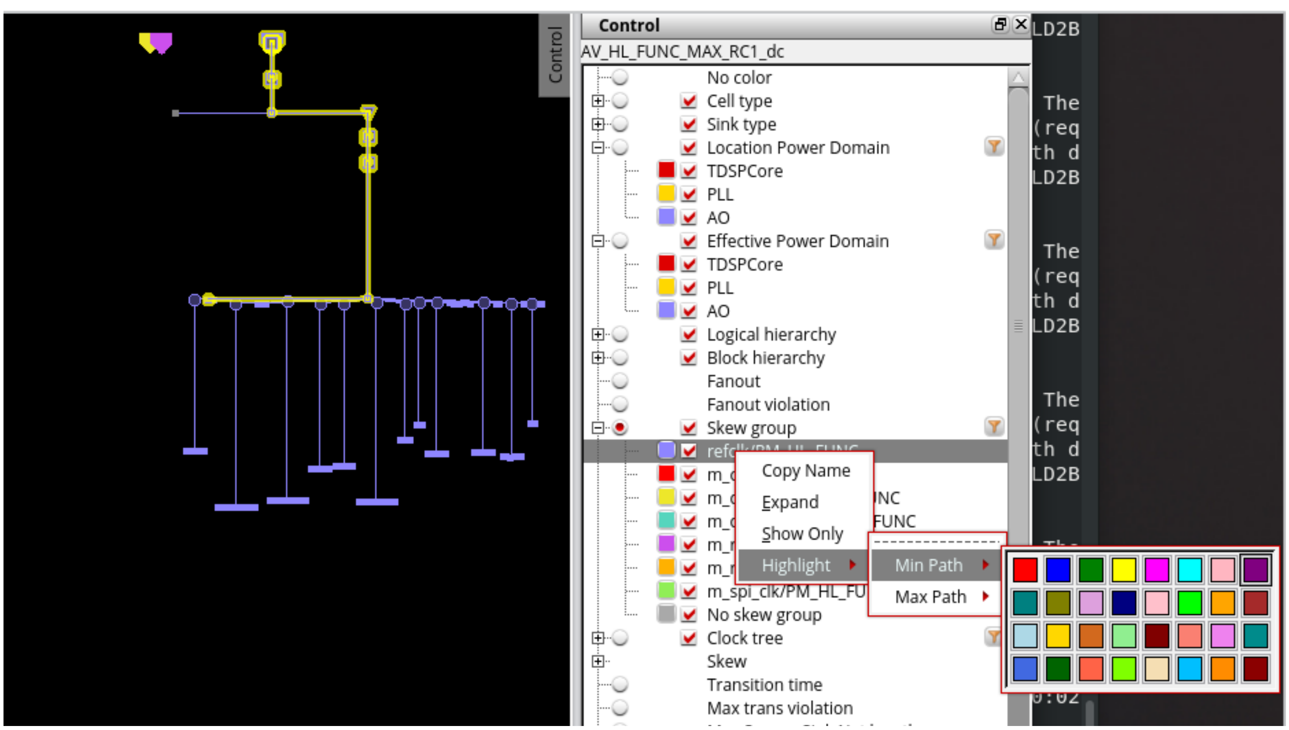
Option Descriptions

- *Copy*: Allows you to copy the name of the skew group.
- *Expand*: Expands all subtrees that pass through the selected skew group in the display area.
- *Show Only*: Displays the selected skew group in the display area of the CTD with all the clock convergence points expanded. In the image below, the skew group with yellow

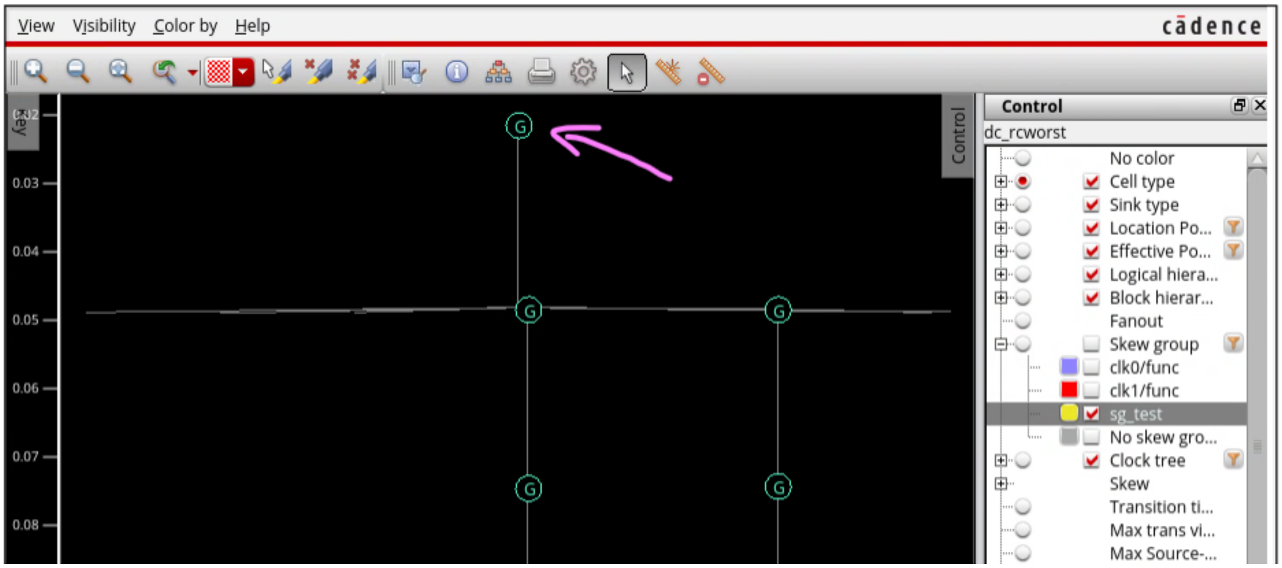
color is selected and the convergence points for this skew groups are expanded in the display area.



■ *Highlight*: Provides the option to highlight the Min Path and Max Path of the selected skew group in a color of your choice from the color picker.



Note: When coloring or displaying by skew group, the source pin is considered part of the skew group if either the input pin it represents or the output pin of the corresponding clock node belongs to that skew group. This ensures that the source pin is visible when other skew groups are hidden.

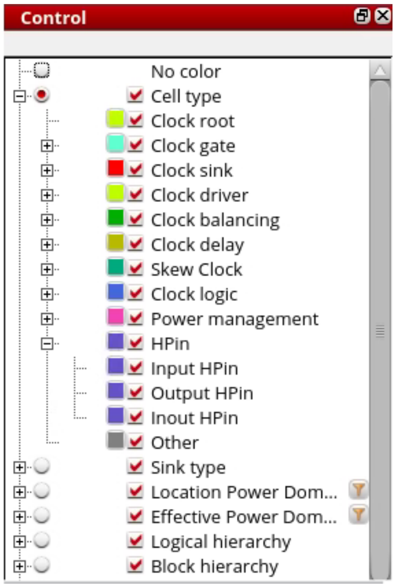


View Preserved Ports

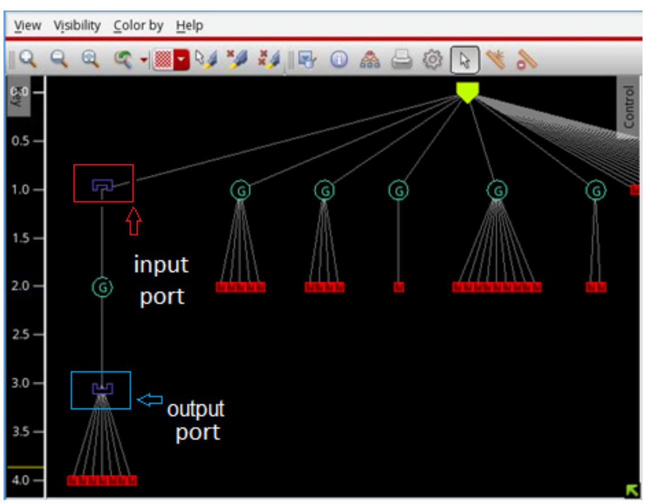
The following options are provided in the Control panel for viewing preserved ports:

- *HPin*
- *Input Hpin*
- *Output HPin*

- *Inout HPin*



The above options are visible in the Control panel only when using the Unit Delay mode. The input and output port bit markers show gates with preserved input and output ports. For example, in the image below, the new input and output port bit markers show that the gate on the left is in a module with a preserved input port before it and a preserved output port after it.

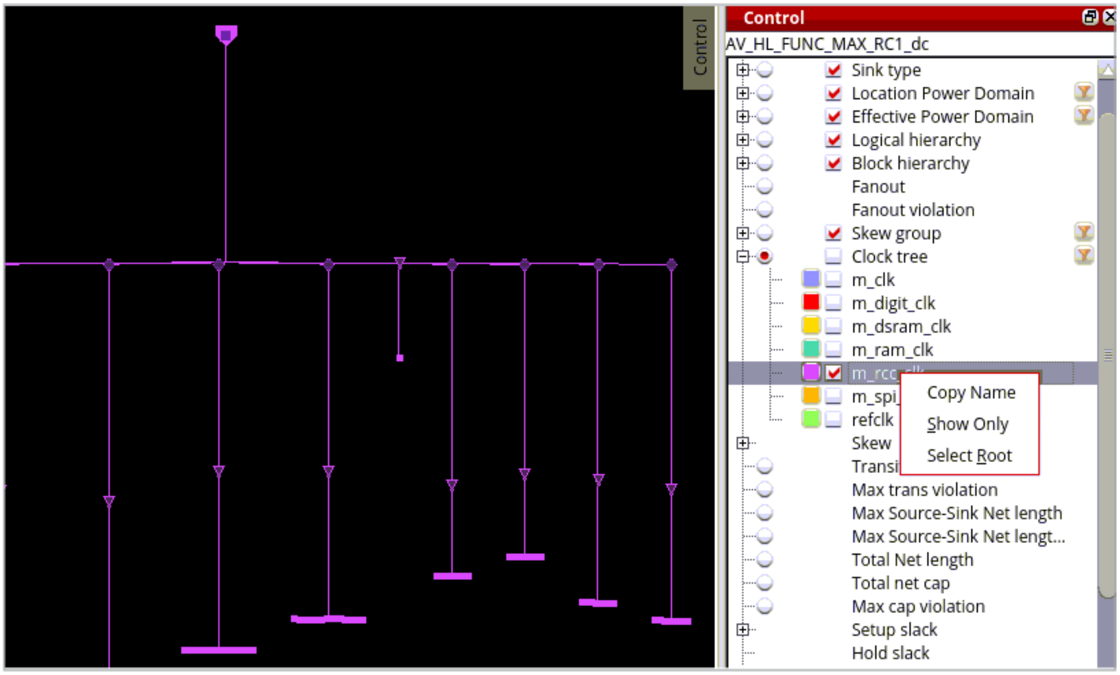


To hide the port markers, use the *View < Simplify < Hide* option.

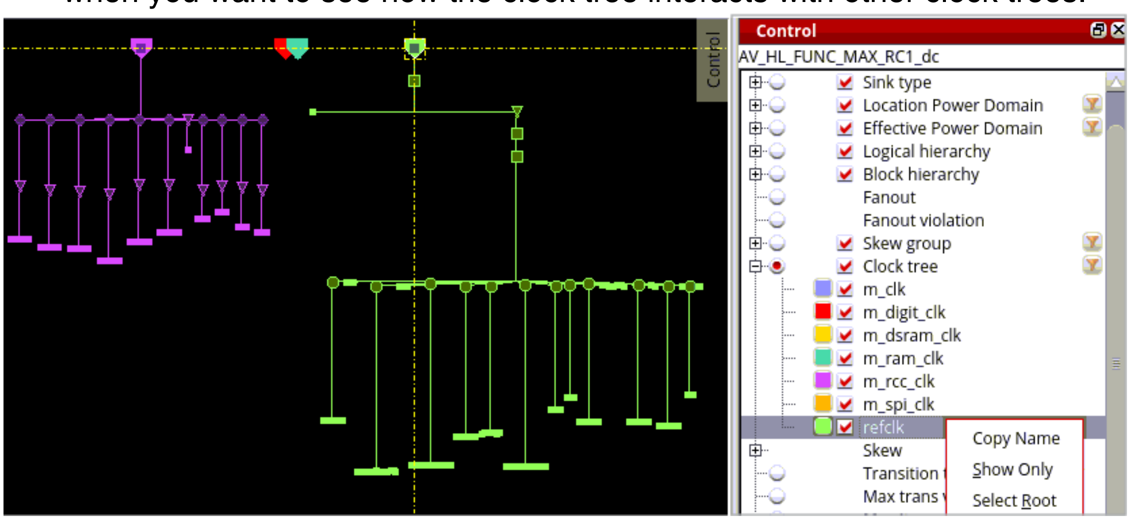
View Specific Clock Trees in the Control Panel

For viewing specific clock trees, the following options are added to the context menu of the clock trees in the Control panel of the CTD.

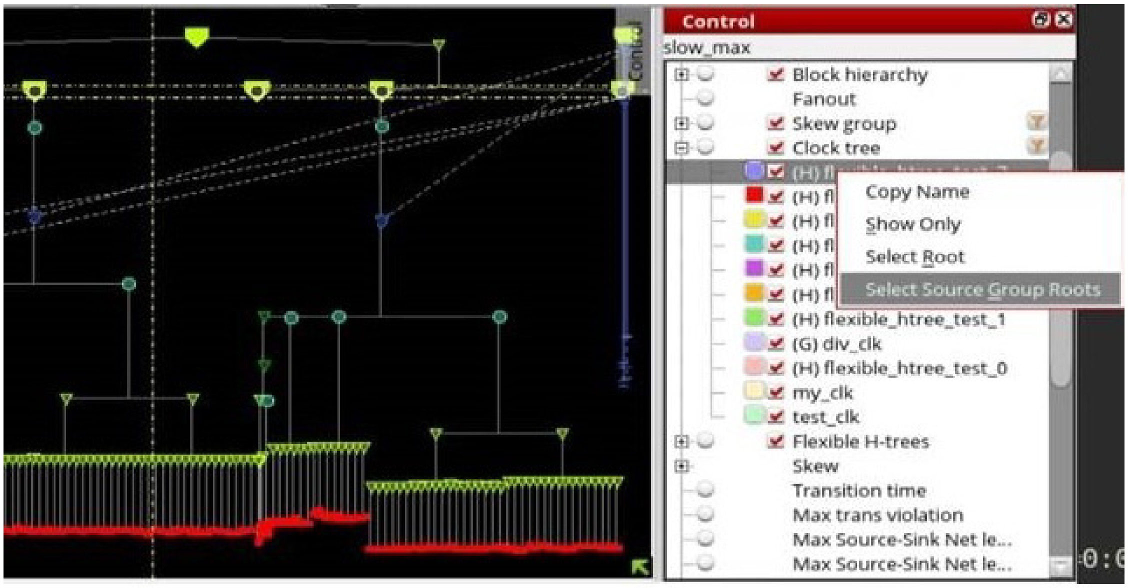
- *Show only*: Hides all the other clock trees and refits the display to show just the one that is selected. This is similar to "Show only" in the skew groups context menu, and is useful when you want to see just that one clock tree.



- *Select root*: Selects the root of the given clock tree in the display area. This helps to find the clock tree in the main CTD area, but it does not hide other clock trees. This is useful when you want to see how the clock tree interacts with other clock trees.

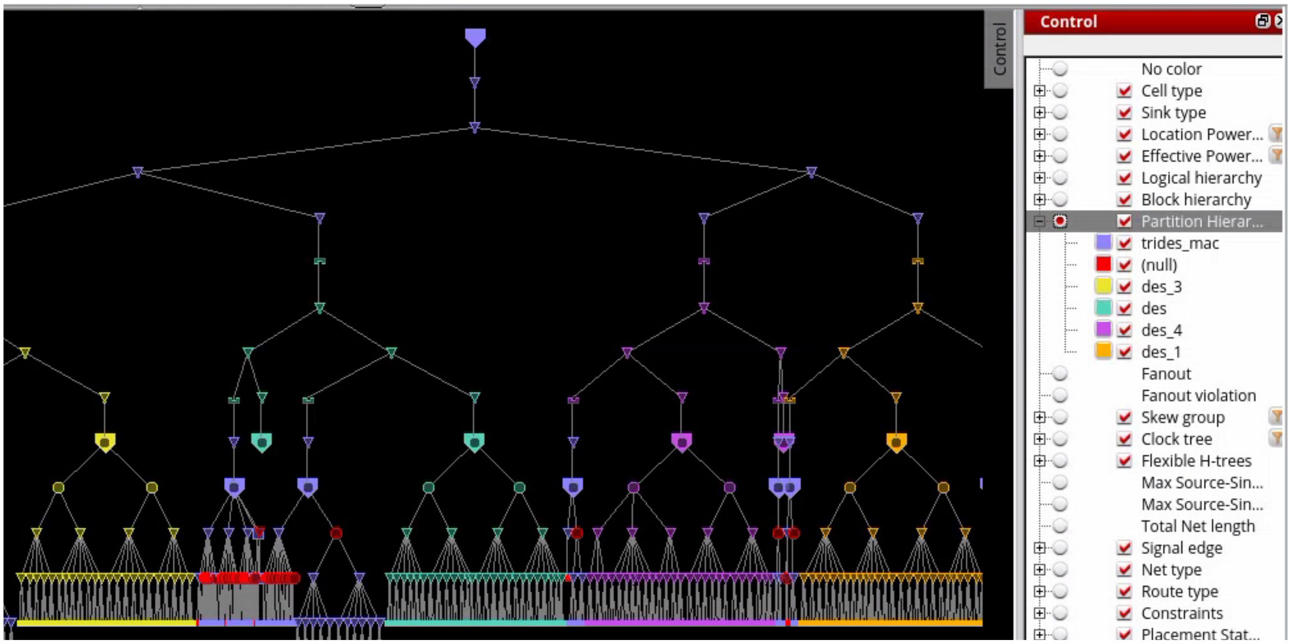


- *Select source group roots*: Selects all the roots of clock trees in the same multitap source group. This is useful for checking multitap sink assignment issues. This option is only available when the selected clock tree belongs to a clock tree source group.



View Partition Hierarchy

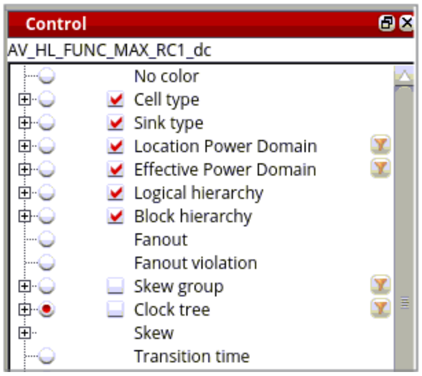
The Partition Hierarchy option, available in the Visibility menu, Color by menu, and Control panel, assigns a distinct color category to each partition in the design. Markers are displayed using the color associated with their respective partition. When partitions are nested, markers use the color of the innermost partition to which they belong.



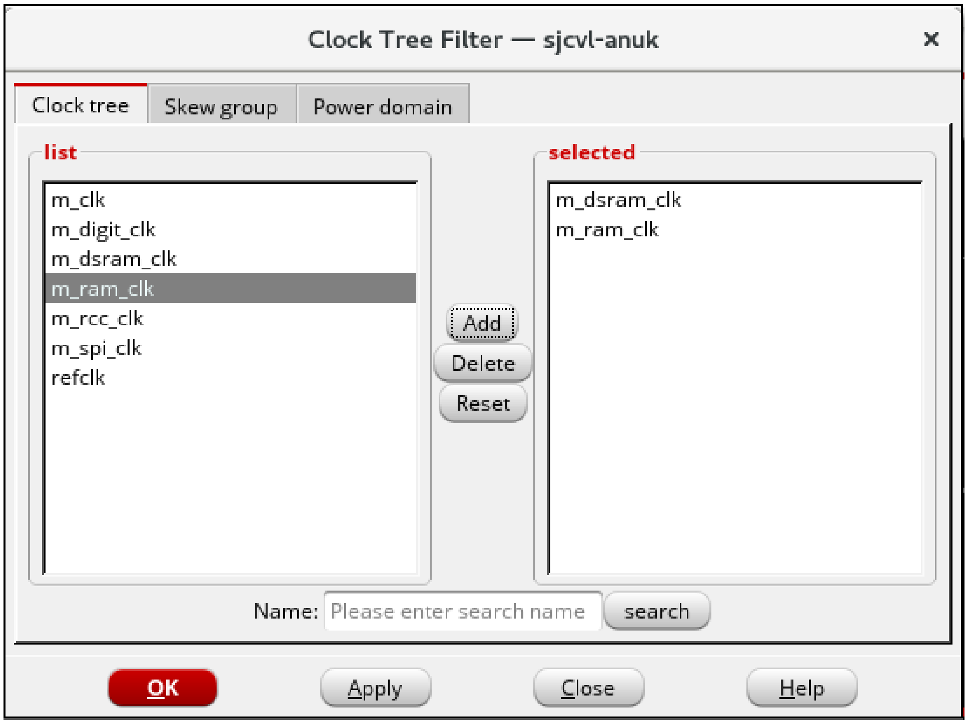
Note: This option is displayed only when partitions are configured in the design. It is available in both unit delay and timing-based modes.

Filter Options in the Control Panel

In the Control panel, filter options are provided for power domains, skew groups, and clock trees. These are available by clicking the filter icons provided next to these items in the Control panel. These options let you view specific clock trees/power domains/skew groups in the display area while filtering out the rest. You can also search for specific clock trees/power domains/skew groups by name to filter from the display.

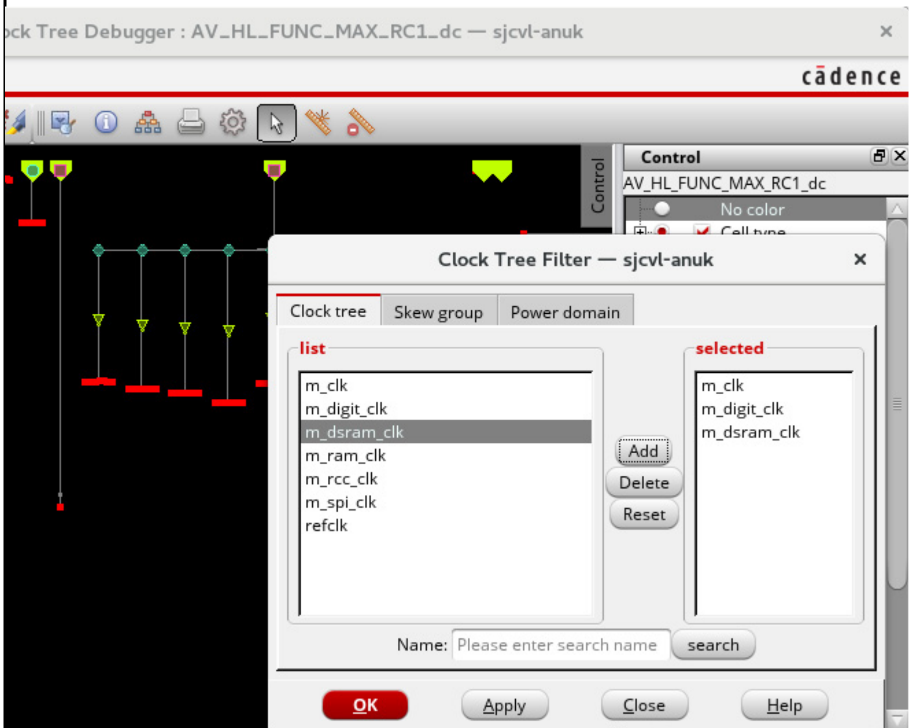


When any of the three filter icons is clicked, the Clock Tree Filter form opens.

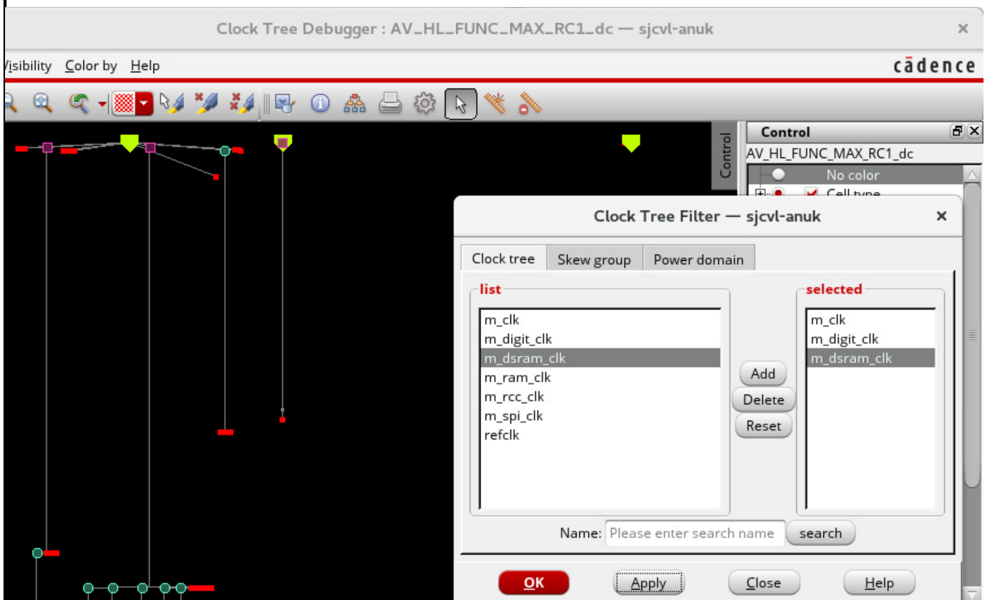


The form has three tabs. Depending upon which filter icon is clicked, the corresponding tab is enabled. The rest are disabled. You can select the clock trees/power domains/skew groups to filter from the display.

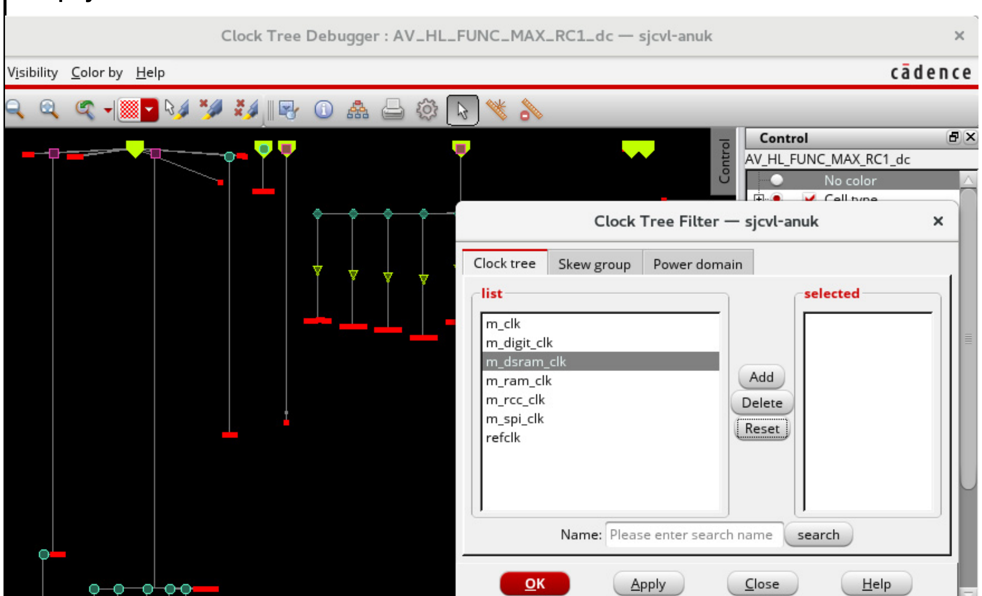
Options	Descriptions
list	Lists all the clock trees/power domains/skew groups.
selected	Lists all the clock trees/power domains/skew groups that are selected and added from the list box.

Add	Adds the clock tree/power domain/skew group selected in the list box to the selected box. The image below shows the clock trees are added to the selected box. 

<table>
<tr>
<td>Delete</td>
<td>Deletes the clock tree/power domain/skew groups in the selected box from the display window. In the image below, when you select a clock tree (mdsram_clk) and click Delete and Apply, the selected clock tree is deleted from the display area.



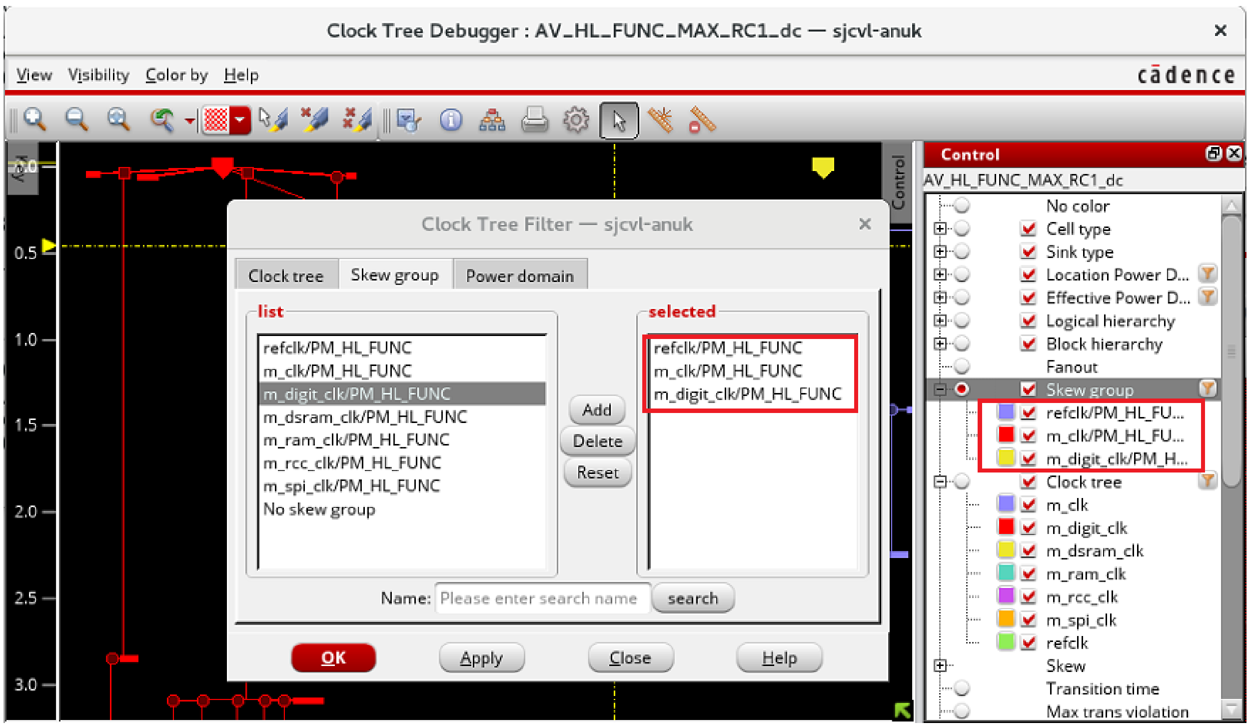
</td>
</tr>
<tr>
<td>Reset</td>
<td>Removes all the clock tree/power domain/skew groups from the selected box and resets the display to the pre-selection state. In the image below, when you click Reset, the clock tree that was deleted from the display area is displayed again and the selected box is empty.



</td>
</tr>
</table>

Name	Specifies the name of the clock tree/power domain/skew group to be searched.
search	Searches the specified clock tree/power domain/skew group.

When you filter the skew group by clicking the filter button on the right of the Skew group option in the Control panel, the list under the Skew option will only show the filtered skew groups instead of all the skew groups.

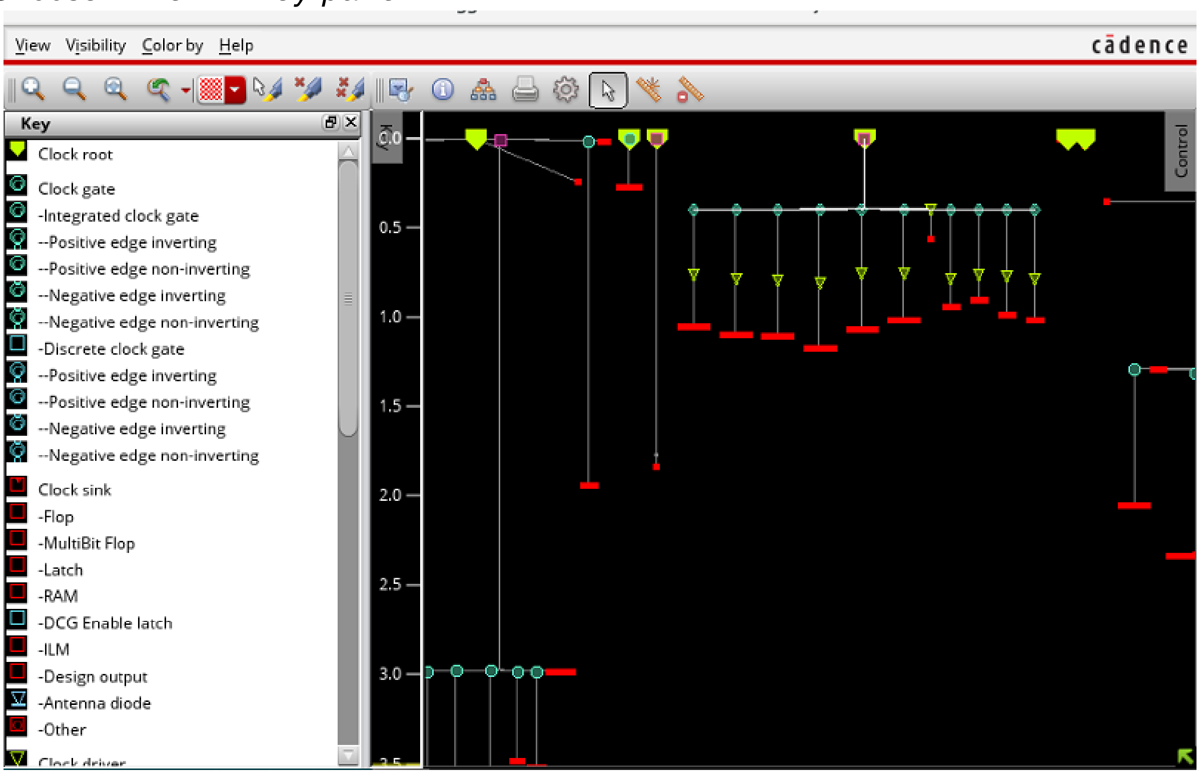


Key Panel of CTD

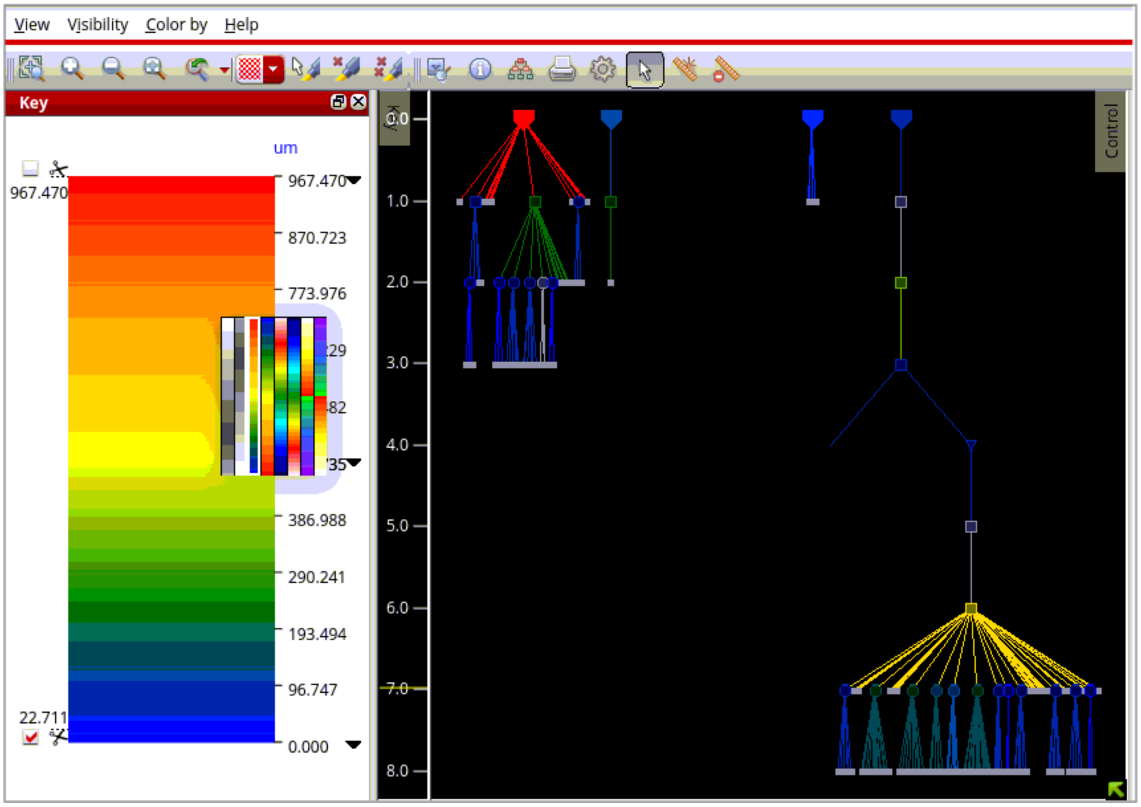
When enumeration or subset coloring is in effect, the Clock Tree Debugger displays a key panel that lists the enumeration categories or subsets along with their corresponding colors. If the Key panel is visible, it updates automatically to reflect the currently selected coloring.

Many enumeration categories are named after database object names, for example, skew groups, and may therefore be lengthy. To avoid introducing a horizontal scroll bar, long names are truncated in the key panel.

Choose – *View* – *Key panel*.



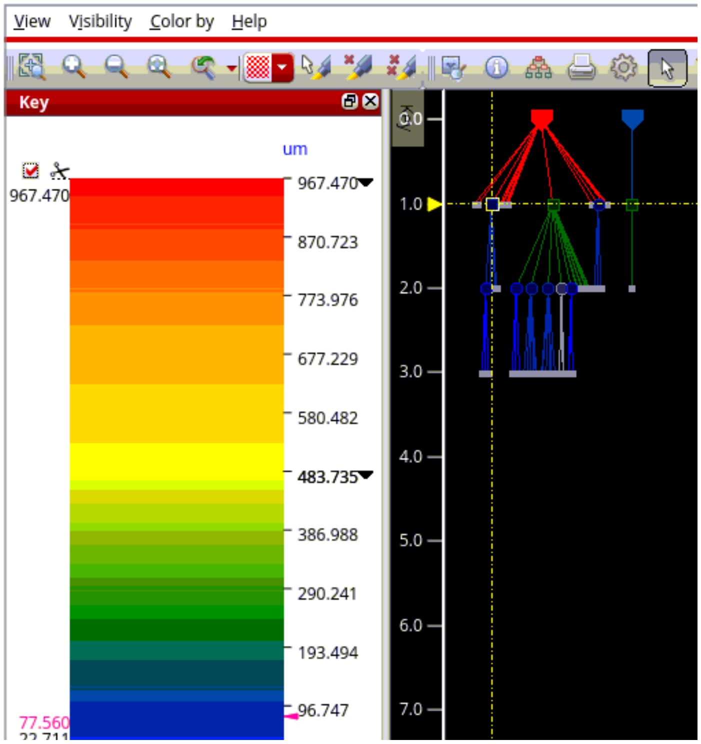
The window has a minimum width but can be resized by dragging it horizontally. When a continuous color scale is in effect, a vertical scale appears on the right side of the visualization. This scale shows the colors in use and provides controls to adjust the scale bounds and to filter objects by value using the scissors. You can right-click the scale to switch between coloring schemes such as grayscale, rainbow, or heat map.



Unlike the enumeration key, which can contain arbitrary text labels, the color-scale key has a fixed, narrow width. The width defines the minimum width of the window. When a color scale is present, a check box is added to the View menu that allows you to show or hide the scale.

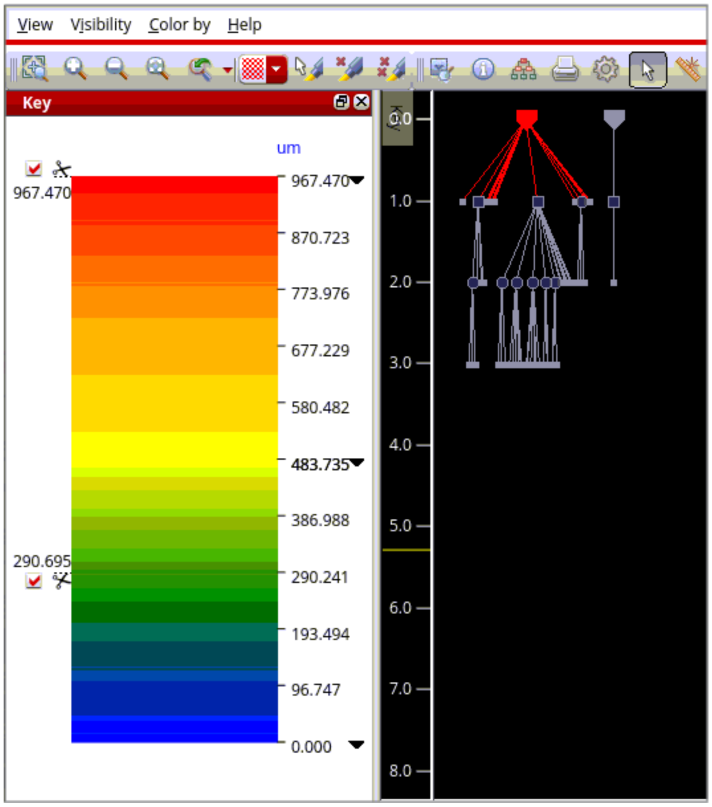
Note: Only one enumeration can be displayed at a time. As a result, the key and color-scale panel are mutually exclusive and share the same window.

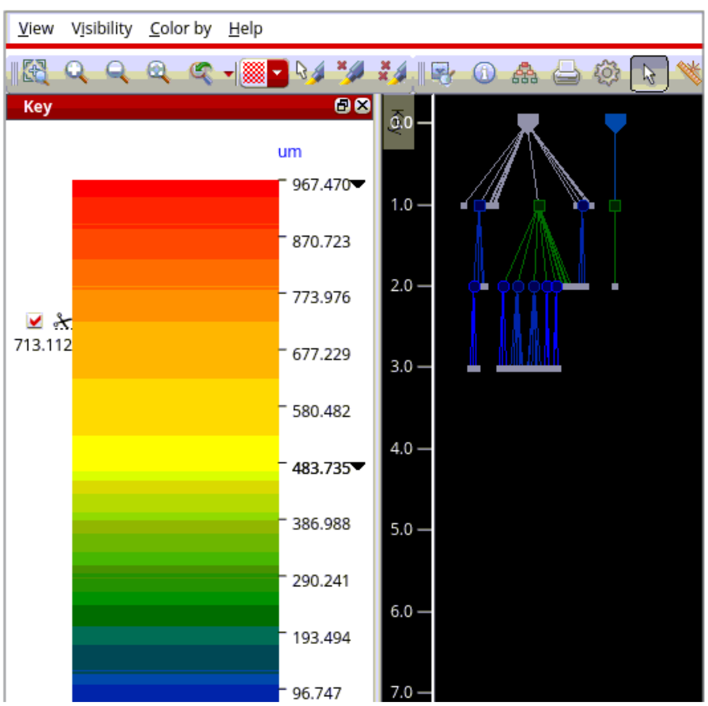
When coloring by a continuous color scale, the values for selected objects are highlighted in red on the scale shown in the Key panel. If a large number of objects are selected, the number of distinct values displayed is limited to reduce clutter and excessive redraw time.



You can use scissors to redraw the CTD display. The upper scissor represents the maximum value of the attribute, and the lower scissor represents the minimum value. As you move the scissors along the color scale, the clock objects with attribute values outside the specified range are hidden. This behavior is illustrated in the images below.

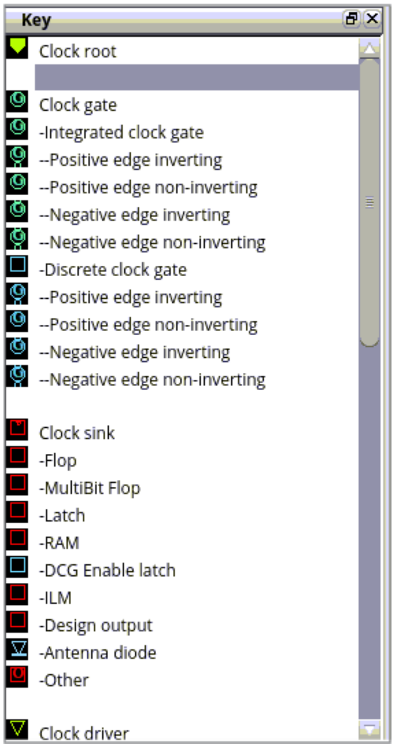
Note: The CTD display area is updated as you drag the scissors up or down.





Key Panel for Cell Type Option

For enumeration of *Cell Type* options, the key panel shows a set of symbols with colors and graphic shapes. This is shown in the image below.



Clock Path Browser

The *Clock Path Browser* window consists of the following:

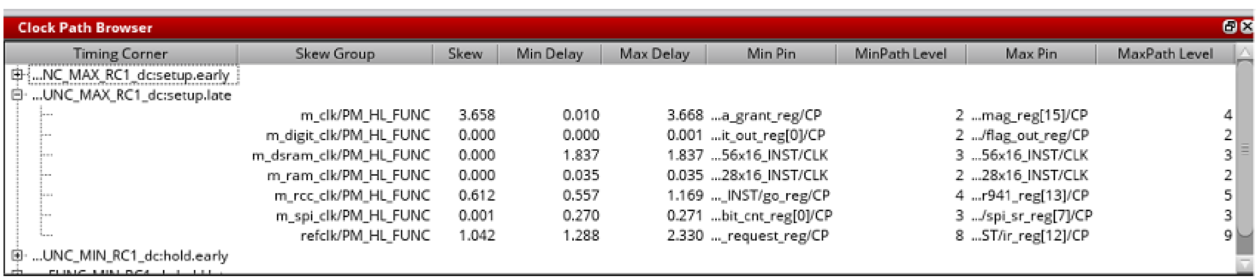
- A menu bar
- A table of path pairs

Choose - *View* - *Clock path browser*. By default, the browser is attached at the bottom of the CTD window. You can choose to display the browser above, below, to the right or left of the main display area of the *Clock Tree Debugger* window.

When you open the CTD main window, the browser does not open by default.

To view the browser in the *Clock Tree Debugger* main window, select *Enable clock browser* in the *View* menu. You can choose to view first, second, or third level of min/max paths for skew groups. For details of above options, see the *Clock Tree Debugger - View section*. You can also update these settings for the Clock Path Browser in the Set Preferences form. For details, see "Set Preferences for CTD GUI Configuration".

To close the browser, click the X at the top-right corner of the browser window.



The *Browser* shows a table of path pairs with one row per minimum (min) or maximum (max) path in a particular skew group for a particular analysis view. Each row provides the following information in the various columns listed below:

- Analysis View
- Skew Group
- Skew
- Min Delay
- Max Delay
- Min Pin
- MinPath Level

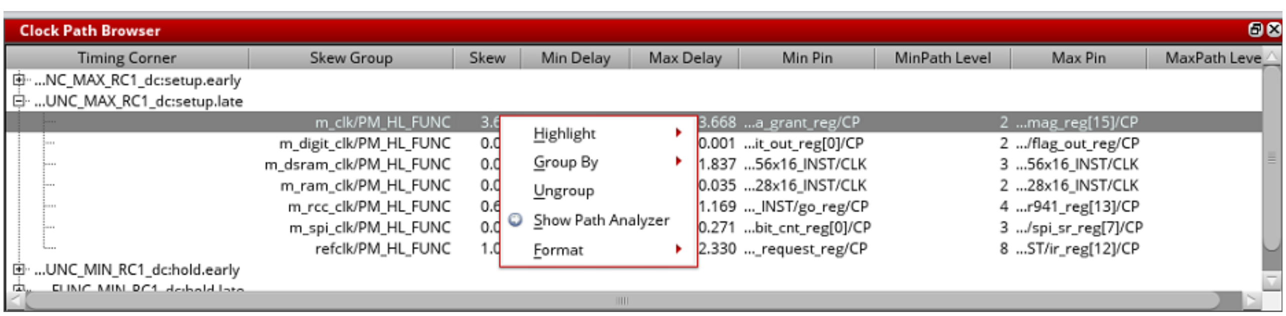
- Max Pin
- MaxPath Level

Cells in the table that correspond to objects such as the skew group, analysis view, and min or max pins have the context menu applicable to that object, allowing the full set of operations such as coloring by a skew group in the clock tree view.

Context Menu of the Clock Path Browser

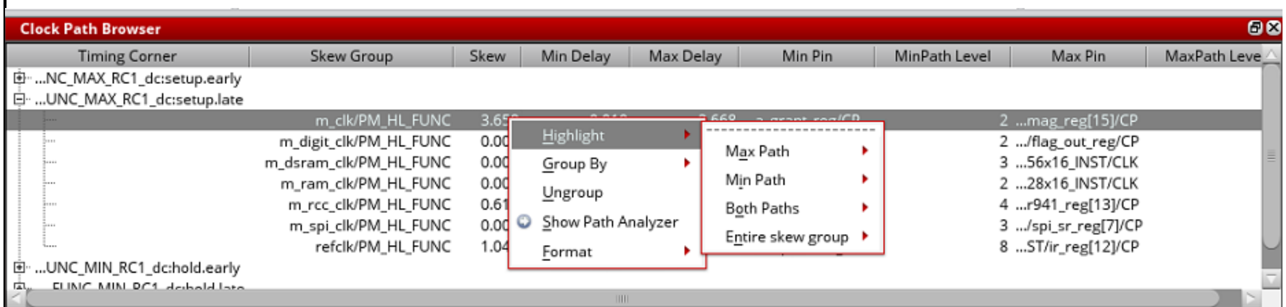
The context menu of the clock path *Browser* provides the following functionalities, which you can access when you right-click any marker or line in the clock tree view:

- Highlight
- Group By
- Ungroup
- Show Path Analyzer
- Format

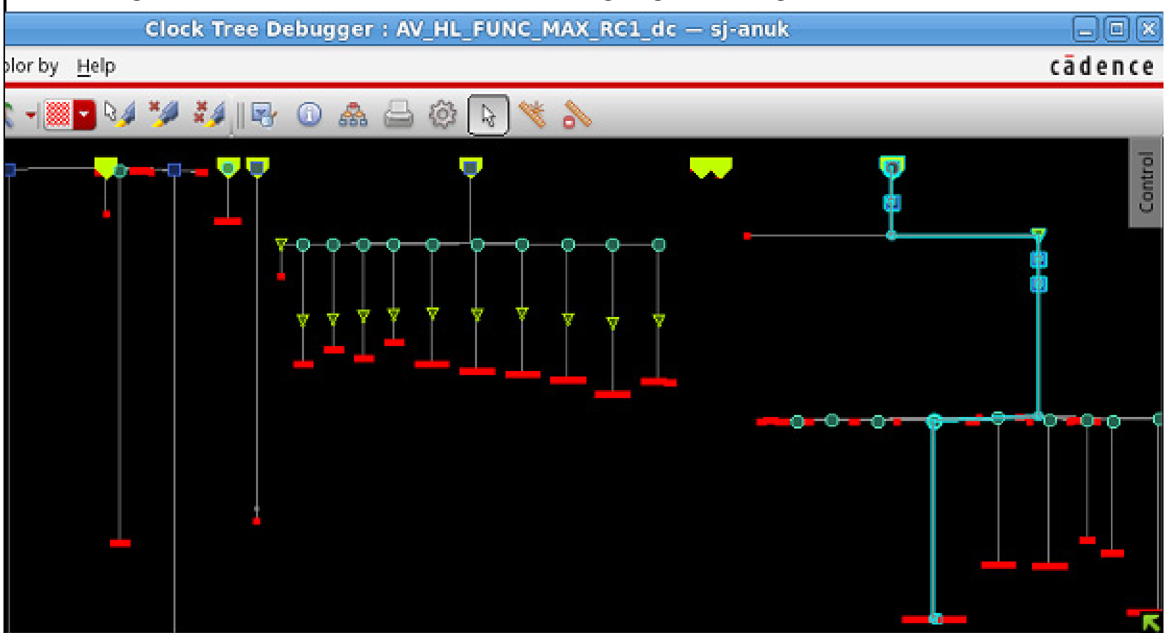


The options are detailed below:

Highlight: Highlights the selected item in the specified color. You can highlight the *Max Path*, *Min Path*, *Both Paths*, and *Entire skew group*. Each of these items has a submenu containing a color picker, as in the *Color by* menu.

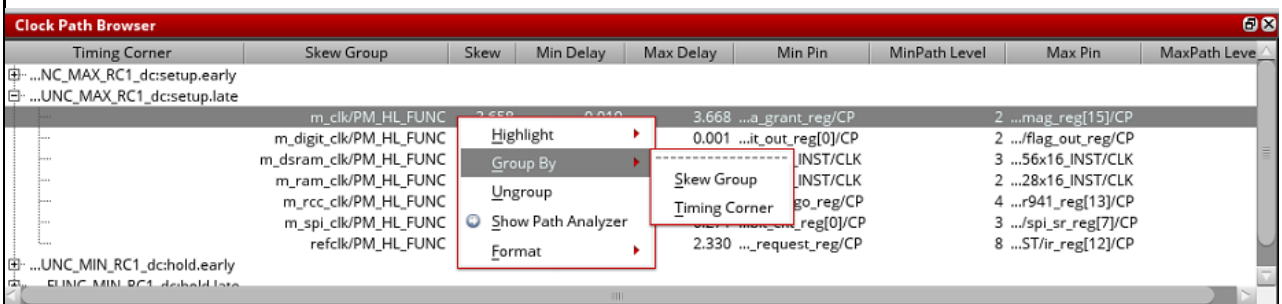


The image below shows the *Max Path* highlighted in green for the selected item.



Group By: You can group the information displayed in the browser by Skew Group or Timing Corner. When Skew Group is selected, the skew group information is displayed in column #1. When Timing Corner is selected, the timing corner information is displayed in column #1.

The image below shows the information grouped by the skew group.

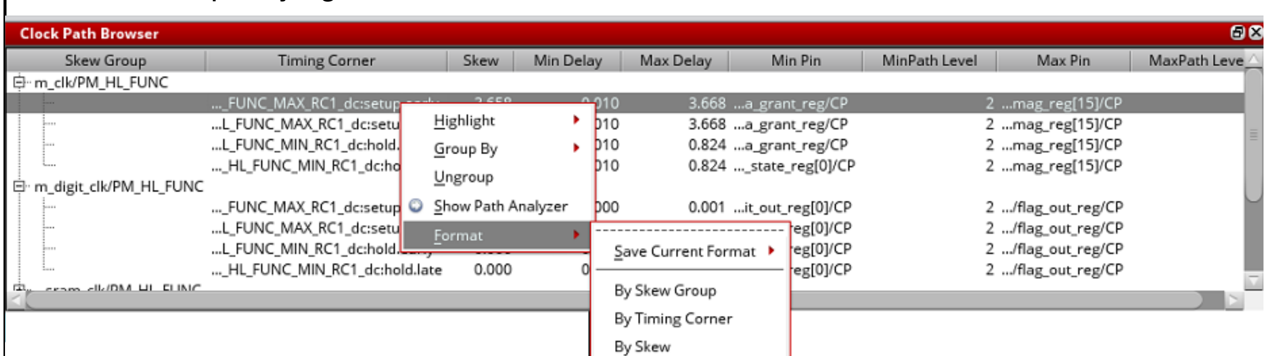


Ungroup: You can ungroup to view all the skew information, which includes both the skew group and analysis view information.

Show Path Analyzer: When you select this option in the context menu, the *Clock Path Analyzer* replaces the *Clock Path Browser* in its window. It provides a tabbed view containing tables for the *min* and m*ax paths*. To go back to the *Clock Path Browser*, click the *Back* button provided in the top-right corner of the window.

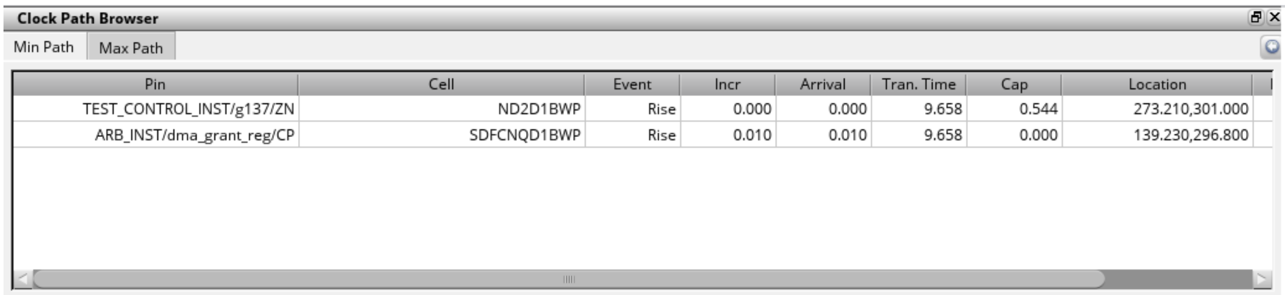
For more information about the analyzer, see the "Clock Path Analyzer on page 352".

Format: Allows you to format the information in the Browser *By Skew Group*, *Timing Corner*, and *By Skew*. When *By Skew Group* is selected, the skew group information is displayed in column #1. When *Timing Corner* is selected, the timing corner information is displayed in column #1.You can also save the current format by selecting *Save Current Format* and specifying the name of the format.



Clock Path Analyzer

Double-click any row in the *Clock Path Browser* to view the *Clock Path Analyzer*. The *Clock Path Analyzer* replaces the *Clock Path Browser* in its window. You can also select the *Show Path Analyzer* option in the context menu of the *Clock Path Browser* to view the *Clock Path Analyzer*.



Each row in the table of the *Clock Path Analyzer* represents a stage in the path in question. The following columns are shown:

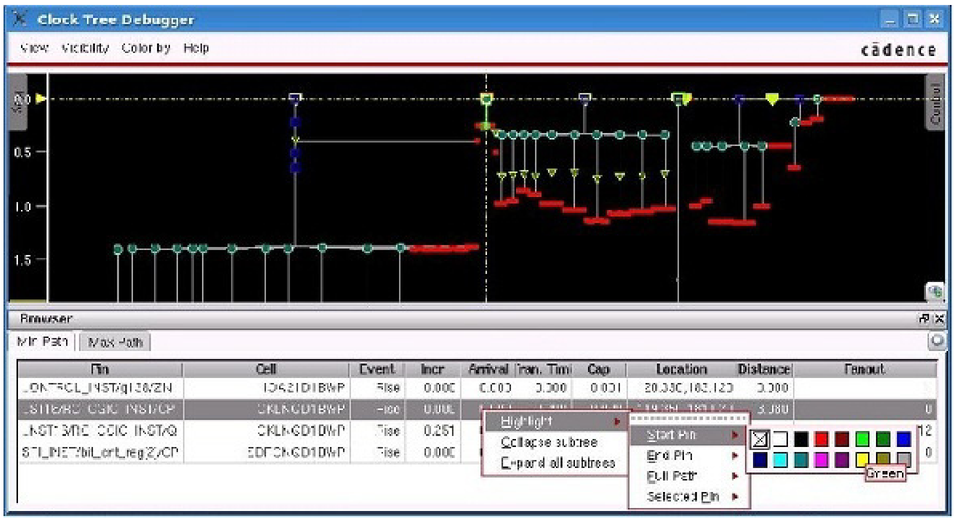
- *Pin* – displays the name of the instance pin to which this row applies
- *Cell* – displays the library cell for the instance
- *Event* – states whether the row relates to a rise or fall event
- *Incr* – displays the delay between this row and the previous one
- *Arrival* – displays the total arrival time of the clock signal at this pin
- *Tran. time* – displays the transition time at this pin
- *Cap* – displays the capacitance at the pin (only for output pins)
- *Location* – displays the location of the pin
- *Distance* – displays the Manhattan distance from the last pin to this one
- *Fanout* – displays the fanout driven by this pin (only for output pins)

Context Menu of the Clock Path Analyzer

Right-click anywhere in the *Clock Path Analyzer* window to view the context menu. The following options are available in this menu:

- Highlight
- Collapse subtree

- Abstract this node
- Unabstract this node
- Expand all subtrees
- Select transitive sink fanout
- Select subtree
- Protect subtree

Options	**Descriptions**
Highlight	You can choose to highlight the *Start Pin*, *End Pin*, *Full Path* and *Selected Pin* in the color of your choice. Each of these items has a submenu containing a color picker, as in the *Color By* menu. The image below shows the *Start Pin* of the selected object highlighted in green. 
Collapse subtree	Collapses the subtree.
Abstract this node	Abstracts the selected node.
Unabstract this node	Unabstracts the selected node.
Expand all subtrees	Expands all the collapsed subtrees.
Select transitive sink fanout	This option is available in the context menu when you right-click objects other than sinks. It selects the transitive sink fanouts for the selected object.

Select subtree	Selects everything under the selected node.
Protect subtree	Protects the subtree from collapsing.

8

Timing & SI Menu

Timing & SI Menu

The Timing & SI Menu has the following submenu options:

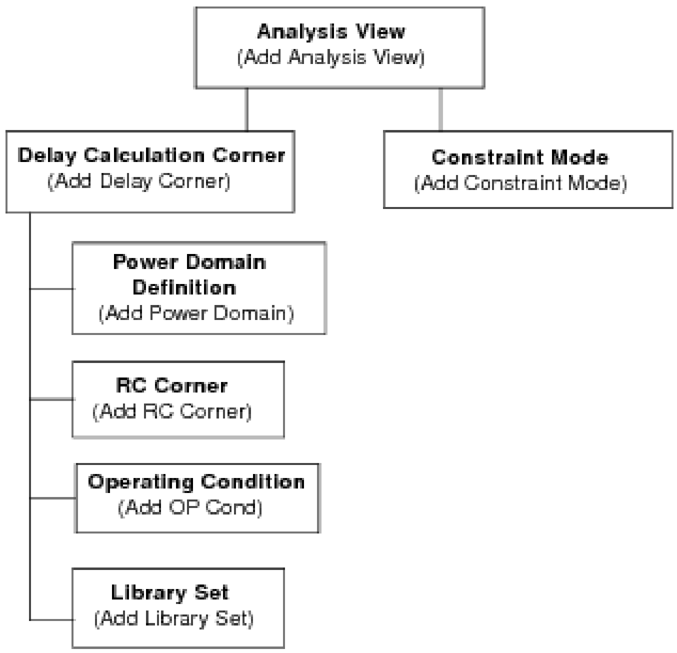
- MMMC Browser
- Extract RC
- Timing
- Debug Timing
- Display Timing Map
- Crosstalk Analysis
- Display Noise Net

These are explained in the following sections.

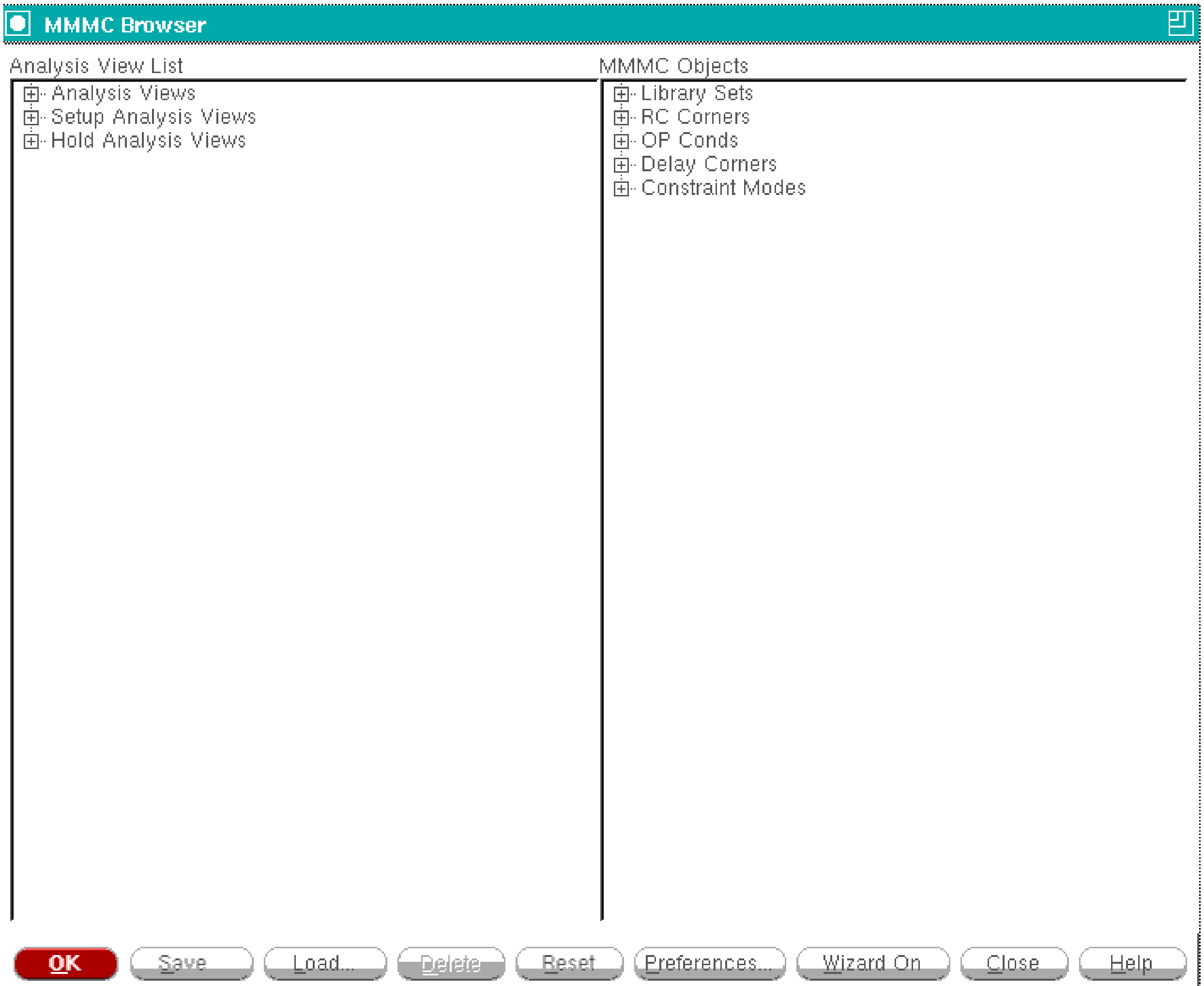
MMMC Browser

Use the MMMC Browser to hierarchically create the view configurations necessary for multi-mode multi-corner analysis. Each top-level set of information is called an analysis view, and is composed of a delay calculation corner and a constraint mode. The active analysis views defined in the software represent the different design variations that will be analyzed.

Hierarchical Analysis View Configuration



- Choose *Design - Configure MMMC*.



You can access the following forms from the MMMC Browser by double clicking on the generic object or analysis view name:

- Add Library Set
- Add OP Cond
- Add RC Corner
- Add Constraint Mode
- Add Delay Corner
- Add Power Domain
- Add Analysis View

- Add Hold Analysis View
- Add Setup Analysis View

You also can access the following editing forms from the MMMC Browser by double clicking on a specific object name:

- Edit Library Set
- Edit OP Cond
- Edit RC Corner
- Edit Constraint Mode
- Edit Delay Corner
- Edit Power Domain
- Edit Analysis View

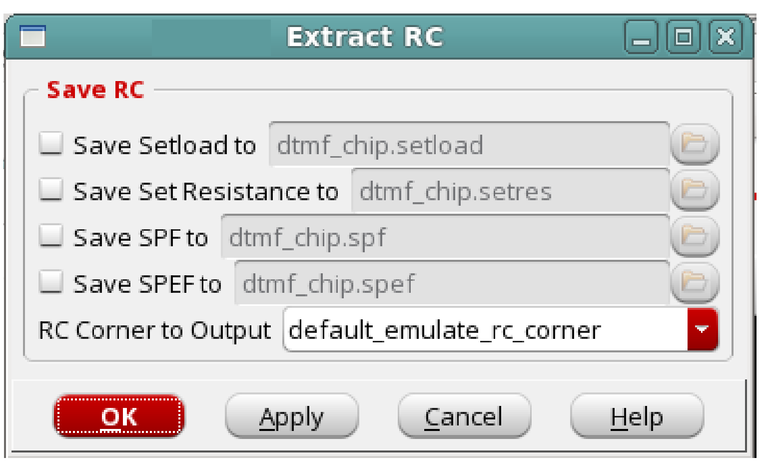
Additionally, you can access the following form to change how the configuration is displayed in the MMMC Browser:

- MMMC Preferences

Extract RC

Use the Extract RC form to run sustainable Quantus extraction in Tempus.

✝ Choose *Timing & SI - Extract RC*.



Fields and Options

Save Setload to	Specifies the name of the `set_load` formatted file in which the value of lump sum capacitance on nets is stored.
Save Set Resistance to	Specifies the name of the `set_resistance` formatted file in which the resistance value on nets is stored.
Save SPF to	Specifies the name of the Standard Parasitic Format file. This format only supports decoupled RC output.
Save SPEF to	Specifies the name of the Standard Parasitic Exchange Format file.
RC Corner to Output	Specifies the RC corner that is used to generate the parasitic output file in the MMMC flow. If you do not specify the corner, the software will use the first corner.

Related Text Commands

The following text commands provide equivalent or additional functionality:

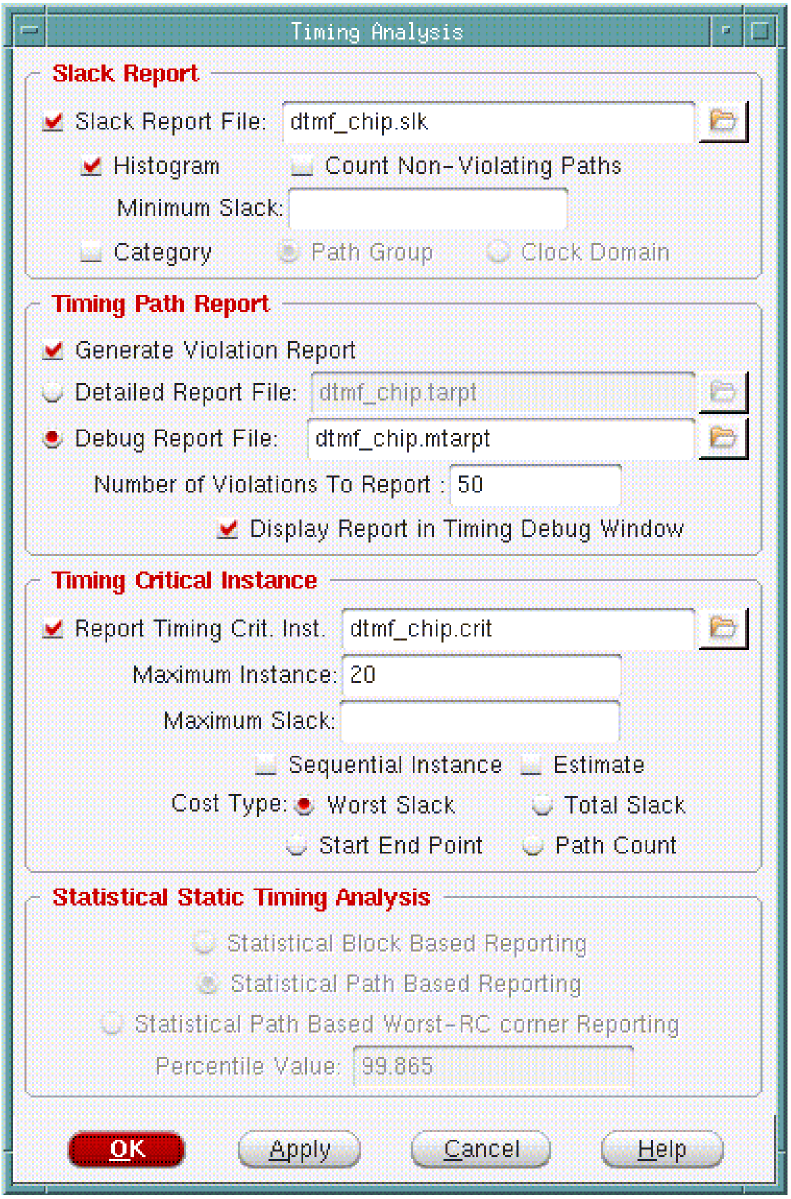
- `extract_rc`
- `rc_out`

For more information, see "RC Extraction Commands" chapter in the *Tempus Text Command Reference*.

Timing

Use the Timing Analysis form to build the timing graph for the design and generate a slack report and a detailed timing violation report.

- Choose *Timing & SI - Timing*.



Fields and Options

Slack Report File	Reports the end point slacks in the design.
Histogram	Generates a histogram of end-point slacks. *Default:* On

Count Non-Violating Paths	
	Generates a histogram consisting of violating and non-violating paths in the design.
Minimum Slack	Specifies the higher bound of the histogram.
Category	Determines on what category the timid report is based.
Path Group	Reports the timing based on the default path groups, including input-to-register, register-to-register, and register-to-output.
Clock Domain	Report the timing based on clock domains.
Generate Violation Report	
	Generates a timing report that provides information about the various violating paths in the design.
Detailed Report File	Generates a detailed timing report with the specified file name.
Debug Report File	Generates a violation report in machine-readable format with the specified name. This report is used for debugging timing results using the timing debug feature.
Number of Violations To Report	
	Specifies the number of worst paths in the design on which to report, regardless of the endpoint. This is useful, but can be time consuming if a large number of paths is specified. The paths are always sorted based on slack.
Display Report in Timing Debug Window	
	Displays the generated .mtarpt file in the *Analysis* tab.
Report Timing Crit. inst.	
	Reports timing critical instance (bottleneck) information to the specified file.
Maximum Instance	Specifies the number of worst cost instances to report. *Default:* 20
Maximum Slack	Specifies the lower bound of the slack range. Only paths with slack worse than the specified slack value are reported. *Default:* 0
Sequential instance	Includes the sequential instances group in the output. *Default:* Off

Estimate	Generates a critical instance report based on the fast algorithm (linear-complexity) to give a slightly pessimistic estimation that may include some false paths. Select this option in early design phase. *Default:* Off
Cost Type:	Specifies the criteria for choosing the critical instance. You can use the *Maximum Slack* and *Cost Type* options together to filter the paths to only those with slack worse than the specified slack value, then report the filtered set of paths based on the specified cost.
Worst Slack	Sets the criteria as the worst slack of all points that pass through an instance. *Default:* On
Total Slack	Sets the criteria as a sum of the slack of the end points. *Default:* Off
Start End Point	Sets the criteria as the sum of the total start and end points connected to the instance. *Default:* Off
Path End Count	Sets the criteria as the sum of the total paths through the instance. *Default:* Off
Statistical Block Based Reporting	
	Reports timing analysis results based on block-based statistical static timing analysis (SSTA).
Statistical Path Based Reporting	
	Reports timing analysis results based on path-based SSTA. By default, the software runs block-based SSTA.
Statistical Path Based Worst-RC corner Reporting	
	Reports timing analysis results based on path-based SSTA with worst RC corner.
Percentile Value	By default, the percentile value is 99.865.

Related Text Commands

The following text command provides equivalent or additional functionality:

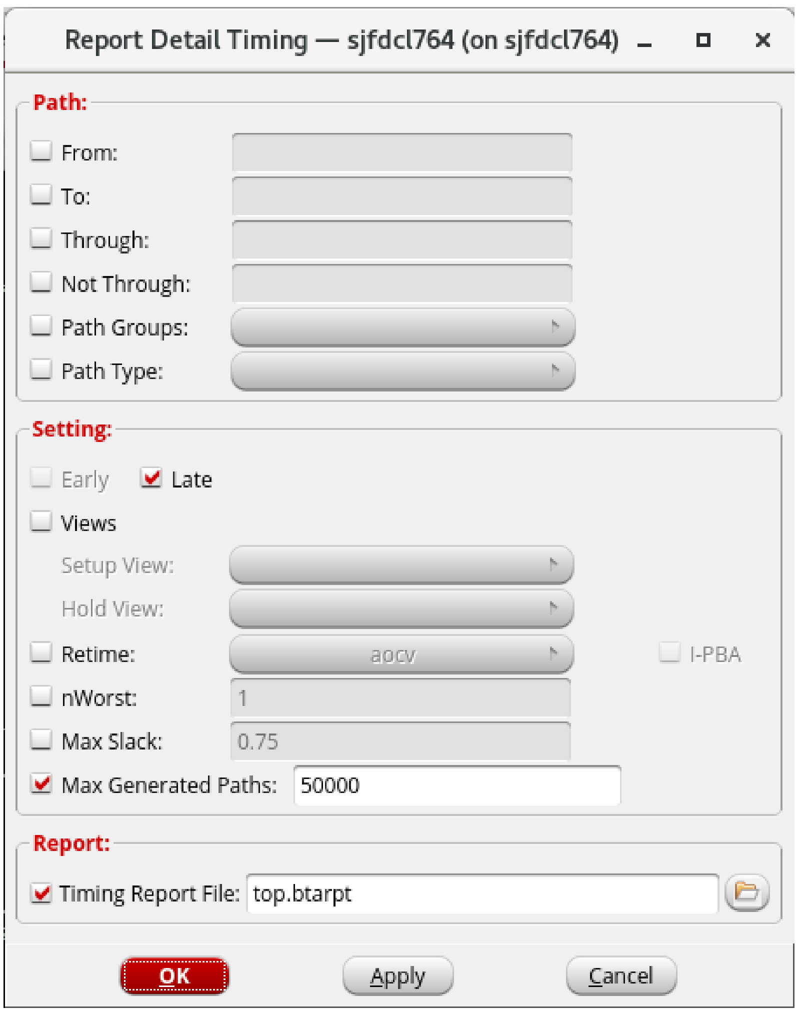
- report_slack_histogram
- report_timing
- report_critical_instance

For more information, see "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Report Detail Timing

Use the Report Detail Timing form to select options for customizing the timing reports.

- Choose *Timing & SI - Report Detail Timing.*



Fields and Options

Path	
From	Selects the paths starting from the object(s). The supported objects are pins and ports.
To	Selects the paths leading to the pin(s).
Through	Selects the paths that traverse through pin(s) or net(s).
Not Through	Selects the paths that do not traverse through the specified nets, ports, or pins of a cell.
Path Groups	Selects the paths that are contained in path groups.
Path Type	Specifies the format of the report by path type. You can choose one of the following options: ■ end ■ summary ■ full ■ full_clock ■ end_slack_only ■ summary_slack_only
Setting	
Early	Generates the timing report for early paths (hold checks).
Late	Generates the timing report for late paths (setup checks).
Views	When you select this option: ■ Select "Setup Views" if you choose Late or Check Type as Setup/ Setup & Hold ■ Select "Hold Views" if you choose Early or Check Type as Hold/ Setup & Hold

Retime	Reanalyzes the specified set of paths using path-based analysis (1 path per endpoint). You can specify the following options: ■ aocv ■ path_slew_propagation ■ aocv_path_slew_propagation
I-PBA	Enables exhaustive path-based analysis mode (IPBA)
nWorst	Specifies the maximum number of paths which can be reported for a particular endpoint. *Default*: 1
Max Slack	Reports only those paths with slack less than the specified value. *Default*: 0.75
Max Generated Paths	Reports the specified number of worst paths in the design. *Default*: 50000
Report	
Timing Report File	Specifies the name of the timing report to be generated.

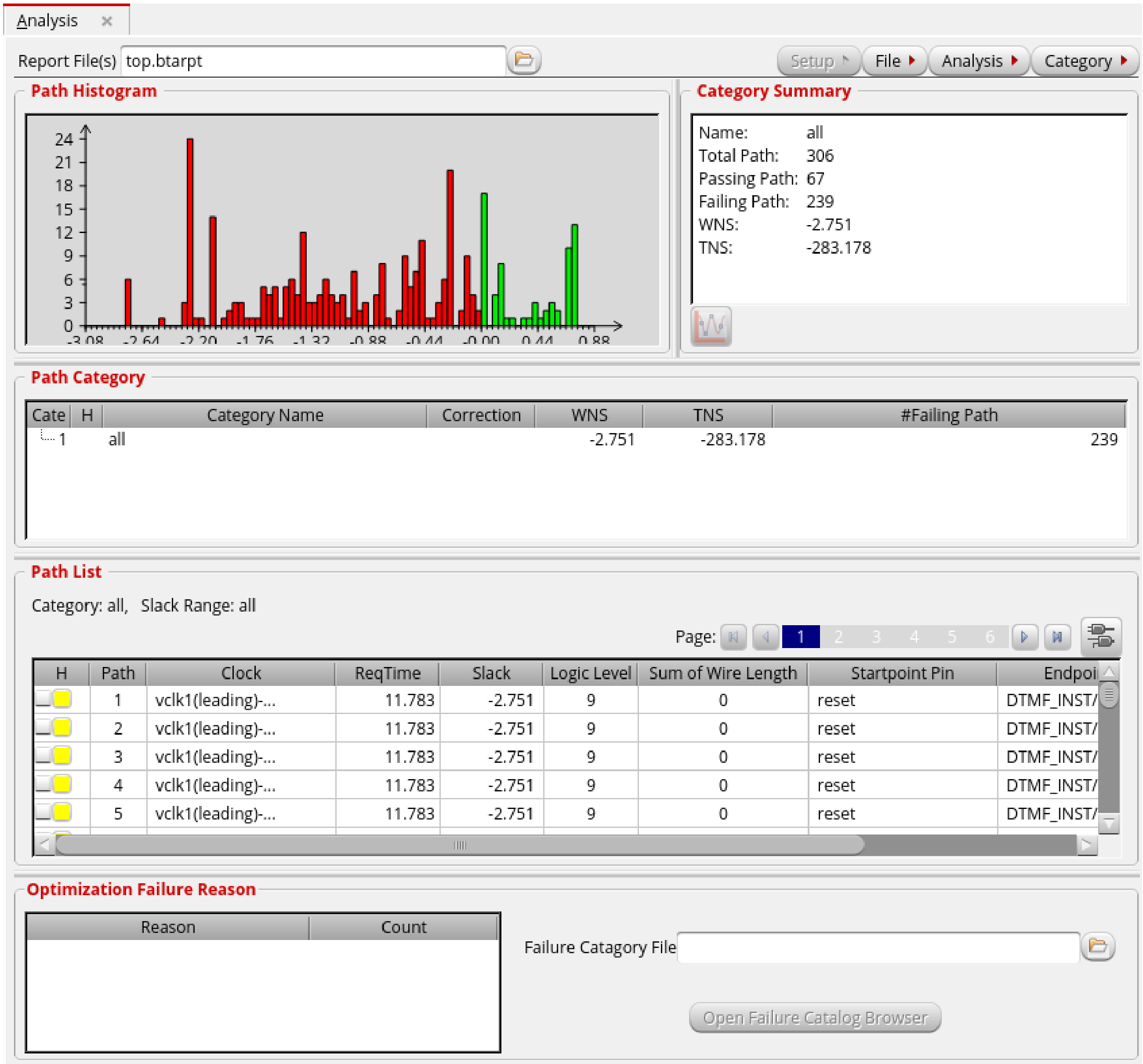
Related Information

- `report_timing`
- "Timing Analysis Commands" in the *Tempus Text Command Reference*

Debug Timing

Use the Timing Debug form to view the debugging timing results.

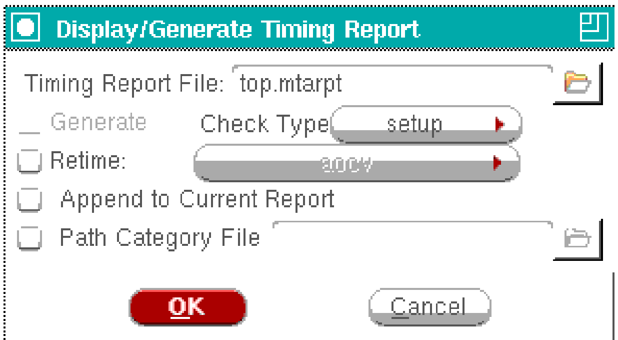
- Choose *Timing & SI - Debug Timing.*



Display/Generate Timing Report

Use the Display/Generate Timing Report form to specify the violation report to be read in for debugging timing results.

- Choose Timing & SI - Debug Timing



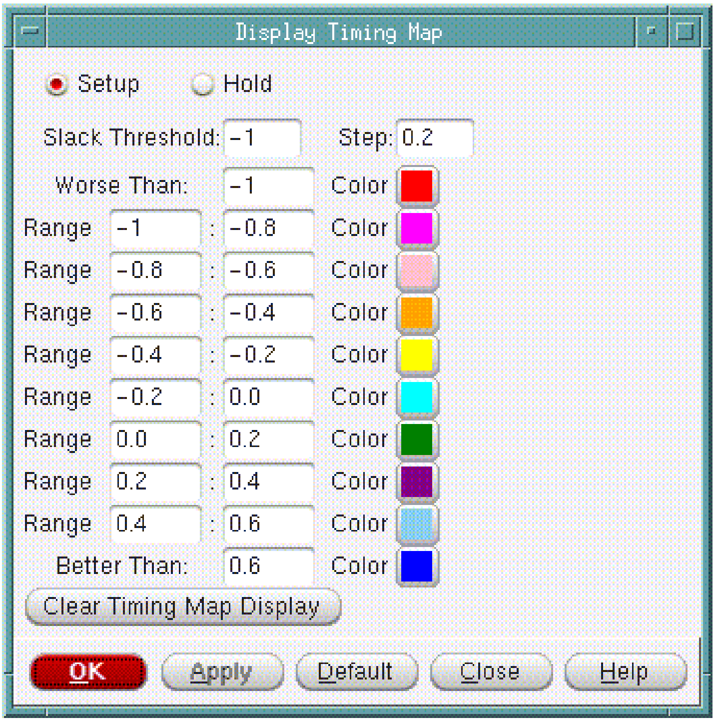
Fields and Options

Timing Report File	Specify a filename, or click the file folder icon to choose a file.
Generate	Generates a violation report file in machine readable format. *Default:* `top.mtarpt`
Check Type	Can be selected for the type of Check: *Setup* or *Hold*. Previously this was available only from the command line.
Retime	Specifies the advanced timing methods (`aocv`, `ssta`, `path_slew_propagation`, and `waveform_propagation`).
Append to Current Report	
	Appends the specified report to the report that you have already loaded in a timing debug session. This feature is useful to collate reports from several views into one timing debug session.
Path Category File	Specifies the path category file. The path category file contains category definitions that you saved in an earlier timing debug analysis.

Display Timing Map

Use the Display Timing Map form to set color-coded slack threshold values for worst timing paths.

- Choose *Timing & SI - Display Timing Map*



Fields and Options

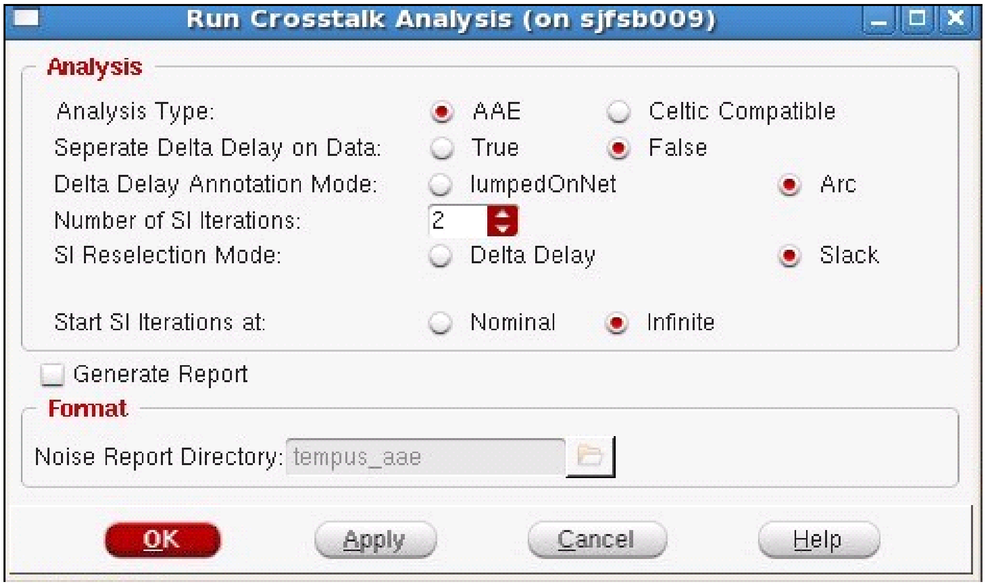
Setup	Specifies to display Setup timing paths.
Hold	Specifies to display Hold timing paths.
Slack Threshold	Sets color-coded slack threshold values for worst timing paths. You can specify a range of threshold values. When set, the instances as seen on the Timing Debug form are color-coded according to the worst timing slack threshold value.
Clear Timing Map Display	Clears the timing slack threshold values.

Tempus will switch between Setup and Hold mode seamlessly and easily without re-calculation, once timing calculation for each mode is done.

Crosstalk Analysis

Use the Run Crosstalk Analysis form to perform signal integrity analysis.

- Choose *Timing & SI - Crosstalk Analysis*



Fields and Options

Analysis Type	Specifies the type of crosstalk analysis to be performed. You can choose to specify the AAE or CeltIC Compatible noise-analysis modes. ■ *AAE:* Specifies the default mode for AAE to run. This will have the best runtime/memory while maintaining correlation to SPICE. ■ *Celtic Compatible:* Specifies the CeltIC-compatibility mode. This option may be slower and use more memory than the default AAE mode.
Separate Delta Delay on Data	Separates the delta delay on data (delta delay is always separated for clock). Works with `-delta_delay_annotation_mode` to determine where the separated delay shows in the timing reports. Default is the delta delay is not separated out.Using this parameter, you can increase runtime/memory. *Default:* `false`

Delta Delay Annotation Mode	Calculates the delay for each arc (cell arc and net arc) in a path. By default, AAE calculates the delay, including SI for each arc and if the user chooses to view the SI delay separately, it would show annotated to each arc. Alternatively, you can choose to annotate the SI delay solely to the net. In this case, the worst-case arc is used to calculated the SI delay and it is annotated solely to the net. ■ *lumpedOnNet:* Lumps the delta delay all onto the net if it is broken out. ■ *Arc:* Separates the delta delay onto each cell arc and net arc as calculated.
Number of SI Iterations	Specifies the maximum number of timing window iterations that should be performed during SI delay analysis. Performing delay analysis results in changes to the timing windows, which can potentially change the victim/attacker net combinations. Timing window iterations enable the software to reach to a point where the delay changes are negligible.
SI Reselection Mode	Specifies the SI reselection criteria during timing window iterations. ■ *Delta Delay:* Reselects a net for calculation if the delta delay on the net is greater than the delay value. ■ *Slack:* Reselects nets that have failing slacks during timing window iterations.
Start SI Iterations at	Instructs AAE where to start the initial SI analysis during timing window iterations. ■ *Nominal:* During timing window iterations, uses nominal timing windows first. ■ *Infinite:* During timing window iterations, uses infinite timing windows first.
Generate Report	Generates glitch noise report when performing noise analysis. When you choose AAE as the type of crosstalk analysis, you need to select this option and the generated glitch noise report gets loaded in the SI tab.
Noise Report Directory	Specifies the noise report directory, default is `tempus_aae`.

Related Text Commands

The following text command provides equivalent or additional functionality:

- report_noise

For more information, see "Signal Integrity Commands" in the *Tempus Text Command Reference*.

Related Topics

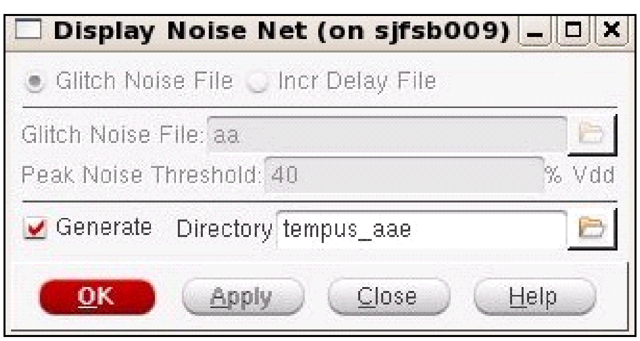
For information on the following topics, see "Performing Signal Integrity Delay Analysis", and "SI Glitch Reporting" in the *Tempus User Guide*.

Display Noise Net

Use the *Display Noise Net* form to browse aggressor information for each of the glitch failures and to locate the glitch nets (aggressor and victim) in the physical layout for ECO fixing. You can generate the required input files for this form by using the write_noise_eco -format edi command. This command generates the following files in the tempus directory (default) that can be loaded in the *Display Noise Net* form:

- aae_<view_name>_max_incDelay
- aae_<view_name>_min_incDelay
- aae_<view_name>

- Choose *Timing & SI - Display Noise Net*

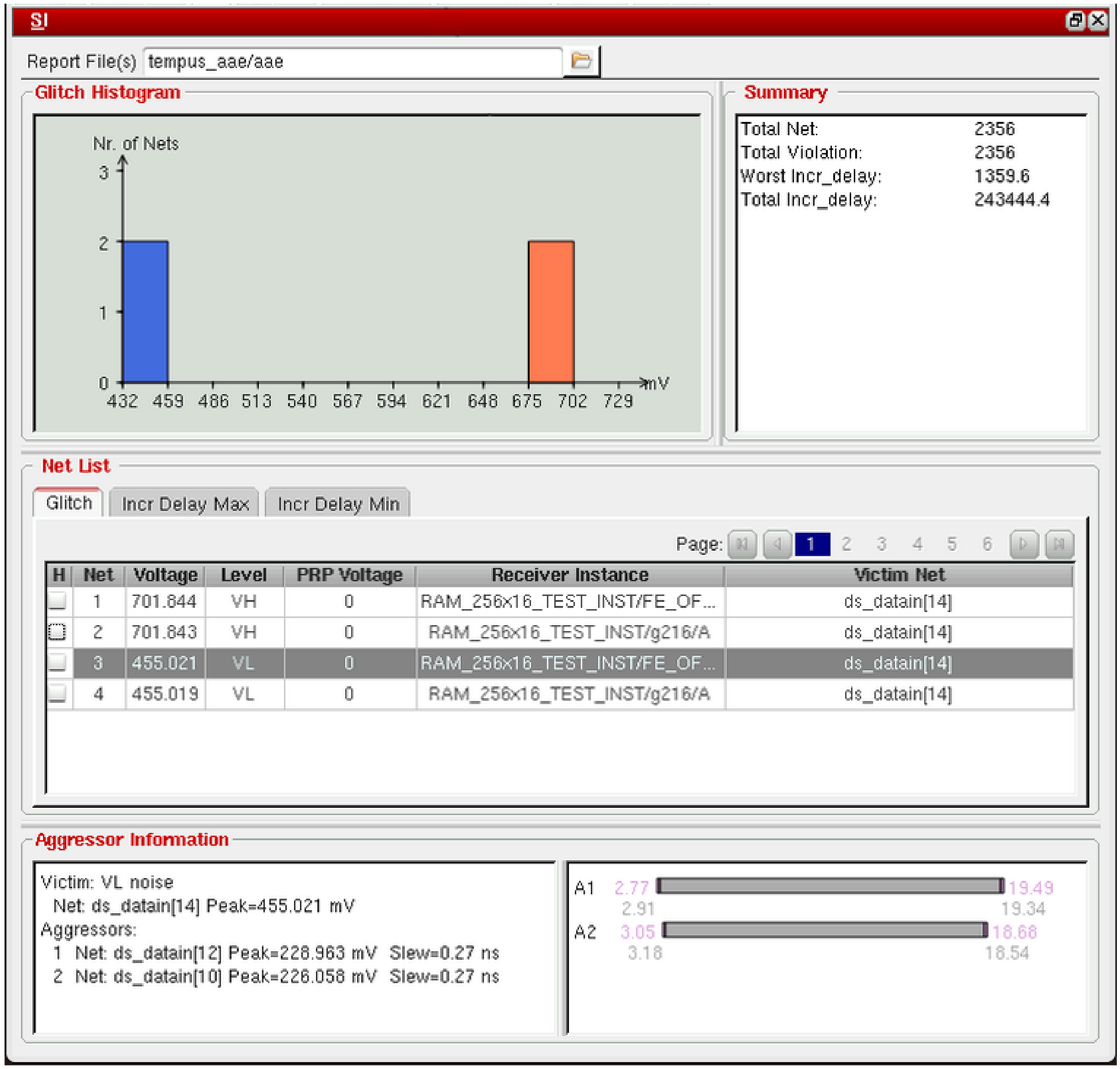


Fields and Options

Glitch Noise File	
Glitch Noise File	Enables you to load the glitch victim noise file (`aae` or `aae_<viewname>` files under the reporting directory). When this option is selected, only the Glitch Noise File field is available.
Peak Noise Threshold	Specifies the peak noise threshold value (as a percentage of VDD) to filter the number of nets to be displayed. *Default:* 40% of VDD
Incr Delay File	
Incr Delay File	Enables you to load the incremental delay file (`aae_<max/min>_incDelay` or `aae_<viewname>_<max/min>_incDelay` files under the reporting directory).
Incr Delay Threshold	Specifies the incremental delay threshold value to filter the number of nets to be displayed. *Default:* 40ps
Generate	Generates noise report when performing noise analysis. For example, if there is only one view, the following files are generated: ■ `aae` ■ `aae_min_incDelay` ■ `aae_max_incDelay` When you select this check box, it enables the *Glitch Noise File* and I*ncr Delay File* fields respectively.
OK	Opens the *SI* tab and closes the *Display Noise Net form*. The *SI* tab shows a list of noise nets.
Apply	Applies the settings.

SI Tab

A sample SI tab is as shown below:



9

Verify Menu

Verify Connectivity

Use the Verify Connectivity form to detect conditions such as opens, unconnected wires (geometric antennas), unconnected pins, loops, partial routing, and unrouted nets. When you verify connectivity, the software generates violation markers in the design window and reports violations. Verifying connectivity does not have database impact unless you save the design, which also saves the violation markers.

Types of Violations Reported

- Antennas (Dangling Wires)
- Unconnected wires (dangling wires).
- Opens: Parts of nets, such as wires or pins, that are connected to each other but are missing a connection to the net as a whole. Marks each part of a net that is missing a connection as an open and displays a violation marker between the parts.
- Loops
- Unconnected pins: Pins that are not connected to any other objects

Checks for Opens, Unconnected Pins, and Unrouted Nets

The following combinations of connectivity checks are not enabled:

Open	*Unconnected Pin*	*Unrouted Net*
Off	On	On
Off	On	Off
Off	Off	On

Choose *Verify - Verify Connectivity.*

.

Verify Connectivity Fields and Options

Net Type	Select one of the following options. *Default: All*	
	All	Checks all nets, including those that have been previously verified.
	Regular Only	Checks only regular nets.
	Special Only	Checks only special nets.
Nets	Select one of the following options. *Default: All*	

	All	Checks all nets of the type specified in the *Net Type* panel.
	Selected	Checks only nets in the selected set that are of the type specified in the *Net Type* panel.
	Named	Checks only nets of the name specified that are of the type specified in the *Net Type* panel. You can use wildcards (*** and *?*) when you specify the net names.
Check	Select one or more of the following options:	
	Open	Checks for open violations. Opens are parts of nets, such as wires or pins, that are connected to each other but are missing a connection to the net as a whole.
	UnConnected Pin	Checks for pins that are not connected to any objects.
	Unrouted Net	Checks for nets that are not routed.
	Connectivity Loop	Checks for connectivity loops in regular wires. The software detects connectivity loops based on the end points of the center line of a regular wire segment or the center of a via. *Connectivity Loop*, *Geometry Loop*, and *Geometry Connectivity* are mutually exclusive.
	DanglingWire Antenna	Checks for violations due to unconnected wires (also called geometrical antennas or dangling wires).
	Weakly Connected Pin	Checks for routing to more than one port of the weakly connected pin ports. When this option is selected, Verify Connectivity marks a Weakly Connected Pin violation in one of the ports of the pin.

	Geometry Loop	Checks for loop violations of regular nets using a geometrical model. The nets do not have to overlap on the center line. When you specify this option, the software does not perform any other connectivity checks. Select this option if you use a third-party router that does not route using the centerline connection routing technique. In this case, *Connectivity Loop* might not detect connectivity loop violations. *Connectivity Loop*, *Geometry Loop*, and *Geometry Connectivity* are mutually exclusive.
	Geometry Connectivity	Checks for connectivity violations of regular wires. Uses a geometrical model instead of the center-line model. In other words, if the wires overlap at any point, they are considered to be connected--they do not have to connect at the center line. Use this parameter if you manually change the routing or use a third-party router that does not route using the center-line connection routing technique. *Connectivity Loop*, *Geometry Loop*, and *Geometry Connectivity* are mutually exclusive.
	Keep Previous Results	Displays incremental results in the Violation Browser. When this option is selected, Verify Connectivity appends violation markers to the previous results instead of overwriting them.
	Divide Power Net	Divides power nets into four subareas for connectivity verification. Use this option to decrease memory usage in 32-bit machines with limited memory. This option might increase or decrease the run time, depending on the design.
	Soft PG Connect	Checks soft Power/Ground connects on masterslice layers. This option is enabled by default. Deselect this option if you want connectivity on masterslice layers to be ignored.

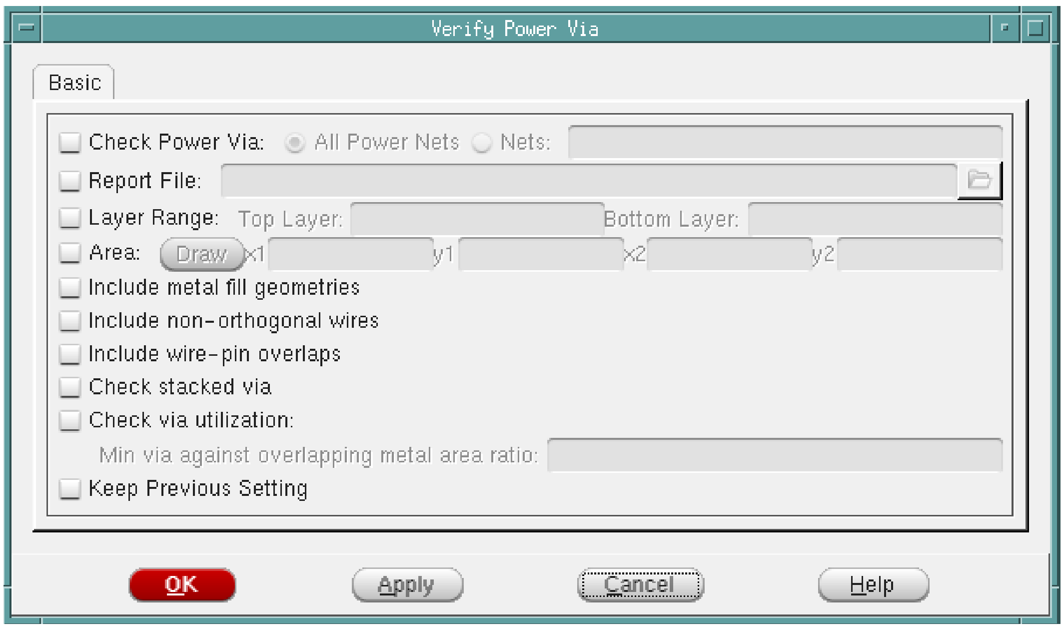
	Raw Violations Mark	Displays violation markers for opens as the bounding box of the island. By default, violation markers for opens are displayed as polygons that include all wires, pins, and vias that connect to the island.
	Use Virtual Connection	Implies a virtual connection for all bumps and external I/O pins of the same net. Select this option to override default behavior of Verify Connectivity in which bumps and external I/O pins of the same net are not considered to be interconnected. It is useful in flip chip designs, where the power bumps are connected outside the chip.
	TSV Die Abstract File	Specifies the input file for TSV die.
Verify Connectivity Report		
	Specifies the report file for connectivity violation data.	
Report Limits	Specifies error and warning limits	
	Error	Specifies the maximum number of errors to report. The software generates a warning message if this number is exceeded. *Default: 1000*
	Warning	Specifies the maximum number of warnings to report. *Default: 50*
Set Multiple CPU	Displays the Multiple CPU Processing form in which you can specify multi-threading settings for verifying connectivity.	

Verify Power Via

Checks for metal geometry overlap on power-grid nets and for missing or open vias in the design.

Note: To verify the design, the power-grid must be routed.

Choose *Verify - Verify - Power Via*.



Fields and Descriptions

Check Power Via	*All Power Nets*: Verifies power vias in all the power nets. This is the default option. *Nets*: Verifies power vias in the specified net(s).
Report File	Specifies the name of the output file for the report.

Layer Range: *Top Layer* *Bottom Layer*	Optional. The layer range checks for missing stacked vias between only bottom and top layers. It also checks for missing vias between all intermediate layer intersections. You can specify standard metal layer names, such as *M1*, *M2*, and so on, or metal layer names as defined in the LEF file. *Default:* All layers. If *Bottom Layer* is specified but *Top Layer* is not , it assumes one layer up.
Area: *Draw x1 y1 x2 y2*	When selected, enables you to draw a box in the display canvas to fill in *x1 y1 x2 y2* values automatically. Optionally, you can also specify *x1 y1 x2 y2* coordinates.
Include metal fill geometries	Specifies that metal fill should also be checked. *Default:* Metal fill will be ignored.
Include non-orthogonal wires	Specifies that non-orthogonal crossing wires should also be checked. *Default:* Non-orthogonal wires will be ignored.
Include wire-pin overlaps	Specifies that wire-pin overlaps should also be checked. *Default:* Wire-pin overlaps will be ignored.
Check stacked via	Checks for missing vias between all non-adjacent as well as adjacent layers. *Default:* Missing stacked vias will be ignored.
Check via utilization	Checks the via utilization.
Min via against overlapping metal area ratio	Specifies the minimum via area against overlapping metal geometry ratio. The value must be between *0* and *1*.
Keep Previous Setting	Specifies that all previous *verifyPowerVia* settings will be kept. *Default:* Previous settings are not kept.

10

Tools Menu

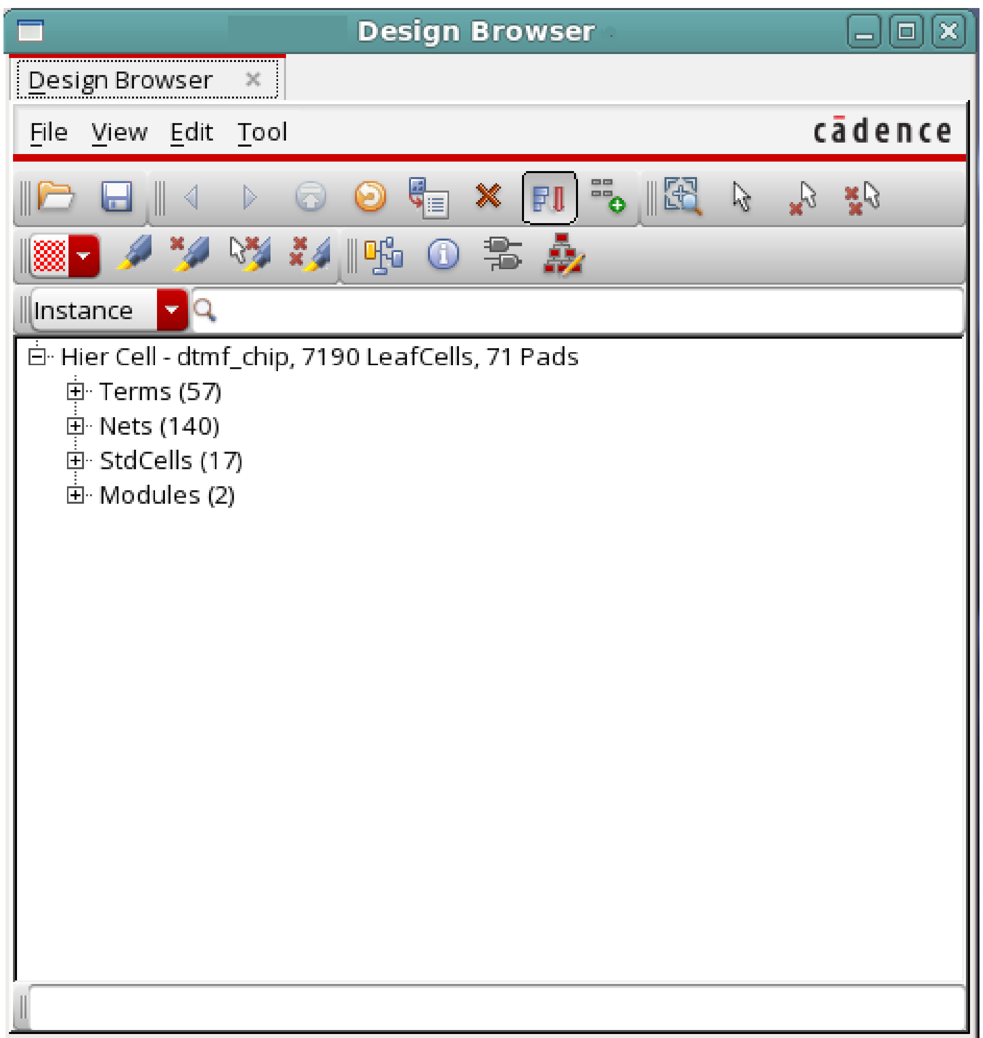
Design Browser

Use the Design Browser to navigate through the chip's design hierarchy. You can use the Design Browser to view the design hierarchy tree at any time after the design is imported. You can use the widgets in the Design Browser to open forms to navigate through displays, and perform actions.

From the Design Browser, you can access the:

- Connectivity Browser: To display the number of nets between instances.
- Attribute Viewer: To display an object's type, name, and attributes.
- Schematic Viewer: To display hierarchical and instance-based schematics.

☨ Choose *Tools - Design Browser*.



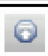
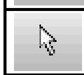
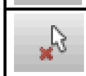
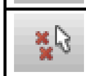
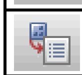
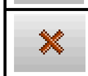
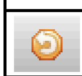
Fields and Options

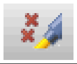
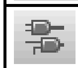
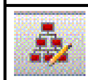
File	Provides the following options:
Load	Displays the Load Design Browser File form. Enter the filename and the format of the placement file.
Save As	Displays the Save Design Browser File form. Specify the filename. The recommended extension is .dbf.
Quit	Closes the Design Browser Window.
View	Provides the following options:
Refresh	Redraws the Design Browser display.
Sort By Size	Displays instances in the order of the number of cells in the instances. Deselect the *View - Sort By Size* option to display instances sorted alphabetically by name.
Accumulate Results	Traces the hierarchical path for the selected object in the design display area.
Edit	Provides the following options:
Delete Group/Group Member	
	Deletes an individual instance from a user-defined group, or to delete an entire user-defined group.
Tool	Provides the following options:
Get Selected	Displays a selected object in the Design Browser
Attribute Viewer	Opens the Attribute Viewer. For more information, see Attribute Viewer on page 390.
Connectivity Browser	Opens the Connectivity Browser. For more information, see Connectivity Browser on page 388.
Find	Enables you to search and display hierarchy information for a specific object. You can view information on the following object types:
Instance	Enables you to search for an instance and display its hierarchy in the Design Browser window. You can traverse to the next hierarchy by double-clicking on the instance name. Along with the instance name, the instance's cell type and number of cells are displayed.

Net	Enables you to search for a net and display its hierarchy in the Design Browser window. Double-clicking the net name displays the net connection.
Group	Enables you to search for an instance group and display its hierarchy in the Design Browser window.
Cell	Enables you to search for a cell and display its hierarchy in the Design Browser window.

Design Browser Tool Widgets

You can use the widgets in the Design Browser to navigate through displays, open tools, make selections, and perform actions.

Widget	Description
	*Previous*--Displays viewable data in a reverse direction.
	*Next*--Displays viewable data in a forward direction.
	*Top Page*--Returns to the top of the design.
	*Connectivity Browser*--Opens the Connectivity Browser.
	*Select*--Selects an object in the design display area.
	*Deselect*--Deselects an object in the design display area.
	*Deselect All*--Deselects all objects in the design display area.
	*Get Selected*--Gets the selected object and displays it in the Design Browser.
	*Attribute Viewer*--Opens the Attribute Viewer for the highlighted instances.
	*Delete Group/Group Member*--Deletes selected floorplan object instance groups.
	*Refresh*--Refreshes the Design Browser display.
	*Zoom Selected*--Zooms in on a highlighted selection.
	*Highlight*--Highlights an object in the design display area.
	*Dehighlight*--Dehighlights an object in the design display area.

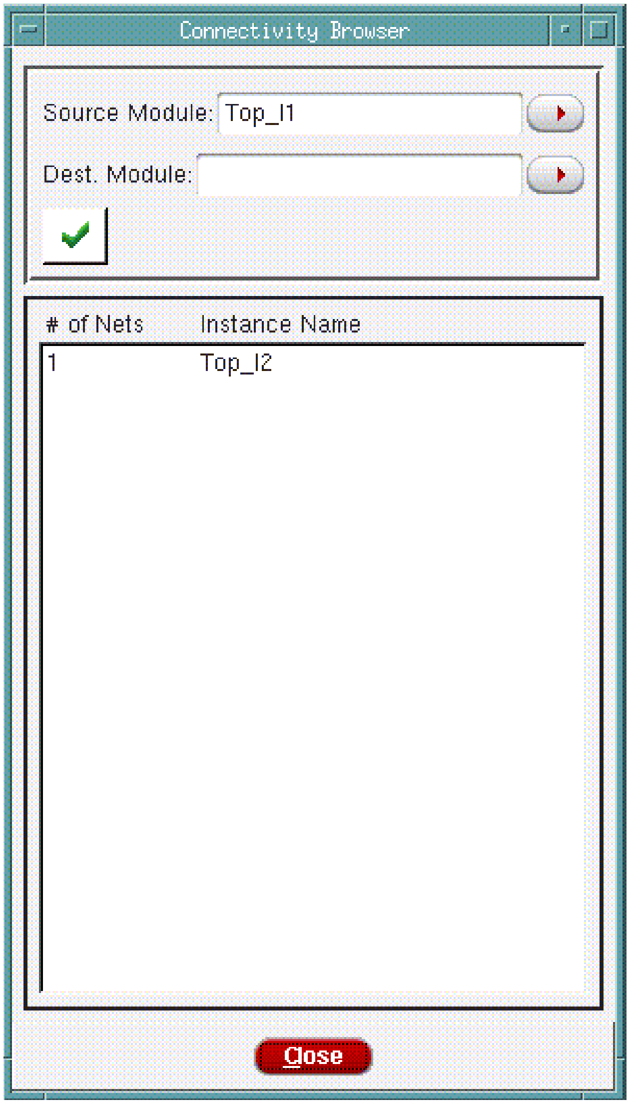
	*Dehighlight All*--Dehighlights all objects in the design display area.
	*Show Instance Based Schematic*--Opens the Schematic Viewer to display schematic in flattened mode for an instance, net, or a pin. For more information, seeSchematic Viewer on page 391.
	*Show Hierarchical Schematic*--Opens the Schematic Viewer to display schematic in hierarchical mode for a module, instance, net, or a pin. For more information, see Schematic Viewer on page 391.

Connectivity Browser

Use the *Connectivity Browser* to display connections from a source module to a destination module, including module guides, blocks, I/O blocks, and standard cells.

- Choose *Tools* - *Design Browser*.

Highlight a module in the Design Browser and click the *Connectivity Browser* widget.



Fields and Options

Source Module Instance	Specify the name of the instance
Destination Module	Use this pull-down menu to select the destination module.
Check	Displays the connection information.

Attribute Viewer

Use the *Attribute Viewer* to display an object's type, name, and attributes. The contents of the Attribute Viewer form change depending on the object selected in the design.

To open the *Attribute Viewer* from the *Layout* window, do the following:

- For a single object, double-left click on an object.

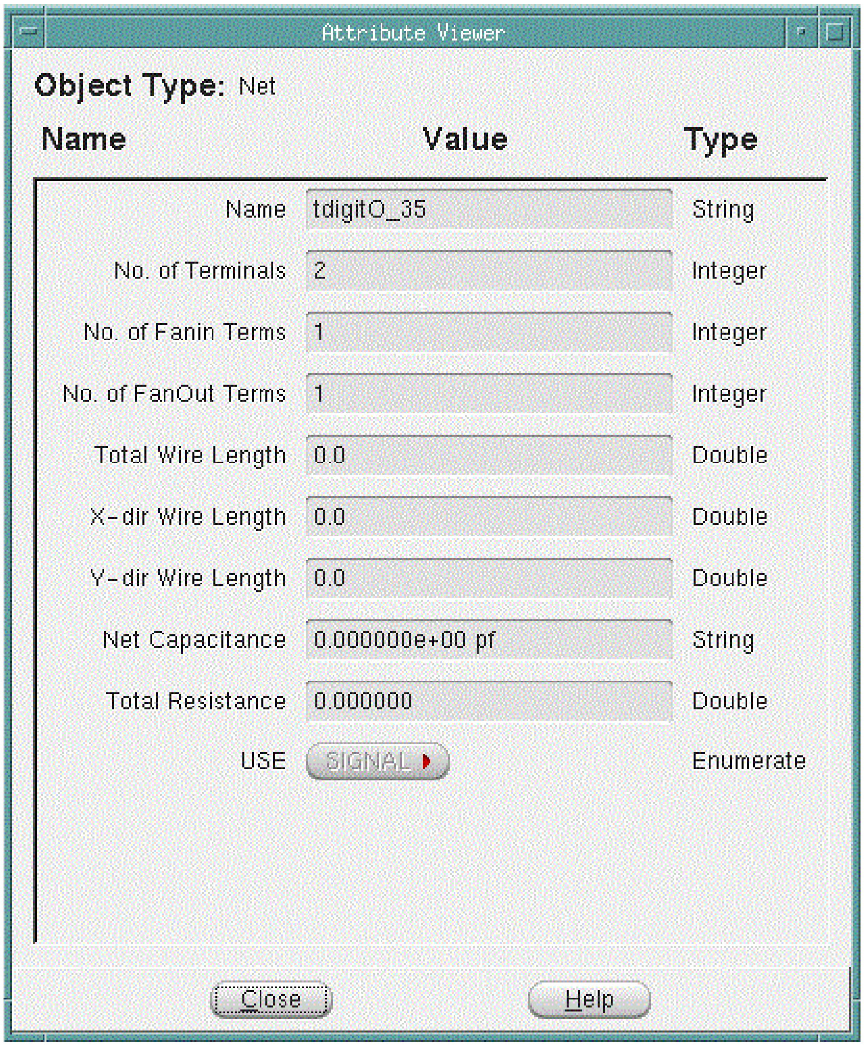
To open the *Attribute Viewer* from the *Design Browser*, do the following:

- Choose *Tools* - *Design Browser*.

In the Design Browser, highlight or left click on an object or a set of objects.

- Click the *Attribute Viewer* from the *Design Browser* tool widgets.

Note: The *Attribute Viewer* displays different attributes, depending on the objects selected.



Fields and Options

Object Type	Displays the type of object currently selected.
Close	Closes the Attribute Viewer window.

Schematic Viewer

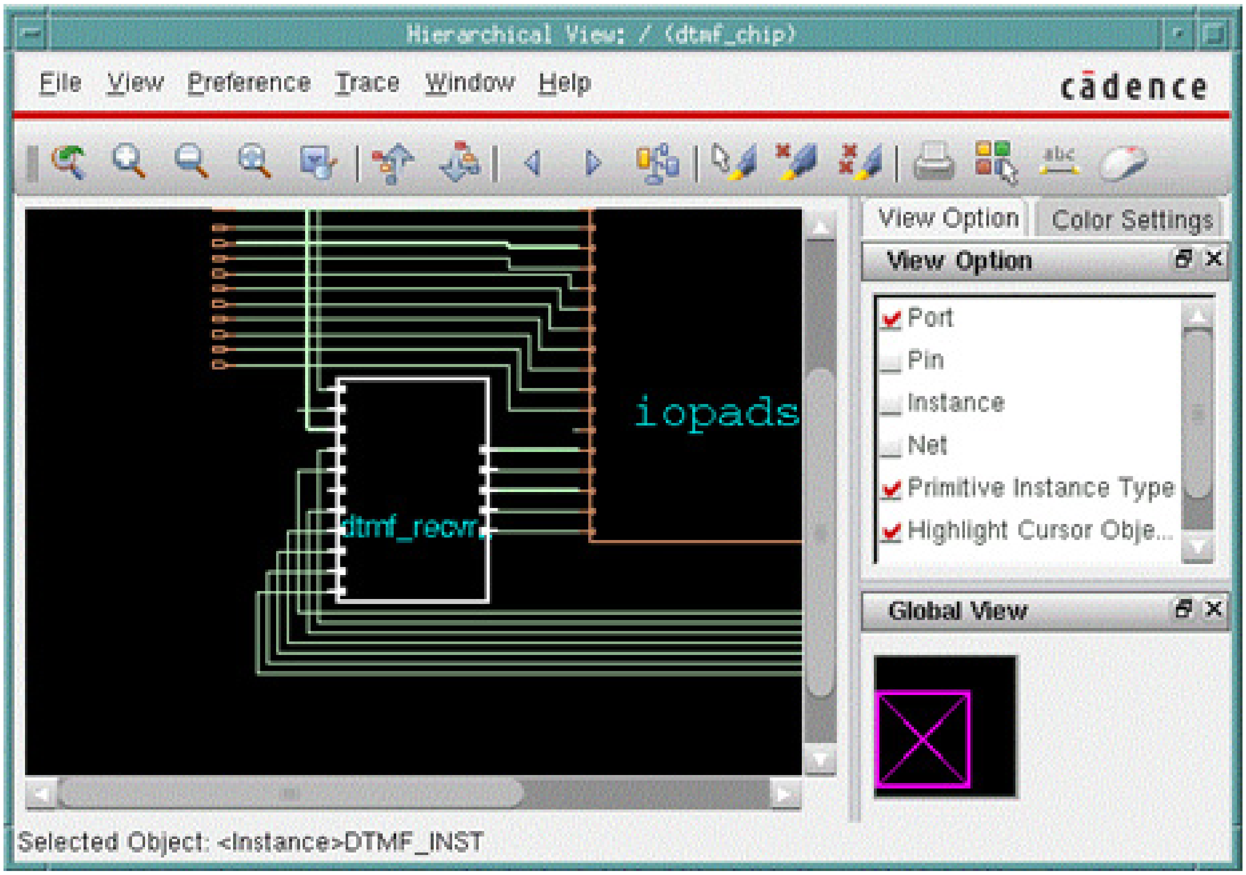
Use the *Schematic Viewer* with the Design Browser to explore the design connectivity, relationship between modules and instances, and design changes, such as CTS and ECO.

You can cross-probe from the Schematic Viewer to the main display to get immediate feedback for debugging and diagnosing your design.

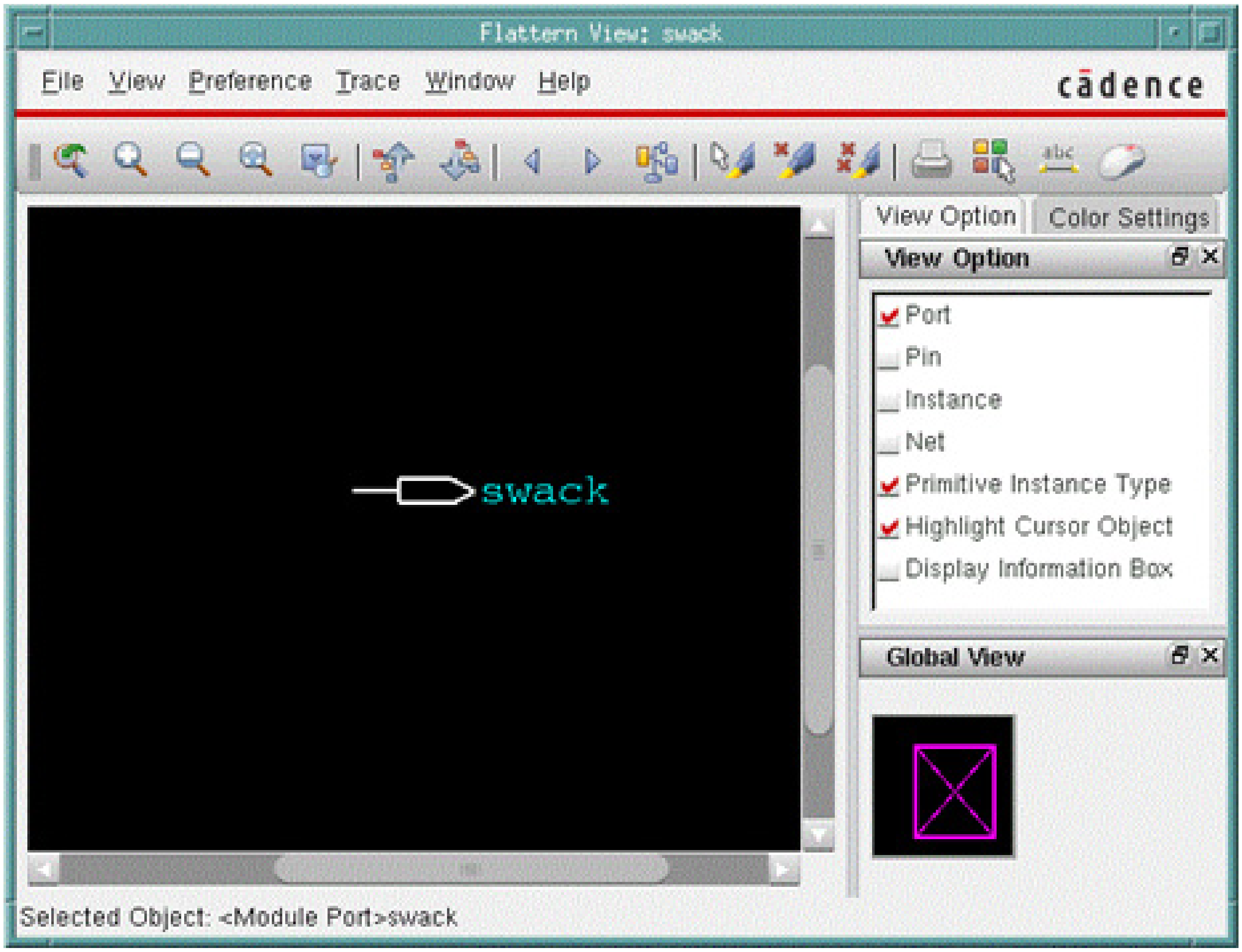
The Schematic Viewer has two main views:

- Hierarchical view - To open the *Schematic Viewer* in Hierarchical view:

Select a module, instance, net, or pin in Design Browser and click the *Show Hierarchical Schematic* widget.



- Flattened view - To open Schematic Viewer in Flattened view, select an instance, net, or pin in Design Browser and click the *Show Instance Based Schematic* widget.



Schematic Viewer Components

In both Flattened and Hierarchical views, the *Schematic Viewer* has the following components:

- Menu Bar
- Toolbar
- Schematic Display Area

Note: Except for the menu bar, the *Schematic Viewer* components are same as that of the Schematic tab in the main Tempus window. For more information on these components, see Schematic Tab Components in the Tab Options chapter.

Menu Bar

- File Menu
- View Menu
- Preference Menu
- Trace Menu
- Window Menu

File Menu

The File menu enables you to print schematics and close the Schematic Viewer window.

Options	Description
Print	Opens the Print Schematic form where you can choose printer options.
Quit	Closes the Schematic Viewer window.

View Menu

The View menu displays various options, which allow you to customize the display in the schematic display area. Many of the options in the View menu are also available on the View Option Pane to the right of the schematic display area.

Options	Description
Port Name	Displays instance/block port names in the schematic display area. De-select this check box if you do not want to display port names. *Default:* `Selected`
Pin Name	Displays I/O pin names in the schematic display area. *Default:* `Unchecked`
Instance Name	Displays instance names in the schematic display area. *Default:* `Unchecked`

Net Name	Displays net names in the schematic display area. *Default:* `Unchecked`
Primitive Instance Type Name	Displays primitive instance type names. *Default:* `Selected`
Highlight Cursor Object	Highlights objects as you move the cursor over them. *Default:* `Selected`
Zoom In	Displays the schematic in greater detail. Each selection zooms in one level.
Zoom Out	Displays more contents of the schematic. Each selection zooms out one level.
Zoom to Full Window	Displays the entire schematic within the display area.
Pan	Pans the current schematic according to the option you select. You can choose to pan: ■ *Left* ■ *Right* ■ *Up* ■ *Down* ■ Horizontal Center ■ Vertical Center
Complete View	Returns to the original full schematic view.
Change Color	Opens the Timing Debug Preferences - Select Color form where you can change the color of the selected object.
Next Color	Changes the selected object's color to the next available color in the palette. You can do this repeatedly.
Display Information Box	Displays an information box that states the name of the object over which the cursor is currently placed and lists pertinent details about that object. *Default:* `Selected`

Preference Menu

Use the Preference menu to customize the color and font settings of the schematic display.

Options	**Description**
Color Settings	Opens the Color Settings form so you can change color and line settings for objects in the schematic display area.
Font Settings	Opens the Font Settings form so you can change the font size and style settings for the schematic display.
Mouse Properties	Opens the *Mouse Properties* form so you can change the font size and style settings for the schematic display.

Trace Menu

Use the Trace menu to trace drivers, loads, and connectivity for the selected object.

Options	**Description**
Trace Driver	Traces all possible drivers of the selected object and highlights them in the schematic display area. The default highlight color for drivers is yellow.
Trace Load	Traces all possible loads of these object and highlights them in the schematic display area. The default highlight color for loads is red.
Trace Connectivity	Traces all possible drivers and loads of the selected object and highlights connectivity from the object's drivers to its loads. The default highlight colors for drivers and loads are yellow and red, respectively.

Window Menu

The *Windows* menu lists all of the active Schematic Viewer windows for your Tempus session. Click on a window name to bring it to the front of your screen.

- Choose *Cascade Window* to refresh your desktop and display the main window on top with all other open windows in a cascading view to the left of the main window.

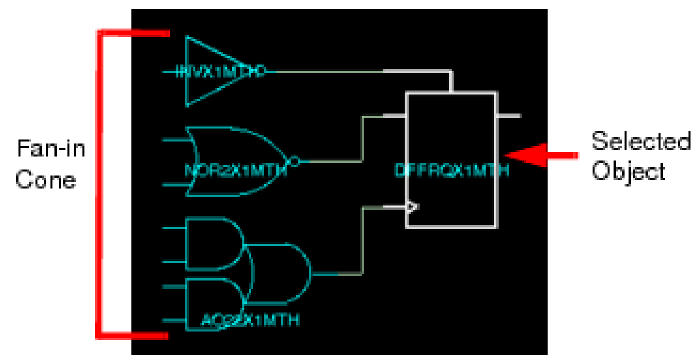
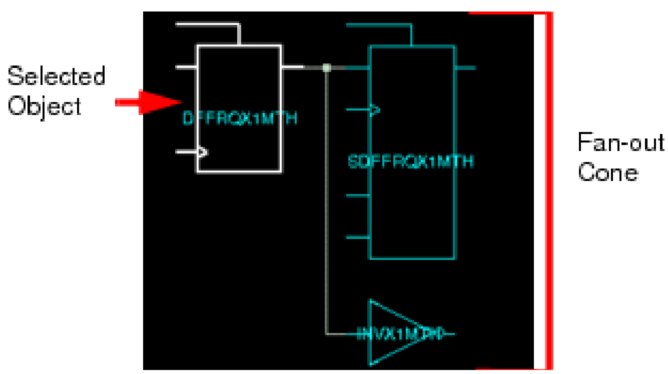
Schematic Viewer Context Menu

To access the context menu, right-click an object in the Schematic Viewer window. The options available in context menu varies depending on the current view of Schematic Viewer:

- Flattened View Context Menu
- Hierarchical View Context Menu

Flattened View Context Menu

The following table describes the options that are available from the Flattened view context menu of Schematic Viewer.

Options	Description
Open Fan-in Cone	Opens the fan-in cone of the selected object. This option is available in the context menu of instances and block ports. 
Close Fan-in Cone	Closes the fan-in cone of an object.
Open Fan-out Cone	Opens the fan-out cone of the selected object. This option is available in the context menu of instances and block ports. 

Close Fan-out Cone	Closes the fan-out cone of an object.
Show Hierarchical Mode	Displays the object schematics in the Hierarchical View.
Highlight in Main Window	Highlights the object that you right-clicked in the schematic display in the main window. This option is useful for cross-probing.
Copy Object Name	Copies the object name.

Hierarchical View Context Menu

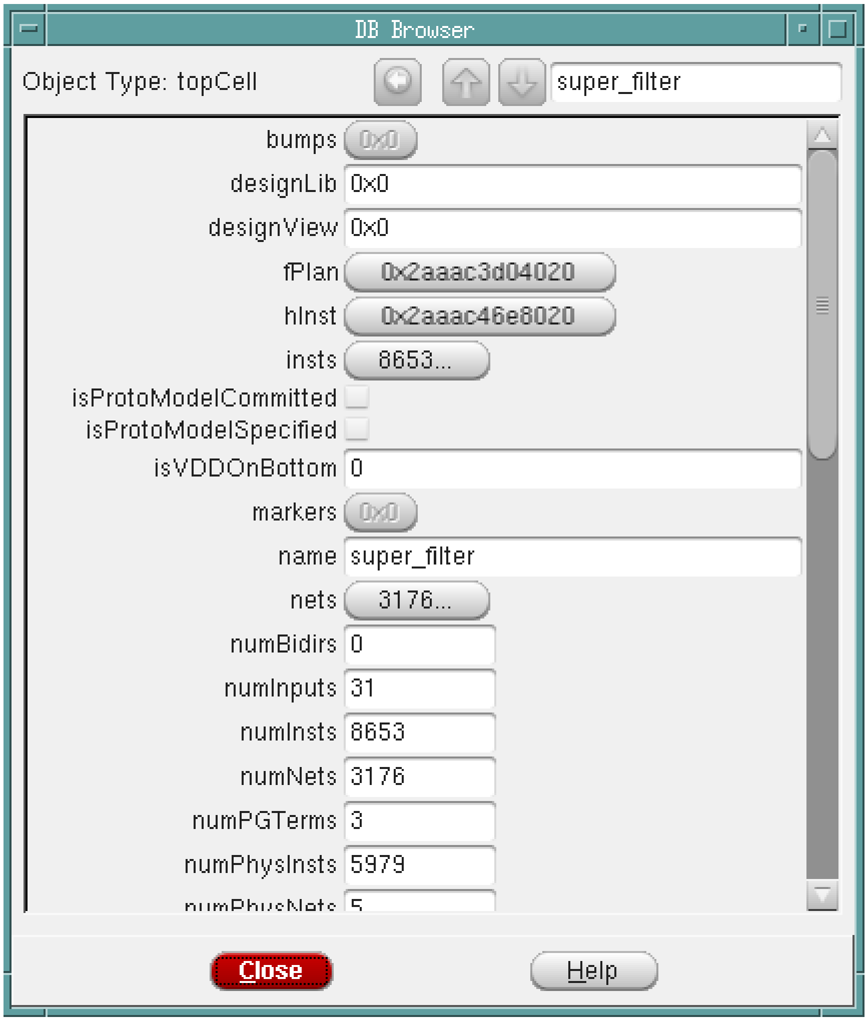
In Hierarchical view, the context menu of the Schematic Viewer is same as the context menu in the Schematic tab. For more information on the options in the Hierarchical view context menu, see Context Menu in the Tab Options chapter.

Object Browser

You can use the object browser (DB Browser) to retrieve the selected object's attributes. The DB Browser displays different attributes, depending on the objects selected in the main window.

To launch the DB browser, select one or more objects in the main window.

- Select *Tools - Object Browser*.



DB Browser Fields and Options

Object Type	Specifies the type of object currently selected.
	Click the *Show next object in list* button to display the attributes of the next object in the sequence. or *Prev* to view the previous object. **Note:** This button is available only when multiple objects are selected in the main window.

	Click the *Show last object in list* button to display the attributes of the previous object in the sequence. **Note:** These buttons are available only when multiple objects are selected in the main window.
	Click the *Back to previous object* button to return to the first object in the list. For example, if you select two instances `i1` and `i2`. Use the *Show next object in list* button to move from `i1` to `i2` as the active instance, then decend to the `i2`'s cell. Now if you select *Back to previous object* button, you return to `i1`.
Close	Closes the DB Browser.

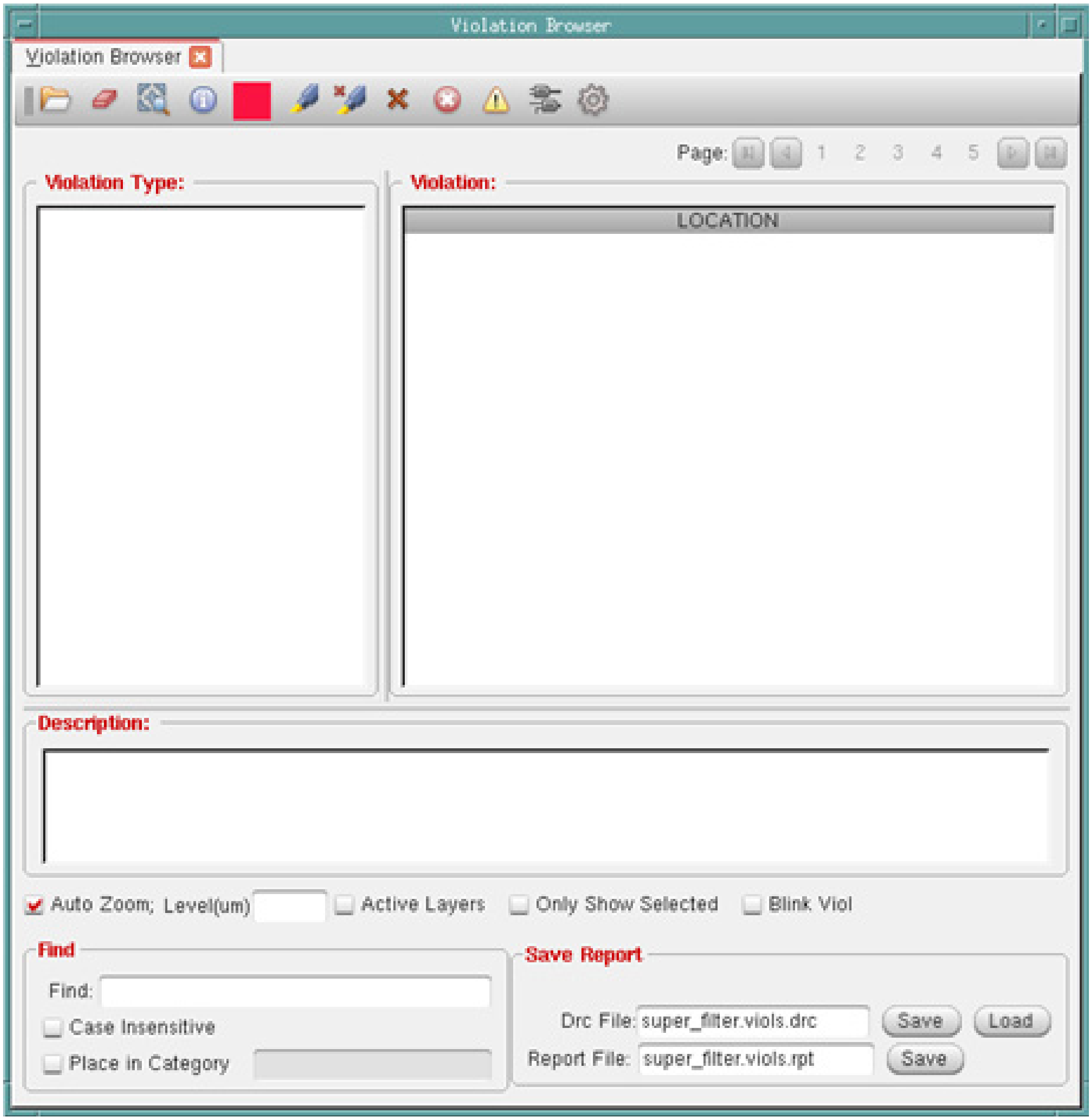
Violation Browser

Use the Violation Browser to display violations and lithography hotspots, mark violation markers as false or true, delete violation markers, and specify a report file.

The Violation Browser also provides access to the following forms:

- Violation Browser Settings
- Load Violation Report
- Clear Violation

- To open Violation Browser, choose *Tools - Violation Browser*.



Violation Browser Fields and Options

	*Load Violation Report* - Opens the Load Violation Report form. Use this form to load a violation marker file and then view the markers in the Violation Browser.

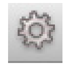
	Clear Violation - Clears all the violation markers in your design.
	Fit Violation - Displays the violation within the design display area.
	Attribute Editor - Opens the *Attribute Editor* and populates it with the highlighted violations. Double click on a violation to update fields in the Attribute Editor.
	Highlight Color - Select this button to specify the color for highlighting violations.
	Highlight Violations - Highlights the selected violation(s) in the design display area as well as in the *Violation* list on the right pane of the Violation Browser. You can select an individual violation in the right pane of the Violation Browser or select violations by type or subtype in the left pane of the browser. To highlight a selected violation or a selected type/subtype of violations, select the highlight color from the color palette and then click the *Highlight Violations* button.
	Dehighlight Violations - Removes the highlight from the selected violation or selected type/subtype of violations in the design display area.
	Delete Violations - Deletes all selected violations from the design database. Markers are also removed from the design display area and the list of violations.
	Mark Violations as False - Marks the selected violation as a false violation. These violations are stored in the database as false violations. To view the violations that you mark as false, click in the *Violation Browser*, select the *View False Violations* check box in the Violation Browser Settings form, and click *OK*.
	Mark Violations as True -- Marks the selected violation as a true violation.
	Show MSV Violation Schematic – Displays schematics associated with the highlighted violation in the Schematic Tab form.
	Setting - Opens the Violation Browser Settings form.

Violation Type	Lists the violation types in a hierarchy window. To show or hide a violation, select it in the *Violation* or *Violation Type* window and click the right mouse button. Select *Show* or *Hide* from the pop-up window. The numbers in parentheses after each violation type represent the number of violations of that type displayed in the main window and the total number of violations of that type. The count includes true and false violations. For example, if *Verify - Connectivity - Open (479/503)* means that there are 503 Open violations and 479 of those violations are shown the main window.
Violation	For the highlighted *Violation Type*, displays the names of the objects with violation markers and their locations.
Description	Displays the layer and bounding box for the violation. **Note:** Layer is not included for placement violations. You can select text in the *Description* text box and copy and paste it to another window. You cannot edit text in the *Description* text box.
Auto Zoom	Automatically zooms in to the selected violations as you browse through the violations. This check box is selected by default.
Zoom Level (um)	Controls the level of zoom for Auto Zoom. When you specify the zoom level, Violation Browser uses the same value for all listed violations so that you do not have to zoom out each time you select a violation.
Active Layers	Keeps only the violation layer(s) visible when you zoom into a violation. If this check box is not selected, the Auto Zoom mode displays layers as per the layer visibility settings on the Layer Control bar.
Only Show Selected	Displays markers for only selected violations. This option is useful if there are many violations in the design. By selecting this option, you can turn off other violation makers in the main window and focus on only the violations you have selected in the browser.
Blink Viol	Makes the marker for the selected violation blink in the main window.
Find	Searches for the specified text and highlights it in the *Violation* window. This field recognizes pattern matching, so you can type `*153*` to search for all objects with the string `153` as part of their *LOCATION*. Press *Enter* to find the specified text.

Case Insensitive	Specifies whether the text in the *Find* text entry box is case sensitive.
Place in Category	Creates a category in the *Violation Type* window and adds the highlighted object to the category. Categories are not saved.
Drc File - Save	Opens a form that allows you to specify the path and name for saving the DRC file.
Drc File - Load	Opens a form in which you can browse and select the DRC file to be loaded.
Report File - Save	Specifies the report file that contains all the violation information. The browser and the report file list the violations in the same order.

Violation Browser Settings

Use the Violation Browser Settings form to specify the settings for displaying violations and lithography hotspots.

- Choose *Tools - Violation Browser* and click the  button.

Violation Browser - Settings Fields and Options

Show Types	Lists the types of violations to display or hide in the Violation Browser. Select one or more of the following options:	
	All	Displays all violations.
	Geometry	Displays geometry violations.
	Process Antenna	Displays process antenna violations.

	Overlap	Displays overlap violations.
	Short	Displays short violations.
	Connectivity	Displays connectivity violations.
	Metal Density	Displays metal density violations.
	AC Limit	Displays AC limit violations.
Show Area	Displays violations in the entire area or a specified portion of the area. Specify the portion of the area by using one of the following methods: ■ Using the *Draw* button and your mouse to create a rectangular area in the design. The software displays the coordinates in the *X1*, *Y1*, *X2*, and *Y2* fields. ■ Manually enter the coordinates in the *X1*, *Y1*, *X2*, and *Y2* fields.	
Other Filters	Specify additional conditions by selecting one or more of the following options:	
	Error Per Type	Specifies a limit for the number of violations parsed by the browser for each violation type. If this check box is not selected, the Violation Browser parses and displays all violations.
	View False Violations	
		Displays violations that you have marked as false.
	Search Description For Filter String	
		Searches for text in the violation message (in the bottom portion of the Violation Browser window) and the violation description (in the top portion of the Violation Browser window).
	Filter Strings	

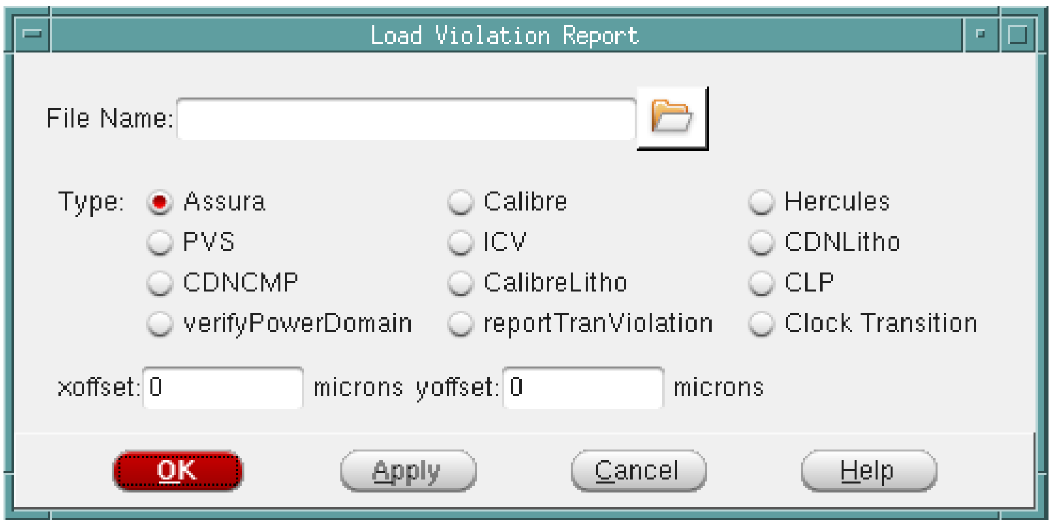
		Specifies a text string for filtering violations. When you specify a text string, the Violation Browser searches the message of each violation and displays only the violations whose messages match the search conditions.
		
		To specify a list of strings, separate each string with a space.
		
		To specify a literal string with space in it, enclose the string in double quotation marks. For example, you can specify strings such as the following:
		
		`M5`
		`(600, 500)`
		`M3 (700, 400)`
		`"Pin of Net"`
		
		You can specify the AND, OR, and NOT conditions with the search strings. Use:
		
		`&&` between strings to specify the AND condition.
		
		`||` between strings to specify the OR condition.
		
		! before a string to specify the NOT condition.

Delete Violations	Enables you to delete all violations or delete violations in the specified area only.	
Delete Violations By Area		
	Enables you to delete violations in the area you specify on this form or the area you select with the mouse.	
	Entire area	Deletes the types of violations specified in the entire core design area.
	Specify	Specifies an area from which to delete violations. Only the types of violations specified by this form are deleted.
	Draw ...	Lets you use the mouse to outline an area from which to delete violations.

Load Violation Report

Use the Load Violation Report form to load a violation marker file or a hotspot interface format (HIF) file and create markers that the Tempus software can interpret. After you load the file, you can view the markers in the Violation Browser. The Auto Query feature displays the rule names for the markers.

- Choose *Tools - Violation Browser* and click the *Load Violation Report* button.



Load Violation Report Fields and Options

Clear Violation

File Name	Specifies the marker file to load.

Type	Specifies the type of marker file. The following file types contain violation markers: ■ *Assura* - For Assura violations ■ *Calibre* - For Calibre violations ■ *Hercules* - For Hercules violations ■ *ICV* - For ICValidator violations ■ *PVS* - For PVS violations ■ *CDNCMP* - For Cadence CMP Predictor (CCP) violations ■ *CLP*, *verifyPowerDomain*, and *Clock Transition* - For low power violations ■ *reportTranViolation* - For DRV transition violations The following types contain hotspot markers, which are used to identify areas susceptible to lithography problems: ■ *CDNLitho* (for LPA and NanoRoute litho HIF files) ■ *CalibreLitho*
xoffset	Specifies the X offset. When you specify the X and Y offsets, Violation Browser displays the violations at the correct coordinates after accounting for the offsets.
yoffset	Specifies the Y offset, which Violation Browser takes into account to displays violations at the correct coordinates.

Use the *Clear Violation* button in the Violation Browser to clear all the violation markers in your design.

- Choose *Tools - Violation Browser* and click the *Clear Violation* button.

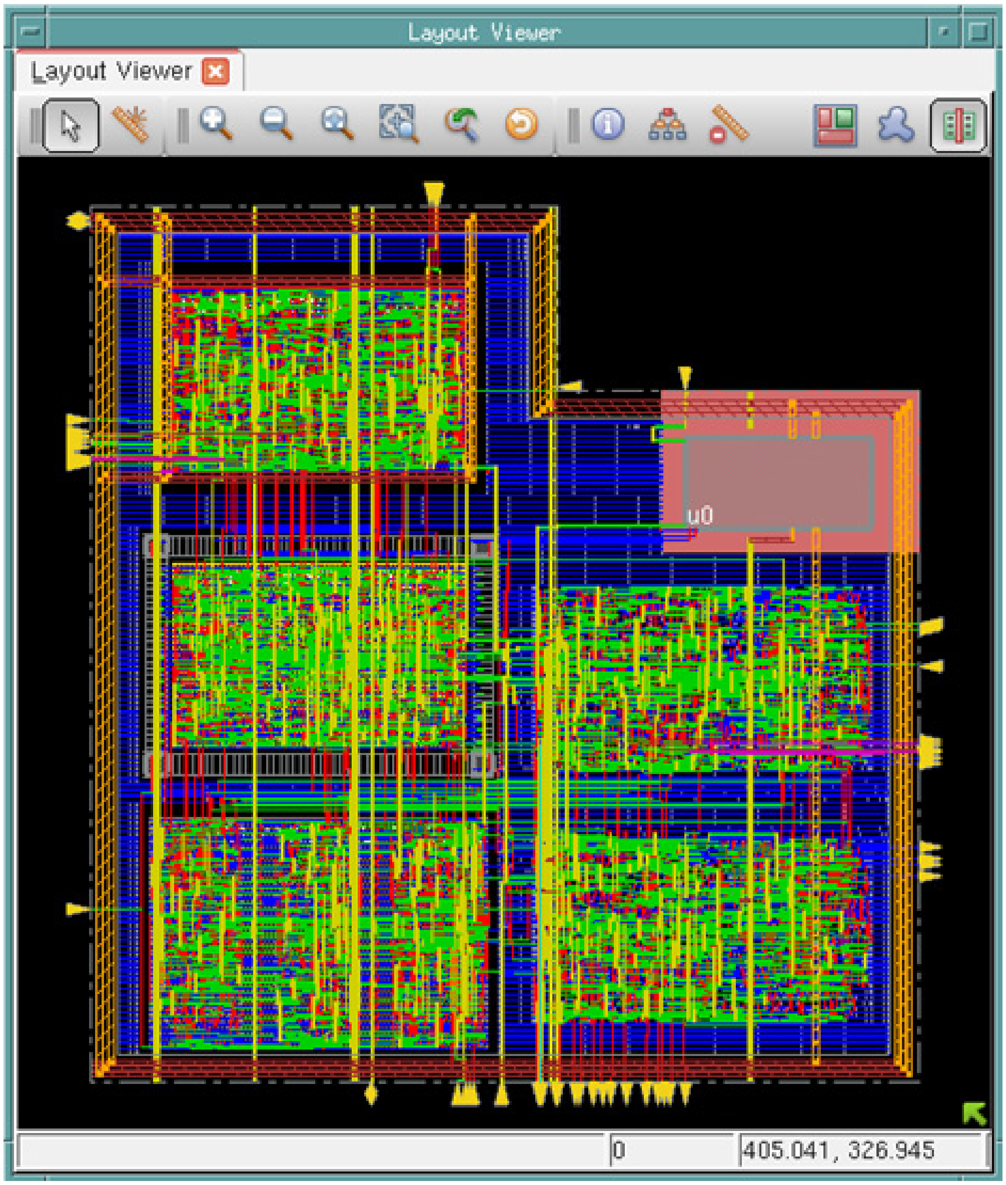
There is no GUI form for this command.

Layout Viewer

Use the *Layout Viewer* menu item to open the layout in a view-only window. You can launch multiple view-only windows in a single Tempus session. This is useful when debugging as it allows you to view and zoom into different portions of a design simultaneously.

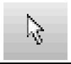
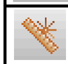
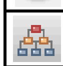
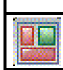
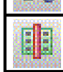
The Layout Viewer window supports all viewing related features, such as *Zoom* and *Pan*. In addition, it supports functions and tools that do not modify the design, such as Attribute Editor and Design Browser.

- Choose *Tools - Layout Viewer*.



Layout Viewer Tool Widgets

You can use the widgets in Layout Viewer to navigate through displays, open tools, and make selections.

Widget	Description
	*Select* - Selects an object in the design display area.
	*Create Ruler* - Click this widget, then left-click in the design display area and drag the mouse to add a ruler (in micrometers), and left-click again to end the ruler. To move the ruler, click and drag it to a new location. Click the ruler again to keep it in the new location.
	*In by 2* - Click this widget to display a smaller area of the design in greater detail. Each click zooms in two levels.
	*Out by 2* - Click this widget to display a larger area of the design in less detail. Each click zooms out two levels.
	*Fit* - Click this widget to display the entire placed design within the design display area.
	*Selected* - Zooms in on a highlighted selection.
	*Previous* - Click this widget to toggle the display between the previous location/ zoom level and the current location/zoom level.
	*Redraw* - Refreshes the Layout Viewer display.
	*Attribute Editor* - Opens the Attribute Editor for the highlighted object.
	*Design Browser* - Opens the Design Browser form.
	*Clear All Rulers* - Clears all rulers from the display area. **Note:** To remove a single ruler, place the mouse cursor over the ruler and press `Delete`.
	*Floorplan View* - Switches to the Floorplan view of the design.
	*Amoeba View* - Switches to the Amoeba view of the design.
	*Physical View* - Switches to the Physical view of the design.

Cell Viewer

The Cell Viewer provides an artwork window for viewing the following objects:

- any LEF library cell
- any OpenAccess database cell view
- abstract view of any partition in the design
- power grid view

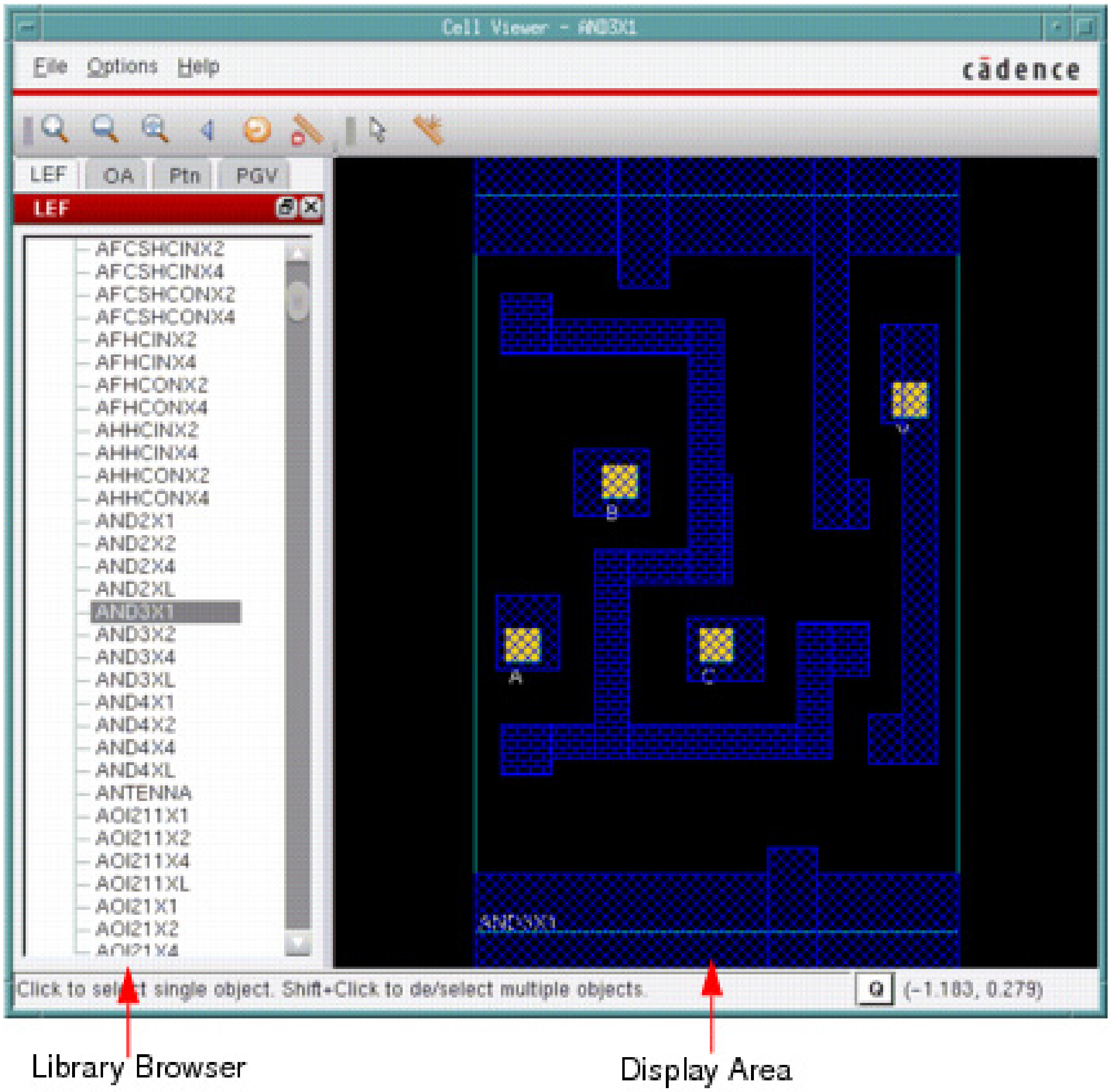
The Cell Viewer displays the content from the Tempus library database. The Cell Viewer provides a convenient way to view the content of a cell, cell view or partition without having to zoom into the design.

The Cell Viewer has the following tabs:

- Cell Viewer - LEF
- Cell Viewer - OA
- Cell Viewer - Ptn
- Cell Viewer - PGV

Cell Viewer - LEF

- Choose *Tools - Cell Viewer*, and click on the LEF page if it is not already selected.

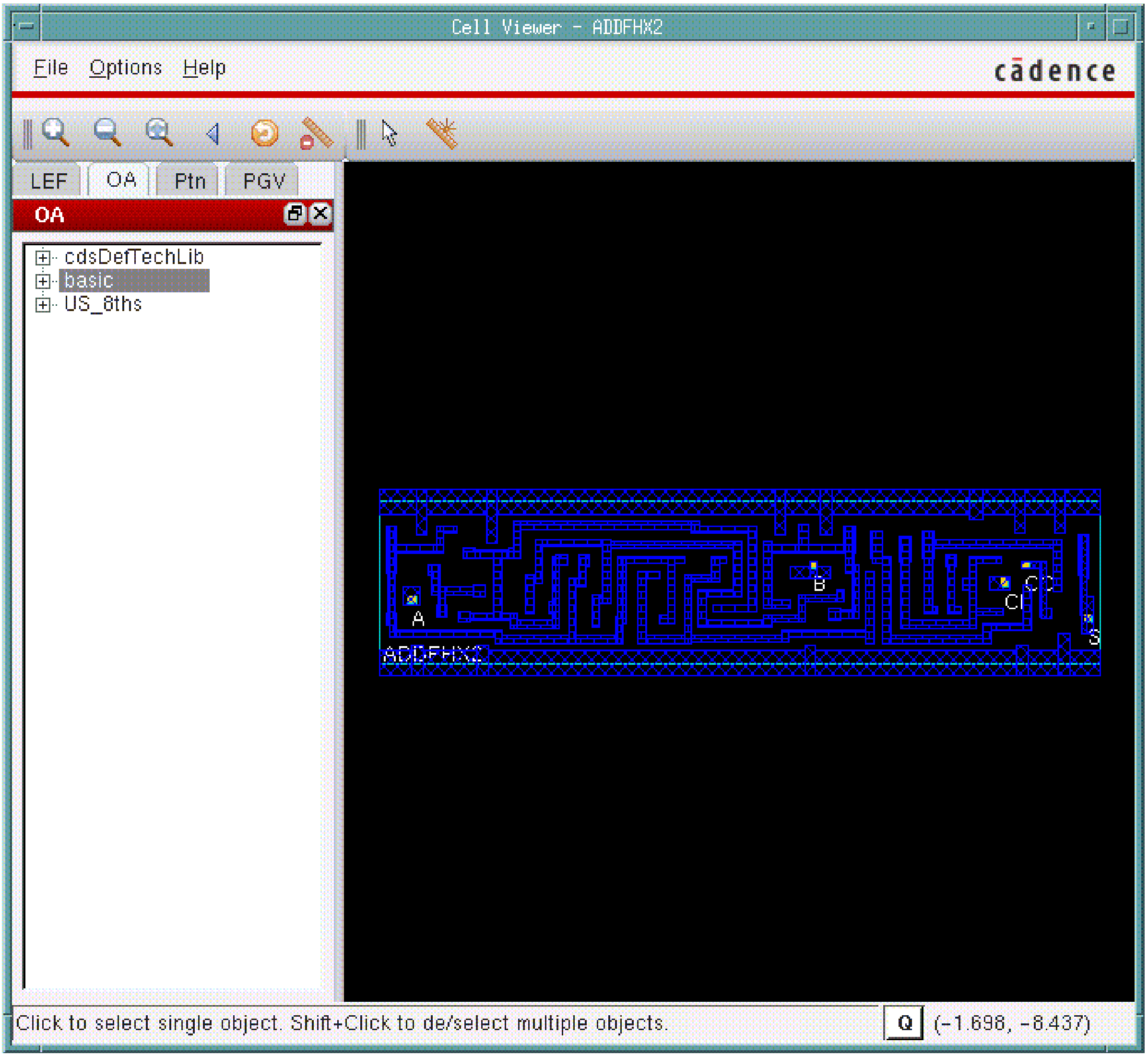


The LEF page displays the details of library cells using the information in the LEF file.

You can navigate to a cell using the Library Browser, which is in the left side of the Cell Viewer. The details of the cell are displayed in the Display Area, which is in the right side of the Cell Viewer.

Cell Viewer - OA

☨ Choose *Tools - Cell Viewer*, and click on the OA page.

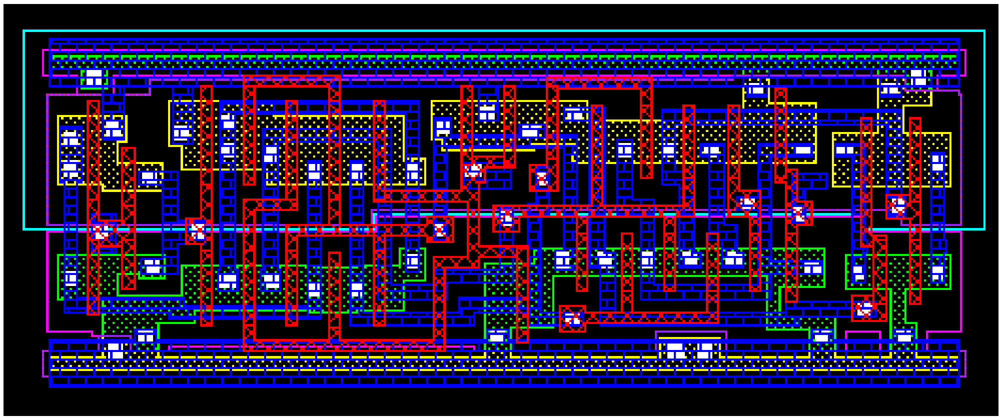


You can view an OA cell in two modes:

- Layout
- Abstract

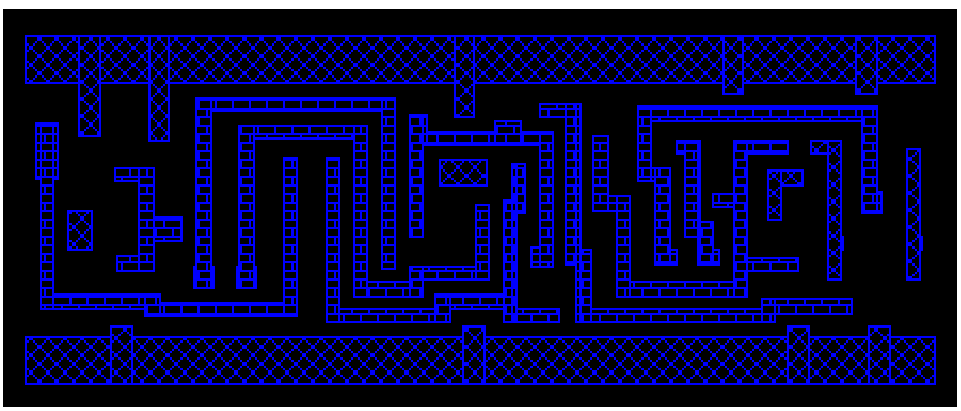
OA-Layout View

In the Library Browser, select the OA cell, and then click *Layout*.



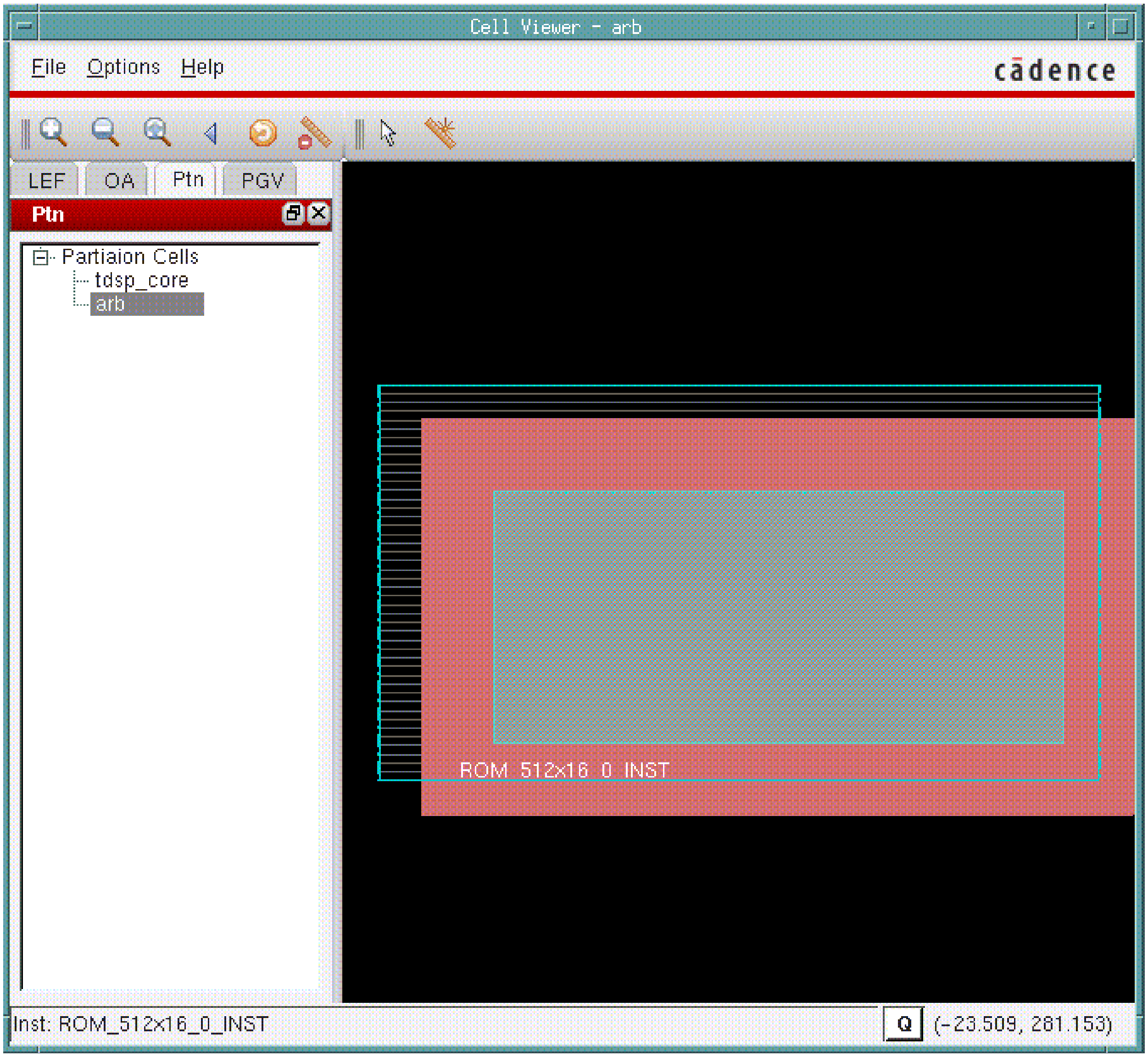
OA-Abstract View

In the Library Browser, select the OA cell, and then click *Abstract*.



Cell Viewer - Ptn

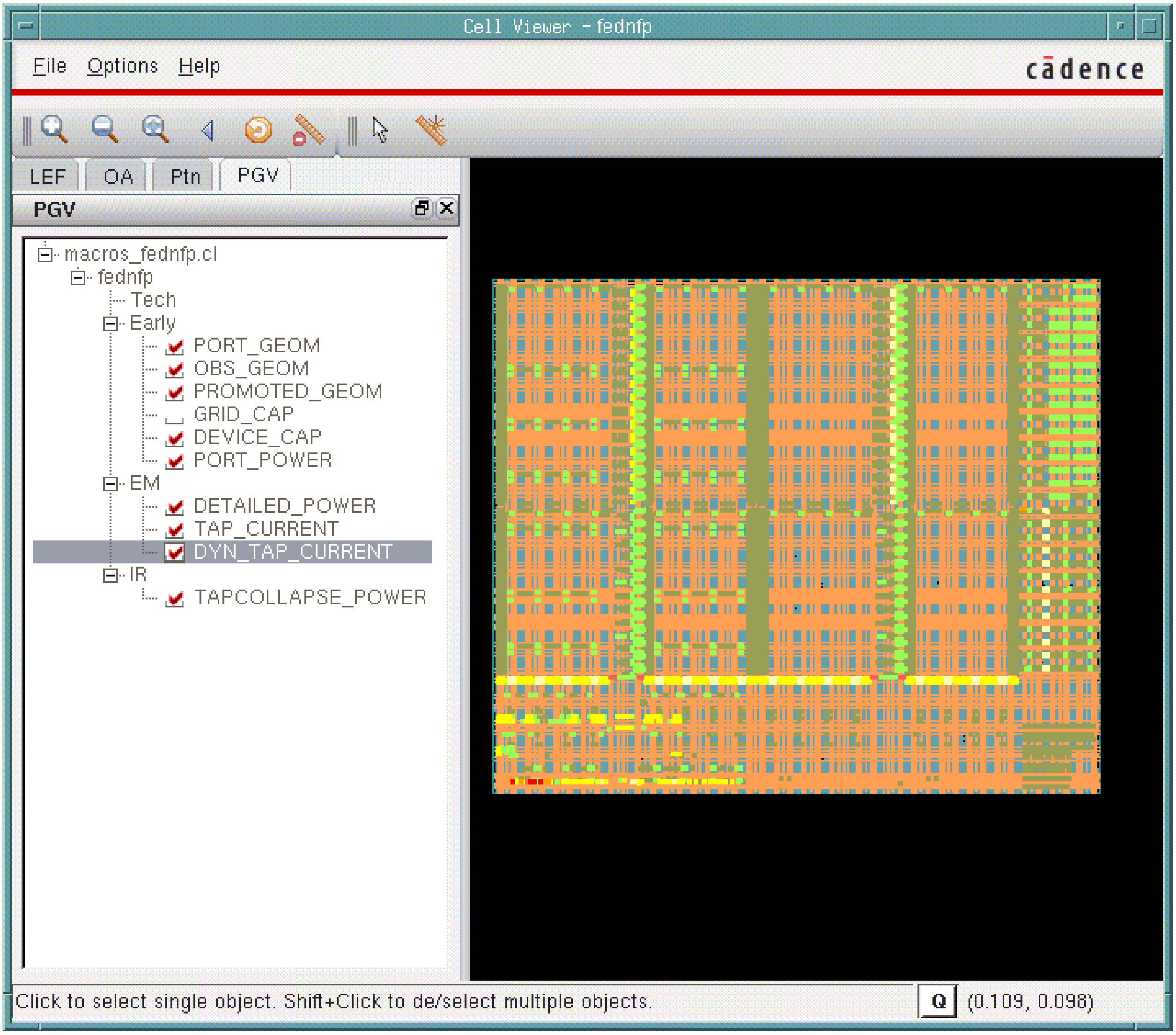
- Choose *Tools - Cell Viewer*, and click on the *Ptn* page.



Use the Ptn page to display the abstract of any partition in the design. You can navigate to a partition through the Library Browser in the left side of the Cell Viewer.

Cell Viewer - PGV

- Choose *Tools - Cell Viewer*, and click on the PGV page.



Use the PGV page to view and debug power-grid libraries. This form allows you to take a physical layout and overlay it with a power-grid resistor network extracted by Power-Grid Library Generator. You can expand a cell to view the default views corresponding to each PGV type (Tech, Early, EM, and IR).

Cell Viewer Widgets

The Cell Viewer supports the following widgets: Zoom In, Zoom Out, Fit, View Last, Redraw, Clear All Rulers, Select and Ruler.

For a description of these widgets, see the Toolbar Widgets on page 456 section in the Tab Options chapter.

Setting Layer Visibility Preferences

Use the Layer Control Bar in the main window to set the visibility of layers in the Cell Viewer. For example, you can turn off the visibility of the layer METAL 3.

Constraint

The Constraint submenu includes three tabs:

- *By Groups*
- *Source*
- *MMMC*

These are explained in the following sections.

- Constraint - By Groups
- Constraint - Source
- Constraint - MMMC

Constraint - By Groups

Use the By Groups tab to select display settings for constraints. The By Groups tab has the following options:

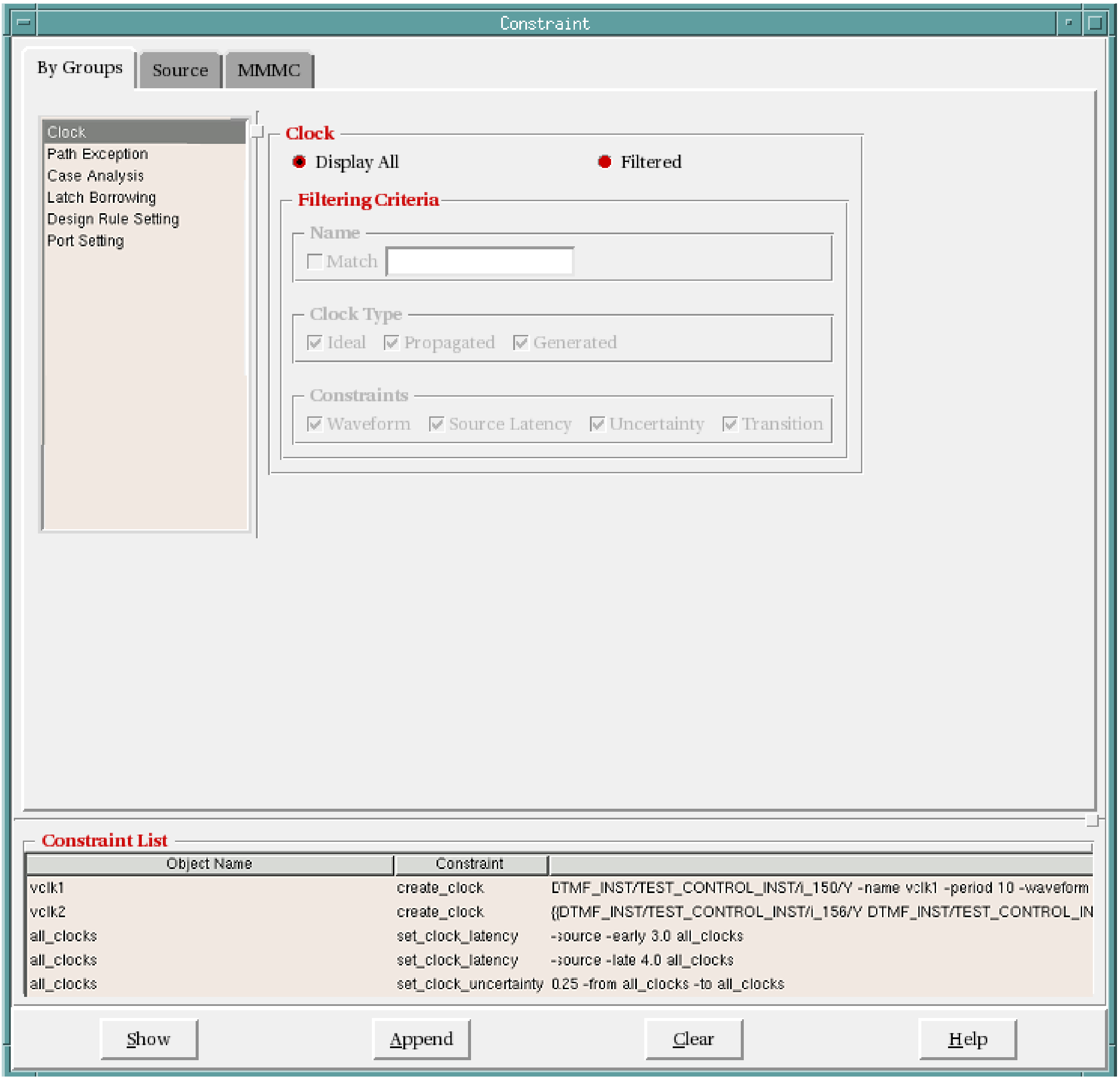
- Clock
- Path Exception
- Case Analysis

- Latch Borrowing
- Design Rule Setting
- Port Setting

Clock

Use the Clock form to set the display settings for constraints related to clocks. The constraints are displayed in the *Constraint List* pane.

- Choose *Constraint - By Groups*, and select *Clock*.



Fields and Options

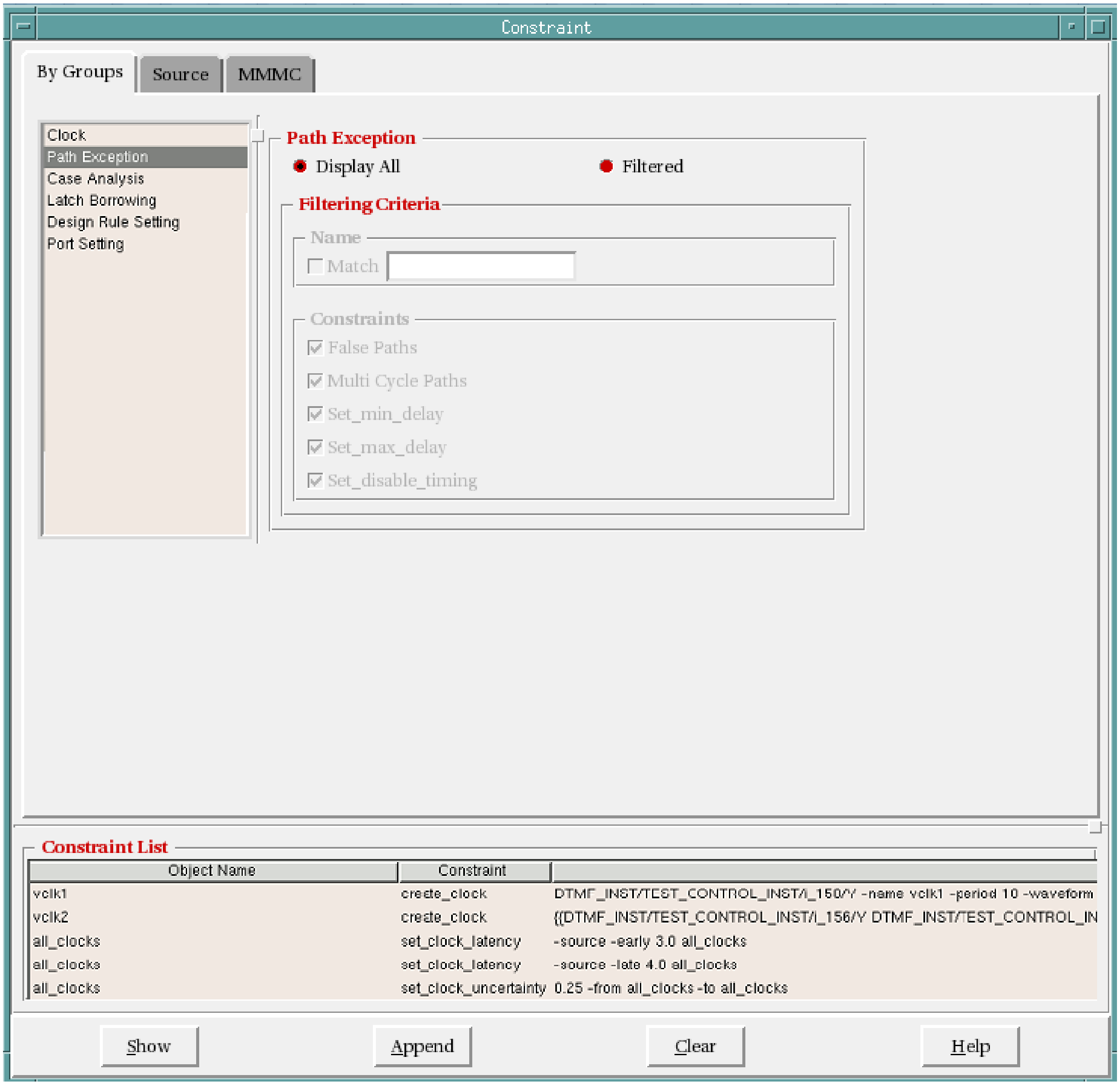
Display All	Displays all constraints in the design.
Filtered	Specifies the filtering criteria for the constraints to be displayed. When you choose this option, the following fields are enabled:

	Match	Displays constraints matching the specified criteria.
	Clock Type	Specifies the clock type for the constraints to be displayed. Choose from *Ideal*, *Propagated* and *Generated*.
	Constraints	Specifies the type of constraints to be displayed. Choose from *Waveform*, *Source Latency*, *Uncertainty*, and *Transition*.

Path Exception

Use the Path Exception form to set the display settings for constraints related to path exception. The constraints are displayed in the *Constraint List* pane.

- Choose *Constraint - By Groups*, and select *Path Exception*.



Fields and Options

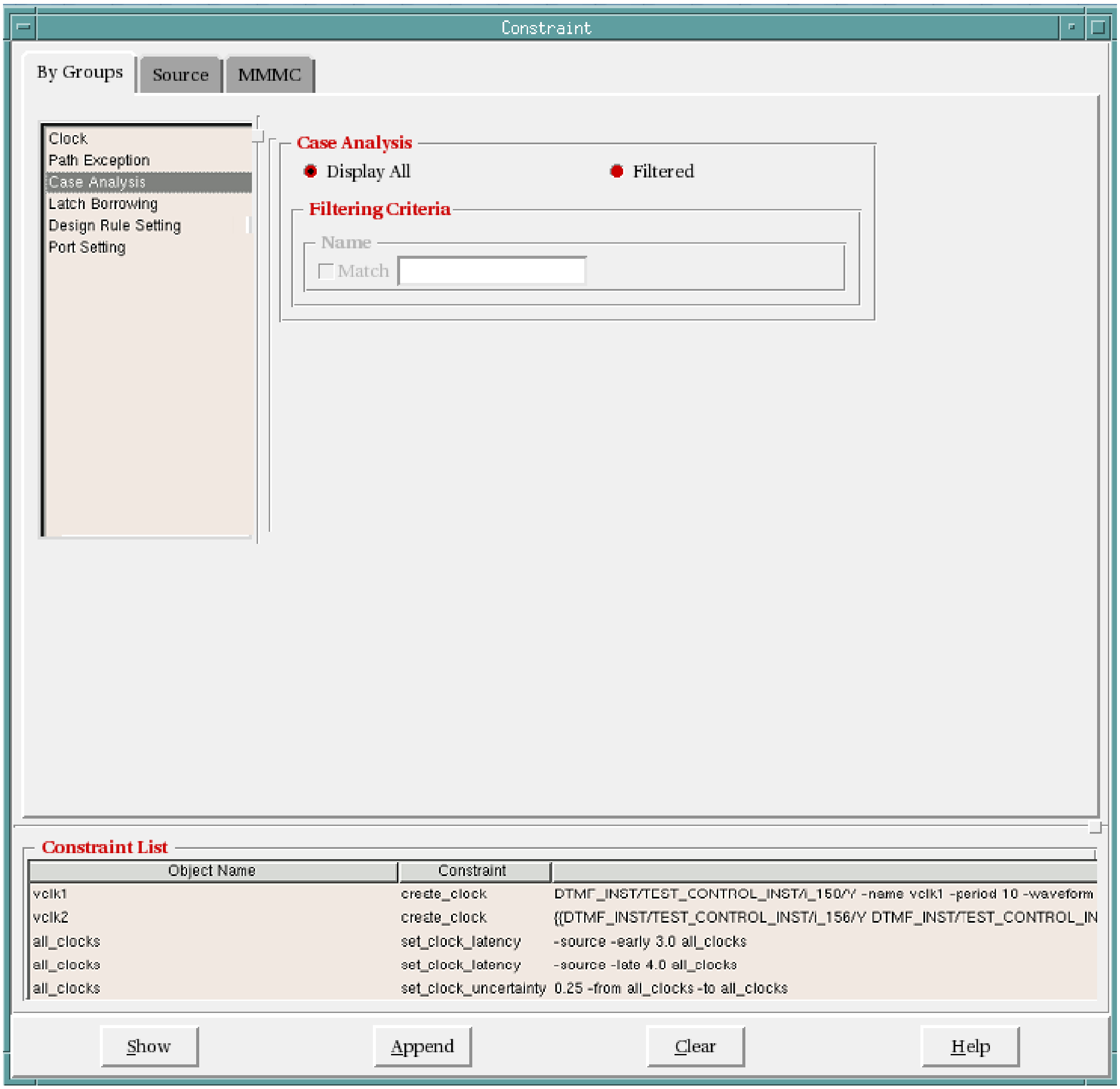
Display All	Displays all constraints in the design.
Filtered	Specifies the filtering criteria for the constraints to be displayed. When you choose this option, the following fields are enabled:

	Match	Displays constraints matching the specified criteria.
	Constraints	Specifies the type of constraints to be displayed. Choose from *False Paths*, *Multi Cycle Path*, *Set_min_delay*, *Set_max_delay*, and *Set_disable_timing*.

Case Analysis

Use the Case Analysis form to set the display settings for constraints related to case analysis. The constraints are displayed in the *Constraint List* pane.

- Choose Constraint - By Groups, and select *Case Analysis*.



Fields and Options

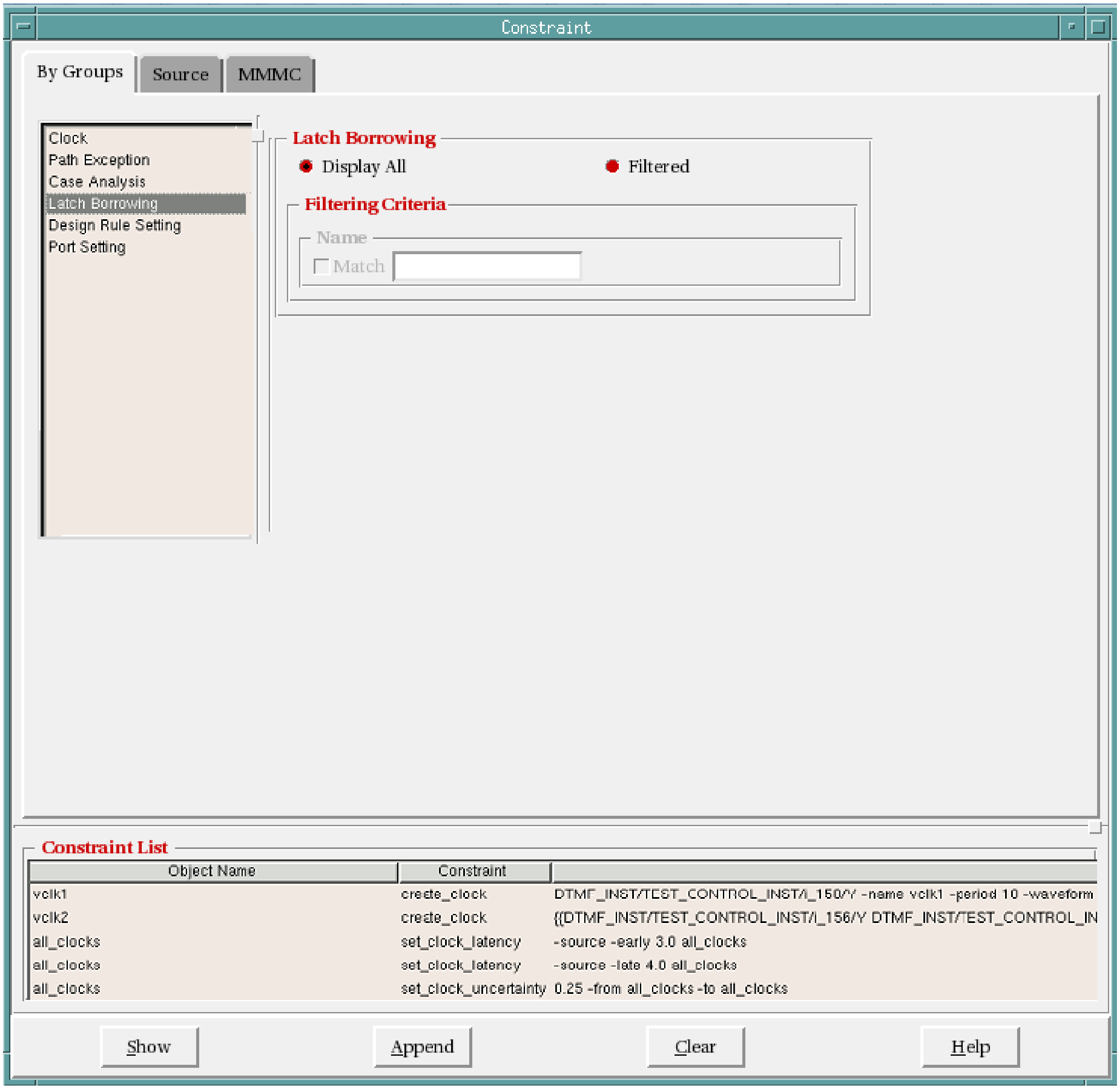
Display All	Displays all constraints in the design.
Filtered	Specifies the filtering criteria for the constraints to be displayed. When you choose this option, the following field is enabled:

	Match	Displays constraints matching the specified criteria.

Latch Borrowing

Use the Latch Borrowing form to set the display settings for constraints related to latch borrowing. The constraints are displayed in the *Constraint List* pane.

- Choose Constraint - By Groups, and select *Latch Borrowing*.



Fields and Options

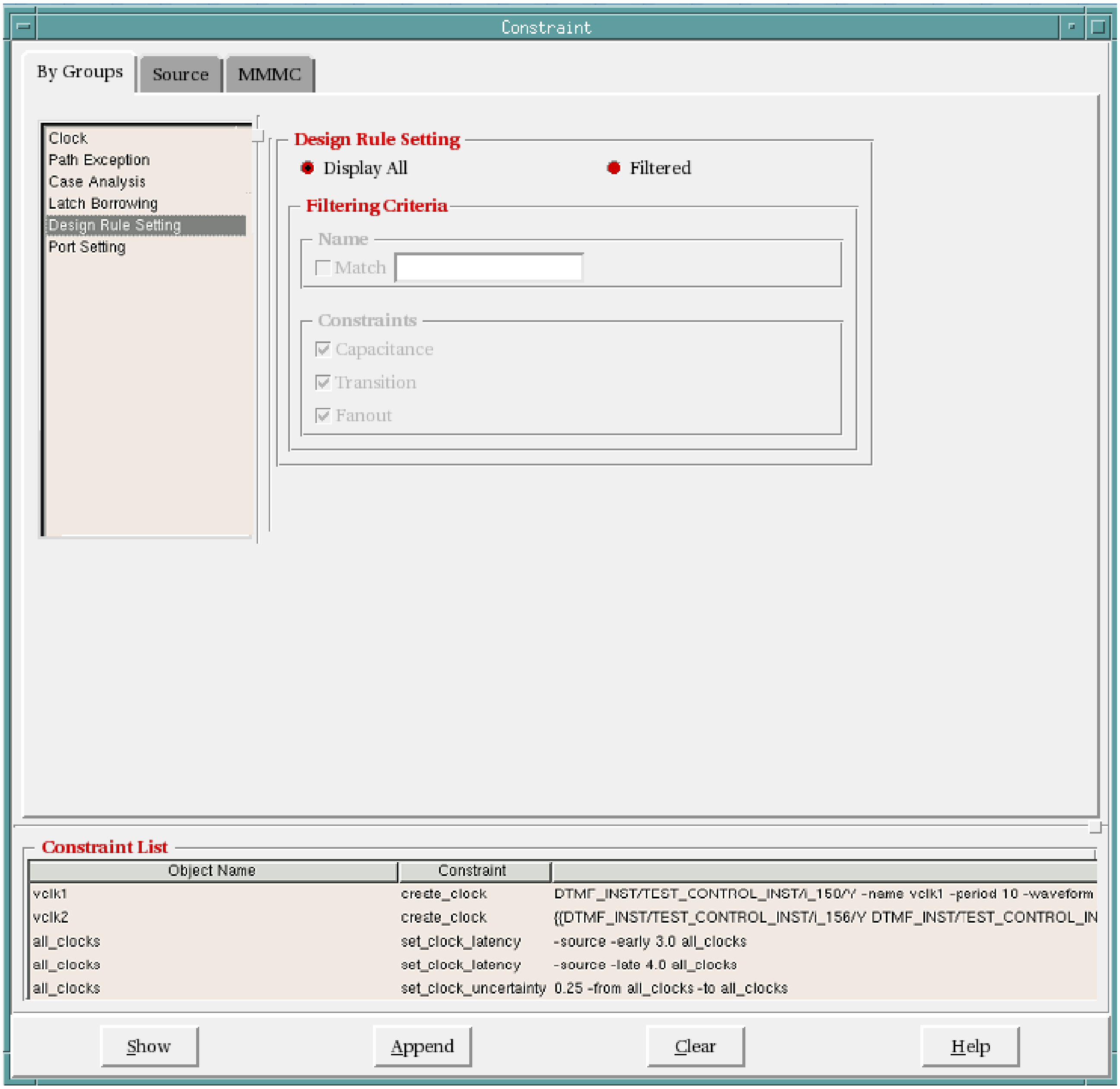
Display All	Displays all constraints in the design.
Filtered	Specifies the filtering criteria for the constraints to be displayed. When you choose this option, the following field is enabled:

	Match	Displays constraints matching the specified criteria.

Design Rule Setting

Use the Design Rule Setting form to set the display settings for constraints related to design rules. The constraints are displayed in the *Constraint List* pane.

- Choose *Constraint - By Groups*, and select *Design Rule Setting*.



Fields and Options

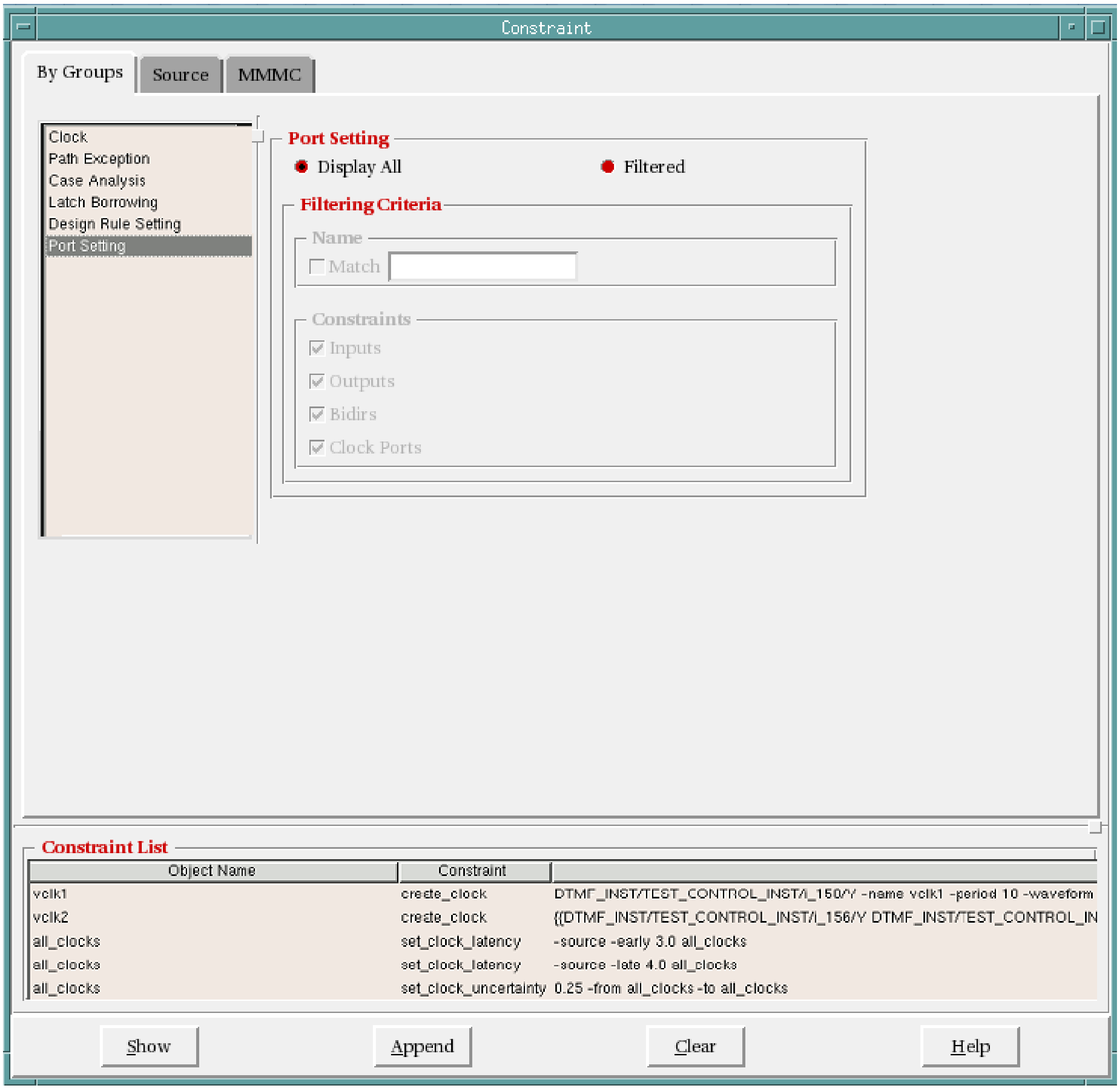
Display All	Displays all constraints in the design.
Filtered	Specifies the filtering criteria for the constraints to be displayed. When you choose this option, the following fields are enabled:

	Match	Displays constraints matching the specified criteria.
	Constraints	Specifies the type of constraints to be displayed. Choose from *Capacitance*, *Transition*, and *Fanout*.

Port Setting

Use the Port Setting form to set the display settings for constraints related to port setting. The constraints are displayed in the *Constraint List* pane.

- Choose *Constraint* - *By Groups*, and select *Port Setting*.



Fields and Options

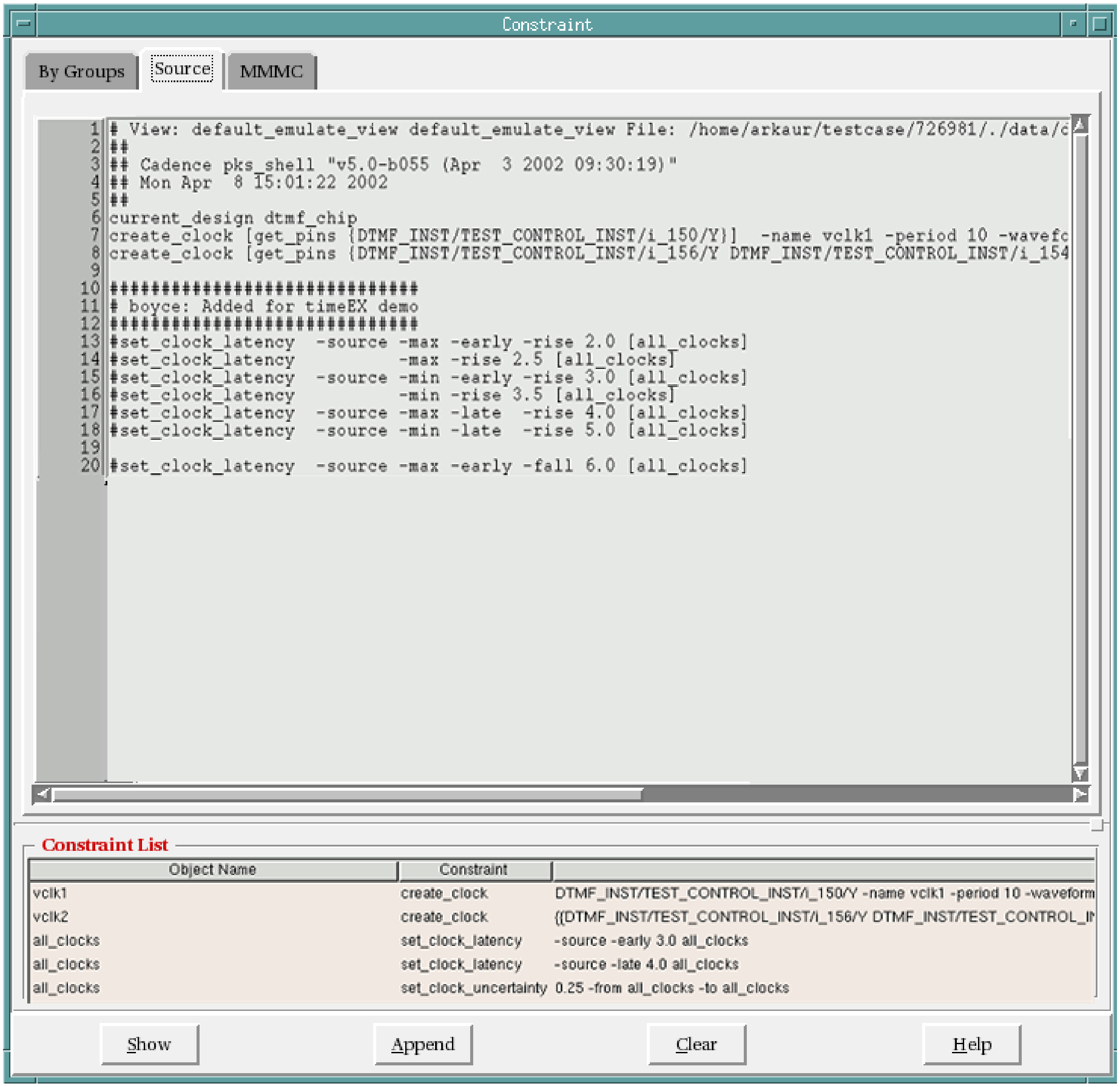
Display All	Displays all constraints in the design.
Filtered	Specifies the filtering criteria for the constraints to be displayed. When you choose this option, the following fields are enabled:

	Match	Displays constraints matching the specified criteria.
	Constraints	Specifies the type of constraints to be displayed. Choose from *Inputs*, *Outputs*, *Bidirs*, and *Clock Ports*.

Constraint - Source

Use the Source tab to view SDC file associated with the constraints that you display using the By Group tab.

- Choose *Constraint* - *Source*.



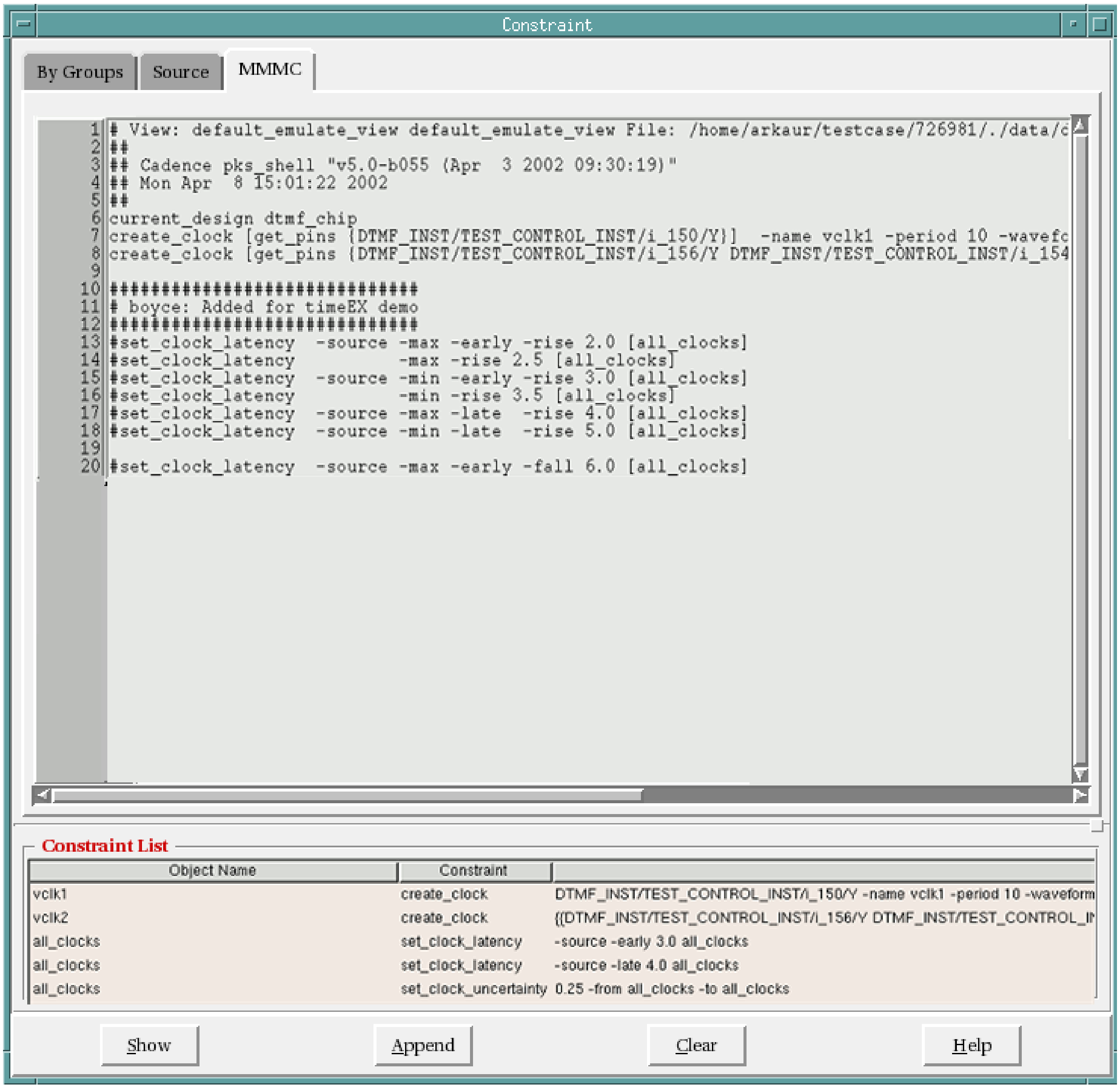
- Double-click on a constraint in the *Constraint List* pane to highlight it in the Source window.

Constraint - MMMC

Use the MMMC tab to view constraints that you used for multi-mode multi-corner (MMMC) analysis.

Note: This tab is not displayed by default. It is displayed when you are in MMMC mode.

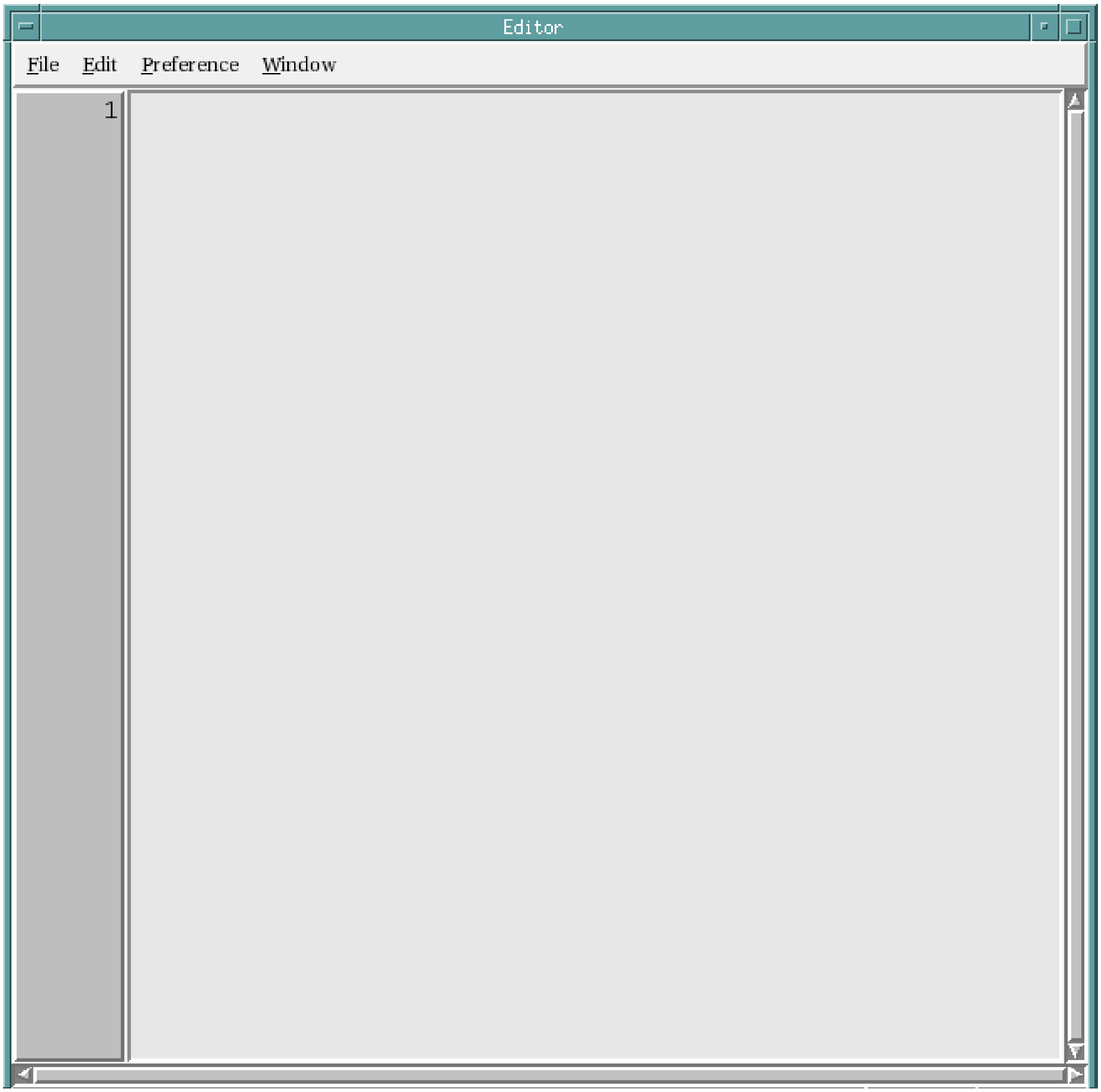
- Choose *Constraint* - *MMMC*.



To highlight a constraint that you used in MMMC mode, double-click on a constraint in the *Constraint List* pane.

Editor

Use the Editor submenu to display and edit a text file.



Fields and Options

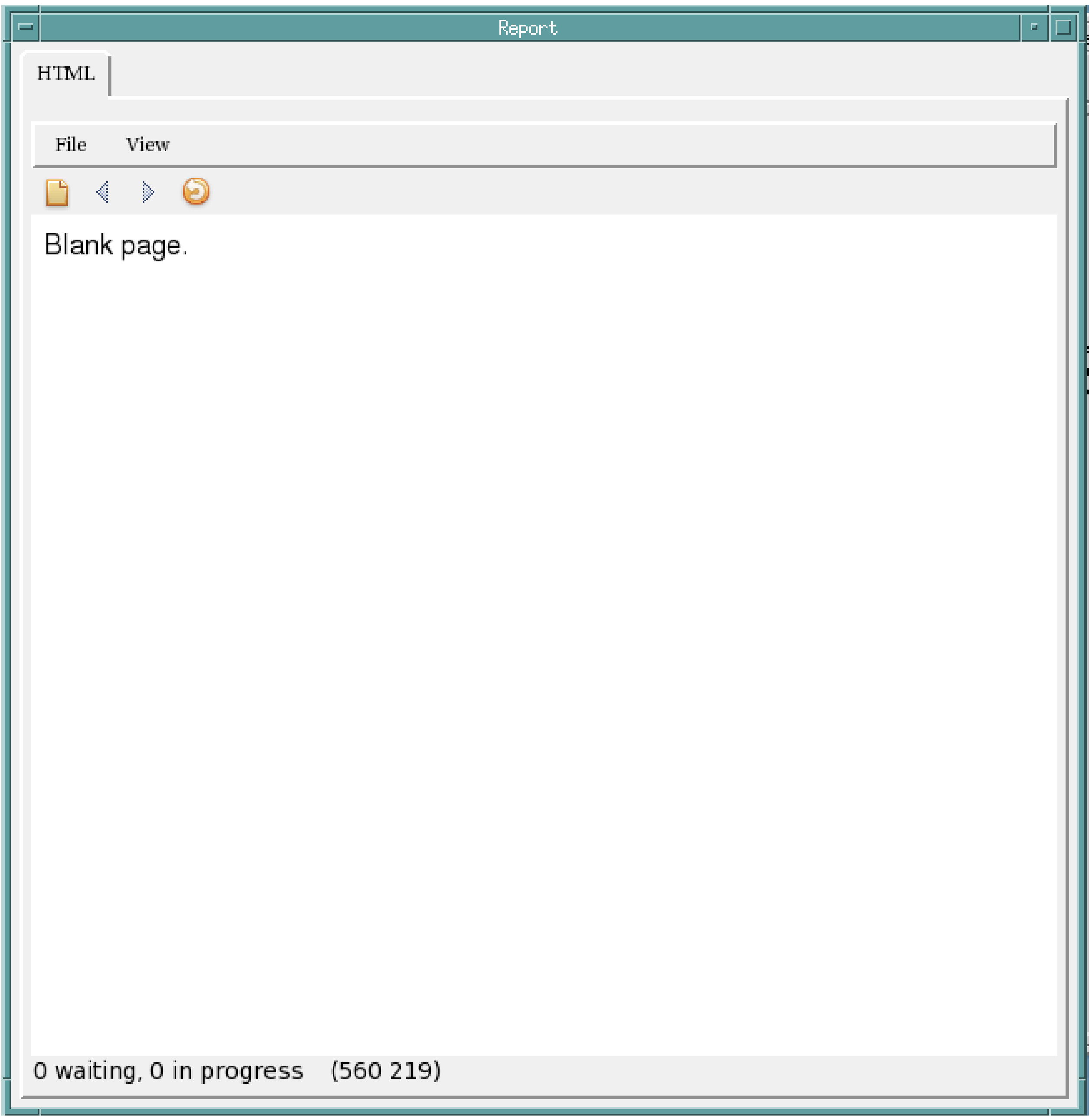
Pulldown	Descriptions

File	Choose from one of the following options:	
	New	Opens a blank file for editing.
	Open	Open an existing file for editing.
	save	Saves a file.
	Save As	Saves a file with a different file name.
	Evaluate	Evaluates the commands in the file.
	Close	Closes the file.
	Exit	Exits the editor window.
Edit	Choose from one of the following options:	
	Undo	Reverses the last operation.
	Cut	Cuts the selected text.
	Copy	Copies the selected text so it can be pasted at another location.
	Paste	Pastes the copied text.
	Find	Opens window to find the specified string in the document.
	Find Next	Finds the next instance of the specified string.
	Find Prev	Finds the previous instance of the specified string.
	Replace	Replaces the specified string with another specified value.
	Goto Line	Go to the specified line number.
	Grep	Searches for all files in the specified directory for the given string.
Preference	Choose form one of the following options:	
	Show Proc Window	
		Displays a right pane with procedure information.
	Show Line Number	
		Shows the line numbers at the left of the screen
	Word Wrap	To wrap words on a line to the next line if they do not fit in window.
	Refresh Highlighting	
		Refreshes highlighting

Window	Provides the ability to switch between any documents that have been opened.

Report

Use the Report submenu to open any HTML reports.



Fields and Options

Menu	Description	
File	Choose from one of the following options:	
	Open File	Opens the specified file.
	Open Tab	Opens a new browser tab.
	Close Tab	Closes the currently selected browser tab.
View	Choose form one of the following options:	
	Gui Font	Specify the GUI font size from a predefined list.
	Browser Zoom	Specify the browser zoom factor from a predefined list.
	Browser Font Scale	
		Specify the browser font size from a predefined list.
	Browser Font Scale Table	
		Specify the browser font size from a predefined list.

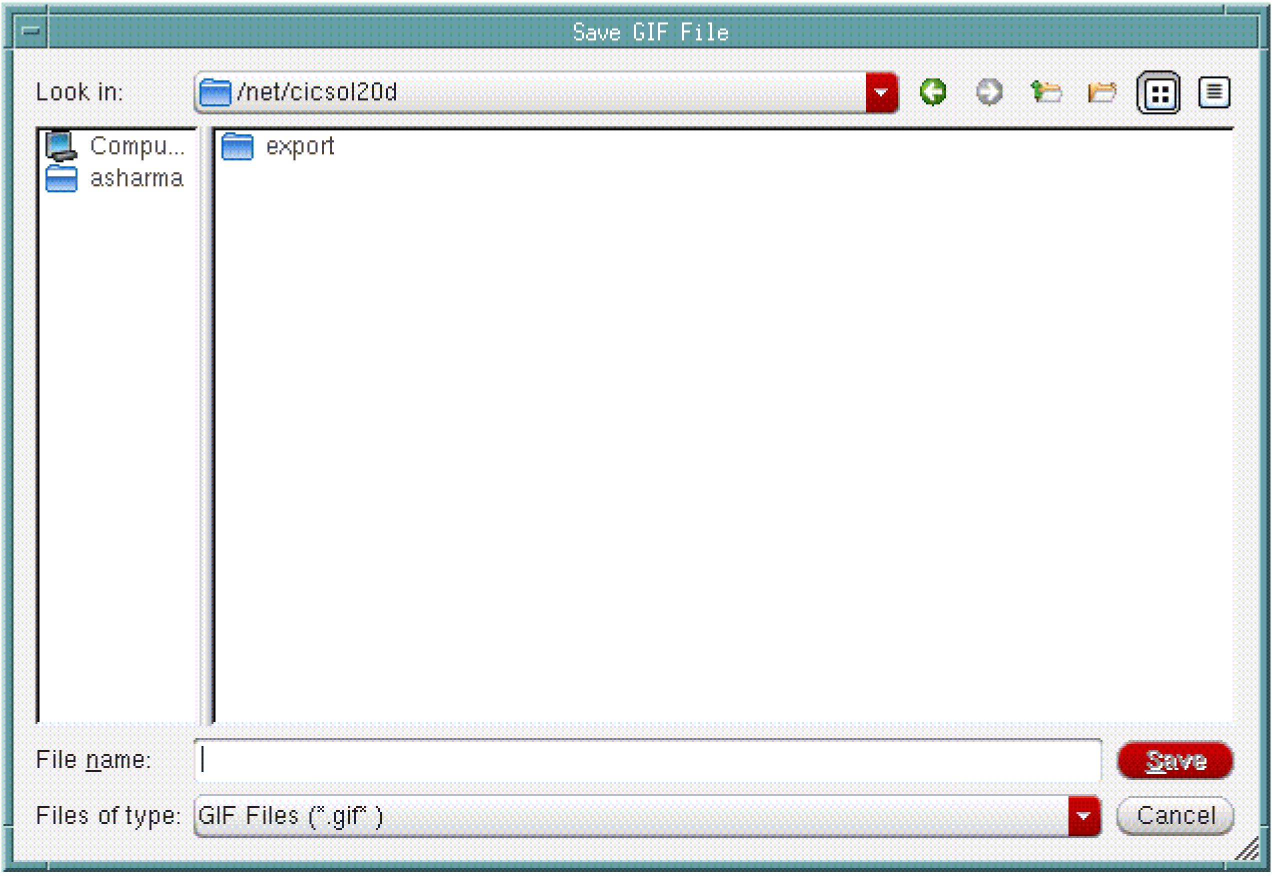
Screen Capture

- Write To GIF File
- Screen Dump
- Display Screen Dump

Write To GIF File

Use the Write To GIF File menu command to save the snapshot of the current screen to a GIF file in the current directory.

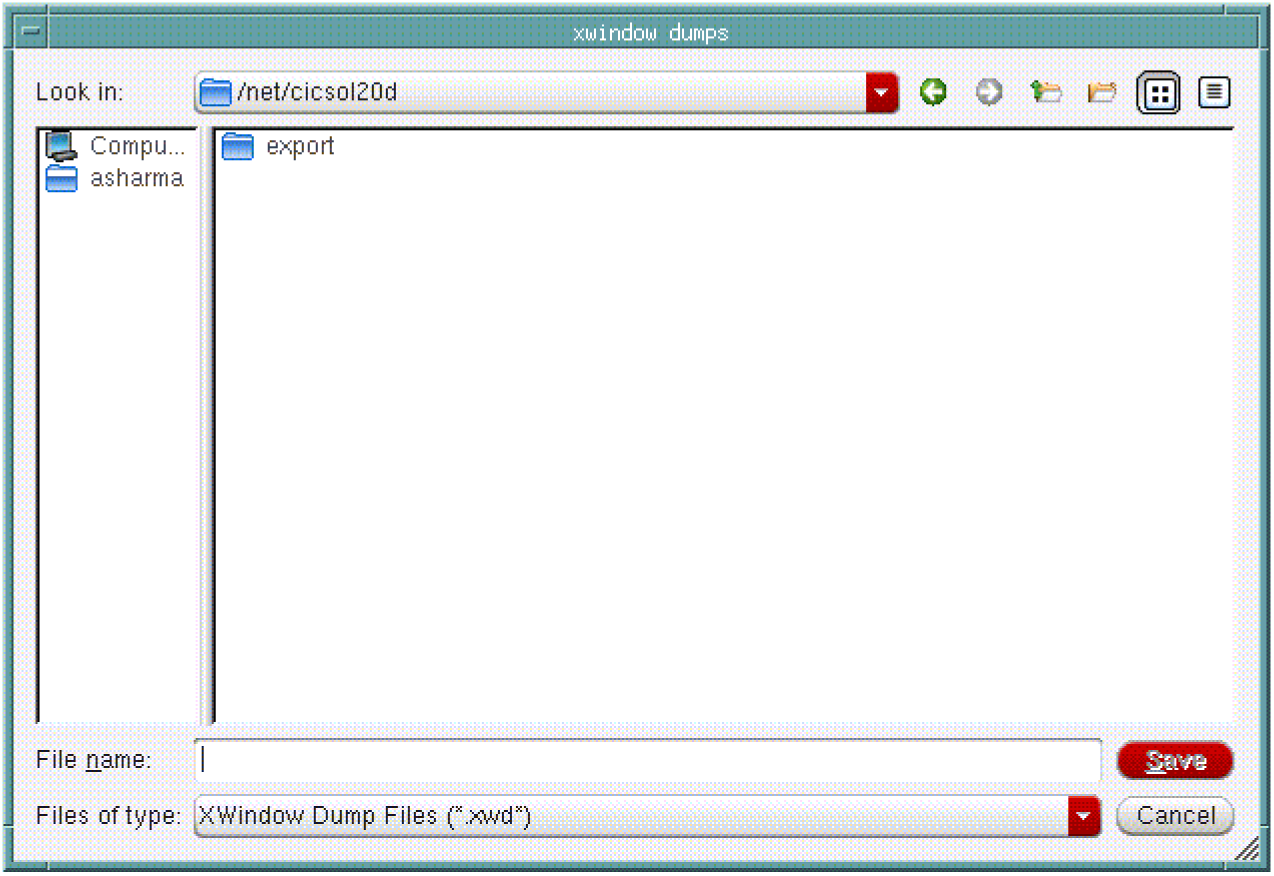
- Choose *Tools - Screen Capture - Write To GIF File*.



Screen Dump

Use the Screen Dump menu command to take a snapshot of the current screen. The file name is in timestamp format that shows when the snapshot was taken. The gzipped file, with the .gz extension, is listed in the text window.

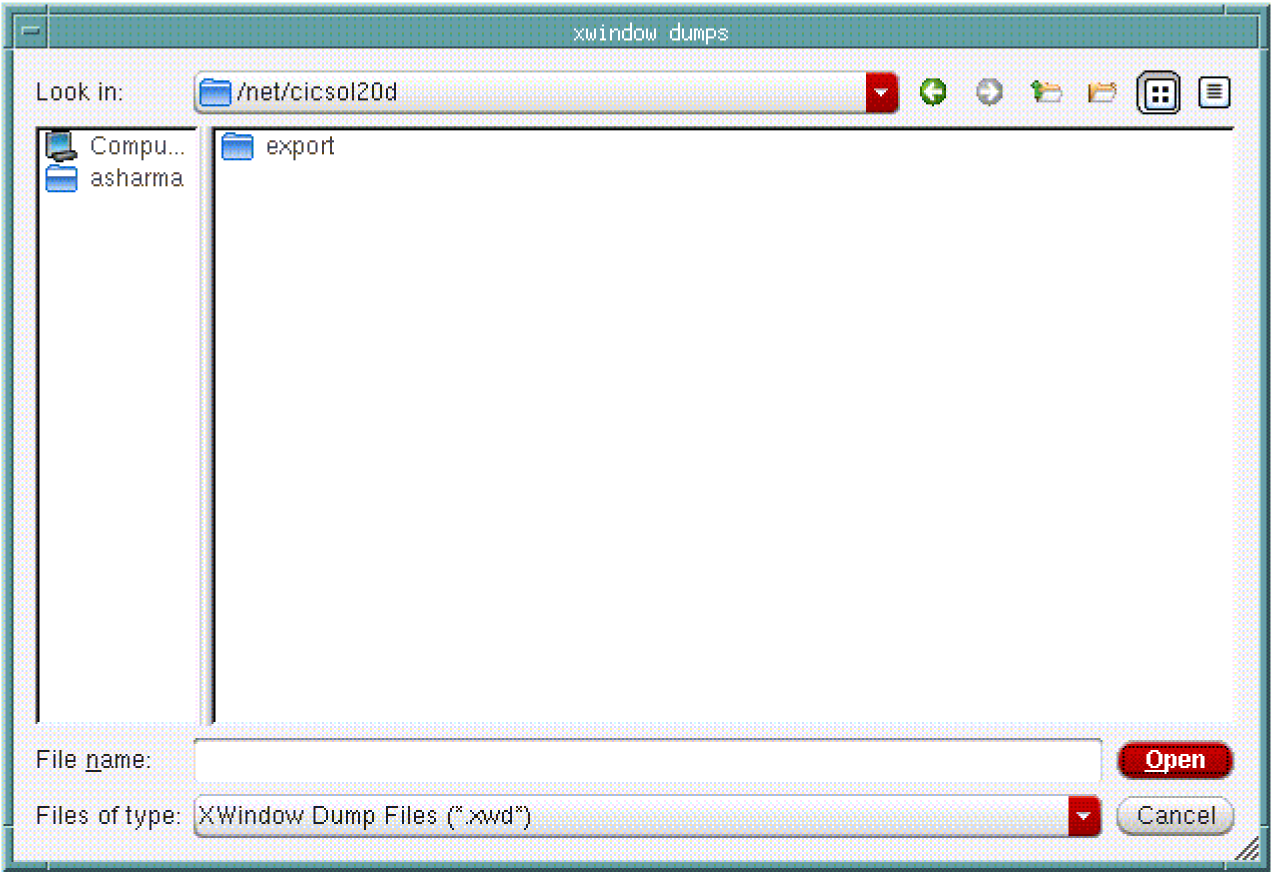
- Choose *Tools - Screen Capture - Screen Dump*.



Display Screen Dump

Use the Display Screen Dump menu command to view snapshots that you have taken earlier.

- Choose *Tools - Screen Capture - Display Screen Dump*.

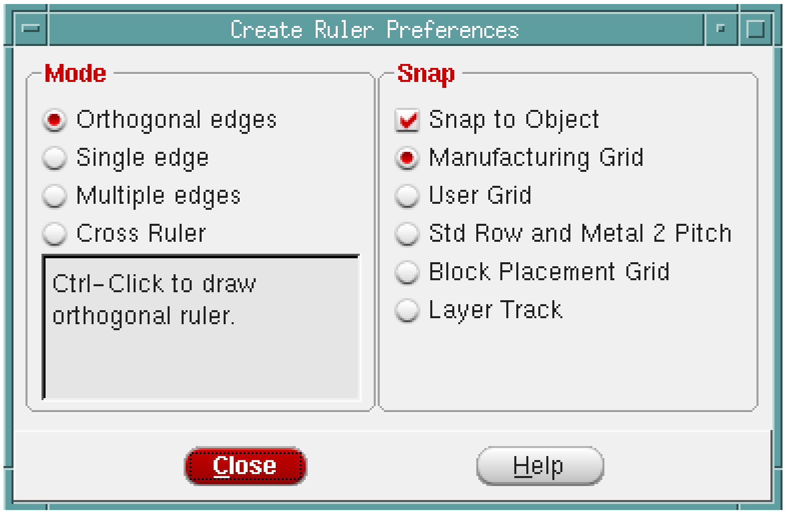


Create Ruler Preferences

Use the Create Ruler Preferences form to set the direction in which ruler lines are drawn.

To open the *Create Ruler Preferences* form:

- Choose *Tools - Create Ruler*. The *Create Ruler Preferences* form is displayed.



Fields and Options

Snap	
Snap to Object	Specifies that the ruler should snap to the nearest floorplan object.
Manufacturing Grid	Specifies that the ruler should snap to the manufacturing grid.
User Grid	Specifies that the ruler should snap to the user specified grid.
Std Row and Metal 2 Pitch	
Block Placement Grid	Specifies that the ruler should snap to the block placement grid.
Layer Track	
Mode	
Orthogonal	Specifies that ruler lines can be drawn in horizontal as well as vertical directions.
Vertical	Specifies that ruler lines will be drawn in the vertical direction.
Horizontal	Specifies that ruler lines will be drawn in the horizontal direction.
X	Specifies that ruler lines can be drawn at a 45 degree angle as well as at a 135 degree angle.
45	Specifies that ruler lines will be drawn at a 45 degree angle.
135	Specifies that ruler lines will be drawn at a 135 degree angle.

Any Angle	Specifies that ruler lines can be drawn in any direction.

Clear All Ruler

Clears all rulers from the display area.

11

Windows Menu

Workspaces

Tempus enables you to save and load your customized window settings easily through workspaces. A workspace contains customized main window settings including:

- Size and placement of window
- Specific orientation of toolbars
- Docking status
- Visibility settings

Use the *Workspaces* submenu to manage your workspaces. To access this submenu:

☨ Choose *Windows - Workspaces*.

The *Workspaces* submenu contains the following menu items:

- Schematic
- Analysis
- Design Browser + Schematic
- Analysis + Physical
- SI + Physical
- Save Workspace
- Delete
- Set Default

Schematic

Use the *Schematic* menu item to switch to the Schematic view of your current workspace. For more information, see Schematic Tab on page 564.

☨ Choose *Windows - Workspaces - Schematic*.

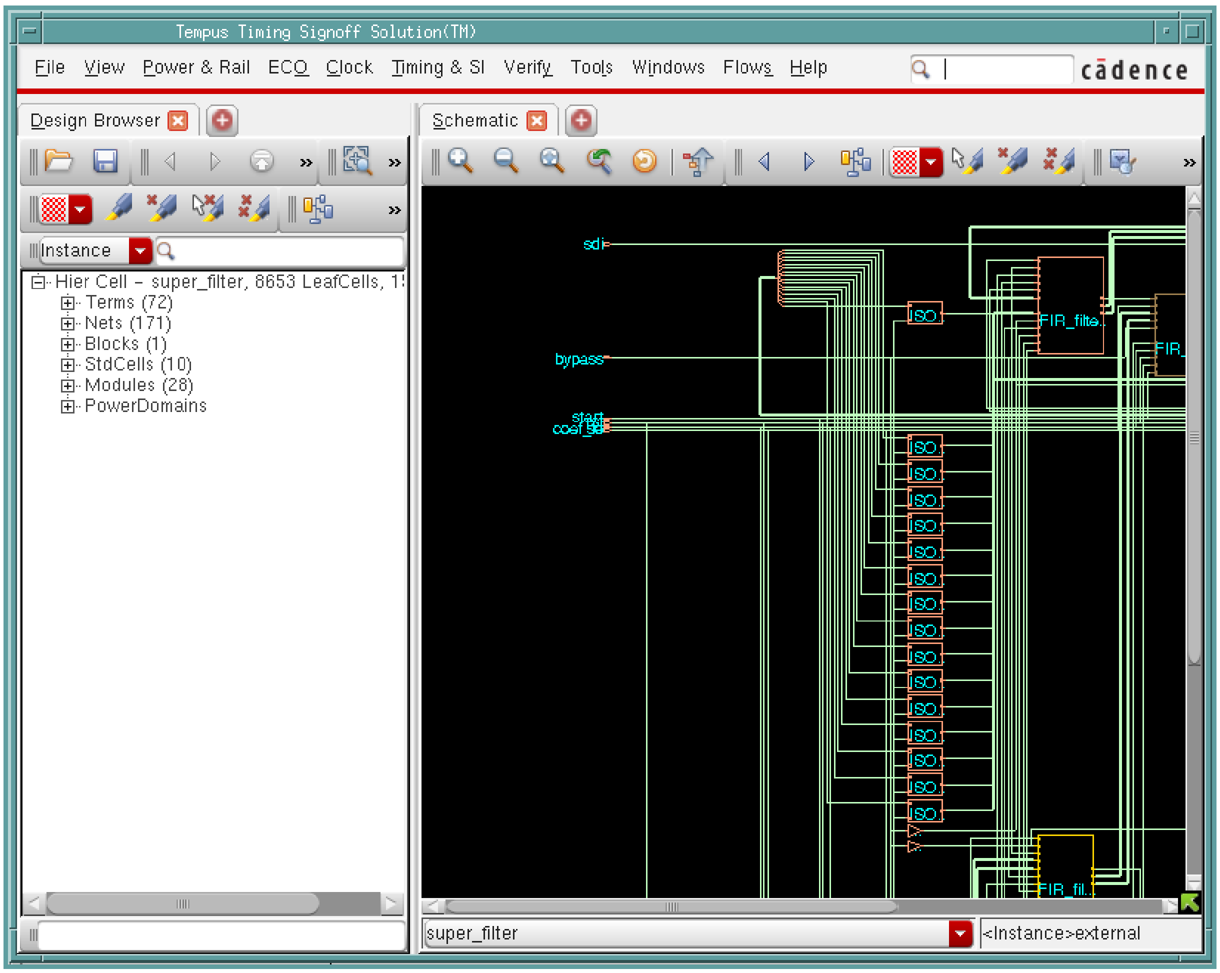
Analysis

Use the *Analysis* menu item to switch to the Analysis view of your current workspace. The analysis window displays the violation information, categories defined to group violation types, graphical display of violations, and category summary. For more information, see Analysis Tab on page 484.

- Choose *Windows - Workspaces - Analysis*.

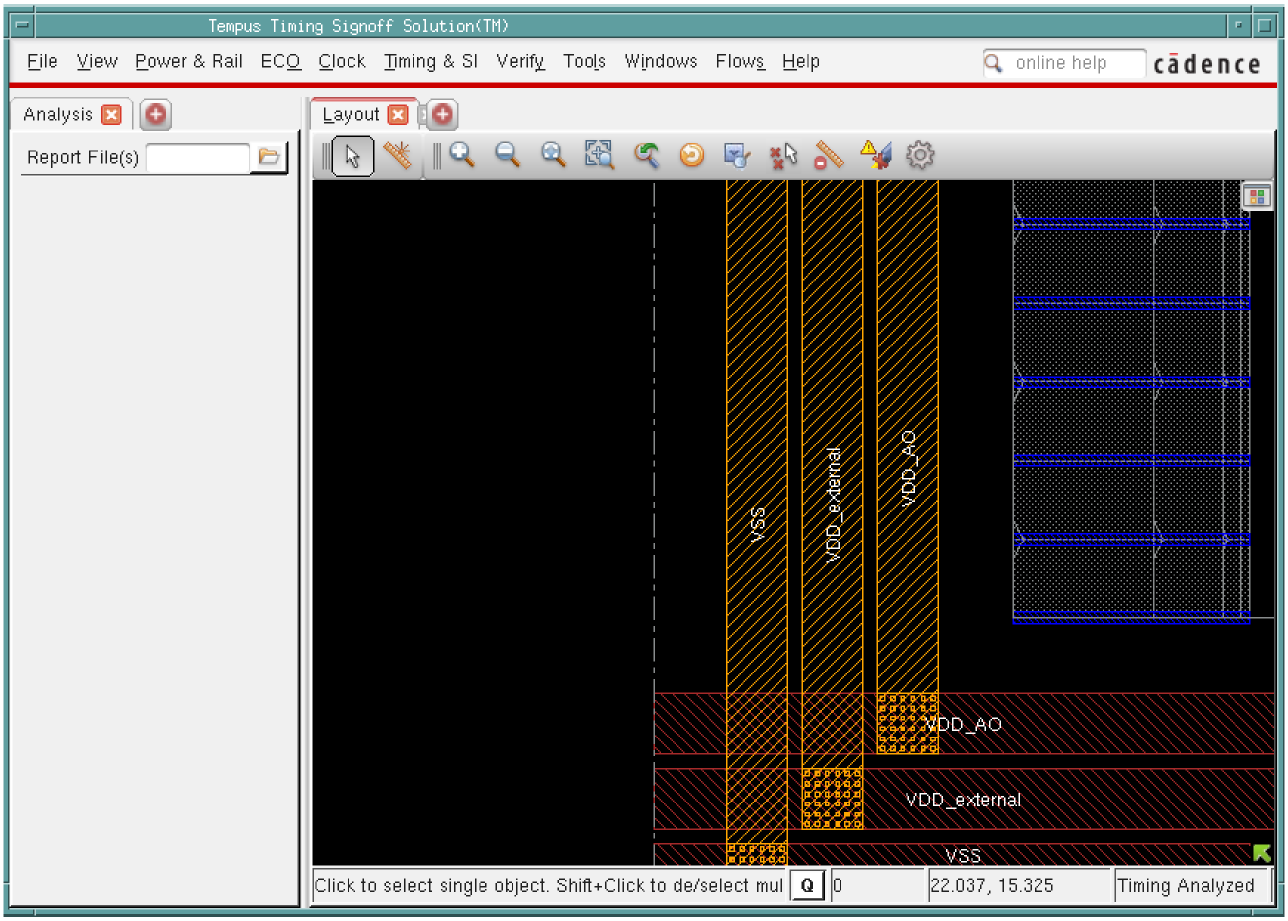
Design Browser + Schematic

Use the *Design Browser + Schematic* menu item to see the Design Browser and the schematic view of the design, side by side.



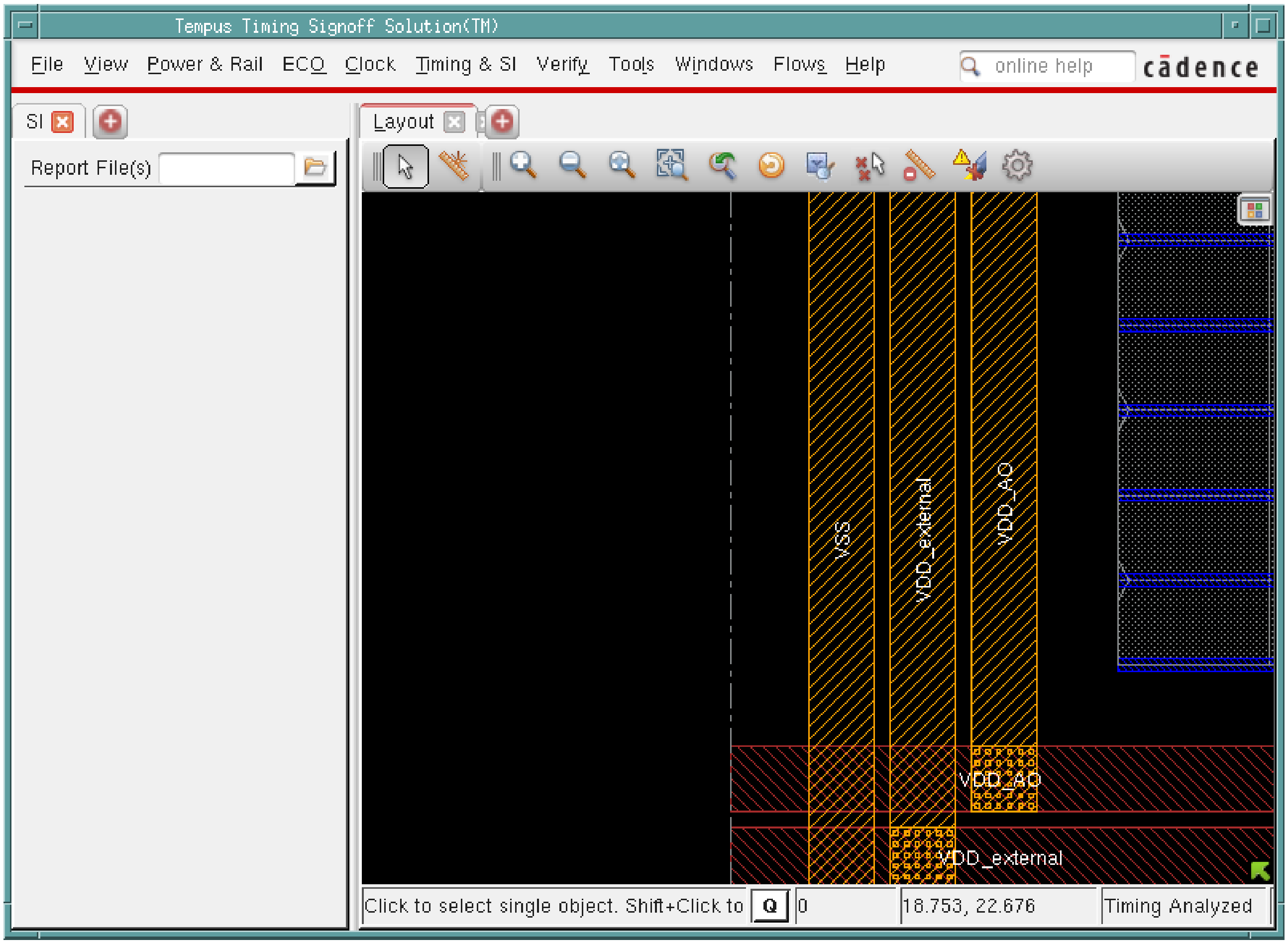
Analysis + Physical

Use the *Analysis + Physical* menu item to see the Analysis tab and the physical view of the design, side by side.



SI + Physical

Use the *SI + Physical* menu item to see the SI tab and the physical view of the design, side by side.



Save Workspace

Use the Save Workspace form to save your workspace to a specific directory. You can then load the workspace whenever you want from the specified directory.

Choose *Windows - Workspaces - Save Workspace*.

Save Workspace Fields and Options

Select workspace name (or enter a new name)	Specifies the name of the workspace you are saving. You can type a new name or, if you have previously saved workspaces, choose an existing name from the drop-down list. The drop-down lists all the workspaces in the selected path.
Select Path	Specifies the location for saving the workspace. You can choose to save the workspace in the current working directory or your home directory. If the workspace is saved to the current directory, it is applied to the current design only. If the workspace is saved to your home directory, it is applicable for all designs. *Default:* Current working directory

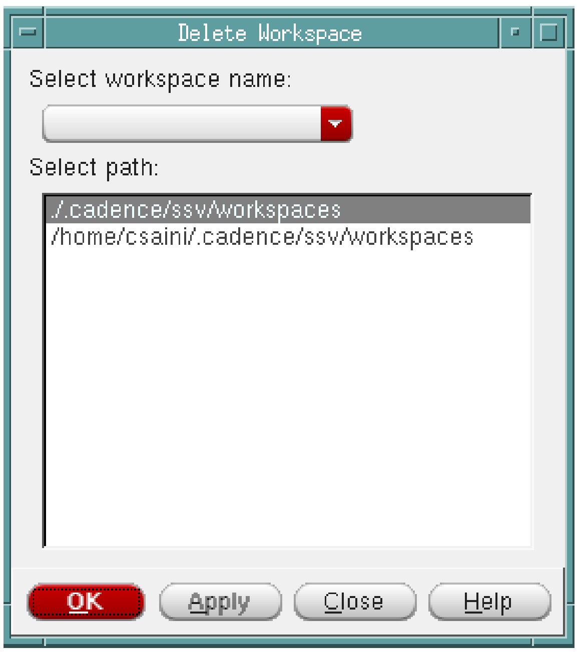
After you have saved a workspace, the saved workspace name appears on the *Workspaces* submenu. Whenever you want to load a workspace, simply click the saved workspace name in the *Workspaces* submenu.

If there are multiple workspaces, the one set as default is loaded automatically. If you have not set any workspace as default, the one that you saved last is loaded automatically. You can then switch to another workspace by selecting the corresponding check box in the *Workspaces* submenu.

Delete

Use the *Delete Workspace* form to delete the selected workspace.

- Choose *Windows - Workspaces - Delete*.



Delete Workspace Fields and Options

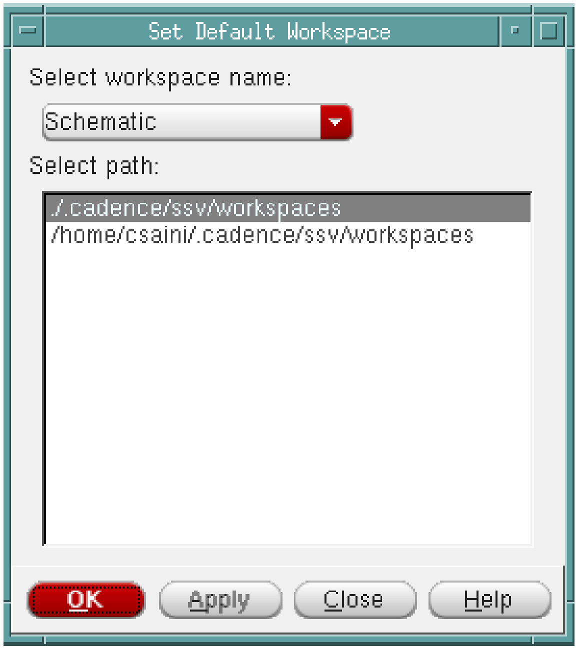
Select workspace name	Specifies the name of the workspace you are deleting. The drop-down lists all the workspaces in the selected path.
Select Path	Specifies the location of the workspace. *Default:* Current working directory

Set Default

Use the Set Default Workspace form to set the selected workspace as default.

- Choose *Windows - Workspaces - Set Default*.

The default workspace is loaded automatically when starting up the software. So if you want Tempus to load a workspace automatically, you need to first save it as default.



Set Default Workspace Fields and Options

Select workspace name	Specifies the name of the workspace you are setting as default. The drop-down lists all the workspaces in the selected path.
Select Path	Specifies the location of the workspace. *Default:* Current working directory

Menus

Use the *Windows - Menus* submenu to show or hide specific menus, as required.

Toolbars

Use the *Windows - Toolbars* submenu to show or hide specific toolbars, as required. For example:

- Schematic Tab: Controls the display of widgets related to the Schematic View in the Toolbar Widget area above the design display area.

- Design Browser: Controls the display of widgets related to the Design Browser in the Toolbar Widget area above the design display area.

12

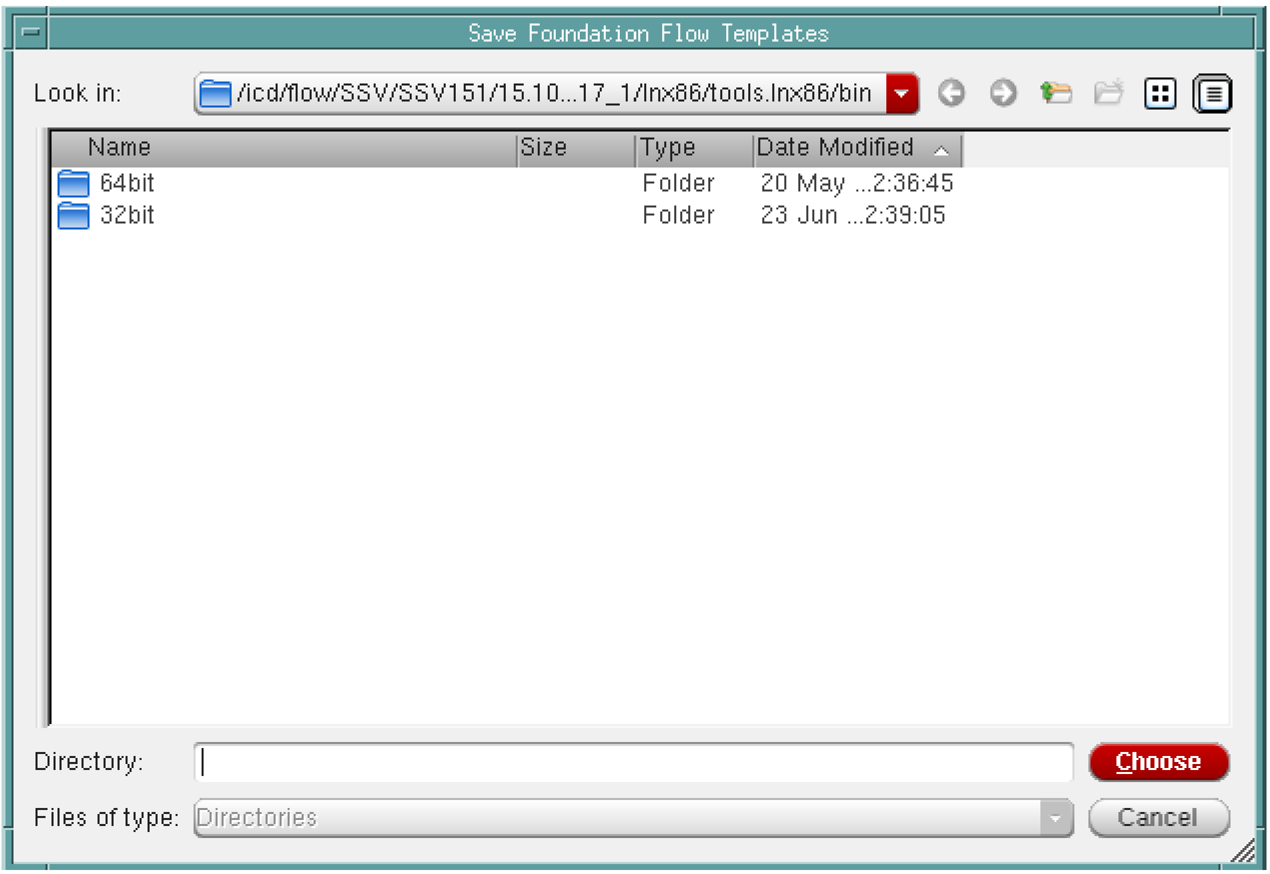
Flows Menu

Create Foundation Flow Template

In Tempus, you can enable customized Foundation Flow templates.

☨ *Choose Flows - Create Template*.

Use this menu item to save the flow templates to a specified directory. The templates consist of a set of Tcl scripts you can customize for your design and technology needs.



Related Topics

Refer to *Tempus Foundation Flows User Guide*.

13

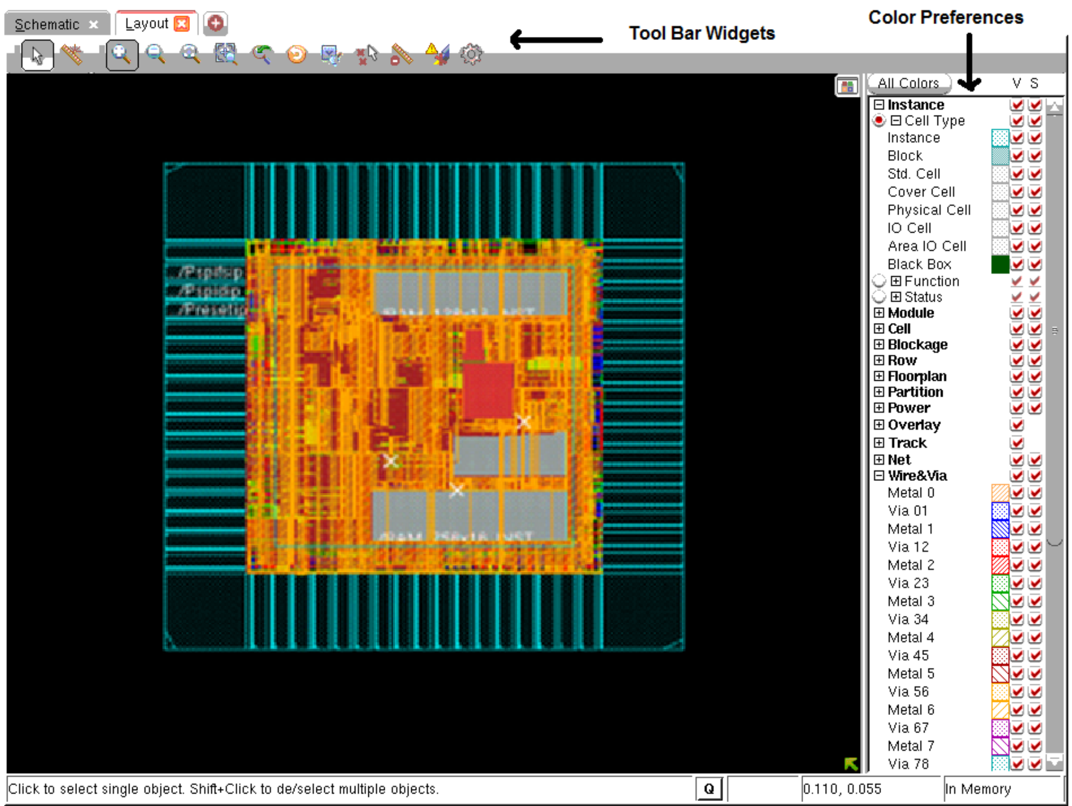
Tab Options

The following tabs are available in Tempus GUI:

Layout Tab

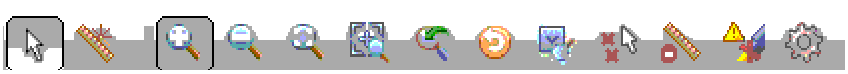
Layout Window Components

Use the Layout window to display the physical view of the design in design display area. The components in the Layout window are identified in the following figure, and described in the sections that follow.

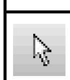


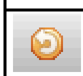
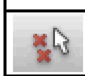
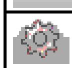
Toolbar Widgets

The following row of widgets, located below the menus and above the design display area, includes shortcuts for some commonly used Innovus commands.



The following table describes the widgets that are located in the Toolbar Widget area:

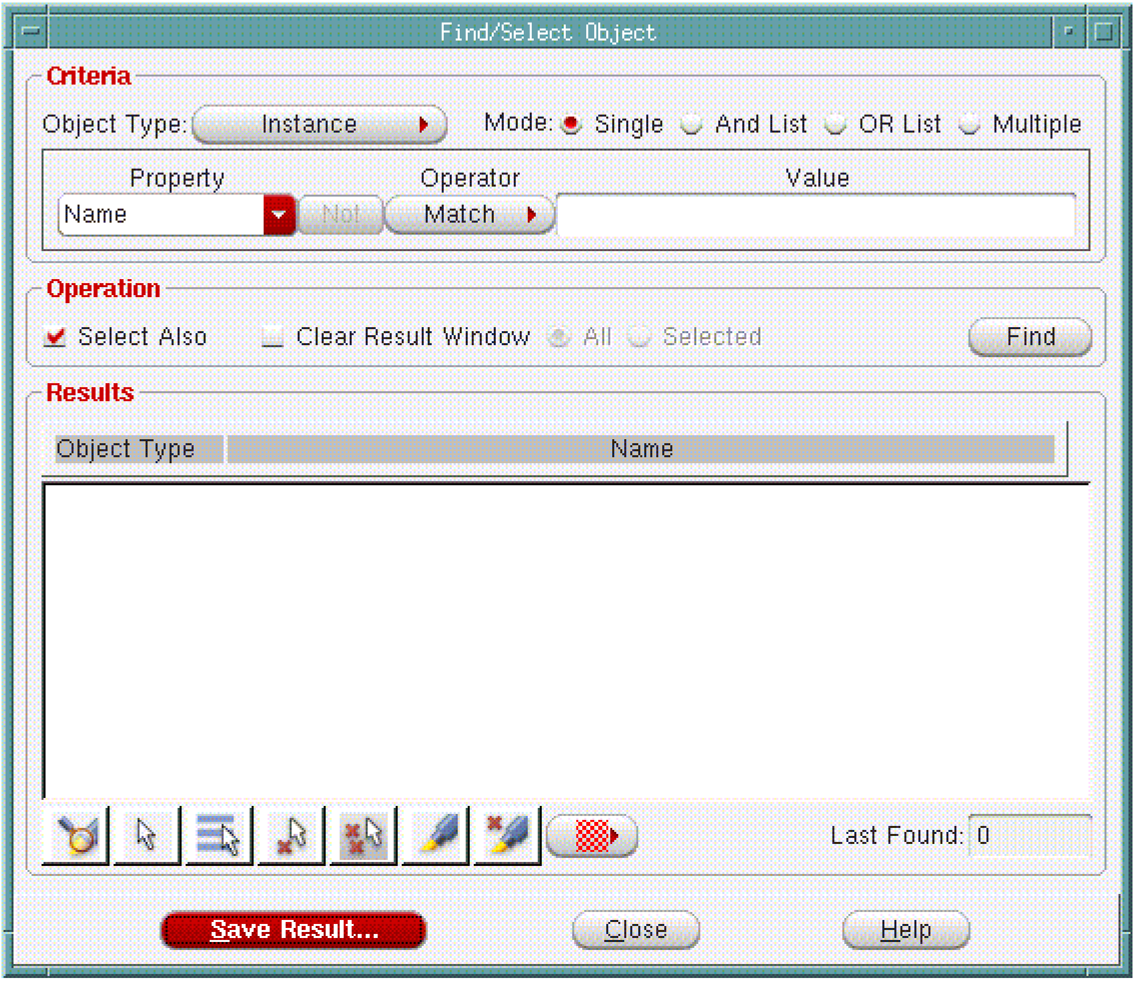
Widget	Description
	*Select*--Click this widget, then click a floorplan object to select it. To select multiple objects, use the *Shift* key or click and drag the mouse.

	*Ruler*--Click this widget, then left-click in the design display area and drag the mouse to add a ruler (in micrometers), and left-click again to end the ruler. To move the ruler, left-click and drag it to a new location. Left-click the ruler again to keep it in the new location. To remove a ruler, left-click on it and press *Esc*. To remove all rulers, press `K` (`Shift-k`). You can set the direction in which rulers can be drawn by clicking the widget, then pressing the `F3` key to display the Set Ruler Mode form. You can choose to draw orthogonal, vertical, horizontal, or diagonal rulers. You can also snap rulers to floorplan objects.
	*Zoom In*--Click this widget to display a smaller area of the design in greater detail. Each click zooms in one level. The equivalent bindkey is `z`.
	*Zoom Out*--Click this widget to display a larger area of the design in less detail. Each click zooms out one level.
	*Fit*--Click this widget to display the entire placed design within the design display area. The equivalent bindkey is `f`.
	*Zoom Selected*--Select one or more design objects, then click this widget to zoom into the area that contains the selected objects.
	*Zoom Previous*--Click this widget to toggle the display between the previous location/zoom level and the current location/zoom level. The equivalent bindkey is `l` (the letter L).
	*Redraw*--Click this widget to update the display with new colors or stipple patterns that you selected from the More Colors form.
	*Clear All Ruler*--Clears all rulers from the display area.
	Find/Select Object--Displays the Find/Select Object form.
	Deselect All Objects- Deselects all the objects.
	Clear DRC Violations--Clears all DRC violations from design display area.
	Opens the Preferences form, which enables you to customize the schematic display settings, color, and font.

Find/Select Object

Use the Find/Select Object form to find and select specific object types in the design display area.

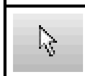
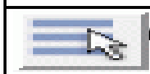
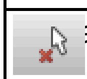
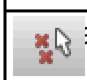
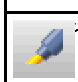
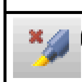
- Choose the *Find/Select Object* widget in the Layout Window.



Fields and Options

Object Type	Specifies the type of object to find. Choose from: *Instance*, *Pin, Net, Module, Instance Group*, or *Bump*. The `Pin` object type pin can be used for I/O pins, partition pins, and instance pins. Names of instance pins and/or partition pins should be specified with the full path name. For example, pin A of instance DTMF_CHIP/alu should be specified as: `DTMF_CHIP/alu/A` You can use the *Net* option to find/display bus names.

Mode	Specifies the type of find/select operation. Select one of the following: ■ *Single*: Performs a simple search for a Property/Value combination. ■ *And List*: Performs a find/select operation that matches an AND list of two or more multiple Property/Value combinations. ■ *OR List*: Performs a find/select operation that matches an OR list of two or more multiple Property/Value combinations. ■ *Multiple*: Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND lists and OR lists.
Property	Displays the available properties for the object that you selected in the Object Type field. You can use the pre-defined property `Connection` to find/select objects by connectivity. The `Connection` property supports connectivity search for the following objects: ■ Instance ■ Pin ■ Net ■ Module ■ Selected objects
NOT	Inverses the value of the operator in the Operator field. For example, If the value in the Operator field is `<`, clicking the NOT button effectively changes the value to `NOT <`.
Operator	Specifies the relationship between the property and its value. This field is dynamically populated based on the specified object type.
Value	Specifies the value of the property. This field is dynamically populated based on the specified object type. In some cases, you may have to manually specify a value, for example, the name of a cell.

Select Also	Finds all objects that meet the specified criteria, reports them in the *Results* window, and selects them in the design display area. *Default:* The objects are reported in the results window but are not selected in the design display area.
Clear Result Window	Clears objects previously listed in the *Results* list, then finds the objects that meet the specified criteria and reports them in the Results list. *Default:* The objects previously listed are retained in the results window along with the results of the current search.
All	Clears all objects previously listed in the Results list.
Selected	Clears those objects previously listed in the Results list that are currently selected in the *Results* list.
Find	Finds all objects that meet the specified criteria and reports them in the *Results* window.
Zoom in  ed	Zooms in the highlighted object(s). You can highlight object(s) by selecting them in the results window and clicking the *Highlight* button.
 *ct*	When you click *Select*, the objects selected in the Results window are selected in the design display area.
 *ll*	When you click *Select All*, all the objects in the Results window are selected in the design display area.
 *elect*	When you click *Deselect*, the objects selected in the Results window are deselected in the design display area.
 *elect All*	When you click *Deselect All*, all the objects in the Results window are deselected in the design display area.
 *hlight*	When you click *Highlight*, the objects selected in the Results window are highlighted in the design display area.
 *ighlight*	When you click *Dehighlight*, the objects selected in the Results window are dehighlighted in the design display area.

Color Preferences

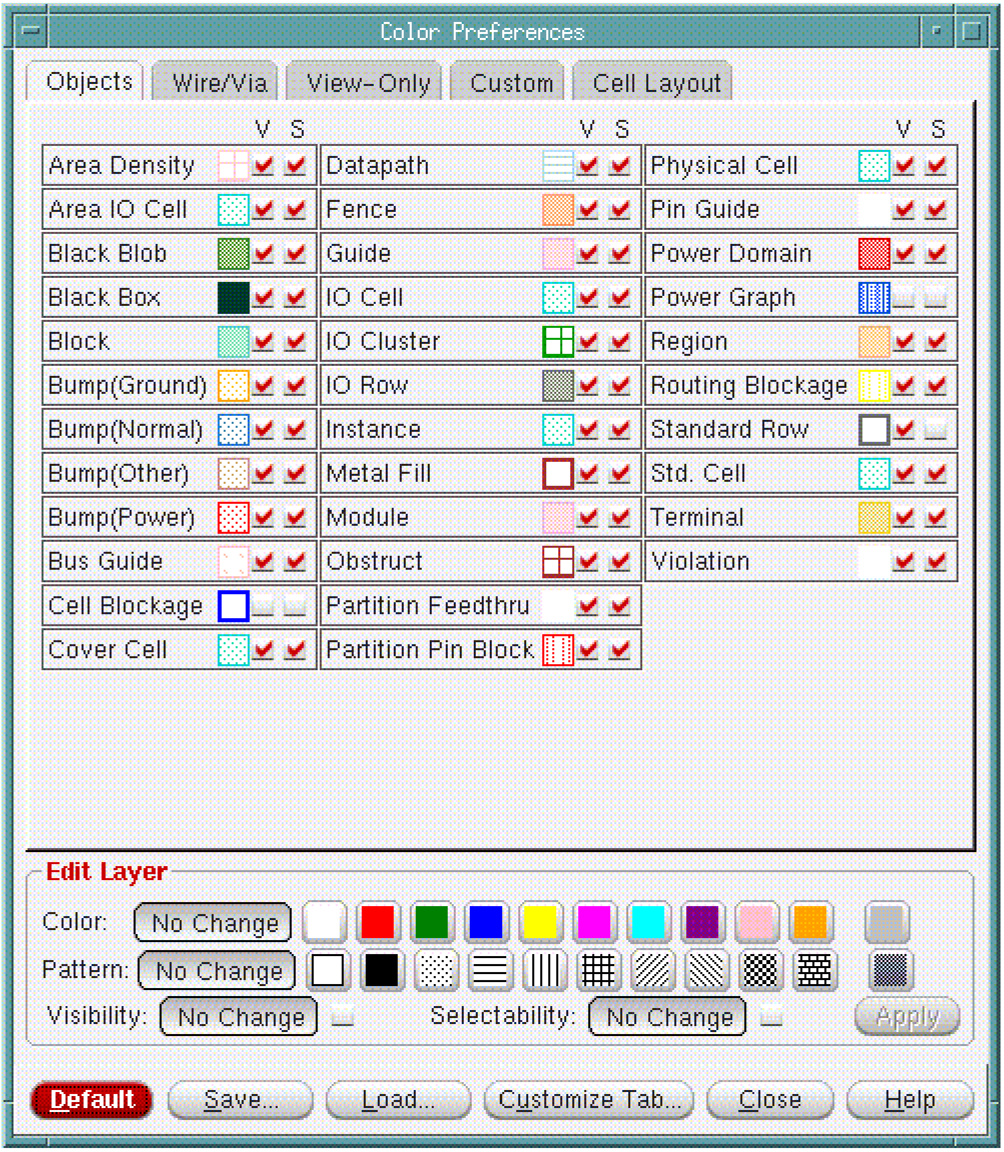
To set advanced color options, click the *All Colors* button in the Colors panel on the Layout window. This opens the Color Preferences form, which has five pages: *Objects*, *Wire/Via*, *View-Only*, *Custom*, and *Cell Layout*

The visibility settings between the pages and the Layout window are synchronized. That is, as soon you change any setting on any page, the corresponding changes are applied in the Layout window.

After making changes to the All Colors form, you can perform the following actions:

- To apply the changes in visibility, selections, color, and stipple pattern to objects in the design display area, click *Close*.
- To save a color preference file, click *Save*. For more information, see Save Color Preference on page 471.
- To load a preference file, click *Load*.

Objects Page



The Objects page includes the visibility and selection controls for objects that can be viewed and selected. Each object type has a color button that displays the current color and stipple pattern used to identify the object in the design display area.

Click a color button to select it, then click a new color and pattern from the *Edit Layer* section of the form. Click *Apply* to update the Color Preferences form.

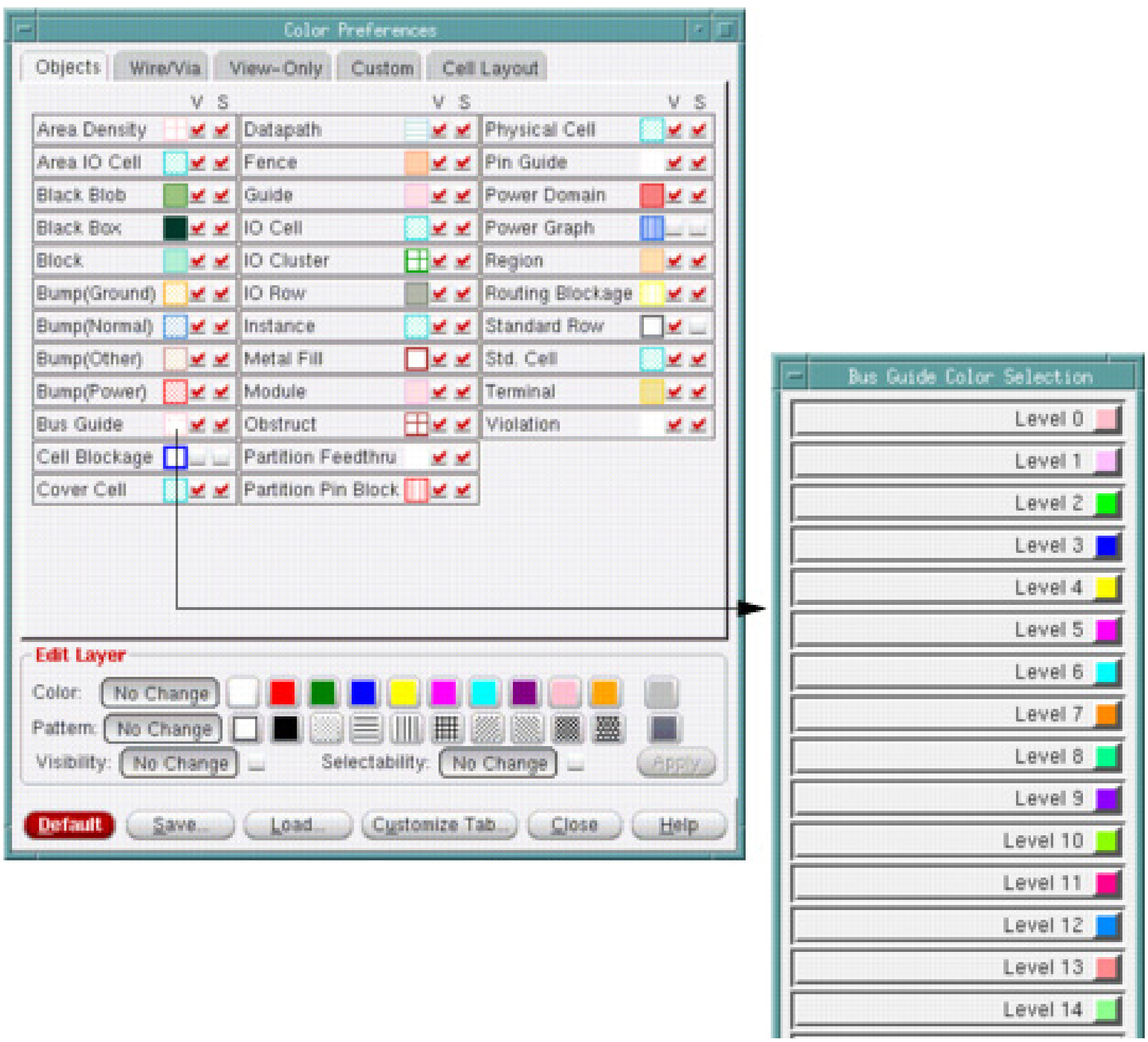
Double-click the color button for an object to open the Color form that shows a check box for indicating whether the object type is visible in the design display area, and another check box for making the object type selectable.

Note: If an object type does not have a Selectability check box, it cannot be selected.

When you move the cursor over an object layer name, the equivalent database object name is displayed.

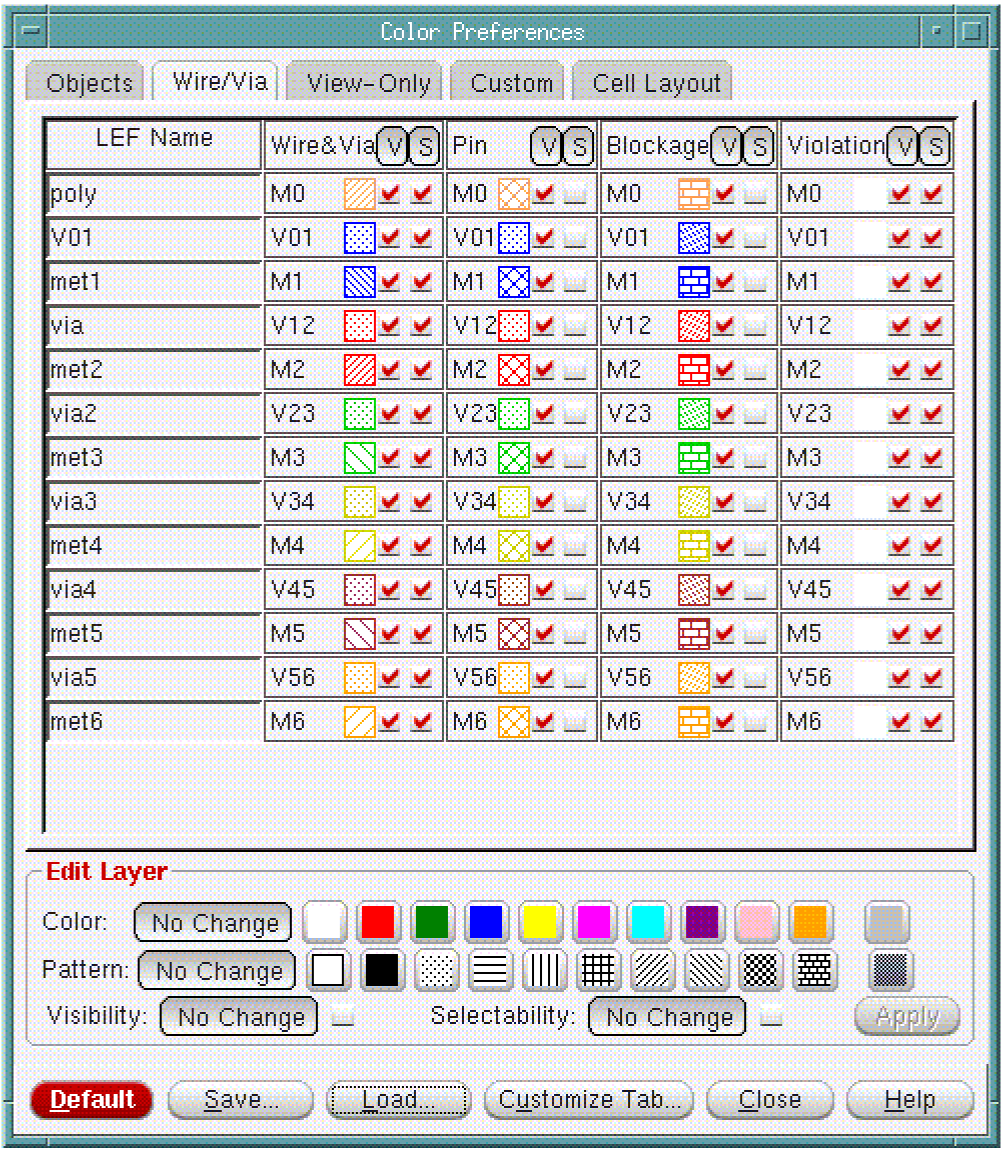
Bus Guide Color Selection

- Use the *Bus Guide Color Selection* form to specify multiple colors for bus guide objects.
- Double-click the *Color Preferences -- Objects -- Bus Guide* object type to display the *Bus Guide Color Selection* form.



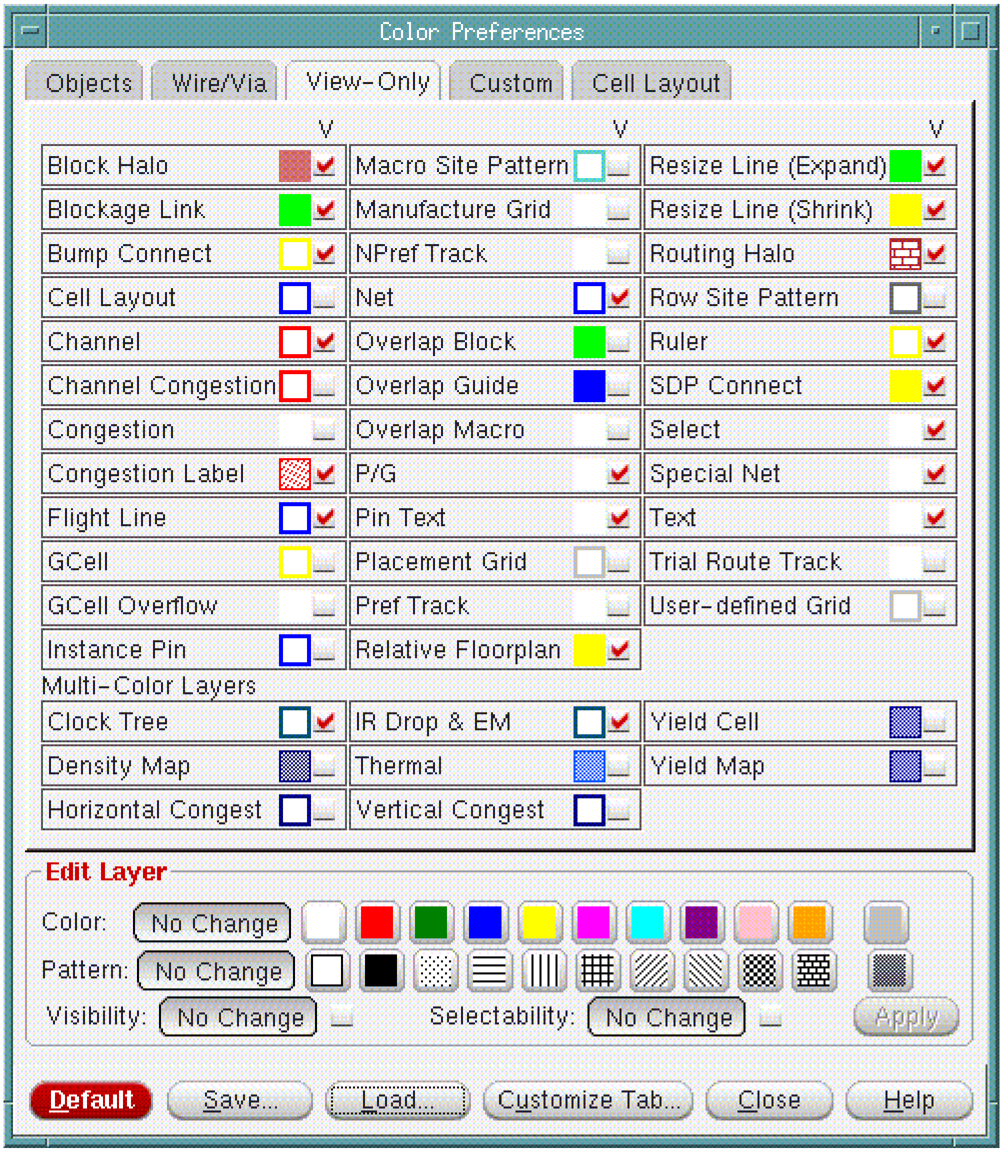
Wire/Via Page

The Wire/Via page includes the visibility and selectability controls for layer-based objects: Wires and Vias, Pins, Blockages, and Violations. You can set global display controls for all objects on the same layer. You can also set controls for individual objects on specific layers.



View-Only Page

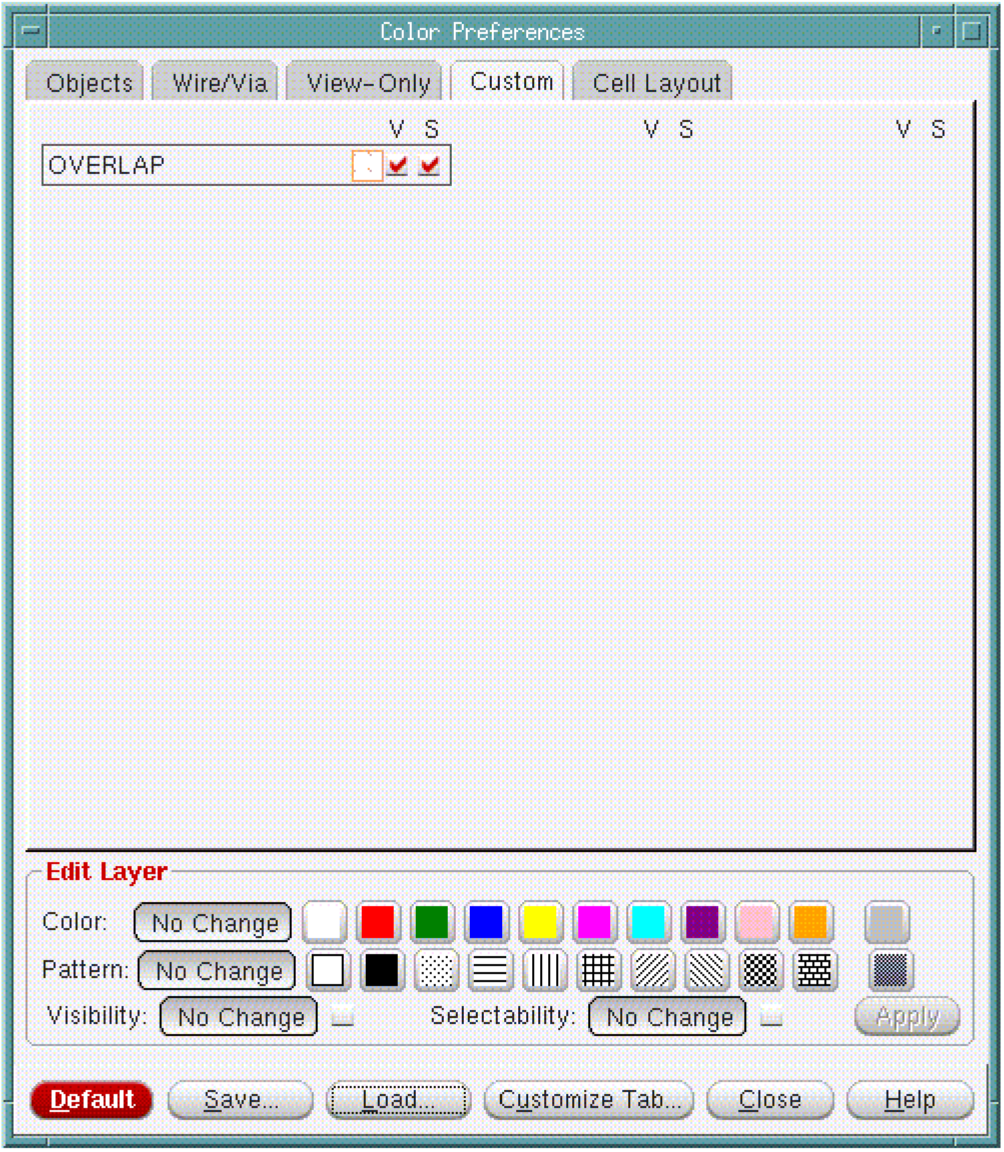
The View-Only page includes the visibility controls for objects that can be viewed but not selected in the Layout window.



Custom Page

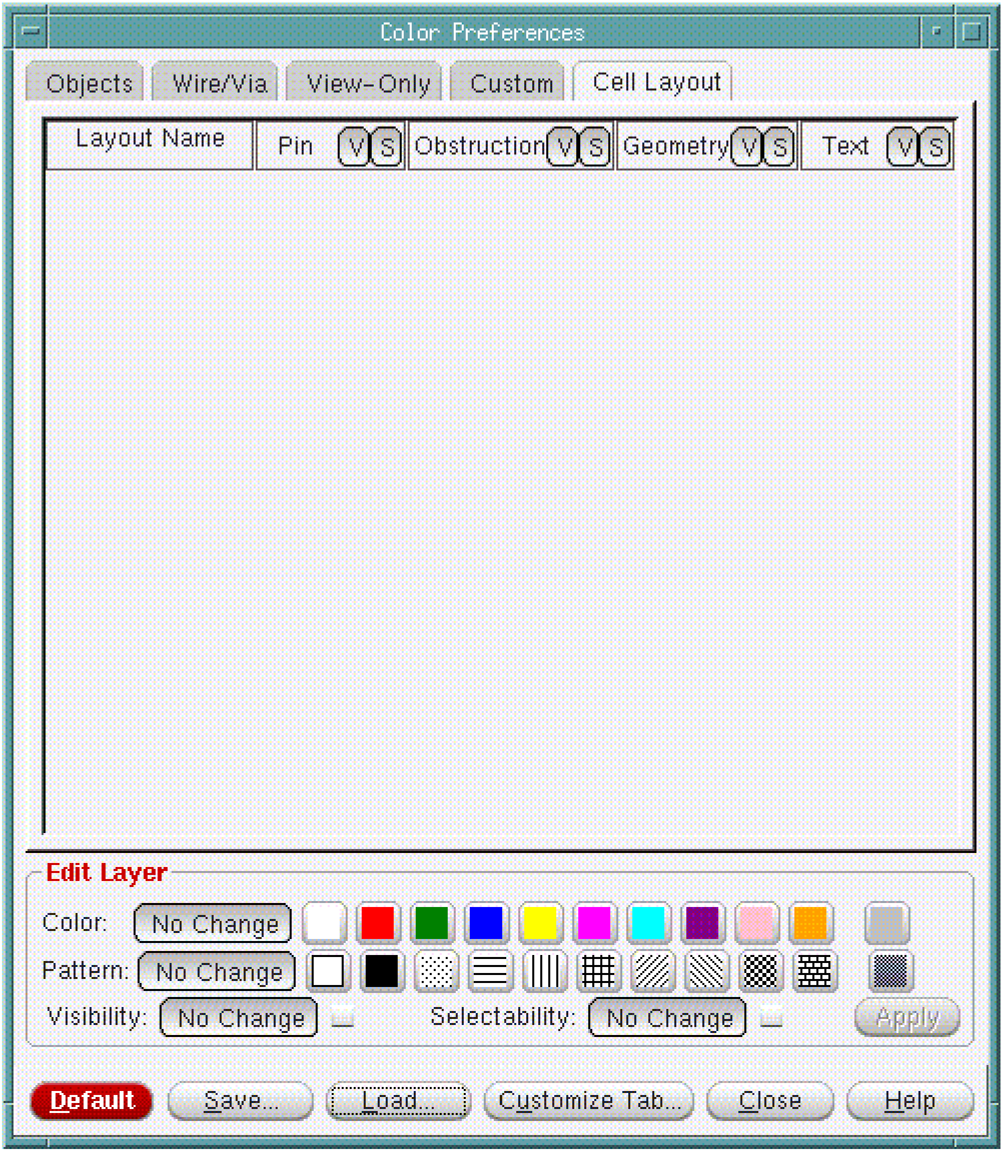
The Custom page contains custom objects.

To select a group of objects that are to have the same color and fill pattern, `Shift`-click the first and last objects in the group, click the new color and the new fill pattern, then click *Close.*



Cell Layout Page

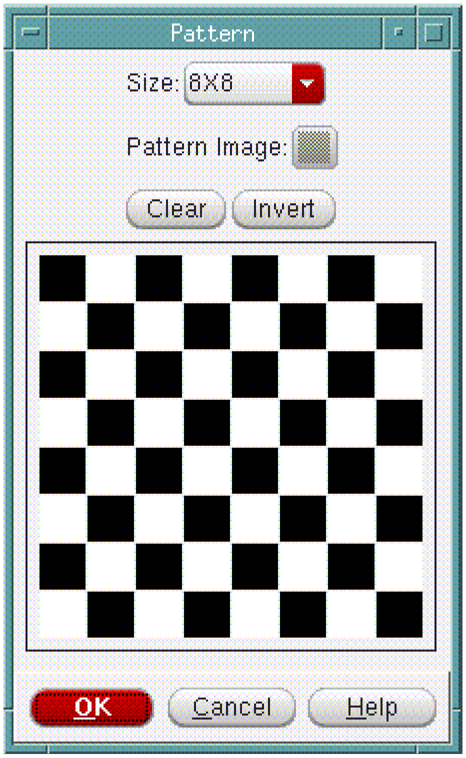
Use the Cell Layout page to set the color preferences for the OA cell layout.



Customizing Layer Color Patterns (Pattern Window)

You can customize the color patterns for the layers by clicking the *Custom pattern* icon in the Edit Layer group. The Custom pattern icon is the rightmost icon in the *Pattern* field.

Click the *Custom* icon to display the Pattern form.



To clear the pattern, click the *Clear* button.

To invert the pattern, click the *Invert* button.

To customize the pattern, do the following:

- To change a square to black color, click on the square with the left mouse button.
- To change a square to white color, click on the square with the right mouse button.

Customize Tab

You can use the *Customize Tabs* form, accessible by clicking the *Customize Tabs* button in any page on the Color Preferences form, to customize, or add additional pages to the Color Bar in the main window.

In any of the pages of the Color Preferences Form, click *Customize Tab*.

Fields and Options

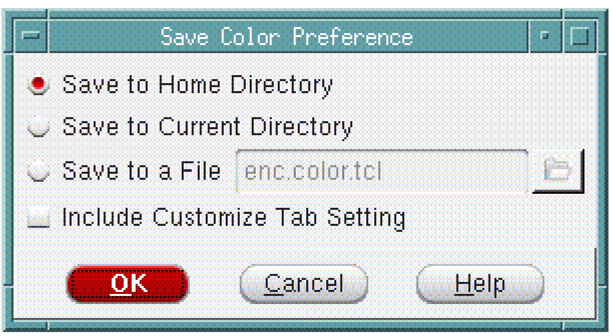
Expand All Pages When in Undock Mode	
	Displays all pages of the Object Color window when you undock the window.
Current Page	Selects the Page in the Color Bar Window that should be customized. By default, the color bar has two pages, but you can create up to four pages.
Groups in Current Page	
	Displays a selectable list of the names of the groups in the current color bar. A group is a collection of objects that are grouped together in the Color Bar. This way you can view the color and visibility controls for related objects together. A page in the color bar can have more than one group. The default first page in the Color Bar displayed in the main window contains the group FPlan View. The default second page contains two groups: Physical View and Layers. You can add, delete, or rename the groups in a Color Bar page.
Group Name	This is an editable field that displays the name of the group currently selected in the *Groups in Current Page* field.
Show Group Name	Displays the name of the group in the Color Bar page. If this option is not selected, the objects in the group are displayed in the Color Bar page, but the name of the group is not displayed. *Default:* The default is *Selected*.
Add	Adds a new group to the page. To add a new group, specify the name of the group in the Group Name field, and then click the *Add* button. You can then add objects to this group by selecting the objects in the *Add Layer* field and then clicking the *Add* button.
Delete	Deletes the group currently selected in the *Groups in Current Page* field.
Updates	Edits the name of the group currently selected in the *Groups in Current Page* field. To edit the name of the group, first edit the name in the Group Name field, and then click Update.
All Layers	Displays a list of all objects that you can add to a group.

Layers in Current Group	
	Displays the objects currently part of the group.
Add	Adds the object currently selected in the All Layers field to the Layers in Current Group field.
Delete	Removes the object currently selected in the *Layers in Current Group* field.

Save Color Preference

Use the Save Color Preference form to save a color preference file. A preference file is a Tcl file that contains color preference information. The recommended filename and extension name is `ets.color.tcl`.

In the Color Preference form, click *Save*.



Fields and Options

Save to Home Directory	
	Saves the color preferences file in your home directory.
Save to Current Directory	
	Saves the color preferences file in the current working directory.
Save to a File	Saves the color preferences file in the specified location.
Include Customize Tab Setting	
	Along with the color preference information, also saves the settings specified through the *Customize Tab* form.

Multicolor Layers

During a design session, the following object types do not have a single color, but each object has a color that shows a level of severity and magnitude:

- *Horizontal Congest*
- *Vertical Congest*
- *IR Drop or EM*
- *Clock Tree*

To see the level indicator colors, double-click the color button for the object type. This opens a color preferences form for that object type.

Metal Layers and Other Layered Components

Click the *Wire/Via* tab to see the metal layers available in the design. *M0 Wire* represents the masterslice layer, *M1 Wire* represents the bottom metal layer, and so on. The vias are also shown, with *V01* representing the via between the masterslice layer and the first metal layer, *V12* representing the via between the first and second metal layers, and so on.

You can set preferences for other layered components in addition to wires and vias. For each metal layer, you can set the color, stipple pattern, visibility, and selections for pins, tracks, blockages, and violations.

Note: Violations are not selectable.

Creating and Editing Color Groups

You can create color groups to customize the color settings in Tempus. A color group can contain custom color settings for various objects, for example, you can set the blockage color to red in one color group and to blue in another color group. You can switch between color views during an Innovus session. The settings specified in a color group are applied in all views.

To create a Color Group, do the following:

- In the Layout window, click the *All Colors* button. The *Color Preferences* form appears.
- Create the custom color settings that you want for the group. For example, you might want to set the block color, or the visibility of blockages.
- Click the *Customize Tab* button. The *Customize Tab* form appears.

- Click the *Color Group* button. The *Edit Color Group* dialog box appears.

Click the *Add* button. The Add Color Group form appears.

In the Group Name field, enter the name of the new color group, and click *OK*. The new color group is created.

To specify or change the color settings for the Color Group, do the following:

- In the Layout window, click the *All Colors* button. The *Color Preferences* form appears.
- Create the custom color settings that you want for the group. For example, you might want to change the instance color, or the visibility of blackboxes.
- Click the *Customize Tab* button. The Customize Tab form appears.
- Click the *Color Group* button. The Edit Color Group dialog box appears.
- Select the name of the *Color Group* to which the changed color settings should be applied.
- Click the *Set* button. The settings are applied to the selected Color Group.

Once you create one or more color groups, the color setting for a particular group can be applied be selecting the group name from a drop-down menu next to the *All Colors* button in the main form. You can thus select the color group from the main window.

Color Bar

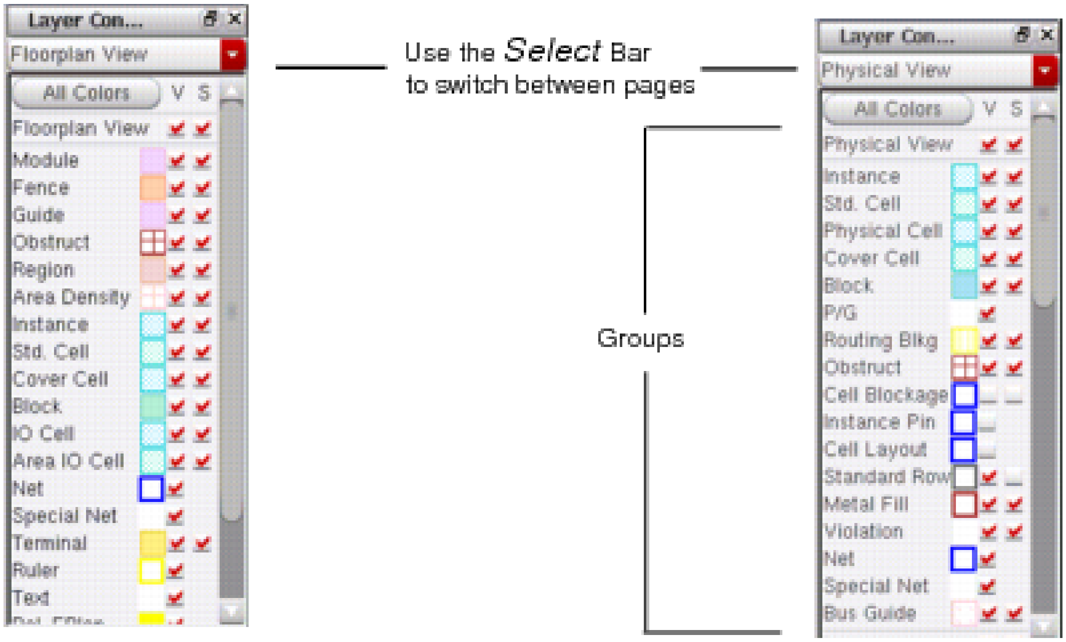
The Layout window includes an Object Color Bar that has color display buttons for objects in the design display area. Each object type has a check box for displaying the object in the design display area and a color button that displays the current color used to identify the object in the design display area.

By default, the Object Color Bar has two pages, but you can create up to four pages. Each page in the color bar can have one or more group. A group is a collection of objects that are grouped together in the Object Color Bar. This way you can view the color and visibility controls for related objects together. The default first page in the Object Color Bar displayed in the Layout window contains the group FPlan View. The default second page contains two groups: Physical View and Layers.

You can use the Customize Tab form to add pages to the Object Color Bar, add or delete the groups, and to add objects in each group. The Customize Tab form is available on all pages of the Color Preferences form.

Note: You can right-click on an individual layer to toggle the visibility/selectability of all objects on the layer.

To switch between the pages, click the *Select* bar.



Page 2 of Color Bar

If you select the *V* (Visibility) check box for an object type, the object type displays in the design display area in the specified color. If you clear the check box, the object type does not display in the design display area.

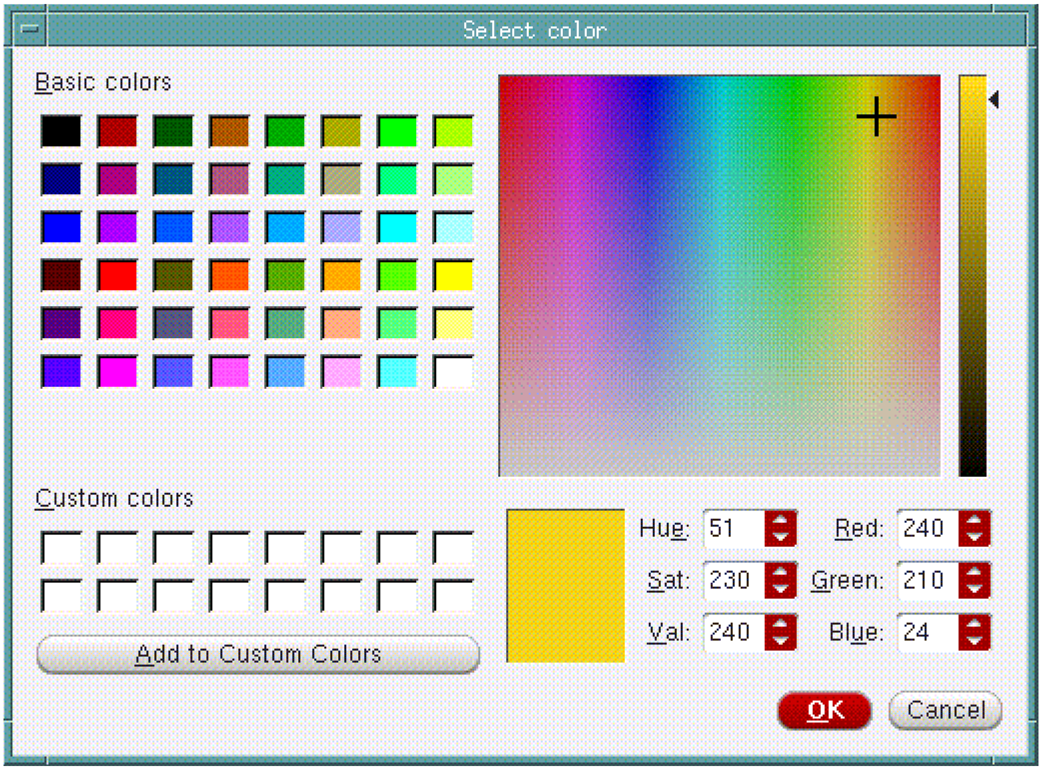
If you select the *S* (Selectability) check box for an object type, the object type can be selected in the design display area when you use the *Selection* icon. If you clear the check box, the object type cannot be selected.

Note: Objects that are not visible are not selectable.

Object Color

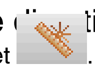
Use the Select color form to customize the display color of an object.

Click on the color indicator of the object type to open the Select color form.

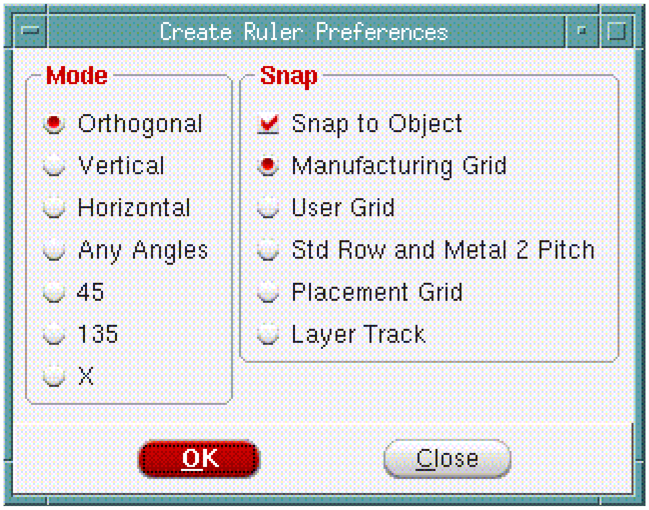


If you use the text fields, you must press the `Return` key after your text entry.

Create Ruler Preferences

Use the Create Ruler Preferences form to set the direction in which ruler lines are drawn.
To open the Create Ruler Preferences form, click the Ruler widget .

- Press the F3 key.

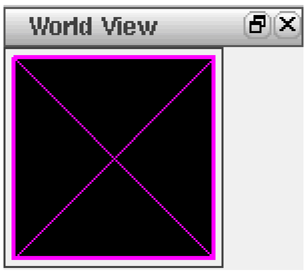


Fields and Options

Snap to Object	Specifies that the ruler should snap to the nearest floorplan object.
Manufacturing Grid	Specifies that the ruler should snap to the manufacturing grid.
User Grid	Specifies that the ruler should snap to the user specified grid.
Std Row and Metal 2 Pitch	
Placement Grid	Specifies that the ruler should snap to the placement grid.
Layer Track	
Orthogonal	Specifies that ruler lines can be drawn in horizontal as well as vertical directions.
Vertical	Specifies that ruler lines will be drawn in the vertical direction.
Horizontal	Specifies that ruler lines will be drawn in the horizontal direction.
X	Specifies that ruler lines can be drawn at a 45 degree angle as well as at a 135 degree angle.
45	Specifies that ruler lines will be drawn at a 45 degree angle.
135	Specifies that ruler lines will be drawn at a 135 degree angle.
Any Angle	Specifies that ruler lines can be drawn in any direction.

Satellite Window

The main window includes a satellite window, which identifies the location of the current view in the design display area, relative to the entire design. The chip area is identified by a yellow box, the satellite view is identified by the pink cross-box. When you display an entire chip in the design display area, the satellite cross-box encompasses the chip area yellow box.



When you zoom and pan through the chip in the design display area, the satellite cross-box identifies where you are relative to the entire chip.

To move to an area in the design display area, click and drag on the satellite cross-box.

To select a new area in the design display area, click and drag on the satellite cross-box.

To resize an area in the satellite window, click with the `Shift` key and drag a corner of the cross-box.

To define a chip area in the satellite window, right-click and drag on an area.

Auto Query

The Layout window includes a *Q* icon located at the bottom of the design display area that enables and disables the auto query. When you enable auto query, text information displays for any object that is under the cursor. You can use auto query for blockages, hand-drawn or inside cell instances, instance pins, special route wires and vias, and cell instances.

If there are overlapping objects under the cursor, text information displays for the object that is on top. Use the `n` key to get information on the next object, and the `p` key to get information on the previous object. You can also use the F8 binding key to copy auto query text from the message bar in the Layout window to the software console. The text is displayed in the software console with the prefix `Message`.

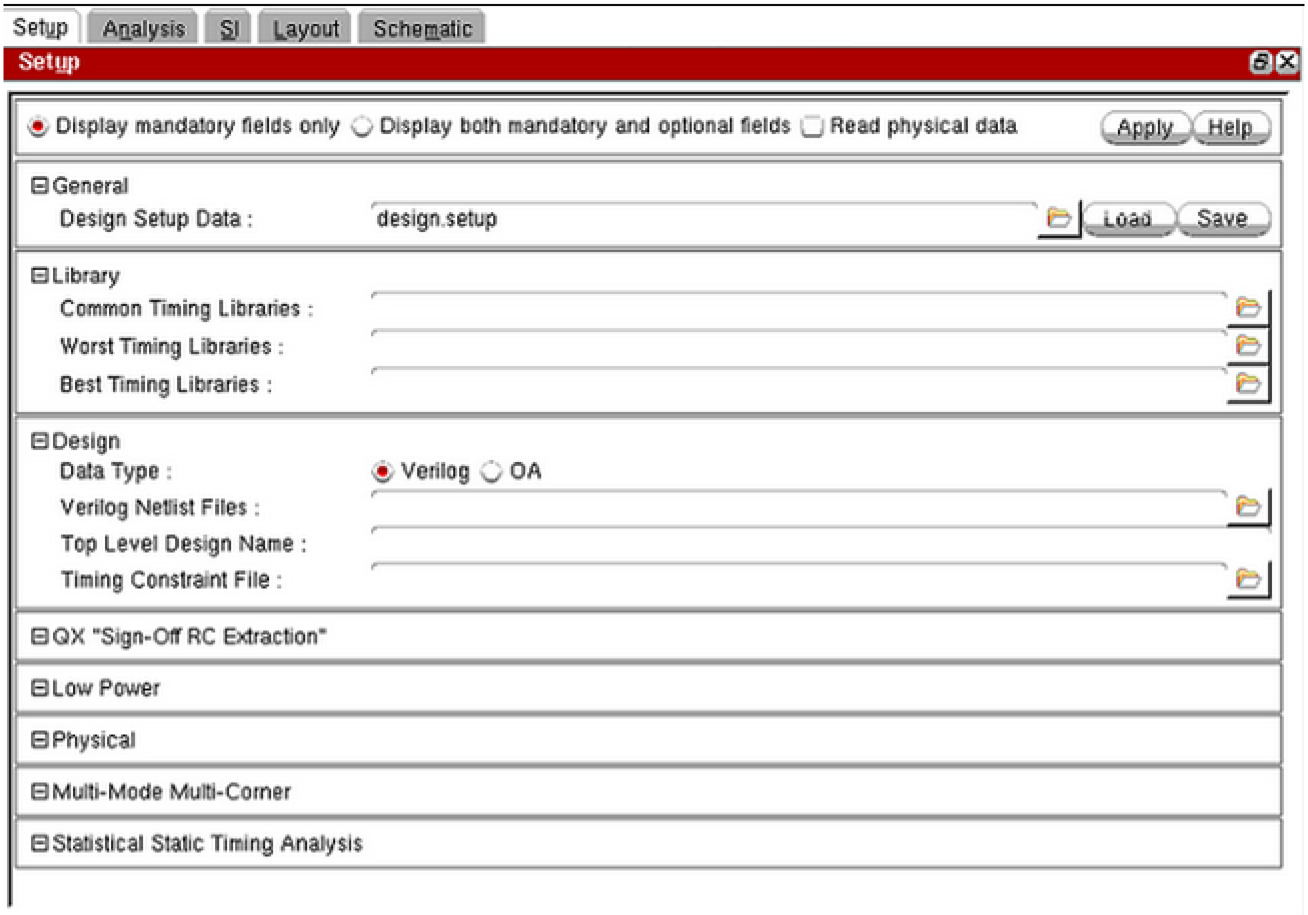
Setup Tab

■ Setup Tab Components on page 479

Setup Tab Components

Use the Setup tab to read the design data.

Click the *Setup* tab.



If you select the design data type as OA:

Fields and Options

Display mandatory fields only

	Displays mandatory fields in the form.
Display both mandatory and optional fields	
	Displays both mandatory and optional fields. The optional fields include Low Power, Physical, Multi-Mode Multi-Corner and Statistical Static Timing Analysis.
Read physical data	Reads physical information such as placement, floorplan and routing that you created using Innovus System in Tempus.
Design Setup Data	Loads a saved Tempus System database. The database consists of configuration file, and all files listed in the configuration file such as verilog, libraries, SDC. The database also consists of the mode file. To restore the timing database that you saved in Tempus, use the *encounter_file* parameter in the `read_design` command. If you use the *config_file* parameter, the database is not restored completely.
Common Timing Libraries	
	Specifies the name of the libraries common to both the *Best Timing Libraries* and the *Worst Timing Libraries*, which are used in both setup and hold analysis.
Worst Timing Libraries	Specifies the name of the timing libraries containing worst-case conditions for setup-time analysis.
Best Timing Libraries	Specifies the name of the timing libraries containing best-case conditions for hold-time analysis.
Worst Noise Libraries	Specifies the noise library (cdB) files used in setup analysis mode. In setup analysis mode, the common cdB files and the max cdB files are used in SI analysis.
Best Noise Libraries	Specifies the noise library (cdB) files used in hold analysis mode. In hold analysis mode, the common cdB files and the min cdB files are used in SI analysis.
Common Noise Libraries	
	Specifies the name of the common noise library (cdB) files used in SI analysis.
Stamp Model Definitions	

	Specifies the Stamp model definition files. Stamp model files contain timing information for a module, such as an instantiated module, block, or partitioned module. You do not need to translate these files before reading them into the software. However, currently you can only read in constant value models. Stamp timing specifications supersede timing information read from the timing library.
Data Type - Verilog	
Verilog Netlist Files	Specifies the gate-level Verilog netlist files to import. If specifying multiple names in the text field, use spaces to separate the names. You can use wildcards (*.*extension*) or directory names to specify multiple files. You can specify compressed files (`.gz` extension).
Top Level Design Name	
	Specifies the top module name, and checks for consistency between verilog netlist and timing libraries. Tempus uses the top module for timing analysis.
Timing Constraint File	Specifies the design constraints (SDC) file to import. You can import one or more constraint files. The constraints can be in the design constraints (SDC) format, or they can be more complicated Tcl constraints embedded in a Tcl script using commands such as `get_property` and `filter_collection`.
Data Type - OA	
Library	Specifies the location of the Open Access database to import. It contains the cell and the view to be loaded.
Cell	Specifies the design cell from the library that is to be loaded.
View	Specifies the view name of the cell that is to be loaded.
SPEF Files	Specifies the name of the SPEF files.
SDF Delay	Specifies the name of the file containing the SDF delays you want to load.
SOCE MSMV File	Specifies the name of the MSMV file. This file uses a format that is similar to the IR-drop file and can be created in Tempus System.

Power Net(s)	Specifies the names of the *Power Nets*. You can enter multiple names by separating the net names with a space, for example, `VDD VDD1`.
Ground Net(s)	Specifies the names of the *Ground Nets*. You can enter multiple names by separating the net names with a space, for example, `GND GND1`.
LEF Files	Specifies the LEF files to import. The LEF file provides the physical data information. You must specify the technology LEF file first, then specify the standard cell LEF and block LEF in any order.
Floorplan File	Specifies the name of the floorplan file to load for the design session.
Placement File	Specifies the name of the placement file to load for the design session.
Routing File	Specifies the name of the routing file to load for the design session.
DEF Options	Choose from one of the following options: ■ *Load All*: Loads all DEF files. ■ *Special Net & Components*: Reads the `SPECIALNETS` and `COMPONENTS` section of the DEF file. ■ *Specify Only*: Loads only the specified files.
DEF Files	Specifies the name of a Design Exchange Format (DEF) file. You can supply hierarchical DEFs to generate a flattened physical database.
View Definition File	Specifies a Tcl command script that contains the analysis view configurations for multi-mode multi-corner timing analysis and optimization. The file contains a list of Tcl commands that create library sets, constraint modes, RC corners, delay corners, and analysis views. Processing the file does not automatically place the software in multi-mode multi-corner analysis mode. If the file also contains a `set_analysis_view` command, then the software is placed in multi-mode multi-corner analysis mode after the file is processed.
MMMC Browser	Opens the Multi-Mode Multi-Corner Browser (MMMC Browser). The browser displays the multi-mode multi-corner configuration, and allows you to create and edit the various objects and analysis views necessary for multi-mode multi-corner analysis.

SPDF File	Specifies the name of a statistical parameter distribution format (SPDF) file. An SPDF file contains statistics of variation of process parameters and is used for statistical static timing analysis (SSTA). You can specify global, random and spatial variations in this file. You can create this file using the data provided by the foundry.
Statistical SPEF File	Specifies the name of the statistical SPEF (S-SPEF) file. The software requires an S-SPEF file to perform sensitivity-based extraction. Sensitivity-based extraction captures variations in interconnects. In sensitivity based extraction, in addition to nominal values of resistances and capacitances, the software also extracts the sensitivities of R and C to different interconnect process parameters. The S-SPEF file is used for statistical static timing analysis (SSTA).

Related Text Commands

The following text commands provide equivalent or additional functionality:

- `read_design`
- `read_def`
- `read_lib`
- `read_power_domain`
- `read_sdc`
- `read_verilog`
- `set_top_module`
- `read_spdf`

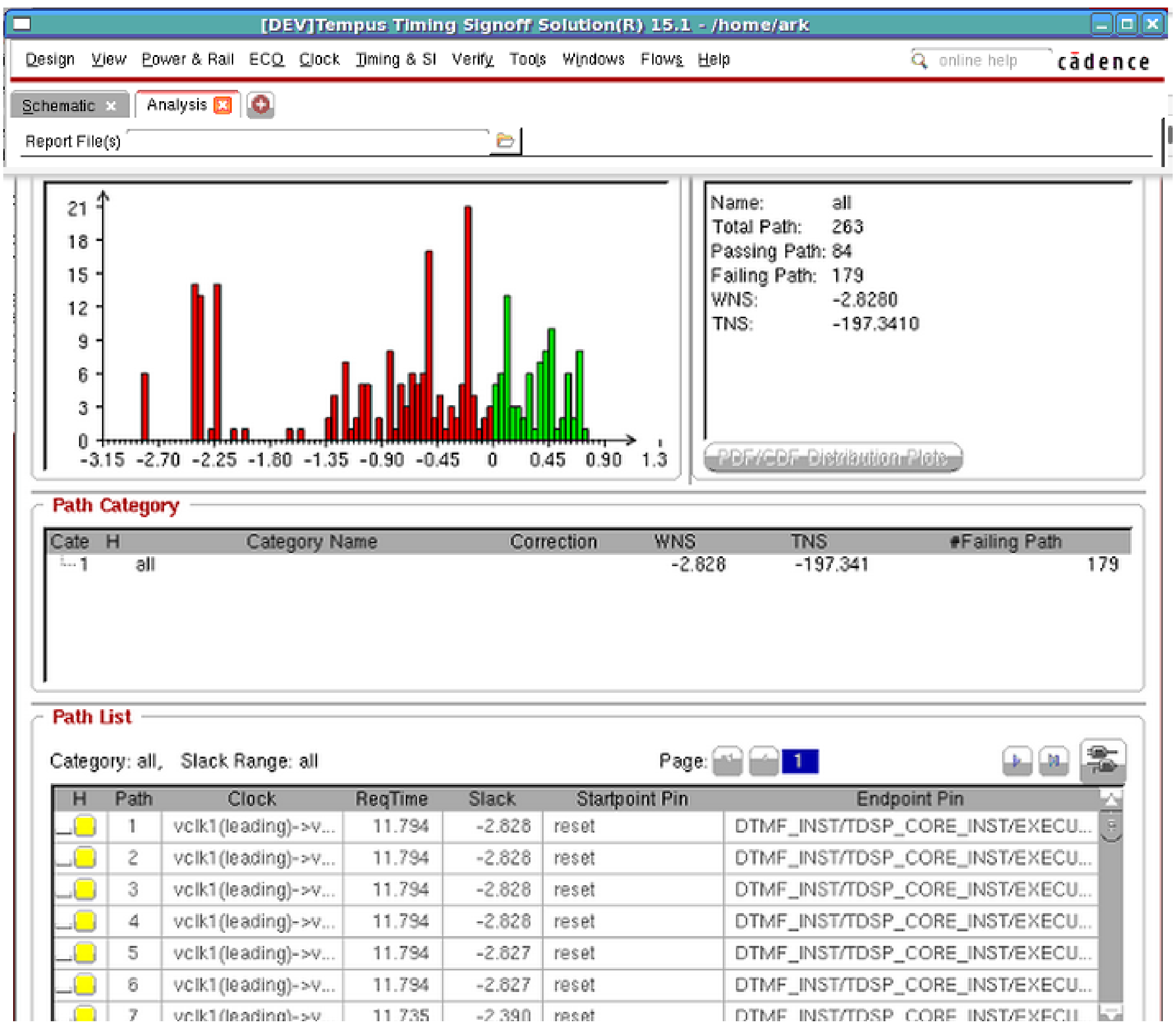
For more information, see the "Design Import Commands" in *Tempus Text Command Reference*.

Analysis Tab

Analysis Window Components

Use the analysis window to display the violation information, categories defined to group violation types, graphical display of violations, and category summary.

☦ Click on the *Analysis* tab.



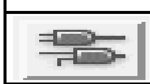
For SSTA analysis, see Analysis Window in Statistical Mode on page 489.

Fields and Options

File	Provides the following menu options:	
	Write Text Report	
		Displays the Write Textual Report (Human Readable) form. Use this form to write a text file containing information about paths that have been loaded from an `.mtarpt` file generated by the Display/Generate Timing Report form. For more information, see Write Textual Report.
	Write Category Summary	
	Displays the Write Category Report File form. Use this form to write a text report containing category summary information. For more information, see Write Category Report File.	
	Preference s	Displays the Timing Debug Preferences form.
Analysis	Displays menu options to create standard categories for timing debug. Following options are displayed:	
	Path Group Analysis	
		Displays Path Analysis form. Use this form to create standard path categories according to basic path groups. For more information, see Path Analysis (Path Groups).
	Clock Analysis	Displays Path Analysis form. Use this form to create categories according to launch clocks - capture clock combinations. For more information, see Path Analysis (Clock Analysis).
	Hierarchical Analysis	
		Displays the Path Analysis form. Use this form to create categories according to the hierarchical characteristics of the paths. The software uses the floorplan file to extract information for guides, fences or Macros. For more information, see "Path Analysis (Hierarchical Analysis) on page 511".

	Clock Matrix Viewer	
		Displays the Clock matrix Viewer form. to show the relationship of source and destination clocks. User can call this tool under the menu of Analysis in Global Timing Debug. For more information, see Clock Matrix Viewer.
	View Analysis	Displays the Path Analysis form. Use this form to create categories according to the views that were defined for multi-mode multi-corner analysis. For more information, see Path Analysis (View Analysis).
	Critical False Path	
		Displays the Path Analysis form. Use this form to create a category containing paths identified as false during the false path analysis by Conformal Constraint Designer (`ccd`). **Note:** The `ccd` should be in the command search path to access this menu.
	Bottleneck Analysis	
		Displays the Path Analysis form. Use this form to create categories according to the bottleneck analysis of the critical paths. For more information, see Path Analysis (Bottleneck Analysis).
	DRV Analysis	Displays the Path Analysis form. Use this form to read or generate a report containing max transition, max capacitance, and max fanout violations. For more information, see Path Analysis (DRV Analysis).
	Hierarchical AnalysisViewer	
		Displays the Hierarchical Analysis Viewer form.
Category	Displays menu options for creating and managing categories. Following options are displayed:	
	Create	Displays the Create Category form. Use this form to specify definitions for categories as per your design requirements. For more information, see Create Path Category.

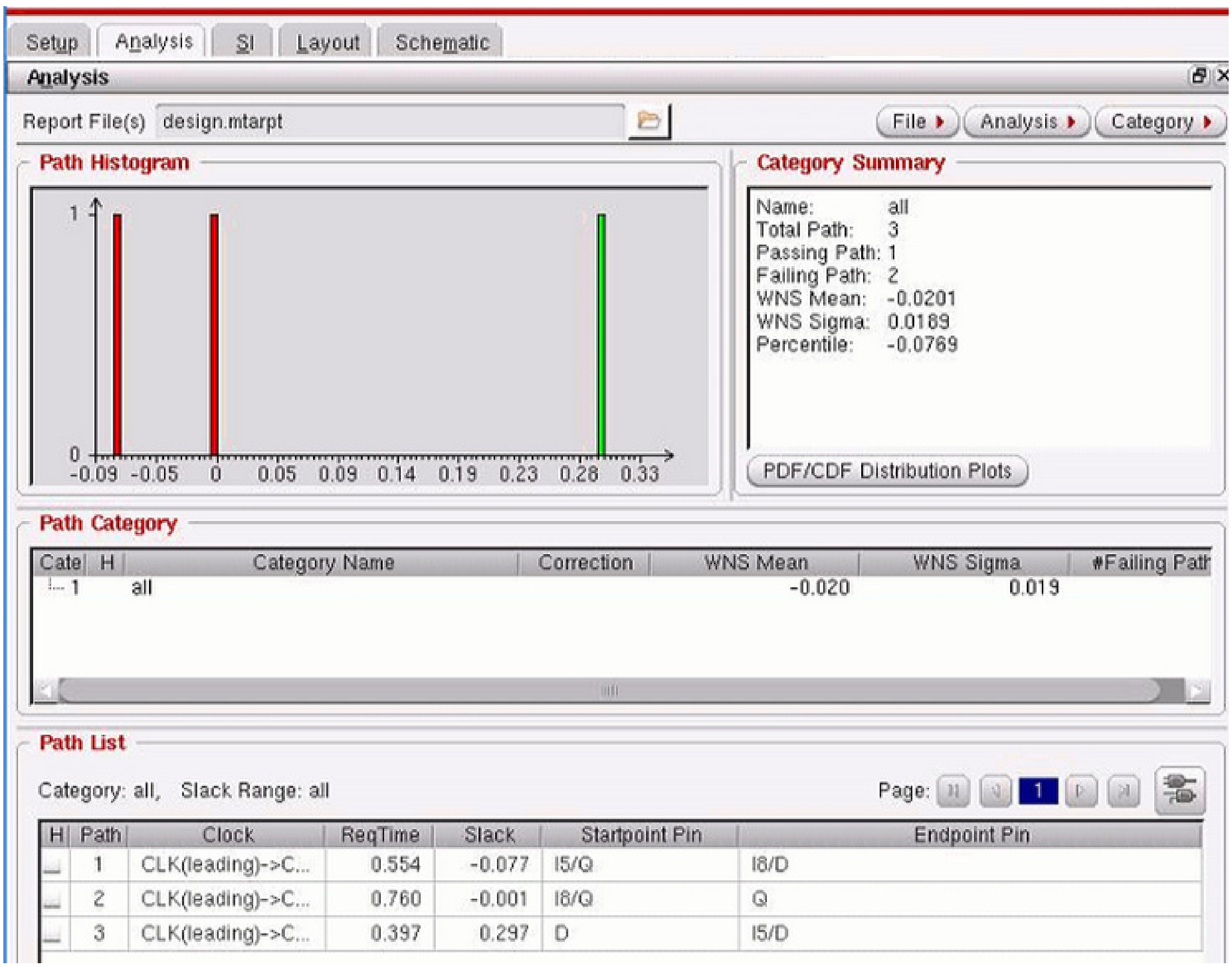
	Save	Saves the category definitions in a file. You can use this file to load a set of predefined categories into a new timing debug session.
	Load	Loads the category definition file.
Path Histogram	Displays the end point histogram. The failing endpoints are displayed in red and the passing endpoints are displayed in green color. Right-click on the histogram to display the following options:	
	Fit	Displays the entire histogram in Path Histogram area.
	Zoom	Displays smaller or larger area of the histogram in greater or lesser detail.
	Pan	Displays areas to the top, bottom, left or right of the displayed histogram.
	Preference	Displays the Timing Debug Preferences form. Use this form to specify the statistics for the histogram and number of pages of report to be displayed in the form.
Category Summary	Displays the summary of a single selected category. To select a category, double-click on the category name in the *Path Category* field.	
Path Category	Displays the category information as nested trees. The columns in this field include, name of the category, worst negative slack (WNS), total negative slack (TNS), and number of failing paths. Right click on the category name to display the following options:	
	List Paths (Category Name)	
		Displays the detailed path information in the *Path List* field.
	Add Category to Histogram	
		Displays the selected category as histogram. The added category is displayed in a different color. Use this option to identify paths grouped under a specific category.
	Hide Category in Histogram	
		Removes the category from the histogram, but does not delete it. Hidden categories are indicated with the letter "H" to the left of the category name.
	Set Category Slack Correction	

		Displays the Set Category Slack Correction form. Use this form to specify the estimated slack correction for the selected category of paths.
	Edit Category	Displays the Create Path Category form. Use this form to change category definition as per your requirements. For more information, see Create Path Category on page 526.
	Delete Category	
		Deletes the selected category.
	Delete All Category	
		Deletes all categories that you have created.
	Change Category Color	
		Displays Color Category Name form. Use this form to specify the color to be used in the Path Histogram for the selected category.
	Copy Cell	Copies the name of the category in the Path Category table. Use this feature to copy and paste the text from Path Category to Create Path Category on page 526.
	Edit Table Column	
		Displays the Edit Table Column form. Use this form to customize columns in the *Timing Debug* and *Timing Path Analyzer* windows.
Path List	Displays the list of paths (endpoint and slack) in the selected category. Double-click on a path to display the Timing Path Analyzer form. Use this form for detailed analysis of a single path. For more information on Timing Path Analyzer form, see Timing Path Analyzer on page 533. When you load multiple debug reports in a single timing debug session, the paths are displayed in different colors corresponding to the report file they are coming from. You can move the cursor over a path to display the name of the report file.	
	Displays the schematic view of the selected path.	

Analysis Window in Statistical Mode

The analysis window displays SSTA details if the statistical static timing analysis (SSTA) mode is enabled.

Click on the *Analysis* tab.



In Statistical mode, the following fields and options are visible and enabled:

PDF/CDF Distribution Plots	
	Displays PDF and CDF plots showing combined distribution of slacks.

Path Category	Reports *WNS Mean* and *WNS Sigma* values for each category.
Path List	Double-click on a path to display the Timing Path Analyzer form. Use this form for detailed analysis of a single path. For more information, see Timing Path Analyzer - Statistical.

Related Text Commands

The following text command provides equivalent or additional functionality:

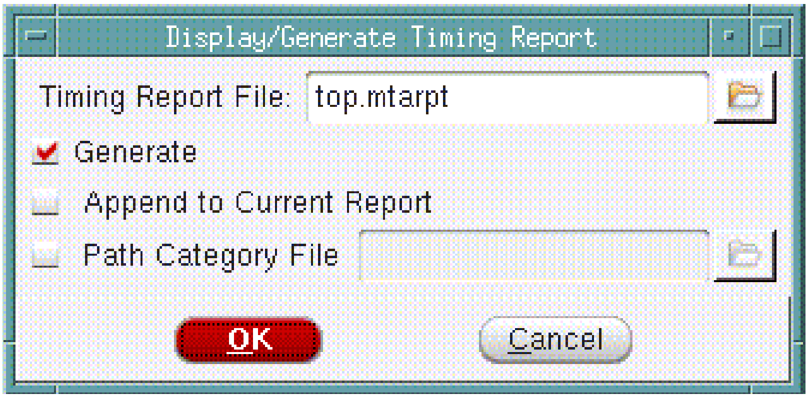
- `analyze_paths_by_basic_path_group`
- `analyze_paths_by_bottleneck`
- `analyze_paths_by_clock_domain`
- `analyze_paths_by_critical_false_path`
- `analyze_paths_by_hierarchy`
- `analyze_paths_by_view`
- `create_path_category`
- `delete_path_category`
- `load_path_categories`
- `load_timing_debug_report`
- `save_path_categories`

For more information, see "Timing Debug Commands" in the *Tempus Text Command Reference*.

Display/Generate Timing Report

Use the Display/Generate Timing Report form to specify the violation report to be read in for debugging timing results.

In the Analysis tab. click the folder icon next to the Report File(s) field.



Display/Generate Timing Report Fields and Options

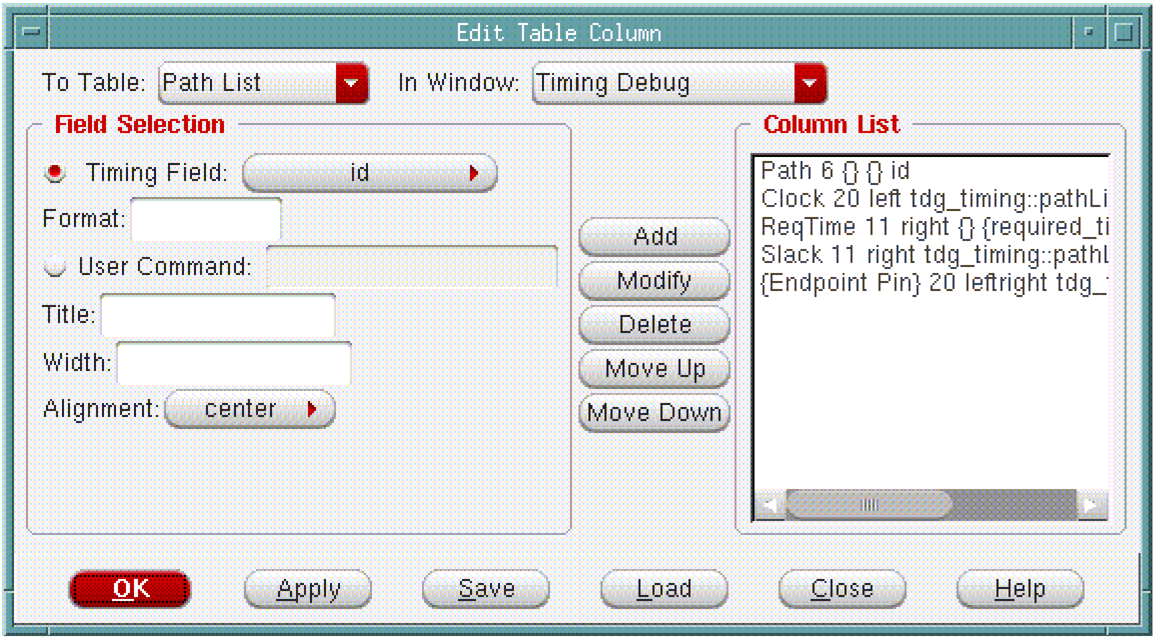
Timing Report File	Specify a filename, or click the file folder icon to choose a file.
Generate	Generates a violation report file in machine readable format. *Default:* `top.mtarpt`
Append to Current Report	
	Appends the specified report to the report that you have already loaded in a timing debug session. This feature is useful to collate reports from several views into one timing debug session.
Path Category File	Specifies the path category file. The path category file contains category definitions that you saved in an earlier timing debug analysis.

Edit Table Column

To edit a table column, follow the steps given below:

- Use the *Edit Table Column* form to customize columns in the *Analysis* and *Timing Path Analyzer* windows.
- Open *Analysis* or *Timing Path Analyzer*, then right-click on a path. A drop-down menu is displayed.

- Select *Edit Table Column*.



Edit Table Column Fields and Options

To Table	Choose the table whose columns you want to modify: For the Timing Path Analyzer window choose *Data Path*, *Capture Clock*, or *Launch Clock*. For the Timing Debug window, choose *Path Category* or *Path List*.
Window Name	Specifies the name of the window that contains the table column you want to edit. Choose *Timing Debug* or *Timing Path Analyzer*.
Timing Field	Specifies the type of information to include in the column. The available options will vary depending upon the table you choose in the *To Table* field.
Format	Specifies a table format. For example, specify `%.3f` if you only want to display three decimal digits.

<table>
<tr><td>User Command</td><td colspan="2">Specifies the procedure you want to use to determine the information you want to include in the column. In the procedure, you must pass the instance as a parameter. You must source the file containing procedure before specifying the procedure here.

For example:
<pre>
Combine fedge (from edge) and tedge (to edge) #information
into a single field

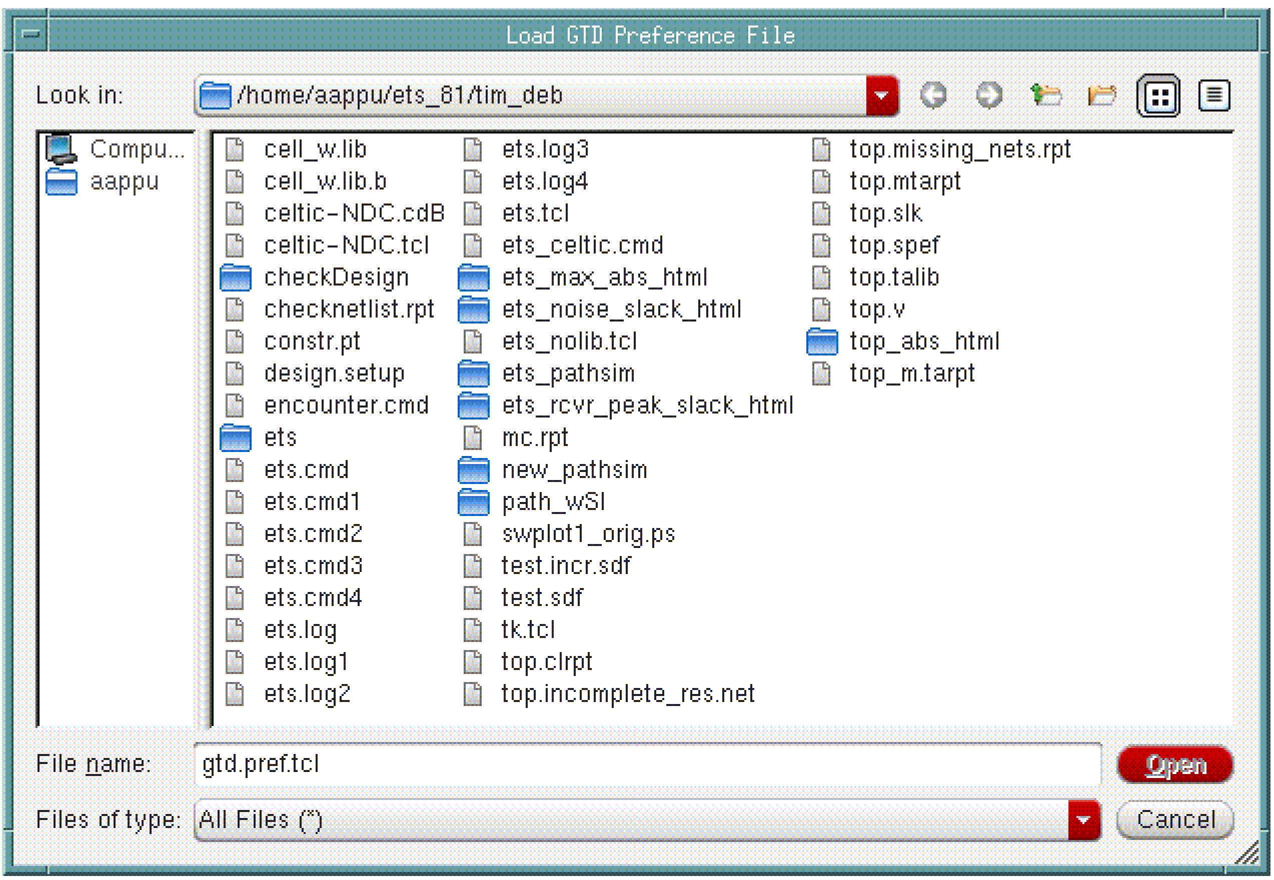
proc my_get_edge {id var} {
     upvar #0 $var p
     if {p(type) == "inst"} {
        return "$p(fedge) -> $p(tedge)"
     } elseif {$p(type) == "port" } {
        return $p(fedge)
     } else {
        return ""
     }
}
</pre></td></tr>
<tr><td></td><td>Title</td><td>Specifies the column title.</td></tr>
<tr><td></td><td>Width</td><td>Specifies the column width.</td></tr>
<tr><td></td><td>Alignment</td><td>Specifies the alignment of the column. Choose from center, left, right, and leftright.</td></tr>
<tr><td>Column List</td><td colspan="2">Lists the columns in the order in which they appear in the table, left to right.

Each entry has the following information:
■ Column title (for example, Name, Arc, Delay)
■ The width of the column
■ The alignment of text in column (center, left, right, leftright)
■ Procedure, if any
■ The existing value from the path information</td></tr>
<tr><td>Add</td><td colspan="2">Adds a column to the column list.</td></tr>
<tr><td>Modify</td><td colspan="2">Lets you modify characteristics. Click on a column in the column list. Edit the information, then click modify.</td></tr>
<tr><td>Delete</td><td colspan="2">Deletes a column from the column list.</td></tr>
<tr><td>Move Up</td><td colspan="2">Moves a column up in the column list. This effectively moves a column to the left in the table.</td></tr>
</table>

Move Down	Moves a column down in the column list. This effectively moves a column to the right table.
Load	Opens the *Load GTD Preferences* form, which you can use to load a tcl command file to use for editing table columns. For more information, see Load GTD Preference File.

Load GTD Preference File

- Use the *Load GTD Preferences* form to load a TCL command file to use for editing table columns.
- On the *Analysis* window or *Timing Path Browser* form, then right-click on a path.
- Select *Edit Table Column*. The Edit Table Column form is displayed.
- Click *Load*.



Load GTD Preferences File Fields and Options

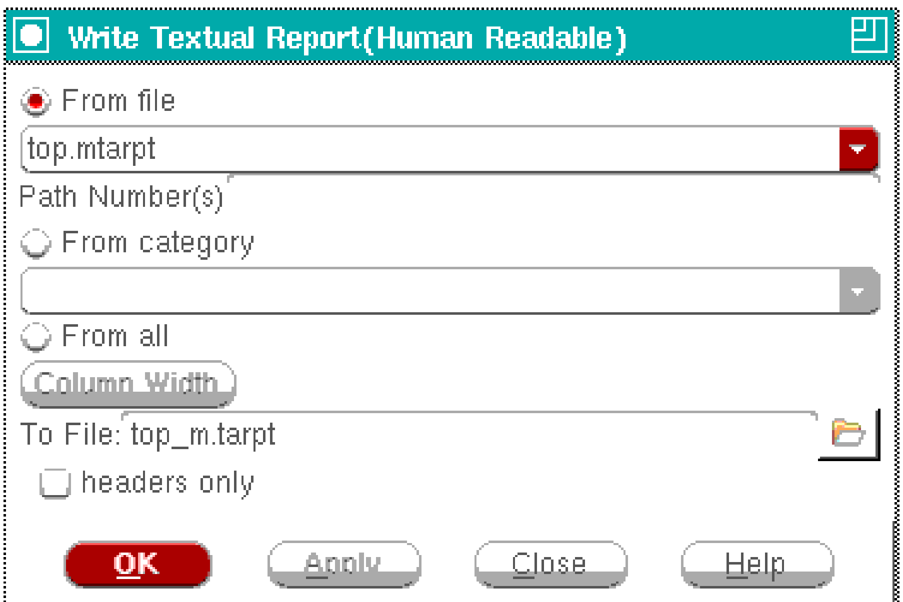
File name	Specifies the file to load.

Write Textual Report

Use the Write Text Report form to write a text file containing information about paths that have been loaded from a `.mtarpt` file generated by the Display/Generate Timing Report.

The generated report cannot be read back into the timing debug environment.

- Choose *Write Text Report* on the *File* menu.



Timing Debug - Write Textual Report File Fields and Options

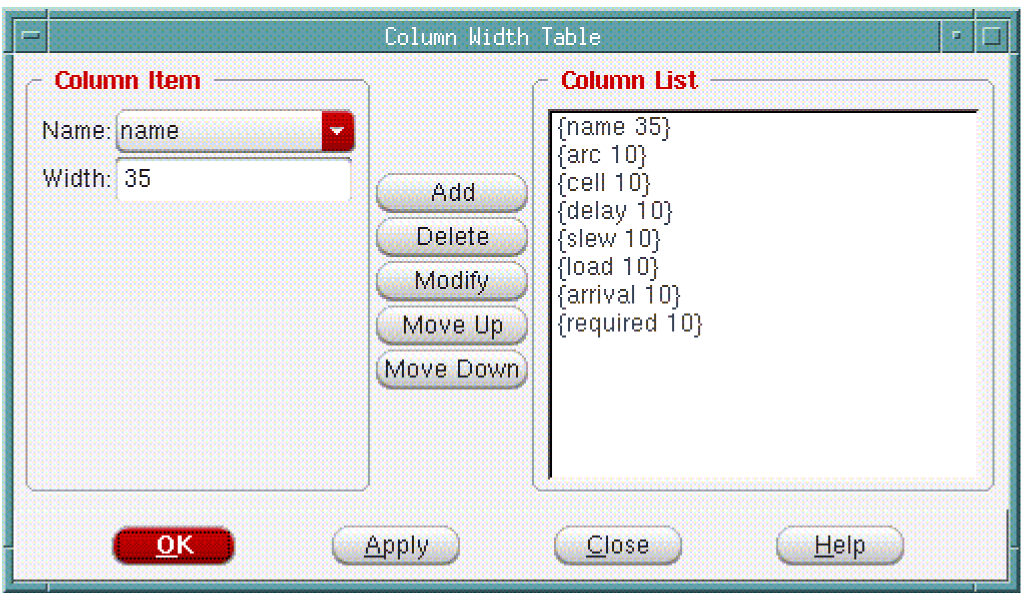
From file	Specifies the name of the `.mtarpt` file to read and convert to a text file. You cannot specify this parameter if you specify *From category*.
Path Number(s)	Writes a single number or path range separated by blank space " "or a comma ",". A path range is two numbers separated by a minus "-" sign. You can use this parameter only if a single file is specified or loaded.
From category	Specifies the category of paths to write to the text file.

From all	Writes all categories to the report file.
Column Width	Displays the Column Width Table form.
To File	Specifies the name of the text output file.
headers only	Writes the file header only. Use this parameter if you do not need a detailed table.

Column Width Table

Use the Column Width Table form to change the width of the columns in the file created by the Write Textual Report form, and to specify the order in which you want to columns to appear.

- Choose *Write Text Report* from the *File* menu, then click on the *Column Width* button.



Column Width Table Fields and Options

Name	Specifies the name of the column to add, modify, or move in the column list.
Width	Specifies the width of the column.
Column List	Specifies a list of columns to create in the report generated by the Write Textual Report form.
Add	Adds a column to the column list.

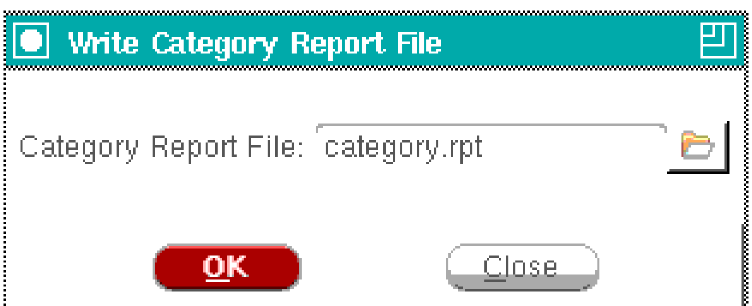
Delete	Deletes a column from the column list.
Modify	Enables you to modify the width of a selected column.
Move Up	Moves the selected column up one position in the column list.
Move Down	Moves the selected column down one position in the column list.

Write Category Report File

Use the Write Category Report File form to write a text file containing the following information:

- Category name
- Total number of paths
- Number of passing paths
- Number of failing paths
- Worst negative slack
- Total negative slack
- TNS

† Choose *File - Write Category Summary*.



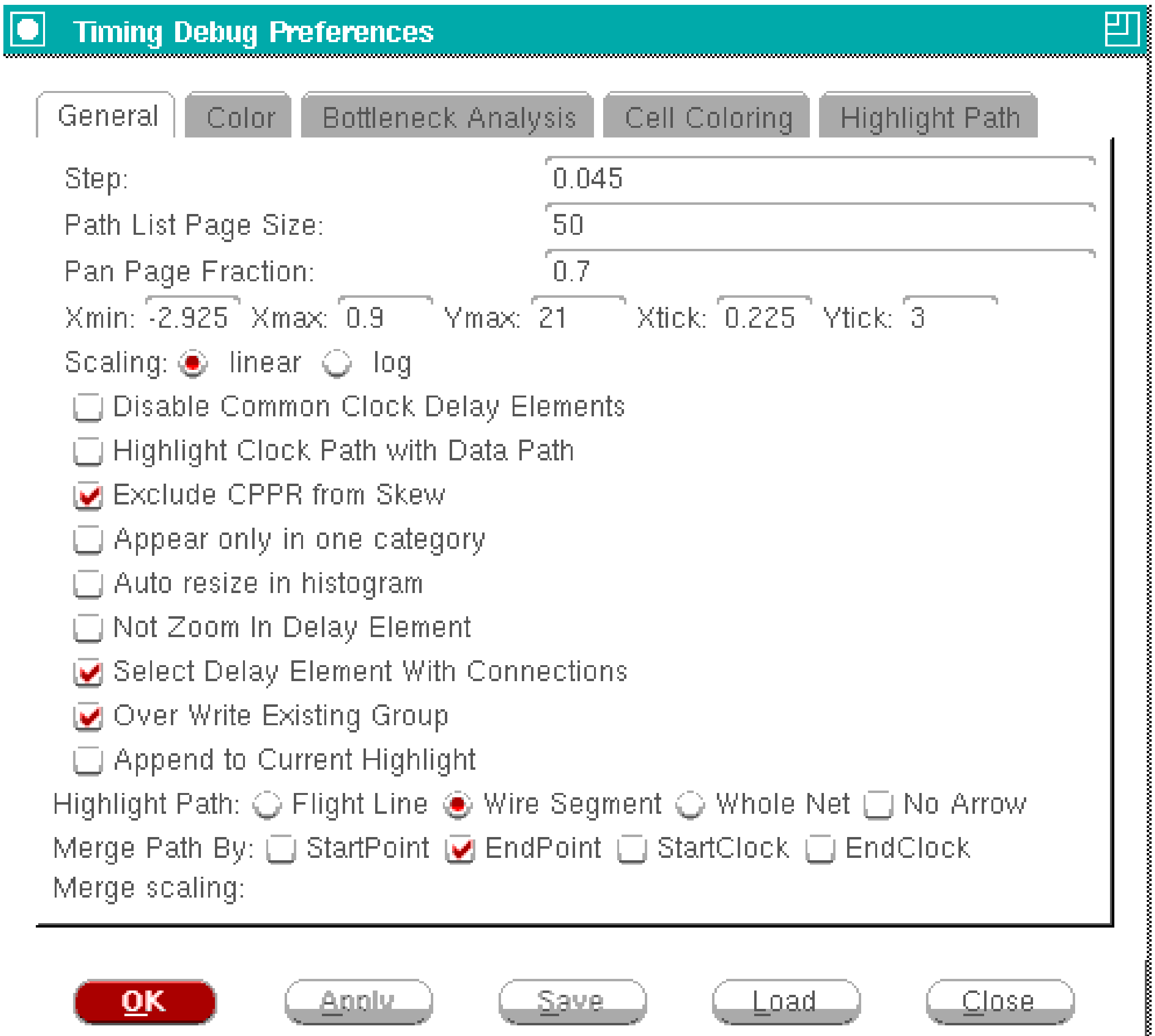
Timing Debug - Write Category Report Fields and Options

Category Report File	Specifies the name of the category output file. *Default:* `category.rpt`

Timing Debug Preferences - General

Use the *General* page of the Timing Debug Preferences form to specify the statistics for the histogram, number of pages of report to be displayed in the form, and highlighting options.

- Choose *Timing - Debug Timing - File - Preferences.*



Timing Debug Preferences Fields and Options

Step	Specifies the step size for the histogram.
Path List Page Size	Specifies number of pages of report to be displayed in the form.

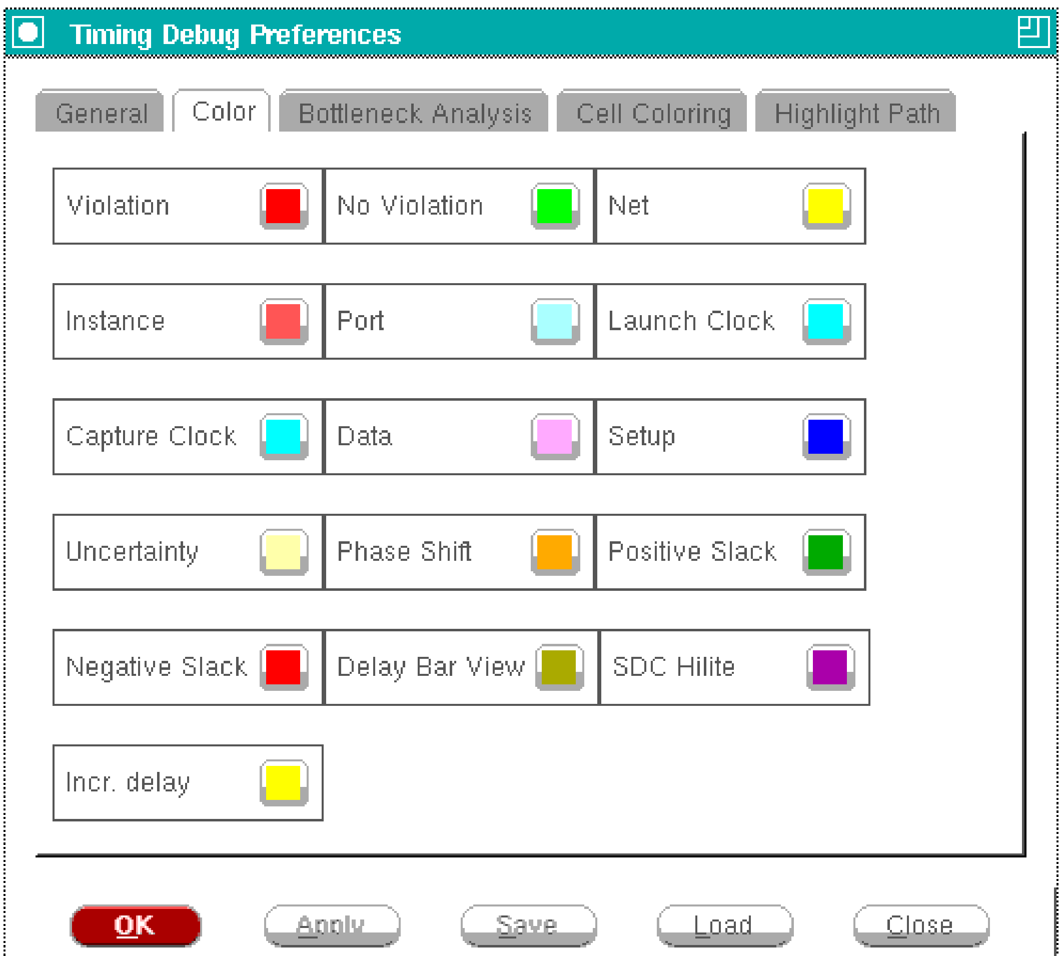
<table>
<tr><td>Pan Page Fraction</td><td colspan="2">Specifies the amount by which the histogram will be moved when you pan-in using the right-click options in the Path Histogram field in the Timing Debug window.</td></tr>
<tr><td colspan="3">Disable Common Clock Delay Elements</td></tr>
<tr><td></td><td colspan="2">Disables paths common to launch and capture clock in the Data Delay field of the Timing Path Analyzer form.</td></tr>
<tr><td colspan="3">Highlight Clock Path with Data Path</td></tr>
<tr><td></td><td colspan="2">Highlights clocks paths along with the data paths in the Design Display Area. By default, only data paths are highlighted.</td></tr>
<tr><td colspan="2">Exclude CPPR from Skew</td><td></td></tr>
<tr><td></td><td colspan="2">To exclude CPPR from Skew</td></tr>
<tr><td colspan="2">Appear Only In One Category</td><td></td></tr>
<tr><td></td><td colspan="2">If selected, the path will only appear in one category.</td></tr>
<tr><td colspan="2">Not Zoom In Delay Element</td><td></td></tr>
<tr><td></td><td colspan="2">Zooms into the clicked instance or wire.</td></tr>
<tr><td colspan="3">Append to Current Highlight</td></tr>
<tr><td></td><td colspan="2">Appends highlights to currently highlighted path.</td></tr>
<tr><td>Highlight Path</td><td colspan="2">Specifies the preference for highlighting various paths. The color code used to highlight path is as follows:
■ Red: Launch clock path
■ Green: Paths specific to capture clock
■ White: Data path
Choose from one of the following options:</td></tr>
<tr><td></td><td>Flight Line</td><td>Creates highlighted lines between the pins of the path.</td></tr>
<tr><td></td><td>Wire Segment</td><td>Highlights the wire segment.</td></tr>
<tr><td></td><td>Whole Net</td><td>Highlights all segments of the routing. The thickness of the highlight represents if the routing is part of the wire segments of the path (thicker) or if it is outside (thinner).</td></tr>
<tr><td>Merger Path By</td><td colspan="2">Specify the method to merge same path in views. Choose from one of the following options:</td></tr>
</table>

	StartPoint	Merge the paths according to StartPoint.
	EndPoint	Merge the paths according to EndPoint.
	StartClock	Merge the paths according to launching clock
	EndClock	Merge the paths according to capturing clock.
Merge Scaling	Specify the scaling number for each view. You can use this option after merging same path in views.	

Timing Debug Preferences - Color

Use the *Color* page of the Timing Debug Preferences form to specify the colors for the slack calculation bar elements in the Timing Path Analyzer form.

- Choose *File - Preferences*, and click on the *Color* tab.



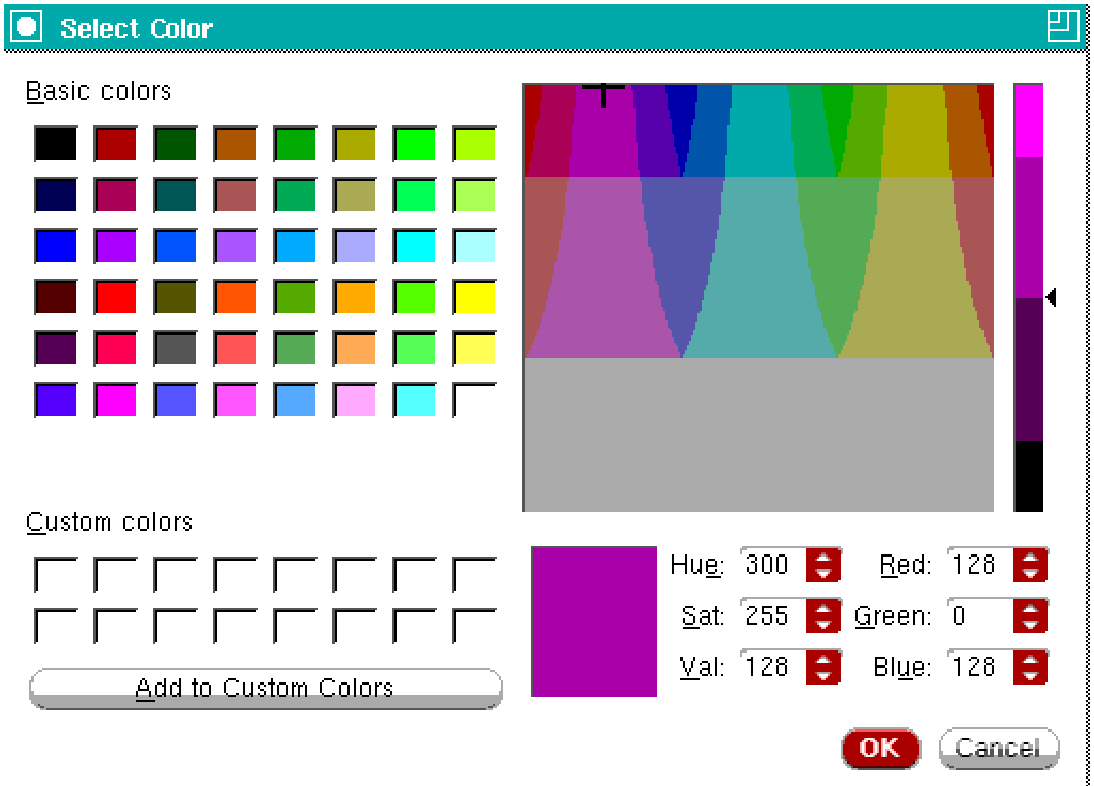
Color Fields and Options

Elements	Displays the element name and the corresponding color that is used in the Slack Calculation field of the Timing Path Analyzer form. Click on a color square to display the *Set Color <Element>* form, which enables you to customize a color.
Update Display	Immediately applies the color selections to the display.
Apply	Stores the selected color information.
Save	Saves the selection.
Load	Loads the color selection in the system.
Close	Closes the form without saving the selection.

Timing Debug Preferences - Color - Select Color

Use the *Select* form to customize the colors displayed on the Timing *Debug Preferences - Color* page.

- Choose *File - Preferences*, click on the *Color* tab, then click on any colored square.



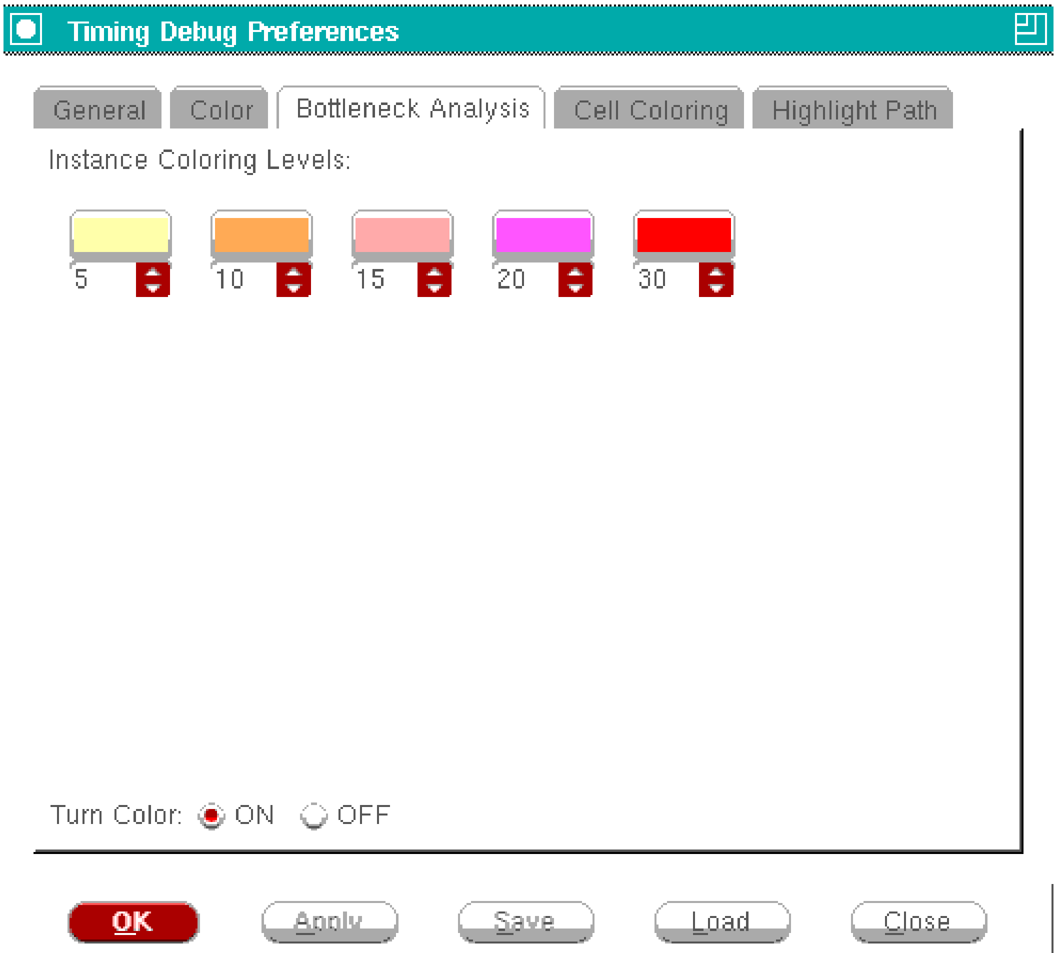
Timing Debug Preferences - Color - Select Color Violation Fields and Options

Basic colors	Displays the available basic colors. Sliders let you set the hue for the colors comprising your custom color.
Custom colors	Displays the custom colors set by you.
Add to Custom Colors	Adds the custom color to the Custom colors section.

Timing Debug Preferences - Bottleneck Analysis

Use the *Bottleneck* page of the Timing Debug Preferences form to colorize cells based on the occurrences of violating paths.

- Choose *File - Preferences*, and click on the *Bottleneck Analysis* tab.



Timing Debug Preferences - Bottleneck Analysis Fields and Options

Instance Coloring Levels

	Specifies color corresponding to the number of times an instance can appear in the violating path. In the delay bar in on the Timing Path Analyzer GUI, the tool colorizes cells accordingly. In the example shown in the previous figure, cells appearing in greater than five and 10 violating paths will be colored yellow. Cells appearing in between 10 and 15 times will be colored orange. Cells appearing in between 15 and 20 times will be colored pink. Cells appearing in between 20 and 30 times will be colored fuchsia and cells appearing more than 30 or more times will be colored red.	
Turn Color	Specifies how the tool colorizes the instances.	
	ON	Displays the colors you selected.
	OFF	Uses the default colors.
Apply	Stores the selected color information	
Save	Saves the selection.	
Load	Loads the selection in the system.	
Close	Closes the form without saving the selection.	

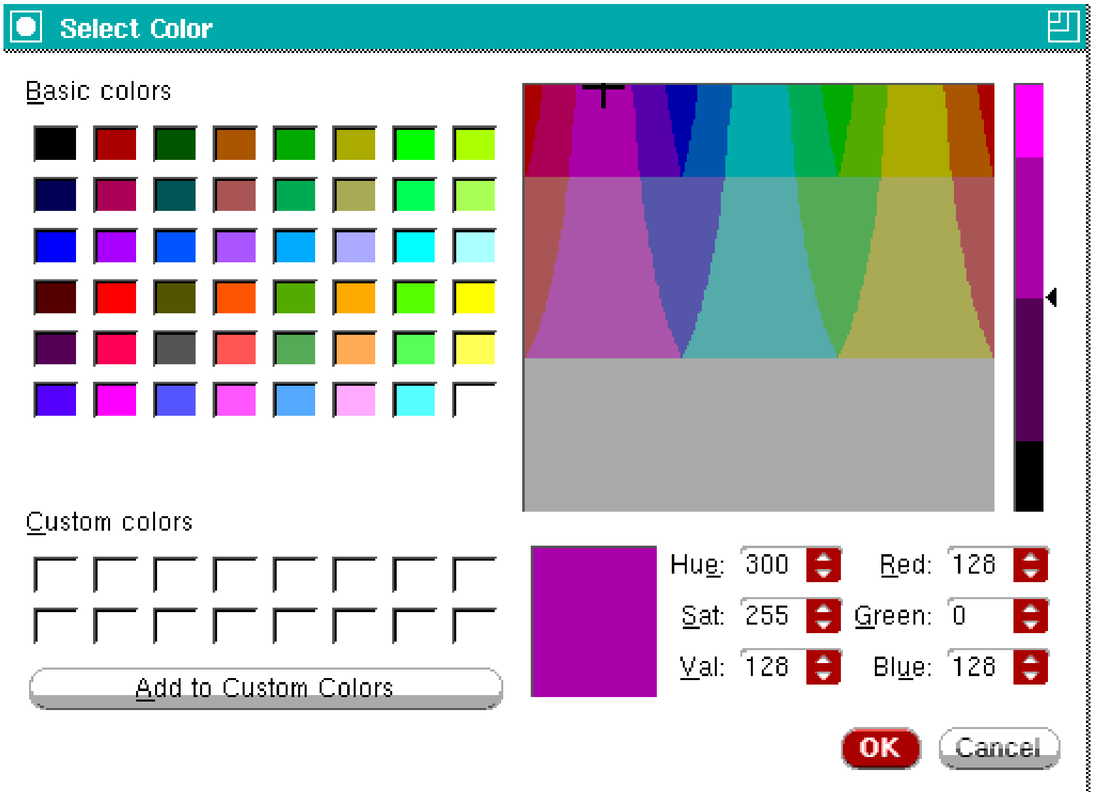
Timing Debug Preferences - Select Color

Use the *Set Color* form to set custom colors for the *Bottleneck Analysis* and *Cell Coloring* form.

☨ Choose *File - Preferences*, click on the *Bottleneck Analysis* tab, then click on a color.

or

- Choose *Debug Timing File - Preferences*, click on the *Cell Coloring* tab, then click on a color.



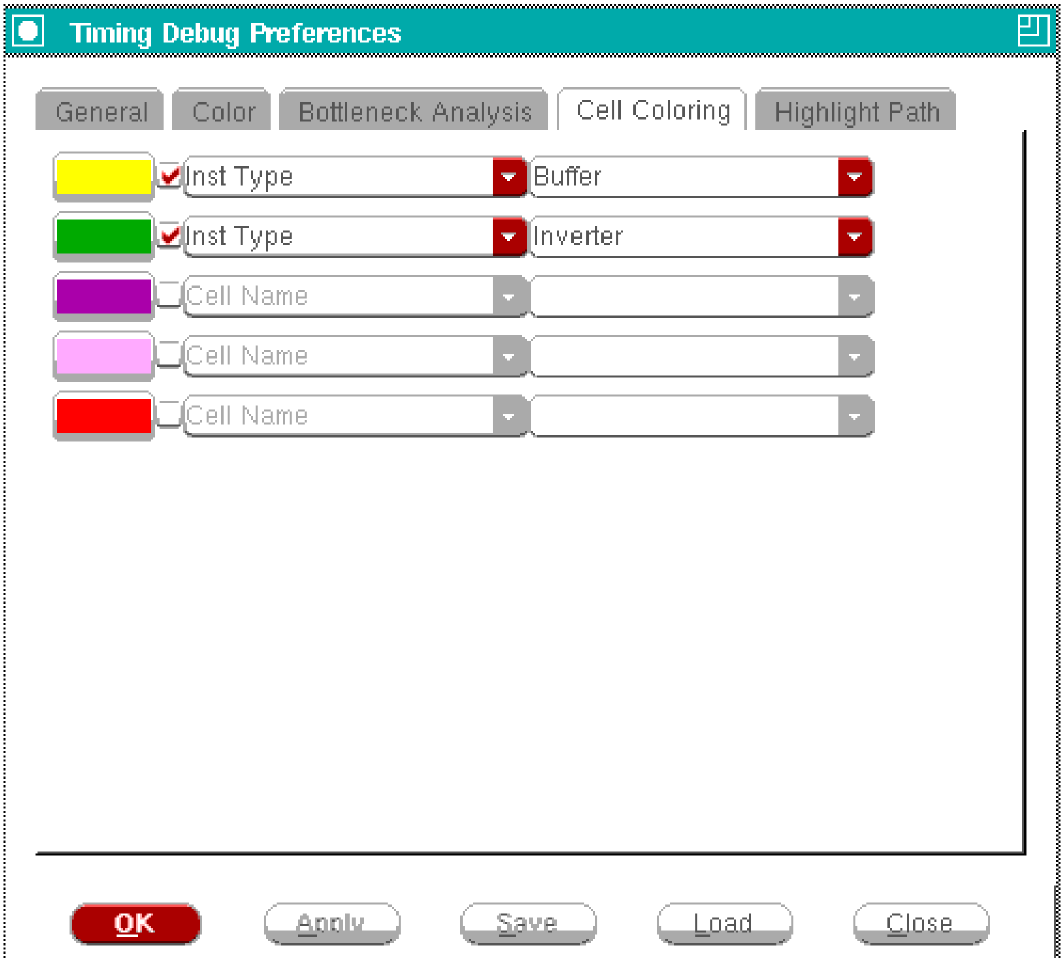
Timing Preferences - Select Color Fields and Options

Basic colors	Displays the available basic colors. Sliders let you set the hue for the colors comprising your custom color.
Custom colors	Displays the custom colors set by you.
Add to Custom Colors	Adds the custom color to the Custom colors section.

Timing Debug Preferences - Cell Coloring

Use the *Cell Coloring* page of the Timing Debug Preferences form to choose colors for specific cells in the delay bar. Color precedence is top to bottom in this form. If you set coloring in a session, the software saves the settings and restores them in the next session.

- Choose *File - Preferences*, and click on the *Cell Coloring* tab.



Timing Debug Preferences - Cell Coloring Fields and Options

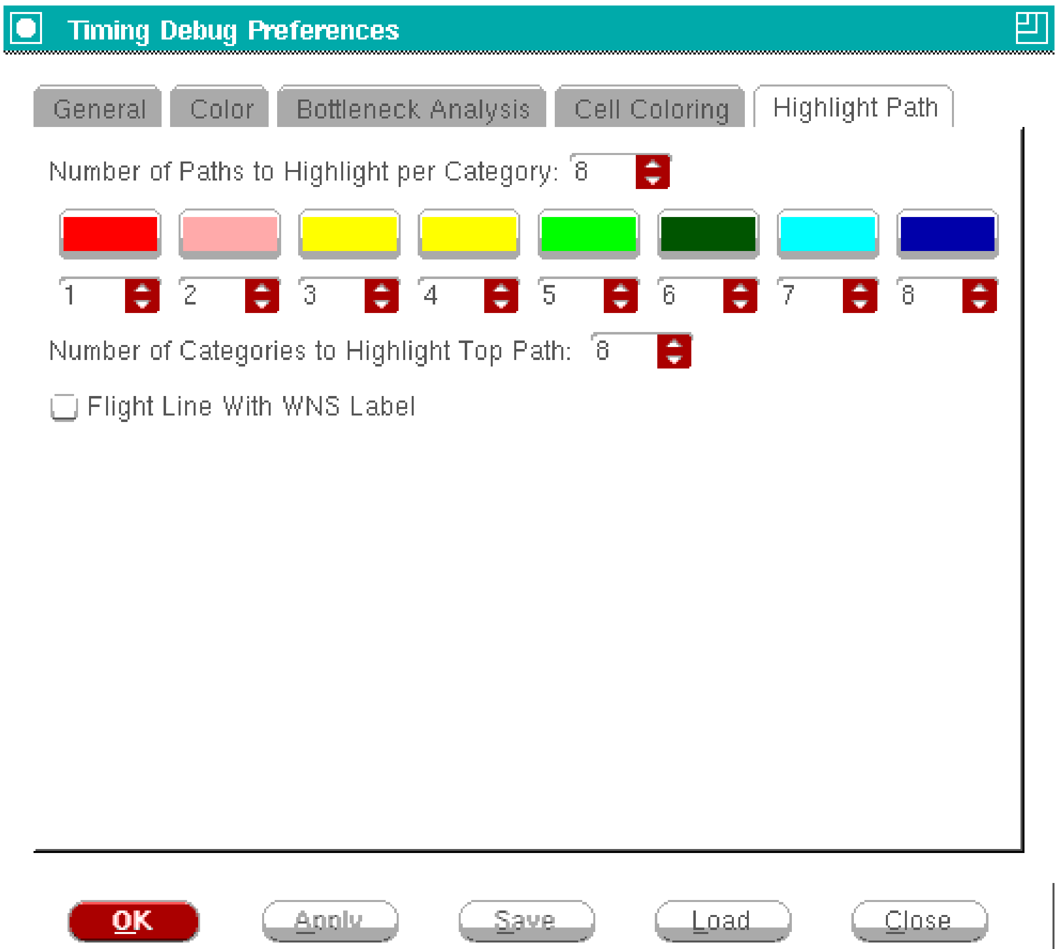
Color elements	Shows the available colors. To customize colors, click on the color. The Timing Debug Preferences - Color - Select Color form is displayed.
Check Box Elements	Indicates whether you want to apply the color to the specified cells.

Cell Name Selection Elements	For each color, you can choose to provide one of the following: ■ Cell name ■ Instance ■ Procedure that you have defined. The procedure is invoked with the full instance name as the argument. You must source the file containing the procedure before you use this feature. For an example, see Timing Debug Preferences - Cell Coloring.
Name Elements	Lets you enter the cell names or procedure name. You can use wildcards when you specify cells, for example: `INV*`, `BUF*`, `DLY*`, `HVT*`.
Save	Saves the selection.
Load	Loads the selection in the system.
Close	Closes the form without saving the selection.

Timing Debug Preferences - Highlight Path

Use the Highlight Path page of the Timing Debug Preferences form to specify the colors and category numbers for color coded critical paths highlighting of categories.

- *Choose Timing - Debug Timing - File - Preferences, and click on the Highlight Path tab.*



Timing Debug Preferences - Highlight Path Fields and Options

Number of Paths of Highlight per category	
	Specify the number of paths per category to be highlighted. The valid value should be from 0 to 30. *Default:* `8` - It means 8 critical paths per category will be highlighted.
Number of Categories to Highlight Top Path	

	Specify the number of categories to be highlighted. The valid value should be from 0 to 30. *Default:* `8` - It means 8 categories will be highlighted.
Flight Line With WNS Label	
	Specify whether to add WNS Label when highlighting a path with the flight line.
Save	Saves the selection.
Load	Loads the selection in the system.
Close	Closes the form without saving the selection.

Path Analysis

You can access the following forms from the Analysis menu:

Path Analysis (Path Groups)

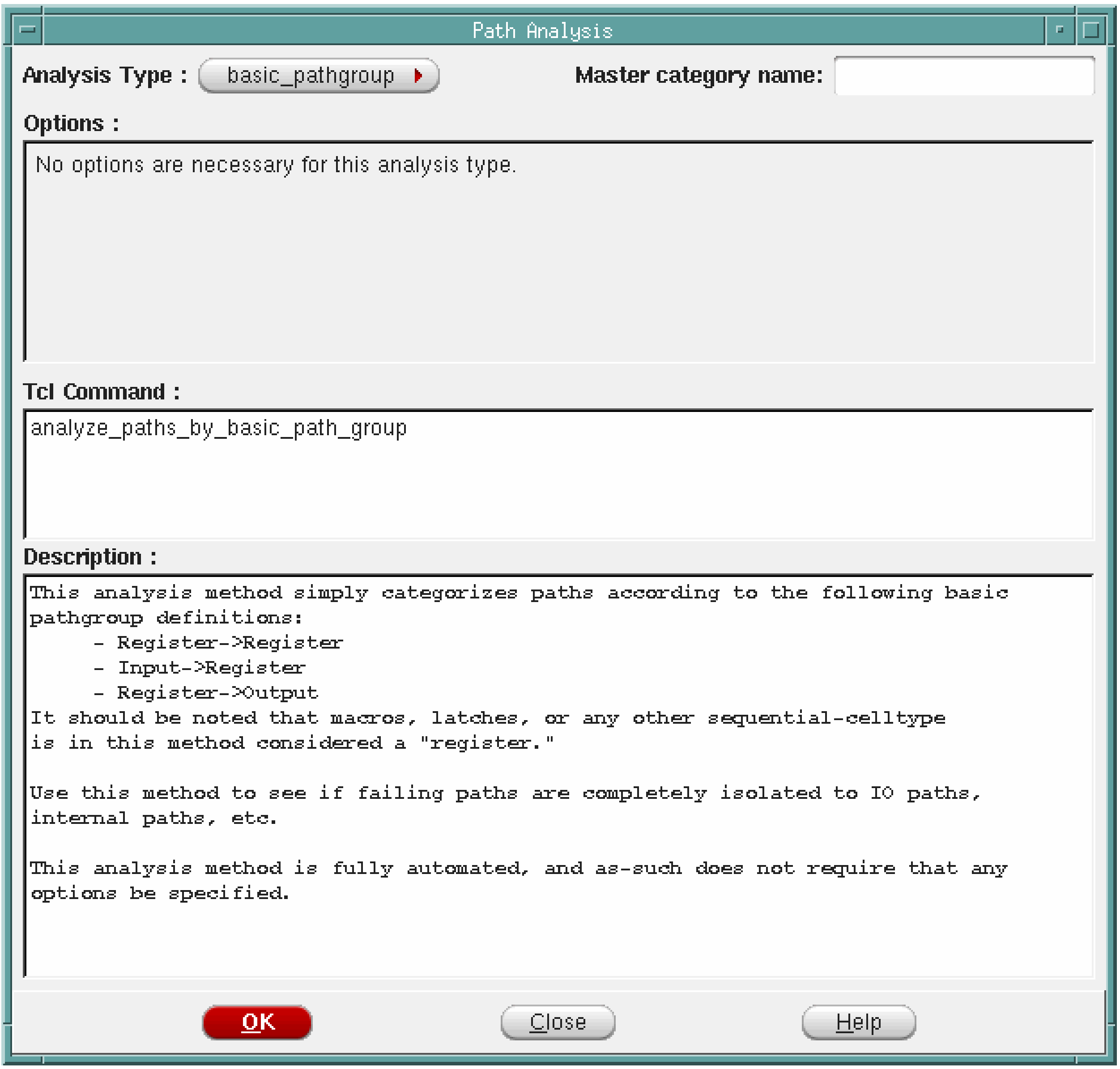
Use the Path Analysis form to create standard path categories according to basic path groups. The basic path groups are:

- Register to register
- Input to register

- Register to output
- Input to Output

Registers can be macros, latches, or sequential cell types.

- Choose *Analysis - Path Group Analysis.*



Path Analysis (Path Groups) Fields and Options

Options	There are no additional options for this analysis.
Tcl Command	Displays the name of the text command that you can specify instead of using the GUI.
Description	Displays the details of the categories that are created when you select the *OK* button.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `analyze_paths_by_basic_path_group`

For more information, see "Timing Debug Commands" in the *Tempus Text Command Reference*.

Path Analysis (Hierarchical Analysis)

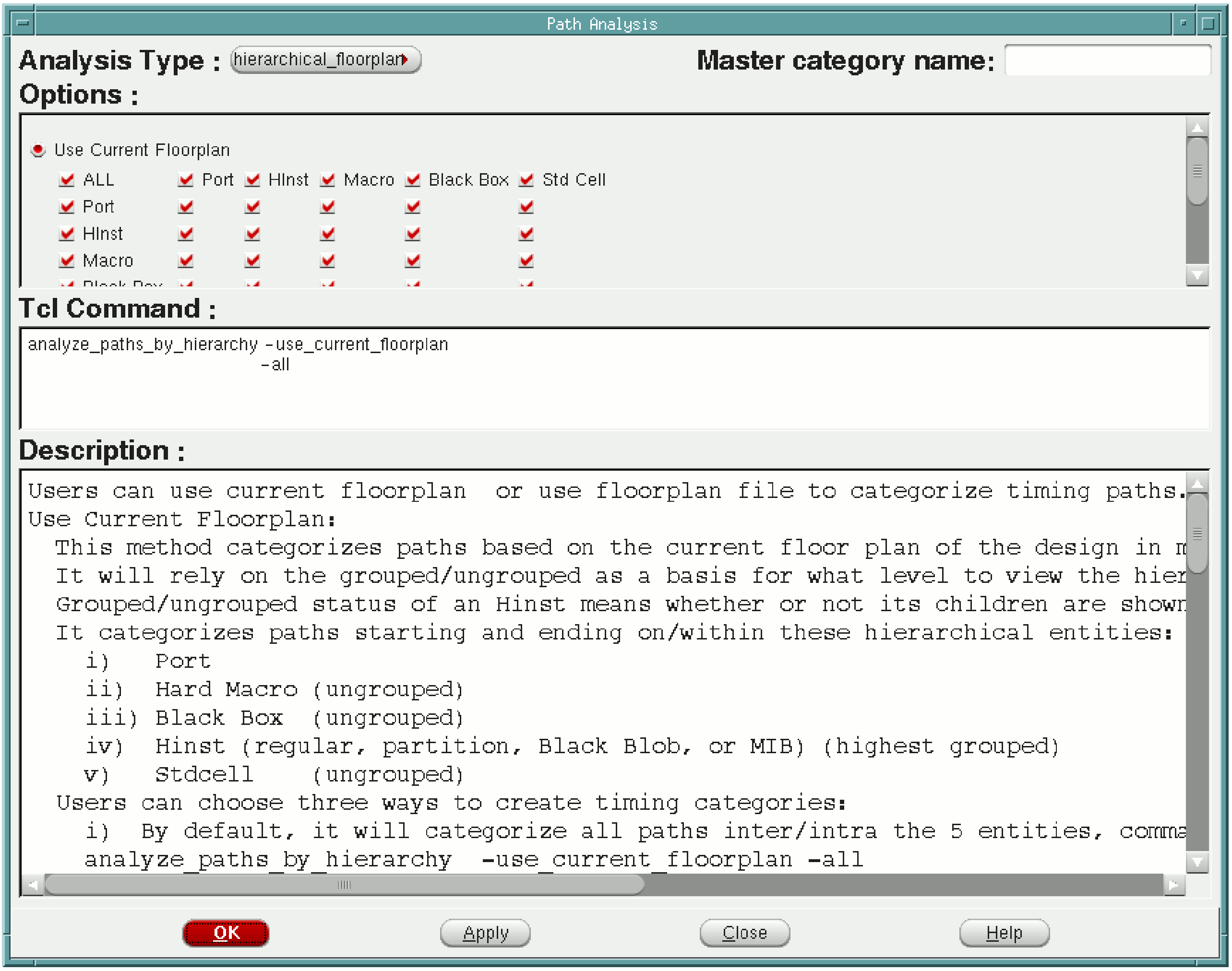
Use the Path Analysis form to create categories according to the hierarchical characteristics of the paths. The software uses the floorplan file to extract information for guides, fences or Macros.

Hierarchical analysis is divided into two types:

- Hierarchical Floorplan Analysis
- Hierarchical Port Analysis

Hierarchical Floorplan Analysis

In the Analysis tab, choose *Analysis - Hierarchical Floorplan.*



Path Analysis (Hierarchical Floorplan Analysis) Fields and Options

Use Current Floorplan	Uses the current floorplan for hierarchical analysis.
Check Box	Specify different types of combination for categories. By default all the types are checked. These options can only be used for the current floorplan.
Floorplan File	Specify the name of the floorplanning file to be used for extracting the object information.

Consider Macros In Category Creation	
	Considers macros in addition to guides and fences for creating categories. By default, macros are not considered. This option can only be used when using floorplan files.
Tcl Command	Displays the name of the text command that you can specify instead of using the GUI.
Description	Displays the details of the categories that are created when you select the *OK* button.

Related Text Commands

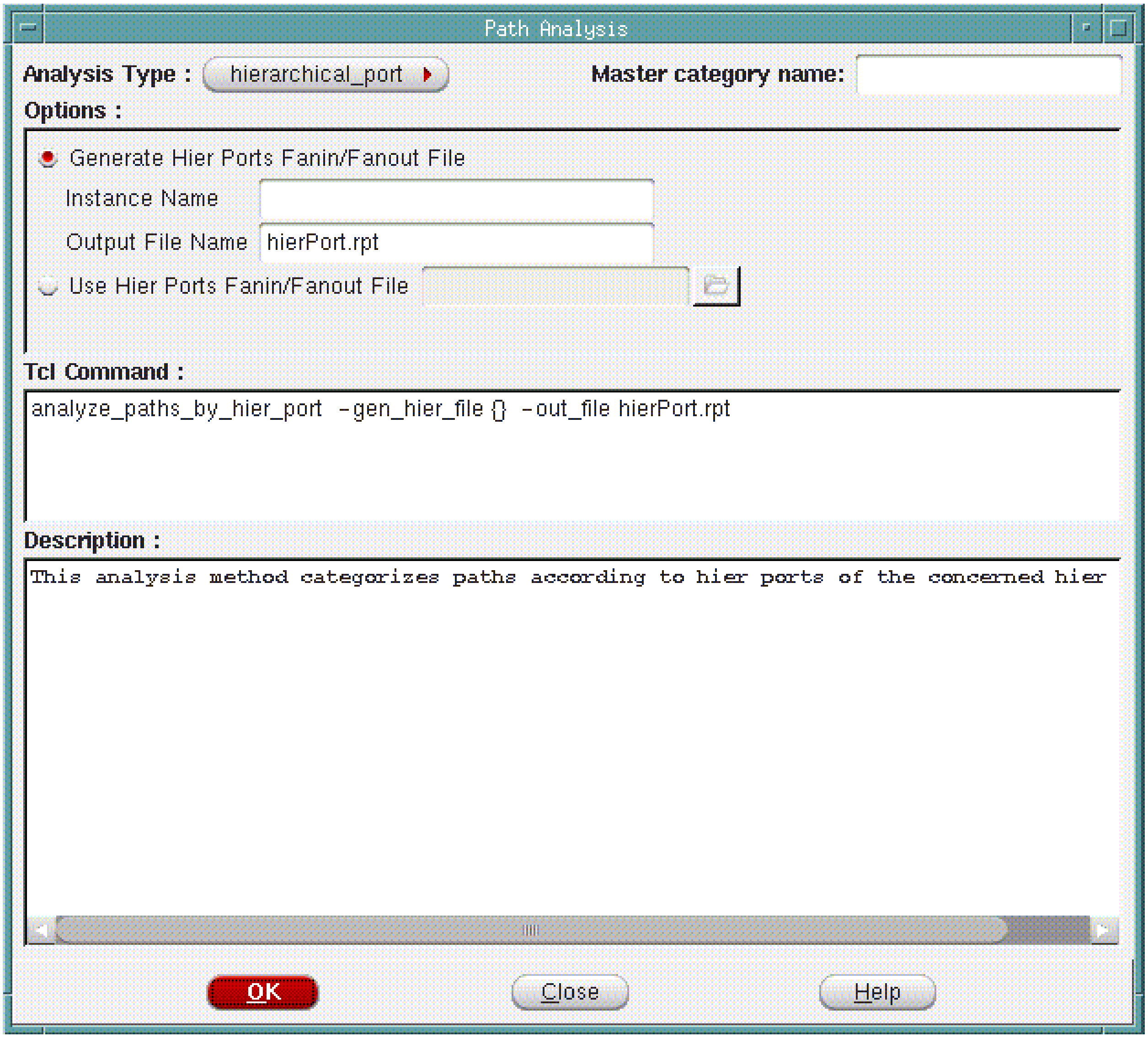
The following text command provides equivalent or additional functionality:

- `analyze_paths_by_hierarchy`

For more information, see "Timing Debug Commands" in the *Tempus Text Command Reference*.

Hierarchical Port Analysis

In the Analysis tab, choose *Analysis - Hierarchical Port.*



Note: The hierarchical ports fanin/fanout file is related to the Timing Debug report.

Related Text Commands

The following text command provides equivalent or additional functionality:

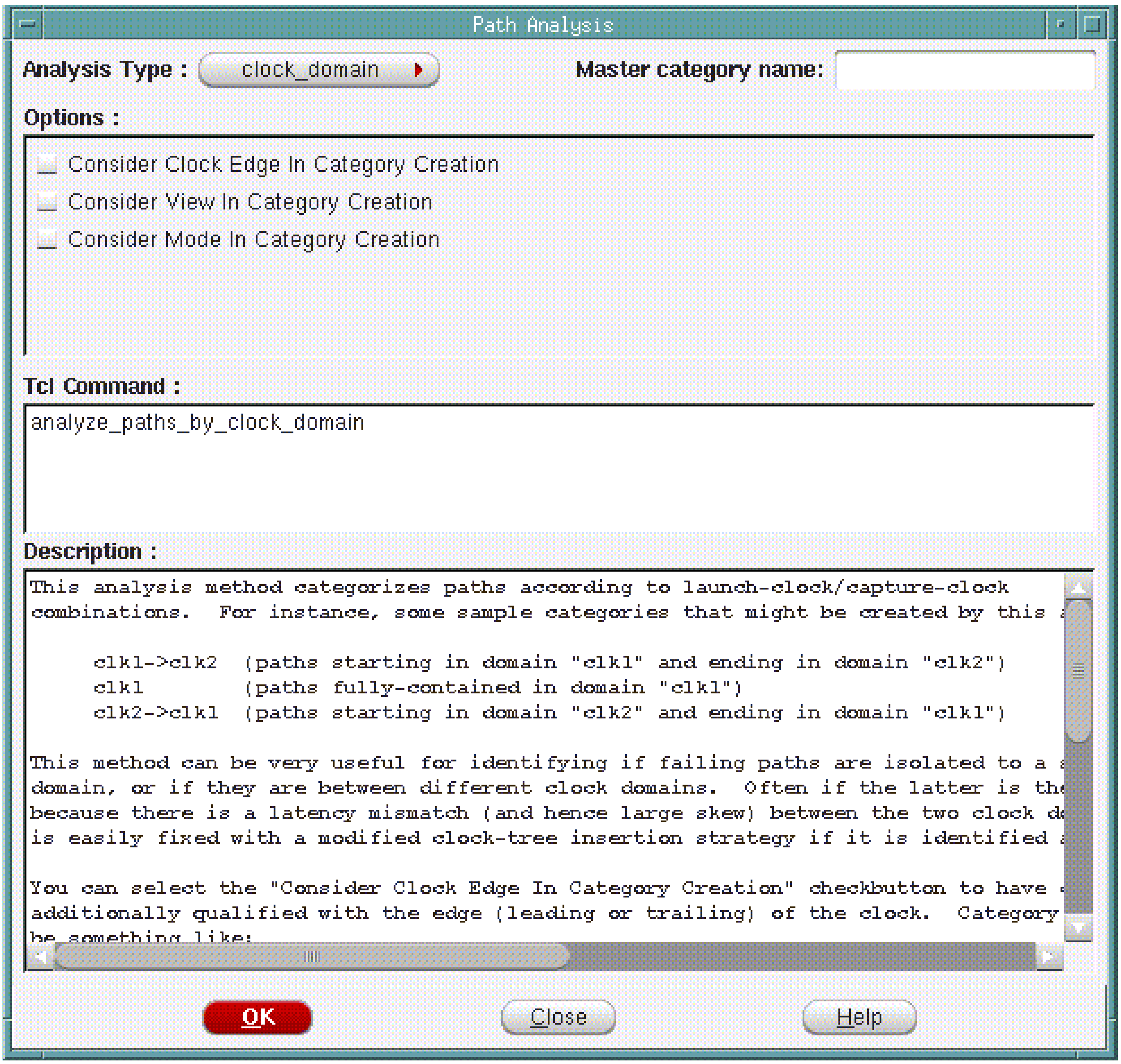
- analyze_paths_by_hierarchy

Path Analysis (Clock Analysis)

Use the Path Analysis form to create categories according to launch clock-capture clock combinations. Following categories are created:

- Paths with clock fully contained in a single domain, clk1.
- Paths with clocks starting one clock domain and ending at another, clk1->clk2.

✝ Choose *Analysis - Clock Analysis.*



Path Analysis (Clock Analysis) Fields and Options

Consider Clock Edge in Category creation (Optional)	
	Considers leading or trailing edge of the clock while defining categories.
Consider View in Category Creation	
	Considers views while defining categories. The paths are sorted according to the clock(s) and the view, for example: Clock1<View1> Clock1<View2> ...
Consider Mode in Category Creation	
	Considers modes while defining categories. The mode is retrieved from the view definition loaded into the design, then the paths are sorted according to the clocks and the mode, for example: Clock1<Mode1> Clock1<Mode2>...
Tcl Command	Displays the name of the text command that you can specify instead of using the GUI.
Description	Displays the details of the categories that are created when you select the *OK* button.

Related Text Commands

The following text command provides equivalent or additional functionality:

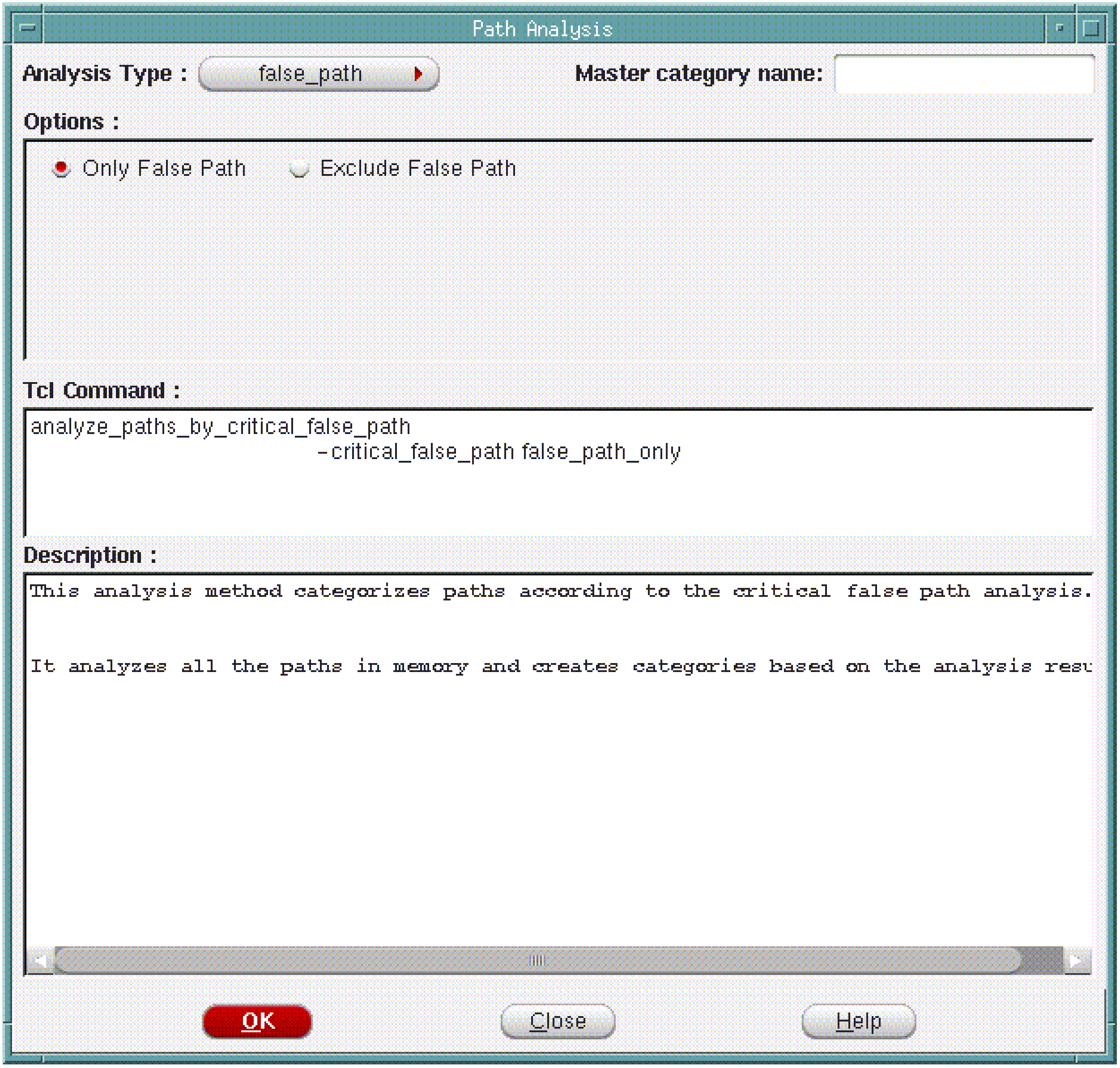
- `analyze_paths_by_clock_domain`

For more information, see "Timing Debug Commands" in the *Tempus Text Command Reference*.

Path Analysis (False Path Analysis)

Use the Path Analysis form to determine whether the reported paths are real critical paths. The analysis is performed by the Conformal Constraint Designer (CCD). As a result of this analysis, the tool creates a category in the *Global Timing Debug* window that contains either the violating paths only, or violating paths except for false paths.

- Choose *Analysis - Critical False Path*



Path Analysis (False Path Analysis) Fields and Options

Only False Path	Creates a category containing violating false paths only.
Exclude False Path	Creates a category containing all paths other than false paths.

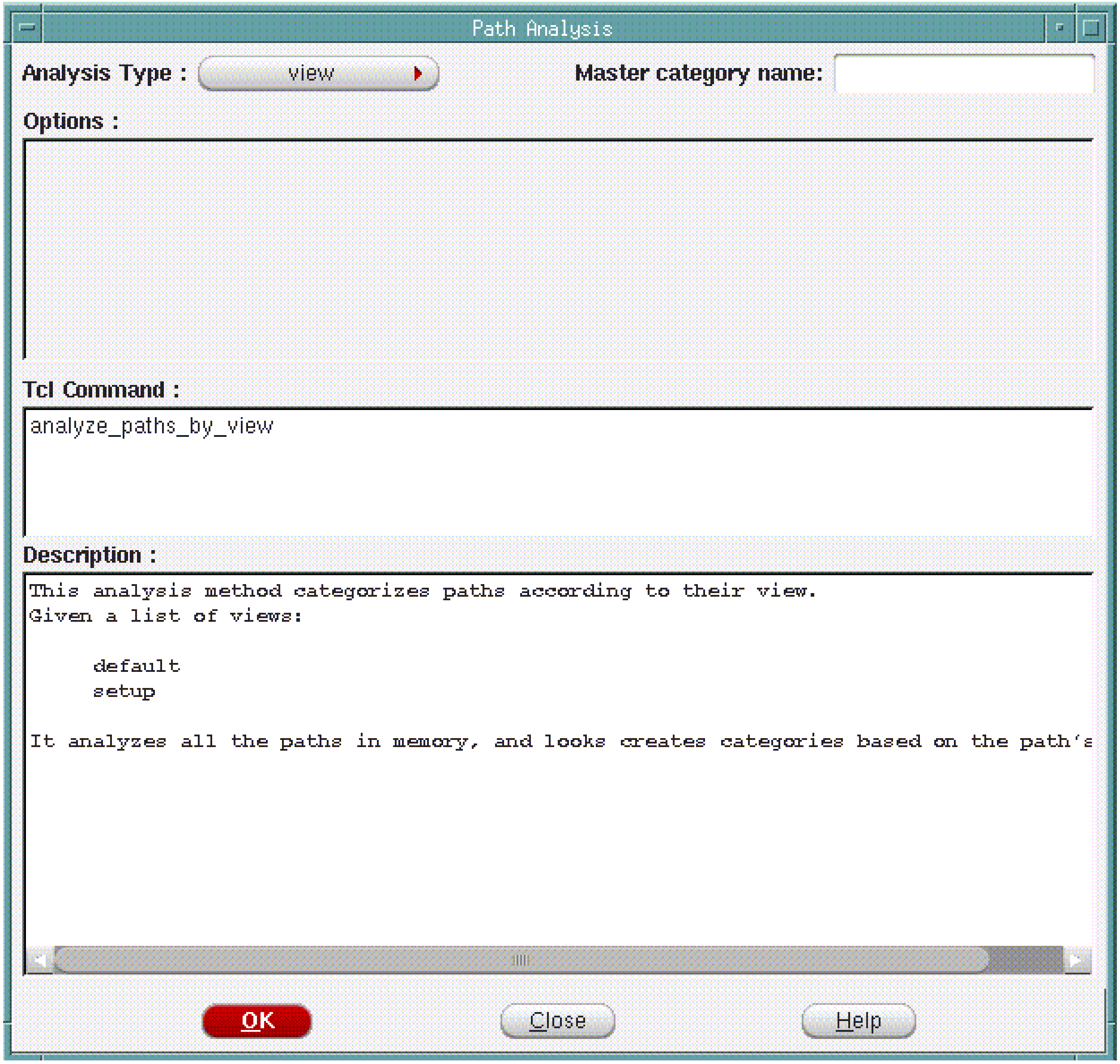
Tcl Command	Displays the name of the text command that you can specify instead of using the GUI.
Description	Displays the details of the categories that are created when you select the *OK* button.

Path Analysis (View Analysis)

Use the Path Analysis form to create categories according to the view that each path is related to. You can analyze the impact of each view on the performance using this category.

Note: In a single report, which contains all the views together, each endpoint is reported once. On the other hand, when you load several reports that were generated for different views, each endpoint may appear in each report with a different slack value.

- Choose *Analysis - View Analysis.*



Path Analysis (View Analysis) Fields and Options

Options	There are no additional options for this analysis.
Tcl Command	Displays the name of the text command that you can specify instead of using the GUI.

Description	Displays the details of the categories that are created when you select the *OK* button.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `analyze_paths_by_view`

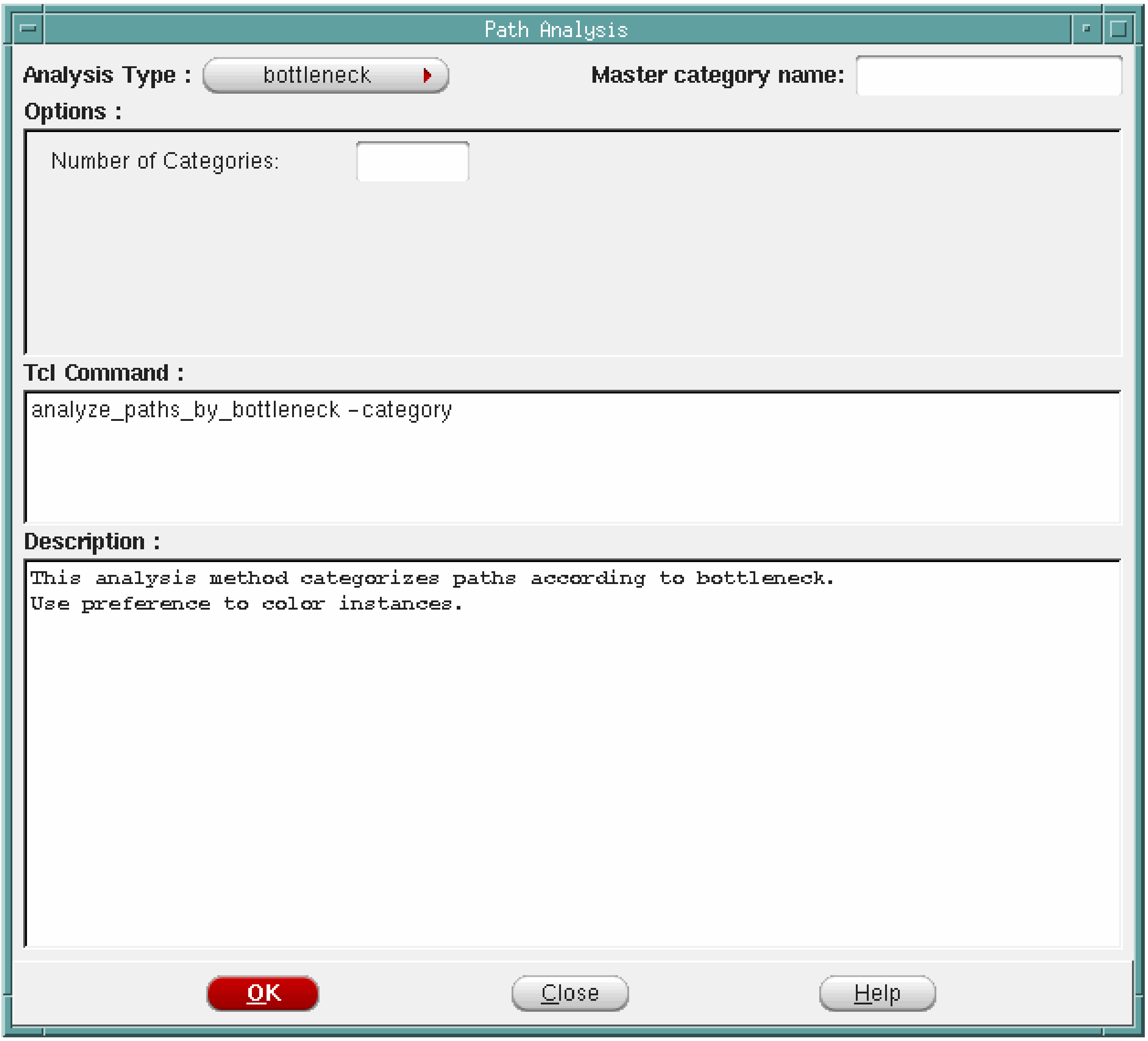
For more information, see "Timing Debug Commands" in the *Tempus Text Command Reference*.

Path Analysis (Bottleneck Analysis)

Use the Path Analysis form to create categories according to the bottleneck analysis of the critical paths. Bottleneck Analysis detects the instances that are often involved in the critical paths in the design. Any change in timing for these instances impacts multiple critical paths in the design. Therefore you can use this information to change your design (floorplan, placement, constraints etc) around these instances. The bottleneck analysis computes the number of occurrences that an instance appears in the paths that you display in the Timing Debug window, and provides the following output:

- Colors most used instance based on number of occurrences.
- Creates *N* number of path categories for the instances with the highest number of occurrences. *N* is the number you specify in the Path Analysis form.

- Choose *Analysis - Bottleneck Analysis.*



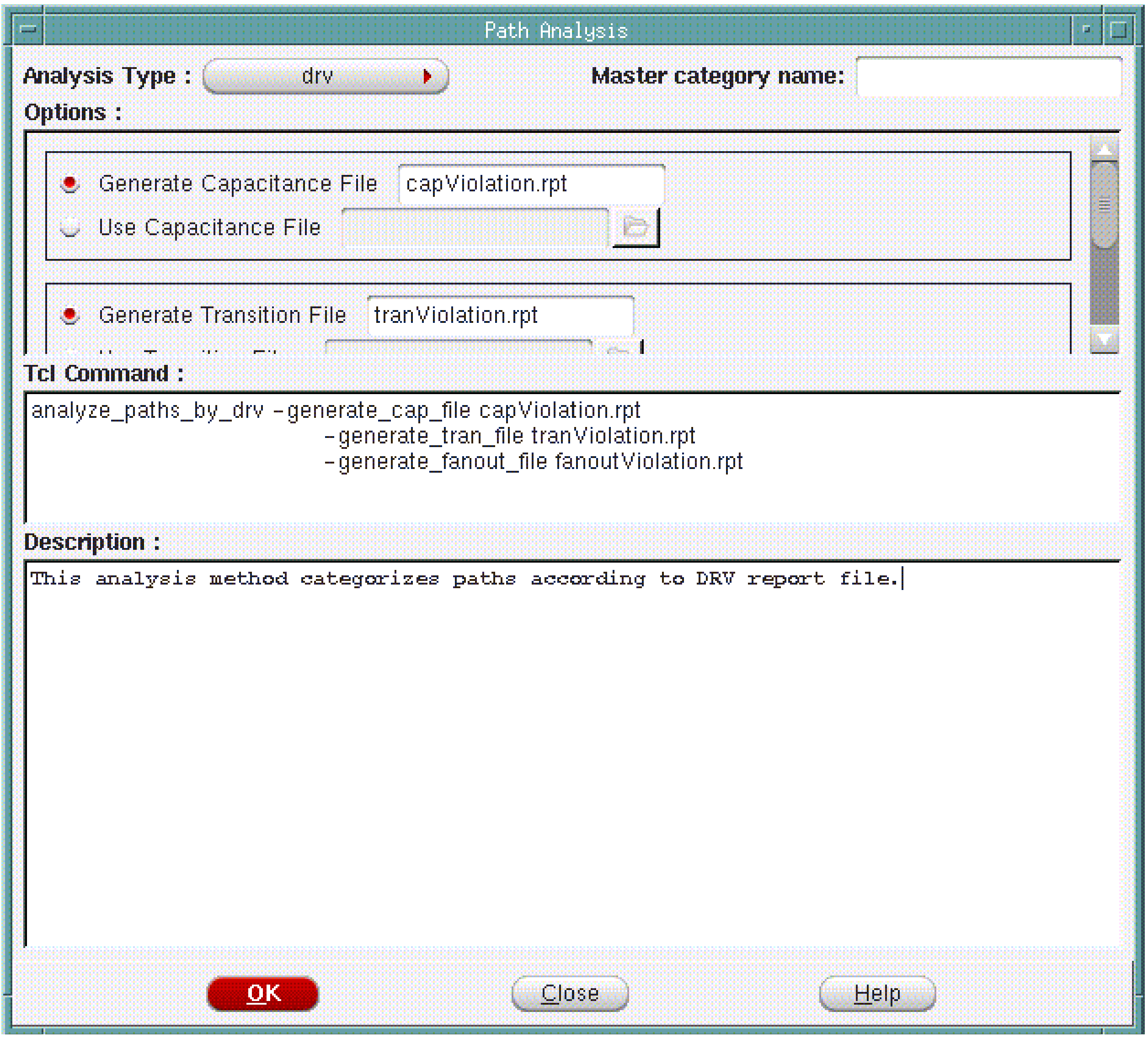
Path Analysis (Bottleneck Analysis) Fields and Options

Number of Categories	Specifies the number of categories to be created for most occurring instances.
Tcl Command	Displays the name of the text command that you can specify instead of using the GUI.
Description	Displays the details of the categories that are created when you select the *OK* button.

Path Analysis (DRV Analysis)

Use the Path Analysis form to read or generate a report containing max transition, max capacitance, and max fanout violations. Paths that are affected by the selected DRV type are grouped into a category.

- Choose *Analysis - DRV Analysis.*



Path Analysis (drv) Fields and Options

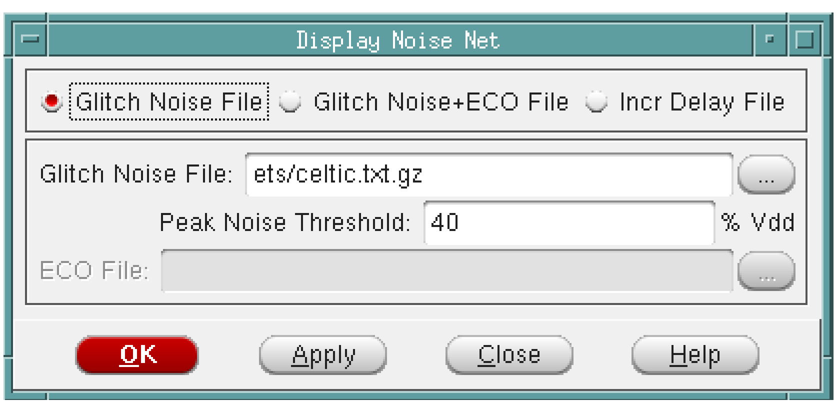
Generate Capacitance File

	Generates a file containing a list of paths containing max capacitance violations.
Use Capacitance File	
	Reads a file containing a list of paths containing max capacitance violations.
Generate Transition File	
	Generates a file containing a list of paths containing max transition violations.
Use Transition File	
	Reads a file containing a list of paths containing max transition violations.
Generate Fanout File	
	Generates a file containing a list of paths containing max fanout violations.
Use Fanout File	Reads a file containing a list of paths containing max fanout violations.

Noise Result Analysis

Use the Display Noise Net form to browse aggressor information for each of the glitch failures and to locate the glitch nets (aggressor and victim) in the physical layout for ECO fixing. You can generate the required input files for this form by using the `write_noise_eco -format edi` command.

- Choose *Analysis - Noise Result Analysis.*



Display Noise Net Fields and Options

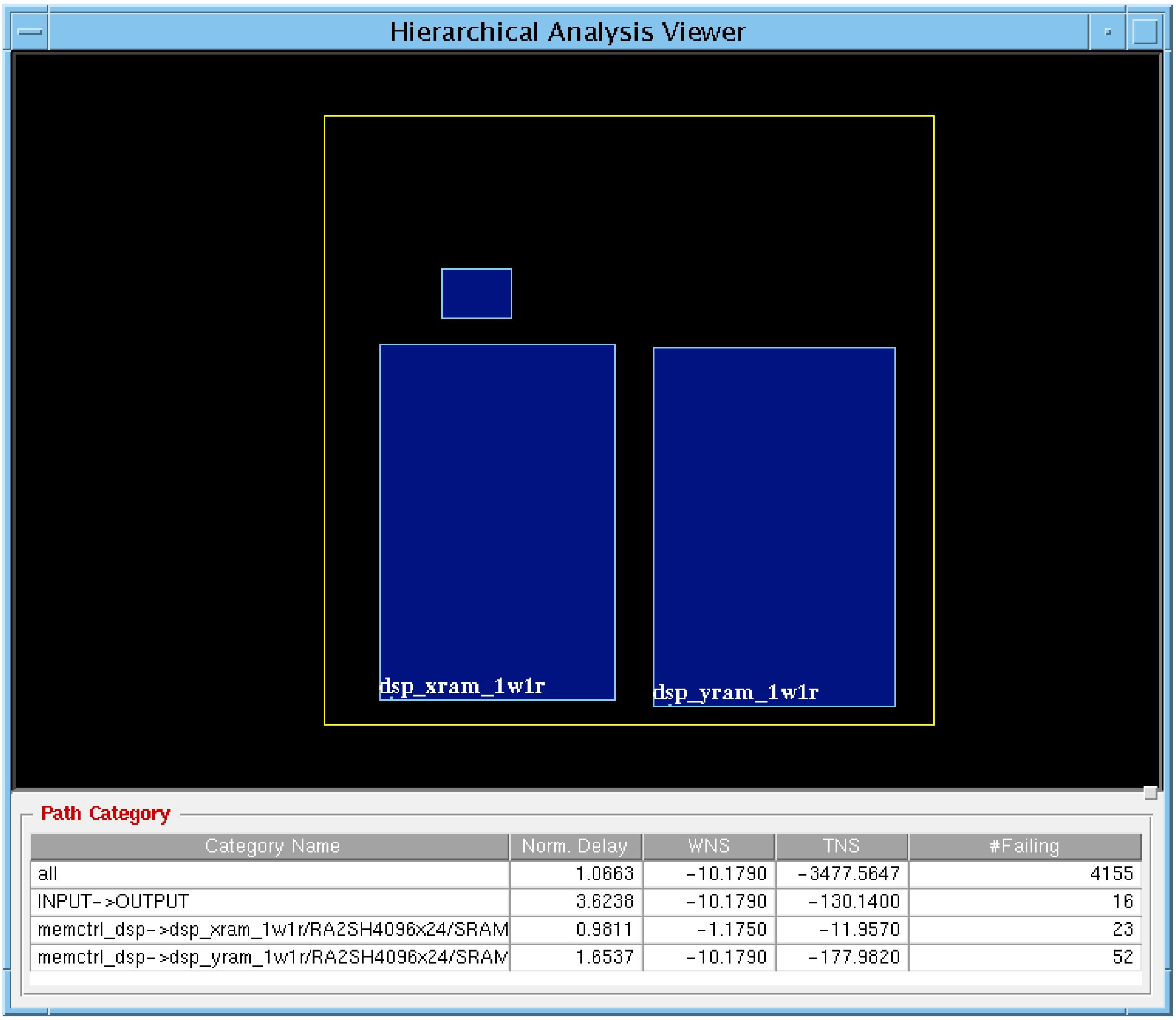
For information on fields and options, see “Display Noise Net”.

Hierarchical Analysis Viewer

Use the Hierarchical Analysis Viewer form to view the results from hierarchical analysis that is done when you create the hierarchical paths category.

For more information on creating the hierarchical paths category, see Path Analysis (Hierarchical Analysis) on page 511".

- Choose *Analysis - Hierarchical Analysis Viewer.*



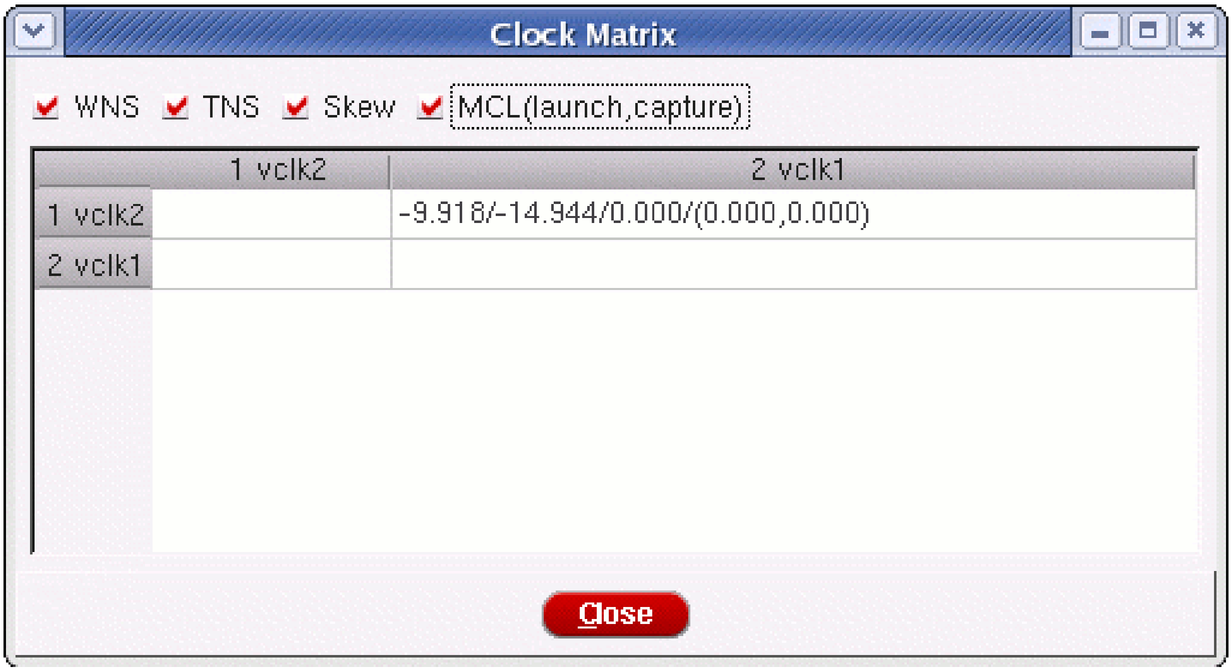
Hierarchical Analysis Viewer Fields and Options

FloorPlan Display Area	Displays the timing connectivity between floorplan modules or partitions.
Path Category	Displays the paths that are grouped in the hierarchical path category.

Clock Matrix Viewer

Use the Clock Matrix Viewer to show the relationship of source and destination clocks. User can call this tool under the menu of Analysis in Global Timing Debug.

- *Choose Timing - Debug Timing - Analysis - Clock Matrix Viewer.*

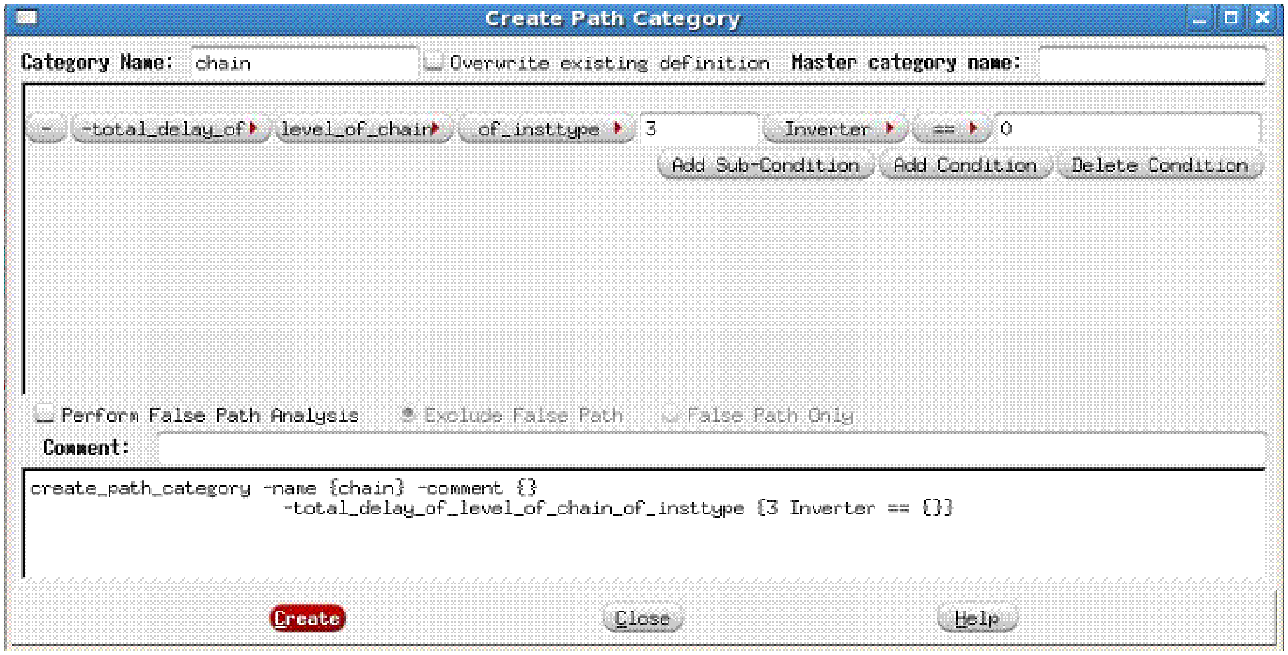


Note: MCL (launch, capture) is the maximum clock latency for launching or capturing a clock path.

Create Path Category

Use the Create Path Category form to define the categories as per your design requirements. You can use the Timing Path Analyzer form to analyze the critical paths and identify possible problems with the design. You can then use the Create Path Category form to define a category or categories to group all paths with same problem.

- Choose *Category - Create Category.*



Create Path Category Fields and Options

Category Name	Specify the name of the category. The values that you specify in this form are reflected as Tcl commands in the display area at the bottom of the form.
Overwrite existing definition	
	Overwrites any predefined category with the same name in your category list.

Definition	Define category using the pull down menu options beginning with *-from* and *inst*. Choose from the following options: ■ *-from* ■ *-not_from* ■ *-through* ■ *-not_through* ■ *-to* ■ *-not_to*
	Choose from the following options: ■ *inst* ■ *pin* ■ *cell* ■ *clock* ■ *port* ■ pd For example, you can choose *-from cell XYZ*.
	■ *-slack* ■ *-uncertainty* ■ *-skew* ■ *-launch_latency* ■ *-capture_latency*

	Choose from the following options: ■ *==* ■ *!=* ■ *>* ■ *<* ■ *>=* ■ *<=* For example, you can choose *-uncertainty >= 0.4*.	
	-check_type	Choose from the following options: ■ *Setup Check* ■ *Hold Check* ■ *Clock Gating Setup Check* ■ *Library Clock Gating Setup Check* ■ *Clock Gating Hold Check* ■ *Library Clock Gating Hold Check* ■ *External Delay Assertion* ■ *Latch Borrowed Time Check* ■ *PulseWidth Check* ■ *ClockPeriod Check* ■ *Recovery Check* ■ *Removal Check*

	-number_of	Specify number of instances with required operator. Optionally, you can choose from the following options: ■ *of_name* ■ *not_of_name* ■ *of_celltype* ■ *not_of_celltype* For example, you can specify *-number_of insts of_name* `INST1` *!=* `4`.
	-delay_of	Specify the value for one of the following options: ■ *any_inst* ■ *any_net* ■ *every_inst* ■ *every_net* For instances, you can optionally choose from the following options: ■ *of_name* ■ *not_of_name* ■ *of_celltype* ■ *not_of_celltype* For example, you can specify *-delay_of any_inst of_name* `INST1` *=* `0.5`. For nets, you can optionally choose from the following option: ■ *of_name* ■ *not_of_name* For example, you can specify *-delay_of any_net of_name* `clk` *=* `0.1`.

	-total_delay_of	Specify the value for one of the following options: ■ path ■ insts ■ nets For instances, you can optionally choose from the following options: ■ *of_name* ■ *not_of_name* ■ *of_celltype* ■ *not_of_celltype* For example, you can specify *-total_delay_of inst of_name* `INST1` = `0.5`. For nets, you can optionally choose from the following options: ■ *of_name* ■ *not_of_name* For example, you can specify *-total_delay_of net of_name* `clk` = `0.1`.
	-view	Specify the name of the view. The paths related to the specified view are grouped under this category.
	-file	Specify the name of the file that contains definitions for creating categories.
	-sdc	Specify the name of the SDC constraint. The paths related to the specified SDC constraint are grouped under this category. You can create a constraint category directly from an SDC constraint by right-clicking on the constraint in the Timing Path Analyzer - Path SDC form. The Create Path Category form displays, showing the line number (from the SDC file) and the name of the constraint.

Master Category name	Creates the category as a nested category for the existing category that you specify in this field.
Add Sub-Condition	Adds a row for definition an OR sub-condition for the named category.
Add Condition	Adds a row for definition an AND sub-condition for the named category.
Delete Condition	Deletes defined condition.
Perform False Path Analysis	
	Click to perform false path analysis
Exclude False Pin	Click to exclude false pins
False Path Only	Click to choose false path
Comment	Allows you to enter any comment to associate with the condition.

Related Text Commands

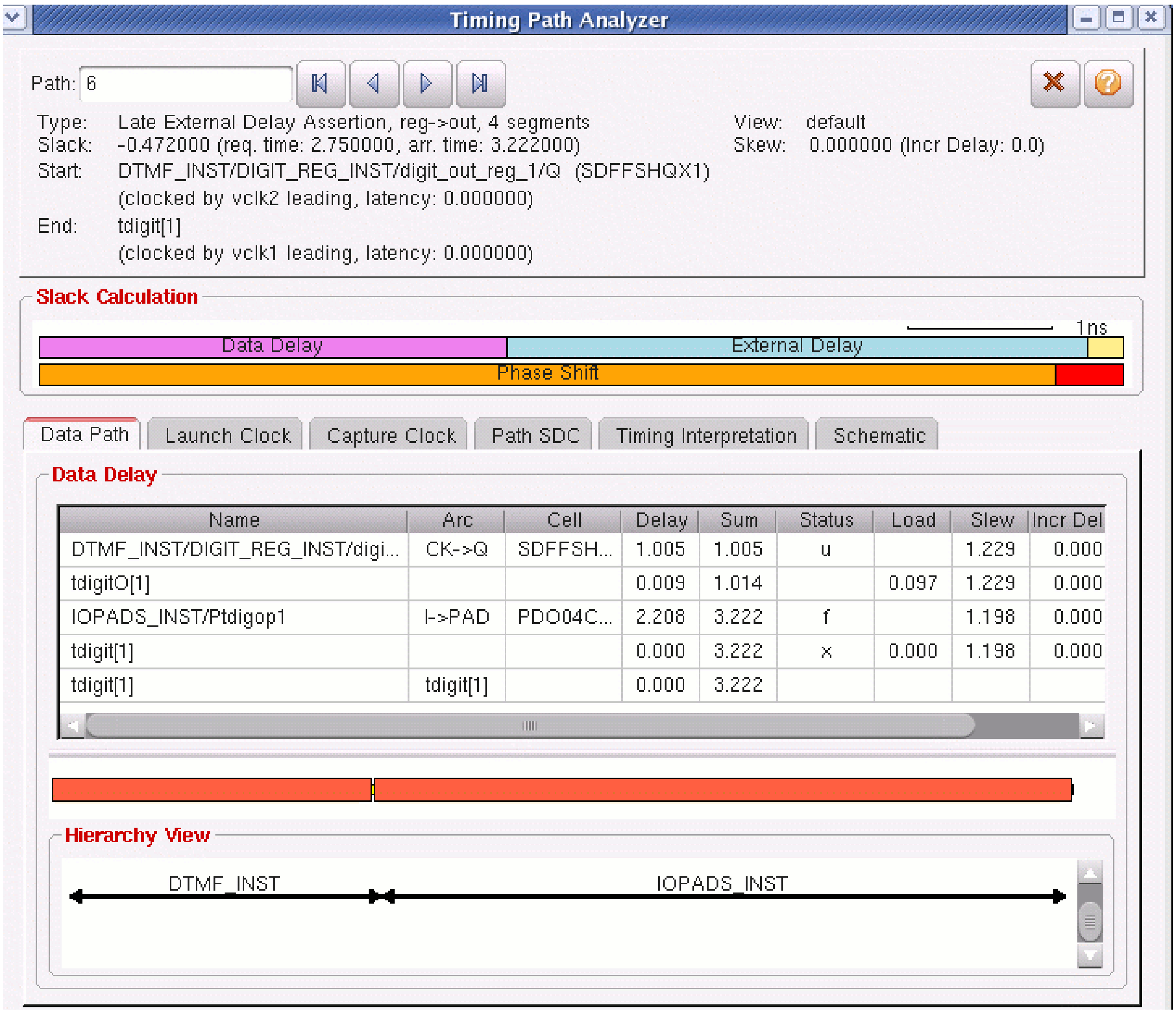
The following text command provides equivalent or additional functionality:

- `create_path_category`

For more information, see "Timing Debug Commands" in the *Tempus Text Command Reference*.

Timing Path Analyzer

Use the Timing Path Analyzer form to identify issues related to a path using slack calculation bars, timing bar, and hierarchy view.



By default, the *Data Path* tab is selected (see Timing Path Analyzer - Data Path).

Timing Path Analyzer - Fields and Options

Path	Specifies the number of path as listed in the Timing Debug window.
Type	Displays the type of selected path.

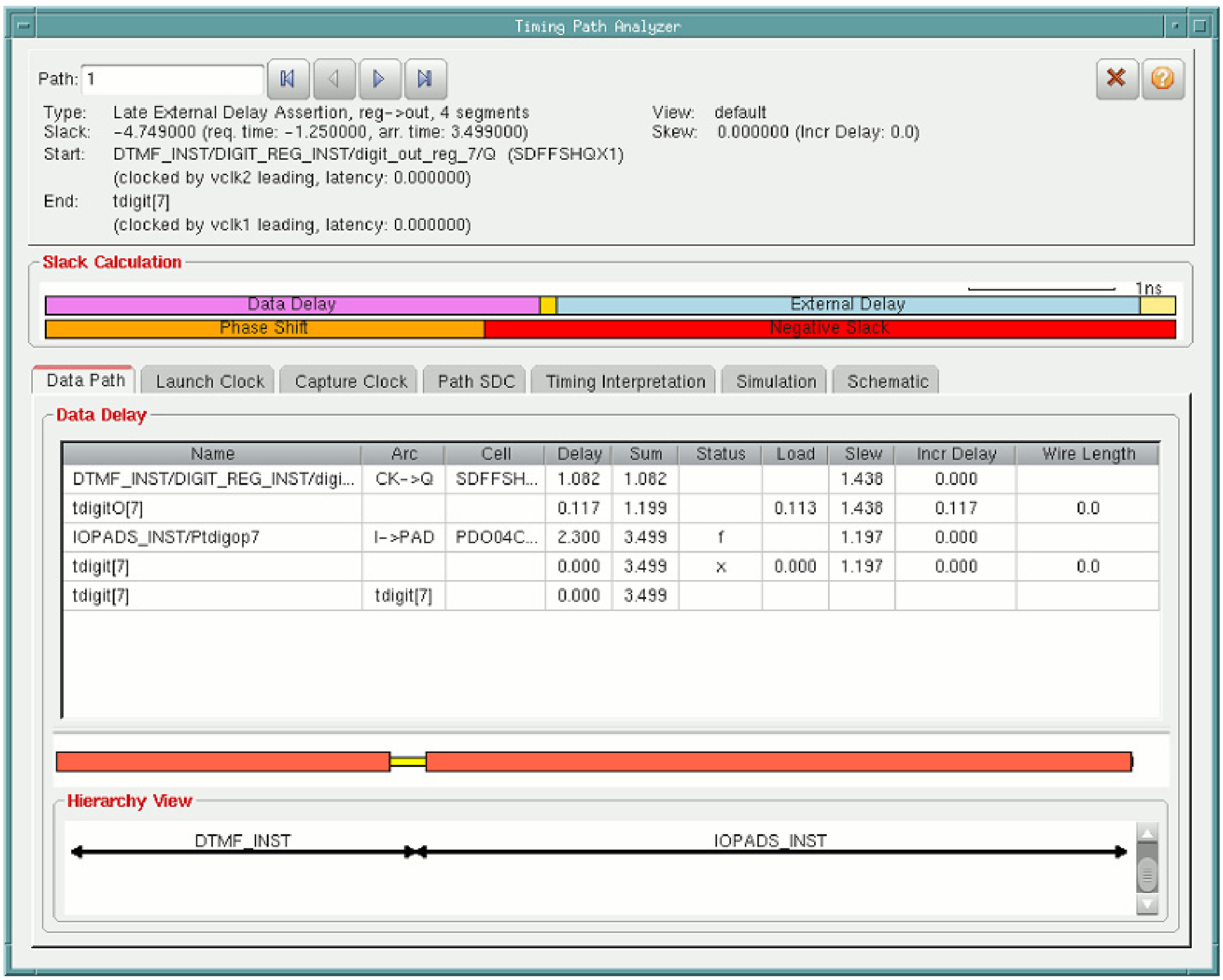
Start	Displays the starting point of the path.
End	Displays the ending point of the path.
Skew	Displays the skew of the path.
Slack	Displays the slack of the path. In statistical mode, the percentile value of the slack is reported. **Note:** Use slack to sort the path list.
Slack Calculation	Displays path arrival time and required time calculations in color bars. The top bar displays the arrival time and the bottom bar displays the required time calculation. For example, arrival time can be represented as consisting of data delay (In pink), Setup time (in blue), and uncertainty (in yellow). You can inspect various aspects of calculation to identify possible problem(s) associated with a path. You can move you cursor over the color bars to display the values associated with each bar element. Use this feature to identify issues related to clock skew, latency balancing or large clock uncertainties. **Note:** The timescale is available in the Slack Calculation section as a horizontal bar where the length of the bar represents the time.
Data Delay	Displays the details of the selected path in the violation reports. You can click on a single element to display it in the design display area. You can select each element consecutively to trace the entire path in the design display area. In statistical mode both the mean and sigma values of delays are reported in the mean delay and sigma delay columns.
Timing Bar	Displays the instance and net delays. The size of the bar indicates the delay associated with that element. Moving the cursor on these elements displays the details of instances or nets. The instance delays are represented in red and the net delays are represented in yellow color. Use this feature to identify issues related to large instance or net delays, repeater chains, paths with large number of buffers, and large macro delays.
Hierarchy View	Displays a path's traversal of logical hierarchy on a time-axis. This is useful for identifying issues related to inter-partition paths or paths that go through a partition.

	Closes the form.

Timing Path Analyzer - Data Path

Use the Timing Path Analyzer - Data Path form to identify issues related to a path using the data path details. The Data Path form displays the details of a complete data path.

- Double-click on a path in the *Path List* in the *Analysis* window.



Timing Path Analyzer - Data Path Fields and Options

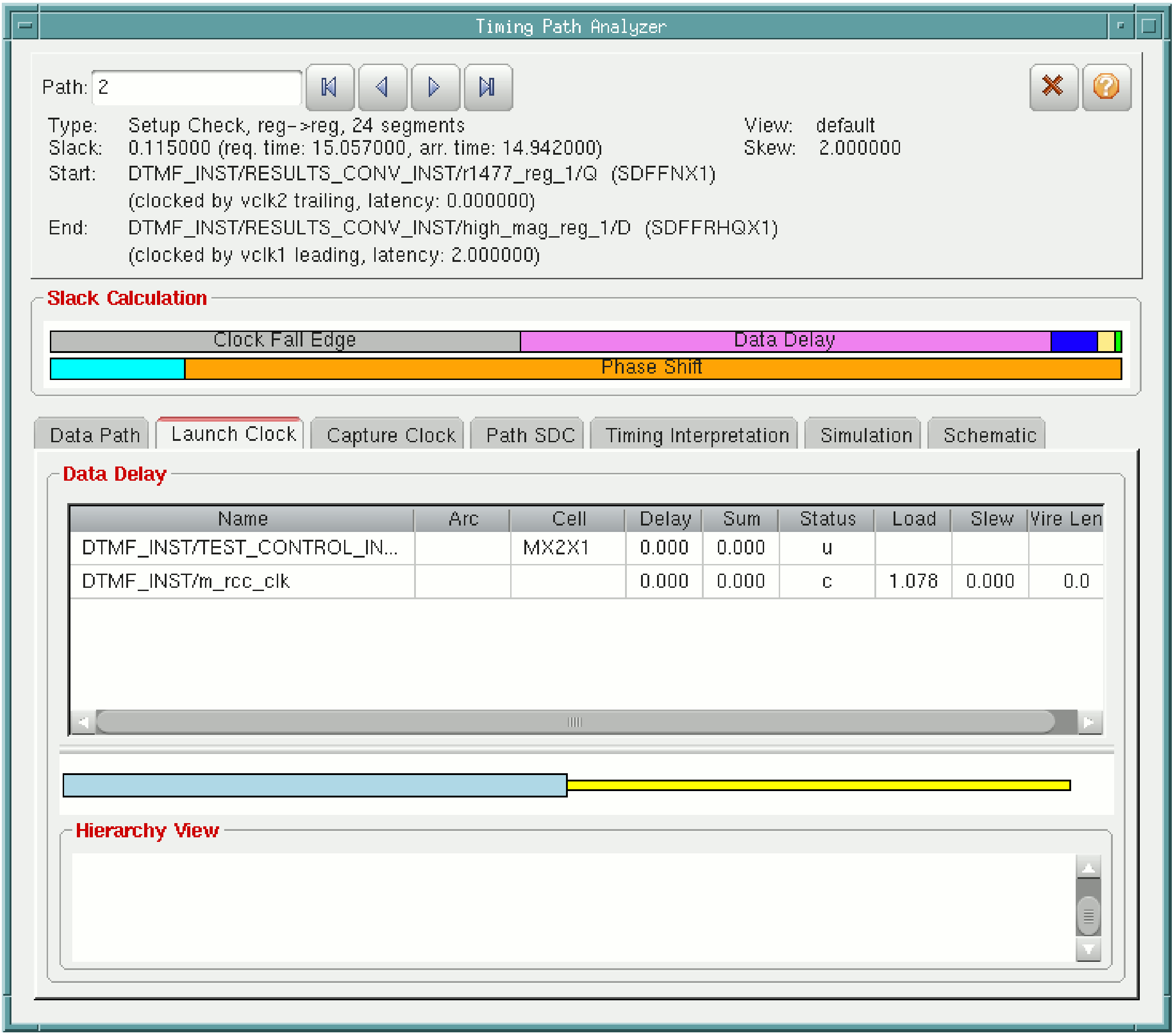
For details on the fields and options see <u>Timing Path Analyzer</u>.

Timing Path Analyzer - Launch Clock

Use the Timing Path Analyzer - Launch Clock form to identify issues related to a path using the launch clock details. The Launch Clock form displays the details of a complete launch clock path.

Double-click on a path in the *Path List* in the *Analysis* window.

- Choose *Launch Clock* tab in the Timing Path Analyzer form.



Timing Path Analyzer - Launch Clock Fields and Options

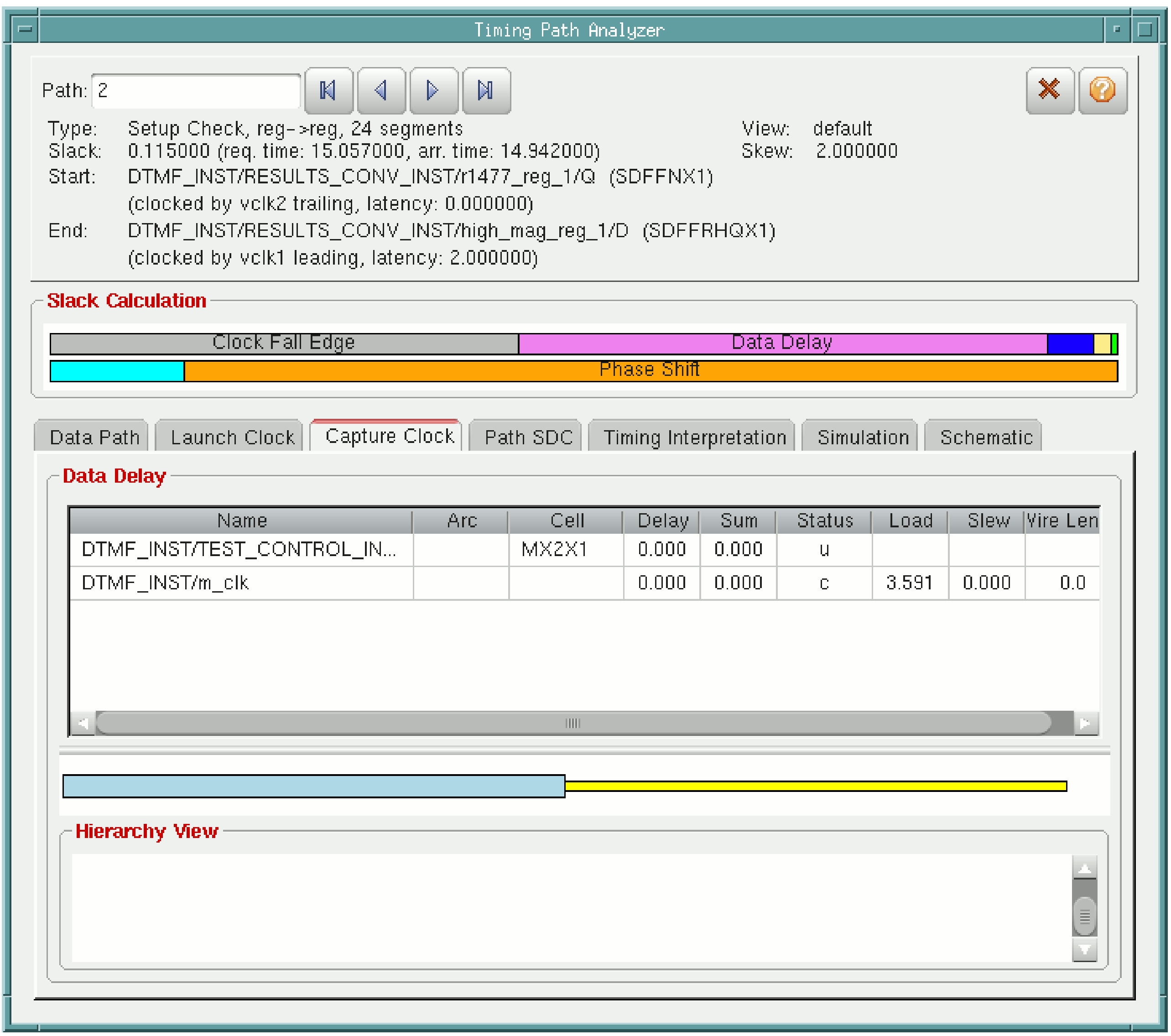
Data Delay	Displays the details of the launch clock of the selected path. You can click on a single item in the table to highlight the clock in the design display area. The *Timing Bar* and *Hierarchy View* display the characteristics of the launch clock path.
	Closes the form.

Timing Path Analyzer - Capture Clock

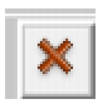
Use the Timing Path Analyzer - Capture Clock form to identify issues related to a path using the capture clock details. The Capture Clock form displays the details of a complete capture clock path.

Double-click on a path in the *Path List* in the *Analysis* window.

☨ Choose *Capture Clock* tab in the Timing Path Analyzer form.



Timing Path Analyzer - Capture Clock Fields and Options

Data Delay	Displays the details of the capture clock of the selected path. You can click on a single item in the table to highlight the clock in the design display area. The *Timing Bar* and *Hierarchy View* display the characteristics of the capture clock path.
	Closes the form.

Timing Path Analyzer - Statistical

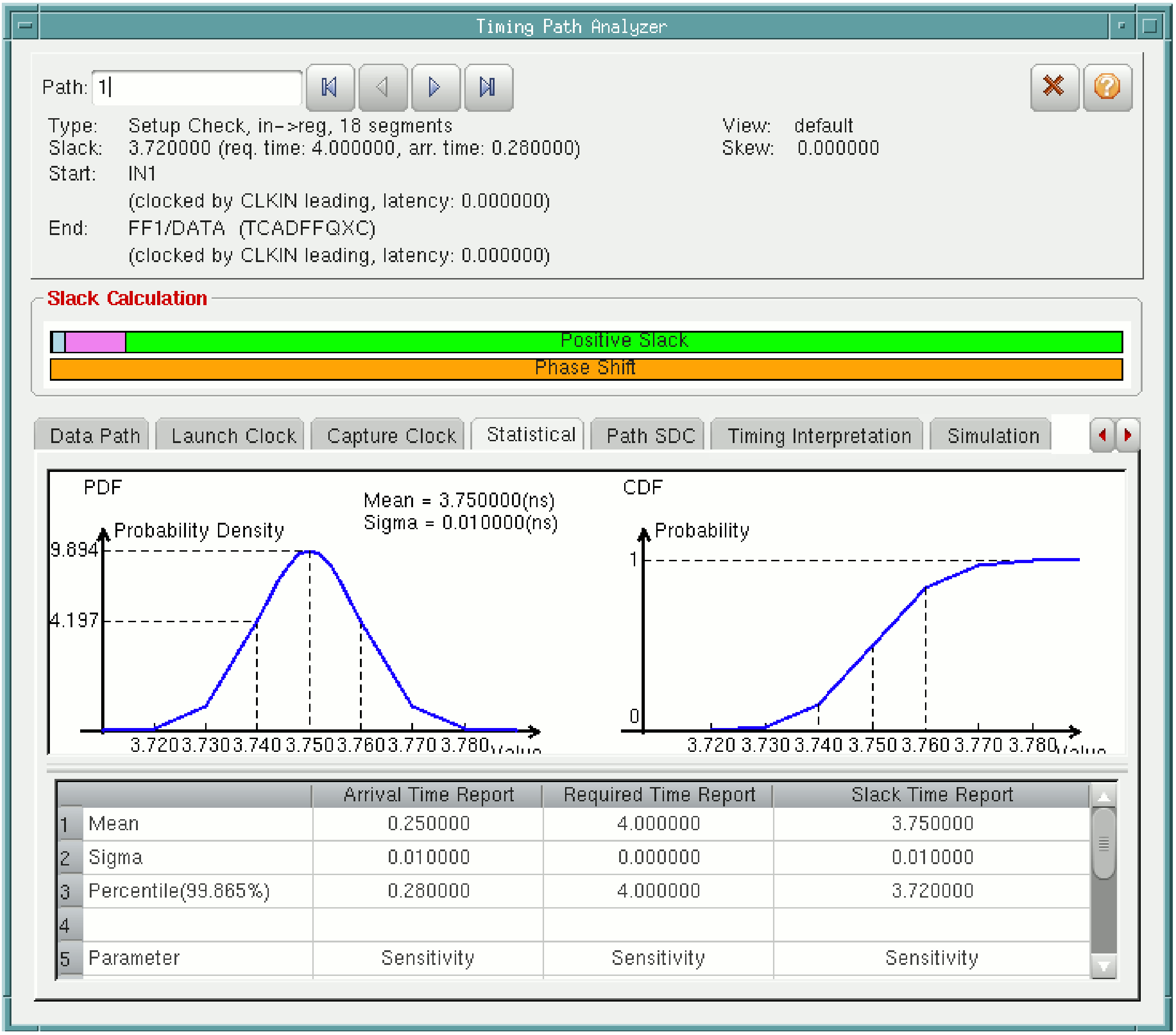
Use the Timing Path Analyzer - Statistical form to view the PDF and CDF plots in statistical mode. The plots show combined distribution of slacks.

To generate a SSTA timing report, see <u>Display/Generate Timing Report</u>.

The *Statistical* tab on the Timing Path Analyzer form is visible only when the SSTA mode is enabled. To enable SSTA mode, see *Specify Analysis Mode - Advanced*.

Double-click on a path in the *Path List* in the *Analysis* window.

- Choose *Statistical* tab in the Timing Path Analyzer form.



Timing Path Analyzer - Statistical Fields and Options

PDF	Displays the probabilistic density function (PDFs) plot. The plot shows the probability that value will occur over a given range.
CDF	Displays the cumulative density function (CDFs) plot. The plot shows Probability vs. Value statistics.

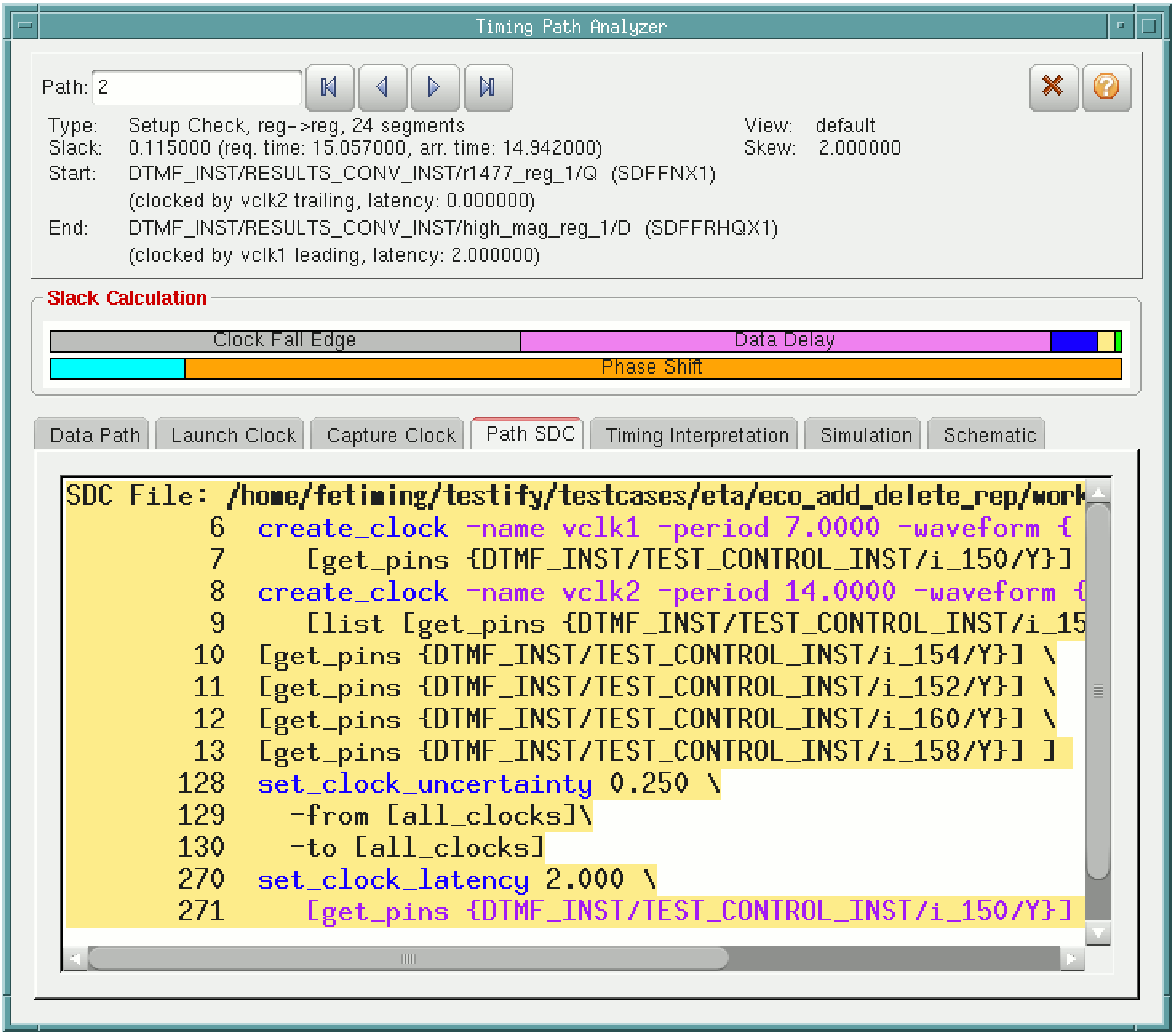
	The mean, sigma, and percentile values of the slack, arrival, and required times are reported in detail along with the sensitivities with respect to each of the process parameters.
	Closes the form.

Timing Path Analyzer - Path SDC

Use the Timing Path Analyzer - Path SDC form to identify issues related to the SDC constraints associated with the selected path.

Double-click on a path in the *Path List* in the *Analysis* window.

- Choose *Path SDC* tab in the Timing Path Analyzer form.



Timing Path Analyzer - Path SDC Fields and Options

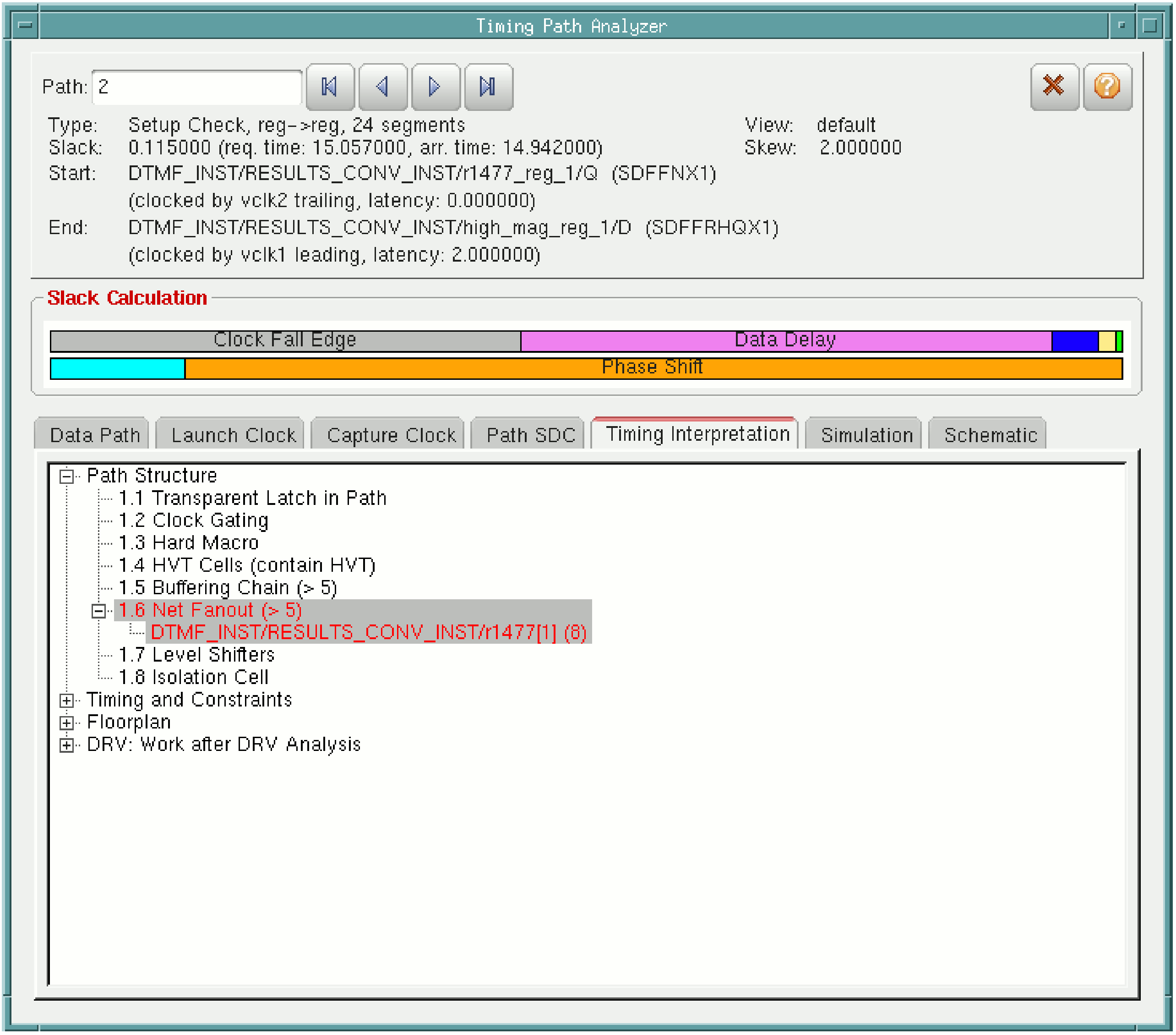
SDC File	Displays the list of SDC constraints associated with the selected path. The list contains the name of the SDC file, the line number that indicates the position of the constraint in the SDC file, and the constraint definition. Use this from to identify issues related to the constraint definition for a path. You can create a path category directly from SDC constraints in the Path SDC form. When you right-click on a constraint, and select *Create Path Category*, the Create Path Category form displays, showing the line number (from the SDC file) and the name of the constraint.
	Closes the form.

Timing Path Analyzer - Timing Interpretation

Use the Timing Path Analyzer - Timing Interpretation form to identify the possible sources of timing problems associated with a selected path. Potential problems are highlighted in red.

Double-click on a path in the *Path List* in the Analysis window.

- Choose the *Timing Interpretation* tab in the Timing Path Analyzer form.



Timing Path Analyzer - Timing Interpretation Fields and Options

Path Structure	Shows the presence of structures that could cause timing issues in the path: ■ *Transparent Latch in Path* ■ *Clock Gating* ■ *Hard Macros* ■ *HVT Cells* ■ *Buffering Chain* ■ *Net Fanout* ■ *Level Shifters* ■ Isolation Cell
Timing and Constraints	
	Shows values that exceed preset limits for timing and constraints: ■ *Large Skew* ■ *Total SI Delay* ■ *SI Delay* ■ *External Delay*
Floorplan	Shows the presence of the following floorplan elements that could be source of problems in the timing path: ■ *Fixed Cells* ■ *Multiple Power Domains* ■ Also shows violations for preset limits for the distances: ■ *Distance from Start to End* ■ *Distance Repeater* ■ *Distance Buffer Tree* ■ *Detour*

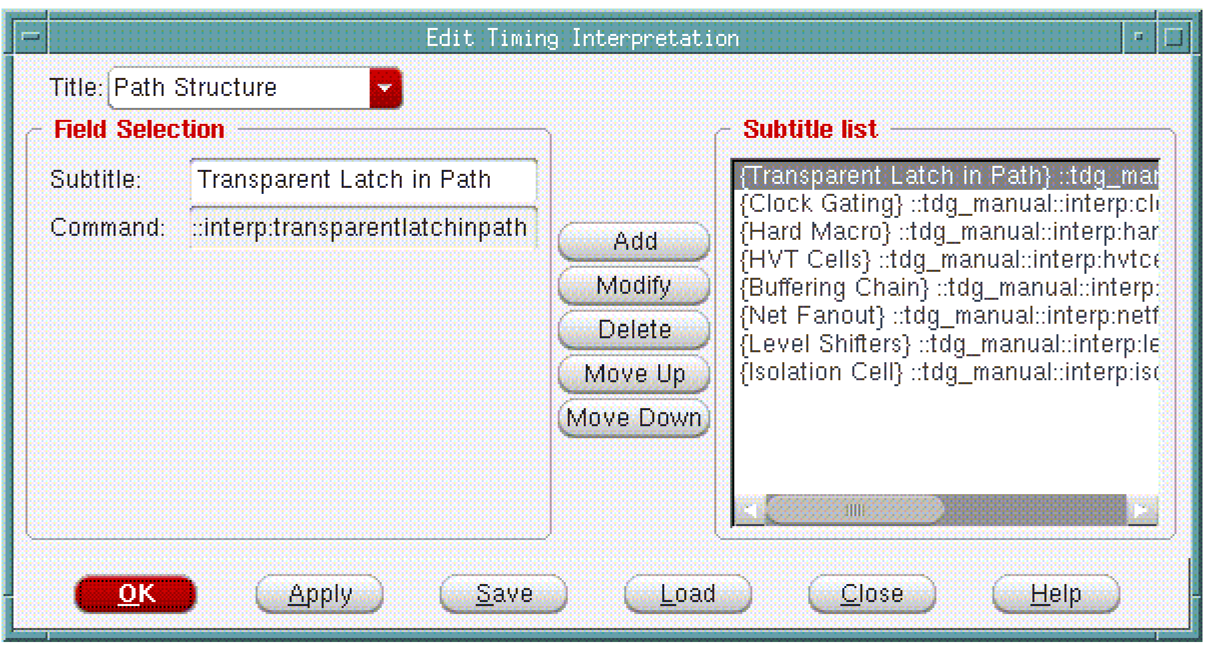
DRVs	Shows max_tran, max_cap, and max_fanout values exceeding the preset limits: ■ *Max Transition* ■ *Max Capacitance* ■ *Max Fanout*

Edit Timing Interpretation

Use the Edit Timing Interpretation form to change the type of information displayed on the Timing Interpretation table of the Timing Path Analyzer.

To open the Edit Timing Interpretation form, open the Timing Path Analyzer form.

- ❑ Click on the *Timing Interpretation* tab.
- ❑ Right-click on any element in the table and select *Edit Timing Interpretation*.



Timing Path Analyzer - Edit Timing Interpretation Fields and Options

Title	Specifies the title of the path element: ■ *Path Structure* ■ *Timing and Constraints* ■ *Floorplan* ■ *DRV*: Work after DRV

Subtitle	■ Specifies the name of the subtitle to modify. *For Path Structure:* ■ *Transparent Latch in Path* ■ *Clock Gating* ■ *Hard Macro* ■ *HVT Cells* ■ *Buffering Chain* ■ *Net Fanout* ■ *Level Shifters* ■ *Isolation Cell* *For Timing and Constraints:* ■ *Large Skew* ■ *Divider in Clock Path* ■ *Total SI Delay* ■ *SI Delay* ■ *External Delay* *For Floorplan:* ■ *Fixed Cells* ■ *Distance from Start to End* ■ *Distance Repeater Chain* ■ *Distance Buffer Tree* ■ *Detour* ■ *Multi Power Domains*
	■ *For DRV: Word after DRV Analysis* ■ *Max Transition* ■ *Max Capacitance* ■ *Max Fanout*

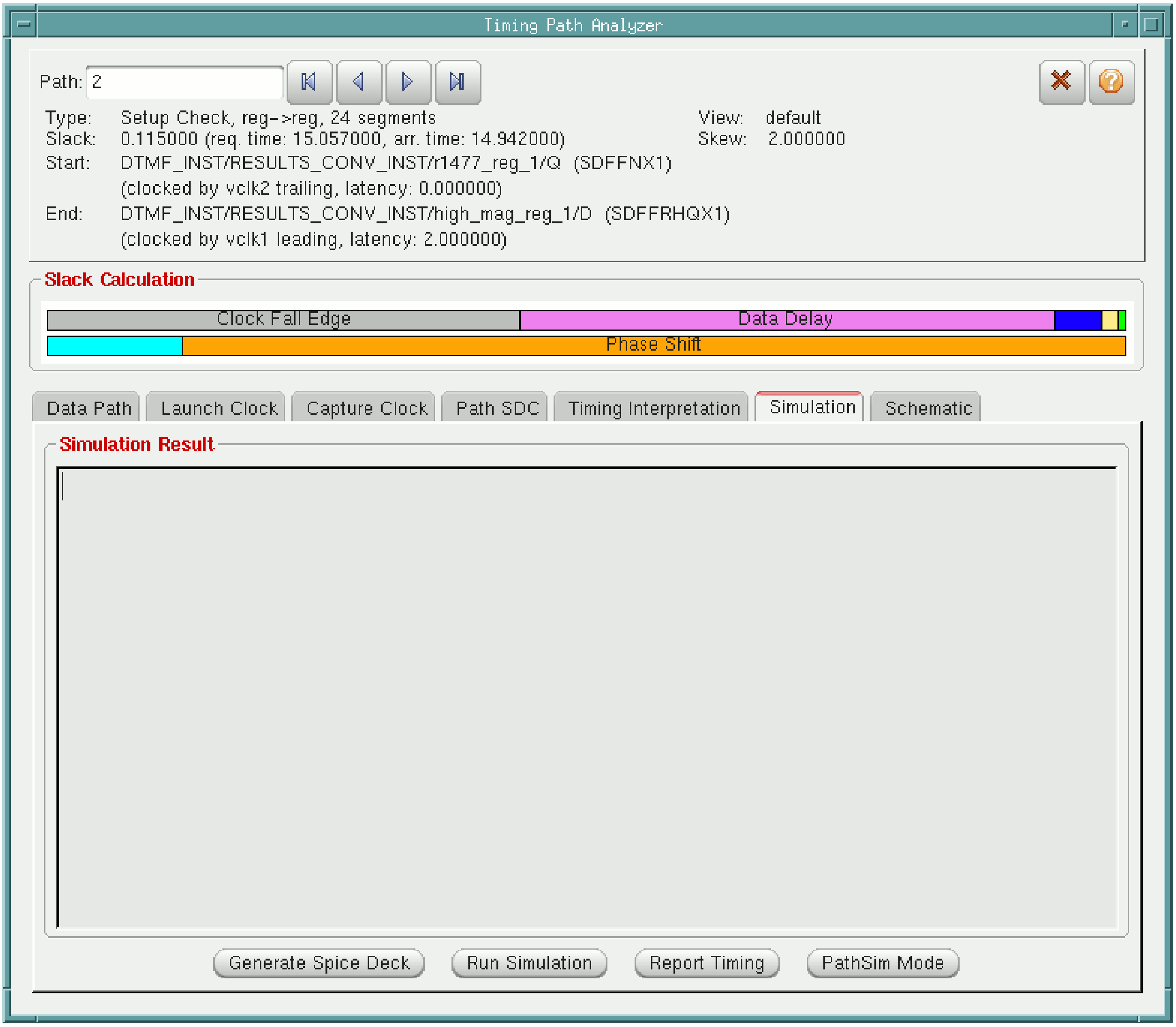
Command	Specifies a procedure to create the subtitle.
Condition	Specifies a condition value.
Subtitle list	Displays a subtitle you selected, or lets you specify the name of the subtitle you would like to add.
Add	Adds the subtitle to the Subtitle list an subtitle that you added or changed in the Field Selection pane.
Modify	Modifies the selected subtitle.
Delete	Deletes the subtitle.
Move Up	Moves the subtitle up the list of subtitles in the Timing Interpretation table.
Move Down	Moves the subtitle down the list of subtitles in the Timing Interpretation table.

Timing Path Analyzer - Simulation

Use the Timing Path Analyzer - Simulation form to identify the possible sources of timing problems associated with a selected path. Potential problems are highlighted in red.

Double-click on a path in the *Path List* in the *Analysis* window.

- Choose the *Simulation* tab in the Timing Path Analyzer form.



Timing Path Analyzer - Simulation Fields and Options

Simulation Result	Displays the path SPICE trace if the Generate Spice Deck option is selected, or displays the timing report if the Report Timing option is selected.

<table>
<tr><td>Generate Spice Deck</td><td>
Generates a path SPICE trace and displays the results in the Simulation Result section.
The output SPICE deck includes active and passive devices, such as field-effect transistors (FETs), capacitors, and resistors. In addition, it includes the initial conditions and voltage sources.
Note: The path SPICE trace can be generated regardless of whether you run signal integrity analysis.
</td></tr>
<tr><td>Run Simulation</td><td>
Performs path simulation and provides a graphical depiction the critical path report in the sWave waveform viewer.
The sWave waveform viewer provides the following features:

Logarithmic scaling of either or both X and Y axes
Origin adjustment for both X and Y axes
Grid adjustments
Marker to denote the point at which the waveform begins to change
Legend for the waveform
Fit to either or both axes
Level 1 Postscript output
View the waveform for selected net(s)
Zoom in or out using a mouse
User-defined ruler measurements using a mouse
Multi-waveform display capability

Note that the SI effect is not considered during path simulation by default. To include the SI effect, select the Crosstalk Effect option in the Set Pathsim Mode form that can be invoked using the Pathsim Mode option.
To run path simulation, you should have the Spectre simulator in your search path. Alternatively, you should specify the path to the Spectre installation using the <code>-spectre</code> parameter of the <code>create_spice_deck</code> command.
</td></tr>
</table>

Report Timing	Generates a timing report that shows the paths for which the SPICE trace was generated. The report appears in the Simulation Result section.
Pathsim Mode	Displays the Set Pathsim Mode form. Use this form to specify path simulation options. For more information, see Set Pathsim Mode form in the “Timing & SI Menu” chapter.

Related Text Commands

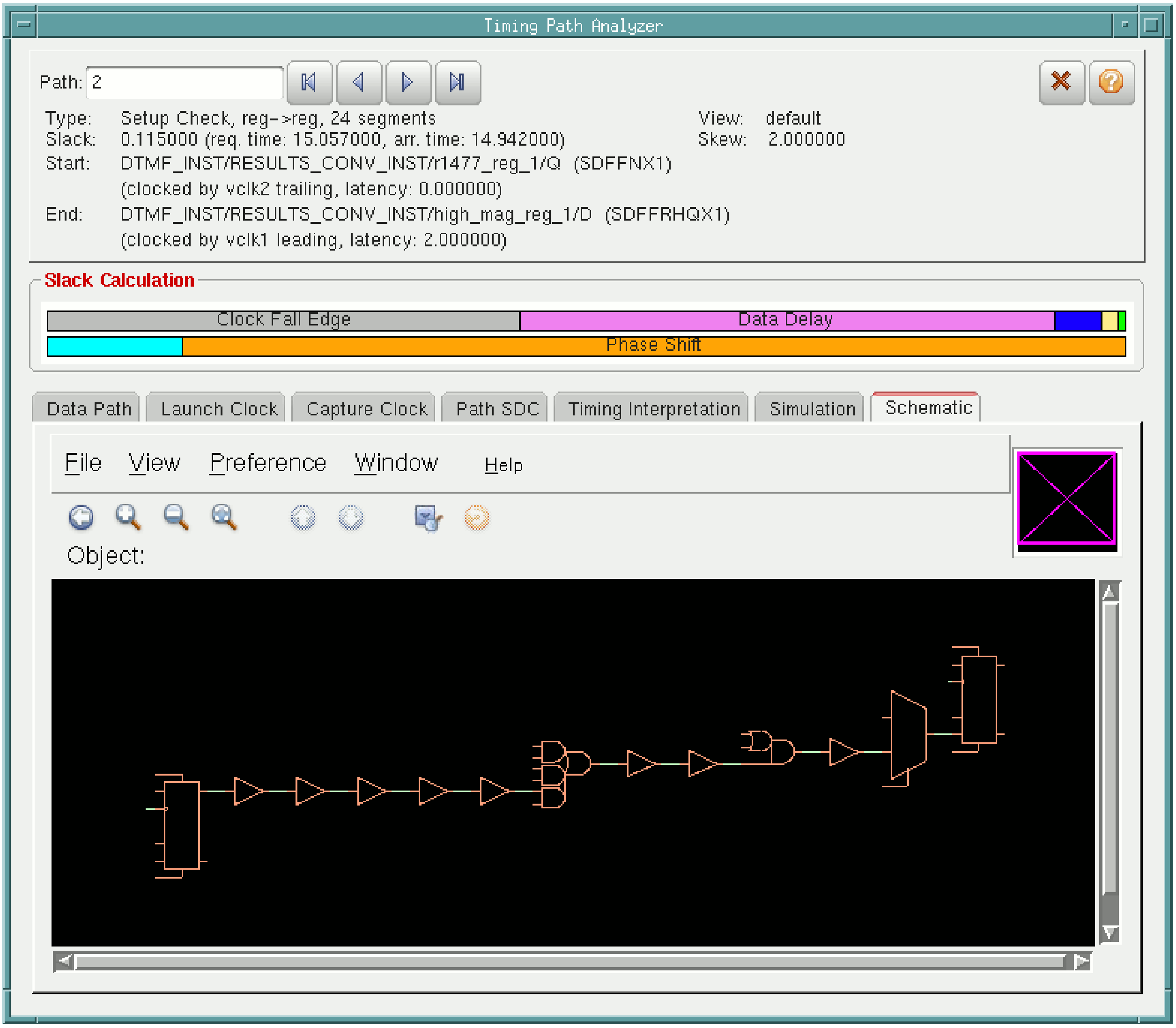
The following text command provides equivalent or additional functionality:

- `create_spice_deck`
- `report_timing`

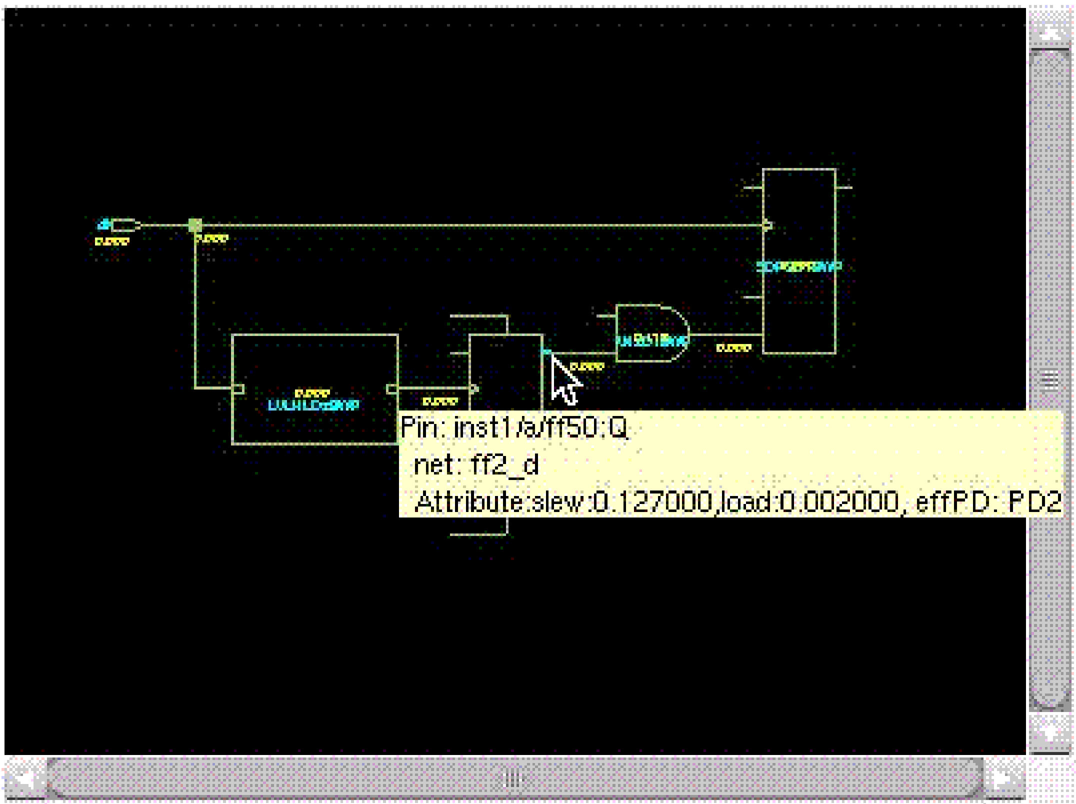
For more information, see "Signal Integrity Commands" and "Timing Analysis Commands" in the *Tempus Text Command Reference*.

Timing Path Analyzer - Schematic

Use the Timing Path Analyzer - Schematics form to display the schematic view of the selected path.



Note: To see timing information (slew and load) of instance output pin move the mouse on the instance's output pin on the Schematic viewer of critical path.



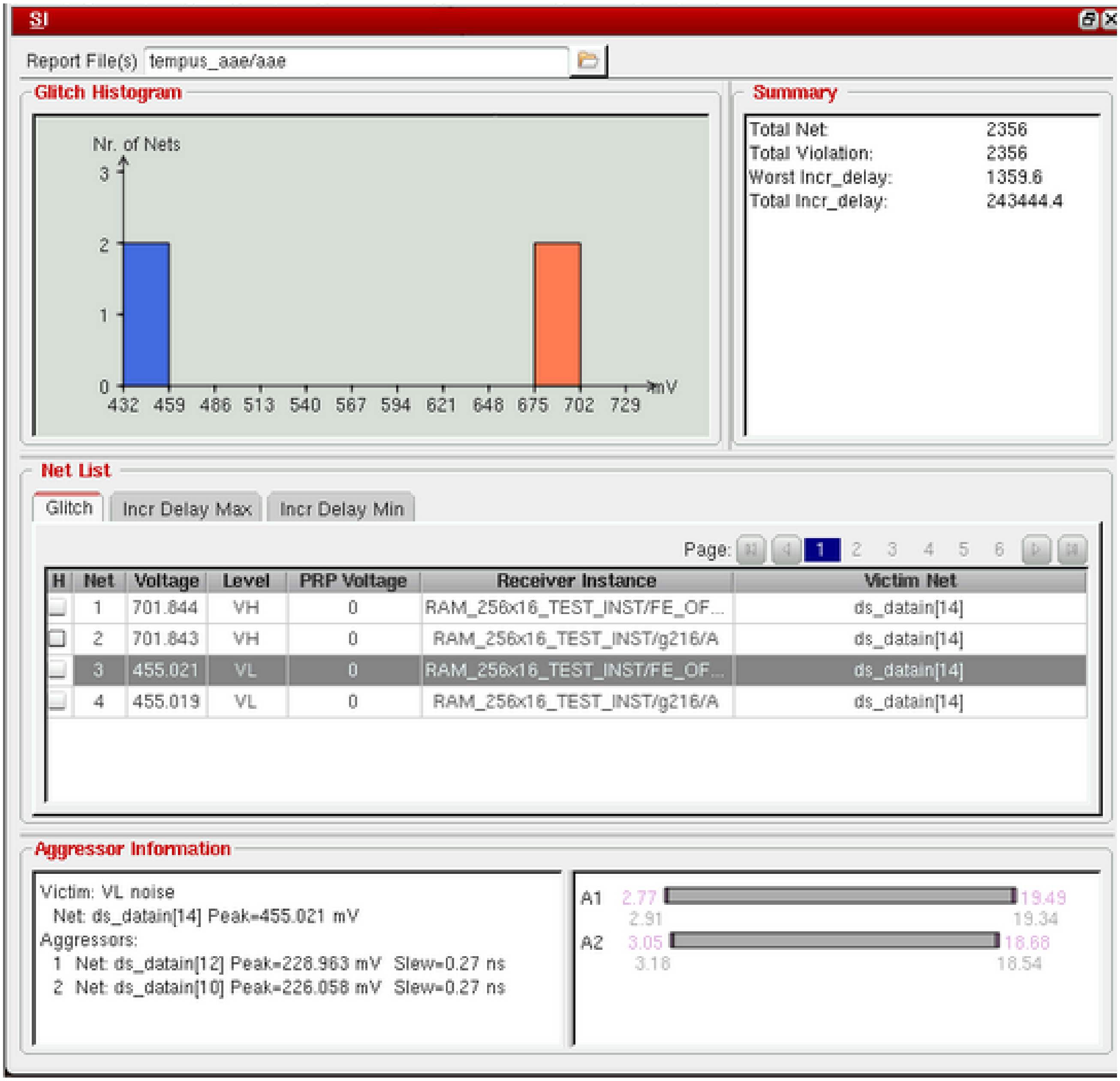
SI Tab

SI Window Components

Use the SI window components to display noise reports after crosstalk analysis.

☨ Click on the SI tab.

The SI tab displays four sections: *Histogram, Summary, Net list,* and *Aggressor Information*.



The SI tab sections are described below:

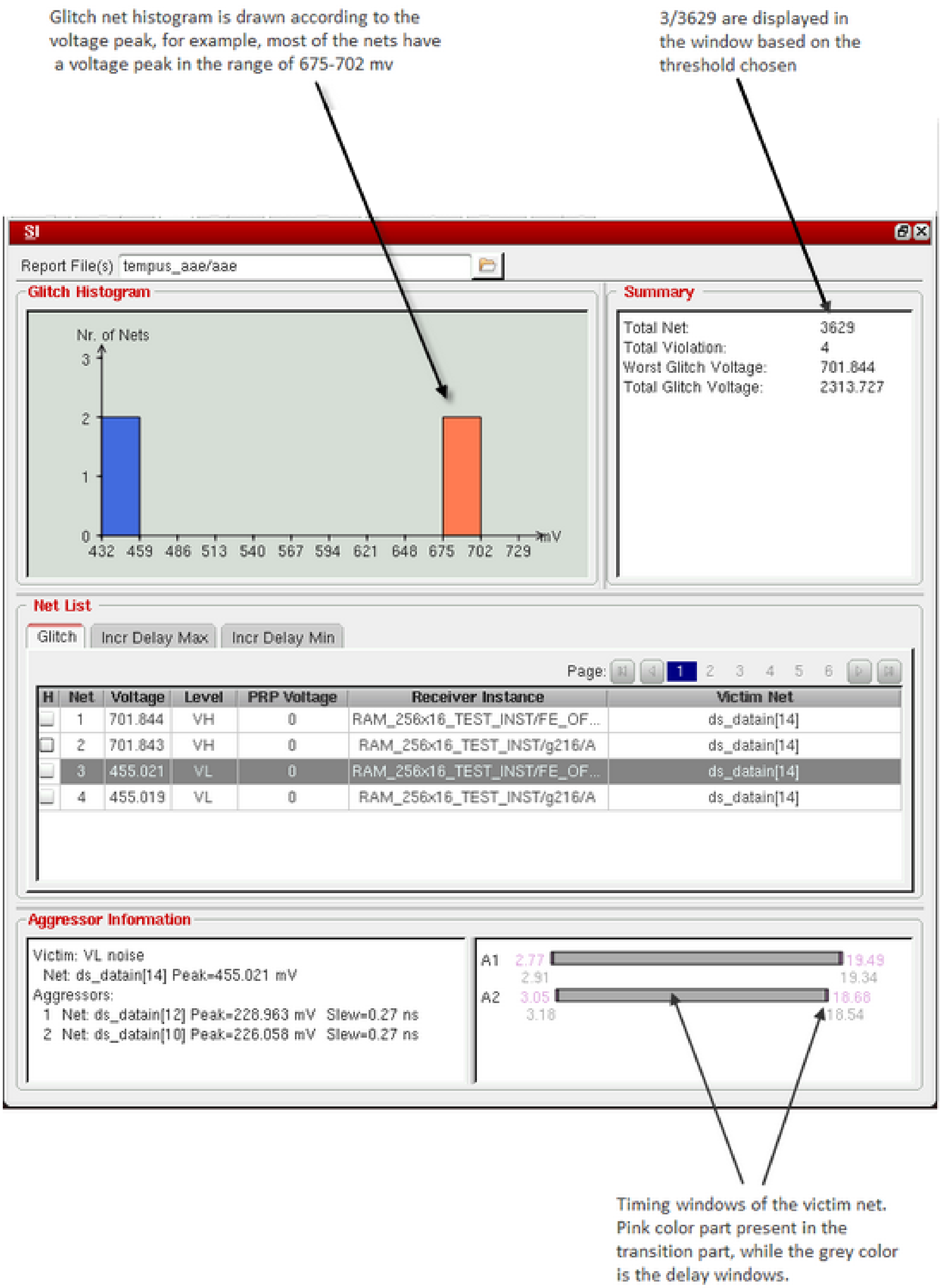
Report File(s)	Specifies the name of the noise report file, which is generated after running crosstalk analysis.

Glitch Histogram	Displays the noise histogram according to the noise type (*glitch* or *incr_delay*). The histogram title is consistent with the noise type. By default, the *Glitch Histogram* is displayed.
Summary	Displays the summary of noise analysis (similar to the information displayed in the Tempus SI log in the `SUMMARY` section). The information includes *Total Net, Total Violation, Worst Glitch Voltage*, *Total Glitch Voltage*, *Worst Incr_delay Voltage*, and *Total Incr_delay Voltage* according to the noise type.
Net List	Includes three tabs: *Glitch*, *Incr Delay Max*, and *Incr Delay Min.* ■ *Glitch*: Represents the glitch noise victims. ■ *Incr Delay Max*: Represents the push out incr delay used for late paths. ■ I*ncr Delay Min*: Represents the push in incr delay used for early paths. By default, the *Glitch tab* is selected if both the *glitch* and *incr_delay* reports are available. As a result, the netlist will be sorted by glitch voltage. If you click the *Incr Delay Max* or *Incr Delay Min tab,* the net list will be sorted by incremental delay.
Aggressor Information	Displays the aggressors information of a victim net. This information includes the victim name, slew, capacitance, peak, XCap, and net name. For example, select a victim net in the *Glitch* tab to display the corresponding aggressor information.

SI Tab - Glitch

Use the Glitch Display tab to display the SI glitch distribution.

In the *SI* tab, click *Glitch*.


Glitch net histogram is drawn according to the voltage peak, for example, most of the nets have a voltage peak in the range of 675-702 mv
3/3629 are displayed in the window based on the threshold chosen
SI
Report File(s) tempus_aae/aae
Glitch Histogram
Nr. of Nets
3
2
1
0
432 459 486 513 540 567 594 621 648 675 702 729
mV
Summary
Total Net: 3629
Total Violation: 4
Worst Glitch Voltage: 701.844
Total Glitch Voltage: 2313.727
Net List
Glitch
Incr Delay Max
Incr Delay Min
Page: 1 2 3 4 5 6
H Net Voltage Level PRP Voltage Receiver Instance Victim Net
1 701.844 VH 0 RAM_256x16_TEST_INST/FE_OF... ds_datain[14]
2 701.843 VH 0 RAM_256x16_TEST_INST/g216/A ds_datain[14]
3 455.021 VL 0 RAM_256x16_TEST_INST/FE_OF... ds_datain[14]
4 455.019 VL 0 RAM_256x16_TEST_INST/g216/A ds_datain[14]
Aggressor Information
Victim: VL noise
Net: ds_datain[14] Peak=455.021 mV
Aggressors:
1 Net: ds_datain[12] Peak=228.963 mV Slew=0.27 ns
2 Net: ds_datain[10] Peak=226.058 mV Slew=0.27 ns
A1 2.77 19.49
2.91 19.34
A2 3.05 18.68
3.18 18.54
Timing windows of the victim net. Pink color part present in the transition part, while the grey color is the delay windows.

Fields and Options

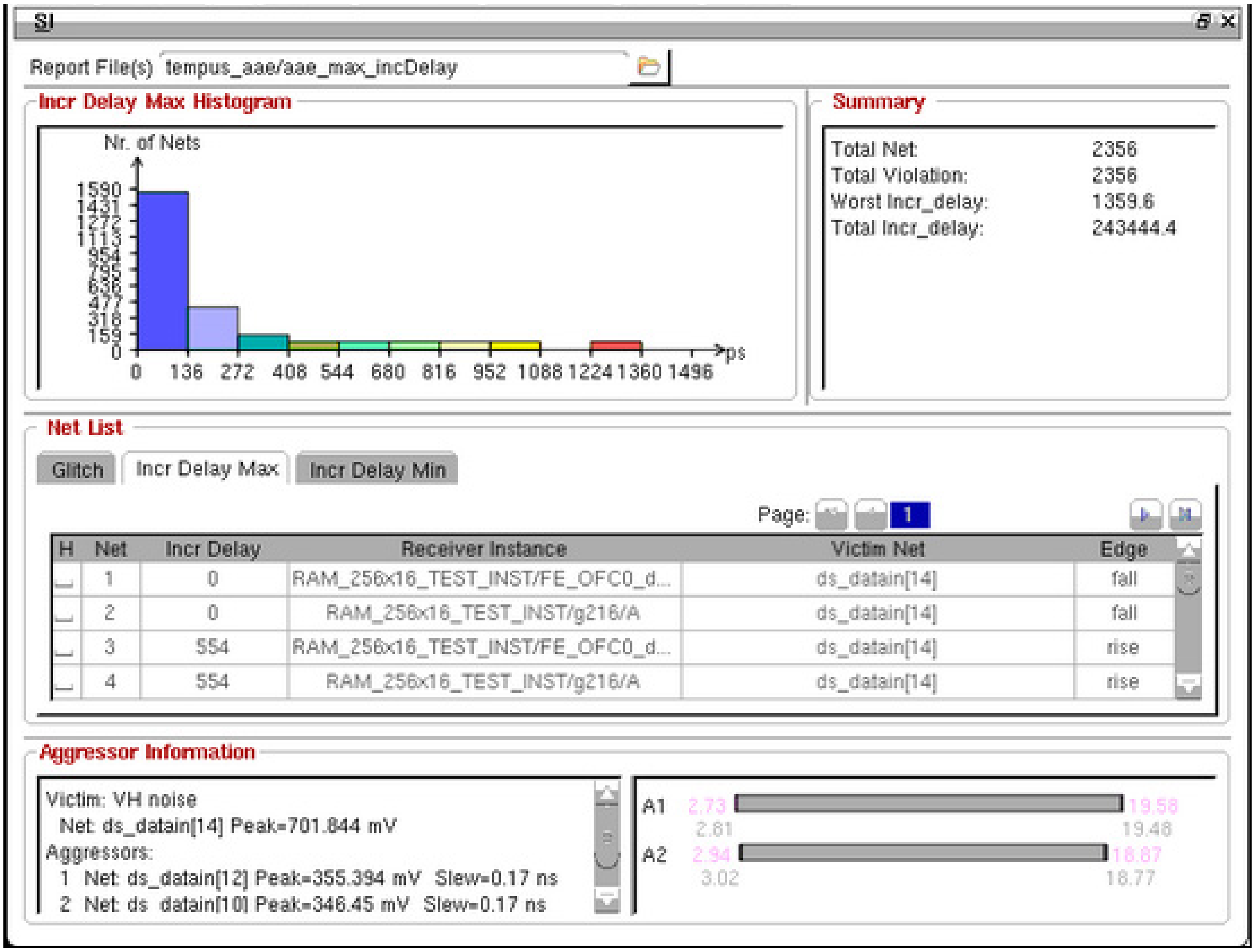
The *Glitch* tab sections are described below:

H	Checkbox to select or deselect a victim net.
Net	Displays the Index.
Voltage	Displays the glitch peak voltage.
Level	Displays the Glitch Voltage Level.
PRP Voltage	Displays the propagated glitch noise voltage.
Receiver Instance	Displays the fanout instance connected to the victim net.
Victim Net	Displays the attacked victim net.
Report File(s)	Specifies the name of the glitch noise report file. By default, `aae` or `aae_<viewname>` will be generated in the reporting directory.
Glitch Histogram	Displays the glitch noise chart, which represents the glitch peak distribution.
Summary	Displays the summary of the *glitch* noise analysis. The information includes: ■ *Total Net:* Is the total number of victim nets. ■ *Total Violation:* Is the total number of victim nets with the glitch peak exceeds the threshold defined during file loading. ■ *Worst Glitch Voltage:* Is the highest glitch peak. ■ *Total Glitch Voltage:* Is the sum of the glitch peak.
Aggressor Information	Displays the slew, capacitance, and timing information for the victim and aggressor nets.

SI Tab - Incremental Delay

Use the *Incr Delay Max* or *Incr Delay Min* tab to display the SI delay distribution.

In the *SI* tab, click *Incr Delay Max*.



Fields and Options

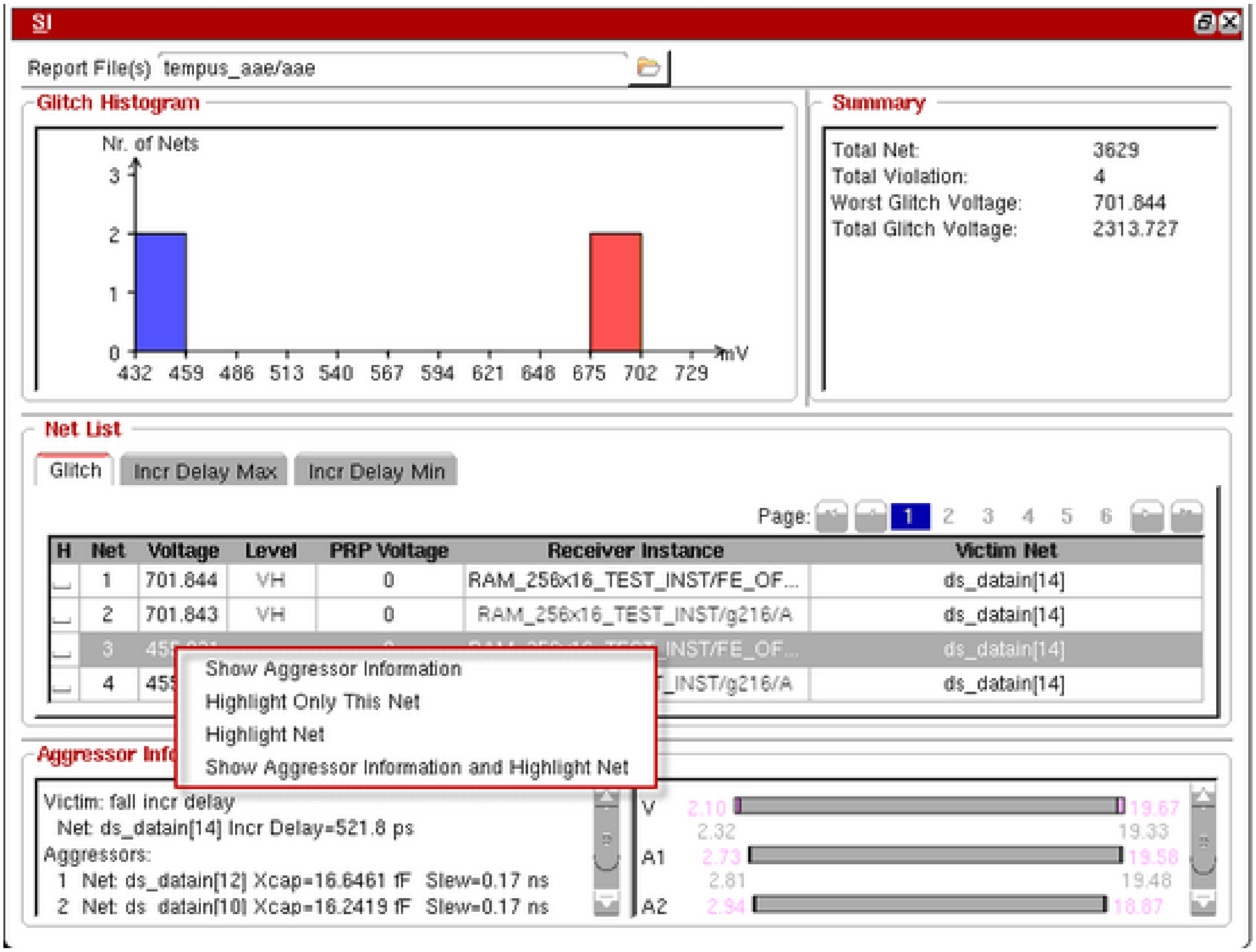
The *Incr Delay Max* tab sections are described below:

H	Checkbox to select or deselect a victim net.
Net	Displays the Index.
Incr Delay	Displays the SI delay, shown in ps.
Receiver Instance	Displays the fanout instance connected to the victim net.
Victim Net	Displays the attacked victim net.

Edge	Displays the SI delay that happens at the rising or fall transition.
Report File(s)	Specifies the name of the incremental delay noise report file. By default, `aae_<max/min>_incDelay` or `aae_<viewname>_<max/min>_incDelay` will be generated in the reporting directory.
Incr Delay Max Histogram	Displays the incremental delay chart, which represents the SI delay distribution.
Summary	Displays the summary of the *incr_delay max* noise analysis. The information includes: ■ *Total Net:* Is the total number of victim nets. ■ *Total Violation:* Is the total number of victim nets with the SI delay that exceeds the threshold defined during file loading. ■ *Worst Incr_delay:* Is the largest SI delay. ■ *Total Incr_delay:* Is the sum of SI delay.
Aggressor Information	Displays the slew, capacitance, and timing information for the victim and aggressor nets.

Shortcut Menu for Net List Section

In the Net List section of the SI tab, right-click on any of the displayed victim net that displays the shortcut menu as shown below.



The following options are displayed:

- *Show Aggressor Information*: Displays the detailed aggressor information of the selected victim net in the *Aggressor Information* section.
- *Highlight Only This Net*: Highlights the net, which gets selected in the "H" column.
- *Highlight Net*: Highlights a set of nets that gets selected in the "H" column.
- *Show Aggressor Information and Highlight Net*: Highlights the victim net and displays the detailed aggressor information of the selected victim net in the *Aggressor Information* section.

Schematic Tab

Schematic Tab Components

Use the *Schematic* tab to explore a design's connectivity, the relationship of modules and instances, and design changes, such as CTS, OPT, and ECO. You can cross-probe from the *Schematic Viewer* to the Layout view to give you flexibility and immediate feedback to debug and diagnose your design.

You can open the Schematic Viewer in the following ways:

From the Main Tempus window:

- Choose the *Schematic* tab. This opens Schematic Viewer in Hierarchical view.

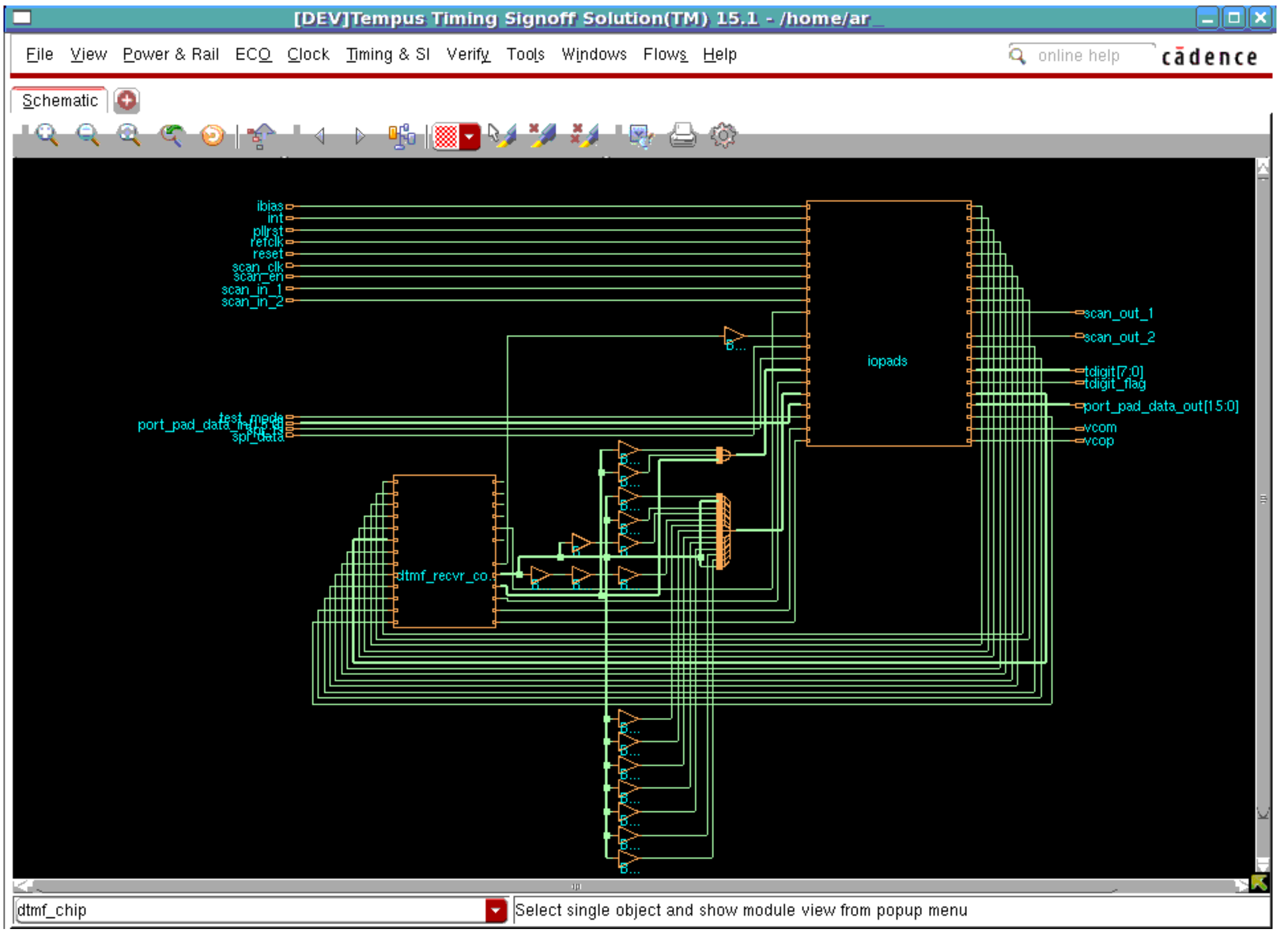
From the *Design Browser*, do one of the following:

- Select a module and click the *Show Hierarchical Schematic* widget. This opens the Schematic Viewer in Hierarchical view.
- Select an instance, pin, or net and click the *Show Instance Based Schematic* widget. This opens the Schematic Viewer in Flattened view.

From the *Violation Browser*:

- Select an MSV violation and click the *Show MSV Violation Schematic* widget.
- From the Timing Debug environment (*Analysis* Tab), do one of the following:

- Double-click to select one of the paths on the timing critical path list to open the Timing Analyzer form and select the *Schematic* tab to change the view to schematic.
- Select one of the paths on the timing critical path list and click the *Schematic* icon at upper right corner of the path list.



Toolbar

Toolbars provide a quick way to access the most commonly used commands. This section describes the toolbar items, which are available for flattened and module schematics, except where noted.

Icon	Action	Description
	Zoom In	Click this widget to display a smaller area of the schematic in greater detail. Each click zooms in one level. The equivalent bindkey is `Z` (`Shift-Z`).

	Zoom Out	Click this widget to display a larger area of the schematic in less detail. Each click zooms out one level. The equivalent bindkey is `z`.
	Zoom to Full Window	Click this widget to display the entire schematic within the display area. The equivalent bindkey is `f`.
	Last View (W)	Click this widget to toggle the display between the previous location or zoom level and the current location or zoom level.
	Reload view	Refreshes the schematic display.
	Find	Click this widget to open the Find Schematic Object form to find and highlight a specific object in the schematic display area.
	Push View Up	Displays the parent-level hierarchy. To push the view up to a higher-level schematic, select an object in the schematic display area and click this widget.
	Trace Driver	Traces all possible drivers of the selected object and highlights them in the schematic display area. The default highlight color for drivers is yellow.
	Trace Load	Traces all possible loads of the seobject and highlights them in the schematic display area. The default highlight color for loads is red.
	Trace Connectivity	Traces all possible drivers and loads of the selected object and highlights connectivity from the object's drivers to its loads. The default highlight colors for drivers and loads are yellow and red, respectively.
	Highlight Selected	Highlights selected object in the schematic display area.
	Dehighlight	Dehighlights an object in the schematic display area.
	Dehighlight All	Dehighlights all objects in the schematic display area.
	Opens the *Find Schematic Object* box	
	Print	Opens the Print Schematic form, which enables you to print the required schematic display.
	Preferences	Opens the Preferences - Display form, which enables you to customize the schematic display settings, color, and font.

Schematic Display Area

The schematic display area constitutes the major portion of this tab. The schematics of the selected object is displayed here in flattened or hierarchical view. To change the default display properties for objects in the display area, use the Color Settings form. You can also customize the display using the quick-access panes to the right of the display area - *View Option* and *Color Settings*.

Context Menu

To access the context menu, right-click an object in the Schematic Display Area The options available in context menu varies depending on the object you have selected in the schematic display area.

The following options are available in the context menu from the Schematic tab, which shows the Hierarchical view of schematics:

Options	**Description**
Attribute Viewer	Opens the Attribute Viewer form from which you can select a different color for the net segment. This option is available only if you right-click a net in the display area.
Design Browser	Displays next-level views for instances. This option is available only if you right-click an instance in the display area. For more details, refer to *Design Browser*.
Highlight	Closes a next-level instance view. This option is available only if you right-click an instance in the display area.
DeHighlight	Traces the previous driver of the selected port or instance pin. This option is available only if you right-click a port or instance pin in the display area.
Copy Name	Traces the next load of the selected port or instance pin. This option is available only if you right-click a port or instance pin in the display area.
Open In	Provides the following options to open the schematic: ■ *Schematic Viewer (new)*: Opens the schematic in a new window. ■ *Schematic Viewer (cone)* ■ *Schematic Viewer (cone append)*

Open Next Level Instance View	Shows the schematic from a different view.
Close Next Level Instance View	Reverts to the previous view.
Show Critical Path	Displays the critical path.
Change Color	Opens the form where you can change the color of the selected object.
Default Color	Resets the selected object type to the default color.
Other Options	
Copy Name	Copies the object name.
Schematic Fan Out	
Trace Previous Driver	Traces the previous driver of the selected port or instance pin. This option is available only if you right-click a port or instance pin in the display area.
Trace Next Load	Traces the next load of the selected port or instance pin. This option is available only if you right-click a port or instance pin in the display area.
Port to Port Trace	Traces all connections to and frTiming Debug Preferences - Select Color form a port and highlights them in the color you select in the Select Color form. This option is available only if you right-click a port in the display area.

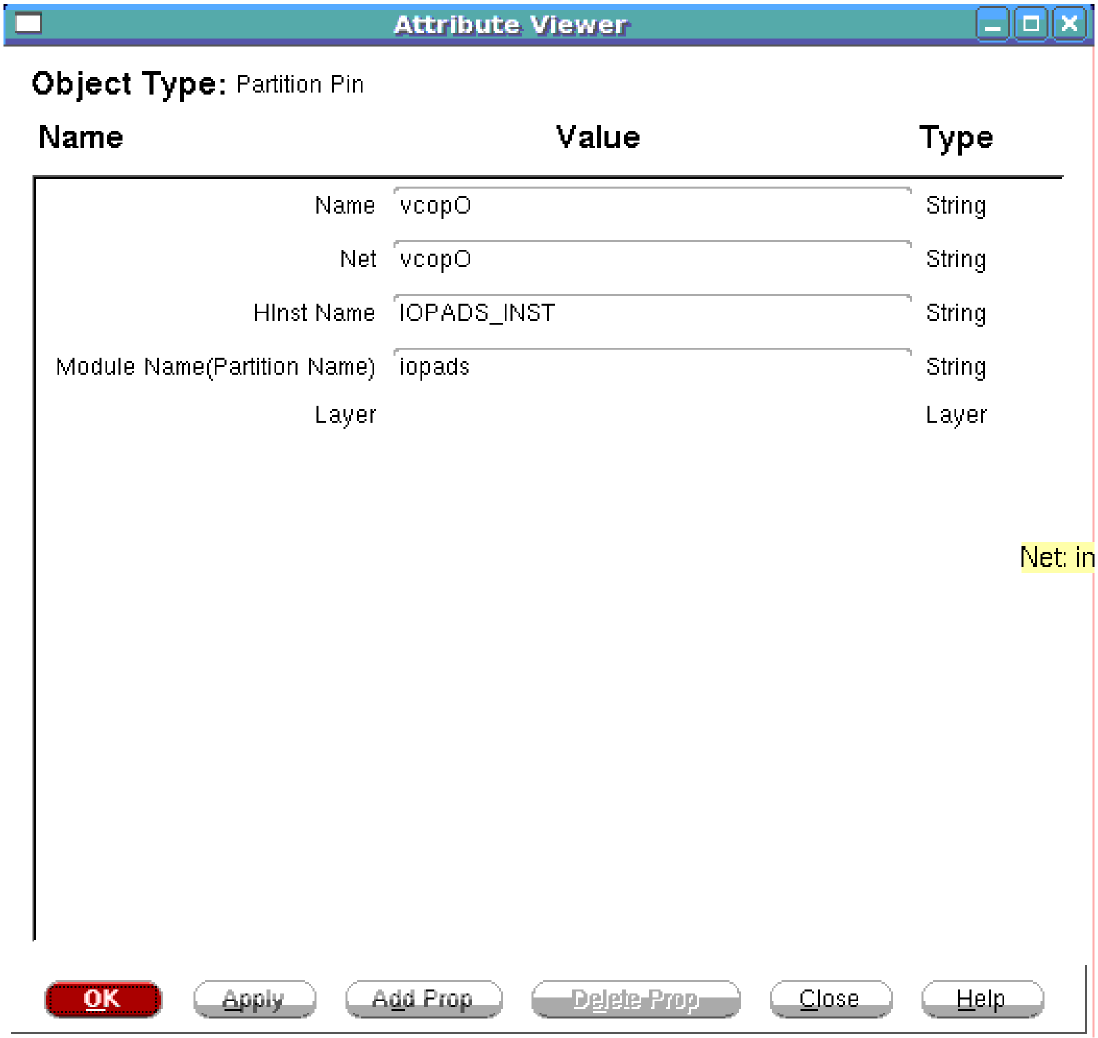
Note: The context menu provides different options in the Flattened view of schematics. If you want to view schematics in the Flattened view, select an instance, pin, or net in Design Browser and click the *Show Instance Based Schematic* widget. For more information, see Flattened View Context Menu on page 397.

Attribute Viewer

Use the Attribute Viewer to display an object's type, name, and attributes. The contents of the Attribute Viewer form change depending on the object selected in the design.

To open the Attribute Viewer from the schematic window, do the following:

For a single object, right click on an object. The following window opens.



Options	Description
Name	Specifies the object name.
Net	Specifies the net name.
Hinst Name	Specifies the name of the hierarchical instance name.
Module Name (Partition Name)	Specifies the module name.
Layer	Specifies the name of the layer.

Color Settings

The Color Settings pane to the right of the Schematic Display Area enables you to change colors of object types quickly.

To change the color of an object type:

- Click the color box next to the required object type on the *Color Settings* pane.
- In the Select Color form, select a new color for the object type. The color of the selected object type changes to the new color in the schematic display area.

Global View Window

The global view window identifies the location of the current view in the schematic display area, relative to the entire area. The global view is identified by the pink crossbox. When you display an entire schematic in the area, the crossbox encompasses the global view window. When you zoom and pan through the schematic in the display area, the crossbox identifies where you are relative to the entire schematic.

To define a new global view, left-click and drag the mouse to create a new crossbox in the global view window.

To pan left, right, up, or down in the schematic display, right-click and drag the crossbox in the global view window.

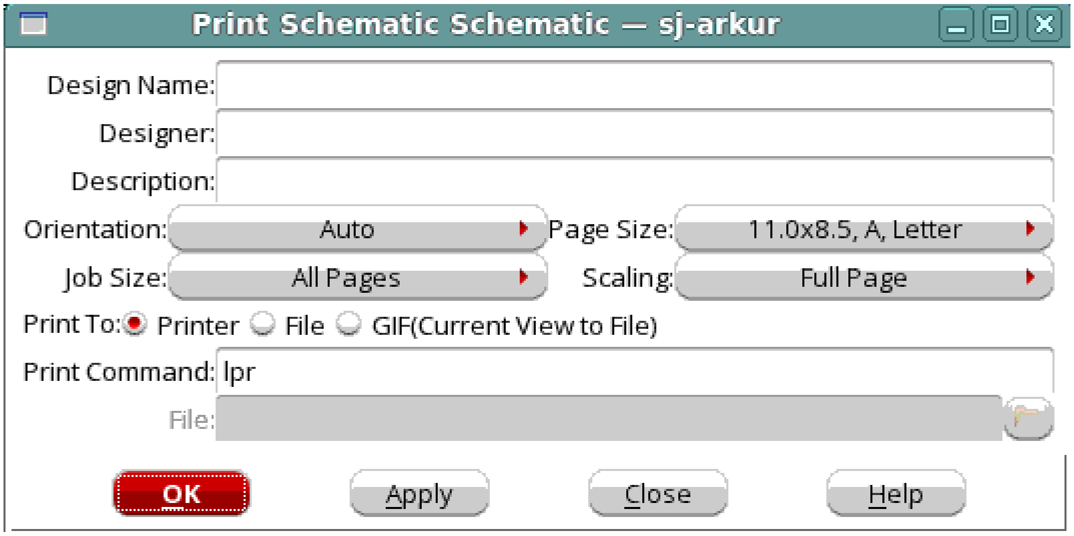
Status Bar

The *Status Bar* is located at the bottom of the Schematic tab. When you select or hover over an object, this bar displays the object type, name, and ID number. If no object is selected or highlighted, the status bar displays the current view name - Hierarchical or Flattened.

Print Schematic

Use the Print Display Area form to print the schematic in the display area.

- Choose *File - Print*.



Print Schematic Fields and Options

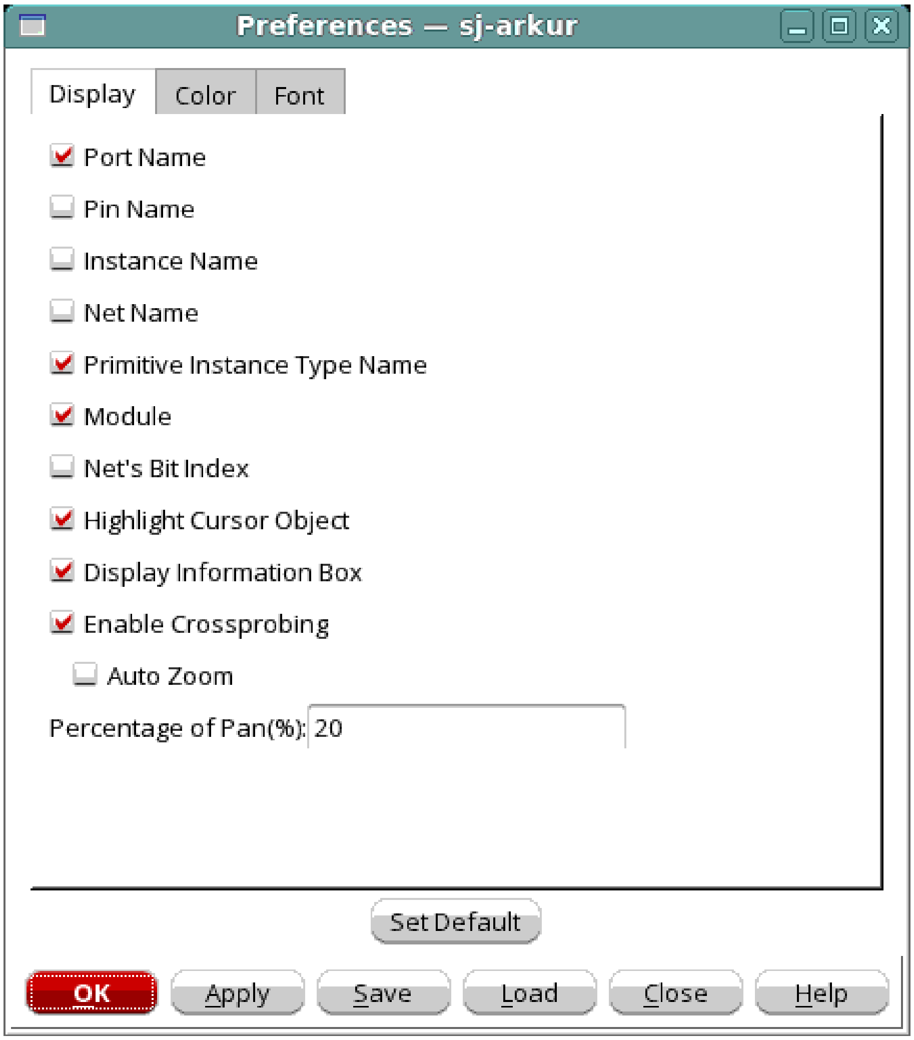
Design Name	Specifies the name of the design. This information appears in the bottom left corner of the printout.
Designer	Specifies the designer name. This information appears in the bottom left corner of the printout.
Description	Specifies a description. This information appears in the bottom left corner of the printout.
Orientation	Specifies the page orientation. Choose *Portrait* or *Landscape*.
Page Size	Specifies the page size. Choose a standard page size of *8.5X11* or use the pull-down menu to specify a custom size.
Job Size	Specifies the job size. Choose to print all the pages or only the current page.
Scaling	Specifies the scaling. Choose *Full Page* or *Current View*.

Print To	Specifies the print output destination. Choose one of the following: *Printer* - If you choose this option, type the appropriate print command in the *Print Command* text box. *File* - If you choose this option, the *File* text box at the bottom is enabled. Type a filename in the *File* text box or use the adjacent folder button to select the appropriate file. *GIF (Current View to File)* - Choose this option to print the current view to the specified file.

Preferences

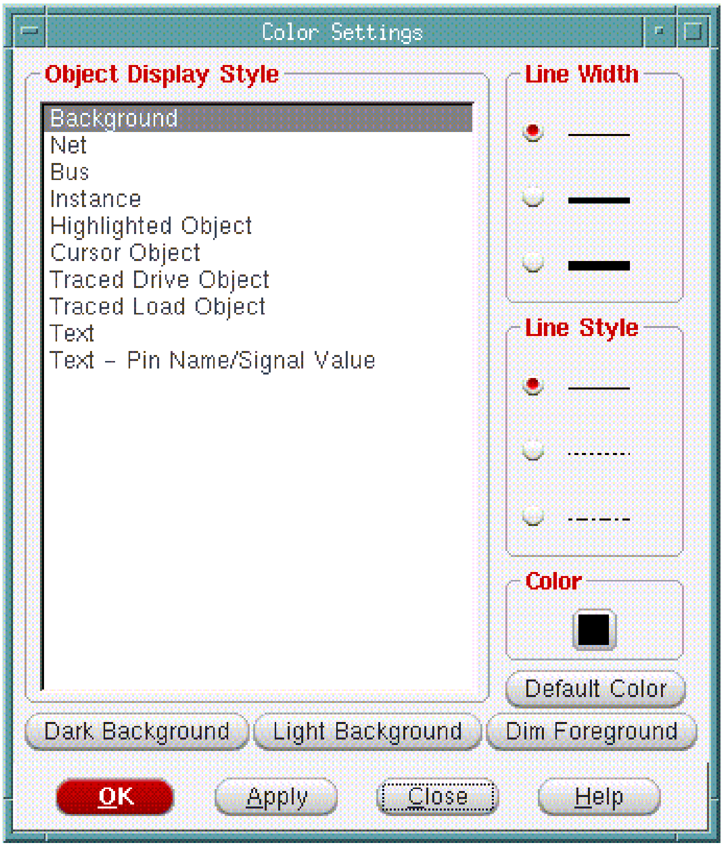
Use the *Preferences* form to change the properties, color, or font of an object in the schematic display.

To open the Preferences form, right-click on an object and click Preferences sub- menu option:



Color Settings

To change color settings, click the *Color* tab in the Preferences screen above. The following window opens.



Color Settings Fields and Options

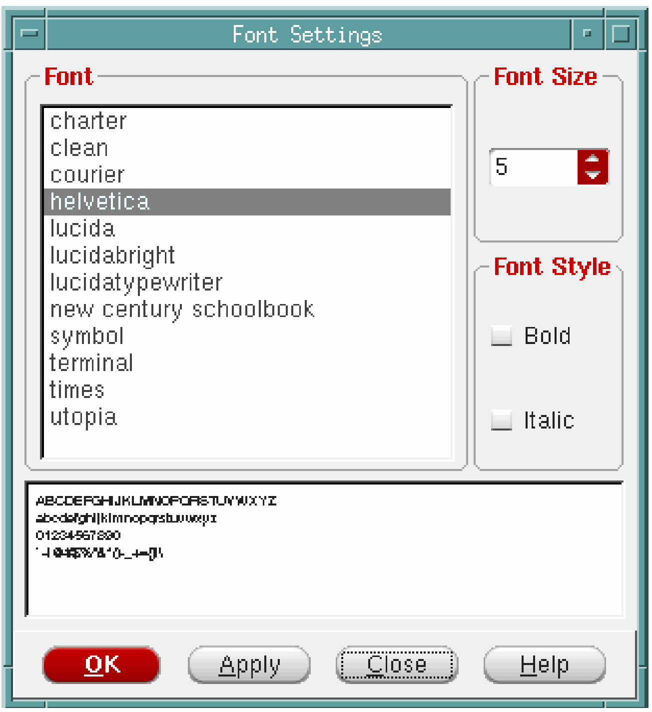
Object Display Style	Specifies the object type for which you want to change color and line settings. Select any object type from this list and choose appropriate color and line settings, as required.
Line Width	Specifies the line width of the object type selected in *Object Display Style* list box.
Line Style	Specifies the line style of the object type selected in *Object Display Style* list box.
Color	Specifies the color of the object selected in *Object Display Style* list box. Click on the color box to open the Timing Debug Preferences - Color - Select Color form and choose a new color for the selected object type.

Default Color	Resets the selected object type to the default color.
Dark Background | *Light Background* | *Dim Background*	
	Specifies the background scheme for the Schematic Display Area. The default is *Dark Background*.

Font Settings

Use the Font Settings form to change the font size and style settings for the schematic display.

To change font settings, click the *Font* tab in the Preferences screen above. The following window opens.



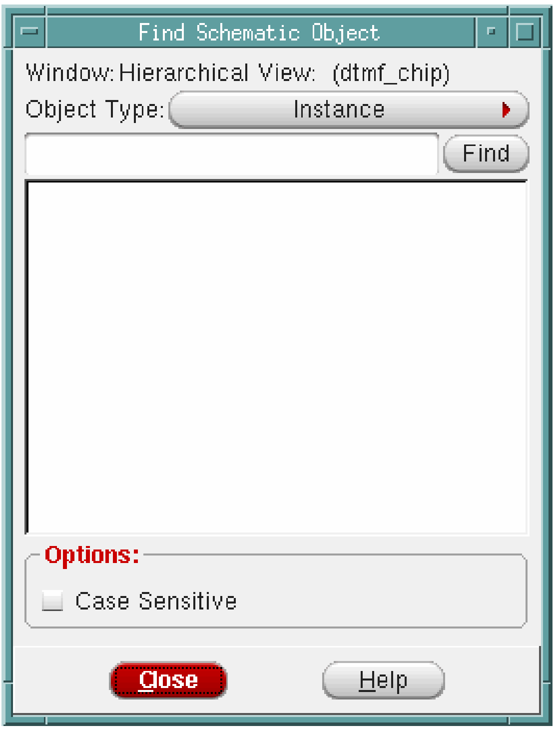
Font Settings Fields and Options

Font	Specifies the font for the text displayed in the schematic display area.
Font Size	Specifies the size of the schematic display font.
Font Style	Specifies the style to be applied to the schematic display font.

Find Schematic Object

Use the Find Schematic Object form to find and highlight a specified object in the schematic display area.

- Click the *Find* widget on the toolbar.



Find Schematic Object Fields and Options

Window	Identifies which schematic window was active when you selected this option.
Object Type	Specifies the object type you want to find.
Find	Specifies the object name or a portion of the name with wildcard characters. Type the object name in the adjacent text box and click *Find*. The search results are displayed in the list box below this field. Double-click a name to highlight the object in the schematic display area.
Case Sensitive	Allows you to make the search case sensitive.

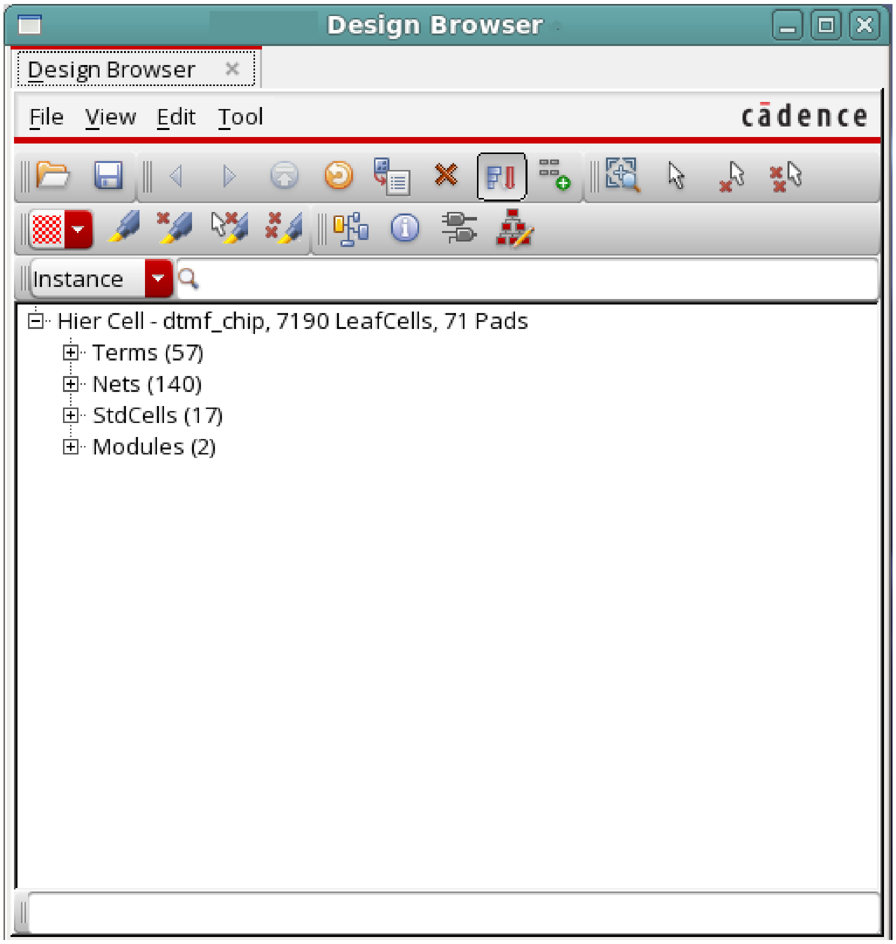
Design Browser

Fields and Options

Design Browser Components

Use the Design Browser to navigate through the chip's design hierarchy. You can use the Design Browser to view the design hierarchy tree at any time after the design is imported. You can use the widgets in the Design Browser to open forms to navigate through displays, and perform actions.

- Choose *Tools - Design Browser*.



Fields and Options

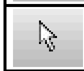
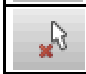
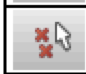
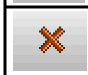
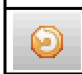
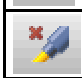
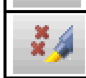
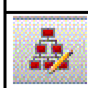
File	Provides the following options:
Load	Displays the Load Design Browser File form. Enter the filename and the format of the placement file.
Save As	Displays the Save Design Browser File form. Specify the filename. The recommended extension is .dbf.
Quit	Closes the Design Browser Window.
View	Provides the following options:
Refresh	Redraws the Design Browser display.
Zoom Selected	Zooms in on the highlighted selection.

Sort By Size	Displays instances in the order of the number of cells in the instances. Deselect the *View - Sort By Size* option to display instances sorted alphabetically by name.
Accumulate Results	Traces the hierarchical path for the selected object in the design display area.
Edit	Provides the following options:
Delete Group/Group Member	
	Deletes an individual instance from a user-defined group, or to delete an entire user-defined group.
Tool	Provides the following options:
Get Selected	Displays a selected object in the *Design Browser*.
Attribute Viewer	Opens the Attribute Viewer. For more information, see Attribute Viewer.
Connectivity Browser	Opens the Connectivity Browser. For more information, see Connectivity Browser.
Find	Enables you to search and display hierarchy information for a specific object. You can view information on the following object types:
Instance	Enables you to search for an instance and display its hierarchy in the Design Browser window. You can traverse to the next hierarchy by double-clicking on the instance name. Along with the instance name, the instance's cell type and number of cells are displayed.
Net	Enables you to search for a net and display its hierarchy in the Design Browser window. Double-clicking the net name displays the net connection.
Group	Enables you to search for an instance group and display its hierarchy in the Design Browser window.
Cell	Enables you to search for a cell and display its hierarchy in the Design Browser window.

Design Browser Tool Widgets

You can use the widgets in the Design Browser to navigate through displays, open tools, make selections, and perform actions.

Widget	Description

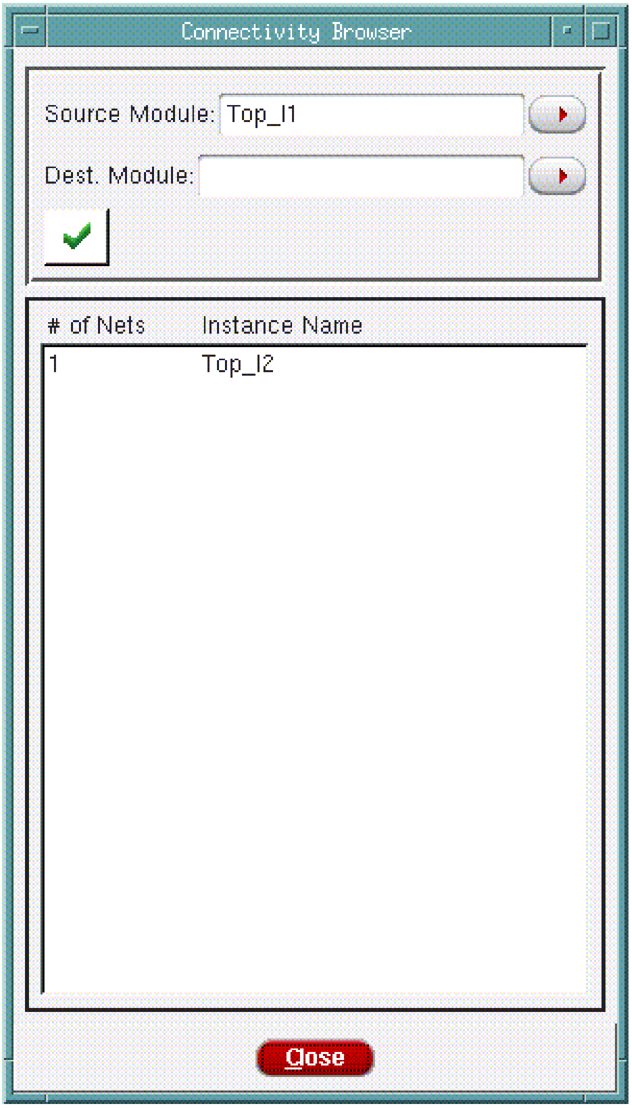
	*Previous*--Displays viewable data in a reverse direction.
	*Next*--Displays viewable data in a forward direction.
	*Top Page*--Returns to the top of the design.
	*Connectivity Browser*--Opens the Connectivity Browser.
	*Select*--Selects an object in the design display area.
	*Deselect*--Deselects an object in the design display area.
	*Deselect All*--Deselects all objects in the design display area.
	*Get Selected*--Gets the selected object and displays it in the Design Browser.
	*Attribute Viewer*--Opens the Attribute Viewer for the highlighted instances.
	*Delete Group/Group Member*--Deletes selected floorplan object instance groups.
	*Refresh*--Refreshes the Design Browser display.
	*Zoom Selected*--Zooms in on a highlighted selection.
	*Highlight*--Highlights an object in the design display area.
	*Dehighlight*--Dehighlights an object in the design display area.
	*Dehighlight All*--Dehighlights all objects in the design display area.
	*Show Instance Based Schematic*--Opens the Schematic Viewer to display schematic in flattened mode for an instance, net, or a pin. For more information, see Schematic Tab.
	*Show Hierarchical Schematic*--Opens the Schematic Viewer to display schematic in hierarchical mode for a module, instance, net, or a pin. For more information, see Schematic Tab.

Connectivity Browser

Use the *Connectivity Browser* to display connections from a source module to a destination module, including module guides, blocks, I/O blocks, and standard cells.

- Choose *Tools - Design Browser*.

Highlight a module in the Design Browser and click the *Connectivity Browser* widget.



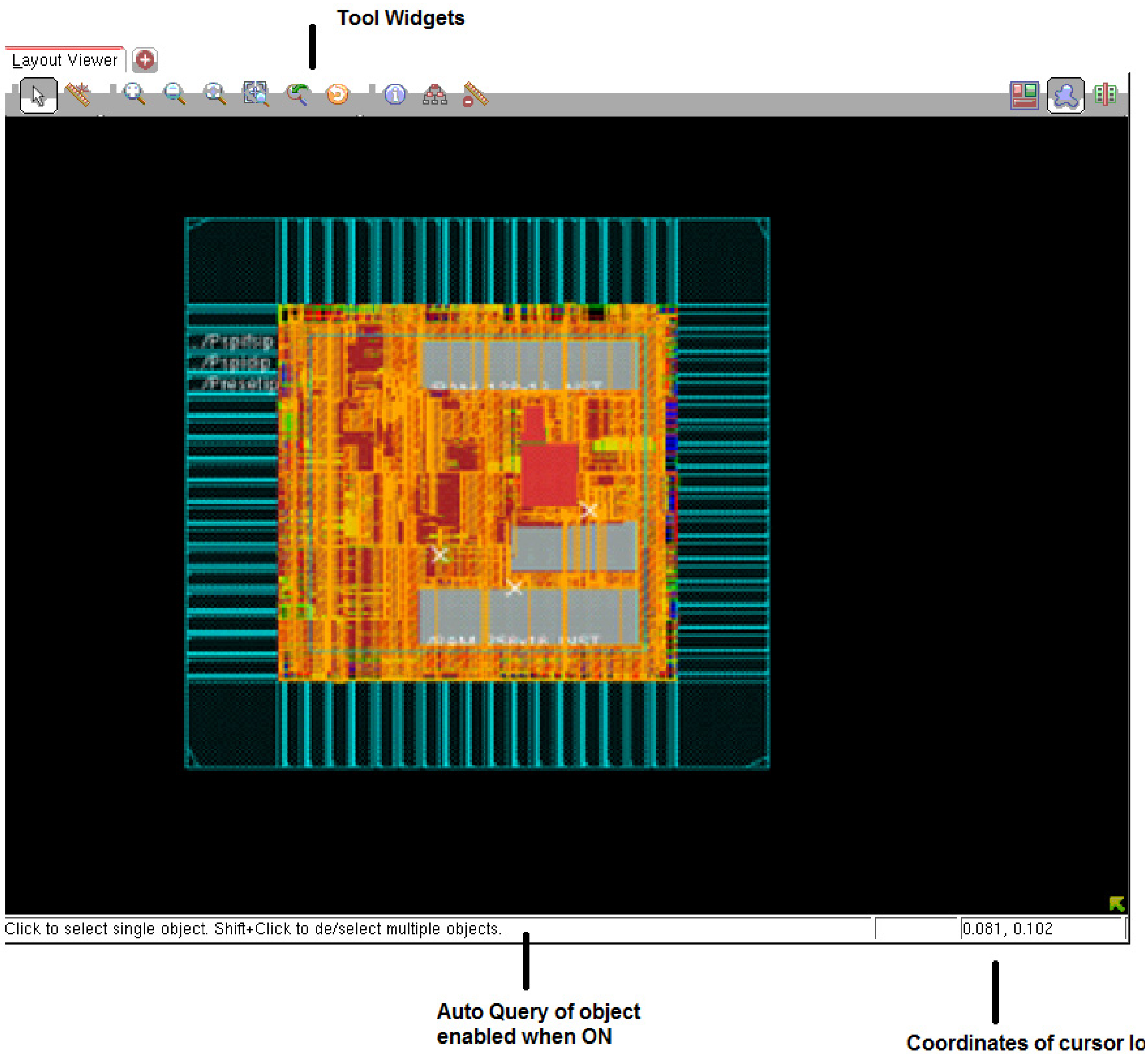
Fields and Options

Source Module Instance	Specify the name of the instance.
Destination Module	Use this pull-down menu to select the destination module.
Check	Displays the connection information.

Layout Viewer

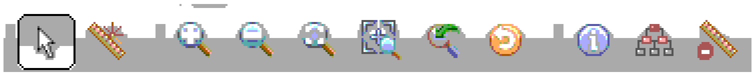
Layout Viewer Components

Use the Layout window to display the physical view of the design in design display area. The components in the Layout window are identified in the following figure, and described in the sections that follow.

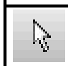
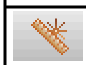
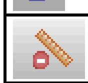


Toolbar Widgets

The following row of widgets, located below the menus and above the design display area, includes shortcuts for some commonly used Innovus commands.



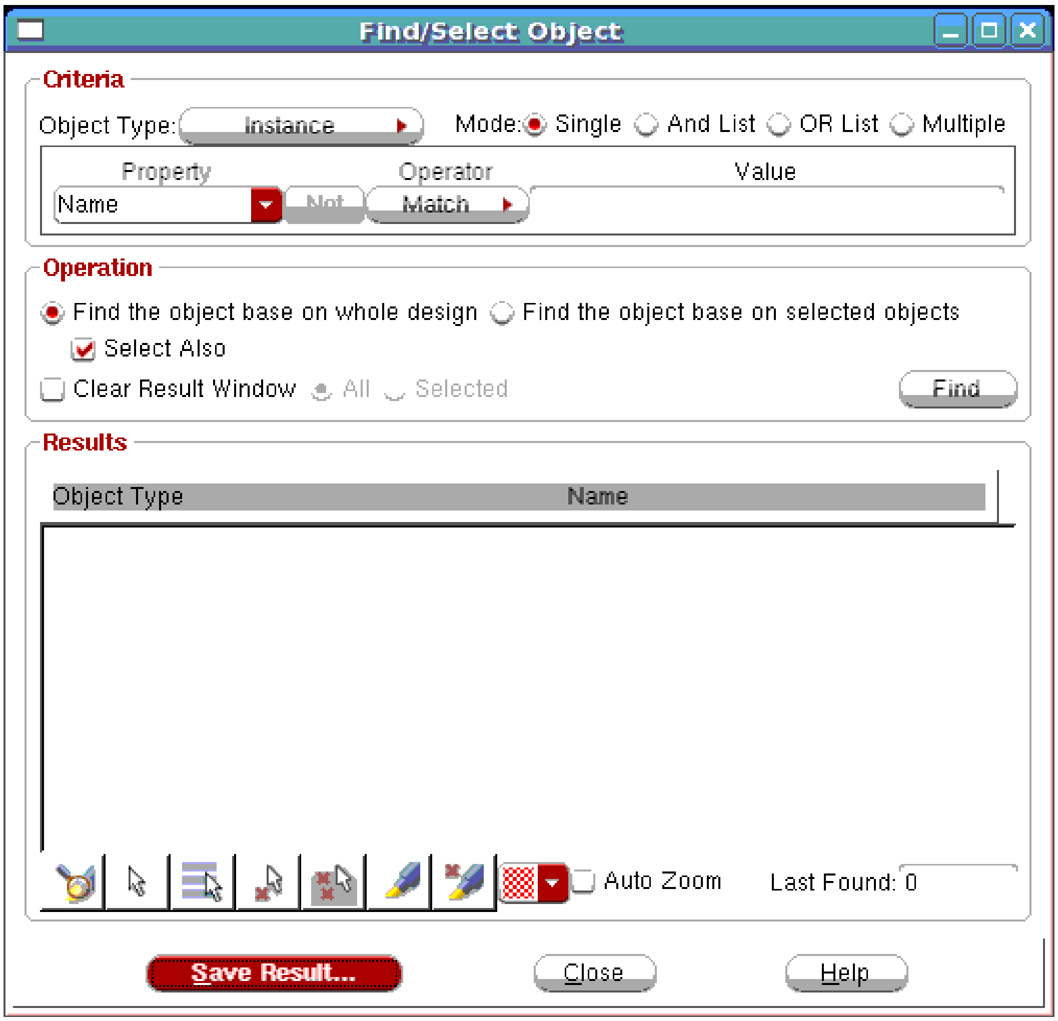
The following table describes the widgets that are located in the Toolbar Widget area:

Widget	Description
	*Select*--Click this widget, then click a floorplan object to select it. To select multiple objects, use the *Shift* key or click and drag the mouse.
	*Ruler*--Click this widget, then left-click in the design display area and drag the mouse to add a ruler (in micrometers), and left-click again to end the ruler. To move the ruler, left-click and drag it to a new location. Left-click the ruler again to keep it in the new location. To remove a ruler, left-click on it and press *Esc*. To remove all rulers, press K (Shift-k). You can set the direction in which rulers can be drawn by clicking the widget, then pressing the F3 key to display the Set Ruler Mode form. You can choose to draw orthogonal, vertical, horizontal, or diagonal rulers. You can also snap rulers to floorplan objects.
	*Zoom In*--Click this widget to display a smaller area of the design in greater detail. Each click zooms in one level. The equivalent bindkey is z.
	*Zoom Out*--Click this widget to display a larger area of the design in less detail. Each click zooms out one level.
	*Fit*--Click this widget to display the entire placed design within the design display area. The equivalent bindkey is f.
	*Zoom Selected*--Select one or more design objects, then click this widget to zoom into the area that contains the selected objects.
	Click this widget to toggle the display between the previous location or zoom level and the current location or zoom level.
	*Redraw*--Click this widget to update the display with new colors or stipple patterns that you selected from the More Colors form.
	Opens the Attribute Editor.
	*Clear All Ruler*--Clears all rulers from the display area.

Find/Select Object

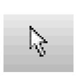
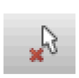
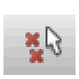
Use the Find/Select Object form to find and select specific object types in the design display area.

- Choose the *Find/Select Object* widget in the Layout Window.



Fields and Options

Object Type	Specifies the type of object to find. Choose from: *Instance*, *Pin, Net, Module, Instance Group*, or *Bump*. The Pin object type pin can be used for I/O pins, partition pins, and instance pins. Names of instance pins and/or partition pins should be specified with the full path name. For example, pin A of instance DTMF_CHIP/alu should be specified as `DTMF_CHIP/alu/A`. You can use the *Net* option to find/display bus names.
Mode	Specifies the type of find/select operation. Select one of the following: ■ *Single*: Performs a simple search for a Property/Value combination. ■ *And List*: Performs a find/select operation that matches an AND list of two or more multiple Property/Value combinations. ■ *OR List*: Performs a find/select operation that matches an OR list of two or more multiple Property/Value combinations. ■ *Multiple*: Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND lists and OR lists.
Property	Displays the available properties for the object that you selected in the Object Type field. You can use the pre-defined property Connection to find/select objects by connectivity. The Connection property supports connectivity search for the following objects: ■ Instance ■ Pin ■ Net ■ Module ■ Selected objects
NOT	Inverses the value of the operator in the Operator field. For example, If the value in the Operator field is <, clicking the *NOT* button effectively changes the value to NOT <.

Operator	Specifies the relationship between the property and its value. This field is dynamically populated based on the specified object type.
Value	Specifies the value of the property. This field is dynamically populated based on the specified object type. In some cases, you may have to manually specify a value, for example, the name of a cell.
Select Also	Finds all objects that meet the specified criteria, reports them in the *Results* window, and selects them in the design display area. *Default:* The objects are reported in the results window but are not selected in the design display area.
Clear Result Window	Clears objects previously listed in the *Results* list, then finds the objects that meet the specified criteria and reports them in the Results list. *Default:* The objects previously listed are retained in the results window along with the results of the current search.
All	Clears all objects previously listed in the Results list.
Selected	Clears those objects previously listed in the Results list that are currently selected in the *Results* list.
Find	Finds all objects that meet the specified criteria and reports them in the *Results* window.
 *Zoom in Highlighted*	Zooms in the highlighted object(s). You can highlight object(s) by selecting them in the results window and clicking the *Highlight* button
Select 	When you click *Select*, the objects selected in the Results window are selected in the design display area.
 *ll*	When you click *Select All*, all the objects in the Results window are selected in the design display area.
Deselect 	When you click *Deselect*, the objects selected in the Results window are deselected in the design display area.
Deselect All 	When you click *Deselect All*, all the objects in the Results window are deselected in the design display area.

Highlight 	When you click *Highlight*, the objects selected in the Results window are highlighted in the design display area.
De highlight 	When you click *Dehighlight*, the objects selected in the Results window are dehighlighted in the design display area.

Set Preferences

To set preferences, right-click and select *Set Preferences* option.

Use the *Set Preference* command in the *View* menu to access the Preferences form. This form allows you to set the design settings for the Tempus session.

You can save custom preference settings to a file, or load a previously saved file or default settings. The default settings are saved in the enc.pref.tcl file; custom settings can be saved with a different filename.

The Preferences form also allows you to create, edit, apply, and save the design keyboard shortcut commands.

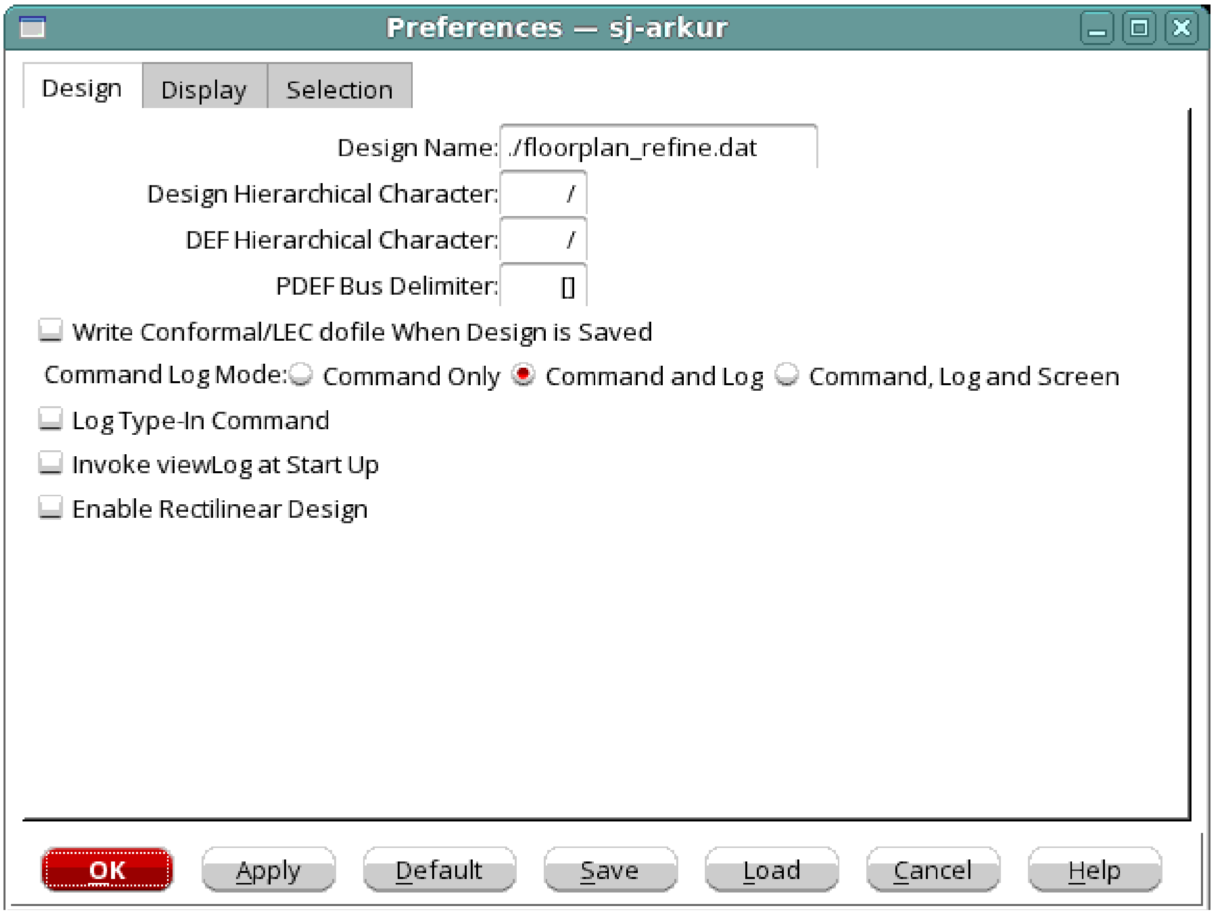
The Preferences form contains the following pages:

- Design Tab
- Preferences - Display
- Preferences - Selection

Design Tab

Use the *Design* page of the Preferences form to set hierarchy levels in the netlist and DEF files, specify bus delimiters for PDEF files, denote output of command logs, and create binding key shortcuts. The default settings are saved in the enc.pref.tcl file; custom settings can be saved with a different filename.

- Choose *View - Set Preference* and click on the *Design* tab.



Preferences - Design Fields and Options

Design Name	Specifies the name of the design for which to set the preferences.
Design Hierarchical Character	
	Denotes the character being used as the hierarchy delimiter in a netlist. If the netlist uses another character, enter it in the text box and import the design. *Default:* `/`

<table>
<tr><td>DEF Hierarchical Character</td><td colspan="2">Denotes the character being used as the hierarchy delimiter in a DEF file. If the DEF file uses another character, enter it in the text box and load the DEF file. When writing a DEF file, the software uses the same character.

Default: <code>/</code></td></tr>
<tr><td>PDEF Bus Delimiter</td><td colspan="2">Denotes the character being used as the hierarchy delimiter in a PDEF file. If the PDEF file uses another character, enter it in the text box and load the PDEF file. When writing a PDEF file, the software uses the same character.

Default: <code>[]</code></td></tr>
<tr><td colspan="3">Write Conformal/LEC dofile When Design is Saved</td></tr>
<tr><td></td><td colspan="2">When saving a netlist, this option saves a dofile (a set of commands contained in a specified file) for the Conformal® Equivalence Checking capabilities.

Default: <code>Off</code></td></tr>
<tr><td>Command Log Mode</td><td colspan="2">Specifies where the command log is written.

Default: Command and Log</td></tr>
<tr><td></td><td>Command Only</td><td>Writes to the command file only.</td></tr>
<tr><td></td><td>Command and Log</td><td>Writes to the command file and log files.</td></tr>
<tr><td></td><td>Command, Log, and Screen</td><td>Writes to the command file, log files, and screen.</td></tr>
<tr><td></td><td colspan="2">Note: When writing to log file and screen, the commands are preceded with <CMD>.</td></tr>
<tr><td>Log Type-In Command</td><td colspan="2">Logs whatever is manually entered (typed), to the command file.

In the case of source commands, only the source command is logged, and not the commands in the sourced file.

Default: <code>Off</code></td></tr>
<tr><td colspan="3">Invoke viewLog at Start Up</td></tr>
</table>

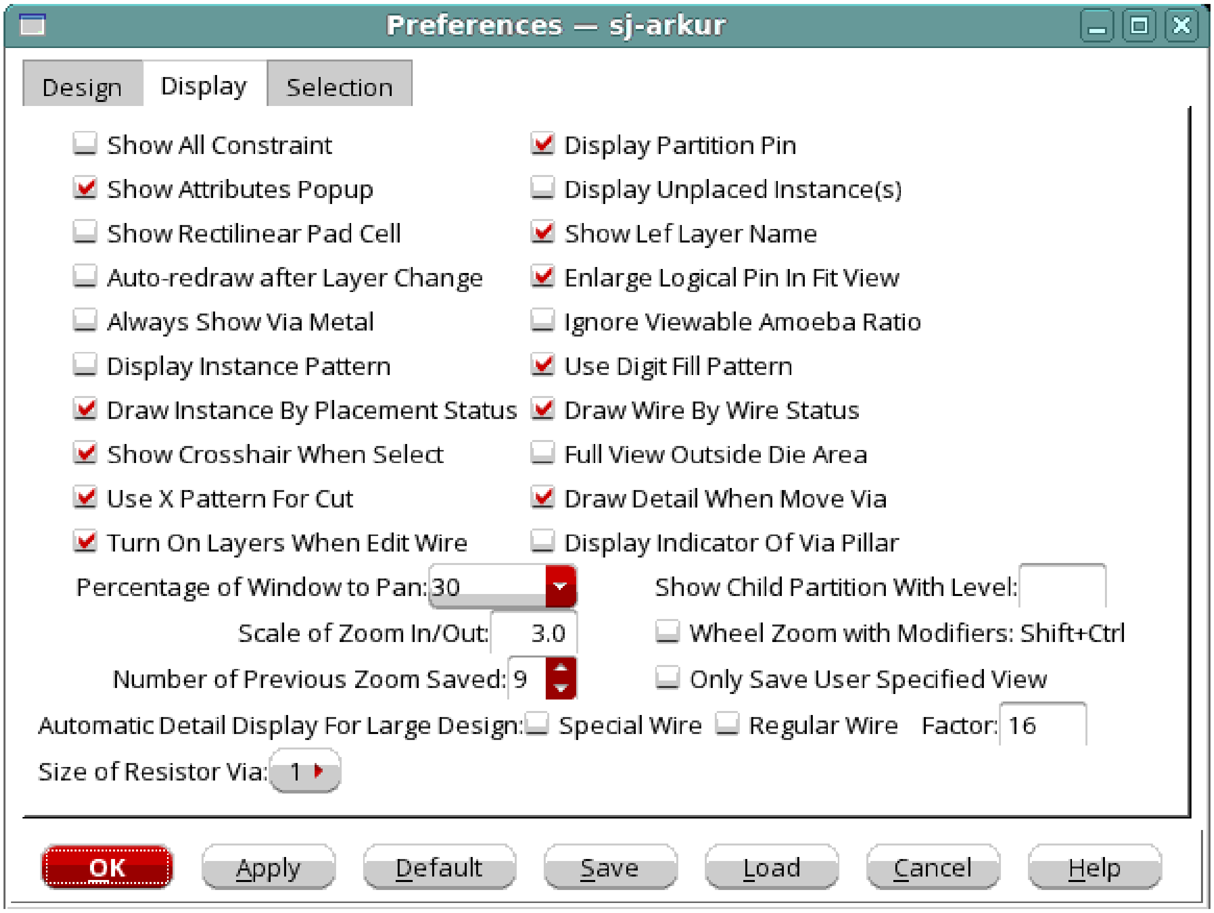
	Displays log viewer at startup. If you select this check box and click Save in the Preferences form, setPreference logviewer 1 is written to the user specified preference file. *Default:* `Off`
Enable Rectilinear Design	
	Enables you to set a rectilinear object or cut a rectilinear area in a design with IO cells.

Preferences - Display

Use the *Display* page of the *Preferences* form to set viewing preferences for standard cells, wires, route congestion ranges, modules, and objects. The default settings are saved in the enc.pref.tcl file; custom settings can be saved with a different file name.

- Choose *View - Set Preference* and click the *Display* tab.

To change display settings, click the *Display* tab:



Preferences - Display Fields and Options

Show Floorplan Objects in Physical View	Enables the display and querying of floorplan objects in the physical view. You can use this feature together with color groups to create a custom view. For example, you can create a color group that has a custom setting for block colors and use this color group while viewing floorplan objects in the physical view. For information on using color groups, see "*Creating and Editing Color Groups*" section. *Default:* `Off`

Show All Constraint	Displays all constraints, such as fences, guides, regions, and soft guides, regardless of the current hierarchy. When this option is selected, you can see the child guides of each fence as you shape the fence. *Default:* `Off`
Display Partition Pin	Displays partition pins. *Default:* `On`
Show Attributes Popup	Specifies that the Context Pop-up Attribute Viewer, which is displayed when you place the mouse cursor over an object, should be displayed.
Display Unplaced Instance(s)	Displays unplaced cells in the Physical view. *Default:* `Off`
Show Rectilinear Pad Cell	Displays bounding box if the pad cell has rectilinear shape. *Default:* `Off`
Show Lef Layer Name	Specifies that the layer name as specified in the LEF file should be displayed in the color bar. *Default:* `On`
Auto-redraw after Layer Change	
	Enables or disables auto-redraw when layers are changed. *Default:* `On`
Enlarge Logical Pin In Fit View	Displays larger symbols for logical pins in Fit view. Select this option to check the pin distribution in a block design. *Default:* `On`

Always Show Via Metal	Keeps the additional metal of a via visible even if you turn off the via cut layer. With the *Always Show Via Metal* option selected: If you turn off the metal layer and keep via on, only the cut via is visible in the main window. If you turn off the via and keep metal layer on, the top/bottom layer of the via is visible. This option makes it easy for you to view the top/bottom layer of the via by turning off the via. If this option is selected, you can select the via metal and view its attributes in Attribute Editor. *Default:* `Off`
Ignore Viewable Amoeba Ratio	Displays all modules in the *Amoeba View* when selected, irrespective of their display size. By default, if the display size of a module is smaller than 1/3rd the area of the current module or 1/20th of the fplan box, it is not displayed in the *Amoeba View*. *Default:* `Off`
Draw Instance Pattern	Marks overlapping instances with X in the display area. This makes it easier to identify overlapping instances as compared to the default fill pattern. *Default:* `Off`
Draw Instance by Placement Status	Controls thickness of instance borders on the basis of their placement status. Select the new *Draw Instance by Placement Status* option if you want fixed instances to be marked with a thick border. *Default:* `On`
Instance Display Threshold	Sets the draw size of the standard cell instance in the physical view. The default value of 0.0 draws all standard cells. A value of 2.0 draws fewer standard cells, thereby yielding faster draw speed. If you change this value, you must redraw the view by choosing *View - Redraw* from the menu.

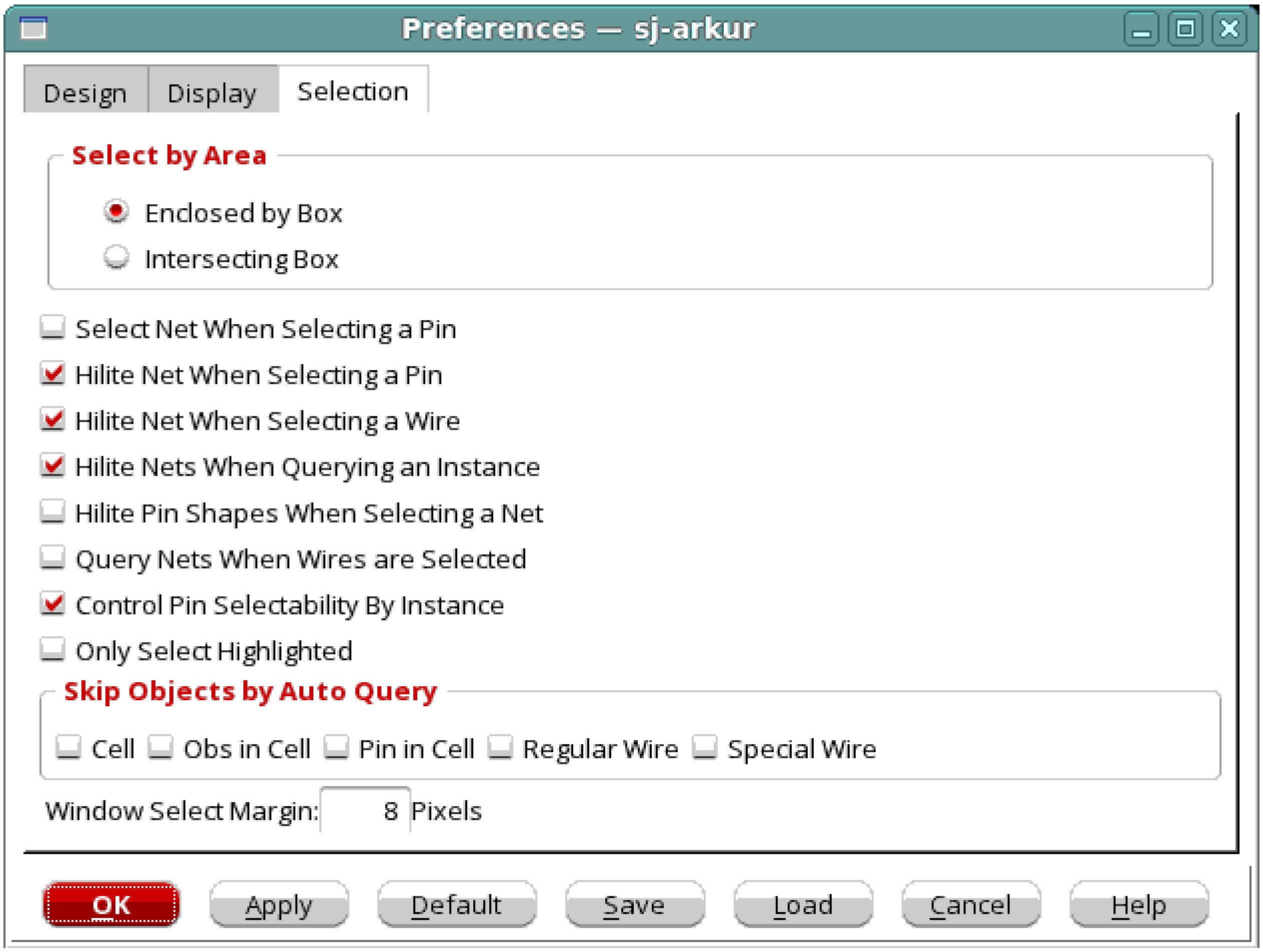
Wire Display Threshold	Sets the minimum number of pixels used to display a wire or via. This value works in conjunction with the *Apply Threshold in 2D* option. At the chip level, set 1 for the *Wire Display Threshold* and turn off *Apply Threshold in 2D*to display all of the wires/vias in the design. If the display speed is too slow, then change either or both variables appropriately.
Apply Threshold in 2D	Checks both width and height of a wire or via when used with *Wire Display Threshold*. When off, the wire or via is displayed if either its width or height meets the threshold. When on, both width or height must meet the threshold for the wire or via to be displayed. *Default:* `Off`
Percentage of Window to Pan	Specifies the percentage of display area that will be panned when you pan the display using the right or the left arrow keys. *Default:* The default value is 95%.
Scale of Zoom In/ Out	Specifies the scale for the zoom in and zoom out feature. You can use a decimal number for the scale.
Wheel Zoom with Modifiers: Shift+Ctrl	Enables use of scroll wheel to zoom in and out, when used along with the `Shift` and `Ctrl` keys.
Number of Previous Zoom Saved	Specifies the number of zoom views that are saved. You can then view the previous zoom views by clicking the Zoom Previous icon or the bindkey W. For example, if you set the number to 4, you can view four previous zoom views by clicking the Zoom Previous icon or the bindkey W. *Default:* `1`
Min. Floorplan Module Size	Filters out modules from the display in the Floorplan view that have an area less than the specified value. For example, a value of 100 displays all the modules that contain 100 or more instances. **Note:** If modules or submodules contain at least one block, they are always displayed, regardless of the specified value. *Default:* `100`
Show Parent Module With Level	Displays regions in hierarchy from the specified level. *Default:* `0`

Show Child Partition With Level	Displays child partitions from the specified level. *Default:* `0`	
Automatic Detail Display for Large Design	Controls the performance of redraw in large designs. If you select *Special Wire* and/or *Regular Wire*, the tool only draws these wires when you zoom in to the specified level (*Factor*). By default, the wires are drawn in full view (no zoom) itself.	
	Special Wire	Draws special wires only when you zoom in to the specified level. *Default:* `Off`
	Regular Wire	Draws regular wires only when you zoom in to the specified level. *Default:* `Off`
	Factor	Specifies the zoom level at which wires will be drawn in large designs. *Default:* `16`

Preferences - Selection

Use the *Selection* page of the Preferences form to select and display objects and determine their behavior within a selected area. The default settings are saved in the enc.pref.tcl file; custom settings can be saved with a different file name.

- Choose *View - Set Preference* and click the *Selection* tab.



Preferences - Selection Fields and Options

Select by Area		
	Enclosed by Box	Selects objects that are fully enclosed inside the specified area. Use the left mouse button to draw and define an area or bounding box. Any objects fully enclosed inside the drawing area are selected. *Default:* `On`

	Intersecting Box	Selects objects that are fully enclosed or intersected within the specified area. Use the left mouse button to draw and define an area or bounding box. Any objects enclosed inside or intersected with the area bounding box are selected. This mode should be used to select wire or row. *Default:* `Off`
Select by Line	Specifies that objects can be selected by drawing a line that crosses the objects. When you select this option and then choose the *Select* widget from the main menu, you can draw a line across several objects in the GUI; all the objects that the line crosses are selected. Alternatively, you can use the `Ctrl + M` key combination to toggle between window and line selection modes quickly: The default selection mode is window. Press the `Ctrl + M` key combination to switch to the line selection mode. After making line selections, press the `Ctrl + M` key combination again to switch back to the window selection mode. *Default:* `Off`	
Sticky Mode	Reverse the object selection behavior. When you select the *Sticky Mode* option, the selection behavior changes as follows: ■ Single click: adds current object to selection. ■ `CTRL` + single click: selects new object (and clears previous selection). This feature is useful when, for example, you want to select multiple objects without having to press the CTRL key while clicking.	
Hilite Net When Selecting a Pin		
	Highlights a net through a pin selection. If this option is set when you select a pin, the corresponding net is highlighted. *Default:* `Off`	
Hilite Nets When Querying an Instance		

	Highlights nets that are connected to an instance when this instance is queried. For example, if you query a selected instance and this option is set, all nets that connect to this instance pin are also highlighted. *Default:* `On`
Hilite Pin Shapes When Selecting a Net	
	Highlights connected cell term shapes when you select a net. *Default:* `Off`
Query Nets When Wires are Selected	
	Queries all nets that connect to the selected wires. *Default:* `Off`
Skip Objects by Auto Query	
	Skips querying of specified objects. You can choose to skip Auto Query of: ■ *Cell* ■ *Obs in Cell* ■ *Pin in Cell* ■ *Regular Wire* ■ *Special Wire*
Window Select Margin	Specifies the number of pixels for the window margin. *Default:* 8
Number of Highlight Color	
	Specifies the number of highlight colors. The available values are 8, 12, and 16. Use highlight colors to highlight sets of objects, each set in a different color. You can select the highlight color through the Edit Highlight Color form (which can be accessed using the *View > Edit Highlight Color* menu item) or directly from the *Highlight* widget in the tool bar. *Default:* `8`

Multicolor Layers

During a design session, the following object types do not have a single color, but each object has a color that shows a level of severity and magnitude:

- *Horizontal Congest*
- *Vertical Congest*
- *IR Drop or EM*
- *Clock Tree*

To see the level indicator colors, double-click the color button for the object type. This opens a color preferences form for that object type.

Metal Layers and Other Layered Components

- Click the *Wire/Via* tab to see the metal layers available in the design.

The *M0 Wire* tab represents the masterslice layer, *M1 Wire* represents the bottom metal layer, and so on. The vias are also shown, with *V01* representing the via between the masterslice layer and the first metal layer, *V12* representing the via between the first and second metal layers, and so on.

You can set preferences for other layered components in addition to wires and vias. For each metal layer, you can set the color, stipple pattern, visibility, and selectability for pins, tracks, blockages, and violations.

Note: Violations are not selectable.

Creating and Editing Color Groups

You can create color groups to customize the color settings in Tempus. A color group can contain custom color settings for various objects. For example, you can set the blockage color to red in one color group and to blue in another color group. You can switch between color views during an Innovus session. The settings specified in a color group are applied in all views.

To create a Color Group, do the following:

- In the Layout window, click the *All Colors* button. The *Color Preferences* form appears.
- Create the custom color settings that you want for the group. For example, you might want to set the block color, or the visibility of blockages.

- Click the *Customize Tab* button. The *Customize Tab* form appears.
- Click the *Color Group* button. The *Edit Color Group* dialog box appears.

- Click the *Add* button. The Add Color Group form appears.

- In the *Group Name* field, enter the name of the new color group, and click *OK*. The new color group is created.

To specify or change the color settings for the Color Group, do the following:

- In the Layout window, click the *All Colors* button. The *Color Preferences* form appears.
- Create the custom color settings that you want for the group. For example, you might want to change the instance color, or the visibility of blackboxes.
- Click the *Customize Tab* button. The Customize Tab form appears.
- Click the *Color Group* button. The Edit Color Group dialog box appears.
- Select the name of the *Color Group* to which the changed color settings should be applied.
- Click the *Set* button. The settings are applied to the selected Color Group.

Once you create one or more color groups, the color setting for a particular group can be applied be selecting the group name from a drop-down menu next to the *All Colors* button in the main form. You can thus select the color group from the main window.

Color Bar

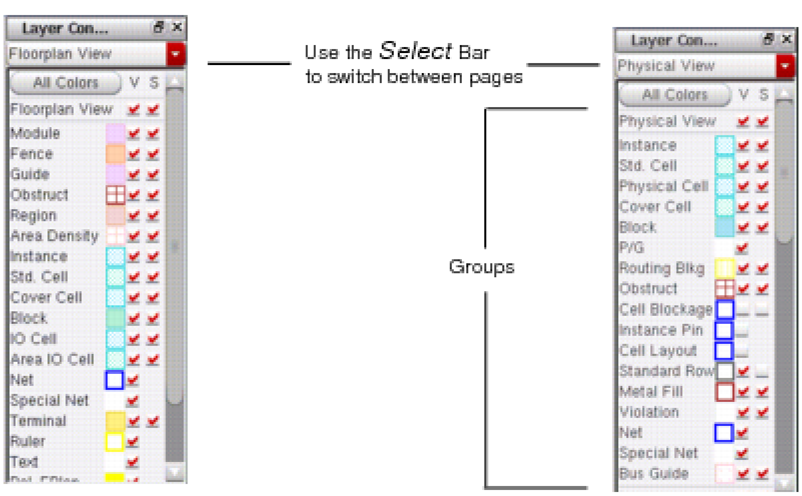
The Layout window includes an Object Color Bar that has color display buttons for objects in the design display area. Each object type has a check box for displaying the object in the design display area and a color button that displays the current color used to identify the object in the design display area.

By default, the Object Color Bar has two pages, but you can create up to four pages. Each page in the color bar can have one or more group. A group is a collection of objects that are grouped together in the Object Color Bar. This way you can view the color and visibility controls for related objects together. The default first page in the Object Color Bar displayed in the Layout window contains the group FPlan View. The default second page contains two groups: Physical View and Layers.

You can use the Customize Tab form to add pages to the Object Color Bar, add or delete the groups, and to add objects in each group. The Customize Tab form is available on all pages of the Color Preferences form.

Note: You can right-click on an individual layer to toggle the visibility/selectability of all objects on the layer.

To switch between the pages, click the *Select* bar.



Page 2 of Color Bar

If you select the *V* (Visibility) check box for an object type, the object type displays in the design display area in the specified color. If you clear the check box, the object type does not display in the design display area.

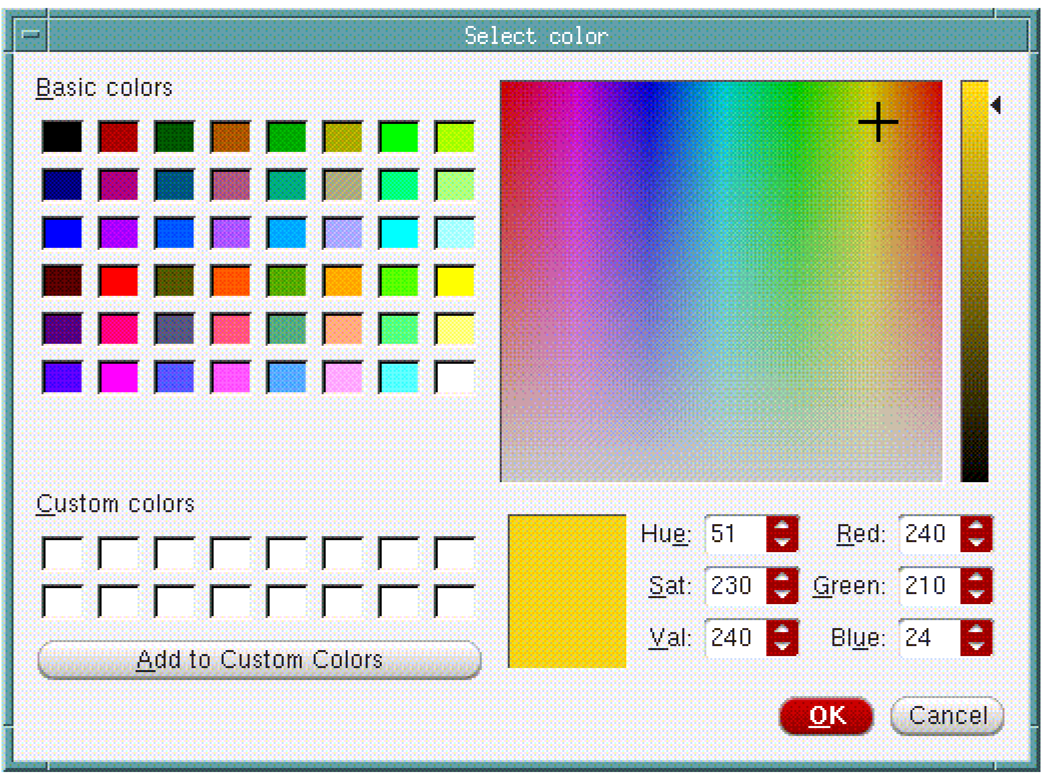
If you select the *S* (Selectability) check box for an object type, the object type can be selected in the design display area when you use the *Selection* icon. If you clear the check box, the object type cannot be selected.

Note: Objects that are not visible are not selectable.

Object Color

Use the Select color form to customize the display color of an object.

Click on the color indicator of the object type to open the Select color form.



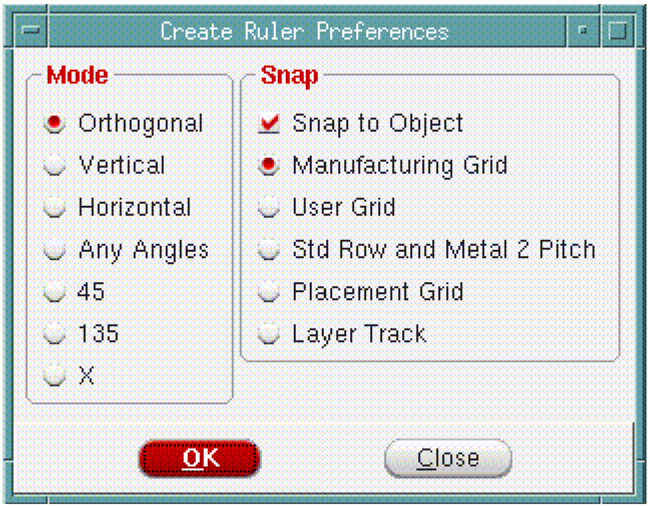
If you use the text fields, you must press the Return key after your text entry.

Create Ruler Preferences

Use the Create Ruler Preferences form to set the direction in which ruler lines are drawn.

To open the Create Ruler Preferences form:

- Click the Ruler widget  and press the F3 key.



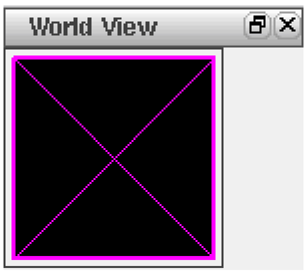
Fields and Options

Snap to Object	Specifies that the ruler should snap to the nearest floorplan object.
Manufacturing Grid	Specifies that the ruler should snap to the manufacturing grid.
User Grid	Specifies that the ruler should snap to the user specified grid.
Std Row and Metal 2 Pitch	
Placement Grid	Specifies that the ruler should snap to the placement grid.
Layer Track	
Orthogonal	Specifies that ruler lines can be drawn in horizontal as well as vertical directions.
Vertical	Specifies that ruler lines will be drawn in the vertical direction.
Horizontal	Specifies that ruler lines will be drawn in the horizontal direction.
X	Specifies that ruler lines can be drawn at a 45 degree angle as well as at a 135 degree angle.
45	Specifies that ruler lines will be drawn at a 45 degree angle.
135	Specifies that ruler lines will be drawn at a 135 degree angle.

Any Angle	Specifies that ruler lines can be drawn in any direction.

Satellite Window

The main window includes a satellite window, which identifies the location of the current view in the design display area, relative to the entire design. The chip area is identified by a yellow box, the satellite view is identified by the pink crossbox. When you display an entire chip in the design display area, the satellite crossbox encompasses the chip area yellow box.



When you zoom and pan through the chip in the design display area, the satellite crossbox identifies where you are relative to the entire chip.

To move to an area in the design display area, click and drag on the satellite crossbox.

To select a new area in the design display area, click and drag on the satellite crossbox.

To resize an area in the satellite window, click with the Shift key and drag a corner of the crossbox.

To define a chip area in the satellite window, right-click and drag on an area.

Auto Query

The Layout window includes a *Q* icon located at the bottom of the design display area that enables and disables the auto query. When you enable auto query, text information displays for any object that is under the cursor. You can use auto query for blockages, hand-drawn or inside cell instances, instance pins, special route wires and vias, and cell instances.

If there are overlapping objects under the cursor, text information displays for the object that is on top. Use the n key to get information on the next object, and the p key to get information on the previous object. You can also use the F8 binding key to copy auto query text from the message bar in the Layout window to the software console. The text is displayed in the software console with the prefix Message.

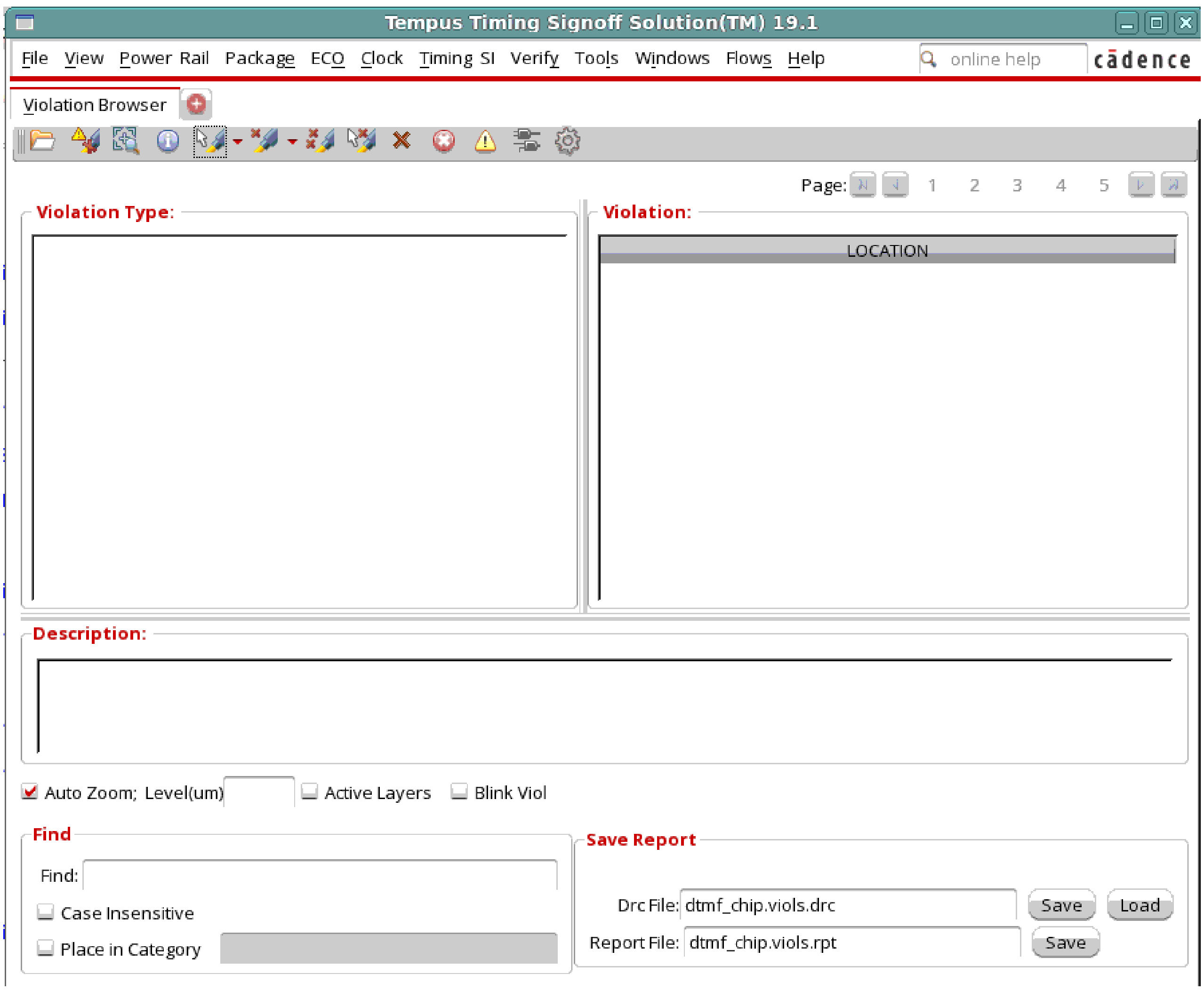
Violation Browser

Use the Violation Browser to display violations and lithography hotspots, mark violation markers as false or true, delete violation markers, and specify a report file.

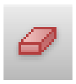
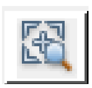
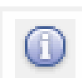
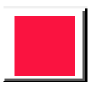
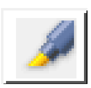
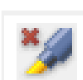
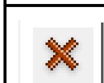
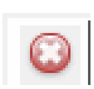
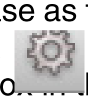
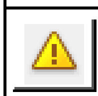
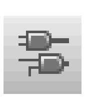
The Violation Browser also provides access to the following forms:

- Violation Browser Settings on page 610
- Load Violation Report on page 614

To open Violation Browser, choose *Tools - Violation Browser*.



Violation Browser Fields and Options

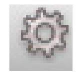
	*Load Violation Report* - Opens the Load Violation Report form. Use this form to load a violation marker file and then view the markers in the Violation Browser.
	*Clear Violation* - Clears all the violation markers in your design.
	*Fit Violation* - Displays the violation within the design display area.
	*Attribute Editor* - Opens the *Attribute Editor* and populates it with the highlighted violations. Double click on a violation to update fields in the Attribute Editor.
	*Highlight Color* - Select this button to specify the color for highlighting violations.
	*Highlight Violations* - Highlights the selected violation(s) in the design display area as well as in the *Violation* list on the right pane of the Violation Browser. You can select an individual violation in the right pane of the Violation Browser or select violations by type or subtype in the left pane of the browser. To highlight a selected violation or a selected type/subtype of violations, select the highlight color from the color palette and then click the *Highlight Violations* button.
	*Dehighlight Violations* - Removes the highlight from the selected violation or selected type/subtype of violations in the design display area.
	*Delete Violations* - Deletes all selected violations from the design database. Markers are also removed from the design display area and the list of violations.
	Mark Violations as False - Marks the selected violation as a false violation. These violations are stored in the database as false violations. To view the violations that you mark as false, click  in the Violation Browser, select the View False Violations check bo[illegible]he Violation Browser Settings form, and click OK.
	*Mark Violations as True* -- Marks the selected violation as a true violation.
	*Show MSV Violation Schematic* – Displays schematics associated with the highlighted violation in the Schematic Viewer form.

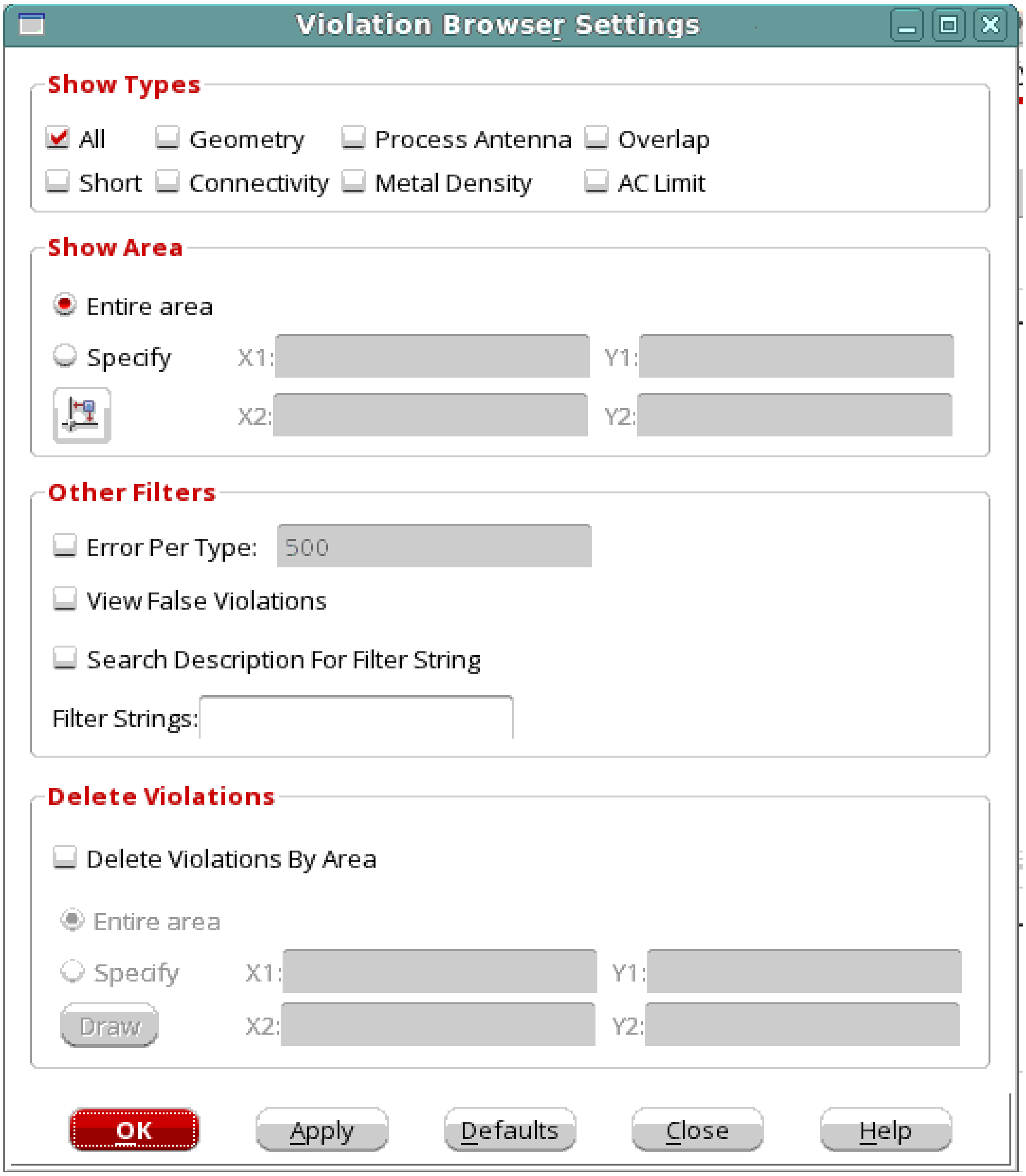
	*Setting* - Opens the Violation Browser Settings form.
Violation Type	Lists the violation types in a hierarchy window. To show or hide a violation, select it in the *Violation* or *Violation Type* window and click the right mouse button. Select *Show* or *Hide* from the pop-up window. The numbers in parentheses after each violation type represent the number of violations of that type displayed in the main window and the total number of violations of that type. The count includes true and false violations. For example, if *Verify - Connectivity - Open (479/503)* means that there are 503 Open violations and 479 of those violations are shown the main window.
Violation	For the highlighted *Violation Type*, displays the names of the objects with violation markers and their locations.
Description	Displays the layer and bounding box for the violation. Layer is not included for placement violations. You can select text in the *Description* text box and copy and paste it to another window. You cannot edit text in the *Description* text box.
Auto Zoom	Automatically zooms in to the selected violations as you browse through the violations. This check box is selected by default.
Zoom Level (um)	Controls the level of zoom for Auto Zoom. When you specify the zoom level, Violation Browser uses the same value for all listed violations so that you do not have to zoom out each time you select a violation.
Active Layers	Keeps only the violation layer(s) visible when you zoom into a violation. If this check box is not selected, the Auto Zoom mode displays layers as per the layer visibility settings on the Layer Control bar.
Only Show Selected	Displays markers for only selected violations. This option is useful if there are many violations in the design. By selecting this option, you can turn off other violation makers in the main window and focus on only the violations you have selected in the browser.
Blink Viol	Makes the marker for the selected violation blink in the main window.

Find	Searches for the specified text and highlights it in the *Violation* window. This field recognizes pattern matching, so you can type `*153*` to search for all objects with the string `153` as part of their *LOCATION*. Press *Enter* to find the specified text.
Case Insensitive	Specifies whether the text in the *Find* text entry box is case sensitive.
Place in Category	Creates a category in the *Violation Type* window and adds the highlighted object to the category. Categories are not saved.
Drc File - Save	Opens a form that allows you to specify the path and name for saving the DRC file.
Drc File - Load	Opens a form in which you can browse and select the DRC file to be loaded.
Report File Save	Specifies the report file that contains all the violation information. The browser and the report file list the violations in the same order.

Violation Browser Settings

Use the Violation Browser Settings form to specify the settings for displaying violations and lithography hotspots.

- Choose *Tools - Violation Browser* and click the  button.



Violation Browser - Settings Fields and Options

Show Types	Lists the types of violations to display or hide in the Violation Browser. Select one or more of the following options:	
	All	Displays all violations.

	Geometry	Displays geometry violations.
	Process Antenna	Displays process antenna violations.
	Overlap	Displays overlap violations.
	Short	Displays short violations.
	Connectivity	Displays connectivity violations.
	Metal Density	Displays metal density violations.
	AC Limit	Displays AC limit violations.
Show Area	Displays violations in the entire area or a specified portion of the area. Specify the portion of the area by using one of the following methods: Using the *Draw* button and your mouse to create a rectangular area in the design. The software displays the coordinates in the *X1*, *Y1*,*X2*, and *Y2* fields. Manually enter the coordinates in the *X1*, *Y1*, *X2*, and *Y2* fields.	
Other Filters	Specify additional conditions by selecting one or more of the following options:	
	Error Per Type	Specifies a limit for the number of violations parsed by the browser for each violation type. If this check box is not selected, the Violation Browser parses and displays all violations.
	View False Violations	
		Displays violations that you have marked as false.
	Search Description For Filter String	
		Searches for text in the violation message (in the bottom portion of the Violation Browser window) and the violation description (in the top portion of the Violation Browser window).
	Filter Strings	

<table>
<tr><td></td><td></td><td>Specifies a text string for filtering violations. When you specify a text string, the Violation Browser searches the message of each violation and displays only the violations whose messages match the search conditions.

To specify a list of strings, separate each string with a space.

To specify a literal string with space in it, enclose the string in double quotation marks. For example, you can specify strings such as the following:

<code>M5 (600, 500)</code>
<code>M3 (700, 400)</code>
<code>"Pin of Net"</code>

You can specify the AND, OR, and NOT conditions with the search strings. Use:

<code>&&</code> between strings to specify the AND condition.

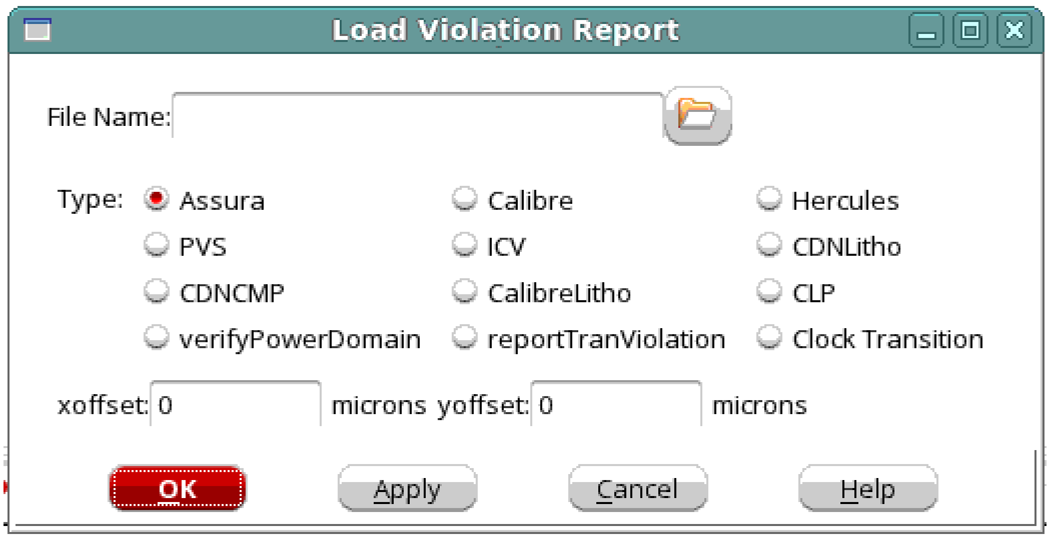
<code>||</code> between strings to specify the OR condition.

! before a string to specify the NOT condition.</td></tr>
<tr><td>Delete Violations</td><td colspan="2">Enables you to delete all violations or delete violations in the specified area only.</td></tr>
<tr><td colspan="3">Delete Violations By Area</td></tr>
<tr><td></td><td colspan="2">Enables you to delete violations in the area you specify on this form or the area you select with the mouse.</td></tr>
<tr><td></td><td>Entire area</td><td>Deletes the types of violations specified in the entire core design area.</td></tr>
<tr><td></td><td>Specify</td><td>Specifies an area from which to delete violations. Only the types of violations specified by this form are deleted.</td></tr>
<tr><td></td><td>Draw ...</td><td>Lets you use the mouse to outline an area from which to delete violations.</td></tr>
</table>

Load Violation Report

Use the Load Violation Report form to load a violation marker file or a hotspot interface format (HIF) file and create markers that the Tempus software can interpret. After you load the file, you can view the markers in the Violation Browser. The Auto Query feature displays the rule names for the markers.

- Choose *Tools - Violation Browser* and click the *Load Violation Report* button.



Load Violation Report Fields and Options

File Name	Specifies the marker file to load.

Type	Specifies the type of marker file. The following file types contain violation markers: ■ *Assura* - For Assura violations ■ *Calibre* - For Calibre violations ■ *Hercules* - For Hercules violations ■ *ICV* - For ICValidator violations ■ *PVS* - For PVS violations ■ *CDNCMP* - For Cadence CMP Predictor (CCP) violations ■ *CLP*, *verifyPowerDomain*, and *Clock Transition* - For low power violations ■ *reportTranViolation* - For DRV transition violations The following types contain hotspot markers, which are used to identify areas susceptible to lithography problems: ■ *CDNLitho* (for LPA and NanoRoute litho HIF files) ■ *CalibreLitho*
xoffset	Specifies the X offset. When you specify the X and Y offsets, Violation Browser displays the violations at the correct coordinates after accounting for the offsets.
yoffset	Specifies the Y offset, which Violation Browser takes into account to displays violations at the correct coordinates.

Clear Violation

Use the *Clear Violation* button in the Violation Browser to clear all the violation markers in your design.

- Choose *Tools - Violation Browser* and click the *Clear Violation* button.